FRM Part II 2026

Liquidity and Treasury Risk Measurement and Management

Gold-source question bank

597 questions, sorted by chapter and learning objective

Scope and LO wording: SchweserNotes 2026 (final authority). Every question solved and independently verified.

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Reading 66. Liquidity Risk

John C. Hull, Risk Management and Financial Institutions, 5th Edition (Hoboken, NJ: John Wiley & Sons, 2018). Chapter 24.

LO	Learning objective	Qs
66.a	explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR).	29
66.b	identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft.	4
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LO 66.a

Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR).

Q1

BT answers P2.T8.20.2.3 · BT notes P2.T8.20.2.3

REBUILT — OCR-damaged numbers restored from the solution arithmetic and the two duplicates merged; the original options were unevenly spaced (a 350k result would be ambiguous) and the source remark about a 400k 'exact' answer was unsupported, so options were rebuilt and z = 1.645 made explicit.

A position worth $1.0 million consists of 50,000 shares of an asset quoted bid $19.00 / offer $21.00. The asset has an expected return of 14.0% per annum and an annual volatility of 26.0%. Geometric returns are assumed normally distributed (lognormal VaR). Use a one-tail z-score of 1.645 for 95% confidence and measure liquidity cost as half of the proportional bid-offer spread times the position value. Which is NEAREST to the one-year liquidity-adjusted lognormal VaR (LVaR)?

A. $250,000
B. $300,000
C. $350,000
D. $420,000

Answer: B

Step by step

1. Lognormal VaR $= V\left[1 - e^{\mu - z\sigma}\right] = 1{,}000{,}000\left[1 - e^{0.14 - 1.645(0.26)}\right] = 1{,}000{,}000\left[1 - e^{-0.2877}\right] \approx 250{,}013$.
2. Mid-price $= (19 + 21)/2 = 20$; proportional spread $s = 2/20 = 0.10$.
3. Cost of liquidation $= sV/2 = 0.10 \times 1{,}000{,}000/2 = 50{,}000$.
4. LVaR $= 250{,}013 + 50{,}000 = 300{,}013 \approx \$300{,}000$.

Why B is correct: LVaR is market-risk VaR plus the cost of unwinding at the bid-offer spread; with a constant spread the cost is half the proportional spread times position value.

Why the others are wrong

- **A:** This is the lognormal VaR alone; the liquidation cost of $50,000 was left out.
- **C:** Charged the full spread (0.10 x $1.0m = $100,000) instead of half, giving 250,013 + 100,000 = 350,013.
- **D:** Used a 99% z-score (2.326), giving VaR ≈ 371,714 plus $50,000 = about $421,700, which does not match the stated 95% confidence.

Q2

AP Mock 1 Q43

REBUILT — Distractors were not traceable to realistic errors; replaced them with omitted-leg and full-spread errors. Original answer (11.75m) was correct.

A company holds the following assets:

Position	Quantity	Bid	Offer
Shares of company A	1,000,000	USD 95.00	USD 96.00
Ounces of a commodity	10,000,000	USD 50.50	USD 51.50
Shares of company Y	10,000,000	USD 49.25	USD 50.50

What is the cost of liquidation in a normal market?

A. USD 5.50 million
B. USD 6.75 million
C. USD 11.75 million
D. USD 23.50 million

Answer: C

Step by step

1. Cost per position $= \frac{1}{2}(\text{offer} - \text{bid}) \times \text{quantity}$ (equivalent to $s\alpha/2$).
2. Shares A: $0.5 \times 1.00 \times 1\text{m} = 0.50\text{m}$.
3. Commodity: $0.5 \times 1.00 \times 10\text{m} = 5.00\text{m}$.
4. Shares Y: $0.5 \times 1.25 \times 10\text{m} = 6.25\text{m}$.
5. Total $= 0.50 + 5.00 + 6.25 = 11.75$ million.

Why C is correct: In a normal market the liquidation cost is half the bid-offer spread on each position, summed across positions.

Why the others are wrong

- **A:** Omitted shares of company Y (0.50 + 5.00).
- **B:** Omitted the commodity (0.50 + 6.25).
- **D:** Used the full spread instead of half (2 x 11.75).

Q3

AP Mock 2 Q2

REBUILT — Distractors were not traceable to errors; replaced with realistic mistakes and stated the perfect-correlation assumption. Original answer (42.93m) was correct.

A bank holds shares and a commodity with mid-market positions of USD 1,235 million and USD 627 million respectively.

Position	Mid-market value (USD m)	Bid-offer spread (USD)	Proportional bid-offer spread
Shares	1,235	mean 1.20, sd 1.40	mean 0.02346, sd 0.01568
Commodity	627	mean 0.14, sd 0.14	mean 0.005678, sd 0.005678

Assume the spreads are normally distributed and perfectly correlated across positions (use a 99% one-tail z of 2.326). What is the cost of liquidation in a stressed market?

A. USD 16.27 million
B. USD 38.79 million
C. USD 85.86 million
D. USD 42.93 million

Answer: D

Step by step

1. Stressed cost $= \frac{1}{2}\sum_i \alpha_i(\mu_i + \gamma\sigma_i)$ using the proportional spreads.
2. Shares: $0.5 \times 1{,}235 \times (0.02346 + 2.326 \times 0.01568) = 0.5 \times 1{,}235 \times 0.05993 = 37.01$.
3. Commodity: $0.5 \times 627 \times (0.005678 + 2.326 \times 0.005678) = 0.5 \times 627 \times 0.018885 = 5.92$.
4. Total $= 37.01 + 5.92 = 42.93$ million.

Why D is correct: The stressed liquidation cost adds $\gamma\sigma$ to the mean proportional spread, halves the result and multiplies by position value; legs are summed (perfect correlation).

Why the others are wrong

- **A:** Used only the mean spreads (normal-market cost: 14.49 + 1.78), ignoring the $\gamma\sigma$ stress term.
- **B:** Applied the stress term to the shares only and the mean spread to the commodity (37.01 + 1.78).
- **C:** Forgot to halve the spread (2 x 42.93).

Q4

AP Mock 2 Q50

REBUILT — One distractor (INR 86,500) was not traceable to any error; replaced with the 95%-z mistake. Original answer was correct.

Samir Khan manages a 10,000-share position in an illiquid stock, XYZ, trading at INR 200 (the midpoint of the bid-ask spread). XYZ's daily return volatility is 2% and the average bid-ask spread is INR 0.80. Assuming normally distributed returns, the liquidity-adjusted daily 99% VaR (z = 2.33) under the constant spread approach is closest to:

A. INR 97,200
B. INR 4,000
C. INR 69,800
D. INR 65,800

Answer: A

Step by step

1. $V = 10{,}000 \times 200 = 2{,}000{,}000$.
2. Market VaR $= V \times 2.33 \times 0.02 = 93{,}200$.
3. Proportional spread $s = 0.80/200 = 0.004$; cost $= \frac{1}{2}sV = 0.5 \times 0.004 \times 2{,}000{,}000 = 4{,}000$.
4. LVaR $= 93{,}200 + 4{,}000 = 97{,}200$.

Why A is correct: Under the constant spread approach LVaR is the ordinary VaR plus half the (fixed) proportional spread times position value.

Why the others are wrong

- **B:** Only the liquidation cost; the market-risk VaR was omitted.
- **C:** Used the 95% z-score (1.645): 65,800 + 4,000 = 69,800.
- **D:** The 95% VaR (z = 1.645) with no liquidity cost added.

Q5

AP Mock 2 Q6

REBUILT — Original relied on undefined labels ('constant spread' vs 'exogenous' approaches) and had a contradictory option D; rewritten around the reading's normal-market (mean spread) versus stressed-market (mean + gamma x sigma) liquidation cost.

Two trading desks hold identical bond portfolios and use the same market-risk VaR. The liquidity-adjusted VaR reported by the Manchester desk is lower than that reported by the London desk. Which is the most likely explanation?

A. Manchester adds a normal-market liquidation cost (mean spread only), while London adds a stressed-market cost using $\mu + \gamma\sigma$ of the spread.

B. Manchester adds a stressed-market cost using $\mu + \gamma\sigma$, while London uses the mean spread only.

C. Manchester uses the full bid-offer spread as the cost of liquidation, while London uses half of it.

D. Manchester uses a higher confidence level for the spread distribution than London.

Answer: A

Step by step

1. LVaR = VaR + cost of liquidation.
2. Normal-market cost $= \frac{1}{2}\sum \alpha_i \mu_i$; stressed-market cost $= \frac{1}{2}\sum \alpha_i(\mu_i + \gamma\sigma_i)$.
3. With identical VaR and holdings, the stressed cost is larger by $\frac{1}{2}\sum \alpha_i \gamma \sigma_i$.
4. So the desk with the lower LVaR is the one that ignores spread volatility (the normal-market measure).

Why A is correct: The stressed measure adds $\gamma\sigma$ to the mean spread, so it always gives a higher liquidation cost and LVaR than the normal-market measure.

Why the others are wrong

- **B:** Reversed: the stressed cost is larger, which would make Manchester's LVaR higher, not lower.
- **C:** Using the full spread instead of half doubles the cost, making Manchester higher.
- **D:** A higher confidence level raises γ and hence the stressed cost, increasing LVaR.

Q6

AP Mock 3 Q72

REBUILT — Distractors were not traceable to realistic errors; replaced. Original answer (58.5m) was correct.

QPR Bank holds 25 million shares of a company and 35 million ounces of a commodity. The shares are quoted bid USD 90.8 / offer USD 92.4 and the commodity bid USD 24.0 / offer USD 26.2. What is the cost of liquidation in a normal market?

A. USD 20.0 million

B. USD 38.5 million

C. USD 117.0 million

D. USD 58.5 million

Answer: D

Step by step

1. Cost $= \frac{1}{2}\sum(\text{offer} - \text{bid}) \times \text{quantity}$.
2. Shares: $0.5 \times 1.6 \times 25\text{m} = 20.0\text{m}$.
3. Commodity: $0.5 \times 2.2 \times 35\text{m} = 38.5\text{m}$.
4. Total = 58.5 million.

Why D is correct: Normal-market liquidation cost is half the dollar bid-offer spread times quantity, summed over

positions.

Why the others are wrong

- **A:** Included only the shares.
- **B:** Included only the commodity.
- **C:** Used the full spread (2 x 58.5) instead of half.

Q7

AnalystPrep Q.2275

REBUILT — Original was a vague definition item with a 'liquid' vs 'perfectly liquid' ambiguity; replaced with a clearer LO 66.a question on managing trading liquidity risk.

A bank wants to reduce the trading liquidity risk of its book. Which action is most consistent with the guidance on trading liquidity risk?

A. Concentrate holdings in one large position to minimise the number of transactions.
B. Hold a larger number of smaller positions and set size limits on individual trades.
C. Plan to liquidate large positions in a single block to limit exposure time.
D. Rely on the mid-market price when estimating liquidation proceeds.

Answer: B

Step by step

1. Large trades widen the bid-offer spread and invite predatory trading.
2. Liquidation cost rises with position size and spread.
3. Splitting exposure into smaller positions and capping trade size limits spread widening.

Why B is correct: Larger positions create more liquidity risk because the spread widens with size; smaller positions and trade-size limits reduce it.

Why the others are wrong

- **A:** A large concentrated position widens the spread when sold; this is the opposite of the recommended approach.
- **C:** Block liquidation of a large position widens the spread and exposes the seller to predatory trading.
- **D:** Sales occur at the bid, not the mid; using mid ignores the liquidation cost.

Q8

AnalystPrep Q.2977

REBUILT — Original options included unexplained values (0.2452, 1.3526, 1.0981); rebuilt with options reflecting realistic mistakes and an explicit z of 1.645.

XYZ stock trades at $40 with a daily standard deviation of 1%. The current bid-ask spread is 2% of the price. Using a constant spread and z = 1.645, the 95% one-day liquidity-adjusted VaR per share is closest to:

A. $0.40
B. $0.66
C. $1.46
D. $1.06

Answer: D

Step by step

1. Market VaR $= 40 \times 1.645 \times 0.01 = 0.658$.
2. Liquidity cost $= \frac{1}{2} \times 0.02 \times 40 = 0.40$.
3. LVaR $= 0.658 + 0.40 = 1.058 \approx \$1.06$.

Why D is correct: Constant-spread LVaR is VaR plus half the proportional spread times price.

Why the others are wrong

- **A:** Only the liquidity cost.
- **B:** Only the market VaR; liquidity cost omitted.
- **C:** Used the full spread (0.80) instead of half: 0.66 + 0.80.

Q9

AnalystPrep Q.3940

REBUILT — Distractors (31.6, 34.6, 28.9) were implausible giveaways, one below the bid; replaced with results of realistic errors. Original answer (30.8) was correct.

A bank holds 50 million shares (bid $80.4, offer $80.8) and 20 million ounces of a commodity with bid $30.6 and unknown offer. The cost of liquidating the portfolio in a normal market is $12.0 million, and the mid-market value

of the commodity position is $614 million. What is the commodity's offer price?

A. $30.7
B. $30.8
C. $31.0
D. $31.8

Answer: B

Step by step

1. Shares: mid = 80.6, $\alpha = 80.6 \times 50 = 4{,}030$m, $s = 0.4/80.6 = 0.004963$; cost = $0.5 \times 0.004963 \times 4{,}030 = 10.0$m.
2. Commodity cost = $12.0 - 10.0 = 2.0\text{m} = \frac{1}{2}s \times 614 \Rightarrow s = 0.006515$.
3. Mid = $614/20 = 30.7$; dollar spread = $0.006515 \times 30.7 = 0.20$.
4. Offer = 30.6 + 0.20 = 30.8 (check: (30.6 + 30.8)/2 = 30.7).

Why B is correct: Subtracting the share cost from the total leaves the commodity cost, which pins down its proportional spread and hence the offer.

Why the others are wrong

- **A:** Forgot the 1/2 in the cost formula (s = 2.0/614 = 0.003257), halving the spread to about 0.10.
- **C:** Doubled the proportional spread when converting to a price (30.6 x (1 + 2 x 0.006515) = 31.0).
- **D:** Attributed the entire $12.0m to the commodity (s = 24/614 = 0.0391), ignoring the share cost.

Q10

AnalystPrep Q.3941

REBUILT — Distractors were arbitrary numbers; replaced with results of realistic errors (ignoring sigma, wrong z, omitting the 1/2).

A bank holds two positions with mid-market values of $1,400 million in shares and $840 million in a commodity. The following information is available:

Item	Mean	Standard deviation
Bid-offer spread, shares	$1.0 million	$1.1 million
Bid-offer spread, commodity	$0.1 million	$0.1 million
Proportional bid-offer spread, shares	0.034	0.018
Proportional bid-offer spread, commodity	0.0044	0.0044

Assume the spreads are normally distributed and perfectly correlated across the two positions. Which is closest to the cost of liquidation in a stressed market at a 95% confidence level (z = 1.645)?

A. $25.65 million
B. $59.25 million
C. $49.42 million
D. $98.83 million

Answer: C

Step by step

1. Stressed-market cost: $\sum_i \frac{\alpha_i(\mu_i + \lambda\sigma_i)}{2}$ using the proportional spread statistics, with $\lambda = 1.645$.
2. Shares: $1{,}400 \times (0.034 + 1.645 \times 0.018)/2 = 1{,}400 \times 0.06361/2 = 44.527$.
3. Commodity: $840 \times (0.0044 + 1.645 \times 0.0044)/2 = 840 \times 0.011638/2 = 4.888$.
4. Total: $44.527 + 4.888 = 49.415 \approx$ $49.42 million.

Why C is correct: The stressed-market cost of liquidation is half of each position value times the worst-case (mean plus λ standard deviations) proportional spread; the dollar spreads in the table are not needed.

Why the others are wrong

- **A:** Ignored the spread volatility (used only μ): $1{,}400 \times 0.034/2 + 840 \times 0.0044/2 = 25.65$, which is the normal-market cost.
- **B:** Used $\lambda = 2.326$ (99%) instead of 1.645 for the 95% level.
- **D:** Forgot to divide by 2 (the position is only crossed half the spread): $2 \times 49.415 = 98.83$.

Q11

AnalystPrep Q.3942

REBUILT — Added units ($ million) to the stem and replaced arbitrary distractors with results of realistic errors.

A bank's liquidity manager holds shares with a mid-market value of $1,426 million and a commodity position with a mid-market value of $814 million. The proportional bid-offer spread for the shares has a mean of 0.0467 and an unknown standard deviation. For the commodity, the mean and standard deviation of the proportional bid-offer

spread are both 0.006857. Spreads are normally distributed, and the cost of liquidation in a stressed market at a 99% confidence level (z = 2.326) is $82.264 million. (Dollar spread statistics for each asset are not needed.) What is the standard deviation of the proportional bid-offer spread for the shares?

A. 0.0557
B. 0.0295
C. 0.0338
D. 0.0239

Answer: D

Step by step

1. Set up $82.264 = 1{,}426(0.0467 + 2.326\,w)/2 + 814(0.006857 + 2.326 \times 0.006857)/2$.
2. Commodity term: $814 \times 0.022806/2 = 9.2822$.
3. Shares: $713 \times 0.0467 = 33.2971$ plus $713 \times 2.326\,w = 1{,}658.44\,w$.
4. $1{,}658.44\,w = 82.264 - 9.2822 - 33.2971 = 39.6847$, so $w = 0.02393 \approx 0.0239$.

Why D is correct: Solving the stressed liquidation-cost equation for the only unknown, the share spread volatility, gives about 0.0239.

Why the others are wrong

- **A:** Forgot the $\lambda = 2.326$ multiplier on w: $39.6847/713 = 0.0557$.
- **B:** Omitted the commodity liquidation cost: $(82.264 - 33.2971)/1{,}658.44 = 0.0295$.
- **C:** Used $\lambda = 1.645$ instead of 2.326: $39.6847/(713 \times 1.645) = 0.0338$.

Q12

AnalystPrep Q.3943

REBUILT — Added units ($ million) and replaced arbitrary distractors with results of realistic errors.

A bank holds shares with an unknown mid-market value of $W million and a commodity position with a mid-market value of $413 million. The proportional bid-offer spread for the shares has mean 0.0721 and standard deviation 0.0675; for the commodity, mean 0.00524 and standard deviation 0.00463. (Dollar spread statistics are not needed.) Spreads are normally distributed, and the cost of liquidation at the 99% confidence level (z = 2.326) in a stressed market is $95.062 million. What is W?

A. $386 million
B. $830 million
C. $801 million
D. $1,002 million

Answer: C

Step by step

1. $95.062 = W(0.0721 + 2.326 \times 0.0675)/2 + 413(0.00524 + 2.326 \times 0.00463)/2$.
2. Shares factor: $(0.0721 + 0.15701)/2 = 0.114553$ per dollar of W.
3. Commodity term: $413 \times 0.016010/2 = 3.3059$.
4. $W = (95.062 - 3.3059)/0.114553 = 800.996 \approx$ $801 million.

Why C is correct: The shares' stressed liquidation cost must equal the total less the commodity cost, and dividing by the per-dollar factor gives W of about 801.

Why the others are wrong

- **A:** Forgot the factor of 1/2 throughout: $(95.062 - 413 \times 0.016010)/0.229105 = 386$.
- **B:** Ignored the commodity's liquidation cost: $95.062/0.114553 = 830$.
- **D:** Used $\lambda = 1.645$ instead of 2.326 for the share spread (a smaller per-dollar cost factor), which makes W too large.

Q13

AnalystPrep Q.3944

REBUILT — Added units ($ million), fixed inconsistent option formatting, replaced arbitrary distractors with results of realistic errors.

A bank holds shares with a mid-market value of $801 million and a commodity with a mid-market value of $413 million. The proportional bid-offer spread is 0.001879 for the shares and 0.002488 for the commodity. What is the cost of liquidation in a normal market?

A. $2.5326 million
B. $1.2663 million
C. $2.6508 million
D. $0.6332 million

Answer: B

Step by step

1. Normal market: $\sum_i s_i \alpha_i / 2$.
2. Shares: $0.001879 \times 801/2 = 0.7525$.
3. Commodity: $0.002488 \times 413/2 = 0.5138$.
4. Total = 1.2663 million.

Why B is correct: Each position costs half its proportional spread times its mid-market value, since a sale occurs at the bid, half a spread below mid.

Why the others are wrong

- **A:** Forgot to divide by 2: 1.5050 + 1.0275 = 2.5326.
- **C:** Added the spreads and applied the sum to total value: $(0.001879 + 0.002488) \times 1{,}214/2 = 2.65$.
- **D:** Divided by 2 twice (divided by 4).

Q14

AnalystPrep Q.3945

REBUILT — Distractors were arbitrary (not tied to any computation); replaced with results of realistic errors.

A bank buys W shares of a company and 19 million ounces of a commodity. The shares are quoted bid \$70.4, offer \$72.8; the commodity is quoted bid \$46.6, offer \$47.2. The mid-market value of the share position is \$2,506 million. Calculate W and the cost of liquidation in a normal market.

A. 35 million shares; \$47.70 million
B. 35 million shares; \$95.40 million
C. 35 million shares; \$42.00 million
D. 34.4 million shares; \$47.70 million

Answer: A

Step by step

1. Share mid = $(70.4 + 72.8)/2 = 71.6$; $W = 2{,}506/71.6 = 35$ million.
2. Share proportional spread = $2.4/71.6 = 0.03352$; cost = $0.03352 \times 2{,}506/2 = 42.00$.
3. Commodity mid = 46.9; value = $46.9 \times 19 = 891.1$; $s = 0.6/46.9 = 0.012793$; cost = $0.012793 \times 891.1/2 = 5.70$.
4. Total = $42.00 + 5.70 = \$47.70$ million.

Why A is correct: Shares are counted at the mid price to get W, and each position's liquidation cost is half the proportional spread times its mid value (equivalently half the dollar spread times units).

Why the others are wrong

- **B:** Forgot to divide by 2, doubling the cost.
- **C:** Included only the share position and left out the commodity (\$5.70 million).
- **D:** Divided the position by the offer price (2,506/72.8 = 34.4) instead of the mid price.

Q15

AnalystPrep Q.3946

REBUILT — Source answer used rounded proportional spreads (3,531,536); exact result is 3,531,330. Options rebuilt so exactly one is correct, with error-based distractors.

A bank holds 1 million shares of company X (mid-market value \$72,000,000) and 1 million units of commodity Y (mid-market value \$47,000,000). The bid-offer spread per unit has mean \$1.30 and standard deviation \$1.03 for the shares, and mean \$1.32 and standard deviation \$0.88 for the commodity. Spreads are approximately normal and perfectly correlated. What is the cost of liquidation in a stressed market at a 99% confidence level (z = 2.326)?

A. \$1,310,000
B. \$2,880,975
C. \$3,531,330
D. \$7,062,660

Answer: C

Step by step

1. Per unit prices: \$72 and \$47. Proportional spread times mid value equals dollar spread times units.
2. Shares: $1{,}000{,}000 \times (1.30 + 2.326 \times 1.03)/2 = 1{,}000{,}000 \times 3.69578/2 = 1{,}847{,}890$.
3. Commodity: $1{,}000{,}000 \times (1.32 + 2.326 \times 0.88)/2 = 1{,}000{,}000 \times 3.36688/2 = 1{,}683{,}440$.
4. Total = \$3,531,330.

Why C is correct: Stressed liquidation cost is $\sum \alpha_i(\mu_i + \lambda\sigma_i)/2$, where the proportional spread times mid value

equals the dollar spread times units.

Why the others are wrong

- **A:** Ignored the spread volatility: $(1.30 + 1.32)/2 \times 10^6$, the normal-market cost.
- **B:** Used $\lambda = 1.645$ (95%) instead of 2.326.
- **D:** Forgot to divide by 2.

Q16

BT Mock Q4 · BT notes P2.T8.20.1.1

REBUILT — Merged two identical items; replaced the weak distractor 0.1538 with the realistic error of dividing by the offer price.

An asset that is not very liquid is quoted bid \$89.00, offer \$96.00. Which is nearest to the proportional bid-offer spread?

A. 0.0378
B. 0.0757
C. 0.0729
D. 7.000

Answer: B

Step by step

1. Dollar spread $p = 96.00 - 89.00 = 7.00$.
2. Mid-market price $= (89 + 96)/2 = 92.50$.
3. $s = p/\text{mid} = 7/92.5 = 0.0757$.

Why B is correct: The proportional spread is the dollar spread divided by the mid-market price.

Why the others are wrong

- **A:** Divided by the sum of bid and offer (185) instead of the mid, giving s/2 = 0.0378, which is the fractional liquidation cost, not the spread.
- **C:** Divided by the offer price (7/96) instead of the mid price.
- **D:** Reported the dollar spread, not the proportional spread.

Q17

BT Mock Q9

REBUILT — Replaced arbitrary distractors with results of realistic errors so only 200,000 is nearest.

A portfolio holds 100,000 shares of a stock; the position is worth \$3.0 million at mid-market. The stock is quoted bid \$29.00, offer \$31.00, and its daily volatility is 1.43%. Assume arithmetic returns are normal with zero expected daily return, and define liquidity-adjusted VaR as VaR plus the cost of liquidation in a normal market. Which is nearest to the one-day 99% LVaR?

A. \$100,000
B. \$150,000
C. \$200,000
D. \$300,000

Answer: C

Step by step

1. VaR $= 3{,}000{,}000 \times 2.326 \times 0.0143 = 99{,}785$.
2. Mid $= 30$; $s = 2/30 = 0.06667$.
3. Liquidation cost $= s\alpha/2 = 0.06667 \times 3{,}000{,}000/2 = 100{,}000$.
4. LVaR $= 99{,}785 + 100{,}000 = 199{,}785 \approx \$200{,}000$.

Why C is correct: LVaR adds the normal-market cost of crossing half the spread to the ordinary VaR; the two pieces are about equal here.

Why the others are wrong

- **A:** Included only VaR (or only the liquidation cost), omitting the other component.
- **B:** Halved the liquidation cost twice (50,000) before adding to VaR.
- **D:** Used the full spread cost (200,000) instead of half the spread, forgetting to divide by 2.

Q18 BT answers P2.T8.20.1.2 · BT notes P2.T8.20.1.2

REBUILT — Original options (750, 1,300) were not produced by any realistic mistake; replaced with options generated by specific errors, answer unchanged.

An investor holds two positions:
- Short shares worth $10,000 where the proportional bid-offer spread is 0.030, and
- Long shares worth $17,000 where the proportional bid-offer spread is 0.040

What is the approximate cost to the investor to unwind this two-position portfolio in a normal market?

A. $245
B. $455
C. $490
D. $980

Answer: C

Step by step

1. Cost of liquidation = $\sum \frac{1}{2} s_i \alpha_i$ (half the proportional spread times position value), regardless of long or short.
2. Short: $0.5 \times 0.030 \times 10{,}000 = 150$
3. Long: $0.5 \times 0.040 \times 17{,}000 = 340$
4. Total = 150 + 340 = $490.

Why C is correct: Unwinding crosses half the bid-offer spread on each position, so cost is $\frac{1}{2} s\alpha$ summed over positions; both long and short positions are costly.

Why the others are wrong

- **A:** $245 halves the spread twice (applies the 1/2 factor again to the total).
- **B:** $455 pairs the spreads with the wrong positions (0.030 with $17,000 and 0.040 with $10,000).
- **D:** $980 is the full spread cost (300 + 680); it omits the one-half factor because only half the spread is crossed on exit.

Q19 BT answers P2.T8.20.1.3 · BT notes P2.T8.20.1.3

REBUILT — Stem clarified that the spreads are proportional and options replaced so every distractor reflects a realistic error; answer unchanged.

An investor holds two positions:
- Long shares worth $25,000 where the proportional bid-offer spread has a mean and standard deviation of 0.050
- Long shares worth $40,000 where the proportional bid-offer spread has a mean and standard deviation of 0.030

Assume the bid-offer spreads are normally distributed and perfectly correlated. Which is nearest to the worst expected cost of unwinding with 95.0% confidence (z = 1.645)?

A. $1,225
B. $2,015
C. $3,240
D. $6,480

Answer: C

Step by step

1. Stressed cost = $\sum \frac{1}{2} \alpha_i (\mu_i + z\sigma_i)$ with z = 1.645.
2. Position 1: $0.5 \times 25{,}000 \times (0.050 + 1.645 \times 0.050) = 0.5 \times 25{,}000 \times 0.13225 = 1{,}653.13$
3. Position 2: $0.5 \times 40{,}000 \times (0.030 + 1.645 \times 0.030) = 0.5 \times 40{,}000 \times 0.07935 = 1{,}587.00$
4. Total = 1,653.13 + 1,587.00 = $3,240.

Why C is correct: The stressed spread is the mean plus z standard deviations, and the cost is half of that spread applied to each position value; the perfect-correlation assumption lets the position costs be summed.

Why the others are wrong

- **A:** $1,225 uses only the mean spread (normal-market cost) and ignores the stress add-on $z\sigma$.
- **B:** $2,015 includes only the $z\sigma$ term and leaves out the mean spread.
- **D:** $6,480 omits the one-half factor, giving the full stressed spread rather than half.

Q20 BT answers P2.T8.20.2.1 · BT notes P2.T8.20.2.1

REBUILT — Original distractors (15,000, 80,000, 1.0 million) were giveaways; replaced with options from realistic errors, answer unchanged.

A portfolio holds 100,000 shares of a stock and this single position has a value of $3.0 million. The stock is quoted bid $29.00, offer $31.00. The stock's daily volatility is 1.43%. Assume normally distributed returns with an expected daily return of zero, and a constant (static) bid-offer spread. Which is NEAREST to the position's one-day

99.0% liquidity-adjusted VaR (LVaR)?

A. $100,000
B. $171,000
C. $200,000
D. $300,000

Answer: C

Step by step

1. Mid = (29 + 31)/2 = 30; proportional spread $s = 2/30 = 0.0667$.
2. Position VaR = $2.326 \times 0.0143 \times 3{,}000{,}000 = 99{,}785$
3. Liquidity cost = $0.5 \times 0.0667 \times 3{,}000{,}000 = 100{,}000$
4. LVaR = 99,785 + 100,000 = $199,785, about $200,000.

Why C is correct: LVaR = VaR + cost of liquidation, i.e. market risk plus the half-spread paid to exit.

Why the others are wrong

- **A:** $100,000 reports only the VaR (or only the liquidity cost) and omits the other component.
- **B:** $171,000 uses the 95% deviate 1.645 for VaR (70,571) instead of 2.326 for 99%.
- **D:** $300,000 omits the one-half factor on the spread (cost = $200,000), giving 99,785 + 200,000.

Q21

BT answers P2.T8.20.2.2 · BT notes P2.T8.20.2.2

REBUILT — Source OCR printed offer as $47.00 (corrected to $41.00 per BT notes); options replaced with realistic-error distractors; answer unchanged.

A position with a value of $2.0 million consists of 50,000 shares of stock. The stock is quoted bid $39.00, offer $41.00. The proportional bid-offer spread has a standard deviation of 2.0%. The stock's daily volatility is 2.00%. Assume normally distributed returns with zero expected return, and measure liquidity cost under stressed conditions using a worst-case (volatile) spread rather than a static spread. Using a normal deviate of 1.65 for both the price and the spread, which is NEAREST to the position's one-day 95.0% LVaR?

A. $116,000
B. $149,000
C. $176,000
D. $232,000

Answer: B

Step by step

1. Mid = 40; mean proportional spread $\mu = 2/40 = 0.05$; $\sigma_s = 0.02$.
2. Position VaR = $1.65 \times 0.02 \times 2{,}000{,}000 = 66{,}000$
3. Stressed liquidity cost = $0.5 \times 2{,}000{,}000 \times (0.05 + 1.65 \times 0.02) = 0.5 \times 2{,}000{,}000 \times 0.083 = 83{,}000$
4. LVaR = 66,000 + 83,000 = $149,000.

Why B is correct: With a volatile spread, LVaR $= \text{VaR} + \frac{1}{2}\alpha(\mu + z\sigma_s)$: the cost uses the worst-case spread rather than the average.

Why the others are wrong

- **A:** $116,000 uses the static spread (cost 50,000) and ignores spread volatility.
- **C:** $176,000 uses 2.326 (99%) for the price VaR (93,040) instead of 1.65.
- **D:** $232,000 omits the one-half factor on the liquidity cost (166,000 + 66,000).

Q22

GARP PE2 Q29 · Mock 4 Q30

A manager at an asset management firm requests that an analyst calculate the cost of liquidation of one of the fund's stock positions. The position consists of 100,000 shares of company ABC and the stock has a current bid price of USD 53.5 and an offer price of USD 54.5. The mean and standard deviation for the stock's proportional bid-offer spread are 0.0185 and 0.0250 respectively. The analyst calculates the cost of liquidation for this entire stock position under a stressed market scenario based on a 99% confidence level. What is the correct cost of liquidation for this stock position?

A. USD 49,950
B. USD 99,900
C. USD 206,955
D. USD 413,910

Answer: C

Step by step

1. Mid = (53.5 + 54.5)/2 = 54.0; position value $\alpha = 100{,}000 \times 54 = 5{,}400{,}000$.
2. Stressed proportional spread = $\mu + 2.326\sigma = 0.0185 + 2.326 \times 0.0250 = 0.07665$
3. Cost = $\frac{1}{2}\alpha(\mu + 2.326\sigma) = 0.5 \times 5{,}400{,}000 \times 0.07665 =$ USD 206,955.

Why C is correct: Stressed-market liquidation cost is half the position value times the 99% worst-case proportional spread.

Why the others are wrong

- **A:** USD 49,950 uses only the mean spread (0.0185) with the one-half factor, ignoring the stress term.
- **B:** USD 99,900 uses the mean spread without the one-half factor (5.4m x 0.0185).
- **D:** USD 413,910 uses the stressed spread but omits the one-half factor.

Q23

Mock 2 Q47

A portfolio manager is attempting to determine the cost of liquidation of a specific holding in a stressed market. The manager holds 50,000 shares of Lear, Inc. (Lear), with bid and offer prices of USD 22.20 and USD 23.10, respectively. The standard deviation of the bid-offer spread (assumed normal distribution) is USD 1.70. The cost of liquidation of Lear at a 95% level of confidence (1.645 z-score for one-tail and 1.96 z-score for two-tail) is closest to:

A. USD 69,900.
B. USD 92,400.
C. USD 105,800.
D. USD 184,800.

Answer: B

Step by step

1. Mean dollar spread = 23.10 − 22.20 = 0.90.
2. Stressed spread = mean + z × σ = 0.90 + 1.645 × 1.70 = 3.6965 (one-tail z, since only a wider spread is adverse).
3. Cost of liquidation = ½ × shares × stressed spread = 0.5 × 50,000 × 3.6965 = USD 92,412.
4. Closest option: USD 92,400.

Why B is correct: In a stressed market the cost of liquidation is half the position times the spread at the chosen confidence level, $\frac{1}{2}\alpha(\mu_s + z\sigma_s)$, using the one-tail z-score because only a wider spread hurts.

Why the others are wrong

- **A:** Used only the spread volatility term, 0.5 × 50,000 × 1.645 × 1.70 = 69,912, omitting the mean spread.
- **C:** Used the two-tail 1.96 instead of the one-tail 1.645: 25,000 × (0.90 + 1.96 × 1.70) = 105,800.
- **D:** Forgot the one-half (a position is liquidated at the bid, i.e. half the spread from mid): 50,000 × 3.6965 = 184,825.

Q24

Mock 4 Q77

REBUILT — Option D (printed 1,354,000) did not match the source's own working (1,356,480); corrected it to the value produced by omitting the one-half.

A risk manager is calculating the liquidity-adjusted VaR (LVaR) of a trading position defined as VaR plus the stressed-market cost of liquidation. The manager gathers the following data:

Item	Value
Position value	USD 20 million
Daily return volatility	2.00%
Mean proportional bid-ask spread	1.20%
Volatility of the proportional bid-ask spread	0.40%
Confidence level (both components)	99% (z = 2.326)

What is the position's liquidity-adjusted VaR?

A. USD 930,400.
B. USD 1,050,400.
C. USD 1,143,440.
D. USD 1,356,480.

Answer: C

Step by step

1. VaR = 20,000,000 × 2.326 × 0.02 = USD 930,400.

2. Stressed proportional spread = 0.012 + 2.326 × 0.004 = 0.021304.
3. Cost of liquidation = ½ × 20,000,000 × 0.021304 = USD 213,040.
4. LVaR = 930,400 + 213,040 = USD 1,143,440.

Why C is correct: LVaR equals ordinary VaR plus the cost of liquidating in a stressed market, $\frac{1}{2}\alpha(\mu_s + z\sigma_s)$, because the spread cost is an additional loss.

Why the others are wrong

- **A:** This is VaR alone; the liquidity cost was not added.
- **B:** Added only the cost at the mean spread (½ × 20m × 0.012 = 120,000), ignoring the spread-volatility term.
- **D:** Omitted the one-half in the liquidation cost (20m × 0.021304 = 426,080), giving 930,400 + 426,080.

Q25

Schweser Module Quiz 66.1 Q1

Rigues Bank owns 5 million shares of a stock with a bid price of $10.50 and an offer price of $12.75. The cost of liquidation in normal market conditions is closest to:

A. $2.813 million.
B. $5.625 million.
C. $6.228 million.
D. $11.250 million.

Answer: B

Step by step

1. Mid price = (10.50 + 12.75)/2 = 11.625; position value α = 5,000,000 × 11.625 = $58,125,000.
2. Proportional spread s = (12.75 − 10.50)/11.625 = 0.19355.
3. Cost = s × α/2 = 0.19355 × 58,125,000 / 2 = $5,625,000.

Why B is correct: Selling at the bid costs half the spread relative to mid, so cost = $s \times \alpha/2$; equivalently 5m × 2.25/2.

Why the others are wrong

- **A:** Halved twice (divided by 4).
- **C:** Divided the dollar spread by the bid (2.25/10.50) rather than the mid price: 58.125m × 0.2143 / 2 = 6.228m.
- **D:** Used the full dollar spread (5m × 2.25) and forgot the one-half.

Q26

AP Mock 1 Q59

REBUILT — Option A said 'almost 27%' though the exact uplift is 27.5%, and D (50%) was not tied to a realistic error; reworded options so exactly one is correct.

A financial manager estimates liquidity-adjusted VaR defined as VaR plus the cost of liquidation at a constant proportional bid-ask spread, with the following data (all in return terms):

Parameter	Value
Mean return, μ	0
Daily volatility, σ	0.49 / √490 (≈ 0.02214)
Proportional bid-ask spread	0.02
Confidence level	95% (z = 1.645)

Which of the following statements is true?

A. The constant spread liquidity adjustment raises VaR by about 27.5%.
B. The constant spread liquidity adjustment reduces VaR by about 27.5%.
C. The constant spread liquidity adjustment raises VaR by about 55%.
D. A small spread cannot translate into a large liquidity adjustment to VaR.

Answer: A

Step by step

1. σ = 0.49/√490 = 0.49/22.136 = 0.022136.
2. VaR = 1.645 × 0.022136 = 0.036414 (μ = 0).
3. Liquidity cost = spread/2 = 0.02/2 = 0.010.
4. LVaR = 0.036414 + 0.010 = 0.046414; uplift = 0.010/0.036414 = 27.5%.

Why A is correct: With a constant proportional spread, LVaR = VaR + spread/2, which always adds to VaR; here the daily VaR is small so a 2% spread (cost 1%) is a large relative increase.

Why the others are wrong

- **B:** The liquidity cost is an additional loss, so it increases rather than reduces VaR.
- **C:** Used the full spread 0.02 instead of half (0.02/0.036414 = 54.9%).

- **D:** False: the adjustment is relative to VaR, and when VaR is small (short horizon, low volatility) even a small spread is material, here 27.5%.

Q27

AnalystPrep Q.2257

REBUILT — Original had a vague 'gilts' distractor and tested exogenous vs endogenous spread outside Schweser wording; rewrote around position size relative to market and how the spread enters cost of liquidation.

I&M Bank holds USD 20 million of US Treasury bonds. The US Treasury has issued bonds with a nominal value of roughly USD 800 billion. When estimating the bank's cost of liquidation, which of the following is the most appropriate assumption?

A. The bid-offer spread is determined by the overall market and does not depend on the size of the bank's own sale, so the cost is $s \times \alpha/2$ with a market-given s.

B. The bid-offer spread will widen materially as the bank sells, so the spread must be modeled as a function of the bank's trade size.

C. The bank can influence the market price of Treasury bonds, so it should add a market-impact cost to the half-spread.

D. The position is too large for the market, so the bank should hold more, smaller positions to reduce liquidity risk.

Answer: A

Step by step

1. Position share of market = 20 million / 800 billion = 0.0025%.
2. A sale this small is far below any market maker's size limit, so it cannot widen the spread or move the price.
3. The spread is therefore given by the market and cost of liquidation is simply $s \times \alpha/2$.

Why A is correct: Spreads widen only when order size exceeds market makers' limits; a position that is a negligible fraction of a huge, liquid market takes the spread as given.

Why the others are wrong

- **B:** Appropriate only for large positions relative to the market; 0.0025% of the market will not widen the spread.
- **C:** Price influence requires a position that is large relative to market depth; this is not the case here.
- **D:** Diversifying into smaller positions helps when positions are large; this one is already trivially small versus the market.

Q28

AnalystPrep Q.2258

REBUILT — Original options (fixed 0.5% or 1% caps vs 'not limited') were vague and gave away the answer; rebuilt around the GARP/Hull facts on the size and variability of bid-offer spreads.

A bank's trader is assessing the trading liquidity of several instruments. Which of the following statements about bid-offer spreads is most accurate?

A. Spreads are fixed by regulation at no more than 1% of the mid-market price for any traded instrument.

B. Spreads can range from roughly 0.05% of the mid-market price for very liquid assets to 5% or even 10% for illiquid ones, and tend to widen when markets become stressed.

C. For an instrument with a market maker, bid and offer quotes diverge from the first unit traded, so the spread widens in proportion to trade size from the outset.

D. Spreads depend only on the instrument type and are unaffected by market conditions or trade size.

Answer: B

Key correction: Source key (AnalystPrep) said D ('not limited'); the idea is right but the stem was too vague, so the item was rebuilt around the quantified GARP range.

Step by step

1. Recall the GARP/Hull description: the bid-offer spread varies from about 0.05% of the mid-market price to as much as 5% or 10%.
2. Spreads are not capped by any fixed rule; they depend on the asset's liquidity, the size of the trade and market conditions.
3. In stressed markets liquidity dries up and spreads widen, so statement B is the only one consistent with this.
4. Market makers quote the same bid and offer up to their size limit, which rules out C.

Why B is correct: Bid-offer spreads are a market-determined trading cost that spans a wide range (about 0.05% to 5-10%) and widen with illiquidity, larger trade size and stress.

Why the others are wrong

- **A:** Invents a regulatory cap (the 1% limit of the original options); no such fixed ceiling exists and spreads of 5-10% occur.
- **C:** Wrong: with a market maker the bid and offer are the same up to the market maker's size limit and only diverge beyond it.
- **D:** Ignores that spreads widen with trade size and in stressed or illiquid markets; spreads are dynamic.

Q29 BT answers P2.T5.404.3

REBUILT — Stem used the non-GARP 'exogenous/endogenous liquidity' jargon and the 2.33 z-value; restated as the GARP cost-of-liquidation formula with 2.326 and replaced weak distractors.

An asset has a mid-market price of \$390.00. Its dollar bid-offer spread is normally distributed with a mean of \$3.90 (1.0% of the mid-price) and a standard deviation of \$0.780 (20 basis points). Using a 99% confidence level for the spread (z = 2.326), which is closest to the per-unit liquidity cost of liquidating the position, i.e. the amount that is added to VaR to obtain a liquidity-adjusted VaR (LVaR)?

A. \$2.86
B. \$5.71
C. \$1.95
D. \$3.90

Answer: A

Step by step

1. Cost of liquidation per unit = half of the worst-case spread: $\frac{1}{2}(\mu + z\sigma)$.
2. Worst-case (99%) spread: $3.90 + 2.326 \times 0.780 = 3.90 + 1.814 = \$5.714$.
3. Half of it: $5.714/2 = \$2.857 \approx \$2.86$.
4. Equivalent in proportional terms: $\frac{1}{2}(1.0\% + 2.326 \times 0.20\%) \times 390 = 0.7326\% \times 390 = \$2.857$.

Why A is correct: A trade is executed at the bid or offer, i.e. half the spread away from mid, and the 99% worst-case spread is $\mu + 2.326\sigma$, so the liquidity cost is half of that.

Why the others are wrong

- **B:** Gives the full 99% worst-case spread (\$5.71) and forgets that only half the spread is paid when trading from the mid-price.
- **C:** Takes half of the mean spread only (3.90/2), ignoring the tail ($z\sigma$) term for stressed spreads.
- **D:** Uses the average spread itself, with neither the tail adjustment nor the one-half factor.

LO 66.b

Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft.

Q30 Mock AP Mock 1 Q52

REBUILT — Original had two defensible answers (A remedy and C failure) and a generic wording; rebuilt to ask for the funding weakness that caused the failure, with one clearly correct option.

Northern Rock, a British mortgage lender, was nationalized after the 2007 crisis. Which of the following best describes the funding weakness that caused its liquidity failure?

A. It funded long-term mortgage lending with short-term wholesale debt and relied on securitization, so it had no fallback when investors stopped lending in 2007.
B. It relied mainly on retail deposits, which are the least stable source of funding, and a retail run began before wholesale markets closed.
C. It sold gold forwards and could not meet margin calls when the gold price rose sharply.
D. It hedged long-term fixed-price supply contracts with short-term futures, so falling prices triggered margin calls.

Answer: A

Step by step

1. Northern Rock was heavily involved in mortgage lending and used short-term debt instruments for funding.
2. After the 2007 subprime crisis, investors became unwilling to lend to banks and the securitization route also

dried up, creating a maturity mismatch with no fallback.
3. Only afterwards did it borrow GBP 3 billion from the Bank of England; the news triggered the retail run (GBP 2 billion withdrawn in a week).
4. Hence the lesson: maintain a more conservative view of funding accessibility.

Why A is correct: The failure came from a funding mismatch between long-term assets and short-term wholesale and securitization funding that vanished in stress; the retail run was a consequence of the emergency borrowing becoming public.

Why the others are wrong

- **B:** Reverses the sequence: the retail run followed the BBC report of emergency borrowing; the root problem was short-term wholesale funding that could not be rolled over. Retail deposits are not described as the main cause.
- **C:** This is Ashanti Goldfields (short gold forwards, margin calls), not Northern Rock.
- **D:** This is Metallgesellschaft (stack-and-roll short-term futures hedging long-dated fixed-price contracts), not Northern Rock.

Q31 AnalystPrep Q.5383 · GARP PE1 Q42 · Mock 3 Q38

A risk manager at a bank is seeking to better understand recent liquidity risk failures. Several real-life cases are reviewed. Which of the following lessons would be best illustrated by the case of Metallgesellschaft in 1993?

A. Negative public perception of emergency borrowing from the central bank can cause a bank run.
B. Positive feedback trading in illiquid instruments can cause excessive losses.
C. Hedging liabilities by rolling forward short-term futures contracts may create cash flow mismatches.
D. Futures provide a better effective hedge for commodities exposure than forwards.

Answer: C

Step by step

1. MG sold 5- to 10-year fixed-price heating oil and gasoline supply contracts.
2. It hedged them with long positions in short-term futures, which must be rolled forward.
3. When oil prices fell, the futures were marked to market daily and generated margin calls (immediate cash outflows).
4. The offsetting gains on the long-term fixed-price contracts would be realized only later, so the cash drain worried senior management and its bankers; MG closed out all hedge positions and abandoned the contracts, a loss of USD 1.33 billion.

Why C is correct: The hedge was economically sound but short-dated futures create daily cash settlement while the hedged liabilities pay out years later, causing a liquidity (cash flow) mismatch.

Why the others are wrong

- **A:** This is the Northern Rock lesson (news of Bank of England borrowing triggered a run).
- **B:** Positive feedback trading relates to liquidity black holes (LO 66.d), not the MG case.
- **D:** Not a lesson of the case; futures and forwards can both hedge, and it was the futures' daily margining that caused the drain, so futures were not 'better' here.

Q32 BT answers P2.T8.20.3.1 · BT notes P2.T8.20.3.1

Liquidity funding risk refers to the firm's ability to meet its cash needs. About liquidity funding risk, which of the following statements is TRUE?

A. Wholesale deposits are the most stable source of funding.
B. A lack of confidence can contribute to an institution's liquidity funding problems.
C. A solvent firm cannot experience a liquidity funding problem; only an insolvent firm can.
D. Case studies such as Northern Rock and Ashanti Goldfields demonstrate that a hedged position is the best protection against liquidity funding problems.

Answer: B

Step by step

1. Liquidity funding risk arises from weak performance, hesitant investors in stressed markets, and short-term funding of longer-term needs.
2. Loss of confidence (e.g., Northern Rock depositors) leads to withdrawals and refusal to roll funding, so B is true.
3. Check the others: wholesale and retail deposits are both somewhat unstable; solvency and liquidity are different; the cases show hedges themselves can drain cash.

Why B is correct: Confidence drives deposit and wholesale funding; when it is lost, funding dries up even for a

solvent firm.

Why the others are wrong

- **A:** GARP states wholesale deposits are a more volatile source of funding than retail deposits and can disappear quickly in stress; even retail deposits are not as stable as they used to be.
- **C:** Confuses liquidity with solvency; a solvent firm can fail from funding problems (e.g., Northern Rock was judged solvent yet could not fund itself).
- **D:** Ashanti was hurt by its hedge (short gold forwards, margin calls) and MG by its futures hedge; hedging does not guarantee liquidity, and Northern Rock was not a hedging case. The lesson is to ensure access to funding for cash flow mismatches.

Q33 Schweser Module Quiz 66.2 Q1

Which of the following statements accurately reflects liquidity management for a financial institution?

A. Investments in Treasury securities help a bank increase its risk and return profile.
B. Stability in retail deposits over the years has enhanced bank liquidity.
C. Central bank borrowings are relatively cheap ways for banks to stay afloat.
D. Stressed markets can limit the effectiveness of trading book liquidation strategies.

Answer: D

Step by step

1. Trading book liquidation raises cash only if the book is liquid.
2. In stressed markets, bid-ask spreads widen and buyers vanish, so liquidation works poorly versus normal conditions.
3. The other statements contradict the reading on Treasuries, retail deposits and central bank borrowing.

Why D is correct: Liquidating the trading book is a liquidity source whose reliability depends on market conditions and fails exactly when needed in stress.

Why the others are wrong

- **A:** Treasuries offer liquidity at low return; they reduce, not increase, risk and return.
- **B:** Retail deposits have become less stable since customers can easily move funds for higher rates.
- **C:** Central bank borrowing is costly: high interest rate, collateral haircut, and a negative signal to the market.

LO 66.c

Evaluate Basel III liquidity risk ratios and BIS principles for sound liquidity risk management.

Q34 AP Mock 2 Q21

REBUILT — Distractors did not correspond to realistic mistakes; replaced them with results of specific errors and kept the data and key.

A bank has the following balance sheet (all figures in millions):

Assets	Amount	RSF factor
Cash	20	0%
Treasury bonds (> 1 yr)	20	5%
Mortgages	50	65%
Small business loans	180	85%
Fixed assets	30	100%
Total	300	

Liabilities & capital	Amount	ASF factor
Retail deposits (stable)	80	90%
Wholesale deposits	100	50%
Tier 2 capital	40	100%
Tier 1 capital	80	100%
Total	300	

Which of the following is closest to the bank's net stable funding ratio (NSFR)?

A. 102%
B. 112%
C. 89%

D. 97%

Answer: B

Step by step

1. ASF = 80(0.90) + 100(0.50) + 40(1.00) + 80(1.00) = 72 + 50 + 40 + 80 = 242.
2. RSF = 20(0) + 20(0.05) + 50(0.65) + 180(0.85) + 30(1.00) = 0 + 1 + 32.5 + 153 + 30 = 216.5.
3. NSFR = ASF/RSF = 242/216.5 = 1.118, i.e. about 112%, above the 100% minimum.

Why B is correct: NSFR is available stable funding divided by required stable funding, each category weighted by its ASF or RSF factor; capital carries a 100% ASF factor, so it adds fully to the numerator.

Why the others are wrong

- **A:** Applying a 100% RSF factor to cash (RSF 236.5) instead of 0% gives about 102%.
- **C:** Inverted the ratio: RSF/ASF = 216.5/242 = 89.5%.
- **D:** Treated stable retail deposits at a 50% ASF factor instead of 90% (ASF 210): 210/216.5 = 97.0%.

Q35

AP Mock 3 Q16

REBUILT — Balance sheet did not balance (assets 117.5 vs funding 100) and options were arbitrary; added an other-liabilities line at 0% ASF (answer unchanged) and replaced distractors with error-based values.

Goodwill Bank's balance sheet contains the following items, with the available stable funding (ASF) and required stable funding (RSF) factors shown:

Funding source	Amount	ASF factor
Retail deposits	35	90%
Wholesale deposits	50	50%
Tier 2 capital	5	100%
Tier 1 capital	10	100%
Other liabilities	17.5	0%

Asset	Amount	RSF factor
Cash	7	0%
Mortgages	38	65%
Treasury bonds	6.5	5%
Small business loans	54	85%
Fixed assets	12	100%

Which of the following is closest to the net stable funding ratio (NSFR)?

A. 116.0%

B. 86.2%

C. 80.2%

D. 74.3%

Answer: B

Step by step

1. ASF = 35(0.90) + 50(0.50) + 5 + 10 + 17.5(0) = 31.5 + 25 + 5 + 10 = 71.5.
2. RSF = 7(0) + 38(0.65) + 6.5(0.05) + 54(0.85) + 12(1.00) = 0 + 24.7 + 0.325 + 45.9 + 12 = 82.925.
3. NSFR = 71.5/82.925 = 86.2%, below the 100% requirement.

Why B is correct: NSFR = ASF/RSF with factor-weighted balances; here required stable funding exceeds available stable funding, so the ratio is below 100%.

Why the others are wrong

- **A:** Inverted the ratio: RSF/ASF = 82.925/71.5 = 116.0%.
- **C:** Applied a 100% RSF factor to Treasury bonds (RSF 89.1) instead of 5%.
- **D:** Applied a 100% RSF factor to mortgages (RSF 96.2) instead of 65%.

Q36

AnalystPrep Q.3950

REBUILT — Source table was garbled (column alignment and LCR/NSFR factors mixed) and unrecoverable; rebuilt as a clean self-contained NSFR calculation with a balanced balance sheet.

RTC Bank reports the following balance sheet (millions) with the Basel III factors indicated.

Assets	Amount	RSF factor
Cash and central bank reserves	300	0%
Government bonds (> 1 yr)	1,000	5%
Residential mortgages	2,200	65%
Loans to small businesses (< 1 yr)	2,000	85%
Fixed and other assets	1,000	100%
Total	6,500	

Liabilities & equity	Amount	ASF factor
Owners' equity	900	100%
Stable retail deposits	1,500	95%
Debt issued (> 1 yr)	1,200	100%
Wholesale funding (< 1 yr)	800	50%
Other liabilities	2,100	0%
Total	6,500	

What is RTC Bank's net stable funding ratio (NSFR)?

A. 106.5%
B. 93.9%
C. 105.3%
D. 72.4%

Answer: B

Step by step

1. ASF = $900(1.00) + 1{,}500(0.95) + 1{,}200(1.00) + 800(0.50) + 2{,}100(0) = 900 + 1{,}425 + 1{,}200 + 400 = 3{,}925$.
2. RSF = $300(0)+1{,}000(0.05)+2{,}200(0.65)+2{,}000(0.85)+1{,}000(1.00) = 0+50+1{,}430+1{,}700+1{,}000 = 4{,}180$.
3. NSFR = $3{,}925/4{,}180 = 93.9\% < 100\%$, so the bank does not meet the NSFR requirement.

Why B is correct: NSFR divides factor-weighted stable funding by factor-weighted required stable funding; the result of 93.9% is below the 100% minimum.

Why the others are wrong

- **A:** Inverted the ratio: RSF/ASF = 4,180/3,925 = 106.5%.
- **C:** Gave 100% weight to deposits and wholesale funding, ignoring the 95% and 50% ASF factors: (900+1,500+1,200+800)/4,180 = 105.3%.
- **D:** Left out owners' equity (ASF 900) although equity has a 100% ASF factor: 3,025/4,180 = 72.4%.

Q37

AnalystPrep Q.3952

REBUILT — Stem did not say which outflow figure is the LCR denominator and used loose terms; clarified HQLA and 30-day stressed net outflows, and replaced arbitrary options with error-based ones.

The following data come from ABC Bank's records. Use them to compute the liquidity coverage ratio (LCR) for 2018 and 2019, respectively.

Item (millions)	2018	2019
High-quality liquid assets	2,100	1,458
Average monthly withdrawals in normal conditions	1,200	2,458
Net cash outflows over a 30-day stress period	1,600	1,700

A. 131% and 86%
B. 175% and 59%
C. 75% and 35%
D. 76% and 117%

Answer: A

Step by step

1. LCR = HQLA/net cash outflows over 30 days of stress; normal-condition withdrawals are irrelevant.
2. 2018: $2{,}100/1{,}600 = 131.3\%$.
3. 2019: $1{,}458/1{,}700 = 85.8\%$, below the 100% requirement.

Why A is correct: The LCR denominator is the net cash outflow over the 30-day stress scenario, not ordinary withdrawals, so the ratios are 131% and 86%.

Why the others are wrong

- **B:** Used average monthly withdrawals as the denominator: 2,100/1,200 = 175% and 1,458/2,458 = 59%.
- **C:** Added normal withdrawals to the stressed outflows: 2,100/2,800 = 75% and 1,458/4,158 = 35%.
- **D:** Inverted the ratio: outflows/HQLA = 1,600/2,100 = 76% and 1,700/1,458 = 117%.

Q38

BT answers P2.T8.20.3.2 · BT notes P2.T8.20.3.2

REBUILT — Two of the options (ROE, NII) were not Basel liquidity ratios, making the item a giveaway; reframed with LCR/NSFR combinations.

Basel III introduced two liquidity requirements, the liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR). In which of these ratios do the bank's regulatory capital instruments (Tier 1 and Tier 2 capital) enter the numerator?

A. The LCR only
B. The NSFR only
C. Both the LCR and the NSFR
D. Neither the LCR nor the NSFR

Answer: B

Step by step

1. LCR = HQLA/net cash outflows in 30 days: the numerator is a stock of liquid assets, so capital is not in it.
2. NSFR = available stable funding/required stable funding: ASF is built by applying ASF factors to funding categories (capital, wholesale deposits, retail deposits, etc.).
3. Capital carries the highest (100%) ASF factor, so it enters the NSFR numerator.

Why B is correct: Capital is a funding category that receives an ASF factor, so it appears only in the NSFR numerator; the LCR numerator is high-quality liquid assets.

Why the others are wrong

- **A:** The LCR numerator is HQLA (an asset measure); capital is not part of it.
- **C:** Capital is not in the LCR; this confuses the two ratios, since only the NSFR weights funding sources.
- **D:** Ignores that the NSFR numerator is weighted funding, which includes capital at a 100% ASF factor.

Q39

Mock 1 Q10

REBUILT — Options were arbitrary; kept data and key and replaced distractors with error-based results.

Bank BCCA has the following balance sheet (figures in millions):

Assets	Amount
Cash	15
Central bank reserves	12
Treasury bonds > 1 year	15
Mortgages	30
Retail loans < 1 year	32
Small-business loans < 1 year	94
Fixed assets	22
Total assets	220

Liabilities & equity	Amount
Retail deposits (less stable)	130
Wholesale deposits	65
Tier 2 capital	5
Tier 1 capital	20
Total liabilities & equity	220

Available stable funding (ASF) factors: 100% Tier 1 and Tier 2 capital, preferred stock, debt > 1 year; 95% "stable" demand and term deposits from individuals/businesses < 1 year; 90% "less stable" demand and term deposits from individuals/businesses < 1 year; 50% wholesale funding (demand and term deposits) < 1 year; 0% all other liability and equity categories.

Required stable funding (RSF) factors: 0% cash and short-term instruments; 5% marketable securities (> 1 year, sovereign 0% risk weight); 15% corporate bonds rated AA- or higher; 50% bonds rated A+ to A-; 65% residential mortgages; 85% loans to small businesses or retail customers < 1 year; 100% all other assets.

What is Bank BCCA's net stable funding ratio (NSFR)?

A. 85.6%
B. 100.1%
C. 121.2%

D. 116.8%

Answer: D

Step by step

1. ASF = 20 + 5 + 130(0.90) + 65(0.50) = 25 + 117 + 32.5 = 174.5.
2. RSF = 15(0)+12(0)+15(0.05)+30(0.65)+32(0.85)+94(0.85)+22(1.00) = 0.75+19.5+27.2+79.9+22 = 149.35.
3. NSFR = 174.5/149.35 = 116.8%.

Why D is correct: Weighting each funding and asset line by its ASF/RSF factor gives ASF 174.5 and RSF 149.35, so NSFR is 116.8% (above 100%).

Why the others are wrong

- **A:** Inverted the ratio: RSF/ASF = 149.35/174.5 = 85.6%.
- **B:** Left out Tier 1 and Tier 2 capital (25) from ASF: 149.5/149.35 = 100.1%.
- **C:** Used the 95% 'stable' factor for retail deposits instead of 90%: (25+123.5+32.5)/149.35 = 121.2%.

Q40

Mock 1 Q50

REBUILT — Original key correct; replaced distractors with explicit error-based values and made the inflow-cap check part of the solution.

A treasury analyst at a mid-size bank is calculating the bank's liquidity coverage ratio (LCR). The analyst assembles the following data:

Item	Amount (USD million)
Level 1 high-quality liquid assets	90
Level 2A high-quality liquid assets (before haircut)	40
Stable retail deposits	500
Unsecured wholesale funding from non-financial corporates	300
Total expected cash inflows over 30 days	60

The applicable run-off rates are 5% on stable retail deposits and 40% on the unsecured wholesale funding. Level 2A assets are subject to a 15% haircut. Cash inflows are recognised up to a cap of 75% of gross outflows. What is the bank's LCR?

A. 85.5%

B. 105.9%

C. 145.9%

D. 152.9%

Answer: C

Step by step

1. HQLA = 90 + 40(1 − 0.15) = 90 + 34 = 124.
2. Gross outflows = 500(0.05) + 300(0.40) = 25 + 120 = 145.
3. Inflow cap = 0.75 × 145 = 108.75; the 60 of inflows is below the cap, so all 60 count.
4. Net outflows = 145 − 60 = 85; LCR = 124/85 = 145.9%.

Why C is correct: LCR = HQLA after haircuts divided by 30-day net outflows (outflows less capped inflows) = 124/85.

Why the others are wrong

- **A:** Ignored the inflows and divided by gross outflows: 124/145 = 85.5%.
- **B:** Excluded the Level 2A assets from HQLA: 90/85 = 105.9%.
- **D:** Forgot the 15% haircut on Level 2A assets: 130/85 = 152.9%.

Q41

Mock 4 Q74

REBUILT — Original key rested on a vague 'not necessarily' statement and a flawed LCR/NSFR option; rebuilt to test the direction of LCR and NSFR changes when HQLA is swapped for loans.

QBCo, a bank, sells a large part of its holdings of government bonds, which are Level 1 high-quality liquid assets with a low (5%) required stable funding factor, and uses the proceeds to extend new one-year corporate loans, which are not HQLA and carry an 85% required stable funding factor. Its funding and its 30-day net cash outflows are unchanged. What is the effect on QBCo's Basel III liquidity ratios?

A. The LCR rises and the NSFR falls

B. The LCR falls and the NSFR rises

C. Both the LCR and the NSFR fall

D. Both the LCR and the NSFR rise

Answer: C

Step by step

1. LCR = HQLA/net 30-day outflows: selling HQLA bonds lowers the numerator while the denominator is unchanged, so the LCR falls.
2. NSFR = ASF/RSF: funding (ASF) is unchanged; each unit moved from bonds (5%) to loans (85%) raises RSF by 0.80, so the NSFR falls.

Why C is correct: The swap removes HQLA (lower LCR numerator) and raises the stable funding the asset side requires (higher RSF), so both ratios deteriorate.

Why the others are wrong

- **A:** Wrongly assumes loans improve the LCR; they are not HQLA, so the LCR falls.
- **B:** Correct that the LCR falls (HQLA is sold), but wrong on the NSFR: loans carry a higher RSF factor (85%) than bonds (5%), so RSF rises and the NSFR falls, not rises.
- **D:** Ignores that the transaction replaces liquid assets with illiquid ones, which hurts both ratios.

Q42 AnalystPrep Q.3948

REBUILT — Option C had a typo ('lines of deposits') and the options were partly meaningless; rebuilt as complete stress-scenario lists matching the GARP description.

The 30-day period in the liquidity coverage ratio (LCR) calculation is one of acute stress that includes a credit downgrade of three notches (e.g., from AA+ to A+). Which of the following lists correctly the other stress events in the scenario?

A. Partial loss of deposits, complete loss of wholesale funding, higher haircuts on secured funding, and drawdowns on lines of credit

B. Complete loss of deposits, partial loss of wholesale funding, higher haircuts on secured funding, and drawdowns on lines of credit

C. Partial loss of deposits, complete loss of wholesale funding, lower haircuts on secured funding, and drawdowns on lines of credit

D. Partial loss of deposits, complete loss of wholesale funding, higher haircuts on secured funding, and full renewal of credit lines with no drawdowns

Answer: A

Step by step

1. The LCR stress scenario consists of: a three-notch downgrade, a partial loss of deposits, a complete loss of wholesale funding, increased haircuts on secured funding, and drawdowns on lines of credit.
2. Compare each option: only A has all four correct; B reverses the deposit and wholesale assumptions, C has lower haircuts, D has no drawdowns.

Why A is correct: The stress is a severe funding shock: deposits partly leave, wholesale funding disappears entirely, secured funding haircuts rise, and customers draw on credit lines.

Why the others are wrong

- **B:** Swaps the severity: deposits are lost only partially and wholesale funding completely, not the reverse.
- **C:** Haircuts increase, not decrease, in stress (lower haircuts would ease secured funding).
- **D:** Drawdowns on lines of credit are part of the stress; assuming no drawdowns understates outflows.

Q43 Schweser Module Quiz 66.2 Q2

Which of the following factors must be taken into account in the net cash outflows in a 30-day period component of the liquidity coverage ratio (LCR) as required by Basel III?

A. Complete losses on deposits.

B. Partial losses of wholesale funding.

C. Larger haircuts on secured funding.

D. Two notch reductions in credit ratings.

Answer: C

Step by step

1. The LCR is $\frac{\text{High-quality liquid assets}}{\text{Net cash outflows over 30 days}} \geq 100\%$.

2. The 30-day denominator assumes a severe stress scenario with specified shocks.
3. Basel III stress assumptions: partial losses of retail/other deposits, 100% loss of wholesale funding, line-of-credit drawdowns, a three-notch credit rating downgrade, and higher haircuts on secured funding.
4. Matching each option: only larger haircuts on secured funding is a stated assumption, so the answer is C.

Why C is correct: The LCR stress scenario explicitly includes higher haircuts on secured funding, along with partial deposit losses, total wholesale funding loss, credit line drawdowns and a three-notch downgrade.

Why the others are wrong

- **A:** Deposits are assumed to suffer only partial (run-off) losses, not complete losses; confuses deposit treatment with wholesale funding.
- **B:** Wholesale funding is assumed to be lost entirely (100%), not partially; reverses the deposit and wholesale treatments.
- **D:** The scenario assumes a three-notch downgrade, not two; a plausible but incorrect number.

LO 66.d

Explain liquidity black holes and identify the causes of positive feedback trading.

Q44

AnalystPrep Q.3949

REBUILT — Original asked for a vague 'reason for positive trading' with a questionable key (predatory trading); rebuilt to test the Schweser causes of positive feedback trading with exactly one correct option.

A risk analyst is reviewing behaviors that can cause positive feedback trading and, in extreme cases, a liquidity black hole. Which of the following is an example of a behavior that gives rise to positive feedback trading?

A. A fund rebalances to its target weights by buying an equity whose price has fallen.
B. A trader sells automatically when an asset's price falls below a pre-set stop-loss level.
C. A holder of a long option position delta-hedges by selling the underlying after a price increase.
D. A market maker narrows its bid-ask spread as trading volume increases.

Answer: B

Key correction: Source key (D, predatory trading) is only one of several causes and the original stem did not isolate it, so the item was rebuilt around an unambiguous cause (stop-loss rules).

Step by step

1. Positive feedback traders buy when prices rise and sell when prices fall, reinforcing the price move.
2. Schweser causes: stop-loss rules, trend/breakout trading, predatory trading, hedging of short option positions, margin calls, leverage/deleveraging.
3. Option B: a stop-loss triggers a sale after a price decline, pushing prices lower: positive feedback.
4. Options A and C trade against the price move (negative feedback); D is a sign of improving liquidity, not a trading rule that amplifies moves.

Why B is correct: A stop-loss rule forces selling after prices fall, which pushes prices lower and triggers further sales, so it amplifies trends and drains liquidity.

Why the others are wrong

- **A:** Buying after a price fall is negative feedback trading, which stabilizes prices; the mistake is treating any re-balancing as destabilizing.
- **C:** Dynamic hedging of a LONG option sells after rises and buys after falls (negative feedback); only hedging of SHORT option positions is positive feedback.
- **D:** Narrower spreads reflect better liquidity and are not a trading behavior that amplifies price moves.

Q45

BT answers P2.T8.20.3.3 · BT notes P2.T8.20.3.3

Peter's startup company has built a new exchange platform that enables participants to trade derivatives on certain cryptocurrencies. Peter knows that market participants have a strong preference for continuous liquidity, and conversely, participants will avoid a market that suffers a lack of liquidity. Peter is preparing a marketing brochure to promote the new exchange. If his goal is to promote the exchange's well-functioning liquidity features, which of the following features is most likely to be promoted in the brochure; i.e., which feature contributes to a stable market with respect to liquidity?

A. Stop loss rules
B. Trend trading

C. Liquidity black hole
D. Negative feedback traders

Answer: D
Step by step
1. A liquid, stable market has a balance of buyers and sellers, dominated by negative feedback traders (buy on declines, sell on rises).
2. Stop-loss rules sell as prices fall: positive feedback.
3. Trend trading buys rising / sells falling prices: positive feedback.
4. A liquidity black hole is the outcome when everyone wants the same side; it is a liquidity failure, not a feature.
5. Only negative feedback traders stabilize prices and supply liquidity: D.

Why D is correct: Negative feedback traders trade against price moves, supplying counterparties on both sides and keeping prices stable.

Why the others are wrong
- **A:** Stop-loss rules force sales into falling markets (positive feedback), draining liquidity.
- **B:** Trend traders follow price direction, amplifying moves and removing counterparties.
- **C:** A liquidity black hole is dried-up liquidity from one-sided trading, the opposite of a well-functioning feature.

Q46 GARP PE2 Q28 · Mock 4 Q29

A national regulator is developing a set of indicators to serve as warning signs that the liquidity of the stock market could be decreasing. The regulator is particularly concerned about the potential for a "liquidity black hole" to develop which could destabilize the stock market. Which of the following observations would be of greatest concern to the regulator as a signal of a potential liquidity black hole occurring?

A. A majority of large investors increasingly follow a trend-following trading strategy.
B. Brokerage firms purchase call options as a hedge against the sale of a large number of put options to institutional investors.
C. Portfolio managers sell stocks that have increased in value and reached their price target.
D. Bid-ask spreads for less liquid stocks become narrower.

Answer: A
Key correction: GARP PE2 key listed B in one place while its own explanation supports A; correct is A because trend following is positive feedback trading.
Step by step
1. A liquidity black hole arises when most participants want the same side of the market, which happens when positive feedback traders dominate.
2. A: if most large investors are trend followers, they all sell into declines and buy into rallies, leaving no counterparties: greatest concern.
3. B: buying call options to offset sold puts is an options-based hedge and does not force one-directional trading of the underlying with the price move.
4. C: selling after a rise is negative feedback and stabilizing.
5. D: narrowing spreads signal improving liquidity.

Why A is correct: Widespread trend following is positive feedback trading, so buyers vanish as prices fall and a black hole can develop.

Why the others are wrong
- **B:** Buying calls to offset sold puts is an option-for-option hedge, not a sale of the underlying after price declines; unlike underlying-based hedging of short options, it does not create trend-following flow.
- **C:** Selling into strength is negative feedback trading and supports liquidity.
- **D:** Narrower spreads indicate better, not worse, liquidity.

Q47 Schweser Module Quiz 66.3 Q1

A highly liquid market tends to result from situations where:

A. stop-loss rules take effect as prices decline.
B. negative feedback traders sell shares as prices rise.
C. positive feedback traders purchase shares as prices rise.
D. breakout trading occurs because of prices moving outside of a range.

Answer: B

Step by step

1. Liquid markets have a balance of buyers and sellers with negative feedback traders dominant.
2. Negative feedback traders sell as prices rise and buy as they fall, keeping prices stable: B.
3. A, C and D describe positive feedback behavior that reinforces price moves.

Why B is correct: Selling into rising prices (and buying into falling prices) provides counterparties and dampens price swings, which sustains liquidity.

Why the others are wrong

- **A:** Stop-loss selling as prices fall amplifies the decline and reduces liquidity.
- **C:** Buying as prices rise is positive feedback and pushes prices further from stability.
- **D:** Breakout trading follows price moves outside a range, reinforcing trends and contributing to illiquidity.

Reading 67. Liquidity and Leverage

Allan Malz, Financial Risk Management: Models, History, and Institutions (Hoboken, NJ: John Wiley & Sons, 2011). Chapter 12.

LO	Learning objective	Qs
67.a	differentiate between sources of liquidity risk and describe specific challenges faced by different types of financial institutions in managing liquidity risk.	16
67.b	summarize the asset-liability management process at a fractional reserve bank, including the process of liquidity transformation.	10
67.c	compare transactions used in the collateral market and explain risks that can arise through collateral market transactions.	3
67.d	describe the relationship between leverage and a firm's return profile (including the leverage effect), and explain the impact of different types of transactions on a firm's leverage and balance sheet.	19
67.e	describe and compare methods to measure and manage funding liquidity risk and transactions liquidity risk.	10
67.f	calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days.	13
67.g	discuss interactions between different types of liquidity risk and explain how liquidity risk events can increase systemic risk.	3

LO 67.a

Differentiate between sources of liquidity risk and describe specific challenges faced by different types of financial institutions in managing liquidity risk.

Q48

AP Mock 1 Q70

Banks borrow short, taking customer deposits, and lend long, issuing mortgages. This is maturity transformation. During the 2007-08 crisis many banks failed because they could not meet the demand of customers wishing to withdraw deposits. From an analyst's perspective, this most likely represents:

A. Balance sheet risk.
B. Transactions liquidity risk.
C. Systematic risk.
D. Maturity transformation risk.

Answer: A

Step by step

1. Depositors withdrawing funds are creditors withdrawing credit from the bank.
2. Creditors withdrawing or changing the terms of credit is funding liquidity risk, also called balance sheet risk.
3. The bank's problem is meeting its liabilities as they fall due, not the price impact of selling an asset, so transactions liquidity risk does not fit.

Why A is correct: Funding (balance sheet) liquidity risk arises when creditors, here depositors, withdraw credit; it is magnified by the maturity mismatch of funding long assets with short liabilities.

Why the others are wrong

- **B:** Transactions liquidity risk is the risk of an adverse price move when trading an asset; the failure here was inability to meet withdrawals, not trading cost.
- **C:** Systematic (undiversifiable market) risk is not a liquidity risk category in this reading; confuses it with systemic risk.
- **D:** Maturity transformation is the activity that creates the exposure, not a risk category in the reading's taxonomy.

Q49

AnalystPrep Q.2255

REBUILT — Original asked for a loose definition of liquidity risk where options B and D were both defensible; rebuilt to test the funding versus transactions liquidity distinction from the reading.

Which of the following statements best distinguishes funding liquidity risk from transactions liquidity risk?

A. Funding liquidity risk is the risk that buying or selling an asset moves its price adversely; transactions liquidity risk is the risk that creditors withdraw credit or tighten its terms.
B. Funding liquidity risk is related to a borrower's creditworthiness and arises when creditors withdraw credit or change its terms; transactions liquidity risk is the risk that the act of buying or selling an asset moves its price adversely.

C. Funding liquidity risk arises only from market-wide stress, whereas transactions liquidity risk arises only from a firm's own credit deterioration.
D. Funding liquidity risk is the risk that the overall financial system is impaired, whereas transactions liquidity risk is specific to one institution.

Answer: B
Key correction: Source key (AnalystPrep) said D; D is a loose definition and B ('inability to meet financial needs') is also defensible, so the item was rebuilt.
Step by step
1. Transactions (market) liquidity risk: buying or selling an asset causes an adverse price move; it concerns assets and markets.
2. Funding (balance sheet) liquidity risk: concerns creditworthiness; creditors may withdraw credit or change terms (e.g., higher collateral).
3. Only B pairs each risk with its correct description.

Why B is correct: Transactions liquidity relates to assets and markets (price impact of trading), while funding liquidity relates to the borrower's credit and its ability to obtain or keep financing.
Why the others are wrong
- **A:** Swaps the two definitions.
- **C:** Funding liquidity risk can arise from the firm's own credit deterioration or from market-wide deterioration; the statement wrongly restricts and reverses the causes.
- **D:** Describes systemic risk, not funding liquidity risk; transactions liquidity risk is not limited to one institution.

Q50 AnalystPrep Q.2276

Transaction liquidity risk can be best defined as:
A. The risk of reducing the demand for an asset by increasing the number of transactions.
B. The risk of moving the price of an asset adversely in the act of buying or selling it.
C. The risk of moving the supply of an asset adversely in the act of buying or selling it.
D. The risk that too many transactions on the trading floor could render the firm unable to meet its day-to-day operation costs.

Answer: B
Step by step
1. The reading defines transactions (market) liquidity risk as the risk that the act of buying or selling an asset results in an adverse price move.
2. B matches this definition; the others describe demand/supply effects or operating costs, not price impact.

Why B is correct: Transactions liquidity risk is the price impact of trading itself; it is low when positions can be unwound quickly, cheaply and without moving the price much.
Why the others are wrong
- **A:** Refers to demand reduction from transaction count, not the adverse price move from trading.
- **C:** Supply is not what moves adversely; the price does.
- **D:** Describes an operational/funding concern about day-to-day costs, not price impact.

Q51 AnalystPrep Q.2277

Nathan James, a trader for Nathan Capital, works on the short-term funding desk at his firm. Over the past few months, markets have been highly volatile but Nathan Capital still enjoys a large capital base and is financially stable. In his monthly report to the liquidity subcommittee of the board of directors, James reports that in the last month, Nathan Capital's chief lender has been steadily increasing collateral requirements to roll over repo contracts. From the perspective of Nathan Capital, this represents:
A. Transactions liquidity risk
B. Balance sheet risk
C. Systematic risk
D. Maturity transformation risk

Answer: B
Step by step

1. A lender raising collateral requirements to roll repo is a creditor changing the terms of credit.
2. The reading lists raising required collateral as a classic example of funding liquidity (balance sheet) risk.
3. No asset sale or price impact is described, so it is not transactions liquidity risk.

Why B is correct: Creditors changing credit terms (higher collateral) is funding liquidity or balance sheet risk, which can force positions to be unwound.

Why the others are wrong

- **A:** No trade price impact is described; this may arise later if positions must be sold, but the event itself is a funding change.
- **C:** Systematic risk is market-wide, undiversifiable risk; the stem describes a lender's behavior toward this firm.
- **D:** Maturity transformation is the activity (funding long with short) that makes the firm exposed, not the risk event described.

Q52 AnalystPrep Q.2278

Yemi Bank was recently ordered to pay a hefty fine as punishment for engaging in certain illegal trades. Following this event, creditors have expressed fears over their investment and would want to introduce more conditions governing the use of funds lent to the bank. This scenario gives an example of:

A. Liquidity credit risk
B. Strategic risk
C. Systemic risk
D. Balance sheet risk

Answer: D

Step by step

1. The bank's perceived creditworthiness has deteriorated after the fine.
2. Creditors respond by changing the terms of credit (more conditions) - the definition of funding liquidity (balance sheet) risk.
3. The problem is specific to Yemi Bank, not the entire financial system, so not systemic risk.

Why D is correct: Perceived credit deterioration leads creditors to tighten or withdraw funding, which is balance sheet risk.

Why the others are wrong

- **A:** Not a risk category in the reading.
- **B:** Strategic risk concerns business-strategy outcomes; the stem is about creditors' reaction on funding.
- **C:** Systemic risk is impairment of the whole system; this is a single-bank event.

Q53 AnalystPrep Q.2279

Elipsa Bank has witnessed a dramatic deterioration of the credit quality of its borrowers in its loan portfolio. As a result, the bank's balance sheet position has worsened. The bank will most likely have to contend with:

A. Systemic risk
B. Funding liquidity risk
C. Reputation risk
D. Strategic risk

Answer: B

Step by step

1. Weaker asset quality worsens the bank's balance sheet and perceived creditworthiness.
2. The reading says funding liquidity risk results when a borrower's credit position deteriorates or is perceived to deteriorate; creditors may withdraw or tighten credit.
3. So the bank most likely faces funding liquidity risk.

Why B is correct: Deteriorating credit position makes funders demand higher costs, more collateral or withdraw, i.e., funding liquidity risk.

Why the others are wrong

- **A:** Only one bank's portfolio is affected; nothing indicates system-wide impairment.
- **C:** Reputation risk is not the direct consequence tested; it is not a liquidity-risk category in the reading.
- **D:** Strategic risk relates to business strategy, not the funding impact of weaker asset quality.

Q54 AnalystPrep Q.2280

Egda Bank has had a rough year in which its financial health and performance substantially deteriorated. An intensive study of the market by its risk management department has attributed the deterioration to bad market conditions. The study also found out that a majority of players in the banking sector have had almost identical problems. This scenario gives an example of:

A. Maturity transformation risk
B. Balance sheet risk
C. Systemic risk
D. Banking sector liquidity risk

Answer: C

Step by step

1. The cause is market-wide conditions and most banks suffer the same problems.
2. Systemic risk is the risk that the overall financial system is impaired by severe stress.
3. A bank-specific balance sheet problem would be idiosyncratic; here it is common to the sector, so systemic risk.

Why C is correct: When distress stems from market conditions and is shared across most institutions, the risk is systemic rather than firm-specific.

Why the others are wrong

- **A:** Maturity transformation is a bank activity, not the shared market-wide risk described.
- **B:** Balance sheet (funding liquidity) risk is the risk that creditors withdraw or tighten credit to a borrower; the stem describes sector-wide distress driven by market conditions, which is best classed as systemic risk.
- **D:** Not a defined risk category in the reading.

Q55 AnalystPrep Q.3875 · BT answers P2.T7.508.1 · BT notes P2.T8.508.1

Which of the following is "the risk of moving the price of an asset adversely in the act of buying or selling it" such that this risk is "low if assets can be liquidated or a position can be covered quickly, cheaply, and without moving the price too much"?

A. Transactions liquidity risk
B. Balance sheet risk
C. Funding liquidity risk
D. Systemic risk

Answer: A

Step by step

1. The quoted definition is the price impact of trading an asset.
2. The reading calls this transactions (market) liquidity risk.
3. Balance sheet and funding liquidity risk are synonyms concerning credit withdrawal; systemic risk concerns the financial system.

Why A is correct: Adverse price movement caused by the act of trading is transactions (market) liquidity risk.

Why the others are wrong

- **B:** Balance sheet risk is the same as funding liquidity risk: creditors withdrawing or changing credit terms.
- **C:** Funding liquidity risk is about creditworthiness and obtaining financing, not trade price impact.
- **D:** Systemic risk is impairment of the overall financial system.

Q56 BT answers P2.T7.511.1 · BT notes P2.T8.511.1

REBUILT — Merged two identical items and replaced the awkward option 'Depth of resiliency' with a clear market-liquidity characteristic that is not a cause.

Each of the following is a source (i.e., cause) of transactions liquidity risk EXCEPT which one?

A. Inventory management by dealers
B. Adverse selection
C. The resiliency of quoted prices after a large trade
D. Differences of opinion among market participants

Answer: C

Step by step

1. The market microstructure sources of transactions liquidity risk are trade processing costs, inventory management, adverse selection, and differences of opinion.
2. Dealers must hold inventory and be compensated for volatility exposure (A is a source).
3. Dealers face informed traders and are compensated through the bid-ask spread (B is a source).
4. Investors disagree on prices and interpretation of news; agreement makes counterparties harder to find (D is a source).
5. Resiliency (how quickly prices recover after a trade) is a characteristic that describes how liquid a market is, not a cause of transactions liquidity risk, so C is the exception.

Why C is correct: Resiliency is an observable dimension of market liquidity (alongside tightness and depth), whereas the four microstructure fundamentals are the underlying causes of transactions liquidity risk.

Why the others are wrong

- **A:** Is a cause: dealers carrying long or short inventory demand price concessions for the volatility exposure. Picking it means forgetting inventory management is a microstructure source.
- **B:** Is a cause: dealers cannot tell informed from noise traders and widen spreads for lemons risk.
- **D:** Is a cause: differences of opinion determine how easily counterparties are found; picking it ignores this fundamental.

Q57

BT answers P2.T8.24.1.1

Henry Hoover, an officer at the Financial Conduct Authority (FCA), is categorizing the liquidity challenges of different types of financial institutions. He has three descriptions but has not matched them to institution types:
1. These institutions borrow short-term to finance longer-term assets and face significant rollover risk, especially when refinancing becomes difficult during stress.
2. These provide highly liquid claims to investors but invest in slightly less liquid short-term securities, which led to runs during the financial crisis.
3. These face complex liquidity risk from the terms under which investors can withdraw funds, and can face redemption pressure even when performing well.
What is the correct order of headings for descriptions 1, 2 and 3?

A. Banks and securities firms; money market mutual funds; hedge funds.
B. Money market mutual funds; hedge funds; banks and securities firms.
C. Money market mutual funds; banks and securities firms; hedge funds.
D. Hedge funds; money market mutual funds; banks and securities firms.

Answer: A

Step by step

1. Description 1: short-term borrowing funding longer-term assets with rollover (cliff) risk is the maturity transformation of banks and securities firms.
2. Description 2: MMMFs offer on-demand redemption at a $1.00 NAV while holding high-quality but less liquid short-term paper; they are subject to runs and can break the buck.
3. Description 3: hedge funds face investor withdrawal terms and redemptions even after good performance.
4. The heading order is banks/securities firms, MMMFs, hedge funds, which is A.

Why A is correct: Each institution type has a distinct liquidity profile: banks (rollover risk), MMMFs (liquid liabilities versus slightly less liquid assets, runs), hedge funds (redemption terms and pressure).

Why the others are wrong

- **B:** Assigns the rollover description to MMMFs and the redemption description to banks; confuses who funds long assets with short borrowing.
- **C:** Matches MMMFs to the rollover description (1) and banks to the run description (2); the headings for 1 and 2 are swapped.
- **D:** Reverses the order entirely, treating the redemption-terms description as rollover risk of hedge funds.

Q58

BT notes P2.T8.508.3 · BT answers P2.T7.508.3

REBUILT — Merged two identical items; the original included mortgage-loan and statistical-arbitrage investors, which are not supported by the Schweser reading, so the item was rebuilt around the strategies the reading names, with a funding-insensitive contrast.

Consider the following investors:
I. Leveraged buyout (LBO) investors
II. Merger arbitrage hedge funds
III. Convertible arbitrage hedge funds

IV. An unlevered investor holding fully equity-funded, long-term Treasury bonds to maturity
Which of these were materially exposed to systematic funding liquidity risk during the 2007-2009 subprime crisis?

A. I and II only
B. II and III only
C. I, II and III only
D. I, II, III and IV

Answer: C
Step by step

1. LBOs: deals fell apart and loans were hung as funding dried up, so I is exposed.
2. Merger arbitrage: mergers were abandoned when financing dried up, causing large losses, so II is exposed.
3. Convertible arbitrage: relies on broker-dealer credit and leverage; financing withdrawal pushed convertible values down, so III is exposed.
4. IV needs no refinancing and has no leverage, so it is not exposed to a funding freeze.
5. Exposed set = I, II and III only.

Why C is correct: Strategies that depend on leverage or continued availability of credit are exposed to systematic funding liquidity risk in addition to their idiosyncratic risks; an unlevered, equity-funded holder is not.

Why the others are wrong

- **A:** Omits convertible arbitrage, which is explicitly leverage-dependent and hit by lost broker-dealer financing.
- **B:** Omits LBOs, which suffered when funding dried up and loans could not be distributed.
- **D:** Includes the unlevered, fully equity-funded investor, who has no rollover or financing exposure.

Q59 BT notes P2.T8.509.1

What is the phenomenon called ”breaking the buck”?

A. When the net asset value (NAV) of a money market mutual fund falls below one dollar ($1.00)
B. When a money market mutual fund cannot borrow dollars at a cost less than its return on assets
C. When so many money market mutual funds borrow from the central bank that interest rates on the dollar rise quickly
D. When a high proportion of shareholders attempt to redeem their shares simultaneously under adverse market conditions

Answer: A
Step by step

1. MMMFs use amortized cost accounting and set a notional share value of $1.00.
2. Credit write-downs or forced sales at a loss can push NAV below $1.00; this is breaking the buck, so A is correct.
3. A run (D) can cause forced sales and lead to breaking the buck, but it is the cause, not the phenomenon itself.

Why A is correct: Breaking the buck is a fall in an MMMF’s NAV below its $1.00 notional value.

Why the others are wrong

- **B:** Describes a borrowing-cost squeeze, unrelated to NAV; invents a funding-spread definition.
- **C:** Describes central bank borrowing pushing up rates; no such definition exists.
- **D:** Describes a run on the fund, which can trigger breaking the buck but is not the same thing; confuses cause with outcome.

Q60 Schweser Module Quiz 67.1 Q1

Jackson Grimes, a trader for Glenn Funds, works on the repurchase agreement (repo) desk at his firm. Markets have been highly volatile, but Glenn Funds has a large capital base and is sound. Grimes reports to the CEO that in the last month, the firm Glenn Funds borrows from has been consistently increasing collateral requirements to roll over repos. From the perspective of Glenn Funds, this represents:

A. systematic risk.
B. transactions liquidity risk.
C. balance sheet risk.
D. maturity transformation risk.

Answer: C
Step by step

1. The lender is changing the terms of credit (higher haircut) at rollover.
2. Funding liquidity risk, also called balance sheet risk, arises when creditors withdraw credit or change terms, even if the borrower is sound.
3. So the risk to Glenn is balance sheet (funding liquidity) risk.

Why C is correct: Tightening collateral terms at rollover is a funding liquidity (balance sheet) risk because the borrower's ability to finance its positions is impaired.

Why the others are wrong

- **A:** Systematic risk is market-wide return risk; the item describes a change in credit terms to one borrower.
- **B:** Transactions liquidity risk concerns the cost of trading assets without moving prices; nothing here involves selling assets.
- **D:** Maturity transformation is the structural mismatch of short funding and long assets; the question is about the rising collateral terms, not the mismatch.

Q61

Schweser Module Quiz 67.1 Q2

Chris Clayton, an analyst for a private equity fund, noticed that merger arbitrage strategies at several hedge funds experienced large losses in late 2007 to early 2008. These losses were likely due to:

A. abandoned merger plans due to a lack of available financing.
B. target prices falling precipitously due to stock market corrections.
C. acquirers filing for bankruptcy as the subprime mortgage crisis unfolded.
D. idiosyncratic risks surrounding the merger arbitrage strategy.

Answer: A

Step by step

1. Merger arbitrage earns the gap between current and announced acquisition prices.
2. In the early subprime crisis, systematic funding liquidity risk caused financing to dry up and mergers to be abandoned.
3. Abandoned deals collapsed the spread, causing large losses: A.

Why A is correct: Losses came from systematic funding risk: deal financing disappeared, so announced mergers were abandoned.

Why the others are wrong

- **B:** Market corrections are not the cited cause; losses arose when deals failed, not generally falling target prices.
- **C:** Acquirer bankruptcies were not the cited mechanism; the issue was lost financing before completion.
- **D:** The reading stresses the losses reflected systematic funding risk, not merely idiosyncratic risk.

Q62

BT answers P2.T7.509.1

What is the phenomenon called "breaking the buck"?

A. When the net asset value (NAV) of a money market mutual fund falls below one dollar ($1.00)
B. When a money market mutual fund cannot borrow dollars at a cost less than its return on assets
C. When so many money market mutual funds borrow from the central bank that interest rates on the dollar rise quickly
D. When a high proportion of shareholders attempt to redeem their shares simultaneously under adverse market conditions

Answer: A

Step by step

1. MMMFs set a notional share value of $1.00 under amortized-cost accounting (SEC Rule 2a-7).
2. Credit write-downs, or forced sales at a loss during a run, can push the NAV below $1.00.
3. That event, NAV below $1.00, is called breaking the buck.

Why A is correct: Breaking the buck is defined by the NAV of a money market fund dropping below its $1.00 notional value.

Why the others are wrong

- **B:** Describes a negative-carry funding situation, not a NAV outcome.
- **C:** Describes central-bank borrowing driving up rates; unrelated to the fund's NAV.
- **D:** A run is a cause that can lead to breaking the buck, but it is not the definition of the term.

Q63 Schweser Module Quiz 67.1 Q3

With respect to the valuation of money market mutual fund (MMMF) assets, funds:

A. are not required to mark-to-market the underlying assets daily.

B. must reflect changes in the values of underlying assets that are the result of changes in credit risks but may ignore value changes that are the result of changes in interest rates.

C. will set the notional values of each of the underlying assets equal to $1.00.

D. are not allowed to invest in any asset with a rating below AAA because asset values must not fluctuate outside of a 10% range around the historical value in order to keep the notional value equal to $1.00.

Answer: A

Step by step

1. MMMFs use amortized-cost accounting under SEC Rule 2a-7.
2. Hence assets need not be marked to market daily, unlike other mutual funds, because very short-term securities rarely revalue much.
3. The $1.00 notional value applies to each fund share, and credit write-downs can still push NAV below $1.00.

Why A is correct: Rule 2a-7 amortized-cost accounting exempts MMMF assets from daily mark-to-market.

Why the others are wrong

- **B:** No such credit-versus-rate split exists in the rule; the actual feature is the exemption from daily marking, though credit write-downs cannot be ignored.
- **C:** The $1.00 notional value is set per fund share, not per underlying asset.
- **D:** Invented AAA/10% restriction; MMMFs hold high-quality short-term paper but no such rule is described.

LO 67.b

Summarize the asset-liability management process at a fractional reserve bank, including the process of liquidity transformation.

Q64 GARP PE2 Q27

REBUILT — Original option B ('issuing securitized mortgages') was ambiguous and arguably described the opposite of liquidity transformation; rewrote options so only one action converts demandable liabilities into illiquid assets.

A regional fractional-reserve bank has the following balance sheet:

Assets (CAD million)	Liabilities and equity (CAD million)
Cash and short-term securities 20	Retail and wholesale deposits 70
Residential mortgages 20	Commercial paper 20
2-year industrial loans 35	2-year notes 25
3-year commercial loans 60	Equity capital 20

Which of the following actions best describes liquidity transformation at the bank?

A. Collecting CAD 10 million in wholesale deposits and holding the proceeds as cash

B. Issuing CAD 10 million in 2-year notes and using the proceeds to extend CAD 10 million in 2-year industrial loans

C. Issuing CAD 10 million in equity and using the proceeds to buy short-term securities

D. Collecting CAD 10 million in retail deposits repayable on demand and using the proceeds to extend CAD 10 million in 3-year commercial loans

Answer: D

Step by step

1. Balance sheet check: assets 20 + 20 + 35 + 60 = 135 = liabilities and equity 70 + 20 + 25 + 20 = 135.
2. Liquidity transformation = funding illiquid assets with liabilities that are liquid (repayable at par on demand), so the bank's liabilities are more liquid than its assets.
3. A: liquid liability to liquid asset (cash), no transformation. B: 2-year funding for 2-year assets, matched. C: equity is permanent, not a liquid claim, and short-term securities are liquid. D: demandable deposits fund illiquid 3-year loans, which is liquidity (and maturity) transformation.

Why D is correct: The bank transforms liquidity when it uses liabilities that depositors can withdraw at par on demand to finance loans that cannot be quickly sold or collected.

Why the others are wrong

- **A:** Liquid liability matched with a liquid asset; this is a reserve position, not transformation.
- **B:** Matched-maturity funding of a 2-year loan; no liquidity or maturity gap is created.
- **C:** Equity is permanent capital rather than a demandable claim and the asset bought is liquid; neither side of the transformation is present.

Q65 AnalystPrep Q.2283

Which of the following best explains why a bank may be incentivized to finance long-term loans with short-term deposits, despite the grave liquidity issues this could create?

A. It is much easier to find long-term deposits than short-term deposits.
B. Interest rates on long-term loans are usually higher than the cost of short-term funding.
C. It is much easier to find clients for long-term financing.
D. Long-term facilities are less risky than short-term ones.

Answer: B

Step by step

1. Maturity mismatching is profitable because short-term investors bear less risk and require a lower return, so short-term funding is cheaper.
2. Long-term assets typically earn higher rates, and the incentive is greatest when the yield curve is upward sloping.
3. The spread (net interest margin) is the reward for taking on rollover risk; only B states this profit motive.

Why B is correct: Borrowing short at low rates and lending long at higher rates widens net interest margin.

Why the others are wrong

- **A:** Reverses reality: short-term (demand) deposits are the abundant, cheap source of funding.
- **C:** Client demand is not the driver of the funding choice; the stem asks about the profit incentive.
- **D:** Long-term facilities are not less risky; they carry more rate, credit and funding-liquidity risk.

Q66 AnalystPrep Q.2284

In a recent discussion between two senior risk managers at a Chinese bank, the topic of rollover risk was raised as a growing concern for their institution. What does the term ”rollover risk” most likely refer to?

A. The bank is increasingly lending to high-risk borrowers, increasing its credit risk exposure.
B. The amount recovered from non-performing loans is declining, increasing the bank’s losses.
C. The bank is increasingly financing long-term loans with short-term deposits, making it hard to repay or refinance its short-term debt.
D. The bank’s portfolio at risk is growing at a relatively higher rate.

Answer: C

Step by step

1. Rollover (cliff) risk is the risk that short-term debt cannot be refinanced, or can only be refinanced at escalating rates.
2. It arises when long-term assets are funded with shorter-term liabilities (maturity mismatch).
3. Only C describes a funding-side refinancing problem; A, B and D are credit-risk concepts.

Why C is correct: Rollover risk is a funding-liquidity risk created by maturity mismatch.

Why the others are wrong

- **A:** Describes credit risk from riskier borrowers, not refinancing risk.
- **B:** Describes a falling recovery rate (loss given default), a credit-risk item.
- **D:** Describes deteriorating asset quality (portfolio at risk), again credit risk.

Q67 BT answers P2.T7.508.2 · BT notes P2.T8.508.2

About the funding liquidity risk of a fractional-reserve bank, Malz asserts each of the following statements as true EXCEPT which statement is not accurate?

A. Funding liquidity risk arises for market participants who borrow short term to finance investments that take longer to become profitable; funding a longer-term asset with a shorter-term liability is called a maturity mismatch.
B. A core function of a commercial bank is to take deposits and lend to firms, transforming long-term illiquid

assets (loans) into short-term liquid liabilities (deposits) that can be used as money.

C. Bank fragility can be mitigated through higher capital (reducing depositors' solvency concerns, the typical trigger of a run) and higher reserves (reducing liquidity concerns).

D. A properly calibrated asset-liability management system can fully immunize a fractional-reserve bank that performs liquidity and maturity transformation, and owes liabilities at par on demand, against a general loss of confidence in its ability to pay depositors.

Answer: D

Step by step

1. A states the definition of maturity mismatch: true.
2. B describes liquidity, maturity and credit transformation by a bank: true.
3. C: higher capital and reserves reduce fragility: true.
4. D is false: a fractional-reserve bank holds only a fraction of deposits as reserves, so if depositors collectively lose confidence and run, no ALM system can fully protect it; it is inherently fragile.

Why D is correct: Because reserves cover only a fraction of demandable liabilities, ALM can reduce but never eliminate vulnerability to a run.

Why the others are wrong

- **A:** True statement (definition of maturity mismatch); not the exception.
- **B:** True statement (bank transformation role); not the exception.
- **C:** True statement (capital addresses solvency fears, reserves address liquidity fears); not the exception.

Q68

GARP PE1 Q20

A treasury manager at a fractional reserve bank is presenting to newly hired analysts on asset-liability management (ALM). The manager explains how liquidity and maturity transformation can be implemented at a bank whose goal is to maximize its net interest margin, and discusses funding-source characteristics and the effectiveness of ALM strategies. Which of the following statements is correct for the manager to make?

A. Short-term wholesale funding is the most reliable and least concentrated source of borrowing available to banks when employing liquidity ALM strategies.

B. Matching the maturity of bank assets with the maturity of bank liabilities is the most appropriate ALM strategy for maximizing the bank's net interest margin.

C. Liquidity and maturity transformation strategies supported by ample buffers of cash and highly liquid assets can be used to maximize the bank's net interest margin.

D. Liquidity and maturity transformation strategies can fully insulate the bank against any non-renewals of short-term debt or depositor runs.

Answer: C

Step by step

1. Net interest margin comes from borrowing short at lower rates and lending long at higher rates, i.e., from transformation.
2. Transformation creates rollover and run risk, so it must be supported by buffers of cash and liquid assets.
3. Check each option: A false (deposits are stickier and more stable than wholesale funding); B false (matching removes the margin); D false (buffers buy time but cannot fully insulate); C true.

Why C is correct: Maximizing margin requires transformation, and prudent transformation is backed by liquid buffers.

Why the others are wrong

- **A:** Reverses reality: wholesale funding is less sticky and more concentrated than retail deposits.
- **B:** Full maturity matching eliminates the term spread, so it minimizes rather than maximizes margin.
- **D:** Overstates ALM: a general loss of confidence can overwhelm any buffer, so full insulation is impossible.

Q69

BT answers P2.T8.24.1.2

REBUILT — Original had two defensible answers (more capital vs stress testing) and a vague regulatory twist; rewrote around a clear liquidity-buffer question with one best answer.

A bank funds 5-year fixed-rate loans mainly with demand deposits and 1-month wholesale borrowing, earning a wide net interest margin. The risk manager wants to reduce vulnerability to deposit runs and non-renewal of wholesale funding while retaining most of the margin from transformation. Which action is most appropriate?

A. Hold a buffer of unencumbered, highly liquid assets sized from stressed funding outflows, and term out part of

the wholesale funding.
B. Replace the 1-month wholesale borrowing with overnight borrowing to lower funding cost.
C. Match every asset and liability maturity exactly, eliminating all term transformation.
D. Raise new equity and invest the proceeds in additional 5-year loans.

Answer: A
Step by step
1. The problem is liquidity risk from transformation: runs and rollover failure.
2. The remedy is to keep the transformation but support it: liquid buffers sized by stressed outflows give time to raise cash, and longer wholesale funding cuts rollover frequency.
3. B shortens funding (more rollover risk); C removes the margin the bank wants to keep; D adds illiquid assets, so liquidity is not improved.

Why A is correct: A liquid-asset buffer plus longer-dated funding reduces run and rollover exposure without eliminating the term spread.
Why the others are wrong
- **B:** Overnight funding must be rolled every day, so it increases rollover risk (it only lowers cost).
- **C:** Would remove liquidity risk but also the net interest margin, contradicting the objective.
- **D:** Equity raises capital, but investing it in illiquid 5-year loans adds illiquid assets and does nothing for liquidity.

Q70 AnalystPrep Q.2281

Aruba Commercial Bank's risk management team is concerned about escalating exposure to funding liquidity risk, the risk of being unable to meet obligations without unacceptable losses. Which of the following scenarios could be the reason for this heightened exposure?
A. Financing short-term loans mostly with short-term deposits.
B. Financing long-term loans mostly with long-term deposits.
C. Financing long-term loans mostly with short-term deposits.
D. Financing short-term loans mostly with long-term deposits.

Answer: C
Step by step
1. Funding liquidity risk rises with maturity mismatch where assets are longer term than liabilities.
2. C funds long-term loans with short-term deposits that can be withdrawn or must be rolled over before the loans are repaid.
3. A and B are matched; D is the reverse mismatch (long liabilities, short assets), which is not a funding-liquidity problem.

Why C is correct: Short-term liabilities funding long-term assets create rollover and withdrawal risk.
Why the others are wrong
- **A:** Maturities are matched; low funding liquidity risk.
- **B:** Maturities are matched; low funding liquidity risk.
- **D:** Liabilities outlast the assets, so cash comes back before it must be repaid; the opposite of the risky mismatch.

Q71 AnalystPrep Q.2282

What is meant by matched funding in the context of lending?
A. Financing long-term loans with borrowed funds.
B. Financing loans with reserve cash saved over a period of time.
C. Financing long-term assets with long-term liabilities.
D. Granting a facility to a qualifying beneficiary after considering their full credit profile.

Answer: C
Step by step
1. Matched funding aligns the maturity of liabilities to that of the assets they finance.
2. So long-term assets are funded with long-term debt, removing the maturity mismatch and rollover risk.
3. A says nothing about maturity; B is self-funding from reserves; D is credit underwriting.

Why C is correct: Matched funding means asset and liability maturities are aligned.
Why the others are wrong

- **A:** Any borrowing qualifies; there is no maturity matching.
- **B:** Describes use of own cash reserves, not maturity alignment.
- **D:** Describes credit assessment, unrelated to funding structure.

Q72 Mock 1 Q66

REBUILT — Original did not say what the bank does with the proceeds, so increase/decrease/either were all defensible; added the proceeds' use and true-sale assumptions.

A retail bank funds 25-year fixed-rate mortgages largely with deposits and short-term wholesale borrowing. It sells a pool of these mortgages in a true sale (no recourse, no liquidity support) to a securitization vehicle, holds the cash proceeds in short-term liquid assets, and originates no new long-term loans. All else equal, what is the effect on the bank's asset-liability maturity mismatch?

A. It increases, because the bank loses stable long-term interest income.
B. It decreases, because a long-dated illiquid asset is replaced by short-term liquid assets.
C. It is unchanged, because the liabilities funding the bank are unchanged.
D. It cannot be determined without knowing the maturity of the deposits.

Answer: B

Step by step

1. Mismatch = long, illiquid assets funded by short, liquid liabilities.
2. The sale removes long-dated mortgages and replaces them with cash and short-term assets, shortening the average asset maturity and raising asset liquidity.
3. Liabilities are unchanged, so the gap between asset and liability maturities narrows (and so does the net interest margin).

Why B is correct: Selling the longest, least liquid assets for cash moves the asset side toward the liability side.

Why the others are wrong

- **A:** Confuses lost income with mismatch; income falls but the maturity gap narrows.
- **C:** The mismatch depends on both sides; a change on the asset side alters it.
- **D:** The direction is determinable: asset maturity falls whatever the deposit maturity.

Q73 AnalystPrep Q.4168

Which of the following is correct about maturity transformation across banks' balance sheets?

A. A maturity mismatch needed to facilitate long-term investment projects should allow banks to earn a spread in a negatively sloped yield curve environment.
B. Banks do not transfer funds from agents in surplus, who demand short-term deposits, to agents in deficit, who have long-term financing needs.
C. Banks might be motivated to increase their maturity mismatch excessively, making themselves vulnerable to funding risk from rolling over short-term liabilities.
D. It mitigates upward pressure and interbank rate volatility on the U.S. dollar.

Answer: C

Step by step

1. Maturity transformation = funding long-term assets with short-term liabilities (borrow short, lend long).
2. The bank earns a spread because long-term rates normally exceed short-term rates (upward-sloping curve).
3. A larger mismatch earns a larger spread, which tempts banks to over-extend; the cost is that short-term liabilities must be rolled over (funding risk).
4. Only statement C describes this trade-off.

Why C is correct: Maturity transformation earns a spread but leaves the bank dependent on rolling over short-term funding, and the spread incentive can push the mismatch too far.

Why the others are wrong

- **A:** The spread is earned when the curve slopes upward; in a negatively sloped (inverted) curve the bank would lose, not earn, the spread.
- **B:** Reverses the intermediation role: banks do channel short-term deposits from surplus agents to deficit agents needing long-term finance.
- **D:** Describes the effect of central bank swap lines on dollar funding stress, not a feature of maturity transformation.

LO 67.c

Compare transactions used in the collateral market and explain risks that can arise through collateral market transactions.

Q74

AnalystPrep Q.3879

REBUILT — Source asked for the haircut with a dollar option as key (25.0); the Schweser convention expresses the haircut as a percentage of collateral, so options were rebuilt as percentages with plausible mis-computations.

An investor borrows $200 in a collateralized loan and posts securities worth $225 as collateral. What is the haircut on this transaction?

A. 11.1%
B. 12.5%
C. 88.9%
D. 25.0%

Answer: A

Key correction: Source key (AnalystPrep) said $25 (the dollar difference); in the Schweser convention the haircut is the percentage of collateral value not lent, 25/225 = 11.1%.

Step by step

1. Collateral value = 225; amount lent = 200.
2. Haircut amount = $225 - 200 = 25$.
3. Haircut (%) = $25/225 = 11.11\%$ of collateral value.
4. Check: fraction lent $1 - h = 200/225 = 88.9\%$, consistent with 'lend $95 against $100' = 5% haircut.

Why A is correct: The haircut is the portion of the collateral's value that is not lent, expressed as a percentage of collateral; it is the cushion by which collateral can fall while the loan remains fully collateralized.

Why the others are wrong

- **B:** Divides the $25 difference by the loan (25/200) instead of by the collateral value.
- **C:** This is the fraction lent (200/225 = 88.9%), i.e. $1 - h$, not the haircut.
- **D:** Reports the dollar difference as a percentage, mixing units ($25 is an amount, not a rate).

Q75

BT answers P2.T7.509.2 · BT notes P2.T8.509.2

In regard to markets for collateral, each of the following is true EXCEPT which definition or statement is false?

A. Margin lending is lending for the purpose of financing a security transaction in which the loan is collateralized by the security.
B. Total return swaps (TRS) are matched pairs of the spot sale and forward repurchase of a security. Both the spot and forward price are agreed now, and the difference between them implies an interest rate.
C. In a securities lending transaction, one party lends a security to another in exchange for a fee, generally called a rebate. The security lender, rather than the borrower, continues to receive dividend and interest cash flows from the security. A common type of securities lending is stock lending, in which shares of stock are borrowed.
D. A haircut ensures that the full value of the collateral is not lent. A haircut of 10.0%, for example, means that if the borrower of cash wants to buy $100.00 of a security, he can borrow only $90.0 from the broker and must put $10.0 of his own funds in the margin account by the time the trade is settled. Similarly, the lender of cash will be prepared to lend $90 against $100 of collateral.

Answer: B

Step by step

1. A: margin loans finance securities purchases and are collateralized by the securities: true.
2. B: a spot sale plus forward repurchase at prices agreed today, with interest implied by the price difference, defines a repurchase agreement (repo), not a TRS: false.
3. C: securities lending is done for a fee (rebate) and the lender continues to receive the dividends/interest: true per the reading.
4. D: a 10% haircut means lending $90 against $100 of collateral: true.

Why B is correct: Statement B gives the definition of a repo; in a TRS one party pays a fee in exchange for the total return on a reference asset.

Why the others are wrong

- **A:** True: this is the standard definition of margin lending.

- **C:** True: the reading states that the securities lender continues to receive dividends and interest and is paid a rebate/fee.
- **D:** True: a 10% haircut means 90 lent against 100 of collateral, with 10 funded by the borrower.

Q76

BT answers P2.T8.24.1.3

REBUILT — Original scenario had a confusing 'avoid vs emphasize' framing with a debatable key (repos carry haircut and rehypothecation risk too); replaced by a self-contained statement question on the Schweser definitions of collateral-market transactions.

Which of the following statements about transactions in the collateral market is correct?

A. In a total return swap, the party providing the total return on the reference asset is effectively short that asset.

B. In a repurchase agreement, the interest on the loan is paid as a separately stated coupon rather than being implied in the spot and forward prices.

C. In a securities lending transaction, the borrower of the securities receives the dividends and interest from the securities and pays the lender a rebate.

D. Regulation T sets the initial margin requirement for securities purchases at 25%.

Answer: A

Step by step

1. TRS: one party pays a fixed fee and receives the total return (income plus capital gains) on a reference asset; the party providing that return is in essence short the asset, so A is correct.
2. Repo: sale at spot with repurchase at a forward price agreed today; interest is implied in the price difference, so B is wrong.
3. Securities lending: the lender continues to receive dividends and interest, so C is wrong.
4. Regulation T: initial margin for securities purchases is 50%, so D is wrong.

Why A is correct: A TRS transfers the asset's total return to the fee payer, which leaves the return provider with the economic exposure of a short position.

Why the others are wrong

- **B:** Confuses a repo with a loan carrying an explicit coupon; the interest is implied by forward minus spot price.
- **C:** Reverses the cash-flow entitlement; the lender keeps receiving dividends and interest.
- **D:** Misremembers the Reg T initial margin, which is 50% (haircut 50%), not 25%.

LO 67.d

Describe the relationship between leverage and a firm's return profile (including the leverage effect), and explain the impact of different types of transactions on a firm's leverage and balance sheet.

Q77

BT answers P2.T8.24.2.2

REBUILT — Stem referred to an unfilled table of question marks and unspecified 'financial conditions'; rewrote it self-contained with the inputs and kept the four financing combinations.

Starling Homes, a real estate investment firm, plans to buy residential property using borrowed funds. House prices are expected to rise 20% over the year. Starling can choose a down payment of 5% or 30% of the purchase price, with the remainder financed by a loan at either 5.5% or 8% per year. Ignore all other costs and income. Which financing option maximizes Starling's return on its own equity?

A. 5% down payment with a loan at 5.5%.

B. 30% down payment with a loan at 8%.

C. 5% down payment with a loan at 8%.

D. 30% down payment with a loan at 5.5%.

Answer: A

Step by step

1. Return on equity = [asset return − (1 − down payment) × loan rate] / down payment.
2. 5% down, 5.5% loan: $(0.20 - 0.95 \times 0.055)/0.05 = 295.5\%$.
3. 5% down, 8% loan: $(0.20 - 0.95 \times 0.08)/0.05 = 248.0\%$.
4. 30% down, 5.5% loan: $(0.20 - 0.70 \times 0.055)/0.30 = 53.8\%$.
5. 30% down, 8% loan: $(0.20 - 0.70 \times 0.08)/0.30 = 48.0\%$. Highest is 5% down at 5.5%.

Why A is correct: The 20% asset return far exceeds either loan rate, so the leverage effect raises ROE as the equity share falls; the lowest down payment and cheapest loan give the largest ROE.

Why the others are wrong

- **B:** Gives the lowest ROE (48.0%); it combines the least leverage with the most expensive debt.
- **C:** Uses maximum leverage (248.0%), but the 8% loan costs more than the 5.5% loan, so it is beaten by A.
- **D:** Gives 53.8%; the 30% down payment means little leverage, so the low rate cannot offset the smaller multiplier.

Q78

AnalystPrep Q.3878

REBUILT — Distractors 25.5%, 27.5%, 33.5% did not follow from realistic errors; replaced with results of common formula mistakes.

ABC Ltd. is a food processing firm with a return on assets of 10%, total assets of $7 and equity capital of $2. The firm's cost of debt is 5%. What is the firm's return on equity?

A. 35.0%
B. 17.5%
C. 22.5%
D. 13.6%

Answer: C

Step by step

1. Leverage $L = 7/2 = 3.5$.
2. ROE $= 3.5 \times 10\% - (3.5 - 1) \times 5\% = 35\% - 12.5\% = 22.5\%$.

Why C is correct: Debt of 5 on equity of 2 costs 12.5% of equity, which is subtracted from the 35% magnified asset return.

Why the others are wrong

- **A:** Ignored the cost of debt (L × ROA).
- **B:** Charged the 5% cost on all assets: 35% − 3.5 × 5% (used L instead of L − 1).
- **D:** Used debt/assets (5/7) as the multiplier: 10% + 0.714 × 5% = 13.6%.

Q79

AnalystPrep Q.2286

Baraba Bank is calculating its leverage ratio. An extract of its balance sheet is as follows:

Item	Amount
Assets	$50 million
Deposits	$30 million
Equity	$20 million

What is the bank's leverage ratio (total assets divided by equity)?

A. 1.00
B. 2.50
C. 1.67
D. 1.50

Answer: B

Step by step

1. Leverage ratio = total assets / equity.
2. $50/20 = 2.5$.

Why B is correct: Leverage in this reading is the multiple of assets to the firm's own equity capital.

Why the others are wrong

- **A:** Treats the balance sheet as unlevered (ratio of 1); ignores the $30 million of deposits.
- **C:** Divided assets by deposits (50/30) instead of by equity.
- **D:** Divided debt by equity (30/20), which is the debt-to-equity ratio and equals leverage minus 1.

Q80

AnalystPrep Q.2288

Barakuda Bank is considering borrowing additional funds to finance an ambitious expansion plan. Under which circumstance is the additional borrowing consistent with the leverage effect, increasing the bank's return on equity?

A. The return on assets equals the cost of the additional borrowing.

B. The cost of the new borrowing equals the cost of the existing borrowing.
C. The cost of the new borrowing is less than the return on assets.
D. The total value of existing assets is less than the total value of existing debt.

Answer: C
Step by step
1. ROE $= L \times \text{ROA} - (L-1) \times r_D$
2. Rewriting, ROE $= \text{ROA} + (L-1)(\text{ROA} - r_D)$, so raising L increases ROE only if ROA $> r_D$.

Why C is correct: Each extra dollar borrowed adds ROA minus the cost of debt to equity returns, which is positive only when ROA exceeds the borrowing cost.
Why the others are wrong
- **A:** If ROA equals the borrowing cost the spread is zero, so leverage adds nothing to ROE.
- **B:** Comparing new borrowing cost with old borrowing cost says nothing about whether the assets earn more than the debt costs.
- **D:** Liabilities exceeding assets means negative equity; this makes borrowing more dangerous, not attractive.

Q81 AnalystPrep Q.3135

REBUILT — Original distractors (2.4, 3.4) did not correspond to realistic mistakes; kept the data but replaced distractors with common errors and clarified the target ROE.

A firm earns a return on assets of 2.5% and its cost of debt is 2%. Its 95% VaR is 2.45% of equity. Management wants a return on equity equal to twice the 95% VaR. What debt-to-equity ratio is required?
A. 0.96
B. 5.80
C. 4.80
D. 8.80

Answer: C
Step by step
1. Target ROE = $2 \times 2.45\% = 4.9\%$.
2. $0.049 = 0.025L - 0.02(L-1) = 0.005L + 0.02$.
3. $L = (0.049 - 0.02)/0.005 = 5.8$.
4. Leverage $L = 1 + D/E$, so $D/E = 5.8 - 1 = 4.8$.

Why C is correct: Solving the leverage-effect formula for L gives assets of 5.8 times equity, which corresponds to debt/equity of 4.8.
Why the others are wrong
- **A:** Ignored the cost of debt (ROE = L × ROA, so L = 1.96) and then subtracted 1.
- **B:** Gave the leverage ratio (assets/equity) of 5.8 instead of debt/equity.
- **D:** Dropped the constant 0.02 in the equation (L = 0.049/0.005 = 9.8) and subtracted 1.

Q82 AnalystPrep Q.3226

A brokerage provides margin loans to U.S. hedge funds. Xenon Hedge Fund wants to take a $3,000,000 equity position while putting up the minimum equity required by the Federal Reserve. Determine the margin loan amount and the leverage ratio of the position.
A. Margin loan $3,000,000; leverage ratio 1.0
B. Margin loan $1,500,000; leverage ratio 1.0
C. Margin loan $1,500,000; leverage ratio 2.0
D. Margin loan $3,000,000; leverage ratio 0.0

Answer: C
Step by step
1. Regulation T sets initial margin at 50%, so the haircut is 50%.
2. Equity = 0.50 × 3,000,000 = 1,500,000; margin loan = 1,500,000.
3. Leverage = assets / equity = 3,000,000/1,500,000 = 2.0.

Why C is correct: A 50% haircut means the fund funds half the position with its own equity, giving leverage of 1/0.5 = 2.

Why the others are wrong

- **A:** Treats the entire position as borrowed with leverage of 1; a 100% loan is not allowed and leverage would not be 1.
- **B:** Correct loan, but leverage 1.0 is the unlevered case; leverage is assets/equity = 2.
- **D:** Assumes a 100% loan and a leverage ratio of zero; 100% financing is not allowed under Regulation T.

Q83

Schweser Module Quiz 67.2 Q2

A broker provides a margin loan to a U.S. hedge fund that puts up the minimum equity required by the Federal Reserve. If the hedge fund wants to take a $250,000 equity position, determine the margin loan amount and the leverage ratio of this position.

A. Margin loan = $125,000; leverage ratio = 1.0.
B. Margin loan = $125,000; leverage ratio = 2.0.
C. Margin loan = $250,000; leverage ratio = 0.0.
D. Margin loan = $250,000; leverage ratio = 1.0.

Answer: B

Step by step

1. The Federal Reserve initial margin is 50% (haircut 50%).
2. Margin loan = $0.50 \times 250{,}000 = 125{,}000$; equity = 125,000.
3. Leverage = $250{,}000/125{,}000 = 2.0$.

Why B is correct: Half the position is funded by equity, so assets are twice equity.

Why the others are wrong

- **A:** Correct loan, but leverage of 1.0 would mean no borrowing at all.
- **C:** Assumes the whole position is financed by the broker, which the 50% minimum equity rule prohibits; leverage cannot be zero.
- **D:** Assumes a 100% loan and leverage of 1.0; both are inconsistent with a 50% equity requirement.

Q84

AnalystPrep Q.3230

REBUILT — Distractors 13.66% and 10.66% were not traceable to realistic errors; replaced with results of common formula mistakes.

A strategist intends to use leverage to increase the returns on a convertible arbitrage strategy. The expected return on assets is 11%. The fund has $16 million invested in the strategy and will finance 60% of it with borrowed funds. The cost of borrowing is 7%. The return on equity is closest to:

A. 6.8%
B. 13.4%
C. 17.0%
D. 10.0%

Answer: C

Step by step

1. Debt = $0.60 \times 16 = 9.6$ million; equity = 6.4 million.
2. Leverage $L = 16/6.4 = 2.5$.
3. ROE $= 2.5 \times 11\% - 1.5 \times 7\% = 27.5\% - 10.5\% = 17.0\%$.

Why C is correct: Equity earns ROA on 2.5 times its capital and pays the 7% cost on the 1.5 times borrowed.

Why the others are wrong

- **A:** Used 11% − 0.6 × 7%, applying the debt share of assets inside a per-equity formula.
- **B:** Used debt/assets (0.6) as the multiplier on the spread: 11% + 0.6 × 4%, instead of debt/equity = 1.5.
- **D:** Charged the 7% cost on all assets: 2.5 × (11% − 7%), i.e., replaced L − 1 with L.

Q85

Schweser Module Quiz 67.2 Q1

Charleston Funds intends to use leverage to increase the returns on a convertible arbitrage strategy. The return on assets (ROA) of the strategy is 8%. The fund has $1,000 invested in the strategy and will finance the investment with 75% borrowed funds. The cost of borrowing is 4%. The return on equity (ROE) is closest to:

A. 4%.
B. 32%.

C. 20%.
D. 12%.

Answer: C
Step by step
1. Debt = 0.75 × 1000 = 750; equity = 250; $L = 1000/250 = 4$.
2. ROE = 4 × 8% − 3 × 4% = 32% − 12% = 20%.
3. Check in dollars: (80 − 30)/250 = 20%.

Why C is correct: Leverage multiplies the 8% ROA by 4 but the 4% interest is paid on 3 times equity.
Why the others are wrong
- **A:** Used the simple spread ROA − cost of debt (8% − 4%) with no leverage multiplier.
- **B:** Multiplied ROA by leverage (4 × 8%) and ignored the cost of borrowing.
- **D:** Added ROA and the cost of debt (8% + 4%) instead of applying the leverage formula.

Q86
AnalystPrep Q.3877

REBUILT — Options lacked units and were random near-misses; rewrote options as percentages tied to common formula errors.
ABC Ltd. is a food processing firm with a return on assets of 10%, total assets of $5 and equity capital of $2. The firm's cost of debt is 3%. What is the firm's return on equity?
A. 25.0%
B. 17.5%
C. 20.5%
D. 14.2%

Answer: C
Step by step
1. Leverage $L = 5/2 = 2.5$.
2. ROE = 2.5 × 10% − (2.5 − 1) × 3% = 25% − 4.5% = 20.5%.

Why C is correct: Debt is 3 on equity of 2, so interest costs 4.5% of equity against a 25% magnified asset return.
Why the others are wrong
- **A:** Ignored the cost of debt (L × ROA).
- **B:** Charged the 3% cost on all assets: 25% − 2.5 × 3% = 17.5% (used L instead of L − 1).
- **D:** Used debt/assets (3/5 = 0.6) as the multiplier: 10% + 0.6 × 7% = 14.2%.

Q87
AnalystPrep Q.3880

A hedge fund has $1,000,000 in cash, corresponding to an initial placement of $1,000,000 in equity by its owners. The hedge fund wants to finance a long position of $1,000,000 worth of stocks at the Reg T margin requirement of 50 percent. A broker facilitates the trade by lending the hedge fund $500,000. Determine the new leverage ratio for the hedge fund (leverage = assets / equity).
A. 0.5
B. 1.0
C. 2.0
D. 1.5

Answer: D
Step by step
1. Before the trade: assets = cash $1,000,000; equity = $1,000,000; leverage = 1.0.
2. The fund pays $500,000 of its own cash and borrows $500,000 from the broker to buy $1,000,000 of stock.
3. After the trade: assets = remaining cash $500,000 + stock $1,000,000 = $1,500,000.
4. Liabilities = margin loan $500,000; equity is unchanged at $1,000,000 (buying at market value creates no gain or loss).
5. Leverage = 1,500,000/1,000,000 = 1.5.

Why D is correct: Margin borrowing grows assets (cash is only partly spent) while equity is unchanged, so leverage rises from 1.0 to Assets/Equity = 1.5.
Why the others are wrong
- **A:** 0.5 is debt-to-equity (500,000/1,000,000), not assets-to-equity.

- **B:** 1.0 is the leverage before the trade; it ignores that the margin loan adds to assets.
- **C:** 2.0 is stock value divided by the equity put into the position (1,000,000/500,000); it ignores the $500,000 of idle cash and the fund's total equity.

Q88

BT Mock Q17

Firm A, Firm B and Firm C are three companies in the bottled water industry. All have a mix of debt and equity. In 2022 all three suffered credit losses that lowered EBIT. Each firm has total assets of USD 1,000 million, borrows at 5%, and pays tax at 25%. Each generated EBIT of USD 100 million before credit losses, and each suffered credit losses of USD 60 million.

Firm	Debt	Equity	Interest expense
A	700	300	35
B	500	500	25
C	300	700	15

Which of the following statements is correct?

A. Firm C has the highest ROE due to its lower equity position.
B. Firm A had the higher ROE because of its large debt position.
C. Without credit losses, Firm A would have had the highest ROE.
D. Without credit losses, Firm B would have had the highest ROE.

Answer: C

Step by step

1. After losses, EBIT = 100 - 60 = 40. ROE = (EBIT − interest)(1 − 0.25)/equity.
2. A: (40 − 35)(0.75)/300 = 1.25%; B: (40 − 25)(0.75)/500 = 2.25%; C: (40 − 15)(0.75)/700 = 2.68%. So C is highest and A lowest after losses.
3. Without losses, EBIT = 100: A: 65(0.75)/300 = 16.25%; B: 75(0.75)/500 = 11.25%; C: 85(0.75)/700 = 9.11%.
4. Before losses ROA = 10% exceeds the 5% cost of debt, so leverage helps and the most levered Firm A is best; after losses ROA = 4% is below 5%, so leverage hurts.

Why C is correct: Leverage amplifies ROE when ROA exceeds the cost of debt (A best at 16.25% without losses) and depresses it when ROA falls below the cost of debt.

Why the others are wrong

- **A:** C does have the highest ROE after losses, but the reason is wrong: C has the largest equity (700), i.e., the least leverage and smallest fixed interest burden.
- **B:** After losses A has the lowest ROE (1.25%), because ROA of 4% is below the 5% cost of debt and its heavy debt magnifies the shortfall.
- **D:** Without losses B (11.25%) is in the middle; the highest ROE is A's 16.25%. This confuses a moderate-leverage firm with the best performer when ROA exceeds the cost of debt.

Q89

BT answers P2.T7.509.3 · BT notes P2.T8.509.3

REBUILT — Merged two copies; OCR-garbled and underivable distractors replaced with options generated by realistic calculation errors (the correct answer 2.6 / 17.0% is unchanged).

Suppose a firm with a simple capital structure has assets of $20.0 million and debt of $10.0 million. Return on assets (ROA) is 9.0% and cost of debt is 4.0%, such that the firm's leverage (assets/equity) is 2.0 and its return on equity (ROE) is 14.0%. If the firm borrows an additional $6.0 million at the same cost of 4.0%, and asset returns are fixed, what is the firm's new leverage and return on equity (ROE)? (Assume the borrowed funds are invested in assets earning the same 9.0% ROA, and equity is unchanged.)

A. Leverage = 1.6 and ROE = 17.0%
B. Leverage = 2.6 and ROE = 23.4%
C. Leverage = 2.6 and ROE = 17.0%
D. Leverage = 2.6 and ROE = 19.4%

Answer: C

Step by step

1. Equity = 20.0 - 10.0 = 10.0 and is unchanged by borrowing.
2. New assets = 20.0 + 6.0 = 26.0; new debt = 16.0; leverage = 26.0/10.0 = 2.6.

3. Asset income = $26.0 \times 9\% = 2.34$; interest = $16.0 \times 4\% = 0.64$; net income = 1.70.
4. ROE = $1.70/10.0 = 17.0\%$. Check: ROE = ROA + (ROA − i) $D/E = 9\% + 5\% \times 1.6 = 17.0\%$.

Why C is correct: Since ROA (9%) exceeds the cost of debt (4%), extra debt on fixed equity raises both leverage (to 2.6) and ROE (to 17.0%) via the leverage effect.

Why the others are wrong

- **A:** 1.6 is debt/equity (16/10), not assets/equity; the ROE is right but the leverage ratio is mis-defined.
- **B:** 23.4% ignores all interest (2.34/10); the interest on the debt must be deducted.
- **D:** 19.4% charges interest only on the old \$10.0 of debt ((2.34 − 0.40)/10) and forgets the interest on the new \$6.0 borrowing.

Q90

BT answers P2.T7.510.1 · BT notes P2.T8.510.3

REBUILT — Merged two copies; OCR-damaged digits repaired and distractors replaced by ones that follow from realistic balance-sheet errors (answer 5.0 unchanged).

On opening day, Lever Brothers Fund LP has a simple economic balance sheet: \$100 in cash and equity of \$100 (an initial placement of \$100 by investors, no debt). Suppose it now adds three derivatives positions:
- A one-month currency forward, in which Lever Brothers is short \$150 against the euro
- An at-the-money (50-delta) three-month long call option on S&P 500 futures, with an underlying index value of \$100
- Short protection on Ford Motor Co. via a five-year credit default swap with a notional amount of \$200

Assume the non-option positions are initiated at market-adjusted prices and spreads (zero NPV), and that the counterparty for all positions is the same prime broker. If the initial margin on the whole derivatives portfolio is \$50, what is the leverage (assets/equity) of the firm's economic balance sheet after the trades?

A. 4.5
B. 5.0
C. 4.0
D. 5.5

Answer: B

Step by step

1. Forward: implicitly a long \$150 euro deposit financed by a \$150 broker loan. Assets +150, liabilities +150.
2. Call option: delta-equivalent = 50% of \$100 = \$50 long index position financed by a \$50 broker loan. Assets +50, liabilities +50.
3. Short CDS protection: equivalent to a long \$200 par FRN financed by a \$200 term loan. Assets +200, liabilities +200.
4. Margin: \$50 of the \$100 cash is posted, so assets show cash \$50 + margin due from broker \$50.
5. Assets = 150 + 50 + 200 + 50 + 50 = 500; liabilities = 400; equity = 100 (zero-NPV positions).
6. Leverage = $500/100 = 5.0$.

Why B is correct: Derivatives are put on the economic balance sheet at their cash-equivalent (delta-equivalent for options) values, each matched by an implicit loan; with margin counted as an asset, assets are \$500 on equity of \$100.

Why the others are wrong

- **A:** 4.5 omits the \$50 margin asset (assets 450), treating posted margin as if it left the balance sheet.
- **C:** 4.0 divides liabilities (\$400) by equity, i.e., debt-to-equity rather than assets-to-equity.
- **D:** 5.5 counts the option at its full \$100 underlying value instead of the \$50 delta equivalent (assets 550).

Q91

BT answers P2.T7.510.2 · BT notes P2.T8.510.1

REBUILT — Merged two copies; OCR-uncertain options replaced with distractors derived from realistic mistakes (answer unchanged).

On opening day, Lever Brothers Multistrategy Master Fund LP has the following economic balance sheet: \$100 in cash, \$20 in debt, and equity of \$80 (an initial placement of \$80 by the owners plus a \$20 loan from a commercial bank). Lever Brothers finances a long position in \$100 worth of an equity at the Reg T margin requirement of 50%. It invests \$50 of its own funds and borrows \$50 from the broker. The broker retains custody of the stock as collateral. If the firm's leverage ratio is defined, per Malz, as assets/equity, what is the change in the firm's leverage ratio?

A. From 1.250 to 1.250
B. From 1.250 to 1.875
C. From 0.250 to 0.875

D. From 1.250 to 2.500

Answer: B

Step by step

1. Initial leverage = 100/80 = 1.250.
2. After the trade: cash falls to $50; stock of $100 is added. Assets = 50 + 100 = 150.
3. Liabilities = $20 bank debt + $50 margin loan = $70; equity is unchanged at $80.
4. New leverage = 150/80 = 1.875.

Why B is correct: Margin borrowing adds $50 to assets (stock $100 less $50 cash used) without changing equity, so leverage rises from 1.250 to 1.875.

Why the others are wrong

- **A:** 1.25 to 1.25 treats the purchase as funded only with own cash (assets stay 100), ignoring the broker loan.
- **C:** Debt-to-equity (20/80 and 70/80) is reported instead of assets-to-equity.
- **D:** 2.5 keeps the full $100 of cash and adds the $100 stock (assets 200), forgetting that $50 of cash was spent.

Q92

BT answers P2.T7.510.2 · BT notes P2.T8.510.2

REBUILT — Merged two copies; one unexplained distractor (1.75) replaced with a mistake-based one (1.875); answer unchanged.

On opening day, Lever Brothers Multistrategy Master Fund LP has the following economic balance sheet: $100 in cash, $20 in debt, and equity of $80. Lever Brothers creates a short position in a stock, borrowing $100 of the security and selling it. This creates a liability equal to the value of the borrowed stock and an asset, equal in value, consisting of the cash proceeds. The proceeds cannot fund other investments: they are collateral held in a segregated short account. Because the stock might rise, Lever Brothers must in addition put up margin of $50. After the trade, what is the leverage (assets/equity) of the firm's economic balance sheet?

A. 1.25
B. 1.50
C. 1.875
D. 2.50

Answer: D

Step by step

1. Cash falls from $100 to $50 after posting $50 margin.
2. Due from broker = $100 short-sale proceeds + $50 margin = $150.
3. Assets = 50 + 150 = 200.
4. Liabilities = $20 debt + $100 borrowed stock = $120; equity = 200 - 120 = 80.
5. Leverage = 200/80 = 2.50.

Why D is correct: A short sale grosses up the balance sheet: the borrowed stock is a liability and the proceeds plus margin are assets, so assets reach $200 on unchanged equity of $80.

Why the others are wrong

- **A:** 1.25 is the pre-trade leverage (100/80); it assumes the short sale is balance-sheet neutral.
- **B:** 1.50 is total liabilities over equity (120/80), not assets over equity.
- **C:** 1.875 leaves the $50 margin deposit out of assets (assets 150), forgetting that margin posted is still an asset due from the broker.

Q93

BT answers P2.T8.24.2.1

REBUILT — Source options gave inconsistent leverage figures and did not specify how proceeds are used, so two answers were defensible; rebuilt with explicit use of proceeds and the question reframed as lowest leverage ratio.

Starwing Pie, a bank, has total assets of $27 billion financed by $11 billion of equity and $16 billion of debt. The bank wants to lower its leverage ratio (total assets / equity) as much as possible. Assume each transaction below is the only one undertaken. Which transaction results in the lowest leverage ratio?

A. Raise $3 billion of equity and hold the proceeds as cash.
B. Issue $3 billion of debt and use the proceeds to buy assets.
C. Sell $2 billion of assets and use the proceeds to repay $2 billion of debt.
D. Raise $3 billion of equity and issue $3 billion of debt, and use all proceeds to buy assets.

Answer: A

Key correction: Source key (A) was right for its version, but its quoted ratios for C (2.78x) and D (2.40x) were inconsistent with the stated balance sheet.

Step by step

1. Current leverage: $L = 27/11 = 2.45$.
2. A: assets $27 + 3 = 30$, equity $11 + 3 = 14$, $L = 30/14 = 2.14$.
3. B: assets 30, equity 11, $L = 30/11 = 2.73$.
4. C: assets $27 - 2 = 25$, equity 11, $L = 25/11 = 2.27$.
5. D: assets $27 + 6 = 33$, equity $11 + 3 = 14$, $L = 33/14 = 2.36$.
6. A has the lowest ratio.

Why A is correct: Leverage is assets/equity; adding equity raises the denominator and the numerator by the same dollar amount, which cuts the ratio more than any transaction that adds debt or only shrinks assets.

Why the others are wrong

- **B:** Debt-financed asset purchase raises assets with equity unchanged, so L rises to 2.73 (above the current 2.45).
- **C:** Repaying debt with asset sales lowers L only to 2.27 because equity is unchanged; people pick it thinking 'less debt = lowest leverage'.
- **D:** A mix of equity and debt lowers L only to 2.36 because the added debt partly offsets the added equity.

Q94

BT notes P2.T6.300.2

REBUILT — Original tested enterprise value and book vs market value definitions not covered by the reading and had a muddled key; rebuilt as an EXCEPT question on the leverage ratio and leverage effect.

Which of the following statements about leverage and a firm's return profile is NOT correct?

A. An all-equity financed firm has a leverage ratio of 1.0, the lowest possible value.

B. If the cost of debt is zero, ROE equals ROA regardless of the leverage ratio.

C. Increasing leverage raises ROE only if ROA exceeds the cost of debt.

D. Buying securities with a 50% haircut, financed by a margin loan for the rest, gives a leverage ratio of 2.0.

Answer: B

Step by step

1. Leverage effect: $\text{ROE} = L \times \text{ROA} - (L - 1) \times c$.
2. With $c = 0$, $\text{ROE} = L \times \text{ROA}$, so ROE is magnified by the leverage ratio and exceeds ROA when $L > 1$. B is false.
3. A: all equity gives $L = A/E = 1.0$, true.
4. C: a change in L changes ROE by $\Delta L\ (\text{ROA} - c)$, positive only if ROA > c, true.
5. D: 50% haircut means equity is half of assets, $L = 1/0.5 = 2.0$, true.

Why B is correct: With free debt the whole leverage multiplier applies to ROA, so ROE equals $L \times$ ROA, not ROA.

Why the others are wrong

- **A:** True: with no debt, assets equal equity, so the ratio is 1.0.
- **C:** True: the marginal effect of leverage on ROE is ROA $- c$, so the sign depends on ROA versus the cost of debt.
- **D:** True: equity of 50% of assets implies assets/equity = 2.0.

Q95

Mock 2 Q77

REBUILT — Clarified that 'double its leverage factor' means the leverage ratio (assets/equity) doubling with assets unchanged; subject header mislabel ignored.

Callahan, Inc. has a return on assets of 7%, total assets of USD 5 million, and equity financing of USD 2 million. Its cost of debt is 3%. Callahan doubles its leverage ratio (assets / equity), holding total assets and ROA fixed. By how much does its return on equity (ROE) increase?

A. 4%

B. 7%

C. 10%

D. 13%

Answer: C

Step by step

1. Initial $L = 5/2 = 2.5$; $\text{ROE} = 2.5(0.07) - 1.5(0.03) = 13\%$.
2. New $L = 5.0$: equity 1, debt 4; $\text{ROE} = 5(0.07) - 4(0.03) = 23\%$.
3. Increase $= 23\% - 13\% = 10\%$.
4. Check: $\Delta L\ (\text{ROA} - c) = 2.5 \times 4\% = 10\%$.

Why C is correct: ROE changes by the change in leverage times the spread between ROA and cost of debt: $2.5 \times (7\% - 3\%) = 10\%$.

Why the others are wrong

- **A:** This is the spread ROA â˜ cost of debt (4%), forgetting to multiply by the change in leverage of 2.5.
- **B:** This is simply ROA; leverage is ignored.
- **D:** This is the starting ROE of 13%, not the increase; the new ROE is 23%.

LO 67.e

Describe and compare methods to measure and manage funding liquidity risk and transactions liquidity risk.

Q96

AnalystPrep Q.2431

REBUILT — Original stock Z had no price data so its liquidity was undefined and the combination options were ambiguous; reframed as single-best-answer with each wrong stock flawed on one liquidity dimension.

A portfolio's investment policy statement requires that it hold only liquid stocks. Based on the information below, which single stock is most consistent with that requirement?

Stock	Information
W	Latest trading information shows the stock could be sold at $12.30 and purchased at $12.32.
X	The majority of shares are held by one family; daily turnover is only 100 shares.
Y	A large-cap stock whose last trade occurred 2 days ago.
Z	Latest trading information shows the stock could be sold at $4.65 and purchased at $8.25.

A. Stock W
B. Stock X
C. Stock Y
D. Stock Z

Answer: A

Step by step

1. Tightness (round-trip cost) is measured by the bid-ask spread; a narrower spread means more liquid.
2. Stock W: spread = 12.32 - 12.30 = $0.02, about 0.02/12.31 ≈ 0.16% of price: very tight.
3. Stock Z: spread = 8.25 - 4.65 = $3.60, about 3.60/6.45 ≈ 56% of mid-price: extremely illiquid.
4. Stock X: concentrated ownership (small free float) and only 100 shares traded daily signal low depth.
5. Stock Y: large cap alone does not make a stock liquid; a last trade 2 days ago means no current evidence of tradability.
6. Only W combines a tight spread with current two-way quotes.

Why A is correct: Liquidity is evidenced by tight bid-ask spreads and active, current trading; only W has both.

Why the others are wrong

- **B:** Ignores that a closely held stock with 100 shares of daily turnover has almost no depth; a large order would move the price sharply.
- **C:** Assumes large cap implies liquid, but no trade for 2 days indicates a stale price and no current market.
- **D:** Reads the $4.65 bid as attractive and overlooks the $3.60 spread (about 56% of mid-price), which makes round-trip cost prohibitive.

Q97

AnalystPrep Q.3884

An investor placed an order to sell a stock when the market price was $50. Because of market volatility, by the time the broker executed the sale the price had fallen to $48. In the market, this phenomenon is known as:

A. bid-ask spread.
B. adverse price impact.
C. slippage.
D. the leverage effect.

Answer: C

Step by step

1. The loss ($50 to $48 = $2 per share) arises from the delay between order and execution, during which the

market moved.
2. A price deterioration in the time it takes to get a trade done is, by definition, slippage.
3. It was not caused by the seller's own order moving the market (adverse price impact), nor by the quoted spread or leverage.

Why C is correct: Slippage is the deterioration in an asset's price during the time it takes to get a trade done.

Why the others are wrong

- **A:** The bid-ask spread is the quoted round-trip cost at a point in time, not a price move during execution delay.
- **B:** Adverse price impact comes from the trader's own actions (e.g. a large order) moving the equilibrium price; here volatility and delay caused the fall.
- **D:** The leverage effect is about borrowing magnifying return on equity and has nothing to do with execution price.

Q98

BT answers P2.T7.512.1 · BT notes P2.T8.512.1

Which of the following refers to "the length of time for which a lumpy order moves the market away from the equilibrium price," which is a characteristic of market liquidity?

A. Depth
B. Tightness
C. Slippage
D. Resiliency

Answer: D

Step by step

1. Market liquidity characteristics: tightness (round-trip cost), depth (order size needed to move price), resiliency (time for the market to recover from a lumpy order).
2. The definition concerns how long the price stays away from equilibrium, i.e. how quickly the market bounces back: resiliency.

Why D is correct: Resiliency measures the length of time lumpy orders keep the market away from equilibrium, i.e. the speed of price recovery.

Why the others are wrong

- **A:** Depth is how large an order must be to move the price adversely: a size measure, not a duration.
- **B:** Tightness is the round-trip cost measured by the bid-ask spread and commissions.
- **C:** Slippage is price deterioration during the time taken to execute a trade, not the time for the market to recover afterwards.

Q99

BT answers P2.T7.512.2 · BT notes P2.T8.512.2

Which of the following refers to "the cost of a round-trip transaction, and is typically measured by the bid-ask spread and brokers' commissions," which is a characteristic of market liquidity?

A. Tightness
B. Depth
C. Adverse price impact
D. Resiliency

Answer: A

Step by step

1. A round-trip cost captured by the bid-ask spread plus commissions is the definition of tightness (also called width).
2. The narrower the spread, the tighter the market and the greater the liquidity.

Why A is correct: Tightness (width) is the round-trip transaction cost, proxied by the bid-ask spread and commissions.

Why the others are wrong

- **B:** Depth concerns how large an order must be to move the price adversely, not the cost of a round trip.
- **C:** Adverse price impact is the price move caused by the trader's own order, not a spread-based cost.
- **D:** Resiliency is the time it takes the market to recover from a lumpy order.

Q100 BT answers P2.T7.512.3 · BT notes P2.T8.512.3

Lack of liquidity manifests itself in observable, yet hard-to-measure ways. Which of the following refers to "the deterioration in the market price induced by the amount of time it takes to get a trade done?"

A. Tightness
B. Bid-ask spread
C. Adverse price impact
D. Slippage

Answer: D

Step by step

1. Transactions liquidity risks arise from fluctuating bid-ask spreads, adverse price impact (the trader's own action moves the price) and slippage (price deteriorates while the trade is being done).
2. The definition refers to the time taken to get the trade done: slippage.

Why D is correct: Slippage is the price deterioration caused by the time it takes to complete a trade.

Why the others are wrong

- **A:** Tightness is the round-trip cost measured by spread and commissions.
- **B:** The bid-ask spread is a point-in-time quote difference, not a price move caused by execution delay.
- **C:** Adverse price impact is caused by the trader's own order affecting the equilibrium price, not by elapsed time.

Q101 Mock 1 Q76

REBUILT — Options C (Tightness) and D (Width) were synonyms, creating an ambiguous pair; replaced Width with Slippage.

A trading manager at a large institution plans a large sale and wants to know whether the market can absorb it without moving the price adversely. In measuring market liquidity, which characteristic is most relevant for the trader?

A. Depth
B. Resiliency
C. Tightness
D. Slippage

Answer: A

Step by step

1. The question is whether the market can absorb the order, i.e. how large an order must be to move the price adversely.
2. That is the definition of depth.
3. Tightness measures round-trip cost and resiliency measures speed of price recovery; neither answers the absorption question.

Why A is correct: Depth describes the order size the market can take before the price moves adversely, which is what a large seller needs to know.

Why the others are wrong

- **B:** Resiliency is how long the market takes to recover after a lumpy order: relevant after the trade, not for capacity to absorb it.
- **C:** Tightness is the bid-ask spread/commission cost, which says nothing about absorbing size.
- **D:** Slippage is price deterioration during execution time, a risk of trading rather than a market-liquidity characteristic measuring capacity.

Q102 Schweser Module Quiz 67.3 Q1

Brett Doninger placed an order to sell a stock when the market price was $42.12. The market was volatile and, by the time Doninger's broker sold the stock, the price had fallen to $41.88. In the market, this phenomenon is known as:

A. adverse selection.
B. transactional imbalance.
C. slippage.
D. the spread risk factor.

Answer: C

Step by step

1. Identify the event: the price moved from $42.12 to $41.88 (a $0.24 fall) during the time needed to execute the trade.
2. Transactions liquidity risk arises from (1) fluctuating bid-ask spreads, (2) the trader's own adverse price impact, and (3) price deterioration while the trade is being completed.
3. Price deterioration between order placement and execution is called slippage.

Why C is correct: Slippage is the adverse price movement that occurs in the time it takes to complete a trade, which is exactly what happened to Doninger's order in a volatile market.

Why the others are wrong

- **A:** Adverse selection is the dealer's risk of trading against better-informed traders (lemons risk), compensated through the bid-ask spread; it is not price movement during execution.
- **B:** 'Transactional imbalance' is not a recognized liquidity-risk term; it is a plausible-sounding invention.
- **D:** The spread risk factor measures the variability of the bid-ask spread used in the liquidity adjustment to VaR; it does not describe a price drift during execution.

Q103

BT answers P2.T8.24.2.3

REBUILT — Original referenced an unspecified fund and had giveaway distractors; rewritten as a self-contained hedge-fund funding liquidity question with realistic distractors.

A hedge fund is exposed to sudden redemption requests from investors and to withdrawal of credit by its prime broker. Which strategy is most appropriate for managing its funding liquidity risk?

A. Shift a large portion of the portfolio into long-term, illiquid bonds that offer an illiquidity premium.

B. Hold a buffer of cash and unpledged short-term Treasury bills that can be used immediately to meet redemptions.

C. Rely mainly on unused borrowing capacity with its brokers as the primary liquidity buffer.

D. Hold unpledged agency bonds and plan to pledge them as collateral for new borrowing when redemptions arrive.

Answer: B

Step by step

1. Redemptions and credit withdrawal force rapid unwinding of positions, so the buffer must be immediately available and not depend on counterparties or market prices.
2. Cash and unpledged T-bills provide unfettered liquidity that does not depend on counterparties or on a sale price in distressed markets.
3. Unused borrowing capacity can be revoked by raising haircuts or refusing collateral; agency bonds were insufficient collateral; illiquid bonds would have to be sold at a loss.

Why B is correct: Cash and unpledged Treasury bills are the most reliable, immediately available liquidity sources under stress.

Why the others are wrong

- **A:** The illiquidity premium compensates for holding illiquid assets but worsens the ability to meet redemptions; fire-sale losses result.
- **C:** Borrowing capacity is not unfettered; counterparties can raise haircuts or decline collateral exactly when stress hits.
- **D:** Agency debt collateral was impaired in the 2007-09 crisis (rising haircuts), so borrowing against it could not be relied on; pledged assets are also subject to revocation.

Q104

Mock 1 Q64

REBUILT — Liquidity duration was not defined in the stem and options C and D were unrealistic; added the definition and rebuilt distractors from realistic slips.

A portfolio manager holds 30,000 shares of SkyTrain, Inc. The daily volume traded in SkyTrain is 70,000 shares, and the manager wishes to trade no more than 18% of daily volume. Liquidity duration is the number of days needed to liquidate the position, i.e. position size divided by the maximum shares tradable per day. Which is closest to the liquidity duration?

A. 0.42 days

B. 2.38 days

C. 5.56 days

D. 238 days

Answer: B
Step by step
1. Permitted daily trading = 0.18 × 70,000 = 12,600 shares.
2. Liquidity duration = 30,000/12,600 = 2.381 days.

Why B is correct: Days to exit equal the position divided by the daily volume the manager is willing to take.
Why the others are wrong
- **A:** Inverted the ratio: 12,600/30,000 = 0.42.
- **C:** Divided 1 by the participation rate (1/0.18) and ignored the position and volume.
- **D:** Treated 18% as 0.18% (0.0018), shrinking capacity to 126 shares a day.

Q105

Mock 2 Q5

REBUILT — Options were loosely rounded (60 vs 60.6) and liquidity duration was undefined; defined it in the stem and rebuilt options as distinct error outcomes.
A manager holds 5 million shares of a security whose average daily trading volume is 825,000 shares. Policy limits liquidation to 10% of daily volume. Liquidity duration is the position size divided by the maximum shares tradable per day. What is the liquidity duration?

A. 0.0165 days
B. 6.06 days
C. 60.6 days
D. 606 days

Answer: C
Step by step
1. Maximum daily trade = 0.10 × 825,000 = 82,500 shares.
2. Liquidity duration = 5,000,000/82,500 = 60.6 days.

Why C is correct: The participation cap limits sales to 82,500 shares per day, so exiting 5 million shares takes about 60.6 days.
Why the others are wrong
- **A:** Inverted the ratio: 82,500/5,000,000 = 0.0165.
- **B:** Ignored the 10% participation limit (5,000,000/825,000 = 6.06).
- **D:** Applied 1% instead of 10% participation (decimal slip).

LO 67.f

Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days.

Q106

AnalystPrep Q.3881

REBUILT — Source asked for the 'expected' cost but its keyed answer was the 95% upper bound, and its distractors were arbitrary; reworded to ask for the 95% upper bound with z = 1.65 stated and rebuilt options from realistic errors.
A stock trades at an ask of $335 and a bid of $331. The sample standard deviation of the proportional bid-ask spread is 0.00019. Assuming daily spread changes are normally distributed with zero mean, what is the 95% upper bound (z = 1.65) on the per-share transaction cost?

A. $2.00
B. $2.05
C. $2.10
D. $4.10

Answer: B
Key correction: Source key (B, labelled 'expected transaction cost') is actually the 95% upper bound; the expected cost is $2.00.
Step by step
1. Midprice $P = (335 + 331)/2 = 333$.
2. Proportional spread $s = 4/333 = 0.012012$.
3. Expected cost $= P\,s/2 = 333 \times 0.006006 = \$2.00$.
4. Spread risk term $= P \times \frac{1}{2} \times 1.65 \times 0.00019 = \$0.0522$.

5. 95% upper bound $= 2.00 + 0.052 = \$2.05$.

Why B is correct: The cost bound is the half-spread plus a normal-based allowance for spread variability, scaled by the midprice and halved: $P \times \frac{1}{2}(s + 1.65\sigma_s)$.

Why the others are wrong

- **A:** Expected (half-spread) cost only; ignores the spread-risk add-on.
- **C:** Did not halve the volatility term: $2.00 + 333 \times 1.65 \times 0.00019 = 2.10$.
- **D:** Omitted the 1/2 factor altogether: $P(s + 1.65\sigma_s) = 4.10$, the full round-trip rather than the half-spread.

Q107

BT answers P2.T8.24.3.2

REBUILT — Exhibit OCR-garbled, the qualitative statement options were a giveaway, and the 8% spread volatility was unverifiable; rebuilt with explicit data and numeric options only.

A trader notes an ask price of \$10.50 and a bid price of \$7.50 for a micro-cap stock. The standard deviation of the proportional spread is 8% (0.08), and daily changes in the spread are assumed normally distributed with zero mean. What are the expected transaction cost and the 99% (z = 2.33) upper bound on the transaction cost per share?

A. Expected \$3.00; upper bound \$3.84
B. Expected \$1.50; upper bound \$3.18
C. Expected \$1.50; upper bound \$2.34
D. Expected \$3.00; upper bound \$4.68

Answer: C

Step by step

1. Midprice $P = 9.00$; $s = 3/9 = 0.3333$.
2. Expected cost $= 9 \times 0.3333/2 = \$1.50$.
3. Risk add-on $= 9 \times \frac{1}{2} \times 2.33 \times 0.08 = \$0.839$.
4. Upper bound $= 1.50 + 0.84 = \$2.34$.

Why C is correct: Expected cost is the half-spread; the 99% bound adds one-half of 2.33 spread standard deviations times the midprice.

Why the others are wrong

- **A:** Used the full spread (no 1/2) for the expected cost, then halved the risk add-on: 3.00+0.84.
- **B:** Correct expected cost but dropped the 1/2 on the risk term: $1.50 + 9 \times 2.33 \times 0.08 = 3.18$.
- **D:** Dropped the 1/2 everywhere: $P(s + 2.33\sigma_s) = 4.68$, which is the full round-trip.

Q108

AP Mock 2 Q19

REBUILT — Stem ambiguous ('expected cost at 99%') and source value depended on 2.326 vs 2.33; clarified as 99% upper bound with z = 2.33 and rebuilt options.

Delight Inc. recently traded at an ask of USD 200.00 and a bid of USD 199.00. The sample standard deviation of the proportional spread is 0.0004. Using z = 2.33, what is the 99% upper bound on the per-share transaction cost?

A. USD 0.5000
B. USD 0.5009
C. USD 0.5930
D. USD 1.1859

Answer: C

Key correction: Source key (B, 0.5927) reflects z = 2.326 and a rounding slip; with the Schweser z = 2.33 the bound is 0.5930 (option C here).

Step by step

1. $P = 199.50$, $s = 1/199.50 = 0.0050125$.
2. $s + 2.33\sigma_s = 0.0050125 + 0.000932 = 0.0059445$.
3. Cost $= 199.50 \times \frac{1}{2} \times 0.0059445 = \$0.5930$.

Why C is correct: The 99% bound is the half-spread inflated by the stressed spread, $P \times \frac{1}{2}(s + 2.33\sigma_s)$.

Why the others are wrong

- **A:** Expected half-spread only, no stress add-on.
- **B:** Added the stress term in proportional terms without scaling by price: 0.5 + 0.0009.
- **D:** Forgot the 1/2: $P(s + 2.33\sigma_s) = 1.186$.

Q109 BT answers P2.T8.24.3.1

REBUILT — Giveaway distractor (decrease to $0.19m) and formula not shown; rebuilt with explicit liquidation assumption and error-based options.

A fund holds a thinly traded position with a 1-day 99% VaR of $5 million. The position can be liquidated in equal parts at the end of each of T = 10 trading days. Using the adjustment factor $\sqrt{(1+T)(1+2T)/(6T)}$, what is the liquidity-adjusted VaR?

A. $5.00 million
B. $9.81 million
C. $15.81 million
D. $19.25 million

Answer: B

Step by step

1. Factor $= \sqrt{(11)(21)/(60)} = \sqrt{3.85} = 1.9621$.
2. Adjusted VaR $= 5 \times 1.9621 = \$9.81$ million (a 96% increase).

Why B is correct: Because the position is sold down gradually, only part of it is held for each extra day, so the adjustment is above 1 but below $\sqrt{T}$.

Why the others are wrong

- **A:** No liquidity adjustment (treats 1-day VaR as sufficient).
- **C:** Applied the square-root-of-time rule, $5\sqrt{10}$, which assumes the whole position is held 10 days.
- **D:** Forgot the square root: $5 \times 3.85 = 19.25$.

Q110 AP Mock 1 Q49 · AP Mock 3 Q73

REBUILT — Two mocks duplicate the same item; option values mixed z = 2.326 rounding and were inconsistent, so rebuilt with z = 2.33 and consistent options.

Shares of Bell Aviation have a bid of USD 32.45 and an ask of USD 32.90. The sample standard deviation of the proportional bid-ask spread is 0.004. Using z = 2.33, what are the 99% transaction cost (per share) and the 99% spread risk factor?

A. Cost USD 0.5295; factor 0.01621
B. Cost USD 0.3773; factor 0.01155
C. Cost USD 0.7545; factor 0.02309
D. Cost USD 0.2250; factor 0.00689

Answer: B

Key correction: Source keys (D for Mock 1, C for Mock 3) agree with this answer up to z rounding (2.326 vs 2.33).

Step by step

1. $P = 32.675$, $s = 0.45/32.675 = 0.013772$.
2. Factor $= \frac{1}{2}(0.013772 + 2.33 \times 0.004) = \frac{1}{2}(0.023092) = 0.011546$.
3. Cost $= 32.675 \times 0.011546 = \$0.3773$.

Why B is correct: The spread risk factor is $\frac{1}{2}(s + 2.33\sigma_s)$ and the 99% cost is that factor times the midprice.

Why the others are wrong

- **A:** Halved s but not the volatility term: factor $0.5s + 2.33\sigma_s = 0.01621$, cost 0.5295.
- **C:** Omitted the 1/2: factor 0.02309, cost 0.7545.
- **D:** Used only the expected half-spread (0.5s = 0.00689), no stress add-on.

Q111 AnalystPrep Q.2731

REBUILT — Source mixed units (sigma in dollars used as proportional) and asked 'expected' cost while keying the 99% bound; clarified and rebuilt options.

Shares trade at a bid of $20.20 and an ask of $20.35. The sample standard deviation of the proportional spread is 0.0003. Using z = 2.33, what is the 99% upper bound on the per-share transaction cost?

A. $0.075
B. $0.082
C. $0.164
D. $0.089

Answer: B

Step by step

1. $P = 20.275$, $s = 0.15/20.275 = 0.0073983$.
2. $s + 2.33\sigma_s = 0.0073983 + 0.000699 = 0.0080973$.
3. Cost $= 20.275 \times \frac{1}{2} \times 0.0080973 = \$0.0821$.

Why B is correct: Cost is the half-spread plus the stressed spread allowance, scaled by midprice.

Why the others are wrong

- **A:** Expected half-spread only (0.075), ignoring the 99% add-on.
- **C:** Forgot the 1/2: P(s+2.33σ) = 0.164.
- **D:** Halved s but not the volatility term: $0.075 + 20.275 \times 2.33 \times 0.0003 = 0.089$.

Q112

AnalystPrep Q.3214

REBUILT — Distractors arbitrary; replaced with error-based options (type normalised to mcq).

Grantson Automobile stock has a bid of \$53.70 and an ask of \$54.10. The proportional bid-ask spread (relative to the midprice) is closest to:

A. 0.37%
B. 0.40%
C. 0.74%
D. 1.48%

Answer: C

Step by step

1. Midprice $= (53.70 + 54.10)/2 = 53.90$.
2. $s = 0.40/53.90 = 0.00742 = 0.74\%$.

Why C is correct: Proportional spread is the dollar spread divided by the midprice.

Why the others are wrong

- **A:** That is the proportional half-spread (0.20/53.90).
- **B:** Reported the dollar spread \$0.40 as a percentage.
- **D:** Doubled the correct answer (e.g., divided by half the midprice).

Q113

AnalystPrep Q.3882

REBUILT — Keyed answer was correct but options were evenly spaced; rebuilt options from error patterns and fixed z = 2.33.

Delight Inc. traded at an ask of \$200 and a bid of \$199. The sample standard deviation of the proportional spread is 0.0004. Using z = 2.33, the 99% spread risk factor is closest to:

A. 0.00251
B. 0.00297
C. 0.00344
D. 0.00594

Answer: B

Step by step

1. $s = 1/199.5 = 0.0050125$.
2. Factor $= \frac{1}{2}(0.0050125 + 2.33 \times 0.0004) = \frac{1}{2}(0.0059445) = 0.00297$.

Why B is correct: Spread risk factor is $\frac{1}{2}(s + 2.33\sigma_s)$.

Why the others are wrong

- **A:** Half-spread only (0.5s), no stress term.
- **C:** Halved s but not the 2.33σ term: 0.0025+0.00093.
- **D:** Forgot the 1/2: s+2.33σ = 0.00594.

Q114

AnalystPrep Q.3883

REBUILT — Distractors were weak; rebuilt around realistic errors.

A trader estimates that a position can be liquidated in equal parts over T = 8 trading days. Using the factor $\sqrt{(1+T)(1+2T)/(6T)}$, by approximately what percentage should the 1-day VaR be increased?

A. 79%
B. 179%
C. 183%

D. 219%

Answer: A

Step by step

1. Factor $= \sqrt{9 \times 17/48} = \sqrt{3.1875} = 1.7854$.
2. Increase $= 1.7854 - 1 = 78.5\% \approx 79\%$.

Why A is correct: The factor exceeds 1 by about 0.79, i.e., a 79% increase, lower than the square-root-of-time increase.

Why the others are wrong

- **B:** Reported the factor 1.79 as a percentage increase (179%) without subtracting 1.
- **C:** Used $\sqrt{8} = 2.83$ (square-root-of-time), giving 183%.
- **D:** Forgot the square root: 3.1875 – 1 = 219%.

Q115

BT answers P2.T7.511.3 · BT notes P2.T8.511.3

REBUILT — Merged duplicate items; garbled formula restored and option C (an unexplained value) replaced by the realistic 'forgot the square root' error.

A position that can be liquidated evenly over T trading days has liquidity-adjusted VaR of $\text{LVaR} = \text{VaR}_{\text{1-day}} \times \sqrt{\frac{(1+T)(1+2T)}{6T}}$. An equity position has a value of \$1,000,000 and annual volatility of 34.0%. Assume 250 trading days per year, i.i.d. normal returns, and a 99% quantile of 2.33. Which is nearest to the 99% liquidity-adjusted VaR if the position requires five trading days to liquidate?

A. \$50,000
B. \$74,200
C. \$110,000
D. \$112,030

Answer: B

Step by step

1. Daily volatility: $0.34/\sqrt{250} = 2.150\%$.
2. 1-day 99% VaR: $1{,}000{,}000 \times 0.02150 \times 2.33 = \$50{,}103$.
3. Liquidation factor, T = 5: $\sqrt{(6)(11)/(30)} = \sqrt{2.2} = 1.4832$.
4. LVaR $= 50{,}103 \times 1.4832 = \$74{,}315$, nearest to \$74,200 (2.326 gives \$74,187).

Why B is correct: Because the position is sold in equal parts, only part of it is exposed for the full T days, so the adjustment (1.48) is above 1 but below $\sqrt{T} = 2.24$.

Why the others are wrong

- **A:** This is just the unadjusted 1-day VaR; it ignores the liquidation-period adjustment.
- **C:** Applied the ratio (1+T)(1+2T)/(6T) = 2.2 without taking the square root.
- **D:** Scaled 1-day VaR by $\sqrt{5}$ (square-root-of-time rule), which overstates VaR because it assumes the whole position is held for five days.

Q116

BT notes P2.T8.511.2

REBUILT — Original distractors were arbitrary and the quantile unspecified; used the Schweser 2.33 quantile and distractors from realistic formula errors.

The bid-ask spread on a share is \$0.35 when the share price (midprice) is \$50.00. The spread, expressed as a proportion of price, is normally distributed with a standard deviation of 0.20% (0.0020). Using a 99% quantile of 2.33, which is nearest to the 99% transactions cost per share (half-spread plus its 99% spread risk)?

A. \$0.175
B. \$0.2915
C. \$0.583
D. \$11.83

Answer: B

Step by step

1. Relative spread: $s = 0.35/50 = 0.0070$.
2. Spread risk factor: $\frac{1}{2}(s + 2.33\sigma_s) = \frac{1}{2}(0.0070 + 2.33 \times 0.0020) = 0.00583$.
3. Cost per share: $P \times 0.00583 = 50 \times 0.00583 = \$0.2915$.

Why B is correct: The 99% cost is the price times the half of (relative spread plus 2.33 spread standard deviations);

only half the spread is paid on a one-way trade.

Why the others are wrong

- **A:** Expected cost only (half-spread, 50 x 0.0035 = $0.175); omits the 2.33Ïƒ spread-risk term.
- **C:** Forgot the factor of 1/2 (round-trip rather than one-way cost): 50 x 0.0116 = $0.583.
- **D:** Entered the spread volatility as 20% (0.20) instead of 0.20%: 25 x (0.007 + 2.33 x 0.20) = $11.83.

Q117

AnalystPrep Q.3225

REBUILT — Formula was missing from stem; added it and replaced implausible distractors with realistic error results.

A pension fund's position has a 1-day VaR of $62 million. The position will be liquidated in equal parts over four trading days. Using $\text{LVaR} = \text{VaR}_{1\text{-day}} \times \sqrt{\frac{(1+T)(1+2T)}{6T}}$, what is the liquidity-adjusted VaR?

A. $124,000,000
B. $84,897,000
C. $77,328,000
D. $116,250,000

Answer: B

Step by step

1. T = 4: $(1+T)(1+2T) = 5 \times 9 = 45$; $6T = 24$.
2. Factor $= \sqrt{45/24} = \sqrt{1.875} = 1.3693$.
3. LVaR $= 62 \times 1.3693 = \$84.897$ million.

Why B is correct: The factor 1.3693 lies between 1 and $\sqrt{4} = 2$, reflecting that only part of the position is held for the full four days.

Why the others are wrong

- **A:** Scaled by $\sqrt{4} = 2$ (square-root-of-time rule), overstating VaR as if the whole position were held four days.
- **C:** Used T = 3 in the formula (factor 1.2472) instead of T = 4.
- **D:** Multiplied by 1.875 without taking the square root.

Q118

BT answers P2.T7.511.2

REBUILT — Source used z = 2.326 giving 0.2913; GARP/Schweser use 2.33 (0.2915); options and distractors rebuilt around the 2.33 formula.

The bid-ask spread on a share is USD 0.350 and the current price is USD 50.00. The relative spread is assumed normally distributed with zero-mean changes and volatility of 20 basis points (0.20%). Using the 99% spread risk factor, which is nearest to the 99% confidence level transactions cost per share?

A. $0.1750
B. $0.2915
C. $0.4080
D. $0.5830

Answer: B

Key correction: Source key (BT) gave $0.29130 using 2.326; with the GARP/Schweser multiplier of 2.33 the answer is $0.2915 (same choice).

Step by step

1. Relative spread $s = 0.350/50.00 = 0.0070$; $\sigma_s = 0.0020$.
2. Spread risk factor $= \frac{1}{2}(s + 2.33\sigma_s) = 0.5(0.0070 + 0.00466) = 0.00583$.
3. Transactions cost $= P \times 0.00583 = 50.00 \times 0.00583 = \$0.2915$.

Why B is correct: The 99% transactions cost is price times half of the spread plus 2.33 spread volatilities, since a trade pays only half the spread.

Why the others are wrong

- **A:** Used only the expected half-spread $50 \times 0.5 \times 0.007$ and omitted the spread-volatility term.
- **C:** Applied the one-half factor only to the expected spread, $50(0.0035 + 0.00466)$; the whole bracket should be halved.
- **D:** Forgot the one-half factor: $50(0.007 + 0.00466)$ is the full round-trip, not one-way, cost.

LO 67.g

Discuss interactions between different types of liquidity risk and explain how liquidity risk events can increase systemic risk.

Q119 AnalystPrep Q.2982

In the period of the global financial turmoil, the collapse of the financial institutions can largely be attributed to both illiquidity and insolvency. Which of the following does NOT describe the sequence of events in the collapse of an intermediary?

A. Issues about the solvency of the company were raised by the reports of losses at the intermediary, or at other institutions.

B. All companies, financial intermediaries, and non-financial companies are still willing to lend to the intermediary.

C. To raise funds, the intermediary is forced to liquidate assets, which might lead to losses in a distressed market.

D. Being aware that the challenges faced by the intermediary are now being compounded by the realized mark-to-market losses, the lenders become more and more reluctant to extend credit to the intermediary.

Answer: B

Step by step

1. Recall the typical sequence in an intermediary's collapse: loss reports raise solvency doubts, funding liquidity tightens, forced asset sales occur at distressed prices, and realized mark-to-market losses deepen the solvency concern.
2. Funding liquidity and transactions liquidity reinforce each other: a funding squeeze forces sales (transactions liquidity risk), and the resulting losses worsen solvency and further reduce funding.
3. Test each option against this sequence: A, C and D are steps in the spiral; B says all lenders remain willing to lend, which contradicts the funding withdrawal at the heart of the spiral.
4. The statement that does NOT describe the sequence is B.

Why B is correct: In a liquidity-solvency spiral, doubts about solvency make lenders withdraw or tighten funding; B asserts the opposite (continued willingness to lend), so it is the one statement not part of the sequence.

Why the others are wrong

- **A:** This is the first step of the sequence (loss reports raise solvency doubts), so it does describe the collapse; choosing it means misreading the NOT in the stem.
- **C:** Forced liquidation of assets in a distressed market is a core link between funding and transactions liquidity risk; it does describe the sequence.
- **D:** Lenders becoming more reluctant as realized mark-to-market losses compound the problem is the reinforcing step of the spiral; it does describe the sequence.

Q120 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst at a regulator is reviewing the tri-party repo market, in which a clearing bank provides intraday credit (a daylight overdraft) to a broker-dealer and holds the dealer's securities as collateral. The analyst is asked how a liquidity risk event at one large dealer could become a systemic risk event through the payments, clearing, and settlement infrastructure (the "plumbing"). Which of the following statements is most accurate?

A. Doubts about whether the dealer's account is undercollateralized can call the clearing bank's own liquidity and solvency into question, and the clearing bank might also decline credit to its other customers, amplifying rollover risk across the market.

B. The clearing bank is exposed to the dealer's counterparty credit risk only, so funding liquidity problems at the dealer cannot reach other market participants.

C. Because tri-party repo is diversified across many small dealers, the failure of any one dealer has no downstream effects through the settlement system.

D. The default of a large dealer would cause the securities in its account to be held rather than sold, which prevents transactions liquidity risk from emerging.

Answer: A

Step by step

1. The clearing bank finances the dealer through a daylight overdraft collateralized by securities, so it bears coun-

terparty credit risk if collateral liquidation would not cover the debt.
2. Even the mere possibility that the dealer's account is undercollateralized calls the clearing bank's liquidity and solvency into question.
3. The clearing bank may decline credit to its customers, and cash lenders may withdraw from repo generally, provoking or amplifying rollover (funding liquidity) events for many participants simultaneously.
4. Tri-party repo is concentrated (three dealers held 38 percent of outstanding tri-party repo in early 2010), and a dealer default would trigger immediate sale of its collateral, adding transactions liquidity risk. Only A is consistent with this.

Why A is correct: Disruptions in the plumbing impact many participants at once; doubts about the collateral coverage threaten the clearing bank itself and can cut credit to others, turning one dealer's liquidity event into a systemic one.

Why the others are wrong

- **B:** Wrong: the clearing bank does bear counterparty credit risk, but that exposure is itself a link through which funding liquidity problems spread; the claim that problems cannot reach others ignores the interaction of credit and liquidity risk.
- **C:** Wrong: tri-party repo is not diversified but concentrated in a few dealers, so the failure of one large dealer would have large downstream effects.
- **D:** Wrong: a large dealer's default would trigger the immediate sale of the collateral securities, not holding them, which brings transactions liquidity risk (price impact) on top of the funding problem.

Q121

BT answers P2.T5.406.3

REBUILT — Original tested the Adrian-Shin/Basel procyclical-leverage argument, which is not in the GARP/Schweser text; rebuilt to test the leverage link between funding and transactions liquidity and systemic risk.

A leveraged investor holds a long position financed with short-term collateralized borrowing. Lenders sharply raise haircuts, and the investor must sell assets quickly. Which statement best describes how this event can increase systemic risk?

A. Forced sales reduce the number of potential holders and depress asset prices, which can impair the solvency of other leveraged holders and trigger further deleveraging (a debt-deflation spiral).

B. Forced sales raise asset prices because the supply of cash in the market increases, which restores funding liquidity for all participants.

C. The event is confined to transactions liquidity risk and has no effect on funding liquidity, so it cannot spread to other institutions.

D. The event reduces systemic risk because a lower leverage ratio makes every other market participant safer.

Answer: A

Step by step

1. Higher haircuts are an increase in funding liquidity risk: the investor must fund the same position with more of its own capital.
2. Unable to roll funding, the investor sells assets, so transactions liquidity risk is realized (adverse price impact).
3. Lower prices reduce collateral values and the solvency of other holders of the same assets, forcing them to delever too.
4. This feedback between funding and transactions liquidity, via leverage, is how a liquidity event becomes a systemic event (debt-deflation crisis).

Why A is correct: Leverage links funding and transactions liquidity: funding stress forces sales, price declines hit other leveraged holders, and the spiral spreads stress across the system.

Why the others are wrong

- **B:** Forced selling lowers, not raises, prices; confuses cash raised by the seller with the market-wide price effect.
- **C:** Funding and transactions liquidity interact in both directions; ignoring the funding-to-market feedback is the error.
- **D:** Individual deleveraging looks safer in isolation, but simultaneous fire sales are destabilizing; ignores price spillovers.

Reading 68. Early Warning Indicators

Shyam Venkat and Stephen Baird, Liquidity Risk Management: A Practitioner's Perspective (Hoboken, NJ: John Wiley & Sons, 2016). Chapter 6.

LO	Learning objective	Qs
68.a	evaluate the characteristics of sound Early Warning Indicators (EWI) measures.	5
68.b	identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve).	10
68.c	discuss the applications of EWIs in the context of the liquidity risk management process.	10

LO 68.a

Evaluate the characteristics of sound Early Warning Indicators (EWI) measures.

Q122 AnalystPrep Q.3844

An indicator that provides information on significant potential stress before the occurrence of an actual event is most likely known as a:

A. Sharp indicator
B. Granular indicator
C. Leading indicator
D. Lagging indicator

Answer: C

Step by step

1. Identify the property described: timing, i.e., a warning given before the event.
2. Indicators that signal before an event are leading; those that signal after are lagging.
3. Sharp/granular describe the level of detail, not the timing.

Why C is correct: A leading indicator gives a warning signal before the event occurs, which is why leading EWIs are much more effective than lagging ones.

Why the others are wrong

- **A:** Sharpness (granularity) concerns detail within the data, not when the signal arrives.
- **B:** Granular means detailed, focused on a subset of data; it says nothing about advance timing.
- **D:** Lagging indicators signal only after the event has occurred, the opposite of the stem.

Q123 AnalystPrep Q.3850

What is a sharp early warning indicator?

A. An indicator that highlights past performance
B. An indicator that depends on another performance measure
C. A highly granular indicator, involving a subset of data
D. An indicator that needs professional experience and quantitative skills to decode

Answer: C

Step by step

1. Schweser: a granular indicator is sharper and is not lost in noise within a dataset.
2. Example: a fall in deposits of one client group is sharper than a fall in total deposits.
3. So a sharp EWI is highly granular, focused on a subset of data.

Why C is correct: Sharpness means granularity: focusing on a specific subset of data so that the signal is not drowned out by noise.

Why the others are wrong

- **A:** Past-performance indicators are lagging, not sharp.
- **B:** Dependence on another measure has nothing to do with sharpness.
- **D:** Interpretation difficulty is not a property of a sharp indicator.

Q124 BT answers P2.T8.20.4.2 · BT notes P2.T8.20.4.2

Venkat writes that "The EWI framework can be summarized as M.E.R.I.T," where the (M) refers to measures. Which of the following is a TRUE characteristic of good early warning indicator (EWI) measures?

A. Only internal but not external measures
B. Forward-looking and sharp (i.e., granular and specific)
C. Measures are disassociated (de-linked) from the escalation process
D. Measures are tracked during business-as-usual environments but should not be affected by stressed environments

Answer: B

Step by step

1. Schweser: EWIs should be leading (forward-looking) and granular (sharp).
2. A is false: both internal and external measures are needed.
3. C is false: measures only help LRM when tied to an escalation plan.
4. D is false: analysis should cover both normal and stressed states.

Why B is correct: Good measures are leading and granular, covering internal and external factors, normal and stressed states and several horizons.

Why the others are wrong

- **A:** Measures should include both internal (balance sheet) and external (macroeconomic) factors.
- **C:** Measures must be linked to escalation; otherwise EWIs have no effect on liquidity risk management.
- **D:** Stressed states must also be examined to test whether liquid resources suffice.

Q125 Schweser Module Quiz 68.1 Q2

Which of the following statements regarding early warning indicators (EWIs) is correct?

A. The more granular the indicator, the more accurate it is.
B. To be effective, EWIs must be tied to a definitive escalation plan.
C. EWIs should focus on external metrics over internal metrics because the former are generally more reliable.
D. Lagging indicators are less prone to error than leading indicators; therefore, they are potentially more useful to liquidity risk managers.

Answer: B

Step by step

1. Recall the Schweser EWI framework: measures should be leading, granular, and include both internal and external metrics.
2. Escalation: EWIs only have a positive effect on liquidity risk management when they are embedded in a clear escalation plan.
3. Test each option against this: A (granularity), C (external over internal), D (lagging over leading) all contradict the reading; B matches it.

Why B is correct: An EWI only produces action if a breach triggers a definitive escalation path to the relevant management personnel; otherwise it is just information.

Why the others are wrong

- **A:** Granularity makes an indicator sharper (harder to lose in noise), not more accurate; this confuses sharpness with accuracy.
- **C:** Sound EWIs combine internal and external measures, and internal measures are valuable because they can reveal stress before it is visible externally; no preference for external ones is stated.
- **D:** Lagging indicators signal only after the event, so they are much less effective than leading indicators; low error does not offset their lateness.

Q126 AnalystPrep Q.3853

In which of the following circumstances is it most optimal to track early warning indicator metrics?

A. After an industry-wide decline in performance
B. During a business-as-usual environment
C. During periods of institution-specific stress
D. After the occurrence of a loss-causing event

Answer: B

Step by step

1. EWIs are leading indicators intended to give advance notice of possible future deterioration.
2. The reading states EWIs should be utilized in normal operating states to warn of potential future worsening of the bank's position.
3. Tracking after an industry decline, during stress, or after a loss is reactive, so too late for early warning.
4. Answer B.

Why B is correct: Early warning only has value if the signal arrives while there is still time to act, which means monitoring continuously in normal conditions.

Why the others are wrong

- **A:** By the time an industry-wide decline has occurred the warning window has passed; the indicator is then coincident or lagging.
- **C:** Stress-state analysis (stress testing) is useful but monitoring only during stress is too late for early warnings.
- **D:** A loss-causing event is the outcome the EWI is meant to anticipate; tracking afterwards is hindsight.

LO 68.b

Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve).

Q127

AnalystPrep Q.5376

REBUILT — Option D used a non-existent 'hedge ratio' metric (giveaway) and the stem was clumsy; replaced D with a plausible but benign indicator and reworded.

John Doe, a risk consultant, is reviewing ABC Bank's liquidity profile. Which of the following developments should he treat as the most prominent warning sign of potential liquidity risk?

A. The spreads on the bank's issued debt and credit default swaps have widened.

B. The bank's assets have grown substantially, and the growth has been financed by an expansion in stable liabilities.

C. The stock prices of the bank's peers have declined, but the bank's own stock price has not declined.

D. The average maturity of the bank's liabilities has lengthened and its funding sources have become more diversified.

Answer: A

Step by step

1. Recall the regulator-listed EWIs: OCC (2012) and Federal Reserve SR 10-6 both list spread increases on fixed-income and swap products; OCC also lists falling stock prices.
2. A: widening spreads on the bank's own debt and CDS signal falling market confidence and rising funding cost, so A is a recognized EWI.
3. B: BCBS (2008) flags a very sharp asset increase, but growth funded by stable liabilities does not signal funding stress.
4. C: the bank's own stock price is not falling, so the stock-price EWI is not triggered.
5. D: longer liability maturity and diversification reduce liquidity risk (BCBS flags lower durations and more concentration as warnings).

Why A is correct: Widening spreads on the bank's own debt and CDS are explicit OCC and Federal Reserve EWIs because they show deteriorating market confidence before funding is lost.

Why the others are wrong

- **B:** Growth is a BCBS warning only when funded by unstable liabilities; stable funding mitigates the risk.
- **C:** Reverses the stock-price EWI: the warning is a falling price of the bank's own stock, not a stable one.
- **D:** Longer maturities and diversification are improvements; the BCBS warning signs are lower durations and more concentration.

Q128

AnalystPrep Q.3848

REBUILT — Original item used operational-risk style indicators (audit scores, staff turnover, trade volumes) not in the liquidity EWI reading, and 'all of the above' was unsupported; rebuilt around the BCBS/OCC EWI lists.

A risk manager is compiling early warning indicators for a bank's liquidity risk dashboard using regulatory guidance (OCC 2012 and BCBS 2008). Which of the following are recognized EWIs of deteriorating liquidity?

I. Falling stock price of the bank

II. Higher margin requirements imposed on the bank
III. Longer average maturity of the bank's liabilities
IV. Frequent breaches or near breaches of internal limits

A. I and II only
B. I, II and IV only
C. II, III and IV only
D. I, II, III and IV

Answer: B

Step by step

1. OCC (2012) EWIs include falling stock prices (I) and higher margins required (II).
2. BCBS (2008) EWIs include frequent breaches or near breaches of limits (IV).
3. BCBS also lists LOWER liability durations as a warning; a longer maturity (III) is a mitigant, not an EWI.
4. So I, II and IV are EWIs: answer B.

Why B is correct: I, II and IV appear in the OCC and BCBS lists; III is the reverse of the BCBS warning sign (lower liability durations).

Why the others are wrong

- **A:** Omits IV; frequent limit breaches are an explicit BCBS (2008) EWI.
- **C:** Includes III (longer maturity is favorable) and omits I (falling stock price).
- **D:** Treats III as an EWI; the BCBS indicator is shorter, not longer, liability duration.

Q129 AnalystPrep Q.3847

Your supervisor asks you to prepare a list of sound early warning indicators for liquidity problems for your bank. Which of the following are sound early warning indicators of a potential liquidity problem?
I. Negative publicity regarding an asset class owned by the institution
II. Stock price declines
III. Widening debt/credit-default-swap spreads
IV. Increase in credit lines
V. Significant deterioration in the bank's financial condition
VI. Increase in the weighted average maturity of liabilities

A. I, II, III, V
B. I, II, IV, VI
C. II, III, IV, V
D. III, IV, V, VI

Answer: A

Step by step

1. Classify each item against the regulatory EWI lists (Fed SR 10-6, OCC 2012, BCBS 2008).
2. I (bad publicity about held assets), II (falling stock prices), III (rising spreads) and V (worsening balance sheet/financial condition) are listed EWIs.
3. IV: an increase in credit lines is a positive development; a reduction in financing from lenders is the warning sign. VI: BCBS 2008 flags lower (shorter) liability durations, so an increase in maturity is favourable.
4. The set containing exactly I, II, III, V is A.

Why A is correct: Sound EWIs are adverse changes: bad publicity, falling equity prices, widening spreads and balance-sheet deterioration; more credit lines and longer liability maturity are favourable.

Why the others are wrong

- **B:** Includes IV and VI, which are favourable changes (more credit lines, longer liability maturity) - direction of the change misread.
- **C:** Includes IV (increase in credit lines) - the EWI is a reduction in financing, not an increase.
- **D:** Includes IV and VI, both favourable, and omits I and II.

Q130 AnalystPrep Q.3852

Early warning signs can be identified using a set of parameters and processes that identify probable risks at a nascent stage. When potential risk events are recognized at an earlier stage, an investment bank is able to prepare itself as the event develops. Which one of the following is LEAST likely an early warning sign for such events?

A. Narrowing of debt or credit default swap spreads

B. Increases in currency mismatches
C. Drop in credit lines
D. Increasing redemptions of CDs before maturity

Answer: A
Step by step
1. Each option is checked for the direction of the adverse signal.
2. Increasing currency mismatches are an EWI under BCBS (2008).
3. A drop in credit lines (reduced financing by lenders, OCC 2012) and rising early CD redemptions (deposit outflows) both signal funding stress.
4. Narrowing spreads signal improving market perception; the EWI is widening/increasing spreads (OCC 2012, SR 10-6). A is the least likely warning sign.

Why A is correct: The warning signal is a widening of debt/CDS spreads; narrowing spreads indicate lower perceived risk.
Why the others are wrong
- **B:** This is a BCBS (2008) EWI, so it is a warning sign - not the answer.
- **C:** Reduced financing from lenders is an OCC (2012) EWI - a warning sign.
- **D:** Early redemption of CDs signals deposit flight - a warning sign.

Q131 AnalystPrep Q.4074

Early warning indicators (EWIs) should act as warning signs to the management to evaluate how changing market conditions and the institution's business strategy may be impacted. This should prompt the management to act in advance of oncoming market disruptions proactively. Which among the following factors is NOT an EWI encompassing market and business factors?

A. A spike in the market volatility
B. An increase in asset quality
C. Real or perceived negative publicity
D. Canceling loan commitments and refusing to renew maturing loans

Answer: B
Step by step
1. EWIs capture adverse changes in market or business conditions.
2. A volatility spike and negative publicity (Fed SR 10-6) are adverse market/business signals.
3. Cancelled commitments and non-renewal of maturing loans reflect reduced financing and tightening funding conditions (OCC 2012).
4. An increase in asset quality is favourable (a CAMELS downgrade would be the warning), so B is not an EWI.

Why B is correct: EWIs flag deterioration; improving asset quality is a positive development and therefore not a warning indicator.
Why the others are wrong
- **A:** A volatility spike is an adverse market condition and is an EWI.
- **C:** Negative publicity, real or perceived, can trigger funding runs and is an EWI (SR 10-6).
- **D:** Pulling commitments and refusing renewals signals tightening of funding conditions - an EWI.

Q132 AnalystPrep Q.5380

ABC Bank employs a collection of early warning indicators to detect emerging risks and possible vulnerabilities in its liquidity position. Which of the following serves as an early warning indicator for a potential liquidity issue?

A. A rapid rise in the leverage ratio and heavy reliance on short-term repo funding.
B. A reduction in the collateral haircuts imposed on the bank's collateralized exposures.
C. An upward revision in the credit rating.
D. Increased diversification of assets.

Answer: A
Step by step
1. BCBS (2008) EWIs include very sharp asset growth, more concentrated funding and lower liability durations.
2. Rapid leverage growth financed by short-term repo is sharp asset growth funded by short-duration, rollover-

dependent liabilities - an EWI.
3. Lower haircuts, a rating upgrade and greater diversification all indicate improving liquidity or funding conditions, not deterioration.

Why A is correct: Rapid balance-sheet growth funded by short-term repo creates rollover risk and matches the BCBS EWIs of sharp asset growth and shortened liability duration.

Why the others are wrong

- **B:** Lower haircuts mean easier funding against collateral; a rise in haircuts would be the EWI.
- **C:** A rating upgrade is favourable; CAMELS or rating downgrades are the EWIs.
- **D:** Diversification reduces concentration; more concentrated assets/liabilities is the EWI.

Q133 BT answers P2.T8.20.4.1 · BT notes P2.T8.20.4.1 · BT Mock Q19

Liquidity early warning indicators (EWI) can be compared to an automobile's dashboard signal system, where the appearance of a red light points attention to something that may become a problem if not addressed. Each of the following is a credible red indicator (red flag) in a liquidity EWI dashboard EXCEPT which is not a red-flag liquidity indicator?

A. Narrowing debt/CDS spreads.
B. Sudden increase in debt costs.
C. Sudden growth in assets accompanied by volatile liabilities.
D. Rapid decline in the weighted average maturity of liabilities.

Answer: A

Step by step

1. Test the direction of each change against the BCBS/OCC/Fed lists.
2. B: higher borrowing costs/spread increases are EWIs (OCC 2012).
3. C: very sharp asset growth (BCBS 2008) funded by volatile liabilities is a red flag.
4. D: lower liability durations are an EWI (BCBS 2008).
5. A: narrowing spreads indicate improving market confidence; widening spreads would be the flag. A is the exception.

Why A is correct: Red flags are adverse movements; narrowing debt/CDS spreads are favourable, whereas widening spreads would signal stress.

Why the others are wrong

- **B:** Rising debt costs are an OCC EWI (higher borrowing rates, spread increases) - a genuine red flag.
- **C:** Sharp asset growth with volatile funding is a BCBS (2008) EWI - a genuine red flag.
- **D:** Shorter liability maturity is a BCBS (2008) EWI - a genuine red flag.

Q134 GARP PE1 Q1 · Mock 3 Q79

A risk consultant is assessing a small bank's liquidity risk profile. While reviewing a presentation produced by the bank, the consultant comes across a list of early warning indicators used to signal potentially heightened liquidity risk. Which of the following trends should the consultant consider as the strongest warning signal for potential liquidity risk at the bank?

A. A decrease in the stock price of some of the bank's peers but not in the stock price of the bank itself.
B. An increase in available credit lines received from other financial institutions.
C. Widening spreads on the bank's issued debt and credit default swaps.
D. Significant asset growth funded by an increase in stable liabilities.

Answer: C

Step by step

1. A bank-specific decline in stock price would be an EWI, but here the bank's own price is unchanged, so the peer fall is not a bank-specific warning.
2. More credit lines means more available funding - favourable, not a warning.
3. Widening spreads on the bank's own debt and CDS are a listed EWI (OCC 2012, SR 10-6), forward-looking and bank-specific.
4. Asset growth is a warning only if funded by volatile liabilities; funding by stable liabilities mitigates the risk.

Why C is correct: Widening spreads on the bank's own debt and CDS show the market is pricing higher bank-specific risk, and they raise funding cost, so they are the strongest signal.

Why the others are wrong

- **A:** The bank's own stock price has not fallen; only peers' prices have, so the signal is not bank-specific.
- **B:** Increased credit lines improve liquidity; the EWI is a reduction in financing.
- **D:** Stable funding offsets the BCBS asset-growth warning; the EWI is sharp growth funded by volatile liabilities.

Q135 Schweser Module Quiz 68.1 Q3

Which supervisory guideline for early warning indicators (EWIs) focuses on intraday liquidity monitoring indicators?

A. BCBS (2008).
B. BCBS (2012).
C. OCC (2012).
D. SR 10-6.

Answer: B

Step by step

1. Recall what each guideline emphasizes: OCC (2012) = EWIs for embedded options and advance notice of negative events; BCBS (2008) = general deterioration indicators (asset growth, concentrations, currency mismatches, shorter liability duration, limit breaches); BCBS (2012) = intraday liquidity indicators; SR 10-6 = EWIs and event triggers tied to the firm's liquidity risk profile.
2. The intraday indicators (daily maximum liquidity, intraday availability, total intraday payments and timing, time-sensitive obligations, payments for financial institution customers, intraday credit lines to such customers) come from BCBS (2012).

Why B is correct: BCBS (2012) is the guideline whose EWI examples are specifically intraday liquidity indicators such as intraday availability and total intraday payments with their timing.

Why the others are wrong

- **A:** BCBS (2008) lists general balance-sheet deterioration indicators (sharp asset growth, concentrations, currency mismatches, shorter liability durations, limit breaches), not intraday indicators; the error is confusing the two BCBS years.
- **C:** OCC (2012) covers EWIs for securities and derivatives with embedded options and advance-notice indicators such as CAMELS downgrades and spread increases; it is not intraday-focused.
- **D:** SR 10-6 (Federal Reserve) stresses EWIs and event triggers consistent with the firm's liquidity risk profile, with examples like bad publicity and balance sheet worsening, not intraday monitoring.

Q136 Schweser Module Quiz 68.1 Q4

Which supervisory guideline for early warning indicators (EWIs) includes a specific provision on EWIs that deals with embedded options?

A. BCBS (2008).
B. BCBS (2012).
C. OCC (2012).
D. SR 10-6.

Answer: C

Step by step

1. Identify the guideline whose EWIs address securities and derivatives with embedded options (e.g., callable debt).
2. This is the OCC (2012) guideline: EWIs should indicate when the options are likely to be exercised and flag any associated contingent liabilities.

Why C is correct: OCC (2012) explicitly requires EWIs for securities and derivatives with embedded options, signalling likely exercise and contingent liabilities.

Why the others are wrong

- **A:** BCBS (2008) focuses on general indicators of liquidity deterioration or increased funding need (asset growth, concentrations, currency mismatches, limit breaches); it has no embedded-option provision.
- **B:** BCBS (2012) focuses on intraday liquidity indicators such as intraday availability and intraday credit lines, not embedded options.
- **D:** SR 10-6 emphasizes EWIs and event triggers matched to the firm's liquidity risk profile and timely communication, not embedded options.

LO 68.c

Discuss the applications of EWIs in the context of the liquidity risk management process.

Q137

AnalystPrep Q.3849

REBUILT — Option A was grammatically inconsistent with the stem and options B/C described other tools; reworded all options to parallel form with the same correct answer.

The primary purpose of putting early warning indicators in place is to:

A. measure precisely how risky each activity of the bank is.
B. ensure that the bank holds enough liquid assets to meet its financial obligations.
C. safeguard asset quality and ensure that liability limits are not breached.
D. prompt management discussion and necessary corrective action.

Answer: D

Step by step

1. Schweser: the main objective of EWIs is to get management to acknowledge the situation and promote dialogue and action (like a dashboard warning light).
2. EWIs signal; they do not themselves hold liquidity, set limits or measure risk.
3. Hence D.

Why D is correct: EWIs are triggers; their value lies in prompting management dialogue, documentation and corrective action through escalation.

Why the others are wrong

- **A:** Describes a risk measure (e.g., VaR), not a warning trigger.
- **B:** Holding liquid buffers is the role of liquidity buffers and ratios, not of EWIs.
- **C:** Asset-quality safeguards and limit frameworks are separate controls; EWIs only alert management.

Q138

BT answers P2.T8.20.4.3 · BT notes P2.T8.20.4.3

REBUILT — Options A and C were vague/unfalsifiable and D asserted unsupported regulatory prescription; reworded distractors so only historical-data calibration is correct.

A bank uses a green/amber/red (stoplight) system to communicate EWI performance against thresholds: green is within normal bounds, amber requires further investigation, and red may warrant immediate response. Which of the following is the BEST way to start calibrating these thresholds?

A. Set them by senior management judgment alone, without reference to data.
B. Use historical data on the EWI, for example its standard deviation over a sufficiently long period including recent events.
C. Abandon thresholds, since stoplight systems are considered obsolete and ineffective.
D. Adopt a single fixed threshold prescribed by BCBS (2008) for all banks.

Answer: B

Step by step

1. Schweser: past data is a frequent source of information for the EWI's standard deviation and for calibrating thresholds.
2. The period should be long enough (e.g., one year) yet include recent events; thresholds are then backtested.
3. A ignores data; C contradicts industry practice; D is false because BCBS (2008) lists indicators but prescribes no threshold values.

Why B is correct: Historical data allows the amber/red cutoffs to be calibrated to avoid both missed signals and excessive false alarms, and then backtested.

Why the others are wrong

- **A:** Judgment without data cannot be backtested or balance missed issues against false alarms.
- **C:** Stoplight thresholds are standard practice; the issue is calibration, not abandonment.
- **D:** Regulators give example indicators, not universal threshold levels; calibration is bank-specific.

Q139 Schweser Module Quiz 68.1 Q1

Which component of the early warning indicator (EWI) framework represents the ultimate desired outcome when measures suggest a potential liquidity problem?

A. Escalation.
B. Integrated systems.
C. Protection.
D. Reporting.

Answer: A

Step by step

1. The framework comprises measures, escalation, reporting, integrated systems and thresholds.
2. Reporting, integrated systems and thresholds support the move from measures to escalation.
3. Escalation brings the issue to relevant management for remedies; this is the desired outcome.

Why A is correct: Measures are only a starting point; they matter when tied to escalation to appropriate personnel who can act.

Why the others are wrong

- **B:** Integrated systems are an enabler giving consistent data, not the end outcome.
- **C:** Protection is not a component of the EWI framework.
- **D:** Reporting is a supporting step that delivers timely information toward escalation.

Q140 AP Mock 1 Q42

Joe Hart is developing early warning indicators to help his bank monitor potential liquidity stress events. He has drafted the following background guidelines:
I. All EWIs must be updated every two months to detect potentially threatening events.
II. The bank should conduct a variety of short-term and protracted, bank-specific and market-wide liquidity stress tests using conservative and regularly reviewed assumptions.
III. The bank should establish a robust escalation policy so that critical decisions and transactions are handled at an appropriate level of management.
Which of these guidelines is incorrect?

A. I.
B. II.
C. III.
D. None of the above.

Answer: A

Step by step

1. Guideline I: Schweser stresses timely reporting, e.g. daily, and intraday (hourly) for banks with heavy trading activity. A two-month refresh is far too slow to give advance warning, so I is incorrect.
2. Guideline II: EWIs are used in normal and stressed states and over different time horizons, and stress tests should be varied and conservative, so II is sound.
3. Guideline III: EWIs only improve liquidity risk management when tied to a clear escalation plan, so III is sound.
4. Only I is incorrect.

Why A is correct: EWIs are valuable only if they give advance notice, which requires frequent (daily or even intraday) reporting rather than a bimonthly refresh.

Why the others are wrong

- **B:** II is consistent with the use of varied stress tests across horizons and states; it is not wrong.
- **C:** III is consistent with the requirement that EWIs be embedded in a clear escalation plan.
- **D:** Wrong because I is incorrect; choosing it means missing that a two-month cycle is too slow.

Q141 BT answers P2.T8.25.2.2

REBUILT — Distractors were giveaways and the best answer was vague; rewrote options so the correct one is escalation per the plan and each wrong one reflects a realistic mistake.

Over the past week a large bank observed: a 12% decline in corporate deposits concentrated among large institutional clients; a 40 bps widening in its senior unsecured spreads and CDS spreads; an LCR fall from 120% to 105%; higher margin demands from prime brokerage counterparties; and media speculation about its liquidity. The bank

has a green/amber/red EWI framework with a defined escalation plan. Which response is most consistent with how EWIs should be applied in the liquidity risk management process?

A. Escalate the multiple deteriorating EWIs to senior management under the escalation plan, run liquidity stress tests and review funding and counterparty positions, with documentation of decisions
B. Take no escalation step because the LCR of 105% is still above the 100% regulatory minimum
C. Immediately sell non-core assets to rebuild liquidity, accepting distressed prices
D. Request central bank emergency liquidity assistance before exploring any other funding measures

Answer: A

Step by step

1. Several independent internal (deposit outflows, LCR) and external (spreads, CDS, margin calls, media) EWIs are deteriorating at once.
2. EWIs are useful only if tied to escalation: the issue goes to relevant management with remedies matched to severity, and management actions are documented.
3. Stress testing and a funding/counterparty review determine whether the buffer is adequate under the now more plausible stress.
4. Therefore A is the response consistent with the EWI framework.

Why A is correct: EWIs exist to trigger timely escalation, dialogue and documented action before a problem becomes acute; a cluster of adverse indicators calls for escalation, analysis and preparation, not inaction or drastic measures.

Why the others are wrong

- **B:** Treats the regulatory minimum as the trigger; EWIs are leading indicators meant to act before a breach, and the LCR has already fallen 15 points alongside other signals.
- **C:** Forced asset sales at distressed prices are a last-resort remedy, not the first response to early warnings; it also crystallises losses and may worsen market perception.
- **D:** Going to the central bank first skips escalation and internal/market funding options and signals distress; it is not the proportionate response at this stage.

Q142 BT answers P2.T8.25.2.3

REBUILT — Original had OCR-damaged numbers (credit line commitments 'increasing' from $7.2bn to $2bn, spreads 750 to 760) and an ambiguous 'most critical' premise; rebuilt as a clear threshold-calibration question on the stoplight approach.

A bank's EWI dashboard uses green, amber and red thresholds. The Chief Risk Officer finds that the amber threshold for short-term wholesale funding reliance is set so tightly that the indicator turns amber several times every month, and in nearly all cases the follow-up shows no real problem. Which action is most appropriate?

A. Recalibrate the green-to-amber threshold using sufficiently long historical data (e.g., one year) that includes recent events, and backtest it, so it limits false alarms without missing emerging issues
B. Remove the amber level and keep only red so that management is contacted only for serious problems
C. Widen the amber threshold as far as possible so that the indicator almost never changes colour
D. Ignore the amber signals, because only a red status requires any management follow-up

Answer: A

Step by step

1. The threshold produces excessive false alarms, so it is poorly calibrated.
2. Per the reading, the shift from green to amber should be precisely calibrated to avoid both missing emerging issues and excessive false alarms.
3. Calibration uses past data (e.g., to compute the indicator's standard deviation) over a period long enough to be reliable (e.g., one year) but including recent events, and thresholds should be backtested and recalibrated if needed.
4. So A is correct.

Why A is correct: Thresholds must balance sensitivity and false alarms; the remedy is data-driven recalibration and backtesting, not eliminating or neutralising the early-warning level.

Why the others are wrong

- **B:** Removing amber eliminates the intermediate 'further follow-up' state and delays warnings until the problem is serious, defeating the point of an early warning.
- **C:** Over-widening swings the error the other way: real emerging issues would be missed.
- **D:** Amber requires additional follow-up by design; ignoring it would be a failure of the escalation process, and the fix lies in calibration.

Q143 AnalystPrep Q.3854

REBUILT — Original 'Blue' option was a giveaway and 'most concerned' was thin; rebuilt to test the meaning of each stoplight status with realistic distractors.

A firm reports EWIs using a green, amber and red stoplight system. One EWI has just moved from green to red. Which interpretation and response is correct?

A. A serious problem exists; the issue requires immediate attention and remedy, escalated to relevant management

B. A potential problem; additional follow-up is required but no urgent action

C. The indicator is satisfactory; no further action is required

D. The indicator is within its normal range, but the threshold should be tightened

Answer: A

Step by step

1. Green means no further action is required.
2. Amber means additional follow-up is needed.
3. Red means immediate attention to remedy a serious problem.
4. Red is the highest-concern status, so A.

Why A is correct: In the stoplight approach, red is the status requiring immediate attention and remediation, and it should be escalated accordingly.

Why the others are wrong

- **B:** Describes amber (additional follow-up), the intermediate status, not red.
- **C:** Describes green; a move to red is the opposite of satisfactory.
- **D:** A red status signals a breach of the threshold, not normal range; recalibration is a separate backtesting matter.

Q144 AnalystPrep Q.3855

Swift Investments, a U.S.-based firm, is a clearing member at the Chicago Mercantile Exchange. It uses a set of early warning indicators to detect liquidity distress at a nascent stage. Given its substantial trading activity, which reporting schedule is most appropriate for the firm's EWI dashboard?

A. Daily

B. Weekly

C. Intraday

D. Monthly

Answer: C

Step by step

1. Timely reporting (e.g., daily) gives time to react to negative events.
2. Firms with a significant trading focus are more likely to use intraday reporting because of their greater exposure to external market conditions.
3. A CME clearing member faces fast-moving margin and market conditions, so intraday reporting is most appropriate: C.

Why C is correct: Heavily trading firms can see liquidity conditions change within a day, so intraday EWI reporting provides the most timely warning.

Why the others are wrong

- **A:** Daily is the standard timely frequency for most banks, but for a trading-intensive clearing member data may be stale within the day.
- **B:** Weekly is too infrequent for fast-moving trading and margin exposures.
- **D:** Monthly is far too slow to serve as an early warning for a trading-focused firm.

Q145 AnalystPrep Q.3845

REBUILT — Original asked a vague 'cost-benefit' question not supported by the reading; rebuilt around the stated objective of EWIs: prompting management acknowledgment, dialogue, action and documentation.

Which of the following best describes the main objective of an early warning indicator trigger in a bank's liquidity risk management process?

A. To get management to acknowledge the situation and promote the necessary dialogue and actions, with documentation of these items

B. To automatically execute asset sales whenever a threshold is breached

C. To replace stress testing by measuring liquidity under hypothetical scenarios
D. To report liquidity metrics to regulators after a liquidity event has occurred

Answer: A
Step by step
1. The reading states the main objective of EWIs is to get management to acknowledge the situation and promote dialogue and actions, together with documentation.
2. EWIs prompt escalation and judgement; they do not mechanically execute remedies (B).
3. They are leading indicators and complement, not replace, stress testing (C), and are forward-looking, not after-the-fact reporting (D).
4. Answer A.

Why A is correct: EWI triggers are an escalation mechanism: they make sure deteriorating conditions reach decision makers who discuss, act and document.

Why the others are wrong
- **B:** Confuses a trigger with an automatic remedy; actions depend on severity and management judgement.
- **C:** Stress testing examines hypothetical stressed states; EWIs monitor actual current metrics in normal states. They are complementary.
- **D:** EWIs are meant to be leading and give advance notice, not report after the event.

Q146 BT answers P2.T8.25.2.1

REBUILT — Stem said 'three banks' but four listed and header order was reconstructed; stated explicitly that there are four banks with clear columns and (start - end) values.

A regulator tracks EWIs for four banks over 12 months. Values are start - end:

Bank	NPL ratio	Loan-to-deposit ratio	CDS spread (bps)	LCR
Bank A	2.5% - 4.8%	85% - 95%	80 - 150	110% - 85%
Bank B	1.8% - 2.1%	78% - 80%	60 - 65	130% - 125%
Bank C	3.2% - 6.0%	90% - 102%	95 - 180	105% - 75%
Bank D	4.0% - 3.5%	88% - 82%	70 - 72	115% - 90%

Based on the EWI trends, which bank shows the strongest signs of financial distress?

A. Bank A
B. Bank B
C. Bank C
D. Bank D

Answer: C
Step by step
1. Compute changes. A: NPL +2.3 pts, LDR +10 pts, CDS +70 bps, LCR -25 pts.
2. B: NPL +0.3, LDR +2, CDS +5, LCR -5.
3. C: NPL +2.8, LDR +12 (now above 100%), CDS +85, LCR -30 (ends at 75%, well below 100%).
4. D: NPL -0.5, LDR -6, CDS +2, LCR -25 (ends at 90%).
5. Bank C deteriorates most on every indicator, breaching 100% on LDR and falling to 75% LCR: C.

Why C is correct: Bank C has the worst deterioration on all four indicators and ends with an LCR far below the 100% minimum and loans exceeding deposits.

Why the others are wrong
- **A:** Deteriorating clearly (LCR 85%, CDS +70) but every indicator is less severe than Bank C's.
- **B:** Almost stable; changes are small and LCR remains 125%.
- **D:** NPLs and LDR improve and CDS is flat; only LCR falls, to 90%, so it is the least concerning after B.

Reading 69. The Investment Function in Financial-Services Management

Peter Rose and Sylvia Hudgins, Bank Management & Financial Services, 9th Edition (New York, NY: McGraw-Hill, 2013). Chapter 10.

LO	Learning objective	Qs
69.a	compare various money market and capital market instruments and discuss their advantages and disadvantages.	16
69.b	identify and discuss various factors that affect the choice of investment securities by a bank.	22
69.c	apply investment maturity strategies and maturity management tools based on the yield curve and duration.	11

LO 69.a

Compare various money market and capital market instruments and discuss their advantages and disadvantages.

Q147

Mock 2 Q10

REBUILT — Original left T-note vs T-bond both defensible for a nine-month horizon; options revised so only the T-note meets all three objectives.

An individual investor wants to invest USD 500,000 for nine months. He requires absolute safety for his entire investment, does not want the lowest possible yield, and wants the potential to earn moderate capital gains during his horizon. Based solely on these objectives, which investment is most suitable?

A. Certificate of deposit (CD).
B. Treasury bill (T-bill).
C. Commercial paper.
D. Treasury note (T-note).

Answer: D

Step by step

1. Absolute safety on USD 500,000: only full-faith-and-credit government debt qualifies. CDs are federally insured only up to USD 250,000, and commercial paper carries issuer credit risk.
2. Not the lowest yield: T-bills are the lowest-yielding government instruments, and their return is only the fixed discount.
3. Potential for moderate capital gains: coupon-bearing T-notes trade at prices that move with rates and typically yield more than T-bills.
4. Only the T-note satisfies all three.

Why D is correct: T-notes combine government backing, a yield above T-bills, and price movement that offers capital gain potential.

Why the others are wrong

- **A:** Fails absolute safety: only the first USD 250,000 is insured, so half the investment is exposed to bank credit risk.
- **B:** Safe, but it is the lowest-yield instrument and offers no coupon or meaningful capital gain potential, violating the second and third objectives.
- **C:** Corporate, uninsured, issued at a discount for short maturities; it carries credit risk and does not give absolute safety.

Q148

AP Mock 3 Q22

REBUILT — Original maturity of exactly one year sits on the T-bill/T-note boundary; changed to six months.

A security issued by the United States federal government that matures six months after the date of issue is most likely a:

A. Treasury note.
B. Treasury bond.
C. Treasury bill.
D. Structured note.

Answer: C

Step by step

1. Maturity at issue is six months, i.e. within one year.
2. Only T-bills are issued with maturities of one year or less; they are zero-coupon, issued at a discount and mature

at par.
3. T-notes mature up to 10 years and T-bonds beyond 10 years at issuance.

Why C is correct: T-bills are the Treasury's short-term discount securities, with original maturity within one year.

Why the others are wrong

- **A:** T-notes are coupon securities issued with maturities of one to ten years, not six months.
- **B:** T-bonds have original maturities beyond 10 years.
- **D:** A structured note is a privately issued derivative-linked security, not a US Treasury instrument.

Q149 AnalystPrep Q.3860

Any securities that reach maturity within one year are most likely regarded as:

A. Money market securities
B. Capital market securities
C. Federal agency securities
D. Certificates of deposit

Answer: A

Step by step

1. Securities are classified by maturity: one year or less = money market; more than one year = capital market.
2. Agency securities and CDs are issuer/instrument types, not maturity-based classes.

Why A is correct: The money-market/capital-market split is made at a one-year maturity, with short-dated securities low in risk and yield.

Why the others are wrong

- **B:** Capital market instruments have maturities greater than one year.
- **C:** Agency securities are defined by the issuer (government-sponsored entities), not by maturity.
- **D:** CDs are one specific instrument and can have terms beyond a year; maturity within a year does not make a security a CD.

Q150 AnalystPrep Q.3864

Which of the following statements is not true?

A. Treasury bills are the long term debt obligations issued by the federal government
B. Inflation risk is the likelihood that rising prices for goods and services will erode the purchasing power of interest income and repaid principal from security or loan
C. Bankers' acceptances are considered to be among the safest of all money market instruments
D. Investment securities are expected to help stabilize financial institutions' income

Answer: A

Step by step

1. Test each statement against the reading.
2. A: T-bills are short-term securities maturing within one year, so calling them long-term obligations is false.
3. B: inflation risk is the erosion of purchasing power of interest and principal; true.
4. C: bankers' acceptances are bank-guaranteed, pure discount money market instruments and are among the safest; true.
5. D: investment securities provide stable income to offset volatile loan income; true.

Why A is correct: T-bills are short-term (maturity of one year or less) zero-coupon securities, so the statement that they are long-term obligations is the only false one.

Why the others are wrong

- **B:** True definition of inflation risk; not the false statement.
- **C:** True: the bank is the primary obligor, so acceptances are very safe.
- **D:** True: securities portfolios are used to stabilize income, so this is not the false statement.

Q151 AnalystPrep Q.3866

Money market instruments which represent a bank's commitment to pay a specified amount of money on a specific future date under specific conditions, are known as:

A. Eurocurrency deposits

B. Government agency securities
C. Bank qualified bonds
D. Bankers' acceptances

Answer: D
Step by step
1. The description is a bank guaranteeing a customer's payment in a trade transaction at a future date.
2. That is the definition of a bankers' acceptance: the bank becomes the primary obligor and the instrument is issued at a discount and matures at par.

Why D is correct: A bankers' acceptance is a bank's payment commitment, mainly used in import/export trade finance.
Why the others are wrong
- **A:** Eurocurrency deposits are uninsured time deposits at large banks, not a payment commitment.
- **B:** Agency securities are debt of government-sponsored or owned agencies with implied support.
- **C:** Bank qualified bonds are a municipal-bond tax concept, not a bank payment guarantee.

Q152
AnalystPrep Q.4239

REBUILT — Original options (18/20/24/36) were arbitrary; rebuilt distractors from realistic formula errors and stated the bank discount convention.
A Treasury bill has a face value of $100 and is purchased for $88. It matures in 120 days. Using the bank discount method (360-day year), the discount rate is:
A. 12.00%
B. 40.91%
C. 36.50%
D. 36.00%

Answer: D
Step by step
1. Discount = 100 - 88 = 12.
2. $DR = \frac{F-P}{F} \times \frac{360}{t} = \frac{12}{100} \times \frac{360}{120}$.
3. $= 0.12 \times 3 = 36.00\%$.

Why D is correct: The bank discount rate expresses the dollar discount as a percentage of face value, annualized on a 360-day year.
Why the others are wrong
- **A:** Did not annualize the 120-day discount (12/100).
- **B:** Divided the discount by price (88) instead of face value: 12/88 × 360/120.
- **C:** Used 365 days instead of 360: 0.12 × 365/120.

Q153
AnalystPrep Q.4240

REBUILT — Stem did not state the 365-day convention (360-day gives a different answer) and one option was unexplained; stated convention and rebuilt options.
A Treasury bill has a face value of $100 and is purchased for $88. It matures in 120 days. Using the YTM (bond-equivalent) yield on a 365-day year, the equivalent yield is:
A. 36.00%
B. 40.91%
C. 13.64%
D. 41.48%

Answer: D
Step by step
1. Return on price: $(100 - 88)/88 = 13.64\%$.
2. Annualize with 365 days: $13.636\% \times 365/120 = 41.48\%$.

Why D is correct: The equivalent yield divides the gain by the purchase price (not face value) and annualizes on 365 days, so it exceeds the discount rate.
Why the others are wrong
- **A:** This is the bank discount rate (divides by face value, 360 days).
- **B:** Used 360 days instead of 365: 12/88 × 360/120.

- **C:** Forgot to annualize the 120-day return.

Q154

AnalystPrep Q.4244

REBUILT — 'Canadian' bill implies a 365-day convention and options were giveaway decimal-place errors; stated the 360-day bank discount method and rebuilt options.

A Treasury bill is priced at \$89.84 per \$100 of par and matures in 124 days. Using the bank discount method (360-day year), the discount rate is closest to:

A. 2.95%
B. 29.50%
C. 33.29%
D. 29.91%

Answer: B

Step by step

1. Discount = 100 - 89.84 = 10.16.
2. $DR = \frac{10.16}{100} \times \frac{360}{124} = 0.10160 \times 2.9032$.
3. $= 29.50\%$ (exact 29.497%).

Why B is correct: Discount over face value, annualized by 360/t.

Why the others are wrong

- **A:** Decimal-place slip: ten times too small.
- **C:** Divided by price: $10.16/89.84 \times 365/124 = 33.29\%$, the equivalent yield rather than the discount rate.
- **D:** Used 365 days: $0.1016 \times 365/124 = 29.91\%$.

Q155

AnalystPrep Q.5420

REBUILT — Distractors were not tied to realistic errors; rebuilt options and kept the irrelevant required return and YTM as deliberate noise.

A treasury analyst is given the following data on a money market security:

Purchase price	\$95
Face value	\$100
Required rate of return	12%
Yield to maturity	8%
Time to maturity	85 days

Using a 360-day year, the bank discount rate is closest to:

A. 5.3%
B. 22.3%
C. 21.2%
D. 21.5%

Answer: C

Step by step

1. Discount = 100 - 95 = 5.
2. $DR = \frac{5}{100} \times \frac{360}{85} = 0.05 \times 4.2353$.
3. $= 21.18\% \approx 21.2\%$. Required return and YTM are not inputs.

Why C is correct: The discount rate uses only face value, price and days to maturity.

Why the others are wrong

- **A:** Not annualized and divided by price: 5/95.
- **B:** Divided by price instead of face value: $5/95 \times 360/85$.
- **D:** Used 365 days: $0.05 \times 365/85 = 21.47\%$.

Q156

AnalystPrep Q.5407 · AnalystPrep Q.5424

Joshua is a portfolio manager considering bankers' acceptances for the firm's short-term portfolio. He knows they are short-term trade-finance instruments in which a bank guarantees a customer's payment, offer yields close to those on T-bills and Eurocurrency deposits, and can be discounted at the Federal Reserve Bank. Based on this, which of the following is the most valid disadvantage of bankers' acceptances?

A. The risk that the bank will default on its guarantee of the acceptance.

B. The availability of bankers' acceptances for certain maturities.
C. The possibility that bankers' acceptances could be subject to negative interest rates.
D. The inability to trade bankers' acceptances in the secondary market.

Answer: B
Key correction: Merged with AnalystPrep Q.5424 (same fact). Q.5424's key said 'issued in odd denominations'; the reading lists limited availability at specific maturities (and taxable income) as the disadvantage. Here B is correct because it states the availability disadvantage.
Step by step
1. Recall the reading's stated disadvantages of bankers' acceptances: limited availability at specific maturities and taxable income.
2. Check each option: only B matches; the others contradict features of the instrument.

Why B is correct: Acceptances arise from specific trade transactions, so supply may not match the maturity an investor wants.
Why the others are wrong
- **A:** The bank is the primary obligor on a very safe instrument; default risk is minor, not the stated disadvantage.
- **C:** Pure discount instruments are issued below par and mature at par; negative interest rates are not a feature.
- **D:** The reading notes an active secondary market; limited resale is not a feature of acceptances.

Q157 AnalystPrep Q.5425

Which of the following securities is considered to carry an implicit guarantee of the U.S. government?
A. Commercial paper.
B. Federal agency securities issued by government-sponsored enterprises
C. International eurocurrency deposits.
D. Short-term treasury notes.

Answer: B
Step by step
1. Identify which securities have government backing: Treasuries have explicit full faith and credit.
2. Agency securities (Fannie Mae, Freddie Mac, etc.) have no direct guarantee but support is implied because Congress would likely rescue them.
3. Commercial paper and Eurocurrency deposits have no government support.

Why B is correct: GSE agency securities are not direct government obligations but are assumed to have implied support.
Why the others are wrong
- **A:** Corporate unsecured discount paper with no government support.
- **C:** Uninsured time deposits at large banks; no guarantee.
- **D:** Treasury notes have an explicit, not implicit, guarantee (full faith and credit).

Q158 BT answers P2.T8.20.5.1 · BT notes P2.T8.20.5.1

Which of the following is insured, at least up to some amount?
A. Treasury bills
B. Municipal bonds
C. Certificates of deposit (CD)
D. All of the above

Answer: C
Step by step
1. Treasury bills are backed by the full faith and credit of the U.S. government, but they are not deposit-insured instruments.
2. Municipal bonds are capital market instruments with no insurance and a low but non-zero default risk.
3. CDs issued by depository institutions carry federal deposit insurance up to $250,000.
4. Only CDs qualify, so 'All of the above' is also wrong.

Why C is correct: CDs are bank deposits and so benefit from federal insurance up to $250,000; T-bills rely on the government's taxing power and municipals on issuer credit, neither is insured.

Why the others are wrong

- **A:** Confuses government backing (taxing power) with insurance; T-bills are very safe but not insured.
- **B:** Municipal bonds are uninsured capital market instruments; tax exemption is mistaken for protection.
- **D:** Follows from wrongly treating A or B as insured.

Q159

GARP PE2 Q25

The treasurer at a US-based bank wants to invest some excess cash on the bank's balance sheet and is comparing different types of money market instruments to use for the potential investment. The treasurer wants the investment to meet the following criteria:
The investment time horizon is 3 months and the bank plans to hold the investment to maturity.
The bank is facing a very low incremental tax rate so is not looking for tax-exempt interest.
The bank does not want to increase its counterparty exposure to other banks but is willing to accept exposure to government entities or industrial firms with a high credit rating.
If the bank wants to earn the highest yield while meeting the above investment criteria, what money market instrument should the bank invest in?

A. Short-term municipal obligations
B. Eurocurrency deposits
C. US Treasury bills
D. Commercial paper

Answer: D

Step by step

1. Low tax rate: the tax exemption on short-term municipal notes has little value, and their pre-tax yield is lower, so A is unattractive.
2. No added bank counterparty exposure: Eurocurrency deposits are time deposits placed with large banks, so B is eliminated.
3. Remaining acceptable issuers: the U.S. government (T-bills) or highly rated industrial firms (commercial paper). Maturity of about 90 days or less fits a 3-month horizon.
4. Commercial paper offers a higher yield than T-bills, which are among the lowest-yielding money market assets, so choose commercial paper.

Why D is correct: Commercial paper is issued by large, highly rated corporations (not banks), matures in 90 days or less, and yields more than T-bills.

Why the others are wrong

- **A:** Tax-exempt yield is only valuable to high-tax investors; with a low tax rate the lower pre-tax yield is a disadvantage.
- **B:** Highest-yielding of several, but it creates exposure to banks, which violates the stated criterion.
- **C:** Meets the criteria but the yield is lowest; choosing it ignores the 'highest yield' requirement.

Q160

Mock 1 Q65

REBUILT — Original wording ('cash drag') was vague and did not make explicit that the instrument must be held for the full six months; reworded so that the six-month maturity and safety priority decide the answer.

An institutional investor has USD 50 million of cash to invest until a known USD payment date six months away and plans to hold to maturity. The investor wants the highest return available but places top priority on safety of principal, and wants a single instrument maturing at the six-month date. Which of the following instruments is most suitable?

A. Banker's acceptance
B. Commercial paper
C. International Eurocurrency deposit
D. Revenue anticipation note

Answer: A

Step by step

1. Maturity fit: commercial paper is typically 90 days or less and Eurocurrency deposits are 30, 60, or 90 days, so neither is available at six months.
2. A banker's acceptance can have a maturity up to six months, is a pure discount instrument, and the bank be-

comes the primary obligor, giving strong safety.
3. Banker's acceptances yield more than T-bills, meeting the return objective.
4. A revenue anticipation note is repaid only from future project revenues and has weakened credit quality, so it fails the safety priority.

Why A is correct: A banker's acceptance combines six-month availability, the accepting bank's obligation to pay, and a yield above T-bills.

Why the others are wrong

- **B:** Typically 90 days or less and unsecured corporate credit; does not match a six-month horizon with safety first.
- **C:** Fixed maturities are 30/60/90 days, and it is an uninsured deposit at a large bank.
- **D:** Repayment depends on future revenues of municipal projects whose credit quality has deteriorated; the tax exemption is irrelevant to the stated objective.

Q161 Schweser 69.1-1

Which of the following characteristics is not an advantage of Treasury bills?

A. They have stable market prices.
B. They can serve as collateral for borrowers.
C. They have higher yields than agency securities.
D. They are backed by the taxing power of the federal government.

Answer: C

Step by step

1. T-bills' short maturities keep prices stable (A is an advantage).
2. They are widely accepted as collateral (B is an advantage).
3. They carry the full backing of the U.S. government including its taxing power (D is an advantage).
4. Agency securities offer higher yields than T-bills, so T-bills have lower, not higher, yields; C is not an advantage.

Why C is correct: T-bills are among the safest instruments and therefore yield less than agency securities.

Why the others are wrong

- **A:** A genuine advantage: short maturity limits price volatility.
- **B:** A genuine advantage: T-bills are commonly pledged as collateral.
- **D:** A genuine advantage: full faith and credit of the government.

Q162 Schweser 69.1-2

Which of the following instruments is most common in trade finance transactions?

A. Commercial paper
B. Bankers' acceptances
C. Tax-anticipation notes
D. Certificates of deposit

Answer: B

Step by step

1. Trade finance involves a bank guaranteeing a customer's payment in an export/import transaction.
2. A banker's acceptance is exactly that: the bank becomes the primary obligor in exchange for a fee.
3. Commercial paper funds corporations generally, TANs are municipal notes, and CDs are bank deposits; none is tied to trade credit.

Why B is correct: Bankers' acceptances are short-term discount instruments created to guarantee payment in import/export transactions.

Why the others are wrong

- **A:** Unsecured corporate funding paper, not tied to specific trade transactions.
- **C:** Municipal notes repaid from anticipated tax revenue.
- **D:** Time deposits issued by banks as a funding source, not a trade guarantee.

LO 69.b

Identify and discuss various factors that affect the choice of investment securities by a bank.

Q163 GARP book Chapter 4 end-of-chapter Q1

A 20-year U.S. Treasury bond with a par value of $1,000 is currently selling for $1,025. The bond carries a 6 percent coupon rate with payments made annually. If purchased today and held to maturity, what is its yield to maturity?

Answer: see model answer — Solve $1{,}025 = \sum_{t=1}^{20} 60/(1+y)^t + 1{,}000/(1+y)^{20}$. Using a financial calculator (N = 20, PV = -1,025, PMT = 60, FV = 1,000) gives $y \approx 5.79\%$. The YTM is below the 6% coupon because the bond is bought at a premium, so the capital loss of $25 at maturity reduces the return.

Step by step

1. Annual coupon = 6% x 1,000 = $60; N = 20; PV = -1,025; FV = 1,000.
2. YTM is the rate that equates the PV of all cash flows to the price: $1{,}025 = \sum 60/(1+y)^t + 1{,}000/(1+y)^{20}$.
3. Solve numerically: $y = 5.786\% \approx 5.79\%$.
4. Check: premium price implies YTM < coupon rate (5.79% < 6%).

Why this is correct: YTM is the discount rate equating the present value of coupons and par to the market price; a premium price forces the YTM below the coupon rate.

Q164 GARP book Chapter 4 end-of-chapter Q3

Calculate the yield to maturity of a 20-year U.S. government bond that is selling for $975 in today's market and carries a 5 percent coupon rate with interest paid semi-annually. Assume $1,000 par and quote the YTM as a nominal annual rate (periodic rate x 2).

Answer: see model answer — Semiannual coupon = $25 and N = 40. Solving $975 = \sum_{t=1}^{40} 25/(1+r)^t + 1{,}000/(1+r)^{40}$ gives $r \approx 2.601\%$ per half-year, so the nominal annual YTM is about 5.20% (effective annual about 5.27%). It exceeds the 5% coupon because the bond sells at a discount.

Step by step

1. Semiannual coupon = 0.05 x 1,000 / 2 = $25; N = 20 x 2 = 40; PV = -975; FV = 1,000.
2. Solve for the periodic rate: $r = 2.601\%$ per half-year.
3. Nominal annual YTM = 2 x 2.601% = 5.20%.
4. Optional: effective annual yield = $(1.02601)^2 - 1 = 5.27\%$.

Why this is correct: With semiannual coupons the periodic yield is solved over 40 half-year periods and doubled; a discount price puts YTM above the coupon rate.

Q165 GARP book Chapter 18 end-of-chapter Q1

A government bond is currently selling for $1,195 and pays $75 per year in interest for 14 years when it matures. If the redemption value of this bond is $1,000, what is its yield to maturity if purchased today for $1,195?

Answer: see model answer — Solve $1{,}195 = \sum_{t=1}^{14} 75/(1+y)^t + 1{,}000/(1+y)^{14}$ (N = 14, PV = -1,195, PMT = 75, FV = 1,000). The result is $y \approx 5.47\%$. It is well below the 7.5% coupon rate (75/1,000) because the bond is bought at a premium and the $195 premium is lost by maturity.

Step by step

1. Annual coupon = $75; N = 14; PV = -1,195; FV = 1,000.
2. Set PV of coupons plus par equal to the price and solve for y.
3. $y = 5.470\% \approx 5.47\%$.
4. Sanity check: premium bond so YTM (5.47%) < coupon rate (7.5%).

Why this is correct: YTM equates discounted coupons and par to the price; paying a premium means the investor earns less than the coupon rate.

Q166 GARP book Chapter 18 end-of-chapter Q2

REBUILT — Original depended on data from the preceding problem; stem made self-contained.

A savings institution buys a bond for $1,195 that pays $75 per year in interest. It sells the bond after five years for $940 (the coupon for year 5 has been received). What average annual yield (holding period yield, HPY) did the institution earn over its five-year holding period?

Answer: see model answer — HPY is the rate equating the purchase price with the cash flows received until sale: $1{,}195 = \sum_{t=1}^{5} 75/(1+h)^t + 940/(1+h)^5$. Solving gives $h \approx 2.19\%$ per year. It is far below the coupon rate because of the $255 capital loss on the sale.

Step by step

1. Cash flows: -1,195 at t=0; +75 at t=1..4; +75 + 940 = 1,015 at t=5.
2. Set $1{,}195 = \sum_{t=1}^{5} 75/(1+h)^t + 940/(1+h)^5$.
3. Solve numerically: $h = 2.19\%$.
4. The capital loss (1,195 - 940 = 255) drags the return well below the 7.5% coupon rate.

Why this is correct: HPY, not YTM, measures return when a security is sold before maturity, using the actual sale price as the terminal cash flow.

Q167 AP Mock 1 Q53

REBUILT — Distractor options were implausible values; replaced with results of realistic errors (current yield, approximation formula, gain not spread properly).

An investor wants to purchase a USD 1,500 par-value Treasury note with a 10% coupon rate maturing in four years. The current price of the note is USD 1,200 and coupons are paid annually. What is the yield to maturity?

A. 12.50%
B. 16.67%
C. 17.34%
D. 18.75%

Answer: C

Step by step

1. Annual coupon = 10% x 1,500 = USD 150.
2. Solve $1{,}200 = \sum_{t=1}^{4} 150/(1+y)^t + 1{,}500/(1+y)^4$.
3. At y = 17%, PV = 1,211.96; at 18%, PV = 1,177.19; exact solution $y = 17.34\%$.
4. Answer: 17.34%.

Why C is correct: YTM is the discount rate that sets the present value of the coupons and par equal to the price of USD 1,200; the discount adds to the coupon yield.

Why the others are wrong

- **A:** Current yield 150/1,200; ignores the USD 300 gain at maturity.
- **B:** Approximation (150 + 300/4)/((1,200 + 1,500)/2); a rough shortcut that understates the true YTM.
- **D:** Computes (150 + 75)/1,200, adding the annual share of the gain to the coupon and dividing by the low purchase price only; overstates YTM.

Q168 AnalystPrep Q.3859

REBUILT — Distractors were values copied from another question; replaced with results of realistic errors.

An investor holds a bond with par value of $2,000. The bond is priced at $1,600, matures in 5 years and pays an annual coupon of 8%. What is the bond's yield to maturity?

A. 8.00%
B. 10.00%
C. 13.80%
D. 15.00%

Answer: C

Step by step

1. Annual coupon = 0.08 x 2,000 = $160.
2. Solve $1{,}600 = \sum_{t=1}^{5} 160/(1+y)^t + 2{,}000/(1+y)^5$.
3. $y = 13.797\% \approx 13.80\%$; check: at 13.80% the price is $1,599.84.

Why C is correct: Both the coupon (8% of par) and the $400 discount gain are earned, so YTM is far above the

coupon rate.

Why the others are wrong

- **A:** Coupon rate (160/2,000); ignores the purchase discount.
- **B:** Current yield 160/1,600; ignores the $400 gain at maturity.
- **D:** Computes (160 + 80)/1,600, adding the annual share of the gain to the coupon and dividing by the low price only; overstates YTM.

Q169

AnalystPrep Q.4234

REBUILT — Source option 'D 25.73%' was an interpolation artefact (exact YTM is 25.72%) and distractors were implausible; recomputed exactly and rebuilt options; remapped to 69.b (YTM).

A bank buys a bond today for $750. The bond pays $150 of interest at the end of each year for 4 years and is redeemed at $1,000 at maturity (the final $150 coupon is paid in addition to the redemption value). The bond's yield to maturity is closest to:

A. 20.00%
B. 24.29%
C. 28.33%
D. 25.72%

Answer: D

Step by step

1. Set price equal to PV of cash flows: $750 = \sum_{t=1}^{4} \frac{150}{(1+y)^t} + \frac{1000}{(1+y)^4}$.
2. Solve numerically (bisection in Python): $y = 25.72\%$.
3. Check at 25.72%: PV of the four coupons is about 349.7 and PV of the principal is about 400.3, total 750.

Why D is correct: YTM is the single discount rate that equates the PV of all promised cash flows to the market price; it is well above the coupon yield because the bond is bought at a deep discount.

Why the others are wrong

- **A:** Current yield $150/750 = 20\%$; ignores the $250 pull-to-par gain.
- **B:** Approximation $(150 + 250/4)/((750 + 1000)/2) = 24.29\%$; an approximate shortcut, not the exact YTM.
- **C:** $(150 + 250/4)/750 = 28.33\%$ divides by the purchase price only; overstated, not the exact YTM.

Q170

GARP book Ch.4 end-of-chapter Q5

Forever Savings Bank is considering buying an $8 million general obligation issue from the York school district, the district's only bond issue this year. The bonds mature in 15 years and carry a nominal annual return of 6.75%. The bank is in the top corporate tax bracket of 35% and borrows at an average 4.25% to fund the purchase. Would you recommend the purchase? Calculate the net after-tax return on this bank-qualified municipal security, and state the tax advantage of being a qualified bond.

Answer: see model answer — The issue is bank-qualified (issued by a small municipality, $10 million or less a year), so the bank may deduct 80% of the interest paid to fund it. Net after-tax return = $6.75\% - 4.25\% \times (1 - 0.80 \times 0.35) = 6.75\% - 3.06\% = 3.69\%$. Interest income is tax-exempt, and 80% of the funding interest is deductible, so the after-tax funding cost falls from 4.25% to 3.06%. The positive 3.69% spread supports buying, subject to credit risk, liquidity risk (small, thinly traded issue) and the bank's capacity to use the tax benefits.

Step by step

1. Qualified bond: issuer sells $\leq$ $10 million a year, so 80% of the interest expense on funds used to buy it is deductible.
2. Deductible interest tax saving = $0.80 \times 0.35 \times 4.25\% = 1.19\%$.
3. After-tax funding cost = $4.25\% - 1.19\% = 3.06\%$, equivalently $4.25\%(1 - 0.28)$.
4. Net after-tax return = $6.75\% - 3.06\% = 3.69\%$ (positive, so attractive).

Why this is correct: Municipal interest is tax-exempt and the bank-qualified status makes 80% of its borrowing cost deductible, lowering the effective cost of funds.

Q171 GARP book Ch.4 end-of-chapter Q6

Forever Savings Bank (35% tax rate, 4.25% cost of funds) is also considering a nonqualified general obligation municipal issue from the city of Richmond, 15-year maturity, nominal return 9.75%. (a) Calculate the net after-tax return on this nonqualified security. (b) What is the difference in net after-tax return versus the bank-qualified security that returns a net 3.69%? (c) Discuss the pros and cons of buying the nonqualified rather than the bank-qualified bond.

Answer: see model answer — (a) For a nonqualified bond none of the funding interest is deductible: net return = 9.75% − 4.25% = 5.50%. (b) 5.50% - 3.69% = 1.81%, so the nonqualified bond earns 1.81 percentage points more. (c) Pros: higher net return. Cons: loses the 80% interest-expense deduction, and a much higher yield usually signals higher credit or business risk. The extra 1.81% must be weighed against the extra risk.

Step by step

1. Nonqualified: interest paid to fund the purchase is not deductible, so the full 4.25% is the cost.
2. (a) 9.75% − 4.25% = 5.50%.
3. (b) Qualified net return (from Q5) = 3.69%; difference = 5.50% − 3.69% = 1.81%.
4. (c) Compare the extra yield against the extra risk and lost tax deduction.

Why this is correct: Without bank-qualified status the borrowing cost gets no tax shield, but the much higher coupon more than compensates here.

Q172 GARP book Ch.4 end-of-chapter Q7

Lakeway Thrift Savings and Trust (35% corporate tax rate, loan revenue up 16%) is considering portfolio shifting. Option (a): sell $4 million of 12-year City of Dallas bonds (7.5% coupon, market value $3,750,000, book value par) and buy $4 million of 12-year Bexar County bonds (8% coupon, bought at par). Option (b): sell $4 million of 12-year U.S. Treasury bonds (12% coupon, book value par) whose market value has risen to $4,330,000. Which shift would you recommend? Is there a good reason not to sell the Treasuries? What other information is needed?

Answer: see model answer — Recommend (a). Selling the Dallas bonds realizes a $250,000 loss, which shelters loan income and saves 0.35 × 250,000 = $87,500 of tax, and the swap lifts the coupon from 7.5% to 8% (+$20,000 a year on $4 million). Option (b) realizes a $330,000 gain, creating 0.35 × 330,000 = $115,500 of extra tax, the opposite of what a bank with high loan income wants; the 12% coupon is also well above market, so the Treasuries are valuable to keep (they also provide liquidity and can be pledged as collateral). Other information needed: credit quality and liquidity of Bexar County versus Dallas, whether both are tax-exempt, the bank's ability to use tax-exempt income, the interest-rate outlook and liquidity needs, and transaction costs.

Step by step

1. (a) Loss on Dallas = 4,000,000 − 3,750,000 = 250,000; tax saving = 0.35 × 250,000 = 87,500.
2. (a) Coupon pickup = (8% − 7.5%) × 4,000,000 = 20,000 a year.
3. (b) Gain on Treasuries = 4,330,000 − 4,000,000 = 330,000; added tax = 0.35 × 330,000 = 115,500.
4. (b) Also gives up an above-market 12% coupon and a liquid, pledgeable asset.

Why this is correct: Tax swapping/portfolio shifting is attractive when securities can be sold at a loss to offset high loan income and replaced by higher-yielding ones.

Q173 GARP book Ch.4 end-of-chapter Q13 (two OCR copies, rids q0012 and q0017)

REBUILT — Foreign debt weight was illegible and the table interleaved; reconstructed it as 4.9% (100% less the listed 95.1%) and separated the mix and maturity tables.

Arrington Hills Savings Bank, a $3.5 billion asset institution, holds the investment portfolio below. It serves a rapidly growing money center into which many businesses are relocating headquarters; the suburbs are also growing quickly as business owners, managers and retired professionals buy new homes. Would you recommend any changes to the portfolio? Explain why.

Security type	Percent of total portfolio
U.S. Treasury securities	38.7%
Federal agency securities	35.2%
State and local government obligations	15.5%
Domestic debt securities	5.1%
Foreign debt securities	4.9%
Equities	0.6%
Maturity bucket	Percent of total portfolio
Under one year	11.3%
One to five years	37.9%
Over five years	50.8%

Answer: see model answer — The portfolio is dominated by safe government paper (73.9% Treasuries plus agencies), but over half of it matures beyond five years and only 11.3% matures within a year. Rapid local growth implies strong loan demand and liquidity needs, so the bank should shorten maturities and raise liquid holdings (for example a ladder with more short-term bills), cutting interest-rate and liquidity risk. Modest additions of municipals (tax benefit, pledging use) and geographic diversification of the small private-sector and foreign holdings can improve yield and reduce business risk without large credit risk.

Step by step

1. Treasuries + agencies = 38.7% + 35.2% = 73.9%: very safe, liquid, low-yield.
2. Maturities: only 11.3% under one year; 50.8% over five years, which exposes the bank to interest rate and liquidity risk.
3. Strong local growth means rising loan demand, so liquidity is needed; shift toward shorter maturities.
4. Consider tax-exempt municipals and geographic diversification for yield and business-risk reduction.

Why this is correct: Investment choice should balance return, tax exposure, liquidity, interest rate and business risk against the bank's growth and funding needs.

Q174

Schweser Module Quiz 12.3 Q3

REBUILT — Original false option and its keyed explanation conflicted (gains are capped as yields fall, not rise) and convexity wording is outside the reading; rewrote options around the Schweser call-risk discussion.

Which of the following statements about call risk on callable bonds is FALSE?

A. When interest rates fall, the issuer may call the bond and the investor must reinvest the proceeds at lower rates.

B. An investor can reduce call risk by buying callable bonds with a longer deferment period.

C. The price appreciation of a callable bond is limited when rates fall because the issuer can repurchase it at a predetermined price.

D. Call risk is greatest when interest rates rise, because issuers then repurchase bonds at the call price.

Answer: D

Step by step

1. Issuers call bonds when rates decline so they can refinance more cheaply; they have no incentive to call when rates rise.
2. A, B and C each match the call-risk description: reinvestment at lower rates, deferment reduces exposure, call price caps upside.
3. D reverses the rate direction, so it is false.

Why D is correct: Call risk arises when rates fall and issuers repurchase at a predetermined price, forcing reinvestment at lower yields.

Why the others are wrong

- **A:** True: this is the definition of call risk and its reinvestment effect.
- **B:** True: a longer deferment period delays when the bond can be called, mitigating call risk.
- **C:** True: a call price acts as a ceiling on price appreciation; choosing it as false confuses the cap with a floor.

Q175

AnalystPrep Q.3865

The risk that a bank must sell part of its investment portfolio before maturity, possibly for a capital loss, is known as:

A. Credit risk

B. Business risk
C. Default risk
D. Liquidity risk

Answer: D
Step by step
1. Liquidity risk is the risk of being unable to sell a security quickly at a reasonable price, creating potential losses.
2. Forced early sale at a loss matches this definition.
3. Credit/default risk concern nonpayment; business risk concerns a weakening local economy.

Why D is correct: Liquidity risk is the risk that a security cannot be sold quickly at a fair price, so an early forced sale may crystallize a capital loss.
Why the others are wrong
- **A:** Credit risk is the risk that the issuer fails to pay principal or interest, not the risk of a forced sale.
- **B:** Business risk is the risk that the local economy weakens and hurts sales and financial position.
- **C:** Default risk is the same family as credit risk: non-payment by the issuer, not a forced early sale.

Q176 AnalystPrep Q.3858

To safeguard public funds, depository institutions in the United States cannot accept deposits from federal, state and local governments unless they post collateral acceptable to those governments. This is most likely known as:
A. Statutory requirements
B. Pledging requirements
C. Investment requirements
D. Operational requirements

Answer: B
Step by step
1. The collateral a bank must post against government deposits is called a pledging requirement.
2. Acceptable collateral is typically high-quality federal and municipal securities, so it affects which securities the bank holds.

Why B is correct: Pledging requirements oblige banks to post acceptable collateral, usually federal and municipal securities, against government deposits.
Why the others are wrong
- **A:** 'Statutory requirements' is a generic term; the reading names this specific factor pledging requirements.
- **C:** 'Investment requirements' is not the reading's term and does not describe collateral for deposits.
- **D:** 'Operational requirements' relates to day-to-day running of the bank, not to collateral on public deposits.

Q177 AnalystPrep Q.3863

REBUILT — Removed irrelevant corporate bond and prime-rate data and replaced an unexplained 6.52% distractor with realistic mistakes.
An Aaa-rated municipal bond has a 5% gross yield to maturity. A taxed financial firm is in the top 35% federal income tax bracket. The tax-equivalent yield (TEY) of the municipal bond is closest to:
A. 3.25%
B. 6.75%
C. 7.69%
D. 14.29%

Answer: C
Step by step
1. TEY = tax-exempt yield / (1 - t).
2. TEY $= 5\%/(1 - 0.35) = 5\%/0.65 = 7.69\%$.
3. A fully taxable bond must yield 7.69% to match the municipal's 5%.

Why C is correct: TEY grosses up the tax-exempt yield by dividing by one minus the tax rate.
Why the others are wrong
- **A:** Multiplied by (1 - t): $5\% \times 0.65 = 3.25\%$, a net-of-tax figure, not a gross-up.
- **B:** Multiplied by (1 + t): $5\% \times 1.35 = 6.75\%$.
- **D:** Divided by the tax rate instead of (1 - t): $5\%/0.35 = 14.29\%$.

Q178 AnalystPrep Q.3868

Loans will be terminated or paid off ahead of schedule. This is a particular problem with residential mortgages and other consumer loans that are pooled and used as collateral in securitized assets. This risk is known as:

A. Liquidity risk
B. Prepayment risk
C. Call risk
D. Business risk

Answer: B

Step by step

1. Early repayment of pooled mortgages or consumer loans means investors do not receive the expected cash-flow schedule.
2. This is prepayment risk, typically when rates fall and borrowers refinance.

Why B is correct: Prepayment risk is the risk that underlying loans in a pool are repaid early, altering expected cash flows and forcing reinvestment.

Why the others are wrong

- **A:** Liquidity risk concerns inability to sell a security quickly at a fair price, not early loan repayment.
- **C:** Call risk applies to issuer redemption of callable bonds, not to borrowers prepaying pooled loans.
- **D:** Business risk concerns weakening of the local economy, not early loan repayment.

Q179 AnalystPrep Q.5423

ABC Bank currently has been experiencing high loan revenues due to a recent surge in the demand for loans. The bank intends to sell lower-yielding securities at a loss in order to reduce its current taxable income. Which of the following tools is the bank most likely utilizing?

A. Tax exposure
B. Tax swap
C. Tax shelter
D. Tax exemption

Answer: B

Step by step

1. The bank has high loan revenue, so current taxable income is high.
2. It sells lower-yielding securities at a loss; the realized loss offsets taxable income.
3. Proceeds are typically reinvested in higher-yielding securities, raising future income.
4. This sell-at-a-loss-and-reinvest strategy is the definition of a tax swap.

Why B is correct: A tax swap realizes losses on low-yielding securities to cut current taxable income while replacing them with higher-yielding ones.

Why the others are wrong

- **A:** Tax exposure is the general term for the tax status of a securities portfolio (taxable vs tax-exempt, TEY, etc.), not a specific loss-selling strategy.
- **C:** A tax shelter is not the named strategy in the text; selling securities at a loss to cut current taxable income and reinvesting in higher-yielding securities is a tax swap.
- **D:** Tax exemption refers to income (e.g., municipal interest) that is not taxed; the bank is realizing losses, not holding exempt income.

Q180 BT Mock Q24 · BT answers P2.T8.20.5.2 · BT notes P2.T8.20.5.2

REBUILT — Three near-identical items merged; distractors (one a typo, 76.67%/16.67%) were implausible giveaways and were replaced with results of realistic formula errors.

A corporation has a marginal tax rate of 30.0% and purchases an A-rated municipal bond with a 4.20% gross yield (yield to maturity). What is the tax-equivalent yield (TEY)?

A. 2.94%
B. 5.46%
C. 6.00%
D. 14.00%

Answer: C

Step by step

1. TEY = municipal yield / (1 - t).
2. TEY $= 4.20\%/(1 - 0.30) = 4.20\%/0.70 = 6.00\%$
3. Check: $6.00\% \times 0.70 = 4.20\%$, so a taxable bond yielding 6.00% matches the muni after tax.

Why C is correct: The TEY is the pretax yield a taxable bond must offer to leave the same after-tax yield as the tax-exempt bond, so the muni yield is grossed up by dividing by (1 - t).

Why the others are wrong

- **A:** Multiplied by (1 - t): 4.20% x 0.70 = 2.94%, which converts in the wrong direction (an after-tax haircut instead of a gross-up).
- **B:** Multiplied by (1 + t): 4.20% x 1.30 = 5.46%, an incorrect gross-up.
- **D:** Divided by the tax rate instead of (1 - t): 4.20% / 0.30 = 14.00%.

Q181

Mock 1 Q68

REBUILT — Answer was correct; distractors C and D were unexplained near-duplicates, so they were replaced with results of realistic mistakes.

A bank purchases a municipal bond with a 5% gross nominal rate of return. The bank's pretax borrowing cost is 3.7%, and it is subject to an income tax rate of 40%. The bond is a bank-qualified bond, but only 75% of the interest expense is tax deductible. The net after-tax return on the municipal bond is closest to:

A. 1.67%
B. 2.41%
C. 2.78%
D. 2.97%

Answer: B

Step by step

1. Municipal interest is tax-exempt, so the gross spread is 5.00% - 3.70% = 1.30%.
2. Deductible interest = 3.70% x 0.75 = 2.775%.
3. Tax shield = 2.775% x 0.40 = 1.11%.
4. Net after-tax return = 1.30% + 1.11% = 2.41%.

Why B is correct: Net return equals the spread between the exempt yield and funding cost plus the tax saving on the deductible portion of the interest expense.

Why the others are wrong

- **A:** Used the non-deductible 25% of interest for the shield: 1.30% + 3.70% x 0.25 x 0.40 = 1.67%.
- **C:** Assumed all interest expense is deductible: 1.30% + 3.70% x 0.40 = 2.78%.
- **D:** Applied (1 - t) = 0.60 instead of t to the deductible interest: 1.30% + 2.775% x 0.60 = 2.97%.

Q182

Schweser 69.2-1

A bank recently purchased a municipal bond with a 5.5% nominal rate of return. To finance the purchase, the bank borrowed funds at a cost of 4.7%. The bank's marginal income tax rate is 34%. The bond is a bank-qualified bond, and 80% of the interest expense is tax deductible. The net after-tax return on the bond is closest to:

A. 0.80%
B. 2.08%
C. 2.40%
D. 4.56%

Answer: B

Step by step

1. Gross spread = 5.50% - 4.70% = 0.80%.
2. Deductible interest = 4.70% x 0.80 = 3.76%.
3. Tax shield = 3.76% x 0.34 = 1.278%.
4. Net after-tax return = 0.80% + 1.278% = 2.08%.

Why B is correct: Municipal interest is untaxed, and 80% of the funding cost is deductible, so the tax shield adds to the pretax spread.

Why the others are wrong

- **A:** Ignored the tax shield and reported only the 0.80% spread.
- **C:** Applied the tax rate to the full interest expense (forgot the 80% limit): 0.80% + 4.70% x 0.34 = 2.40%.

- **D:** Forgot to multiply the deductible interest by the tax rate: 0.80% + 3.76% = 4.56%.

Q183 Schweser 69.2-2

An investment firm that is prohibited from holding speculative grade securities would have to sell a security when its rating changes from:

A. AAA to AA.
B. BBB to BB.
C. BB to B.
D. B to CCC.

Answer: B

Step by step

1. Investment grade is BBB/Baa or higher; speculative grade is BB/Ba or lower.
2. The firm may only hold investment grade, so it must sell when a security crosses from BBB to BB.
3. Other changes either stay investment grade (AAA to AA) or occur within the speculative category.

Why B is correct: The forced sale is triggered at the investment-grade boundary, BBB to BB.

Why the others are wrong

- **A:** AAA to AA is a move within investment grade; the security stays eligible.
- **C:** BB to B is a downgrade within speculative grade; the security was already ineligible, so no new trigger.
- **D:** B to CCC is also within speculative grade; the firm should already not hold it.

Q184 GARP PE1 Q11 · Mock 3 Q69

A bank treasurer is choosing the most appropriate investment strategy given the bank's balance sheet and economic conditions. The treasurer gathers the following information:
- The bank faces a high interest rate environment with a flat yield curve.
- The bank's asset mix includes a high proportion of US Treasury bonds purchased years ago when interest rates were significantly lower.
- The bank has a high proportion of revenues from loans and anticipates a record high level of profitability this year.
Which strategy is most appropriate to maintain the bank's current level of total income?

A. Barbell strategy
B. Carry trade
C. Riding the yield curve
D. Tax swap

Answer: D

Step by step

1. High rates now versus low rates when the Treasuries were bought means the old bonds trade below cost (unrealised losses).
2. Record profitability means a large taxable income that realised losses can shelter.
3. Tax swap: sell the low-coupon bonds, realise the loss for tax relief, and reinvest in bonds at today's higher yields, which supports total income.
4. A flat curve means no term spread, so carry trade and riding the yield curve offer no pickup.

Why D is correct: Selling underwater low-coupon bonds in a high-profit year realises tax-deductible losses and lets proceeds earn the higher current yield.

Why the others are wrong

- **A:** A barbell addresses liquidity versus return mix; none of the facts mention a liquidity or return-mix problem.
- **B:** Carry trade borrows short to buy long; with a flat curve the long yield is no higher than the borrowing cost, so no carry.
- **C:** Riding the yield curve needs a steep upward slope so securities roll down to higher prices; a flat curve gives no roll-down gain.

LO 69.c

Apply investment maturity strategies and maturity management tools based on the yield curve and duration.

Q185 GARP book Ch.4 end-of-chapter Q2

REBUILT — Open-ended GARP problem with no answer converted to a numeric MCQ with computed key and plausible distractors.

A bond has a $1,000 face value, a 6% annual coupon (paid annually), 10 years to maturity and a yield to maturity of 5%. What is the bond's (Macaulay) duration?

A. 5.10 years
B. 7.52 years
C. 7.89 years
D. 8.31 years

Answer: C

Step by step

1. Price = $\sum 60/1.05^t + 1000/1.05^{10} = 1077.22$.
2. Weight each cash flow's PV by its time t: $\sum t \cdot PV_t = 8{,}501.6$.
3. Duration = $8501.6/1077.22 = 7.89$ years.
4. Duration is below the 10-year maturity because coupons are received earlier.

Why C is correct: Duration is the present-value-weighted average time to receive the cash flows, discounted at the 5% yield.

Why the others are wrong

- **A:** Included only the coupon stream and omitted the principal repayment from the weights.
- **B:** Computed modified duration 7.89/1.05 instead of Macaulay duration.
- **D:** Weighted times by undiscounted cash flows rather than present values.

Q186 GARP book Ch.4 end-of-chapter Q4

REBUILT — Open-ended GARP problem with no answer converted to a numeric MCQ with computed key and plausible distractors.

A bond matures in 20 years, pays a 7% annual coupon, has a $1,000 face value and a yield to maturity of 10%. What is its (Macaulay) duration?

A. 9.09 years
B. 10.00 years
C. 11.34 years
D. 14.46 years

Answer: B

Step by step

1. Price = $\sum_{t=1}^{20} 70/1.10^t + 1000/1.10^{20} = 744.59$.
2. $\sum t \cdot PV_t = 7{,}447.3$.
3. Duration = $7447.3/744.59 = 10.00$ years.
4. Duration is about half the maturity: coupons are large relative to the yield and discounting shrinks distant flows.

Why B is correct: Discounting at the 10% yield shrinks distant cash flows, so the PV-weighted average time is about 10 years, well below 20.

Why the others are wrong

- **A:** Reported modified duration 10.00/1.10 instead of Macaulay duration.
- **C:** Discounted the cash flows at the 7% coupon rate instead of the 10% yield.
- **D:** Used undiscounted cash flows as weights.

Q187 GARP book Ch.4 end-of-chapter Q8

REBUILT — Open-ended (draw curve, discuss) converted to a self-contained MCQ testing the same yield-curve interpretation.

Treasury yields by maturity: 3-month 1.90%, 6-month 2.10%, 1-year 2.25%, 2-year 2.51%, 3-year 2.82%, 5-year 3.28%, 7-year 3.56%, 10-year 3.98%, 20-year 4.69%, 30-year 5.25%. An institution holds 75% of its investment portfolio in 7- to 30-year Treasury bonds and 25% in bills and notes maturing in under one year. Which assessment and recommendation is most appropriate?

A. The curve is upward sloping; the long-heavy portfolio earns higher yield but faces large capital losses if rates rise and has limited liquidity, so management should shorten maturities, for example toward a ladder or barbell with a shorter duration.
B. The curve is upward sloping, which signals falling rates; management should extend further into 20- to 30-year

bonds to lock in capital gains.

C. The curve is downward sloping, signalling falling short-term rates; the long-heavy portfolio is well positioned and no change is needed.

D. The curve is flat, so maturity choice does not matter and the portfolio carries no interest rate risk.

Answer: A

Step by step

1. Yields rise steadily from 1.90% (3 months) to 5.25% (30 years): an upward-sloping curve with a spread of about 335 bp.
2. An upward slope offers more yield for longer maturity but compensates for greater price risk, and may reflect expectations of rising rates.
3. 75% in 7- to 30-year bonds means high duration: a rate rise causes large capital losses, and only 25% is in liquid short-term instruments.
4. Recommendation: reduce duration and raise liquidity, for example by moving to a ladder or barbell with a shorter long end.

Why A is correct: An upward-sloping curve rewards extending maturity but at the cost of more price risk; a long-concentrated portfolio is exposed to rate increases and is illiquid.

Why the others are wrong

- **B:** An upward slope does not signal falling rates (a downward slope does); extending further would raise exposure to losses.
- **C:** Misreads the shape: the curve slopes upward, and a long-heavy portfolio is the one hurt by rate increases.
- **D:** Yields clearly differ by maturity, and longer bonds have higher interest rate risk.

Q188

GARP book Ch.4 end-of-chapter Q9 · GARP book Ch.4 end-of-chapter Q9 (repeat)

REBUILT — Open numeric problem converted to a MCQ; two identical copies merged.

A bond has a duration of 8.89 years. Market rates on comparable bonds fall from 7.5% to 7.25%. What is the approximate percentage change in the bond's value?

A. +2.07%

B. +2.22%

C. -2.07%

D. +20.7%

Answer: A

Step by step

1. $\Delta P/P \approx -D \times \Delta y/(1+y)$.
2. $\Delta y = 7.25\% - 7.50\% = -0.0025$; $1 + y = 1.075$.
3. $\Delta P/P = -8.89 \times (-0.0025)/1.075 = +0.02067$, i.e. about +2.07%.

Why A is correct: Rates fall, so price rises; the Macaulay duration must be divided by (1 + y) to get the price sensitivity.

Why the others are wrong

- **B:** Forgot to divide by $1 + y$ ($8.89 \times 0.0025 = 2.22\%$).
- **C:** Sign error: a rate decrease raises the bond price.
- **D:** Decimal slip: used 0.025 (25 bp entered as 2.5%) instead of 0.0025.

Q189

GARP book Ch.4 end-of-chapter Q10 · GARP book Ch.4 end-of-chapter Q10 (repeat)

REBUILT — Open question rebuilt as MCQ with explicit one-year horizon; two copies merged.

A savings institution's bond portfolio has an average duration of 4.5 years. Its investments officer wants to protect the portfolio's return (immunize) over a planned one-year holding period. What should the officer do?

A. Reduce the portfolio's duration to about one year.

B. Lengthen the portfolio's duration to above 4.5 years.

C. Leave the duration at 4.5 years, because duration already measures interest rate risk.

D. Raise the duration to about 9 years so that price gains offset reinvestment losses.

Answer: A

Step by step

1. Immunization requires portfolio duration = length of the planned holding period.
2. Horizon is one year, current duration is 4.5 years, so duration exceeds the horizon.
3. Shift toward shorter-maturity or higher-coupon securities to bring duration down to about 1 year.
4. Then price and reinvestment effects of rate changes offset each other over the year.

Why A is correct: Setting duration equal to the holding period balances price risk against reinvestment risk.

Why the others are wrong

- **B:** Longer duration increases price sensitivity and moves further from the one-year horizon.
- **C:** A duration of 4.5 years matches a 4.5-year horizon, not one year.
- **D:** A duration above the horizon leaves the portfolio exposed to price losses if rates rise.

Q190 GARP book Ch.4 end-of-chapter Q11 · GARP book Ch.4 end-of-chapter Q11 (repeat)

REBUILT — Open question referred to an unreproduced table (Table 4.3); rebuilt as a self-contained MCQ testing the rate expectations approach.

A bank's economics department forecasts GDP growth of 4.5% a year for at least two years, which management expects to push interest rates higher. Under a rate expectations approach, what should the investments officer do?

A. Shift toward shorter-maturity or floating-rate securities and consider selling long-term fixed-rate bonds, accepting that sales may realize losses.
B. Extend the portfolio to longer maturities to lock in current yields before they fall.
C. Do nothing, since rate changes affect only the bank's liabilities.
D. Sell short-term securities and buy long-term bonds to maximize capital gains.

Answer: A

Step by step

1. Strong growth implies rising rates.
2. Bond prices fall when rates rise, and longer maturities (higher duration) fall most.
3. So shorten maturities or move to floating-rate assets; sell long-term fixed-rate holdings, realizing losses that tax management can partly offset.

Why A is correct: Rising-rate expectations call for shortening the portfolio to limit capital losses.

Why the others are wrong

- **B:** Extending maturity is the response to expected falling rates.
- **C:** Asset prices also respond to rates; a bond portfolio is directly exposed.
- **D:** This is the falling-rate strategy and would magnify losses if rates rise.

Q191 GARP book Chapter 4 end-of-chapter Problem 12 · GARP book Chapter 4 end-of-chapter (sub 4, duplicate)

REBUILT — Duplicate GARP problem records merged; the OCR-interleaved fragment about a 75%/25% portfolio was removed because its lead-in is missing.

A bank's economics department forecasts a significant recession: output and employment are projected to decline significantly over the next 18 months. What are the implications for an investment portfolio manager? What is the outlook for interest rates and inflation? Which types of investment securities would you add to the portfolio during the recession, and why? What other information about the bank's balance sheet and earnings would you want?

Answer: see model answer — A recession implies falling loan demand, lower inflation and falling interest rates, so bond prices should rise. Under the rate expectations approach the manager would lengthen maturities (back-end load or long-duration Treasuries and other high-quality securities) to lock in current yields and earn capital gains, while avoiding lower-quality credits whose default risk rises in a downturn and keeping enough liquid assets for deposit and loan needs. Useful additional information: the current maturity/duration structure and unrealised gains or losses on the portfolio, loan quality and expected loan demand, liquidity needs, the bank's tax position and earnings, and its interest-rate gap/duration exposure.

Q192 AnalystPrep Q.3867

The most aggressive investment maturity strategy, which requires a bank to regularly shift the maturities of its securities in response to fluctuations in interest rates, is called the:

A. Front-end load maturity policy
B. Back-end load maturity policy
C. Barbell strategy
D. Rate expectations approach

Answer: D

Step by step

1. Identify the strategy that relies on interest rate forecasts and actively changes maturities.
2. Rate expectations approach: shorten maturities when rates are expected to rise (avoid price declines) and lengthen when rates are expected to fall (capture gains).
3. The other strategies fix a maturity structure and do not require forecast-driven shifts.

Why D is correct: Only the rate expectations approach changes the maturity structure in response to forecasts of interest rate movements, requiring frequent trading and sophisticated forecasting.

Why the others are wrong

- **A:** Front-end load puts funds in short maturities (e.g. three years or less) for liquidity; it is a static structure, not forecast-driven.
- **B:** Back-end load puts funds in long maturities for higher return; static, and may require borrowing for liquidity.
- **C:** Barbell holds short and long maturities with little in the middle; the mix is not regularly shifted with rate forecasts.

Q193 AnalystPrep Q.3871

The practice of protecting the return on purchased securities from changes in interest rates, regardless of which direction rates move, is known as:

A. Portfolio shifting
B. Duration
C. Portfolio immunization
D. Securitization

Answer: C

Step by step

1. Immunization sets portfolio duration equal to the planned holding period.
2. Then price risk and reinvestment risk offset: rates up lowers prices but raises reinvestment income; rates down does the reverse.
3. Total return is therefore protected whichever way rates move.

Why C is correct: Portfolio immunization matches duration to the holding period so the opposing price and reinvestment effects offset.

Why the others are wrong

- **A:** Portfolio shifting means selling securities (often at a loss, for tax purposes) and substituting higher-yielding ones; it does not lock in a return whichever way rates move.
- **B:** Duration is the sensitivity measure used to build an immunized portfolio, not the protective practice itself.
- **D:** Securitization pools loans into tradable securities; it is unrelated to interest rate protection.

Q194 BT answers P2.T8.20.5.3 · BT notes P2.T8.20.5.3

The Acme Investment Firm wants to re-position one of its bond portfolios and can choose among the Ladder Policy, Front-end Load Maturity Policy, Back-end Load Maturity Policy, Barbell Strategy, or Rate Expectations Approach. Its goal for the portfolio is neither to maximize income nor to maximize the upside potential for earnings; instead, the portfolio is to be used primarily as a source of liquidity. Which strategy is best?

A. Ladder Policy
B. Front-end Load Maturity Policy
C. Back-end Load Maturity Policy
D. Rate Expectations Approach

Answer: B

Step by step

1. Goal: liquidity first, no emphasis on income or upside.
2. Front-end load invests in short maturities (e.g. three years or less), giving significant liquidity.
3. Longer maturities (back-end load) stress income and weaken liquidity; the ladder aims at stable income and rate expectations at capital gains.

Why B is correct: Front-end load concentrates holdings in short maturities, so the portfolio is highly liquid at the cost of lower yield.

Why the others are wrong

- **A:** Ladder frees cash only as each rung matures and part of the funds sit in longer maturities; it targets stable income and simplicity, not primary liquidity.
- **C:** Back-end load holds long maturities for higher return and may require borrowing to meet short-term liquidity needs, the opposite of the goal.
- **D:** Rate expectations seeks capital gains from forecast rate moves, which is an upside/earnings objective the firm has ruled out.

Q195 Schweser Module Quiz 69.3 Q1

Which of the following investment maturity strategies would most likely reduce income fluctuation in the medium and long term, while generating sufficient ongoing liquidity for potential investment opportunities?

A. Ladder
B. Barbell
C. Back-end load
D. Front-end load

Answer: A

Step by step

1. Ladder places an equal share in each maturity up to the maximum, so income is spread across the term structure.
2. As each rung matures, cash is freed for new opportunities, giving ongoing liquidity.

Why A is correct: The ladder diversifies reinvestment timing across maturities (smoothing income) and releases cash regularly as rungs mature.

Why the others are wrong

- **B:** Barbell concentrates in short and long ends; liquidity comes only from the short end and income is exposed to rates at the extremes.
- **C:** Back-end load holds long maturities, giving little ongoing liquidity and possibly requiring borrowing.
- **D:** Front-end load is highly liquid but all funds reprice quickly, so medium/long-term income fluctuates with short rates.

Reading 70. Liquidity and Reserves Management: Strategies and Policies

Peter Rose and Sylvia Hudgins, Bank Management & Financial Services, 9th Edition (New York, NY: McGraw-Hill, 2013). Chapter 11.

LO	Learning objective	Qs
70.a	calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank.	10
70.b	compare strategies that a bank can use to meet demands for additional liquidity.	9
70.c	estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators).	18

LO 70.a

Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank.

Q196

GARP book Chapter 5 end-of-chapter (GARP-LR-6 sub 1)

Ocean View State Bank estimates that over the next 24 hours the following cash inflows and outflows will occur (all figures in millions of dollars):

Deposit withdrawals	$100
Deposit inflows	95
Scheduled loan repayments	90
Acceptable loan requests	60
Borrowings from the money market	80
Sales of bank assets	40
Stockholder dividend payments	150
Revenues from sale of nondeposit services	95
Repayments of bank borrowings	60
Operating expenses	50

What is this bank's projected net liquidity position in the next 24 hours? From what sources can the bank cover its liquidity needs?

Answer: see model answer — Supply of liquidity = 95 + 90 + 80 + 40 + 95 = $400 million; demand = 100 + 60 + 150 + 60 + 50 = $420 million. Net liquidity position $L = 400 - 420 = -\$20$ million, a liquidity deficit. The bank can cover it by converting assets (selling liquid securities or other assets, the asset conversion strategy), by raising additional borrowed funds (money market borrowings, repos, fed funds, CDs, discount window; liability management), or by a balanced mix of both, e.g., cutting dividends or attracting more deposits.

Step by step

1. Supplies of liquidity: deposit inflows 95 + loan repayments 90 + money market borrowings 80 + asset sales 40 + nondeposit service revenues 95 = 400.
2. Demands for liquidity: withdrawals 100 + acceptable loan requests 60 + dividends 150 + repayment of borrowings 60 + operating expenses 50 = 420.
3. L = Supply − Demand = $400 - 420 = -20$ (million).
4. Demand exceeds supply, so there is a $20 million liquidity deficit that must be funded.
5. Funding options: asset conversion, liability (purchased liquidity) management, or a balanced strategy.

Why this is correct: Net liquidity position is total liquidity supply minus total liquidity demand; a negative result means additional funds must be raised from asset sales and/or new borrowings.

Q197

GARP book Chapter 5 end-of-chapter (GARP-LR-6 sub 2)

REBUILT — Open numeric problem converted to a 4-option MCQ with distractors based on sign and omission errors.

Mountain Top Savings is projecting a net liquidity deficit of $10 million next week. Expected demands for liquidity are quality loan demand of $32 million, repayments of previous borrowings of $15 million, stockholder dividends of $10 million, and other operating expenses of $15 million, in addition to deposit withdrawals. Expected supplies of liquidity are deposit inflows of $26 million, revenues from nondeposit services of $18 million, scheduled customer loan repayments of $23 million, asset sales of $10 million, and money market borrowings of $15 million. What must Mountain Top's expected deposit withdrawals be for the coming week?

A. $10 million

B. $20 million

C. $30 million
D. $12 million

Answer: C
Step by step
1. Supply = 26 + 18 + 23 + 10 + 15 = 92.
2. Known demands excluding withdrawals = 32 + 15 + 10 + 15 = 72.
3. $L = 92 - (72 + W) = -10$.
4. $20 - W = -10 \Rightarrow W = 30$ million.

Why C is correct: Setting net liquidity (supply minus demand) equal to the stated deficit of -10 and solving for the one unknown demand component gives withdrawals of 30.
Why the others are wrong
- **A:** Treats the deficit as a surplus of +10 (20 - W = +10), giving W = 10.
- **B:** Equates supply minus the known demands (92 - 72 = 20) to withdrawals, ignoring the 10 deficit.
- **D:** Omits the 18 of nondeposit service revenues from supply (74 - 72 + 10 = 12).

Q198

AnalystPrep Q.3885

REBUILT — Original distractors (-$23, $23) did not correspond to any realistic error; replaced with distractors from plausible mistakes and stated units in millions.

ABC Bank predicts its cash inflows and outflows over the next day (in $ millions) as follows:

Cash inflows	Amount	Cash outflows	Amount
Deposit inflows	119	Deposit withdrawals	129
Borrowings from the money market	110	Acceptable loan requests	90
Sales of bank assets	61	Stockholder dividend payments	172
Revenues from sale of non-deposit services	118	Repayments of bank borrowings	84
Scheduled loan repayments	122	Operating expenses	70

What is the bank's projected net liquidity position for the next day (in $ millions)?

A. +15
B. -15
C. +157
D. -125

Answer: B
Step by step
1. Net liquidity position L = supply of liquidity − demand for liquidity.
2. Supply: $119 + 110 + 61 + 118 + 122 = 530$.
3. Demand: $129 + 90 + 172 + 84 + 70 = 545$.
4. $L = 530 - 545 = -15$: a liquidity deficit of $15 million that must be funded.

Why B is correct: Demand for liquidity (545) exceeds supply (530), so the net liquidity position is a deficit of 15; all listed inflows (including money market borrowings) are supplies and all outflows (including dividends and repayments of borrowings) are demands.
Why the others are wrong
- **A:** Sign reversed: computed demand minus supply (545 - 530) and reported a surplus.
- **C:** Omitted the $172 stockholder dividend payments from demand: 530 - 373 = +157.
- **D:** Left the 110 of money market borrowings out of supply (530 - 110 = 420) and computed 420 - 545 = -125; borrowings are a source of liquidity.

Q199

AnalystPrep Q.3886

REBUILT — OCR-garbled supply items (loan repayments shown as $5/$28) did not sum to the printed total of 122; set scheduled loan repayments to 33 so the table sums to 122, and rebuilt distractors.

Prime Savings has a projected net liquidity surplus of $31 million in the coming week. Using the supply and demand of liquidity below ($ millions), calculate the total expected stockholder dividend payments for the week.

Supply of liquidity	Amount
Deposit inflows	31
Revenues from nondeposit service sales	23
Scheduled repayments of previous customer loans	33
Asset sales	15
Money market borrowings	20
Total supply of liquidity	122
Demand for liquidity	Amount
Expected quality loan demand	31
Necessary repayments of previous borrowings	26
Deposit withdrawals	22
Disbursements to cover operating expenses	10
Total demand excluding stockholder dividends	89

A. $33 million
B. $2 million
C. $64 million
D. $120 million

Answer: B
Step by step
1. Supply: $31 + 23 + 33 + 15 + 20 = 122$; demand excluding dividends: $31 + 26 + 22 + 10 = 89$.
2. Net liquidity position $L = 122 - (89 + D) = 31$, where D is dividends.
3. $33 - D = 31 \Rightarrow D = 2$.

Why B is correct: Net liquidity equals supply minus total demand; the surplus of 31 means total demand is 91, so dividends are 91 - 89 = 2.

Why the others are wrong
- **A:** Computed 122 - 89 = 33, the surplus before dividends, and forgot to subtract the stated surplus of 31.
- **C:** Added the surplus to the gap: 33 + 31 = 64, instead of subtracting it.
- **D:** Added the surplus to demand excluding dividends (89 + 31 = 120), ignoring total supply.

Q200

AnalystPrep Q.5379

REBUILT — Source inflow/outflow classification was garbled and distractors were weak; presented a clean table and replaced distractors with realistic misclassification errors.

ABC Bank's estimated cash flows for the coming week are below ($ millions).

Cash flow	Amount
Scheduled loan repayments	150
Acceptable loan requests	110
Borrowings from money market	120
Operating expenses	50
Deposit withdrawals	80
Deposit inflows	70
Stockholder dividend payments	30
Repayments of bank borrowings	50

What is ABC Bank's net liquidity position at the end of the coming week (in $ millions)?

A. -100
B. -20
C. +20
D. +50

Answer: C
Step by step
1. Supply: loan repayments 150 + money market borrowings 120 + deposit inflows 70 = 340.
2. Demand: loan requests 110 + operating expenses 50 + withdrawals 80 + dividends 30 + repayments of borrowings 50 = 320.
3. $L = 340 - 320 = +20$: a liquidity surplus of $20 million.

Why C is correct: Supply of liquidity (340) exceeds demand (320), leaving a surplus of 20 that should be invested to cover future needs.

Why the others are wrong

- **A:** Treated money market borrowings as not a supply of liquidity and dropped them: 220 - 320 = -100.
- **B:** Sign reversed (320 - 340).
- **D:** Omitted the $30 dividend from demand: 340 - 290 = +50.

Q201

BT Mock Q29

Over the next 24 hours, Greenlux State Bank estimates the following cash flows (USD millions):

Item	Amount
Incoming deposits	90
Customer loan repayments	25
Sales of bank assets	20
Non-deposit service fee income	15
Deposit withdrawals	60
Acceptable loan requests	45
Repayment of bank borrowings	5

What is the bank's projected net liquidity position?

A. -30.0 million.
B. +40.0 million.
C. +90.0 million.
D. +150.0 million.

Answer: B

Step by step

1. Supplies of liquidity: deposits 90 + loan repayments 25 + asset sales 20 + fee income 15 = 150.
2. Demands for liquidity: withdrawals 60 + acceptable loan requests 45 + repayment of borrowings 5 = 110.
3. L = supply − demand = 150 − 110 = +40 million, a liquidity surplus.

Why B is correct: Net liquidity position is total supplies of liquidity minus total demands; loan repayments, asset sales and fee income add to supply while withdrawals, loan requests and debt repayment are demands.

Why the others are wrong

- **A:** Treats asset sales (20) and fee income (15) as outflows: 90 + 25 - 60 - 45 - 5 - 20 - 15 = -30.
- **C:** Subtracts only deposit withdrawals from total supply (150 - 60 = 90), ignoring loan requests and borrowing repayment.
- **D:** Reports gross supply only (150) without deducting any demands.

Q202

BT answers P2.T8.20.6.1 · BT notes P2.T8.20.6.1

REBUILT — Merged two identical BT items (OCR-damaged figures restored from the worked solution) and replaced distractors that did not correspond to realistic errors.

Over the next 24 hours, Greenlux State Bank estimates that the following cash inflows and outflows (all figures in USD millions) will occur:

Deposit withdrawals	70
Deposit inflows	100
Scheduled loan repayments	60
Acceptable loan requests	90
Borrowings from the money market	80
Sales of bank assets	30
Stockholder dividend payments	20
Revenues from sale of nondeposit services	10
Repayment of bank borrowings	50
Operating expenses	40

What is the bank's projected net liquidity position?

A. -70 million
B. +10 million
C. +30 million
D. +50 million

Answer: B

Step by step

1. Supplies: deposit inflows 100 + loan repayments 60 + money market borrowings 80 + asset sales 30 + nondeposit revenue 10 = 280.
2. Demands: withdrawals 70 + loan requests 90 + dividends 20 + repayment of borrowings 50 + operating expenses 40 = 270.
3. $L = 280 - 270 = +10$ million (surplus).

Why B is correct: Money market borrowings are a source of spendable funds (supply), while repayment of borrowings, dividends and operating expenses are demands; the difference is +10.

Why the others are wrong

- **A:** Leaves out money market borrowings (supply 200): 200 - 270 = -70.
- **C:** Counts only deposit flows: 100 - 70 = +30, ignoring all other items.
- **D:** Omits operating expenses from demands: 280 - 230 = +50.

Q203

Mock 3 Q60

A treasurer at a small regional bank is assessing the bank's liquidity position. The treasurer estimates that the following cash inflows and outflows will occur in the next week:

Cash flows	Amount (USD million)
Deposit withdrawals	30
Deposit inflows	70
Scheduled loan repayments	80
Acceptable loan requests	50
Borrowings from money market	60
Operating expenses	40
Stockholder dividend payments	20
Repayment of bank borrowings	30

Which of the following is the correct amount (in millions of USD), at the week's end, for the bank's net liquidity position?

A. -80.

B. -20.

C. 40.

D. 100.

Answer: C

Step by step

1. Supplies: deposit inflows 70 + loan repayments 80 + money market borrowings 60 = 210.
2. Demands: withdrawals 30 + loan requests 50 + operating expenses 40 + dividends 20 + repayment of borrowings 30 = 170.
3. $L = 210 - 170 = +40$ million.

Why C is correct: Net liquidity = supplies minus demands; borrowings raised are supply and borrowings repaid are demand.

Why the others are wrong

- **A:** Treats money market borrowings (60) as a demand: 70 + 80 - 170 - 60 = -80.
- **B:** Omits money market borrowings from supply: 150 - 170 = -20.
- **D:** Treats deposit withdrawals as a supply instead of a demand: 210 + 30 - 140 = 100.

Q204

Mock 4 Q68

REBUILT — Replaced distractor -80 (no realistic error produces it) with -120 (omitting money market borrowings).

A treasurer at a regional bank is assessing the bank's liquidity position. The treasurer estimates that the following cash inflows and outflows will occur in the next week:

Cash flows	Amount (USD million)
Deposit withdrawals	50
Deposit inflows	80
Scheduled loan repayments	120
Acceptable loan requests	100
Borrowings from money market	80
Operating expenses	70
Stockholder dividend payments	40
Repayment of bank borrowings	60

Which of the following is the correct amount (in millions of USD), at the week's end, for the bank's net liquidity position?

A. USD -200.
B. USD -120.
C. USD -40.
D. USD 80.

Answer: C

Step by step

1. Supplies: deposit inflows 80 + loan repayments 120 + money market borrowings 80 = 280.
2. Demands: withdrawals 50 + loan requests 100 + operating expenses 70 + dividends 40 + repayment of borrowings 60 = 320.
3. $L = 280 - 320 = -40$ million: a liquidity deficit that must be covered by raising funds.

Why C is correct: Supplies (280) fall short of demands (320), giving a deficit of 40.

Why the others are wrong

- **A:** Treats money market borrowings as a demand: 200 - 400 = -200.
- **B:** Omits money market borrowings from supply: 200 - 320 = -120.
- **D:** Treats repayment of bank borrowings (60) as a supply: -40 + 120 = +80.

Q205

Schweser Module Quiz 70.1 Q1

REBUILT — Source omitted its exhibit (list reconstructed from the explanation) and distractor 3,405,000 had no realistic error, so it was replaced.

Genny Richards is the liquidity manager for Legend Bank. In evaluating the bank's net liquidity position, Richards anticipates the following amounts:

Asset sales	$1,325,000
Incoming deposits	$2,500,000
Nondeposit services revenue	$950,000
Deposit withdrawals	$1,015,000
Dividend payments	$470,000
Loan requests	$845,000

Legend Bank's net liquidity position is closest to:

A. $2,330,000.
B. $2,445,000.
C. $2,915,000.
D. $3,385,000.

Answer: B

Step by step

1. Supplies: 1,325,000 + 2,500,000 + 950,000 = 4,775,000.
2. Demands: 1,015,000 + 470,000 + 845,000 = 2,330,000.
3. $L = 4{,}775{,}000 - 2{,}330{,}000 = 2{,}445{,}000$, a surplus.

Why B is correct: Asset sales, new deposits and fee income supply liquidity; withdrawals, dividends and loan requests demand it.

Why the others are wrong

- **A:** This is total demand (2,330,000), not the net position.
- **C:** Omits dividend payments from demands: 4,775,000 - 1,860,000 = 2,915,000.
- **D:** Treats dividends as a supply: 2,445,000 + 2 x 470,000 = 3,385,000.

LO 70.b

Compare strategies that a bank can use to meet demands for additional liquidity.

Q206 GARP book Ch.5 Q6 (q0023) · GARP book Ch.5 Q6 (q0032)

The Bank of Your Dreams has a simple balance sheet (figures in $ millions):

Assets	Liabilities and Equity		
Cash	$100	Deposits	$4,000
Securities	1,000	Other liabilities	500
Loans	4,000	Equity	600
Total assets	5,100	Total liabilities and equity	5,100

Depositors withdraw $500 million.

a. If the asset conversion method is used and securities are sold to cover the deposit drain, what happens to the size of the bank?

b. If liability management is used to cover the deposit drain, what happens to the size of the bank?

Answer: see model answer — a. Asset conversion: the bank sells $500 million of securities to pay the withdrawals. Securities fall to $500 million and deposits fall to $3,500 million, so total assets fall from $5,100 million to $4,600 million: the bank shrinks by $500 million. b. Liability management: the bank replaces the lost deposits with $500 million of borrowed funds (for example federal funds, repos, jumbo CDs or discount window loans). Deposits fall by 500 and borrowings rise by 500, and the asset portfolio is untouched, so total assets stay at $5,100 million: the bank's size is unchanged.

Step by step

1. Deposit drain: liabilities fall by $500 million (deposits $4,000 to $3,500).
2. a. Asset conversion: sell $500 million of securities for cash and pay out depositors; assets fall by 500, so 5,100 - 500 = $4,600 million.
3. b. Liability management: borrow $500 million in the money market to pay out depositors; deposits down 500, borrowings up 500, assets unchanged at $5,100 million.

Why this is correct: Asset conversion funds the drain by liquidating assets, so the balance sheet shrinks; liability management replaces one liability with another and leaves the asset portfolio and total size unchanged.

Q207 GARP book Ch.5 Q7 (q0024) · GARP book Ch.5 Q7 (q0033)

The liquidity manager of the Bank of Your Dreams needs cash to meet unanticipated loan demand: the loan officer wants to make $600 million in loans. Balance sheet ($ millions): Cash 100; Securities 1,000; Loans 4,000; Total assets 5,100; Deposits 4,000; Other liabilities 500; Equity 600.

a. If asset conversion is used and securities are sold to provide money for the loans, what happens to the size of the bank?

b. If liability management is used to provide funds for the loans, what happens to the size of the bank?

Answer: see model answer — a. Asset conversion: selling $600 million of securities (1,000 to 400) funds the new loans (4,000 to 4,600). One asset is swapped for another, so total assets remain $5,100 million: the bank's size is unchanged. b. Liability management: the bank borrows $600 million and lends it out. Loans rise to 4,600 with securities unchanged, and borrowings rise by 600, so total assets and liabilities grow to $5,700 million: the bank gets bigger by $600 million.

Step by step

1. New loans use $600 million of funds; loans rise from $4,000 to $4,600 million.
2. a. Asset conversion: sell $600 million of securities (1,000 to 400); asset composition changes but total assets = 5,100.
3. b. Liability management: borrow $600 million; assets rise by 600 to 5,100 + 600 = $5,700 million, with liabilities rising equally.

Why this is correct: Selling securities only reshuffles assets, whereas borrowing adds a new liability and so enlarges the balance sheet.

Q208

AnalystPrep Q.3947

REBUILT — Original stem was garbled (unnamed 'RCP' staff, awkward 'not likely' wording); rewrote it as a clean 'which is NOT a source of additional liquidity' question with the same options.

A bank's liquidity manager is looking for ways to raise additional liquidity. Which of the following is NOT a way for the bank to obtain additional liquidity?

A. Liquidating trading book positions
B. Holding cash and Treasury securities that can be converted to cash
C. Increasing investments in real estate
D. Borrowing money at short notice

Answer: C

Step by step

1. Liquidity strategies are asset-based (convert liquid assets), liability-based (borrow), or a mix.
2. Trading book positions are marketable and can be sold quickly: asset conversion.
3. Cash and Treasuries are the classic liquid assets (ready market, stable price, reversible): asset conversion.
4. Short-notice borrowing is purchased liquidity: liability management.
5. Real estate is illiquid (no ready market, price may fall on a quick sale), and buying it consumes cash, so it does not provide liquidity.

Why C is correct: Real estate is illiquid and increasing it uses up funds, whereas the other three are standard asset-conversion or liability-management sources.

Why the others are wrong

- **A:** A valid asset-conversion source: trading positions are marketable and can be sold quickly.
- **B:** A valid asset-conversion source: cash and Treasuries are the most liquid assets (ready market, stable price, reversible).
- **D:** A valid liability-management source: the ability to borrow at short notice supplies purchased liquidity.

Q209

AnalystPrep Q.3890

REBUILT — Original option A (advertising for NOW deposits) is not a listed Schweser/GARP strategy and the stem was vague; replaced it with jumbo negotiable CDs and clarified the stem.

A bank faces a liquidity deficit and wants to raise cash using asset conversion or liability management. Which of the following actions would NOT help meet the deficit?

A. Issuing jumbo negotiable certificates of deposit
B. Selling some of its Treasury bills
C. Aggressively making new loans
D. Selling securities under agreements to repurchase

Answer: C

Step by step

1. Asset conversion raises cash by selling liquid assets (Treasury bills).
2. Liability management raises cash by borrowing: jumbo negotiable CDs, repurchase agreement sales, federal funds, discount window, etc.
3. Making new loans is a use of funds that turns liquid funds into illiquid assets and increases the deficit.

Why C is correct: New lending absorbs cash, so it worsens rather than meets a liquidity deficit; the other actions all supply cash.

Why the others are wrong

- **A:** Jumbo CDs are a standard borrowed-liquidity source; they bring in cash.
- **B:** Selling Treasury bills is asset conversion and generates cash.
- **D:** Selling securities under repurchase agreements is a borrowing (purchased liquidity) that brings in cash now; confusing it with buying securities under repo (an asset) would wrongly mark it as a use of funds.

Q210

AnalystPrep Q.3900

REBUILT — Original options (increase/decrease/no change/cannot establish) included a giveaway 'cannot establish'; rebuilt with quantified options and stated how the withdrawal is funded.

Loud Bank has the following balance sheet:

Assets	Amount ($ millions)
Cash	48
Securities	940
Loans	3,932
Total assets	4,920
Liabilities and equity	Amount ($ millions)
Deposits	3,948
Other liabilities	440
Equity	532
Total liabilities and equity	4,920

Depositors withdraw $200 million. Loud Bank uses the asset conversion (asset liquidity management) strategy and sells $200 million of securities to pay the withdrawal. What is the effect on the size of the bank's balance sheet?

A. Total assets increase to $5,120 million
B. Total assets decrease to $4,720 million
C. Total assets are unchanged at $4,920 million
D. Total assets decrease to $4,520 million

Answer: B

Step by step

1. Withdrawal: deposits fall from 3,948 to 3,748, a −200 change on the liability side.
2. Asset conversion: securities are sold for cash and the cash is paid out, so securities fall from 940 to 740, a −200 change on the asset side.
3. New total assets = 4,920 − 200 = 4,720 million; the balance sheet shrinks.

Why B is correct: Under asset conversion, liquidity is created by liquidating assets, so the cash leaving the bank to depositors removes both an asset and a liability and the balance sheet shrinks by the amount withdrawn.

Why the others are wrong

- **A:** Treats the withdrawal as adding to the balance sheet; deposits actually fall, and nothing is borrowed in this strategy.
- **C:** This is the result under liability management (borrowing replaces deposits); here the assets are sold, not replaced.
- **D:** Double counts the $200 million, deducting it from assets once for the sale and again for the deposit drain.

Q211

AnalystPrep Q.3901

REBUILT — Replaced the vague 'cannot establish' option set with quantified options.

Loud Bank has the following balance sheet:

Assets	Amount ($ millions)
Cash	48
Securities	940
Loans	3,932
Total assets	4,920
Liabilities and equity	Amount ($ millions)
Deposits	3,948
Other liabilities	440
Equity	532
Total liabilities and equity	4,920

Depositors withdraw $200 million. Loud Bank uses the liability management (borrowed liquidity) strategy and borrows $200 million in the money market to pay the withdrawal, leaving its asset portfolio unchanged. What is the effect on the size of the bank's balance sheet?

A. Total assets increase to $5,120 million
B. Total assets decrease to $4,720 million
C. Total assets are unchanged at $4,920 million
D. Total assets decrease to $4,520 million

Answer: C

Step by step

1. Deposits fall by 200 (3,948 to 3,748).

2. New borrowings add 200 to liabilities, so liabilities plus equity remain 4,920 million.
3. Assets are unchanged because the borrowed cash is paid straight out to depositors; total assets stay at 4,920.

Why C is correct: Borrowed liquidity replaces one liability (deposits) with another (purchased funds), so the asset side and total balance sheet size do not change.

Why the others are wrong

- **A:** Adds the new borrowings without deducting the lost deposits.
- **B:** Ignores the new borrowings; this is the asset conversion outcome.
- **D:** Deducts the withdrawal twice and ignores the offsetting borrowing.

Q212

AnalystPrep Q.3902

REBUILT — Stem mixed the earlier deposit drain with the loan funding, making the baseline ambiguous; fixed the baseline and used quantified options.

Loud Bank has the following balance sheet:

Assets	Amount ($ millions)
Cash	48
Securities	940
Loans	3,932
Total assets	4,920
Liabilities and equity	Amount ($ millions)
Deposits	3,948
Other liabilities	440
Equity	532
Total liabilities and equity	4,920

Ignore any deposit withdrawals. The bank's loan officer wants to make $400 million of new loans, and management funds them entirely by purchasing liquidity (liability management), e.g., by issuing jumbo CDs. What is the effect on the size of the bank's balance sheet?

A. Total assets increase to $5,320 million
B. Total assets decrease to $4,520 million
C. Total assets are unchanged at $4,920 million
D. Total assets increase to $5,720 million

Answer: A

Step by step

1. Borrowing raises liabilities by 400, so liabilities plus equity become $4{,}920 + 400 = 5{,}320$.
2. New loans raise assets by 400 (loans 3,932 to 4,332); no asset is sold.
3. Total assets $= 4{,}920 + 400 = 5{,}320$ million; the balance sheet grows.

Why A is correct: Liability management adds funding without selling any assets, so new lending expands both sides of the balance sheet by the amount borrowed.

Why the others are wrong

- **B:** Applies the asset conversion logic (funding by selling assets shrinks the bank), which is not used here.
- **C:** Assumes the borrowing is offset by something else; nothing is sold or repaid, so the bank grows.
- **D:** Adds the $400 million twice (once for the loans and again for the borrowing).

Q213

BT answers P2.T8.20.6.2 · BT notes P2.T8.20.6.2

Experienced liquidity managers tend to employ one of three strategies: asset liquidity management (asset conversion), borrowed liquidity (purchased liquidity or liability management), or balanced liquidity management. About these strategies, each of the following statements is true EXCEPT which is false?

A. A liquid asset has a ready market, a reasonably stable price and is reversible.
B. If a bank's primary goal is to avoid shrinking its balance sheet (and weakening the appearance of the balance sheet), then a borrowed liquidity strategy is better than an asset liquidity strategy.
C. If a bank's primary goal is to minimize risk, then a borrowed liquidity strategy is better than an asset liquidity strategy.
D. A balanced liquidity strategy balances the opportunity cost associated with storing liquidity in assets against the risks of interest rate volatility and credit availability associated with borrowing liquidity.

Answer: C

Step by step

1. Test A: a liquid asset has a ready market, stable price and is reversible, which matches the standard definition (true).
2. Test B: borrowing leaves the asset portfolio unaltered while asset sales shrink the balance sheet, so borrowing is better for that goal (true).
3. Test C: borrowed liquidity is risky because of interest rate volatility, uncertain borrowing costs and credit availability, so asset liquidity is the lower-risk approach; the statement is false.
4. Test D: balanced management trades off the opportunity cost of holding liquid assets against borrowing risks (true).

Why C is correct: Borrowed liquidity exposes the bank to interest rate and credit availability risk, so it is the riskier, not the safer, strategy.

Why the others are wrong

- **A:** True: it is the textbook three-part definition of a liquid asset.
- **B:** True: asset sales shrink the balance sheet whereas borrowing leaves assets intact.
- **D:** True: this is exactly the trade-off a balanced strategy manages.

Q214 Schweser Module Quiz 70.1 Q2

A bank utilizing the liability management strategy is most likely to use which of the following sources of liquidity?

A. U.S. Treasury securities.
B. Federal agency securities.
C. Repurchase agreement sales.
D. Municipal bond investments.

Answer: C

Step by step

1. Liability management means borrowing funds to meet liquidity demands.
2. Borrowed sources include jumbo CDs, repurchase agreement sales (selling securities with an agreement to repurchase), Eurocurrency issuance, fed funds borrowings, discount window and FHLB advances.
3. Of the choices, only repo sales is a borrowing; the others are liquid assets.

Why C is correct: A repo sale is a collateralized borrowing, a liability, while the other choices are assets held under asset liquidity management.

Why the others are wrong

- **A:** Treasuries are liquid assets used in asset conversion, not borrowed funds.
- **B:** Agency securities are also liquid assets held for asset liquidity.
- **D:** Municipal bonds are liquid assets held under asset liquidity management.

LO 70.c

Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators).

Q215 AnalystPrep Q.3904 · GARP book Chapter 5 end-of-chapter (Abigail Savings Bank)

REBUILT — Relabeled to LO 70.c (expected liquidity requirement across probability-weighted scenarios); options lacked units and distractors were weak; the open GARP version tests the same method with different numbers and is folded in as a source.

Grand Bank's liquidity manager estimates the following liquidity outcomes for next month. Deficits are negative and surpluses positive.

Scenario	Probability	Liquidity position ($ millions)
1	45%	-500
2	20%	-575
3	25%	+350
4	10%	+100

What is the bank's expected liquidity requirement?

A. -$156.25 million
B. -$242.50 million

C. +$242.50 million
D. -$340.00 million

Answer: B
Step by step
1. Expected requirement $= \sum p_i L_i$.
2. $0.45(-500) = -225$; $0.20(-575) = -115$; $0.25(350) = +87.5$; $0.10(100) = +10$.
3. Sum: $-225 - 115 + 87.5 + 10 = -242.5$.

Why B is correct: Weighting each scenario by its probability gives an expected deficit of $242.5 million, so management should plan to raise about that much liquidity (borrow or sell assets).
Why the others are wrong
- **A:** Simple average of the four outcomes ((-500-575+350+100)/4), ignoring probabilities.
- **C:** Correct magnitude but sign flipped; the net result is a deficit, not a surplus.
- **D:** Included only the two deficit scenarios (-225 - 115) and ignored the offsetting surpluses.

Q216
GARP book Ch.5 Q3

First National Bank of Belle Mead has forecast its checkable deposits, time and savings deposits, and commercial and household loans over the next eight months. The resulting estimates (in millions) are shown below. Use the sources and uses of funds approach to indicate which months are likely to result in liquidity deficits and which in liquidity surpluses if these forecasts turn out to be true. Explain carefully what you would do to deal with each month's projected liquidity position.

Month	Checkable Deposits	Time and Savings Deposits	Commercial Loans	Consumer Loans
January	$120	$550	$650	$160
February	115	500	650	230
March	100	500	700	210
April	90	485	700	175
May	105	465	710	160
June	80	490	700	200
July	90	525	700	175
August	100	515	675	150

Answer: see model answer — Estimated liquidity surplus (+) / deficit (-) = change in total deposits minus change in total loans (millions): February -125, March -45, April +10, May 0 (balanced), June -30, July +70, August +50. For deficit months (Feb, Mar, Jun) uses of funds exceed sources, so line up funding in advance: convert liquid assets (securities, fed funds sold) or borrow (fed funds, repos, CDs, discount window), sized to each deficit; February is the largest and needs the most advance planning. For surplus months (Apr, Jul, Aug) invest the excess in liquid short-term assets or repay borrowings, keeping a cushion for the next deficit. May is essentially balanced, so monitor only.
Step by step
1. Total deposits = checkable + time and savings: Jan 670, Feb 615, Mar 600, Apr 575, May 570, Jun 570, Jul 615, Aug 615.
2. Total loans = commercial + consumer: Jan 810, Feb 880, Mar 910, Apr 875, May 870, Jun 900, Jul 875, Aug 825.
3. Monthly change in deposits: Feb -55, Mar -15, Apr -25, May -5, Jun 0, Jul +45, Aug 0. Change in loans: Feb +70, Mar +30, Apr -35, May -5, Jun +30, Jul -25, Aug -50.
4. Surplus/(deficit) = ΔDeposits − ΔLoans: Feb -125, Mar -45, Apr +10, May 0, Jun -30, Jul +70, Aug +50.
5. Deficit months need funding (asset sales or liability management); surplus months should be invested short term or used to repay borrowings and build a cushion.

Why this is correct: Deposit growth supplies liquidity and loan growth uses it, so each month's change in deposits less change in loans gives the projected surplus or deficit.

Q217
AnalystPrep Q.3897

No financial institution can confidently tell if it has sufficient liquidity until it has passed the market's test. Which of the following is a signal that management should take note of?
A. Stock price behavior
B. Public confidence

C. Borrowings from the central bank
D. All of the above

Answer: D
Step by step
1. Under the market signals (discipline of the marketplace) approach, management monitors public confidence, stock price behavior, risk premiums on CDs and borrowings, loss sales of assets, ability to meet loan commitments, and central bank borrowings.
2. A, B and C are each listed signals, so all of the above.

Why D is correct: Each of the three items is a market signal that liquidity may be inadequate; a yes to any warrants a review of liquidity policy.
Why the others are wrong
- **A:** Stock price decline is a signal, but so are B and C; choosing a single signal is incomplete.
- **B:** Loss of public confidence (withdrawals) is a signal, but not the only one listed.
- **C:** Larger and more frequent central bank borrowing is a signal, but A and B also qualify.

Q218
GARP book Ch.5 Q4

Queen Savings is attempting to determine its liquidity requirements today (the last day in August) for the month of September. September is usually a month of heavy loan demand. Management holds an 85 percent reserve in liquid assets or borrowing capacity for each dollar of hot money deposits, a 25 percent reserve behind vulnerable deposits, and a 5 percent reserve for core funds. Time and savings deposits carry a zero percent reserve requirement and all checkable deposits carry a 3 percent reserve requirement. Queen currently has total loans outstanding of $2,500 million, which two weeks ago were as high as $2,550 million. Annual loan growth over the past three years has been about 6 percent. Prepare low and high estimates for Queen's total liquidity requirement for September.

Millions of Dollars	Checkable Deposits	Savings Deposits	Time Deposits
Hot money funds	$10	$5	$1,200
Vulnerable funds	65	152	740
Stable (core) funds	85	450	172

Answer: see model answer — Using the structure of funds method, the liability liquidity reserve is about $1,306.5 million: 0.85 x 1,214.7 + 0.25 x 955.05 + 0.05 x 704.45. Loan requirement: low estimate uses current loans of $2,500 million, potential loans 2,500 x 1.06 = 2,650, so 150; high estimate uses the recent peak of $2,550 million, potential loans 2,703, so 2,703 - 2,500 = 203. Total liquidity requirement is about $1,456.5 million (low) to $1,509.5 million (high), held in liquid assets and borrowing capacity.
Step by step
1. Total by category: hot money 10 + 5 + 1,200 = 1,215; vulnerable 65 + 152 + 740 = 957; stable 85 + 450 + 172 = 707.
2. Subtract legal reserves (3% of checkable only): hot 1,215 - 0.03(10) = 1,214.7; vulnerable 957 - 0.03(65) = 955.05; stable 707 - 0.03(85) = 704.45.
3. Liability liquidity reserve = 0.85(1,214.7) + 0.25(955.05) + 0.05(704.45) = 1,032.50 + 238.76 + 35.22 = 1,306.48.
4. Loan requirement = potential loans - actual loans. Low: 2,500(1.06) − 2,500 = 150. High: 2,550(1.06) − 2,500 = 203.
5. Total requirement: low 1,306.48 + 150 = 1,456.48; high 1,306.48 + 203 = 1,509.48 (millions).

Why this is correct: The structure of funds approach sets aside a percentage of each funds category (net of legal reserves) plus the amount needed to fund potential good loans above current loans.

Q219
GARP book Ch.5 Q5

Using the following financial information for Wilson National Bank (figures in thousands of dollars), calculate as many of the liquidity indicators discussed in the chapter as you can. Do you detect any significant liquidity trends? Which trends should management investigate?

Item	Most Recent Year	Previous Year
Cash and due from depository institutions	$345,000	$358,000
U.S. Treasury securities	176,000	178,000
Other securities	339,000	343,000
Pledged securities	287,000	223,000
Federal funds sold and reverse repurchase agreements	175,000	131,000
Loans and leases, net	2,148,000	1,948,000
Total assets	3,500,000	3,250,000
Demand deposits	600,000	556,000
Savings deposits	730,000	721,000
Time deposits	1,100,000	853,000
Total deposits	2,430,000	2,130,000
Core deposits	850,000	644,000
Brokered deposits	58,000	37,000
Federal funds purchased and repurchase agreements	217,000	237,000
Other money market borrowings	25,000	16,000

Answer: see model answer — Recent year / previous year: cash position 9.86% / 11.02%; liquid securities 5.03% / 5.48%; core deposit ratio 24.29% / 19.82%; net fed funds and repo position -1.20% / -3.26%; capacity ratio 61.37% / 59.94%; pledged securities ratio 35.79% / 29.97% (pledged / [Treasury + other + pledged securities]); deposit composition (demand / time) 54.55% / 65.18%; deposit brokerage index 2.39% / 1.74%. The hot money and loan commitments ratios cannot be computed (volatile liabilities such as large CDs and unused commitments are not given). Adverse trends: lower cash and Treasury ratios, higher capacity and pledged securities ratios, more brokered deposits; management should investigate loan growth outpacing liquid assets, rising pledged securities and brokered-deposit reliance. Favorable: higher core deposit ratio, better net fed funds position, lower demand-to-time ratio.

Step by step

1. Cash position = 345,000 / 3,500,000 = 9.86% (prior 358,000 / 3,250,000 = 11.02%); liquid securities = 176,000 / 3,500,000 = 5.03% (5.48%).
2. Core deposit ratio = 850,000 / 3,500,000 = 24.29% (644,000 / 3,250,000 = 19.82%); net fed funds position = (175,000 - 217,000) / 3,500,000 = -1.20% (prior (131,000 - 237,000) / 3,250,000 = -3.26%).
3. Capacity ratio = 2,148,000 / 3,500,000 = 61.37% (1,948,000 / 3,250,000 = 59.94%).
4. Pledged securities ratio = 287,000 / (176,000 + 339,000 + 287,000) = 35.79% (223,000 / 744,000 = 29.97%).
5. Deposit composition = 600,000 / 1,100,000 = 54.55% (556,000 / 853,000 = 65.18%); deposit brokerage index = 58,000 / 2,430,000 = 2.39% (37,000 / 2,130,000 = 1.74%).
6. Compare changes: falling stored liquidity (cash, Treasuries) and rising capacity, pledged and brokered ratios signal tightening liquidity; offsets are a higher core deposit ratio and better fed funds position.

Why this is correct: Each liquidity indicator is a balance sheet ratio, and the change from one period to the next is more informative than the level.

Q220 GARP book Ch.5 Q8

Suppose Abigail Savings Bank's liquidity manager estimates that the bank will experience a $375 million liquidity deficit next month with a probability of 15 percent, a $200 million liquidity deficit with a probability of 35 percent, a $100 million liquidity surplus with a probability of 35 percent, and a $250 million liquidity surplus bearing a probability of 15 percent. What is this savings bank's expected liquidity requirement? What should management do?

Answer: see model answer — Expected liquidity position = 0.15(-375) + 0.35(-200) + 0.35(+100) + 0.15(+250) = -56.25 - 70 + 35 + 37.5 = -$53.75 million, an expected deficit of $53.75 million. Management should arrange about $54 million of funding (sell liquid assets or borrow in money markets, e.g. fed funds, repos, CDs) and prepare a contingency plan for the worst case ($375 million deficit).

Step by step

1. Weight each outcome by probability: $0.15(-375) = -56.25$; $0.35(-200) = -70.00$; $0.35(100) = 35.00$; $0.15(250) = 37.50$.
2. Sum: $-56.25 - 70 + 35 + 37.5 = -53.75$ million, a deficit.
3. Action: line up roughly $53.75 million of funding (asset conversion or borrowing) and plan for the worst-case deficit.

Why this is correct: The expected liquidity requirement is the probability-weighted average of the scenario sur-

pluses (+) and deficits (-).

Q221 AnalystPrep Q.3888

XYZ Bank has the following forecasts for its checkable deposits, time and savings deposits, commercial loans, and consumer loans over the next eight months (US$).

Month	Checkable Deposits	Time and Savings Deposits	Commercial Loans	Consumer Loans
January	219	639	729	209
February	216	591	731	267
March	203	593	783	265
April	189	580	785	213
May	210	562	797	211
June	187	589	789	234
July	189	626	791	232
August	209	618	768	230

Employ the sources and uses of funds approach to determine the month which is likely to have a liquidity surplus.

A. February
B. March
C. June
D. August

Answer: D

Step by step

1. Surplus/(deficit) = ΔDeposits - ΔLoans, with total deposits = checkable + time and savings and total loans = commercial + consumer. Totals Jan to Aug: deposits 858, 807, 796, 769, 772, 776, 815, 827; loans 938, 998, 1,048, 998, 1,008, 1,023, 1,023, 998.
2. Changes in deposits: Feb -51, Mar -11, Jun +4, Aug +12. Changes in loans: Feb +60, Mar +50, Jun +15, Aug -25.
3. Surplus/(deficit): Feb $-51-60=-111$; Mar $-11-50=-61$; Jun $4-15=-11$; Aug $12-(-25)=+37$.
4. Only August is positive among the options.

Why D is correct: In August deposits rise while loans fall, so sources exceed uses by 37, a liquidity surplus.

Why the others are wrong

- **A:** February is a deficit of 111: deposits fell 51 and loans rose 60.
- **B:** March is a deficit of 61: deposits fell 11 and loans rose 50.
- **C:** June is a deficit of 11: deposits rose only 4 while loans rose 15; looking at the deposit increase alone gives the wrong sign.

Q222 AnalystPrep Q.3889

XYZ Bank has the following forecasts for its checkable deposits, time and savings deposits, commercial loans, and consumer loans over the next eight months (US$).

Month	Checkable Deposits	Time and Savings Deposits	Commercial Loans	Consumer Loans
January	219	639	729	209
February	216	591	731	267
March	203	593	783	265
April	189	580	785	213
May	210	562	797	211
June	187	589	789	234
July	189	626	791	232
August	209	618	768	230

Using the sources and uses of funds approach, which of the following months is likely to result in a liquidity deficit?

A. February
B. April
C. July
D. August

Answer: A

Step by step

1. Surplus/(deficit) = ΔDeposits - ΔLoans, with total deposits = checkable + time and savings and total loans = commercial + consumer. Totals Jan to Aug: deposits 858, 807, 796, 769, 772, 776, 815, 827; loans 938, 998, 1,048, 998, 1,008, 1,023, 1,023, 998.
2. Changes in deposits: Feb -51, Apr -27, Jul +39, Aug +12. Changes in loans: Feb +60, Apr -50, Jul 0, Aug -25.
3. Surplus/(deficit): Feb $-51-60=-111$; Apr $-27-(-50)=+23$; Jul $39-0=+39$; Aug $12+25=+37$.
4. Only February is negative, a deficit of 111.

Why A is correct: In February deposits fall by 51 while loans grow by 60, so uses exceed sources by 111.

Why the others are wrong

- **B:** April is a surplus of 23: deposits fell 27 but loans fell more (50); looking only at the deposit decline misses this.
- **C:** July is a surplus of 39: deposits rose 39 and loans were unchanged.
- **D:** August is a surplus of 37: deposits rose 12 and loans fell 25.

Q223

AnalystPrep Q.3894

The following financial information pertains to Daylight Bank:

Assets	Amount (US$ millions)
Cash and due from depository institutions	546
U.S. Treasury securities	382
Other securities	550
Pledged securities	503
Federal funds sold and reverse repurchase agreements	396
Net loans and leases	2,374
Total Assets	3,431

Liabilities |

Demand deposits	741
Savings deposits	976
Time deposits	1,351
Total deposits	2,686
Core deposits	1,111
Brokered deposits	214
Federal funds purchased and repurchase agreements	488
Other money market borrowings	186

Calculate Daylight Bank's net federal funds and repurchase agreements position.

A. -11.54%
B. -2.68%
C. 2.68%
D. 11.54%

Answer: B

Step by step

1. Net position = (fed funds sold and reverse repos - fed funds purchased and repos) / total assets.
2. $(396-488)/3{,}431=-92/3{,}431=-2.68\%$.

Why B is correct: The bank borrows more overnight (488) than it lends (396), giving a negative net position of 2.68% of total assets (a lower liquidity reading).

Why the others are wrong

- **A:** -11.54% uses fed funds sold alone (396/3,431 = 11.54%) and attaches the wrong sign.
- **C:** 2.68% reverses the subtraction (purchased minus sold), flipping the sign.
- **D:** 11.54% = 396/3,431 uses only fed funds sold and ignores fed funds purchased.

Q224

AnalystPrep Q.3895

The following financial information pertains to Daylight Bank:

Assets	Amount (US$ millions)
Cash and due from depository institutions	546
U.S. Treasury securities	382
Other securities	550
Pledged securities	503
Federal funds sold and reverse repurchase agreements	396
Net loans and leases	2,374
Total Assets	3,431

Liabilities |	
Demand deposits	741
Savings deposits	976
Time deposits	1,351
Total deposits	2,686
Core deposits	1,111
Brokered deposits	214
Federal funds purchased and repurchase agreements	488
Other money market borrowings	186

Calculate Daylight Bank's cash position indicator.

A. 7.97%
B. 11.13%
C. 15.91%
D. 32.38%

Answer: C

Step by step

1. Cash position indicator = (cash and due from depository institutions) / total assets.
2. 546/3,431 = 15.91%.

Why C is correct: The cash position indicator compares cash and due from depository institutions with total assets; a higher value means greater liquidity.

Why the others are wrong

- **A:** 7.97% = 214/2,686 is the deposit brokerage index (brokered / total deposits).
- **B:** 11.13% = 382/3,431 is the liquid securities indicator (Treasuries / total assets).
- **D:** 32.38% = 1,111/3,431 is the core deposit ratio.

Q225

AnalystPrep Q.3896

REBUILT — Original odd-one-out item had a doubtful key (it treated the hot money ratio as a negative indicator); rebuilt to ask which listed ratio is a negative (higher = less liquid) indicator, with exactly one correct option.

Among liquidity indicators, a higher value of which of the following ratios signals LESS liquidity (a negative liquidity indicator)?

A. Liquid securities indicator
B. Hot money ratio
C. Pledged securities ratio
D. Net federal funds and repurchase agreements position

Answer: C

Key correction: Source key (C, hot money ratio as odd one out) conflicts with the notes: capacity ratio and pledged securities ratio are negative indicators, while a higher hot money ratio signals more liquidity.

Step by step

1. Higher = more liquidity: cash position, liquid securities, core deposit, hot money (money market assets / volatile liabilities), net fed funds and repo position.
2. Higher = less liquidity: capacity, loan commitments, pledged securities, deposit composition, deposit brokerage index.
3. Only the pledged securities ratio is in the second group.

Why C is correct: A larger share of securities pledged against government deposits leaves fewer securities available to sell for liquidity, so a rise signals greater future liquidity needs.

Why the others are wrong

- **A:** U.S. government securities / total assets: more marketable securities means more liquidity (positive indicator).
- **B:** Hot money ratio = money market assets / volatile liabilities; a higher ratio means volatile funding is covered by quickly saleable assets (positive). Confusing it with risky "hot money liabilities" leads here.
- **D:** A higher net fed funds position means the bank is a net overnight lender, which improves liquidity (positive).

Q226

AnalystPrep Q.5382 · Mock 3 Q45

A credit analyst is evaluating the liquidity of a small regional bank for a credit committee. Using quarterly financial statements, the analyst calculates liquidity indicators over the past three years. Which of the following trends should the analyst be most concerned about?

A. The bank's average net federal funds and repurchase agreements position has been increasing.
B. The bank's capacity ratio has been increasing.
C. The bank's pledged securities ratio has been decreasing.
D. The bank's loan commitments ratio has been decreasing.

Answer: B

Step by step

1. Classify each indicator: a higher value means more liquidity for the net federal funds and repo position; a higher value means less liquidity for the capacity ratio, pledged securities ratio and loan commitments ratio.
2. Capacity ratio = net loans and leases / total assets. Loans are the least liquid assets, so a rising ratio means the balance sheet is becoming less liquid.
3. A rising net fed funds/repo position (overnight lending exceeds overnight borrowing) is favourable; falling pledged securities (more securities free to sell) and falling unused commitments (fewer future drawdowns) are also favourable.
4. Only the rising capacity ratio is an adverse trend.

Why B is correct: The capacity ratio is a negative liquidity indicator: an increase signals more illiquid loans relative to total assets and therefore a worse liquidity position.

Why the others are wrong

- **A:** A rising net federal funds and repo position is a positive indicator (the bank is a net overnight lender); reading it as a warning reverses its sign.
- **C:** A falling pledged securities ratio frees more securities to sell or repo, so liquidity improves; the mistake is treating any change in a negative indicator as bad rather than its direction.
- **D:** A falling loan commitments ratio means fewer potential takedowns, which reduces future liquidity demand; treating a decrease as adverse confuses the direction of a negative indicator.

Q227

BT answers P2.T8.20.6.3 · BT notes P2.T8.20.6.3

REBUILT — Cleaned a stray bank name ('Acme'), added the sources/uses convention (monthly changes versus the prior month) and shortened the giveaway-length option D.

First City Bell Bank has forecast its deposits and loans (in millions) for the next six months. Using the sources and uses of funds approach, the monthly liquidity gap equals the change in total deposits minus the change in total loans, measured against the prior month (so January is the base month and the first gap is for February).

Month	Checkable Deposits	Time and Savings Deposits	Commercial Loans	Consumer Loans
January	120	400	500	140
February	110	490	640	220
March	90	490	700	200
April	80	475	710	175
May	95	475	700	150
June	75	480	720	210

Which of the following statements is TRUE?

A. During February, sources of liquidity equal zero.
B. During the first quarter (February and March), the bank expects a positive liquidity gap.
C. Over the cumulative period February through June, the bank expects a positive liquidity gap.
D. During April, sources of 15 are more than offset by uses of 25, giving a negative gap of 10.

Answer: D

Step by step

1. Total deposits by month: 520, 600, 580, 555, 570, 555. Total loans: 640, 860, 900, 885, 850, 930.
2. Monthly gap = Δdeposits − Δloans: Feb $80 - 220 = -140$; Mar $-20 - 40 = -60$; Apr $-25 - (-15) = -10$; May $15 - (-35) = +50$; Jun $-15 - 80 = -95$.
3. February: deposits rose 80, so sources are 80, not zero (A false). Q1 (Feb+Mar) gap = $-140 - 60 = -200$, negative (B false). Cumulative Feb-Jun gap = $-200 - 10 + 50 - 95 = -255$, negative (C false).
4. April: deposits fell 25 (a use) and loans fell 15 (a source), so sources 15 versus uses 25 gives −10 (D true).

Why D is correct: Deposit increases and loan decreases are sources; deposit decreases and loan increases are uses. April nets to $15 - 25 = -10$, a deficit, while the other statements contradict the computed gaps.

Why the others are wrong

- **A:** February deposits rose by 80 (an inflow), so sources are 80; zero sources would result from ignoring the deposit increase.
- **B:** Q1 gaps are −140 and −60, a deficit of 200; a positive result comes from reversing the sign (counting loan growth as a source).
- **C:** The cumulative gap is −255; a positive answer comes from reversing the sign, treating loan growth as a source and deposit growth as a use.

Q228

BT answers P2.T8.20.7.1 · BT notes P2.T8.20.71

REBUILT — Source distractors (notably 177.0) could not be tied to realistic errors; replaced with options generated by specific mistakes.

Kingstreet Savings is using the structure of funds approach to estimate its liquidity requirement. Its deposits are classified as hot money, vulnerable and stable (core) funds (millions of dollars):

Millions of dollars	Hot money funds	Vulnerable funds	Stable (core) funds
Checkable deposits	30.0	20.0	30.0
Savings deposits	10.0	40.0	60.0
Nonpersonal time deposits	70.0	70.0	50.0
Desired reserve percentage	80%	50%	20%

The legal reserve requirement is 10% on checkable and savings deposits and 0% on nonpersonal time deposits. The desired percentage applies to each category net of the legal reserves held against it. Ignoring the loan liquidity requirement, what is the liquidity reserve required for the vulnerable funds?

A. 13.0 million
B. 30.0 million
C. 62.0 million
D. 65.0 million

Answer: C

Step by step

1. Vulnerable funds total $20 + 40 + 70 = 130$.
2. Legal reserves held: $0.10 \times (20 + 40) + 0 \times 70 = 6.0$.
3. Funds net of legal reserves: $130 - 6 = 124$.
4. Required reserve: $0.50 \times 124 = 62.0$ million.

Why C is correct: The 50% desired reserve is applied to vulnerable funds after deducting legal reserves already held: $0.5 \times (130 - 6) = 62$.

Why the others are wrong

- **A:** 13.0 equals 10% × 130: it applies the legal reserve rate to all vulnerable funds and never applies the 50% desired reserve.
- **B:** 30.0 equals $0.5 \times (20 + 40)$: it leaves out the 70 of nonpersonal time deposits (which carry no legal reserve but are still vulnerable funds).
- **D:** 65.0 equals 0.5×130: it forgets to deduct the 6.0 of legal reserves already held.

Q229

BT answers P2.T8.20.7.3 · BT notes P2.T8.20.7.3

REBUILT — Source defined the pledged securities ratio over total assets; corrected to pledged securities over total securities per the GARP text.

Consider five of the liquidity indicators used in the liquidity indicator approach:
- Capacity ratio = net loans and leases / total assets
- Pledged securities ratio = pledged securities / total securities holdings
- Hot money ratio = money market assets / volatile liabilities

- Core deposit ratio = core deposits / total assets
- Deposit composition ratio = demand deposits / time deposits

A positive liquidity indicator is one where an increase signals greater liquidity (and an unexpected decrease is a warning sign); a negative indicator is the reverse. The core deposit ratio is a positive indicator. For which other indicator above would an unexpected decrease be a cause for concern?

A. Capacity ratio
B. Hot money ratio
C. Pledged securities ratio
D. Deposit composition ratio

Answer: B

Step by step

1. Hot money ratio = money market (short-term, saleable) assets / volatile liabilities; it shows whether volatile funding is matched by quick-to-sell assets. A higher ratio means more liquidity, so a drop is a warning.
2. Capacity ratio: more loans relative to assets means less liquidity, so a fall is favourable.
3. Pledged securities ratio: more pledged securities means fewer securities available to sell, so a fall is favourable.
4. Deposit composition ratio: more demand deposits relative to time deposits means a less stable funding base, so a fall is favourable.

Why B is correct: Only the hot money ratio is a positive indicator among the options; its unexpected decrease means volatile liabilities are less well covered by liquid assets.

Why the others are wrong

- **A:** The capacity ratio is a negative indicator (loans are illiquid); it is its increase, not its decrease, that signals concern.
- **C:** The pledged securities ratio is negative: a decrease frees securities for sale. Treating all ratios as positive gives this error.
- **D:** The deposit composition ratio (demand/time) is negative: a decrease means more stable time deposits. Mistaking volatile demand deposits as a liquidity source gives this answer.

Q230

Mock 1 Q34

REBUILT — Source key (D) answered a different quantity than the stem asked; stem clarified to exclude the loan requirement and distractors rebuilt around realistic errors.

A bank has the following deposit and nondeposit liabilities broken down into three categories:

Category	Amount (USD million)	Desired reserve
Hot money	50	80%
Vulnerable funds	55	30%
Core funds	100	20%

Legal reserves are 3% of each category, and the desired reserve percentages are applied after deducting legal reserves. Actual loans are USD 165 million with the potential to reach USD 225 million. The bank wishes to be able to meet all good loans. Based on the information listed, the bank's liability liquidity reserve (excluding the loan requirement) is closest to:

A. USD 60 million.
B. USD 74 million.
C. USD 77 million.
D. USD 134 million.

Answer: B

Key correction: Source key (Mock 1) said D; D is the total liquidity requirement (liability reserve plus loan requirement), but the stem asks for the liability liquidity reserve, which is B.

Step by step

1. Reserve before legal-reserve adjustment: $0.80 \times 50 + 0.30 \times 55 + 0.20 \times 100 = 40 + 16.5 + 20 = 76.5$.
2. Deduct 3% legal reserves from each category: $76.5 \times 0.97 = 74.2$ million.
3. This is the liability liquidity reserve, about USD 74 million.
4. The loan requirement $225 - 165 = 60$ is added only for the total liquidity requirement ($74.2 + 60 = 134.2$).

Why B is correct: Liability liquidity reserve $= \sum$ (desired % $\times$ category funds net of legal reserves) $= 0.97 \times 76.5 = 74.2$.

Why the others are wrong

- **A:** 60 is only the loan requirement 225 − 165; it ignores the deposit-side reserve altogether.
- **C:** 77 (76.5) applies the desired percentages to gross funds and forgets to deduct the 3% legal reserves.
- **D:** 134 is the total liquidity requirement 74.2 + 60; it adds the loan requirement that the question excludes.

Q231 Schweser Module Quiz 70.2 Q1

Which of the following indicators would create concern for a liquidity manager looking to stabilize liquidity and create confidence in the bank's position?

A. An increasing hot money ratio.
B. An increasing deposit composition ratio.
C. Increases in reverse repurchase agreements.
D. An excess of federal funds sold over federal funds purchased.

Answer: B

Step by step

1. Deposit composition ratio = demand deposits / time deposits. A rise means more volatile demand deposits relative to time deposits that have fixed maturities and early-withdrawal penalties.
2. A higher ratio therefore indicates a less stable funding base and greater liquidity needs - a negative indicator.
3. Hot money ratio (money market assets / volatile liabilities), reverse repos and net fed funds sold all add liquid assets, so they are positive indicators.

Why B is correct: An increasing deposit composition ratio shifts funding toward demand deposits that can leave at any time, raising liquidity needs.

Why the others are wrong

- **A:** The hot money ratio is a positive indicator: an increase means more money market assets cover volatile liabilities. Confusing it with the volatile liabilities themselves gives this error.
- **C:** Reverse repos are short-term money market assets that can be quickly converted to cash, so increases improve liquidity.
- **D:** Federal funds sold exceeding federal funds purchased is a net overnight lender position, which is a positive liquidity indicator.

Q232 Mock 1 Q49

REBUILT — Original distractors (overnight borrowings to total assets, noncore funding to long-term assets) are not liquidity indicators in the Schweser reading; replaced options with the reading's actual indicators.

A risk manager is estimating the bank's liquidity needs using the liquidity indicator approach. Which of the following ratios is she least likely to use as a liquidity indicator?

A. Total debt to equity.
B. Capacity ratio.
C. Hot money ratio.
D. Loan commitments ratio.

Answer: A

Step by step

1. The liquidity indicator approach uses balance-sheet ratios whose changes signal liquidity strength or weakness.
2. Ratios where a higher value means more liquidity: cash position indicator, liquid securities indicator, core deposit ratio, hot money ratio, net fed funds and repo position.
3. Ratios where a higher value means less liquidity: capacity ratio (net loans and leases / total assets), loan commitments ratio (unused commitments / total assets), pledged securities ratio, deposit composition ratio, deposit brokerage index.
4. Total debt to equity is a leverage/solvency measure, not in either list, so it is the one least likely to be used.

Why A is correct: Debt to equity measures capital structure and leverage, not the ability to meet cash needs, whereas the other three are standard liquidity indicators in the reading.

Why the others are wrong

- **B:** Capacity ratio (net loans / total assets) is a liquidity indicator; a higher value means less liquidity. Wrongly excluded by thinking only liquidity-positive ratios count.
- **C:** Hot money ratio (money market assets / volatile liabilities) is a liquidity indicator; higher means more liquidity. Mistake: not recognizing the term.
- **D:** Loan commitments ratio (unused commitments / total assets) is a liquidity indicator; higher means less

liquidity. Mistake: ignoring off-balance-sheet claims on liquidity.

Reading 71. Intraday Liquidity Risk Management

Shyam Venkat and Stephen Baird, Liquidity Risk Management: A Practitioner's Perspective (Hoboken, NJ: John Wiley & Sons, 2016). Chapter 4.

LO	Learning objective	Qs
71.a	identify and explain the uses and sources of intraday liquidity.	11
71.b	discuss the governance structure of intraday liquidity risk management.	4
71.c	differentiate between methods for tracking intraday flows and monitoring risk levels.	12

LO 71.a

Identify and explain the uses and sources of intraday liquidity.

Q233

Mock AP Mock 1 Q50

REBUILT — Original asked an open-ended 'result' with two defensible answers (severe trading decrease vs bankruptcy); rebuilt to test how a bank manages outgoing payments when intraday credit is withdrawn.

Molino Bank's clearing bank stops processing Molino's daylight overdrafts, so Molino can no longer count on intraday credit. Molino starts the day with limited cash at the central bank and has large outgoing client wire transfers scheduled. Which of the following is the most appropriate way for Molino to manage its intraday position?

A. Delay or throttle non-critical outgoing payments so that payments are released only up to cash balances plus incoming funds received.
B. Release all payments on schedule, because any overdraft will be cleared by the end of the business day.
C. Declare insolvency, because loss of intraday credit means obligations cannot be met.
D. Take no action until the end-of-day PCS settlement, because intraday shortfalls are resolved at that point.

Answer: A

Key correction: Source key (C, bankruptcy) is not supported: the reading treats loss of intraday credit as a constraint to be managed, not as automatic failure.

Step by step

1. Intraday credit is a source of liquidity; without it, available funds = opening cash balances + incoming funds flow (the largest source).
2. Outgoing wire transfers are the largest use of intraday liquidity and can be slightly delayed or controlled until incoming receipts arrive.
3. So Molino should prioritise critical payments and throttle the rest until receipts rebuild its balance.
4. Releasing everything, waiting for settlement, or declaring insolvency all ignore this lever.

Why A is correct: With intraday credit removed, the only way to stay within the available balance is to control the timing of outgoing payments until incoming receipts arrive.

Why the others are wrong

- **B:** Treats an overdraft as harmless; the clearing bank has refused to allow it, and hard controls prevent exceeding limits.
- **C:** Overstates the consequence: losing one source of intraday liquidity does not by itself mean failure; the bank can manage the constraint by controlling payment timing.
- **D:** PCS settlement happens once, near day end, so waiting leaves payments unmade throughout the day.

Q234

AnalystPrep Q.3905

A bank treasurer at ABC bank realizes that the bank is experiencing a liquidity shortage during the day and is unable to pay XYZ bank for foreign transaction services owed. The treasurer suggests the use of intraday liquidity as a remedy. Which of the following uses of intraday liquidity best describes the treasurer's use of the funds?

A. Asset purchasing and funding.
B. Funding of nostro accounts.
C. Collateral pledging.
D. Outgoing wire transfers.

Answer: B

Step by step

1. Identify the payment: cash owed to another bank for foreign transaction services.
2. Per the reading, banks fund correspondent (nostro) accounts abroad as compensation for services provided.
3. Match the use: funding of nostro accounts.

Why B is correct: A cash transfer to a correspondent bank in compensation for its services is the definition of funding a nostro account.

Why the others are wrong

- **A:** Asset purchases cover securities, fixed assets and client loans, not payments for services.
- **C:** Collateral pledging is earmarking collateral for margin or deposits, not paying a fee.
- **D:** Outgoing wires are payments on LVPS for the bank or clients; a plausible confusion but the payment here is to a correspondent for services.

Q235

AnalystPrep Q.3908

A bank treasurer has been given an intraday credit whose use was determined from the following characteristics:
1. There is a single settlement per day, mainly in the late afternoon.
2. It may be either a source or a use of funds depending on a participant's net position on the day.
3. It can be forecast for securities that have multi-day settlements, but is more difficult to forecast for same-day settlement activity.

Which of the following uses of intraday liquidity do these characteristics describe?

A. Outgoing wire transfers.
B. Funding of nostro accounts.
C. Collateral pledging.
D. Settlements at payment, clearing and settlement (PCS) systems.

Answer: D

Step by step

1. One settlement per day, late afternoon: PCS systems.
2. Net position makes it a source or use: net settlement with clearinghouse.
3. Easier to forecast for multi-day settlements than same-day: PCS settlements.
4. All three match settlements at PCS systems.

Why D is correct: Net, once-a-day, late-afternoon settlement with multi-day predictability is the profile of PCS system settlements.

Why the others are wrong

- **A:** Wires occur throughout the day in a steady pattern and are a use only.
- **B:** Nostro funding can be a source or use, but is not a single late-afternoon settlement.
- **C:** Collateral pledging is a use only, forecast from trading volumes and price changes.

Q236

AnalystPrep Q.3919

REBUILT — Original stem was garbled (debts to one bank, services owed next morning); rewrote so the source and use are unambiguous.

XYZ Bank borrows federal funds in the afternoon to cover a funding shortfall and returns the funds the next morning. The borrowed funds are used to pay a correspondent bank for services it provides. Which of the following correctly identifies the source of intraday liquidity and the use to which XYZ puts the funds?

A. Cash balances; outgoing wire transfers.
B. Incoming funds flow; settlements at PCS systems.
C. Overnight borrowings; funding of nostro accounts.
D. Liquid assets; outgoing wire transfers.

Answer: C

Step by step

1. Fed funds borrowed now and repaid the next morning = overnight borrowing (source).
2. Overnight borrowings differ from intraday credit because they are repaid only the next day.
3. Payment to a correspondent for services = funding of nostro accounts (use).

Why C is correct: Funds borrowed overnight are a source; cash transferred to a correspondent for its services is nostro funding.

Why the others are wrong

- **A:** Cash balances are opening balances, not borrowed funds; and the payment is to a correspondent, not a wire.
- **B:** Incoming funds flow comes from receipts and FMU settlements, not borrowing; PCS settlement is not a payment for services.
- **D:** Liquid assets are near-cash holdings sold or repoed, not borrowed funds; use is again mis-identified.

Q237

BT answers P2.T8.20.8.1 · BT notes P2.T8.20.8.1

REBUILT — Merged two identical items and corrected the typo 'Income funds flow' to 'Incoming funds flow'.

Which of the following is a USE of intraday liquidity?

A. Incoming funds flow.
B. Term repo (as the repo seller).
C. Funding of nostro accounts at a correspondent bank outside the home market.
D. Intraday credit (e.g., Federal Reserve unsecured committed line of credit).

Answer: C

Step by step

1. Uses: outgoing wires, PCS settlements, nostro funding, collateral pledging, asset purchases/funding.
2. Sources: cash balances, incoming funds flow, intraday credit, liquid assets, overnight borrowings, other term funding.
3. Only nostro funding is a use.

Why C is correct: Funding nostro accounts is a cash outflow to correspondents and is a listed use; the other three are sources.

Why the others are wrong

- **A:** Incoming funds flow is the largest source of intraday liquidity.
- **B:** Selling via term repo raises cash, i.e. other term funding, a source.
- **D:** Intraday credit lines such as the Fed net debit cap are a source.

Q238

BT answers P2.T8.25.3.1

REBUILT — Replaced option D (uncommitted lines, defensibly weak) with a clearly effective measure so only one option is least effective.

You are the treasurer of a large bank during an intraday liquidity crunch: a counterparty defaults, delaying expected inflows; the central bank's RTGS system has a temporary outage; large client wire transfers have used much of your liquidity; and most high-quality liquid assets are already pledged as collateral. Which of the following actions would be LEAST effective in mitigating intraday liquidity stress?

A. Prioritise critical payments and delay non-essential outgoing transfers while monitoring liquidity in real time.
B. Sell long-term illiquid assets immediately at market prices to raise cash for same-day transactions.
C. Borrow federal funds or other overnight funding in the interbank market to cover the shortfall, returning it the next morning.
D. Request intraday credit from the central bank or correspondent banks to bridge the shortfall.

Answer: B

Step by step

1. Same-day tools: control outgoing payments (A), overnight borrowing (C), intraday credit (D).
2. Illiquid assets cannot be sold quickly; fire-sale prices and market signalling make it costly and unreliable intraday.
3. B is least effective.

Why B is correct: Illiquid assets are not near-cash; unlike payment throttling, borrowing or intraday credit, they cannot reliably convert to same-day cash.

Why the others are wrong

- **A:** Effective: outgoing payments can be delayed until receipts arrive.
- **C:** Effective: overnight borrowings are a source of intraday funds, repaid next morning.
- **D:** Effective: intraday credit is a standard source for temporary overdrafts.

Q239 GARP PE1 Q50

A junior treasury analyst at a US-based regional bank evaluates the bank's intraday transactions and their effect on its liquidity profile. Which of the following observations would be correct?

A. Intraday overdrafts are prohibited by regulators, but the bank can borrow from the Federal Reserve before the opening of business to resolve expected liquidity needs.

B. The bank's clients can serve as a source of intraday liquidity, which allows the bank to convert liquid assets such as US Treasury bills to cash.

C. The balance of money market assets in the bank's investment portfolio tends to be volatile and is therefore difficult to forecast on any given day.

D. Funding of nostro accounts at correspondent banks is typically initiated by clients entering into loans or repo transactions.

Answer: B

Step by step

1. Liquid assets can be converted to cash with clients, e.g. a repo, so clients can be a source.
2. Intraday overdrafts/credit are permitted (Fed net debit cap).
3. Liquid-asset balances have low volatility and are predictable.
4. Nostro funding is not directly initiated by clients.

Why B is correct: A repo with a client turns Treasury bills into same-day cash, so clients can supply intraday liquidity.

Why the others are wrong

- **A:** Intraday credit and daylight overdrafts are permitted (Fed charges interest on its line).
- **C:** Liquid asset balances have low volatility and are predictable.
- **D:** Nostro funding is initiated by the bank, not directly by clients.

Q240 GARP PE2 Q22

A junior treasury analyst at a large US bank assesses how the bank manages intraday liquidity. In a scenario the bank has an intraday overdraft of its account at a financial market utility (FMU). Which of the following correctly describes a cause of an intraday overdraft and the bank's most likely response?

A. The bank pledges cash collateral against a purchase of OTC securities, which leads it to borrow intraday funds from the US Federal Reserve to eliminate the overdraft.

B. Customers transfer funds to the bank in advance of an expected purchase of real estate, which leads the bank to deposit funds into a special account to eliminate the overdraft.

C. Interest rates fall while the bank has a positive duration gap, which leads the bank to sell liquid assets to eliminate the overdraft.

D. The FMU increases the bank's net debit cap, which triggers an overdraft and leads the bank to initiate a reverse repurchase agreement to provide the necessary intraday liquidity.

Answer: A

Step by step

1. Collateral pledging is a use of intraday liquidity (cash outflow) that can cause an overdraft.
2. Intraday credit from the central bank is a source that can cover it.
3. B is an inflow; C is an interest-rate event; D reverses the role of the cap.

Why A is correct: Pledging collateral is an outflow that can create an overdraft, and intraday central bank credit is a standard source to cover it.

Why the others are wrong

- **B:** Incoming customer funds are a source, so they would reduce, not cause, an overdraft.
- **C:** Falling rates with a positive duration gap is interest-rate risk, not an intraday cash outflow.
- **D:** A higher cap permits more overdraft; it does not trigger one, and a reverse repo would use cash rather than provide it.

Q241 Schweser Module Quiz 71.1 Q1

A new bank treasurer is attempting to obtain a better understanding of the bank's daily cash inflows and outflows. Which of the following transactions is the most significant use of intraday liquidity?

A. Asset purchases/funding.

B. Funding of nostro accounts.
C. Outgoing wire transfers.
D. Settlements at payment, clearing, and settlement (PCS) systems.

Answer: C
Step by step
1. Uses of intraday liquidity: outgoing wires, PCS settlements, nostro funding, collateral pledging, asset purchases/funding.
2. Outgoing wire transfers (for the bank itself or its clients) are the largest and most continuous outflow, paid steadily through the day.
3. Therefore outgoing wire transfers are the most significant use.

Why C is correct: Outgoing wires occur throughout the day in large volume and must be managed against the intraday credit limit, making them the most significant use.
Why the others are wrong
- **A:** Asset purchases (securities, fixed assets, client loans) are a genuine use but smaller and more forecastable; confuses 'a use' with 'the most significant use'.
- **B:** Nostro funding gives a net inflow or outflow depending on activity and is typically smaller than total wire payments.
- **D:** PCS systems often settle once daily, near day end, on a net basis, so the flow is smaller and less continuous than wires.

Q242 Schweser Module Quiz 71.1 Q2

A bank treasurer is attempting to minimize liquidity funding costs. Which of the following sources of liquidity is most likely to have the greatest explicit financial cost?
A. Cash balances.
B. Incoming funds flow.
C. Intraday credit.
D. Liquid assets.

Answer: C
Step by step
1. Cash balances and liquid assets are owned by the bank: only implicit opportunity cost, no explicit charge.
2. Incoming funds flow is received from counterparties and FMUs: no explicit cost.
3. Intraday credit from the central bank may carry explicit interest or require high-quality collateral to be pledged, so it has the greatest explicit cost.

Why C is correct: Intraday credit is borrowed, so the central bank may charge interest or require collateral, an explicit cost that owned or received funds do not carry.
Why the others are wrong
- **A:** Cash balances are the bank's own funds; the cost is only the forgone return, not explicit.
- **B:** Incoming payments are a free source of liquidity, though timing is uncertain.
- **D:** Liquid assets are owned; holding them has opportunity cost only, so confusing implicit with explicit cost leads here.

Q243 Schweser Module Quiz 71.1 Q3

Which of the following factors is least likely to increase a bank treasurer's challenge in managing cash in order to stay within its intraday credit limits?
A. Cash flow volatility.
B. Credit quality of assets.
C. Insufficient real-time data.
D. Securities price volatility.

Answer: B
Step by step
1. The reading names three key problems: volatile cash flows, macroeconomic factors (securities price volatility causing margin calls, monetary policy), and lack of real-time balances.

2. A, C and D map directly to these problems.
3. Credit quality of the bank's liquid assets is high and does not add to the intraday cash-management challenge, so B is least likely.

Why B is correct: Volatile flows, price-driven margin calls and missing real-time balances all complicate staying within credit limits; high credit quality of liquid assets does not.

Why the others are wrong

- **A:** Sudden inflows/outflows make the intraday overdraft hard to control; it is a stated challenge.
- **C:** Without real-time balances the treasurer cannot see current cash or remaining needs; it is a stated challenge.
- **D:** Price moves trigger margin calls and collateral outflows; it is a stated challenge.

LO 71.b

Discuss the governance structure of intraday liquidity risk management.

Q244

AnalystPrep Q.3910

REBUILT — Original keyed 'All of the above' although the reading lists four key areas (the item omitted risk measurement and monitoring), making it a giveaway; rebuilt as a NOT-one-of-the-four-areas item.

A strong governance structure for managing intraday liquidity risk covers four key areas. Which of the following is NOT one of those four key areas?

A. Active risk management
B. Integration with risk governance
C. Risk measurement and monitoring
D. Elimination of all intraday credit extended to customers

Answer: D

Step by step

1. The reading lists four key areas: (1) active risk management, (2) integration with risk governance, (3) risk assessment, (4) risk measurement and monitoring.
2. A, B and C match items (1), (2) and (4) respectively.
3. Eliminating all intraday credit is not a governance area; the framework aims to measure and monitor intraday credit provided and used, not to abolish it.

Why D is correct: Governance rests on active management, integration with the three lines of defense, risk assessment and measurement/monitoring; banning intraday credit is not part of that structure.

Why the others are wrong

- **A:** It is one of the four areas (proactive approach, e.g. settlement and systemic risks recognized in risk appetite); wrongly treated as not belonging.
- **B:** It is one of the four areas (duties embedded in the three lines of defense).
- **C:** It is one of the four areas (monitoring intraday credit provided to customers and used by the firm); easily overlooked because many candidates recall only three.

Q245

BT Mock Q39 · BT answers P2.T8.20.8.2 · BT notes P2.T8.20.8.2

Venkat explains that all risk management frameworks start with a governance structure defining the roles and responsibilities of bank employees and committees in overseeing risk-related activities, and that effective governance includes oversight of intraday liquidity risk. Regarding the governance structure of intraday liquidity risk management, which of the following statements is TRUE?

A. Intraday liquidity risk should be accepted as a cost of doing business.
B. Roles and responsibilities should be defined within the eight lines (two times four) of defense model.
C. Payment and settlement systems should only be a source of funds; if they become a use of funds, a yellow flag should be triggered.
D. Intraday liquidity risk should be incorporated in the risk taxonomy and is a component of risk self-assessment, including settlement risks.

Answer: D

Step by step

1. Active risk management means intraday liquidity risk is not treated as a passive given; settlement and systemic risks are explicitly recognized and classified in the risk appetite.
2. Risk assessment focuses on settlement risks: identification, measurement and evaluation, with internal-control analysis by business managers and risk management.
3. Duties are integrated into three lines of defense (treasury, corporate risk management, internal audit).
4. Only D is consistent with these points.

Why D is correct: Governance requires proactive treatment: intraday risk is classified in the risk framework and assessed, including settlement risk, rather than accepted passively.

Why the others are wrong

- **A:** Reflects the passive view the reading rejects; a proactive approach is desirable.
- **B:** Invents an 'eight lines' model; the framework uses three lines of defense (2 x 4 is a fabricated distractor).
- **C:** Payment systems are both sources and uses of intraday liquidity; the one-way rule and 'yellow flag' are not part of the reading.

Q246

BT answers P2.T8.25.3.2

A large international bank is experiencing a surge in outgoing payments due to heightened market volatility. The bank's three lines of defense are structured as follows:

Line of Defense	Department	Responsibility
1st	Treasury	Manages intraday funding and ensures liquidity is available for payments.
2nd	Corporate Risk Management	Establishes policies and monitors intraday credit exposure.
3rd	Internal Audit	Independently assesses the effectiveness of the bank's risk management framework.

During a sudden liquidity squeeze, the bank's net debit cap at the central bank is nearly exceeded and daylight overdrafts are rising. Treasury considers delaying some payments, but corporate risk management warns that delayed settlements could create reputational and counterparty risks. In this scenario, which department should ensure that any deviations from standard liquidity procedures are within risk limits and approved by senior management?

A. Treasury Department
B. Corporate Risk Management
C. Internal Audit
D. Treasury Department and Internal Audit

Answer: B

Step by step

1. Identify the function: oversight of policy compliance, limits and approvals is a second-line role.
2. The second line is corporate risk management, which establishes policies, monitors intraday credit exposure and is the focus of the governance framework.
3. Treasury (first line) executes funding and payment prioritization; Internal Audit (third line) reviews independently after the fact.
4. Answer: corporate risk management.

Why B is correct: Corporate risk management is the second line of defense that polices limits and policy deviations and escalates to senior management.

Why the others are wrong

- **A:** Treasury is the first line that manages funding; it proposes the deviation and cannot also police it.
- **C:** Internal audit is independent and reviews ex post; it does not approve real-time deviations.
- **D:** Combines the two lines that are not responsible for limit oversight, omitting the second line.

Q247

Schweser Module Quiz 71.2 Q1

In the context of characteristics of an effective governance structure in controlling intraday liquidity within a bank, there is emphasis on expertise in which line of defense?

A. Treasury.
B. Internal audit.
C. Information technology.
D. Corporate risk management.

Answer: D

Step by step

1. The framework adopts the standard three lines of defense: treasury, corporate risk management, internal audit.
2. Emphasis on expertise is placed on the second line, corporate risk management.
3. Information technology is not a line of defense.

Why D is correct: The reading places the emphasis on expertise in the second line, corporate risk management.

Why the others are wrong

- **A:** Treasury is the first line, but the emphasis is not on it.
- **B:** Internal audit is the third line, not the focus.
- **C:** IT is not a line of defense; it only sits on oversight committees alongside treasury.

LO 71.c

Differentiate between methods for tracking intraday flows and monitoring risk levels.

Q248

AnalystPrep Q.3846

REBUILT — Original EXCEPT item mixed metrics with a made-up 'daily minimum liquidity requirement' giveaway and a misworded option; rebuilt as a clean flow-vs-risk classification item.

Which of the following is NOT classified as a measure for quantifying and monitoring intraday risk levels?

A. Client intraday credit usage
B. Payment throughput
C. Time-sensitive obligations
D. Intraday credit relative to Tier 1 capital

Answer: C

Step by step

1. The risk-level measures are: daily maximum intraday liquidity usage, intraday credit relative to Tier 1 capital, client intraday credit usage and payment throughput.
2. Time-sensitive obligations (payments that must settle at a specific time, e.g., margin calls) are one of the measures for understanding intraday flows.
3. Therefore C is the only item outside the risk-level group.

Why C is correct: Time-sensitive obligations are tracked to understand intraday flows (amounts and deadlines), not to quantify risk levels.

Why the others are wrong

- **A:** Is a risk-level measure (client's peak overdraft relative to its credit line); wrongly excluded by confusing it with credit-line flow data.
- **B:** Is a risk-level measure (outgoing payments relative to time of day); excluded by mistaking it for a payment-flow statistic.
- **D:** Is a risk-level measure (available unsecured intraday credit to Tier 1 capital, a proxy for systemic risk); excluded by mistaking it for a capital ratio unrelated to intraday risk.

Q249

AnalystPrep Q.3911

REBUILT — Stem was vague ('measures they think are accurate') and the explanation weak; reworded to ask for the pair made up only of risk-level measures.

Which of the following pairs consists ONLY of measures for quantifying and monitoring intraday risk levels?

A. Daily maximum intraday liquidity usage and client intraday credit usage
B. Client intraday credit usage and total bank intraday credit lines available and usage
C. Total bank intraday credit lines available and usage and total intraday credit lines to clients and counterparties
D. Time-sensitive obligations and payment throughput

Answer: A

Step by step

1. Risk-level measures: daily maximum intraday liquidity usage, intraday credit relative to Tier 1 capital, client intraday credit usage, payment throughput.
2. A: both items are on this list.
3. B: total bank credit lines available and usage is a flow-tracking measure.
4. C: both items are flow-tracking measures.

5. D: time-sensitive obligations is a flow-tracking measure.

Why A is correct: Only A pairs two items from the risk-level group.

Why the others are wrong

- **B:** Pairs one risk-level measure with a flow-tracking measure (total bank credit lines available and usage).
- **C:** Both items are flow-tracking measures about credit lines, not risk-level ratios.
- **D:** Payment throughput is a risk-level measure, but time-sensitive obligations is a flow measure.

Q250

AnalystPrep Q.3913

REBUILT — Options mixed items (credit status, past credit position) that are not defined in the reading, making the keyed pair debatable; rebuilt as an 'except' on the data stored per payment.

For total payments, a bank stores information on each payment to understand intraday flows. Which of the following is NOT one of the items stored for each payment?

A. Payment amount and time received or originated
B. Routing information and payment system used
C. Payer and payee
D. The counterparty's long-term credit rating

Answer: D

Step by step

1. The per-payment data warehouse record includes amount, time received/originated, times for each processing step, routing information, payer and payee, payment system used and any suspensions.
2. A, B and C are all on that list.
3. A credit rating of the counterparty is not a payment attribute, so D is the exception.

Why D is correct: Payment-level data describe the payment and its processing, not the counterparty's credit quality.

Why the others are wrong

- **A:** Amount and time are core payment-level data.
- **B:** Routing and payment system are stored for each payment.
- **C:** Payer and payee are stored for each payment.

Q251

AnalystPrep Q.3914

A bank keeps, for each electronic payment, the payer and payee, payment system used, any suspensions, time received or originated, and routing information. Under which measure for understanding intraday flows do these statistics most accurately fall?

A. Other cash transactions
B. Total payments
C. Settlement positions
D. Total intraday credit lines to clients and counterparties

Answer: B

Step by step

1. Payment-level details (payer, payee, system, suspensions, timing, routing) are the data stored under total payments.
2. They allow summaries such as payments sent and received by client type and net settlement-account position through the day.

Why B is correct: These are the per-payment records that define the total payments measure.

Why the others are wrong

- **A:** Other cash transactions covers intraday and end-of-day FMU settlement positions and collateral snapshots, not payment-level records.
- **C:** Settlement positions are deadline-specific FMU positions, not individual payment attributes.
- **D:** Credit lines to clients concern potential and actual credit extended, not payment details.

Q252 AnalystPrep Q.3915

Which of the following measures is derived by comparing a client's peak daily intraday overdraft to its established credit line?

A. Payment throughput
B. Daily maximum intraday liquidity usage
C. Total bank intraday credit lines available and usage
D. Client intraday credit usage

Answer: D

Step by step

1. Client intraday credit usage = client's peak daily intraday overdraft relative to its committed or uncommitted credit line.
2. Aggregated, it shows the liquidity needed to support client activity.

Why D is correct: The definition of client intraday credit usage is peak overdraft divided by the credit line.

Why the others are wrong

- **A:** Payment throughput measures outgoing payments as a percentage of total payments over the day.
- **B:** Daily maximum usage uses the bank's own largest net negative balance relative to its credit line, not a client's.
- **C:** Total bank credit lines compares normal intraday credit use with maximum credit available; it is a flow measure.

Q253 AnalystPrep Q.3916

REBUILT — Stem was ambiguous with 'settlement positions' (source key A was correct but disputed); added the securities-settlement-network collateral snapshot detail that uniquely identifies other cash transactions.

A bank tracks its intraday and end-of-day settlement positions at every financial market utility (FMU) in which it participates, captures as much transaction-level detail as possible, and for securities settlement networks takes snapshots of its account and collateral positions through the day. Which measure for understanding intraday flows does this describe?

A. Total payments
B. Total bank intraday credit lines available and usage
C. Other cash transactions
D. Settlement positions

Answer: C

Key correction: Source key (AnalystPrep) said A (other cash transactions) in its option order; this is correct, and the item was reworded to remove the ambiguity with settlement positions.

Step by step

1. Tracking intraday and end-of-day positions at all FMUs, maximizing transaction-level detail, and taking collateral snapshots at securities settlement networks is the definition of other cash transactions.
2. Settlement positions is the narrower fallback: minimum data on deadline-specific FMU positions if full account reconstruction is unavailable.

Why C is correct: These activities (position tracking at all FMUs plus SSN account and collateral snapshots) are the described content of other cash transactions.

Why the others are wrong

- **A:** Total payments is payment-level data on electronic payments, not FMU position and collateral snapshots.
- **B:** Concerns the amount of intraday credit used versus available, not FMU positions.
- **D:** Settlement positions is the minimum fallback of deadline-specific FMU data when complete account reconstruction is unavailable; it lacks the full-detail and collateral-snapshot focus.

Q254 AnalystPrep Q.3917

REBUILT — Original asked for a meaningless category name with no substantive content; rebuilt as a correct classification item.

Which of the following correctly classifies a measure as one for understanding intraday flows?

A. Intraday credit relative to Tier 1 capital
B. Other cash transactions
C. Daily maximum intraday liquidity usage
D. Payment throughput

Answer: B

Step by step

1. Flow-understanding measures: total payments, other cash transactions, settlement positions, time-sensitive obligations, total credit lines to clients and counterparties, total bank credit lines available and usage.
2. Risk-level measures: daily maximum intraday liquidity usage, credit relative to Tier 1 capital, client credit usage, payment throughput.
3. Only 'other cash transactions' is in the flow group.

Why B is correct: Other cash transactions (FMU position and collateral tracking) is a flow-tracking measure; the other three are risk-level measures.

Why the others are wrong

- **A:** Risk-level measure (systemic risk indicator).
- **C:** Risk-level measure (largest negative balance relative to the credit line).
- **D:** Risk-level measure (share of outgoing payments by time of day); easily confused with payment flow data.

Q255 AnalystPrep Q.3918

ABC Bank is a direct participant in an FMU and actively measures the flow of its outgoing payments relative to total payments or time markers (payment throughput). Which of the following is NOT a reason for doing so?

A. To track volume patterns and help ensure that all of the day's payments are processed on time
B. To meet FMU requirements for submitting a target percentage of payments by a deadline
C. To identify peak periods over time and their correlation with intraday liquidity on hand and intraday credit usage
D. To determine the best means of acquiring intraday liquidity

Answer: D

Step by step

1. Payment throughput is tracked for three reasons: timely processing, FMU target-percentage requirements, and identifying peak periods and their link to liquidity and credit usage.
2. A, B and C correspond to these three reasons.
3. Choosing the best funding source for intraday liquidity is not among them.

Why D is correct: The three stated purposes of payment throughput are timeliness, FMU compliance and peak-period/liquidity correlation.

Why the others are wrong

- **A:** Is the first stated reason (timely processing).
- **B:** Is the second stated reason (FMU target percentage by deadline).
- **C:** Is the third stated reason (peak periods vs liquidity and credit use).

Q256 BT answers P2.T8.20.8.3 · BT notes P2.T8.20.8.3

Each of the following is a measure for quantifying and monitoring risk levels EXCEPT one, which is a measure for understanding intraday flows. Which one?

A. Client intraday credit usage
B. Intraday credit relative to Tier 1 capital
C. Total payments
D. Daily maximum intraday liquidity usage

Answer: C

Step by step

1. Risk-level measures: daily maximum intraday liquidity usage, intraday credit relative to Tier 1 capital, client intraday credit usage, payment throughput.
2. Total payments is a flow-tracking measure (payment amounts, timing, net settlement-account position).

Why C is correct: Total payments is the only listed item that belongs to the intraday-flow group.

Why the others are wrong

- **A:** Risk-level measure (client peak overdraft to credit line).
- **B:** Risk-level measure (unsecured available intraday credit to Tier 1 capital).
- **D:** Risk-level measure (largest negative balance to credit line).

Q257

BT answers P2.T8.25.3.3

REBUILT — Original was a vague 'best approach' item with giveaway distractors and a mismatched explanation; rebuilt to test the Schweser distinction between a risk-monitoring metric (payment throughput) and tracking metrics.

During a market-wide liquidity crunch, a bank's incoming payments are delayed and it is concerned that its own outgoing payments may be sent late in the day, risking breach of a financial market utility's (FMU) requirement to settle a minimum share of payments by set times. Which risk-level monitoring metric is most directly designed to address this concern?

A. Payment throughput, the proportion of outgoing payments made and when they occur during the day.
B. Intraday credit relative to Tier 1 capital.
C. Client intraday credit usage.
D. Total intraday credit lines extended to clients and counterparties.

Answer: A

Step by step

1. The concern is the timing of outgoing payments and compliance with an FMU's timing stipulation.
2. Schweser: payment throughput metrics look at the proportion of outgoing payments and when they occur in the day, to monitor payment patterns, ensure no payments are late and confirm compliance with the FMU's requirement.
3. The other three metrics measure credit exposure or capital-relative credit, not payment timing.

Why A is correct: Payment throughput tracks what share of outgoing payments is completed by what time of day, which directly shows whether timing obligations to the FMU are being met.

Why the others are wrong

- **B:** This ratio of unsecured and available intraday credit to Tier 1 capital gauges intraday settlement (systemic) risk, not payment timing.
- **C:** This is the clients' highest intraday borrowing relative to their limits; it shows liquidity provided to clients, not the bank's own payment timing.
- **D:** This is a flow-tracking measure of potential credit exposure to clients and counterparties, unrelated to when payments are released.

Q258

Mock 2 Q24

A treasury manager at a large clearing bank is establishing indicators to monitor the bank's intraday liquidity position. The bank settles a high volume of payments through a real-time gross settlement system and relies on incoming payments to fund a portion of its outgoing obligations. Which of the following best describes an appropriate intraday liquidity monitoring indicator?

A. The bank's end-of-day net settlement position across all payment systems.
B. The bank's daily maximum intraday liquidity usage, measured as the largest net cumulative negative position during the day relative to the credit line.
C. The bank's liquidity coverage ratio calculated at the close of each business day.
D. The proportion of the bank's total assets that qualify as Level 1 high-quality liquid assets.

Answer: B

Step by step

1. Intraday risk arises from the path of balances during the day, so the metric must capture the intra-day extreme.
2. Schweser's daily maximum intraday liquidity usage takes the most negative balance of the day (relative to the credit line limit), viewed across all cash accounts.
3. Only option B captures that peak; the others are end-of-day or stock measures.

Why B is correct: Daily maximum intraday liquidity usage measures the peak funding need during the day, which is what intraday risk monitoring requires.

Why the others are wrong

- **A:** End-of-day net position can be flat even if the bank was deeply overdrawn mid-day, so it hides intraday exposure.
- **C:** The LCR is a 30-day stress measure calculated at a point in time and does not capture intraday peaks.
- **D:** The Level 1 HQLA share is a balance-sheet composition stock measure, not a measure of intraday cash-flow usage.

Q259 Schweser Module Quiz 71.2 Q2

Which of the following metrics is most appropriate in determining a bank's systemic risk?

A. Client intraday credit usage.
B. Intraday credit relative to Tier 1 capital.
C. Daily maximum intraday liquidity usage.
D. Total intraday credit lines to clients and counterparties.

Answer: B

Step by step

1. Schweser: the ratio of unsecured and available (not used) intraday credit to Tier 1 capital indicates the bank's intraday settlement risk, i.e. the systemic risk it contributes.
2. Client usage, maximum usage and client credit lines describe client borrowing, the bank's own peak usage and exposure, not systemic risk relative to capital.

Why B is correct: Comparing unsecured, available intraday credit to Tier 1 capital shows how large the bank's intraday settlement exposure is relative to its loss-absorbing capital, the systemic-risk gauge.

Why the others are wrong

- **A:** Highest client intraday borrowing relative to limits shows how much liquidity must be provided to clients, not systemic risk.
- **C:** Peak negative balance relative to the credit limit measures the bank's own liquidity usage, not its contribution to systemic risk.
- **D:** Total credit lines to clients and counterparties is a flow-tracking metric of maximum exposure, not scaled to capital.

Reading 72. Monitoring Liquidity

Antonio Castagna and Francesco Fede, Measuring and Managing Liquidity Risk (United Kingdom, John Wiley & Sons, 2013). Chapter 6.

LO	Learning objective	Qs
72.a	differentiate between deterministic and stochastic cash flows and provide examples of each.	5
72.b	describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process.	3
72.c	describe and apply the concepts of liquidity risk, funding cost risk, liquidity generation capacity, expected liquidity, and cash flow at risk.	5
72.d	interpret the term structure of expected cash flows and cumulative cash flows.	6
72.e	discuss the impact of available asset transactions on cash flows and liquidity generation capacity.	5

LO 72.a

Differentiate between deterministic and stochastic cash flows and provide examples of each.

Q260

AnalystPrep Q.3920

In monitoring liquidity, cash flows can be classified along two dimensions. A group of FRM students were asked to state those two dimensions, and four groups responded as follows. Group A: time and amount. Group B: time and credit scores. Group C: credit potential and amount. Group D: term structure and time. Which group stated the correct dimensions?

A. Group A
B. Group B
C. Group C
D. Group D

Answer: A

Step by step

1. Schweser: cash flows at a point in time are identified on two dimensions, time and amount.
2. In each dimension a cash flow is either deterministic (known) or stochastic (random).
3. Credit scores, credit potential and term structure are not classification dimensions.
4. Only Group A names both correct dimensions.

Why A is correct: Cash flows are classified by when they occur (time) and how large they are (amount), each being deterministic or stochastic.

Why the others are wrong

- **B:** Time is correct but credit scores are not a dimension; credit is only one source of stochastic amount.
- **C:** Amount is correct but 'credit potential' is not a dimension; it confuses a driver of stochastic amount with a classification axis.
- **D:** Time is correct but term structure describes the maturity profile of cash flows, not a classification perspective.

Q261

AnalystPrep Q.3921

Which of the following statements accurately describes deterministic cash flows?

A. They occur at random future instants, and their amount cannot be fully determined.
B. They occur at predictable future instants and in an amount known with certainty.
C. Their amount cannot be predicted, but they occur at predictable future instants.
D. They are unpredictable in both amount and timing.

Answer: B

Step by step

1. Deterministic means known: both timing and amount are known in advance.
2. A and D describe cash flows stochastic in time and amount.
3. C describes known timing with unknown amount, which is stochastic in amount only.

Why B is correct: A cash flow is deterministic only when both its timing and its amount are known with certainty.

Why the others are wrong

- **A:** Describes cash flows that are stochastic in both timing and amount.
- **C:** Mixes dimensions: known timing but unknown amount is stochastic in amount (e.g., a floating coupon).
- **D:** Describes fully stochastic cash flows (e.g., recovery on a default).

Q262

AnalystPrep Q.3923

A tutor describes a set of cash flows as follows: (a) they relate to contracts such as fixed-rate bonds, fixed-rate mortgages and loans, and bonds issued or loans received held in the bank's liabilities; (b) they result from periodic interest payments and periodic capital repayments if the asset is amortizing. Which type of cash flow is described?

A. Deterministic cash flows
B. Behavioral cash flows
C. Stochastic cash flows
D. Credit-related cash flows

Answer: A

Step by step

1. Fixed-rate contracts carry contractual schedules of interest and principal.
2. Timing and amount are therefore known at the reference date.
3. Known timing and known amount means deterministic.

Why A is correct: Contractual fixed-rate payment schedules fix both date and amount in advance, which is the definition of deterministic cash flows.

Why the others are wrong

- **B:** Behavioral flows depend on client decisions (prepayments, withdrawals), not on contractual schedules.
- **C:** Stochastic flows have uncertain timing or amount; fixed-rate schedules have neither.
- **D:** Credit-related flows arise from credit events such as default or recovery, not scheduled payments.

Q263

BT answers P2.T8.25.4.1

REBUILT — OCR-damaged dollar figures and an inconsistent withdrawal range (75 to 30 around an average of 20) were replaced with consistent values; answer unchanged.

A commercial bank classifies its cash flows over the next year to assess potential liquidity shortfalls:
1. Fixed-rate bond coupon payments: $10 million due every six months.
2. Floating-rate loan interest receipts: about $5 million per quarter, with the rate indexed to 3-month SOFR.
3. Customer deposit withdrawals: on average $20 million per month, but fluctuating between $15 million and $30 million depending on economic conditions.
4. Committed corporate credit lines of $100 million, with an estimated drawdown of 25% to 50% in periods of economic stress.

Which of the above cash flows is deterministic?

A. 1
B. 2
C. 3
D. 4

Answer: A

Step by step

1. Item 1: fixed coupon, fixed dates; amount and timing both known, so deterministic.
2. Item 2: the SOFR index makes the amount uncertain; indexed/contingent, stochastic in amount.
3. Item 3: depends on depositor behavior and the economy; behavioral, stochastic.
4. Item 4: drawdowns are client decisions driven by stress; behavioral, stochastic in timing and amount.

Why A is correct: Only the fixed-rate coupon has contractually known timing and amount; the other three depend on market indices or client behavior.

Why the others are wrong

- **B:** Timing is known but the amount floats with SOFR, so it is stochastic in amount (indexed).
- **C:** Withdrawals vary in size and timing with customer behavior and conditions; behavioral and stochastic.
- **D:** Drawdown of a committed line is a client-exercised liquidity option; amount (25% to 50%) and timing are uncertain.

Q264 Schweser Module Quiz 72.1 Q1

Each of the following cash flows is stochastic in both timing and amount EXCEPT:

A. new debt issued by a bank.
B. a bank's recovery on a client default.
C. the payout of an American-style call option.
D. mortgage prepayments due to interest rate declines.

Answer: A

Step by step

1. Test each flow on both dimensions.
2. Recovery on default: date and amount both uncertain, so stochastic in both.
3. American option payout: exercise date and payoff both uncertain; mortgage prepayment depends on rates and borrower choice, so stochastic in both.
4. New debt issuance is planned for a known date (time deterministic) but its amount depends on new business (amount stochastic), so it is the exception.

Why A is correct: New bond issuance is scheduled, so its timing is deterministic even though the amount is stochastic; the others are stochastic in both dimensions.

Why the others are wrong

- **B:** Both timing and size of a recovery are uncertain, so it is stochastic in both and not the exception.
- **C:** American-style exercise can occur any time and the payoff depends on the market, so stochastic in both.
- **D:** Prepayment timing and amount depend on rates and borrower behavior, so stochastic in both.

LO 72.b

Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process.

Q265 AnalystPrep Q.3924

A bank's liquidity options (for example, credit lines and demand deposits) are exercised by customers. Which of the following correctly describes the two primary effects that the exercise of a liquidity option has on the bank?

A. An impact on the balance sheet from the amount repaid or withdrawn, and a (positive or negative) financial impact.
B. A (positive or negative) financial impact and an effect on the contract deposit rate.
C. An effect on the contract deposit rate and an impact on the balance sheet.
D. An effect on available cash reserves and a financial impact.

Answer: A

Step by step

1. Exercise of a liquidity option means the holder gives cash to, or receives cash from, the bank.
2. Effect 1: the balance sheet changes by the amount repaid or withdrawn.
3. Effect 2: a positive or negative financial impact, driven by the spread between the original contract terms (rate, spread, market values) and those at exercise.
4. Only option A lists both effects.

Why A is correct: Exercise changes the balance sheet by the amount moved and creates a gain or loss from the difference between contract terms and current market terms.

Why the others are wrong

- **B:** Omits the balance sheet effect; the contract deposit rate is a contract term, not an effect of exercise.
- **C:** Omits the financial impact and cites the contract deposit rate, which is an input to the financial impact, not an effect of exercise in itself.
- **D:** Cash reserves are a tool used to hedge liquidity impacts, not a primary effect of exercise; the balance sheet effect is missing.

Q266

Schweser Module Quiz 72.1 Q2

REBUILT — Original option B (financial options align exercise with profitability) is also a true statement per the reading, giving two defensible answers; B was reworded into a false statement about liquidity options.

Which of the following statements comparing liquidity options to standard financial options is most accurate?

A. Financial options typically produce negative cash flows after exercise.
B. Liquidity options are exercised only when exercise is financially profitable to the holder.
C. Both types of option are typically exercised only when they are in the money.
D. Liquidity options may or may not be financially profitable upon exercise.

Answer: D

Step by step

1. Standard financial options are exercised when profitable; post-exercise cash flows are often positive.
2. Liquidity options are exercised because the holder needs (or wants to supply) liquidity; the decision depends on cash flows after exercise, which may be independent of financial profitability.
3. Hence exercise of a liquidity option may or may not be profitable: D.

Why D is correct: Liquidity options are exercised to meet a liquidity need, so the exercise decision is not tied to profitability.

Why the others are wrong

- **A:** Reverses the reading: financial options often produce positive cash flows.
- **B:** Applies the financial-option rule to liquidity options; liquidity options are exercised for liquidity needs, even when unprofitable.
- **C:** Only financial options are exercised when in the money; liquidity options are driven by liquidity needs.

Q267

BT answers P2.T8.25.4.2

REBUILT — Source data was OCR-damaged (coupon and credit-line sizes, mortgage inflow range) and asked for a stress net total; rebuilt as a clean incremental-outflow question on liquidity options (sight deposits and committed credit lines) under stress.

A bank's base-case behavioral assumptions are: customer deposit withdrawals of $25 million per month and drawdowns of 30% on $120 million of committed credit lines over the year. Both are liquidity options with stochastic amounts. Under a stress scenario, withdrawals rise to $35 million per month and credit line drawdowns rise to 55%. All other cash flows are unchanged. What is the incremental cash outflow over the 12 months relative to the base case?

A. $150 million
B. $120 million
C. $486 million
D. $186 million

Answer: A

Step by step

1. Extra deposit withdrawals: $(35 - 25) \times 12 = 120$.
2. Extra credit line drawdown: $(0.55 - 0.30) \times 120 = 30$.
3. Incremental outflow $= 120 + 30 = 150$ million.

Why A is correct: Only the change from the base case matters; unchanged flows cancel, leaving the extra withdrawals plus the extra drawdown share.

Why the others are wrong

- **B:** 120 counts only the extra deposit withdrawals and ignores the larger credit line drawdown.
- **C:** 486 is the total stressed outflow (35 x 12 + 0.55 x 120), not the incremental amount over the base case.
- **D:** 186 = 120 + 0.55 x 120 adds the full stressed drawdown instead of only the increase from 30% to 55%.

LO 72.c

Describe and apply the concepts of liquidity risk, funding cost risk, liquidity generation capacity, expected liquidity, and cash flow at risk.

Q268

AnalystPrep Q.2262

REBUILT — Original keyed 'worst outcome is a $15M outflow', which confuses LaR with the minimum cash flow; rebuilt with numbers so LaR is the shortfall relative to the expected cash flow, as in the text.

Bilco Bank measures liquidity at risk (LaR) at a 95% confidence level over a 1-day horizon. Its expected net cash flow for tomorrow is an inflow of USD 10 million, and the 5th percentile of its net cash flow distribution is an outflow of USD 5 million. Which of the following is the correct LaR and interpretation?

A. LaR is USD 15 million: with 95% confidence, tomorrow's net cash flow will not fall more than USD 15 million below its expected level.

B. LaR is USD 5 million: with 95% confidence, the bank's worst net cash outflow tomorrow is USD 5 million.

C. LaR is USD 10 million: with 95% confidence, the bank's net cash flow tomorrow will be at least USD 10 million.

D. LaR is USD 15 million: with 95% confidence, the bank's worst net cash outflow tomorrow is USD 15 million.

Answer: A

Key correction: Source key (AnalystPrep) said D (worst outcome is a 15M outflow); LaR is defined relative to expected cash flow (expected minus minimum), so that reading is incorrect.

Step by step

1. LaR is the unexpected cash flow: the difference between the expected (average) cash flow and the minimum cash flow at the chosen confidence level.
2. Expected cash flow = +10; minimum cash flow at 95% (5th percentile) = -5.
3. LaR = 10 - (-5) = USD 15 million.
4. Interpretation: with 95% confidence the net cash flow will not be more than USD 15 million below expectation (a shortfall of 15 relative to +10, i.e. an outflow of 5).

Why A is correct: LaR measures the adverse deviation of cash flow from its expected value, so it nets the minimum cash flow against the expected cash flow.

Why the others are wrong

- **B:** Reports the minimum cash flow itself (outflow of 5) and ignores that LaR is measured relative to the expected cash flow.
- **C:** Reports the expected cash flow; it is neither a risk measure nor a lower bound.
- **D:** Correct number but wrong meaning: the worst outflow at 95% is 5, not 15; the 15 is the shortfall relative to the expected inflow of 10.

Q269

BT answers P2.T8.20.9.2 · BT notes P2.T8.20.9.2

Which of the following best describes the term structure of expected liquidity, TSL(e)?

A. TSL(e) is the cumulative change in the term structure of available assets (TSAA)

B. TSL(e) is a combination of the term structures of cash flow at risk (CFaR) and liquidity at risk (LaR)

C. TSL(e) is a combination of the term structure of expected cash (TSEC), change in working capital (CIWC), and change in deposits (CID)

D. TSL(e) is a combination of the term structures of cumulative expected cash flows (TSECCF) and cumulated liquidity generation capacity (TSCLGC)

Answer: D

Step by step

1. TSL(e) shows whether the bank can cover negative cumulated cash flows at each future date.
2. At each date: TSL(e) = TSECCF + TSCLGC.
3. Only option D lists these two building blocks.

Why D is correct: Expected liquidity is the expected cumulated cash flow plus the cumulated liquidity the bank can generate from its sources.

Why the others are wrong

- **A:** TSAA tracks available assets and feeds the LGC, but TSL(e) is not its cumulative change.
- **B:** CFaR and LaR are the unexpected (tail) measures used for the at-risk version of the term structure, not the expected one.
- **C:** These terms are not components of the framework; they are invented distractors.

Q270 BT answers P2.T8.20.9.3 · BT notes P2.T8.20.9.3

Consider the following four definitions related to liquidity risk:
- Liquidity risk: The event that in the future the bank receives smaller than expected amounts of cash flows to meet its payment obligations.
- Funding cost risk: The event that in the future the bank has to pay greater than expected cost (spread) above the risk-free rate to receive funds from sources of liquidity that are available.
- Liquidity generation capacity: The ability of a bank to generate positive cash flows, beyond contractual ones, from the sources of liquidity available in the balance sheet and off the balance sheet at a given date.
- Cash flow at risk (CFaR): The amount of economic losses due to the fact that on a given date the algebraic sum of positive and negative cash flows and of existing cash available at that date is different from some (desired) expected level.

About these definitions, which of the following statements is TRUE?

A. Liquidity risk is inaccurate, but the other three are correct
B. Funding cost risk is inaccurate, but the other three are correct
C. Cash flow at risk (CFaR) is inaccurate, but the other three are correct
D. All four definitions are correct

Answer: C

Step by step

1. Liquidity risk (quantitative): receiving smaller than expected cash flows to meet obligations - correct.
2. Funding cost risk: paying a greater than expected spread over the risk-free rate - correct.
3. LGC: ability to generate positive, non-contractual cash flows from sources of liquidity - correct.
4. The CFaR definition is actually the comprehensive definition of liquidity risk (economic loss when cash flows plus cash differ from expectation). CFaR is the maximum positive (or minimum negative) cash flow at a confidence level, netted with the expected cash flow on that date.

Why C is correct: CFaR is a statistical tail measure of cash flow relative to expectation, not the economic-loss definition of liquidity risk.

Why the others are wrong

- **A:** The liquidity risk definition matches the quantitative dimension.
- **B:** The funding cost risk definition matches the cost-of-liquidity dimension.
- **D:** Ignores that the CFaR statement describes liquidity risk, not CFaR.

Q271 BT answers P2.T8.25.4.3

REBUILT — Source had OCR-damaged options and an inconsistent buffer figure; reconstructed data and options with exactly one correct answer.

A bank assesses whether it can cover cash flow shortfalls over the next year using the term structure of expected liquidity, TSL(e) = TSECCF + TSCLGC. Under normal conditions it expects annual inflows of USD 795 million, outflows of USD 230 million, and has starting cash of USD 50 million. In the stress scenario, inflows fall 25% and outflows rise 40%. The bank can generate an additional USD 100 million of liquidity from its buffer (cumulated LGC). What is the bank's TSL(e) at the end of the year under stress?

A. USD 424.25 million
B. USD 324.25 million
C. USD 715.00 million
D. USD 374.25 million

Answer: A

Step by step

1. Stressed inflows = 795 x 0.75 = 596.25.
2. Stressed outflows = 230 x 1.40 = 322.00.
3. TSECCF = 50 + 596.25 - 322.00 = 324.25.
4. TSL(e) = 324.25 + 100 = USD 424.25 million.

Why A is correct: TSL(e) adds the liquidity the bank can generate (cumulated LGC) to the cumulated expected cash flow, here under stressed flows.

Why the others are wrong

- **B:** Omits the USD 100 million liquidity buffer (TSECCF only).
- **C:** Uses normal-scenario flows (50 + 795 - 230 + 100), ignoring the stress.
- **D:** Omits the starting cash of USD 50 million.

Q272 Schweser Module Quiz 72.2 Q1

If a bank manager is concerned about the high cost of liquidity risk, she is most likely concerned about:

A. selling assets quickly at fair market prices.
B. receiving lower than expected cash inflows in the future.
C. generating positive cash flows from its sources of liquidity.
D. a high interest rate for borrowing relative to the risk-free rate.

Answer: D

Step by step

1. The cost of liquidity risk is funding cost risk: paying a higher than expected spread over the risk-free rate to obtain funds.
2. This matches option D.

Why D is correct: Funding cost risk is the risk of an adverse (wide) spread over the risk-free rate on funds.

Why the others are wrong

- **A:** Inability to sell assets at fair prices and with immediacy is market liquidity risk, part of quantitative liquidity risk.
- **B:** Lower than expected inflows is quantitative liquidity risk, not the cost dimension.
- **C:** Generating positive cash flows from liquidity sources describes liquidity generation capacity.

LO 72.d

Interpret the term structure of expected cash flows and cumulative cash flows.

Q273 AP Mock 3 Q74

The table gives the cash flows of institution XYZ, ordered by maturity. Assets bear no default risk and no liquidity options are embedded in the deposits.

Expiry	Notional +	Interest +	Notional -	Interest -	Net
1	20	6	0	-4	+22
2	0	5	-10	-4	-9
3	0	5	0	-3	+2
4	0	5	0	-2	+3
5	50	5	0	-3	+52
6	0	2	0	-3	-1

What is the term structure of expected cumulated cash flows (TSECCF) at the 6-year expiry?

A. 70
B. 22
C. 18
D. 69

Answer: D

Step by step

1. Net flow each expiry = sum of the four columns: $+22,\ -9,\ +2,\ +3,\ +52,\ -1$.
2. TSECCF is the running total of the net flows.
3. Cumulated: $22,\ 13,\ 15,\ 18,\ 70,\ 69$.
4. At the 6-year expiry the TSECCF is 69.

Why D is correct: The TSECCF at a date is the sum of all net cash flows (notional and interest, inflows and outflows) from the start through that date, so the year-6 net outflow of 1 must be included.

Why the others are wrong

- **A:** 70 is the cumulated figure at year 5; it ignores the net outflow of 1 at year 6.
- **B:** 22 is just the year-1 cumulated (and net) flow, not the year-6 balance.
- **C:** 18 is the cumulated figure at year 4; it stops cumulating too early.

Q274 AnalystPrep Q.3930

REBUILT — Original answer option merely negated option B (giveaway); replaced with a plausible false statement about the expected-value nature of the TSECCF.

Which of the following statements about the term structure of expected cash flows (TSECF) and the term structure of expected cumulated cash flows (TSECCF) is least accurate?

A. Behavioral models are used to adjust cash flows for liquidity options embedded in products such as sight deposits and prepayable mortgages.

B. The TSECF includes the cash flows of all current contracts; many are stochastic because they are linked to market indices such as Libor or Euribor.

C. Cash flows from new business are included; they are typically stochastic in both amount and timing and are handled with models that capture the related risks.

D. Because the TSECF is built from expected values, a TSECCF that is positive at every date guarantees that the bank faces no liquidity shortfall.

Answer: D

Step by step

1. Check each statement against the TSECF/TSECCF framework.
2. A: behavioral models adjust cash flows for liquidity options - accurate.
3. B: all current contracts are included and index-linked flows are stochastic - accurate.
4. C: new business flows are included and modelled as stochastic - accurate.
5. D: the TSECCF captures only expected values; realized flows can differ, so a positive expected TSECCF is no guarantee - inaccurate.

Why D is correct: The TSECF and TSECCF forecast expected values only, so a positive expected cumulated balance does not rule out shortfalls when realized flows are worse than expected (a negative balance is a warning sign, but a positive one is not a guarantee).

Why the others are wrong

- **A:** Accurate: behavioral models are the standard way to adjust for liquidity options. Choosing it reflects thinking options are ignored.
- **B:** Accurate: all existing contracts are included and index links make many flows stochastic. Choosing it reflects thinking only deterministic flows count.
- **C:** Accurate: new business is part of the model and is stochastic in amount and time. Choosing it reflects thinking only existing contracts are included.

Q275 AnalystPrep Q.3931

Given the assets and liabilities of a financial institution below, ordered by expiry regardless of contract type, assume the assets bear no default risk and no liquidity options are embedded in the deposits. Use the table to build the TSECF.

Expiry	Notional on Positive Cashflows	Interest on Positive Cashflows	Notional on Negative Cashflows	Interest on Negative Cashflo
1	25	7	0	-5
2	0	5	-20	-6
3	0	5	0	-3
4	0	5	0	-3
5	60	8	-35	-3
6	0	2	0	-3
7	0	2	-20	-3
8	0	2	0	0
9	0	2	0	0
10	15	2	0	0
>10	0	-	-25	0

Calculate the TSECCF at the 2-year expiry.

A. -21

B. 6

C. 7

D. 27

Answer: B

Step by step

1. Year 1 net flow: $25 + 7 + 0 - 5 = 27$.

2. Year 2 net flow: $0 + 5 - 20 - 6 = -21$.
3. TSECCF at year $2 = 27 + (-21) = 6$.

Why B is correct: The TSECCF is the cumulated total of net flows (notional plus interest) through the date, here 27 plus the year-2 net of -21.

Why the others are wrong

- **A:** -21 is the year-2 net flow alone; it forgets to add the year-1 balance.
- **C:** 7 comes from using the year-1 interest outflow of -5 instead of the year-2 outflow of -6 (27 + 5 - 20 - 5).
- **D:** 27 is the year-1 TSECCF; it stops before year 2.

Q276

AnalystPrep Q.3932

Given the assets and liabilities of a financial institution below, ordered by expiry regardless of contract type, assume the assets bear no default risk and no liquidity options are embedded in the deposits. Use the table to build the TSECF.

Expiry	Notional on Positive Cashflows	Interest on Positive Cashflows	Notional on Negative Cashflows	Interest on Negative Cashflo
1	25	7	0	-5
2	0	5	-20	-6
3	0	5	0	-3
4	0	5	0	-3
5	60	8	-35	-3
6	0	2	0	-3
7	0	2	-20	-3
8	0	2	0	0
9	0	2	0	0
10	15	2	0	0
>10	0	-	-25	0

Calculate the TSECCF for the term of more than 10 years to expiry.

A. 10
B. 14
C. 27
D. 39

Answer: B

Step by step

1. Net flows by expiry: $27, -21, 2, 2, 30, -1, -21, 2, 2, 17, -25$.
2. Cumulated: $27, 6, 8, 10, 40, 39, 18, 20, 22, 39, 14$.
3. The TSECCF in the >10 bucket is 14 (the 39 at year 10 less the final notional outflow of 25).

Why B is correct: The >10-year bucket contains the last outflow of 25, which must be added to the year-10 cumulated balance of 39.

Why the others are wrong

- **A:** 10 is the cumulated figure at year 4; it comes from stopping the running total far too early.
- **C:** 27 is the year-1 balance, not the final cumulated total.
- **D:** 39 is the year-10 TSECCF; it omits the -25 notional outflow in the >10 bucket.

Q277

BT answers P2.T8.20.9.1 · BT notes P2.T8.20.9.1

REBUILT — Original relied on four missing bar-chart exhibits (Series A-D); rebuilt as a self-contained numeric question on the same balance sheet, with equity repaid beyond 10 years per the GARP convention.

Geofinancial Bank has the following simplified balance sheet (in millions):

Assets	Liabilities
Bonds 30	Deposits 30
Loans 70	Bonds 50
Equity 20	

Assets expire as follows: bonds (30) in 1 year; loans: 40 in 5 years, 10 in 7 years, 20 in 10 years. Liabilities expire as follows: deposits: 10 in 1 year, 20 in 2 years; bonds: 10 in 5 years, 30 in 7 years, 10 beyond 10 years. Equity (20) is repaid to shareholders only at the end of the business, i.e. beyond 10 years. Ignore interest and assume the assets bear no default risk and no liquidity options exist.

What are the net cash flow at the 2-year expiry and the TSECCF at the 7-year expiry?

A. Net -20; TSECCF +10
B. Net -20; TSECCF -20
C. Net 0; TSECCF +10
D. Net -20; TSECCF +30

Answer: A
Step by step
1. Net flow = expiring assets minus expiring liabilities: year 1: 30-10 = +20; year 2: 0-20 = -20; year 5: 40-10 = +30; year 7: 10-30 = -20; year 10: +20; >10: -(10+20) = -30.
2. Net cash flow at the 2-year expiry = -20.
3. TSECCF: year 1 = 20; year 2 = 0; year 5 = 30; year 7 = 30 - 20 = 10.
4. So net at year 2 is -20 and TSECCF at year 7 is +10.

Why A is correct: TSECF entries are period net flows while the TSECCF is their running total; year 7 cumulates +20, -20, +30 and -20 to +10.
Why the others are wrong
- **B:** -20 as the TSECCF is the year-7 net flow, not the cumulated balance.
- **C:** 0 for the net flow is the year-2 cumulated figure (20 - 20), confusing the TSECCF with the period net flow.
- **D:** +30 is the year-5 TSECCF; it omits the year-7 net outflow of 20.

Q278 Schweser Module Quiz 72.2 Q2

Which of the following statements is least accurate regarding the TSECF and the TSECCF calculations and the models used to estimate them?
A. Rollovers of maturing liabilities should be included.
B. Behavioral components are often adjusted for liquidity options.
C. Default risk and correlations between bank counterparties need to be captured.
D. Only deterministic cash flows are counted, with stochastic cash flows captured in other models.

Answer: D
Step by step
1. The models must include rollovers, new business, behavioral adjustments for liquidity options and credit risk with correlations.
2. Many cash flows are stochastic because of links to market indices, and these are modelled within the TSECF/TSECCF.
3. D claims stochastic flows are excluded, which is false.

Why D is correct: Stochastic flows (for example index-linked and behavioral flows) are an integral part of the TSECF and TSECCF models, not left to other models.
Why the others are wrong
- **A:** Accurate: rollovers of maturing liabilities and new issuance to fund asset growth should be included.
- **B:** Accurate: behavioral models adjust cash flows for liquidity options.
- **C:** Accurate: credit models must reflect default risk and counterparty correlations.

LO 72.e

Discuss the impact of available asset transactions on cash flows and liquidity generation capacity.

Q279 AnalystPrep Q.3935

REBUILT — Stem did not state the coupon or its dates and the key accrued interest over 10 months instead of since the last coupon; I specified a 10% semiannual bond bought on a coupon date and corrected the accrual to 4 months.

A bank bought a bond with a notional of 1,000,000 on a coupon date at time 0. The bond pays a 10% annual coupon semiannually. After 10 months (0.833 years) the bank sells a notional of 300,000 at a clean price of 99.95 to generate liquidity. Using the term structure of cumulated liquidity generation capacity (TSCLGC), what inflow is recorded for this sale?
A. 299,850
B. 309,850

C. 314,850
D. 324,850

Answer: B
Key correction: Source key (A, 324,840) added accrued interest for the full 0.833 years; the coupon was last paid at month 6, so accrued interest covers only 4 months (0.333 years) and the correct inflow is 309,850.
Step by step
1. The TSCLGC inflow equals notional sold times the dirty price (clean price plus accrued interest).
2. Coupons are paid at months 6 and 12, so at month 10 accrued interest runs for 4 months = 0.3333 years: 10% × 0.3333 = 3.333% of par.
3. Dirty price = 99.95% + 3.333% = 103.283%.
4. Inflow = 300,000 × 1.03283 = 309,850.

Why B is correct: The sale proceeds include interest accrued since the last coupon date, not since purchase; only 4 months have accrued.
Why the others are wrong
- **A:** 299,850 = 300,000 x 99.95% ignores accrued interest (uses the clean price).
- **C:** 314,850 accrues 6 months (0.5 years), i.e. a full coupon period, instead of 4 months.
- **D:** 324,850 accrues the full 0.833 years since purchase, forgetting that the coupon paid at month 6 reset accrued interest.

Q280

AnalystPrep Q.3939

REBUILT — Original options were implausible numbers not tied to realistic errors; I replaced them with distractors from specific mistakes.
A liquidity division lends a notional of 600,000 of a bond 3 months after purchase for a period of 6 months. The TSECF records no cash flow at inception, and the TSAA shows a reduction of the available quantity of 600,000. After 6 months the bond is returned to the bank, which receives a lending fee of 3.5% p.a. What is the fee received?
A. 5,250
B. 21,000
C. 10,500
D. 8,925

Answer: C
Step by step
1. In security lending only possession passes; no cash is paid at inception, and the bank's only cash flow is the fee at expiry.
2. Fee = notional × annual fee rate × term in years.
3. Fee = 600,000 × 3.5% × 0.5 = 10,500.

Why C is correct: The fee accrues on the notional lent for the 6-month contract life, giving 10,500, which then increases the TSECF at expiry.
Why the others are wrong
- **A:** 5,250 uses a 3-month term (0.25 years) instead of 6 months.
- **B:** 21,000 forgets to scale the annual rate to the 6-month life (uses a full year).
- **D:** 8,925 wrongly applies a 15% haircut; haircuts apply to repo-type cash lending, not to a lending fee.

Q281

AnalystPrep Q.5412

REBUILT — Answer key used 6 months of accrued interest instead of 3 and two distractors were unexplained; I corrected the key and rebuilt the options from realistic errors.
Three months after purchasing a bond on its coupon payment date, ABC Bank enters into a repurchase agreement (repo) on the bond to generate immediate liquidity. Bond and repo details:

Notional (GBP)	10,000,000
Annual coupon (paid semiannually)	3.5%
Current price of bond (GBP, per 100)	97
Repo interest rate	2%
Repo haircut	3.5%

What is the cash outflow for ABC Bank when the 6-month repo contract expires?
A. 9,539,387

B. 9,624,669
C. 9,885,375
D. 9,454,105

Answer: A
Key correction: Source key (B, 9,624,669) accrued 6 months of coupon (0.035 x 1/2); only 3 months have elapsed since the last coupon, so accrued interest is 0.035 x 0.25 and the correct answer is 9,539,387.
Step by step
1. Accrued interest after 3 months = $3.5\% \times 0.25 = 0.875\%$ of par, so dirty price = 97% + 0.875% = 97.875%.
2. Cash received at start = $10{,}000{,}000 \times 0.97875 \times (1 - 0.035) = 9{,}444{,}937.5$.
3. Interest on the repo for 6 months: $9{,}444{,}937.5 \times 2\% \times 0.5 = 94{,}449.4$.
4. Outflow at expiry = $9{,}444{,}937.5 \times 1.01 = 9{,}539{,}387$.

Why A is correct: The repo repays the cash received (dirty price less haircut) plus 6 months of repo interest.
Why the others are wrong
- **B:** 9,624,669 uses 6 months of accrued coupon (1.75%) instead of the 3 months elapsed.
- **C:** 9,885,375 omits the haircut and so overstates the initial cash and repayment (dirty price x notional x 1.01).
- **D:** 9,454,105 uses the clean price 97% without accrued interest (0.97 x 10m x 0.965 x 1.01).

Q282

GARP PE1 Q26 · Mock 3 Q54

REBUILT — Merged duplicate versions (one said Eurobond); reworded to ask for the cumulative net effect on expected cash flows up to month 7, since the 19.6m decrease only holds cumulatively (TSECCF).

A bank plans to purchase a newly issued 1-year bond with a face value of GBP 20 million. The bond has an annualized coupon rate of 8%, paid semi-annually, and is priced at 102% of par. A treasury analyst must describe the impact of the purchase on the cumulative expected cash flows (TSECCF) and the term structure of available assets (TSAA). Which of the following correctly describes the change in both, measured at a 7-month time horizon from the purchase date?

A. The cumulative cash flows will decrease by GBP 20 million and the TSAA will increase by GBP 20.4 million.
B. The cumulative cash flows will decrease by GBP 19.6 million and the TSAA will increase by GBP 20 million.
C. The cumulative cash flows will increase by GBP 19.6 million and the TSAA will decrease by GBP 20 million.
D. The cumulative cash flows will increase by GBP 20 million and the TSAA will decrease by GBP 20.4 million.

Answer: B
Step by step
1. Purchase outflow = $20 \times 1.02 = 20.4$ million (price times notional).
2. Coupon at month 6 = $20 \times 8\%/2 = 0.8$ million, inside the 7-month horizon.
3. Cumulative cash flow at month 7 = $-20.4 + 0.8 = -19.6$ million.
4. The TSAA rises by the notional (face value) of 20 million until maturity, when it drops to zero.

Why B is correct: A purchase creates an outflow of price x notional, offset by coupons received, and increases the TSAA by the notional held.
Why the others are wrong
- **A:** Uses the face value for the cash flow and the price-based amount for the TSAA; the amounts are swapped and the coupon ignored.
- **C:** Signs reversed: a purchase is a cash outflow and the TSAA increases.
- **D:** Signs reversed and amounts swapped (20.4m is the cash outflow, not the TSAA change).

Q283

Schweser Module Quiz 72.3 Q1

Gerard Bank enters into a reverse repo transaction for six months. Which of the following statements is most accurate regarding the impact of the transaction?

A. The TSLGC will not be impacted.
B. Both the TSAA and the TSECCF are impacted.
C. The TSECF is impacted, but the TSECCF will not be impacted.
D. Possession and ownership of the asset belong to the counterparty.

Answer: B
Step by step

1. In a reverse repo the bank pays cash and receives the asset: possession passes to the bank, ownership stays with the counterparty.
2. Possession raises the TSAA by the notional amount.
3. The cash paid and the interest received at the end are contract cash flows, so both the TSECF and the TSECCF are impacted.
4. The TSLGC is impacted only if the asset is subsequently repoed, so it is not unconditionally unaffected.

Why B is correct: A reverse repo changes the TSAA (higher) and the expected and cumulative cash flows (outflow, then inflow with interest).

Why the others are wrong

- **A:** The TSLGC is not impacted by the reverse repo itself, but it is impacted if the asset is then repoed, so the flat statement is not the most accurate; the TSAA and TSECCF are certainly impacted.
- **C:** The TSECCF is just the cumulation of the TSECF, so it is impacted too.
- **D:** The counterparty owns the asset but the bank has possession; possession drives availability.

Reading 73. The Failure Mechanics of Dealer Banks

Darrell Duffie, "The Failure Mechanics of Dealer Banks," Journal of Economic Perspectives (24:1, pp. 51–72, 2010).

LO	Learning objective	Qs
73.a	compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business.	9
73.b	identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks.	10
73.c	assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.	5

LO 73.a

Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business.

Q284

AnalystPrep Q.2296

Arrenberg Bank is a large dealer bank performing many over-the-counter trades. In what manner are its over-the-counter trades negotiated?

A. Trades are negotiated both publicly or privately.
B. Trades are negotiated publicly.
C. Trades are negotiated privately.
D. Trades are negotiated publicly with some portions of the contract kept secret.

Answer: C

Step by step

1. Over-the-counter (OTC) markets have no exchange order book; dealers deal bilaterally with counterparties.
2. Per Duffie, OTC trades are privately negotiated, which is why OTC derivatives can be customized to a client's needs.
3. Hence the correct description is private negotiation.

Why C is correct: OTC trades are bilateral and privately negotiated, in contrast with exchange trading where orders are publicly displayed.

Why the others are wrong

- **A:** Hedges between public and private; an OTC trade is not negotiated publicly at all.
- **B:** Describes exchange trading, where bids and offers are publicly displayed.
- **D:** Invents a hybrid; OTC contract terms are agreed privately in full.

Q285

AnalystPrep Q.2294

REBUILT — Original options B and D were vague ("Sellers and buyers", "Corporate clients"); reworded so the primary/secondary contrast is unambiguous.

Bens Bank, a dealer bank, operates in both primary and secondary markets. In the primary market, the bank most likely intermediates between:

A. issuers and investors of securities.
B. investors who sell and investors who buy existing securities.
C. other banks.
D. corporate clients and their lenders.

Answer: A

Step by step

1. In the primary market, new securities are created and sold for the first time.
2. The dealer, often as underwriter, buys the equities or bonds from the issuer and resells them over time to investors.
3. So the primary-market intermediation is between issuers and investors.

Why A is correct: Primary-market dealing links issuers of new securities to investors, with the dealer acting as underwriter.

Why the others are wrong

- **B:** This is secondary-market intermediation of existing securities.
- **C:** Interdealer trading occurs but is not the defining primary-market role.
- **D:** Lending to corporates is commercial banking, not primary-market securities dealing.

Q286

AnalystPrep Q.2295

REBUILT — Original options C and D were vague or overlapping; reworded so only the secondary-market role fits.

Kralingen Bank, a dealer bank, operates in both primary and secondary markets. In the secondary market, the bank most likely intermediates between:

A. issuers of securities and investors.
B. investors who sell and investors who buy existing securities.
C. its commercial banking depositors and borrowers.
D. corporate clients and merger targets.

Answer: B

Step by step

1. The secondary market trades securities already issued.
2. The dealer stands ready to have its bid prices hit by sellers and its ask prices hit by buyers (Duffie).
3. So it intermediates among investors, between sellers and buyers.

Why B is correct: In secondary markets the dealer quotes bid and ask prices and intermediates among investors who sell and buy existing securities.

Why the others are wrong

- **A:** This is the primary-market role (underwriting new issues).
- **C:** Deposit-taking and lending is commercial banking, a different line of business.
- **D:** M&A advice is an investment banking service, not secondary-market dealing.

Q287

AnalystPrep Q.2298

REBUILT — Original asked how dealers 'benefit' and option B (offsetting positions) was also defensible; rewritten around the matched book so exactly one option is correct.

A dealer bank runs a "matched book" in over-the-counter derivatives. Which statement best describes how this practice works for the dealer?

A. It keeps client-initiated positions unhedged and profits from favorable market moves.
B. It charges each client a fixed monthly premium regardless of trading activity.
C. It lays off the risk of client trades through offsetting trades and earns the difference between bid and offer terms.
D. It delivers the underlying commodities to counterparties at a markup.

Answer: C

Step by step

1. For most OTC derivatives one counterparty is a dealer.
2. The dealer lays off much or all of the risk of a client trade by entering offsetting trades with other counterparties (a matched book).
3. Its profit is the difference between the bid and offer terms on the matched trades.

Why C is correct: A matched book offsets client-initiated trades so the dealer earns the bid-offer difference while bearing little market risk.

Why the others are wrong

- **A:** That is proprietary trading, a separate activity; a matched book aims to remove market risk.
- **B:** Dealers earn spreads on trades, not monthly premiums.
- **D:** OTC derivatives dealing earns bid-offer spreads on risk transfer; it does not consist of delivering underlying commodities at a markup.

Q288

AnalystPrep Q.2299

Utvin Bank, a dealer, trades over-the-counter derivatives. What is the primary source of profit for the bank in this business?

A. Transaction commissions

B. Differences between bid and offer terms
C. Interest rate movements
D. Changes in the value of securities held

Answer: B
Step by step
1. A dealer runs a matched book, aiming for offsetting trades.
2. Because the risk is laid off, profit does not come from price moves.
3. It comes from the difference between the bid and offer terms on the offsetting trades.

Why B is correct: Matched-book dealers earn the bid-offer spread rather than relying on market direction.
Why the others are wrong
- **A:** Dealers are principals earning a spread, not agents charging commissions.
- **C:** Rate exposure is what the matched book hedges away; it is not the primary profit source.
- **D:** Gains from changes in value are proprietary-trading outcomes, not the core dealing profit.

Q289
BT answers P2.T7.809.1 · BT notes P2.T8.809.1

Duffie describes the main business lines of large dealer banks as (1) securities dealing, underwriting and trading, (2) over-the-counter derivatives, and (3) prime brokerage and asset management. Each of the following statements about these business lines is true EXCEPT:

A. A tri-party repo agreement, in which a clearing bank holds the collateral and returns cash to the creditor, helps mitigate counterparty risk and eases collateral management.
B. For most OTC derivatives, one counterparty is a dealer who typically lays off much or all of the risk of client-initiated trades by running a matched book.
C. The best measure of the systemic risk of OTC derivatives is the total market value of all outstanding contracts, and master swap agreements do not reduce actual risks, only administrative burden.
D. Some large dealers are prime brokers offering securities management, clearing, cash management, securities lending, financing and reporting services to clients.

Answer: C
Step by step
1. A is true: the clearing bank in a tri-party repo holds collateral and insulates the lender from borrower default.
2. B is true: matched book, profiting on bid-offer differences.
3. D is true: this is Duffie's list of prime brokerage services.
4. C is false: the total market value of all derivative contracts must net to zero (positive = negative positions), so it is not a measure of systemic risk, and master swap agreements net exposures across trades and lower both exposures and collateral requirements (e.g., A's exposure of 100 to B and B's exposure of 80 to A net to 20 of collateral).

Why C is correct: Statement C contradicts Duffie: netting under master swap agreements reduces actual exposures, and aggregate market value sums to zero.
Why the others are wrong
- **A:** True statement, so it is not the exception; a clearing bank holds the collateral.
- **B:** True statement; matched book is how dealers lay off client risk.
- **D:** True statement; it lists the standard prime brokerage services.

Q290
GARP PE2 Q24 · Mock 4 Q25

A treasury manager presents to new analysts on the activities of dealer banks, comparing their major lines of business and the risks in each. Which of the following would be a correct statement for the manager to make?

A. Dealer banks often intermediate large equity block trades and trades in municipal and corporate bonds.
B. Dealer banks fund their own trading positions through repos but typically refrain from providing repo financing to fund investors' purchases of securities.
C. Dealer banks usually prefer to support external hedge fund counterparties over internally managed hedge funds in times of financial stress.
D. Dealer banks are prohibited by regulators from operating asset-management activities for institutional and wealthy clients.

Answer: A

Step by step

1. Duffie: dealers dominate OTC intermediation of bonds issued by corporations and municipalities and often intermediate large equity block trades, so A is correct.
2. B is wrong: a hedge fund can finance securities by placing them in repos with a dealer, i.e., dealers do provide repo financing to clients.
3. C is wrong: dealers have an incentive to support their own internal hedge funds to protect reputation and franchise value.
4. D is wrong: dealers often have large asset-management divisions for institutional and wealthy clients.

Why A is correct: Intermediating block trades and corporate and municipal bonds is a core dealer securities function described by Duffie.

Why the others are wrong

- **B:** Reverses the facts: dealers do lend against securities to clients such as hedge funds via repos.
- **C:** Dealers support internal funds (reputation, franchise value), not external ones, when distressed.
- **D:** Asset management for institutional and wealthy clients is a normal dealer business line; no such prohibition.

Q291 Schweser Module Quiz 73.1 Q2

Banks are most likely to diversify their exposure to a specific asset class such as mortgages by grouping these assets together and selling them to:

A. hedge funds.
B. government agencies.
C. the U.S. Federal Reserve.
D. special purpose entities.

Answer: D

Step by step

1. A bank sponsors a financial corporation or trust (special purpose entity) set up for the purpose.
2. The SPE pays the bank for the pooled loans with the proceeds of debt issued to third-party investors.
3. The SPE's debt is serviced from the pooled asset cash flows, so the bank transfers the exposure.

Why D is correct: Pooled loans are sold to a special purpose entity that funds the purchase by issuing debt, moving the exposure off the bank.

Why the others are wrong

- **A:** Hedge funds are prime brokerage clients, not the standard securitization vehicle.
- **B:** Government agencies are not the vehicle described for dealer bank securitizations.
- **C:** The Fed is a lender of last resort, not a buyer of pooled loans in this context.

Q292 Schweser Module Quiz 73.1 Q3

The formation of large bank holding companies results in diseconomies of scope with respect to:

A. risk management.
B. technology.
C. marketing.
D. financial innovation.

Answer: A

Step by step

1. Dealer banks operate as holding companies spanning commercial banking, prime brokerage, investment banking, asset management and off-balance-sheet activities.
2. Combining these activities is argued to produce economies of scope in information technology, marketing and financial innovation.
3. The 2007-2009 crisis showed that executive management and boards did not fully understand or control risk taking across the firm, i.e., diseconomies of scope in risk management (and corporate governance).
4. Therefore the answer is risk management.

Why A is correct: Large, complex holding companies were too broad to be effectively overseen, so risk management and governance suffered diseconomies of scope, while technology, marketing and innovation are the areas where scope is argued to help.

Why the others are wrong

- **B:** Technology is cited as a source of economies of scope (shared IT infrastructure), not diseconomies; confuses a claimed benefit with the crisis-revealed weakness.
- **C:** Marketing (cross-selling to a common client base) is an argued economy of scope.
- **D:** Financial innovation is argued to benefit from combined activities, so it is an economy of scope.

LO 73.b

Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks.

Q293

GARP PE1 Q29 · Mock 3 Q51

Large dealer banks have often financed significant fractions of their assets using short-term (overnight) repurchase agreements in which creditors hold bank securities as collateral against default losses. The table below shows the quarter-end financing of four A-rated broker-dealer banks (USD billion).

Financial instruments	Bank P	Bank Q	Bank R	Bank S
Owned	656	750	339	835
Pledged as collateral	258	472	139	209
Not pledged	398	278	200	626

If repo creditors become equally nervous about each bank's solvency, which bank is most vulnerable to a liquidity crisis?

A. Bank P
B. Bank Q
C. Bank R
D. Bank S

Answer: B

Step by step

1. Nervous repo creditors will not roll over, and the bank must replace the cash with unencumbered assets, so the key measure is the share of owned instruments already pledged.
2. $P = 258/656 = 39.3\%$; $Q = 472/750 = 62.9\%$; $R = 139/339 = 41.0\%$; $S = 209/835 = 25.0\%$.
3. Bank Q has the highest pledged share and the smallest unpledged cushion relative to its balance sheet (278 of 750).

Why B is correct: A run by repo creditors removes financing secured by pledged assets; the more of the balance sheet that is already encumbered, the less collateral remains to raise replacement cash, so the highest pledged fraction (Q) is most exposed.

Why the others are wrong

- **A:** P is only 39.3% pledged, well below Q; choosing it ignores the ratio.
- **C:** R is 41.0% pledged; its small absolute figures are irrelevant to the proportion.
- **D:** S has the largest unpledged amount (626) and the lowest share (25.0%), so it is the least vulnerable; choosing it comes from focusing on the largest balance sheet rather than the pledged share.

Q294

AP Mock 1 Q60

REBUILT — Replaced the 'All of the above' format with explicit numbered statements and options so the choice is self-contained.

Capital Bank is a dealer bank with investment banking, deposit-taking and corporate lending segments. The CRO is wary of liquidity problems triggered by a run arising from poor performance or an external shock. Which of the following steps could the bank take to mitigate the risk of losing liquidity from a run by short-term creditors?

I. Establishing lines of bank credit
II. Holding a buffer stock of cash and liquid securities
III. Laddering the maturities of its liabilities so that only a small fraction must be refinanced in a short period

A. I only
B. II and III only
C. I and II only
D. I, II and III

Answer: D

Step by step

1. Duffie lists three ways to mitigate loss of liquidity from a run by short-term creditors: lines of bank credit, a dedicated buffer of cash and liquid securities, and laddering liability maturities.
2. Statements I, II and III each match one of the three, so all are valid.

Why D is correct: Credit lines and a liquid buffer supply cash when funding flees, and laddering limits how much debt must be refinanced at once, so all three reduce run risk.

Why the others are wrong

- **A:** Credit lines alone omit the buffer and laddering, both of which are recognised measures.
- **B:** Leaves out credit lines, which are explicitly named as a mitigant.
- **C:** Leaves out laddering, which limits the amount of debt maturing in any short window.

Q295

AP Mock 2 Q1

REBUILT — Original table had six banks but only four options, with the true highest-ratio Bank D (44.3%) nearly tied with C (44.4%) and omitted from the options; changed Bank C's pledged amount to 300m and set options so exactly one is best.

Large dealer banks finance a significant part of their assets with short-term solutions such as overnight repurchase agreements in which creditors hold bank assets as collateral. The table shows year-end financing of six broker-dealer banks (GBP).

Bank	A	B	C	D	E	F
Instruments owned	896m	745m	630m	700m	962m	1,230m
Pledged as collateral	229m	121m	300m	310m	151m	280m

Suppose repo creditors raise questions about the solvency of all of these banks. Which bank is most vulnerable to a liquidity crisis?

A. Bank A
B. Bank D
C. Bank C
D. Bank F

Answer: C

Step by step

1. Vulnerability depends on the share of owned instruments already pledged: pledged/owned.
2. A: $229/896 = 25.6\%$; B: $121/745 = 16.2\%$; C: $300/630 = 47.6\%$; D: $310/700 = 44.3\%$; E: $151/962 = 15.7\%$; F: $280/1230 = 22.8\%$.
3. Bank C has the highest share, so the smallest unencumbered base for replacement financing.

Why C is correct: The more of its inventory is already encumbered, the less collateral a bank can use to replace repo funding that creditors refuse to roll.

Why the others are wrong

- **A:** A is 25.6% pledged; it is not close to the maximum.
- **B:** D has the largest absolute pledged amount (310m), but its share (44.3%) is lower than C's 47.6%; the error is using the amount instead of the ratio.
- **D:** F has the largest balance sheet (1,230m) but only 22.8% pledged; size is not the measure.

Q296

AnalystPrep Q.2305

REBUILT — Original options included a giveaway ('Hiding some transactions') and vague wording; rewrote the options around Duffie's point that a dealer cannot refuse counterparties trades at prevailing market terms.

A dealer bank is perceived to face a solvency crisis. Its OTC derivatives counterparties are trying to reduce their exposure, in part by trading with the dealer in ways that make it pay out cash. If the dealer wants to avoid sending an adverse signal of weakness to the market, what should it do?

A. Refuse to quote competitive bid prices on options so that no cash is paid out
B. Continue to let counterparties trade at terms prevailing elsewhere in the market, even though this drains cash
C. Widen its bid-ask spreads sharply to discourage counterparties from trading with it
D. Decline novation and offsetting-trade requests from smaller counterparties only

Answer: B

Step by step

1. Counterparties reduce exposure by borrowing from the dealer, entering offsetting trades that make it pay cash, harvesting cash from in-the-money positions, or seeking novation.
2. These actions drain the dealer's cash, but refusing them or quoting uncompetitively signals liquidity concerns and can accelerate the run.
3. Duffie: the dealer cannot afford to refuse these trades at terms prevailing elsewhere in the market, so B.

Why B is correct: Any visible refusal or off-market pricing reveals weakness and triggers further flight, so the dealer must keep accepting trades on market terms despite the cash drain.

Why the others are wrong

- **A:** Refusing to quote competitive bids preserves cash but signals liquidity concerns to the market.
- **C:** Off-market spreads are an effective refusal and are read as a sign of weakness.
- **D:** Selective refusal still signals weakness and the news spreads to counterparties of all sizes.

Q297

AnalystPrep Q.2984

Dealer banks, like other financial institutions, can collapse for a variety of reasons. Which one of the following is NOT typically associated with the collapse of a dealer bank?

A. The flight of short-term creditors
B. The departure of prime brokerage clients
C. Cash-draining actions by derivatives counterparties designed to lower their exposures to the dealer bank
D. The granting of additional clearing-bank privileges

Answer: D

Step by step

1. Duffie's four failure mechanisms: flight of short-term creditors, departure of prime brokerage clients, cash-draining actions by derivatives counterparties, and the loss of clearing-bank privileges.
2. A, B and C are on the list; D is the opposite of the fourth (a gain, not a loss, of privileges).

Why D is correct: Collapse is driven by losing clearing-bank privileges; additional privileges would support, not undermine, the dealer's liquidity.

Why the others are wrong

- **A:** Repo and other short-term creditors refusing to roll is the first listed mechanism.
- **B:** Prime brokerage client exits remove cash and collateral the dealer relied on.
- **C:** Counterparties borrowing from the dealer, trading offsetting contracts or harvesting cash drain liquidity.

Q298

AnalystPrep Q.5381

The table shows quarter-end repo financing for four A-rated broker-dealer banks (GBP billion).

Financial security	Bank A	Bank B	Bank C	Bank D
Owned	240	189	320	280
Pledged as security	150	110	180	170
Not pledged	90	79	140	110

If repo creditors become equally nervous about every bank, which bank is the most susceptible to a liquidity crisis?

A. Bank A
B. Bank B
C. Bank C
D. Bank D

Answer: A

Step by step

1. Compute pledged/owned: A 150/240 = 62.5%; B 110/189 = 58.2%; C 180/320 = 56.3%; D 170/280 = 60.7%.
2. Bank A has the highest encumbered share and thus the least unpledged collateral relative to its balance sheet.

Why A is correct: A larger pledged share means less unencumbered collateral to replace funding withdrawn by repo creditors.

Why the others are wrong

- **B:** B is 58.2% pledged, below A and D.
- **C:** C has the largest absolute pledged amount (180) but the lowest share (56.3%); the error is using amounts, not ratios.

- **D:** D is 60.7%, second highest; close but below A.

Q299 BT answers P2.T7.809.2 · BT notes P2.T8.809.2

Duffie explains the similarities between a classic depositor run at a commercial bank and the failure mechanisms for dealer banks. The key mechanisms are the flight of short-term creditors, departures of prime-brokerage clients, cash-draining actions by derivatives counterparties designed to lower their exposures, and, most decisively, the loss of clearing-bank privileges. In regard to these failure mechanisms, which of the following statements is TRUE?

A. The contractual "right of offset" guarantees that a dealer bank will retain cash settlement privileges

B. During the financial crisis, the exit of prime brokerage clients reduced the stress on major dealer banks like Morgan Stanley by shrinking their balance sheets and therefore their exposure

C. A dealer bank that finances long-term assets with overnight repos can experience funding liquidity problems if haircuts suddenly decrease

D. In the case of Lehman, OTC derivative contracts were exempted by law as "qualifying financial contracts" from the automatic stay at bankruptcy that holds up other creditors, which resulted in a large post-bankruptcy drain on the defaulting dealer

Answer: D

Step by step

1. A: the right of offset lets the clearing bank stop intraday payments that would overdraw the account, which leads to loss of settlement privileges.
2. B: client exits remove cash and re-pledgeable collateral (Morgan Stanley's fell 69%), increasing stress.
3. C: liquidity problems arise when haircuts rise, not fall.
4. D: matches Duffie: the QFC exemption means unwinding the derivatives portfolio drains the defaulted dealer, with priority to derivative counterparties.

Why D is correct: Derivatives counterparties are exempt from the automatic stay, so they can unwind and drain cash after default, which increases other creditors' incentive to run first.

Why the others are wrong

- **A:** The right of offset is what lets a clearing bank withdraw settlement privileges; it does not guarantee them.
- **B:** Prime broker client flight removes liquidity and collateral, increasing rather than reducing stress.
- **C:** A fall in haircuts would increase borrowing capacity; problems arise when haircuts increase.

Q300 Mock 1 Q15

An institutional trader is a counterparty to numerous OTC derivatives transactions with a dealer bank. In recent months the trader has lost confidence in the solvency of the dealer bank. Which of the following actions is the trader LEAST likely to take?

A. Borrow funds from the dealer

B. Request a novation of the contract to another dealer

C. Enter into new offsetting derivatives contracts with the dealer

D. Request to have at-the-money options revised to in-the-money strike prices

Answer: D

Step by step

1. A worried counterparty wants to reduce its net exposure to the dealer, ideally by receiving cash from it.
2. A, B and C are standard techniques: borrowing from the dealer, novating to another dealer, and entering new trades that make the dealer pay cash.
3. The standard option technique runs the other way: revising in-the-money options to at-the-money strikes to harvest cash. Moving at-the-money options to in-the-money strikes raises what the dealer owes without any cash being received, increasing exposure.

Why D is correct: Exposure-reducing actions pull cash out of the dealer; revising an option to an in-the-money strike increases the dealer's obligation with no cash change, the opposite of what a nervous counterparty wants.

Why the others are wrong

- **A:** Borrowing creates an offsetting payable owed to the dealer, reducing net exposure.
- **B:** Novation moves the exposure to another dealer, a standard exposure-reduction step.
- **C:** New offsetting trades can be structured so the dealer pays cash, reducing exposure.

Q301

Schweser Module Quiz 73.1 Q1

REBUILT — Original keyed 'depositors withdrawing' as least likely, but a deposit run is a genuine liquidity threat (only not among the Schweser 73.b accelerators); replaced it with a clearly non-accelerating option so exactly one answer is defensible.

A large dealer bank's counterparties begin to doubt its solvency. Which of the following actions by counterparties or clients is least likely to accelerate a liquidity crisis at the dealer bank?

A. Repurchase agreement creditors refuse to roll over their short-term funding.
B. Prime brokerage clients withdraw their cash and securities from the dealer.
C. A hedge fund requests a novation of its derivative to another dealer bank.
D. A hedge fund agrees to leave additional unencumbered cash balances in its prime brokerage account.

Answer: D

Key correction: Source key (Schweser) said D (depositors removing savings); that option was ambiguous because a deposit run also drains liquidity, so the item was rebuilt.

Step by step

1. Identify the accelerators in the reading: OTC derivatives counterparties reducing exposure (restructuring, offsetting contracts, novation), flight of repo creditors, flight of prime brokerage clients; loss of cash settlement privileges is the final collapse.
2. A: repo creditors not renewing removes funding, so it accelerates the crisis.
3. B: prime brokerage clients leaving removes cash and securities the dealer was using as a liquidity source.
4. C: a novation can transfer the cash collateral to the second dealer, draining liquidity.
5. D: clients leaving more cash with the dealer adds liquidity, so it does not accelerate the crisis.

Why D is correct: Crises are accelerated when counterparties and clients withdraw funding, collateral or cash; a client leaving extra cash with the dealer does the opposite.

Why the others are wrong

- **A:** Repo creditor flight is a named accelerator; choosing it confuses a funding run with a benign action.
- **B:** Flight of prime brokerage clients removes a source of liquidity and is a named accelerator.
- **C:** Novation moves collateral to another dealer and signals distrust (Bear Stearns' refusal added stress), so it accelerates the crisis.

Q302

Schweser Module Quiz 73.2 Q1

One potential solution for mitigating the liquidity risk caused by derivatives counterparties exiting their large dealer bank exposures is most likely the:

A. use of central clearing.
B. use of a novation through another dealer bank.
C. requirement of dealer banks to pay out cash to reduce counterparty exposure.
D. creation of new contracts by counterparties.

Answer: A

Step by step

1. Counterparties flee a doubtful dealer by novating, entering offsetting contracts, or taking cash out of in-the-money positions.
2. Central clearing novates OTC derivatives to a central counterparty standing between the original parties, so the dealer's solvency matters less and counterparties have less reason to run.
3. Caveat: it works only for standardized contracts (not AIG's customized credit derivatives).
4. B, C and D are the exit channels themselves and add liquidity pressure.

Why A is correct: A central counterparty interposes itself between the original parties, removing the incentive to exit the dealer's exposures.

Why the others are wrong

- **B:** Novation to another dealer is one of the ways counterparties exit; it can move collateral away from the original dealer.
- **C:** Paying out cash to reduce exposure drains the dealer's liquidity, worsening the problem.
- **D:** New offsetting contracts with the dealer reduce counterparty exposure but require the dealer to pay out cash, adding pressure.

LO 73.c

Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.

Q303

AnalystPrep Q.2983

REBUILT — Original tested generic bank-failure tools with an obvious giveaway option; rewritten around the dealer-bank policy measures in the reading.

Which of the following is NOT among the policy measures discussed for alleviating firm-specific and systemic risks of large dealer banks?

A. Creating a tri-party repo utility with less discretion in rolling over a dealer's repo positions

B. Central clearing of standardized OTC derivatives

C. Higher capital requirements that include off-balance-sheet positions

D. Excluding off-balance-sheet positions from capital requirements to preserve the dealer's cash-management flexibility

Answer: D

Step by step

1. A tri-party repo utility is a proposed fix for unstable repo funding: it would have fewer conflicting incentives and less discretion in rolling over repos.
2. Central clearing reduces the threat of OTC derivatives counterparties fleeing a questionable dealer.
3. Capital requirements are expected to rise and to cover off-balance-sheet positions to reduce leverage.
4. Excluding off-balance-sheet positions would do the opposite of what is proposed, so D is the exception.

Why D is correct: The proposals tighten capital to include off-balance-sheet exposures; excluding them would allow more hidden leverage.

Why the others are wrong

- **A:** A proposed measure; not the exception.
- **B:** A proposed measure for standardized OTC derivatives; not the exception.
- **C:** A proposed measure; not the exception.

Q304

BT answers P2.T7.809.3 · BT notes P2.T8.809.3

Duffie examines several policy measures that might alleviate firm-specific and systemic risks related to large dealer banks. With which of the following statements would he most DISAGREE?

A. The threat posed by the flight of over-the-counter derivatives counterparties can be lowered by central clearing.

B. Distress-contingent convertible debt is likely to be destabilizing during financial distress, and regulators should consider curtailing its use.

C. Short-term tri-party repos are a particularly unstable source of financing; potential remedies include a tri-party repo utility, central-bank insurance of tri-party repo transactions, or an emergency bank financed by repo market participants.

D. Perhaps the most important source of systemic risk is the impact of dealer-bank fire sales on market prices and investor portfolios; in the crisis this risk was significantly mitigated by lender-of-last-resort financing and capital injections.

Answer: B

Step by step

1. Test each statement against Duffie's proposals.
2. A, C and D restate his views (central clearing, repo remedies, fire sales mitigated by central-bank financing and capital injections).
3. Duffie proposes distress-contingent convertible debt, with mandatory rights offerings of equity triggered by leverage or liquidity thresholds, to recapitalize before a run begins.
4. So B, which calls them destabilizing, is the statement he would disagree with.

Why B is correct: Contingent convertibles are a proposed tool to recapitalize a dealer before a run and curb excessive risk taking, not a destabilizing one.

Why the others are wrong

- **A:** Consistent with Duffie: central clearing reduces exposure to fleeing derivatives counterparties.
- **C:** Consistent with Duffie's discussion of tri-party repo remedies.
- **D:** Consistent with Duffie's fire-sale discussion.

Q305

BT answers P2.T8.24.3.3

REBUILT — Original had subjectively defensible options (A, C, D all reasonable); rebuilt as a single-best question on the tri-party repo remedy.

In a crisis of confidence in a dealer bank's solvency, tri-party clearing banks have an incentive to limit their exposure to the dealer by restricting its repo financing and clearing functions. Which proposed policy measure most directly addresses this vulnerability?

A. A tri-party repo utility with less discretion in rolling over the dealer's repo positions
B. Central clearing of standardized OTC derivatives
C. Distress-contingent convertible debt
D. Giving clearing banks full discretion to cut repo financing to weakened dealers

Answer: A

Step by step

1. The problem is clearing-bank discretion and conflicting incentives when rolling repos.
2. A repo utility is designed with fewer conflicting incentives and less discretion, addressing this directly.
3. Central clearing addresses OTC derivatives counterparties, not tri-party repo.
4. Convertible debt recapitalizes the dealer but does not change clearing-bank behaviour; D worsens the problem.

Why A is correct: A utility removes the discretion and conflicts that cause clearing banks to pull repo funding from a weakened dealer.

Why the others are wrong

- **B:** Targets OTC derivatives counterparty flight, a different channel.
- **C:** Helps recapitalize before a run but does not alter clearing-bank incentives.
- **D:** Increases discretion, the opposite of the proposed remedy.

Q306

Schweser Module Quiz 73.2 Q2

Which of the following is not a policy objective of the U.S. Treasury Department's 2008 Troubled Asset Relief Program to help dealer banks recover from the subprime market crisis?

A. Provide below-market financing rates for bidders of "toxic" assets.
B. Absorb losses beyond a pre-specified level.
C. Force the sale of illiquid assets in order to better determine the "true" value.
D. Mitigate the effect of adverse selection.

Answer: C

Step by step

1. The program subsidized bidders for toxic assets through below-market financing (A) and absorbing losses beyond a set level (B).
2. It aimed to mitigate adverse selection (D).
3. Forcing sales of illiquid assets would deepen losses and worsen liquidity, so it is not an objective.

Why C is correct: Forced sales into thin markets produce deep discounts and losses; the policy instead subsidized buyers.

Why the others are wrong

- **A:** Is an actual TARP/PPIP feature.
- **B:** Is an actual feature.
- **D:** Is an actual objective.

Q307

AnalystPrep Q.2292

Coliseu Bank is a large dealer bank considered "too big to fail" and can expect support from the government and central bank in distress. This implicit support is associated with behaviour that may have disastrous consequences. Which of the following is an example?

A. Manipulation of financial statements.
B. Taking more inefficient risks.
C. Offering products at exorbitant prices.
D. Only doing business with high net-worth firms and individuals.

Answer: B

Step by step

1. Implicit support lowers the bank's funding cost and shifts downside to taxpayers.
2. This creates moral hazard: excessive, socially inefficient risk taking.
3. The other options are not linked to the too-big-to-fail guarantee.

Why B is correct: Expected rescue creates moral hazard, encouraging more risk than is socially efficient.

Why the others are wrong

- **A:** Fraud is not caused by a support guarantee.
- **C:** Pricing is unrelated to the guarantee.
- **D:** Client selection is unrelated to moral hazard.

Reading 74. Liquidity Stress Testing

Shyam Venkat and Stephen Baird, Liquidity Risk Management: A Practitioner's Perspective (Hoboken, NJ: John Wiley & Sons, 2016). Chapter 3.

LO	Learning objective	Qs
74.a	differentiate between various types of liquidity, including funding, operational, strategic, contingent, and restricted liquidity.	5
74.b	estimate contingent liquidity via the liquid asset buffer.	3
74.c	discuss liquidity stress test design issues such as scope, scenario development, assumptions, outputs, governance, and integration with other risk models.	16

LO 74.a

Differentiate between various types of liquidity, including funding, operational, strategic, contingent, and restricted liquidity.

Q308

AnalystPrep Q.4127

Axon, a financial institution, uses its available liquidity to fund its day-to-day business and to ensure the orderly clearing of payment transactions. Under which category of funding liquidity does this use fall?

A. Contingent liquidity
B. Operational liquidity
C. Strategic liquidity
D. Restricted liquidity

Answer: B

Step by step

1. Identify the purpose: day-to-day funding and orderly clearing of payments.
2. Operational liquidity is the cash needed to run the business daily and ensure orderly payment clearing.
3. Match: operational liquidity.

Why B is correct: Operational liquidity is defined as the cash required to fund the business on a daily basis and to ensure orderly clearing of payment transactions.

Why the others are wrong

- **A:** Contingent liquidity is the buffer of high-quality liquid assets and facilities for general obligations under stress, not routine payment flows.
- **C:** Strategic liquidity is held for future investment needs such as acquisitions or capital expenditure, not daily operations.
- **D:** Restricted liquidity is earmarked for specific predetermined uses such as collateral, not for daily payment clearing.

Q309

BT Mock Q49 · BT answers P2.T8.20.10.1 · BT notes P2.T8.20.10.1

Which of the following types of liquidity is available in the form of the institution's liquidity asset buffer (LAB) and represents liquidity available to meet general financial obligations under a stress scenario?

A. Operational
B. Strategic
C. Contingent
D. Restricted

Answer: C

Step by step

1. The LAB holds unencumbered, very high-quality liquid assets (and facilities).
2. Its role is to meet general obligations when stress materializes.
3. That is the definition of contingent liquidity; it is what the LAB measures and what the stress test sizes.

Why C is correct: Contingent liquidity is the liquidity available to meet general financial obligations under stress, held through the liquidity asset buffer.

Why the others are wrong

- **A:** Operational liquidity funds day-to-day business and payment clearing and is not available for general stressed obligations.
- **B:** Strategic liquidity is held for future business needs (acquisitions, capex) and is not designed for stress.
- **D:** Restricted liquidity is available only for specifically defined purposes (e.g., collateral) and cannot meet general obligations.

Q310

GARP PE1 Q75

A junior treasury analyst at a bank classifies assets into types of liquidity to assess funding sources for a liquidity stress test, covering both on- and off-balance sheet assets. Which of the following correctly describes an example of restricted liquidity?

A. High-quality liquid assets required to fund the bank's day-to-day operations
B. Cash and money-market instruments allocated to finance the bank's expansion into new geographic regions
C. A position in mortgage-backed securities held within a special purpose vehicle established by the bank
D. US Treasury notes pledged as collateral against a large wholesale loan taken by the bank

Answer: D

Step by step

1. Restricted liquidity = liquid assets usable only for specifically defined purposes, e.g., collateralizing wholesale borrowing.
2. D: pledged Treasuries are committed to the wholesale loan and unavailable for general obligations: restricted.
3. A is operational, B is strategic; C is an SPV holding that is not defined by a use restriction.

Why D is correct: Restricted liquidity is liquid assets available only for specifically defined purposes such as collateral for wholesale borrowing; pledged Treasuries fit exactly.

Why the others are wrong

- **A:** Funds for day-to-day operations are operational liquidity.
- **B:** Funds held for expansion are strategic liquidity (future business needs).
- **C:** Assets in an SPV raise a consolidation/transfer question but are not an example of assets reserved for a defined purpose such as collateral.

Q311

Mock 2 Q32

A treasury analyst is categorizing the funding liquidity available to the bank. A substantial portion of the bank's securities portfolio is pledged as collateral against secured borrowings and derivative exposures and is therefore unavailable for other funding needs. Which category of funding liquidity is most directly affected by this collateralization?

A. Contingent liquidity
B. Operational liquidity
C. Restricted liquidity
D. Strategic liquidity

Answer: C

Step by step

1. Pledged assets are committed to a specific purpose (collateral).
2. Assets limited to a predefined use and unavailable for general needs are restricted liquidity.
3. Hence the securities move out of usable buffers into restricted liquidity.

Why C is correct: Collateralized securities can be used only for their defined purpose and cannot meet general obligations in a stress test, which is restricted liquidity.

Why the others are wrong

- **A:** Contingent liquidity is the unencumbered buffer; pledging reduces it rather than defining it.
- **B:** Operational liquidity is cash for daily operations and payment clearing, unrelated to pledging.
- **D:** Strategic liquidity is held for future investments such as acquisitions or capex.

Q312 Schweser Module Quiz 74.1 Q1

Which type of liquidity is meant specifically to fund capital asset purchases?

A. Contingent
B. Funding
C. Restricted
D. Strategic

Answer: D

Step by step

1. Strategic liquidity is held for potential investment opportunities such as fixed asset purchases or mergers and acquisitions.
2. Match: strategic.

Why D is correct: Strategic liquidity is held to meet future business needs such as capital expenditure and acquisitions.

Why the others are wrong

- **A:** Contingent liquidity is for general obligations under stress, not planned investment.
- **B:** Funding liquidity is the umbrella category (the four types sit under it), not a specific use.
- **C:** Restricted liquidity is reserved for predefined uses such as collateral.

LO 74.b

Estimate contingent liquidity via the liquid asset buffer.

Q313 BT answers P2.T8.20.10.2 · BT notes P2.T8.20.10.2

Acme Bank has a normal liquidity asset buffer (LAB) of $60.0 million. To estimate its contingent liquidity, it runs stress scenarios. Stressed outflows (a mix of 60% contractual and 40% contingent outflows, covering retail deposit outflows, unsecured wholesale funding outflows, secured funding runoff and drawdowns of credit and liquidity facilities) produce a net estimate of $28.0 million. Stressed inflows (secured funding maturities, customer loan repayments and drawdowns on liquidity facilities available to Acme) produce a net estimate of $15.0 million. Acme also holds $16.0 million of operational liquidity. What is the bank's estimate of contingent liquidity (the stressed liquidity asset buffer)?

A. $32.0 million
B. $47.0 million
C. $63.0 million
D. $103.0 million

Answer: B

Step by step

1. Stressed LAB = normal LAB â^' stressed cash outflows + stressed cash inflows.
2. $60.0 - 28.0 + 15.0 = 47.0$ million.
3. Operational liquidity ($16.0 million) funds day-to-day operations and cannot satisfy general liabilities, so it is neither added nor subtracted.
4. The 60/40 contractual/contingent mix is already inside the $28.0 million net outflow and does not change the calculation.

Why B is correct: Contingent liquidity is the normal LAB reduced by stressed outflows and offset by stressed inflows; operational liquidity is separate and excluded.

Why the others are wrong

- **A:** Ignored stressed inflows: $60.0 - 28.0 = 32.0$. Inflows offset outflows and must be added.
- **C:** Added operational liquidity to the correct result: $47.0 + 16.0 = 63.0$. Operational liquidity cannot be used for general liabilities and is excluded.
- **D:** Added the outflows instead of subtracting them: $60.0 + 28.0 + 15.0 = 103.0$.

Q314 Schweser Module Quiz 74.1 Q2

REBUILT — Original added back the unused $10 million of a facility, but GARP treats facility drawdowns as stressed inflows (so $0.96 billion was also defensible); removed the facility and added an operational-liquidity twist.

Firm A has a normal liquidity asset buffer of $1,000 million in highly liquid assets. Separately, it holds $120 million of operational liquidity that is not part of the buffer. In a sudden stress scenario, retail customers withdraw $150 million of deposits and make $80 million of loan repayments. The firm must also meet $60 million of margin and collateral calls on its derivatives because of falling collateral values and higher volatility. What is the estimate of Firm A's stressed liquidity asset buffer?

A. $0.71 billion
B. $0.87 billion
C. $0.93 billion
D. $0.99 billion

Answer: B

Key correction: Source key (Schweser Module Quiz 74.1 Q2) said B ($0.88 billion) by adding back the unused $10 million of a liquidity facility; under GARP, facility drawdowns are stressed inflows, so the source item was ambiguous and has been rebuilt with different data.

Step by step

1. Stressed outflows = deposit withdrawals + margin/collateral calls = 150 + 60 = 210 million.
2. Stressed inflows = loan repayments = 80 million.
3. Stressed LAB = normal LAB â˚' outflows + inflows = 1,000 − 210 + 80 = 870 million = $0.87 billion.
4. Operational liquidity (120 million) is outside the buffer and cannot meet general liabilities, so it is ignored.

Why B is correct: The stressed buffer subtracts all stressed outflows (including derivative margin calls) and adds stressed inflows; operational liquidity is excluded.

Why the others are wrong

- **A:** Subtracted the loan repayments as if they were outflows: 1,000 − 150 − 60 − 80 = 710. Inflows must be added.
- **C:** Omitted the margin calls: 1,000 − 150 + 80 = 930. Derivative margin calls are stressed outflows.
- **D:** Added operational liquidity: 870 + 120 = 990. Operational liquidity is not available for general stressed liabilities.

Q315 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A treasury analyst at a regional bank is estimating the bank's contingent liquidity using the liquid asset buffer. The bank's normal liquid asset buffer is $240 million. Under the stress scenario, modeled outflows are retail deposit withdrawals of $90 million, unsecured wholesale funding losses of $70 million, secured funding runoff of $35 million, and drawdowns of credit and liquidity facilities of $25 million. Contractual inflows of $30 million of customer loan repayments and $20 million of secured funding maturities are expected, but the scenario assumes market conditions limit realized inflows to 60% of these amounts. What is the bank's stressed liquid asset buffer?

A. $20 million
B. $50 million
C. $70 million
D. $158 million

Answer: B

Step by step

1. Stressed outflows = 90 + 70 + 35 + 25 = 220 million.
2. Contractual inflows = 30 + 20 = 50 million; limited to 60%: 0.60 × 50 = 30 million.
3. Stressed liquid asset buffer = normal buffer - stressed outflows + stressed inflows = 240 − 220 + 30 = 50 million.

Why B is correct: The stressed buffer equals the normal buffer less stressed outflows plus stressed inflows, and inflows must be reduced to the amount that market conditions allow the bank to realize, which is $30 million.

Why the others are wrong

- **A:** Adds nothing back for inflows: 240 − 220 = 20. Ignores that stressed inflows partially offset stressed outflows.
- **C:** Uses the full contractual inflows of $50 million: 240 − 220 + 50 = 70. Ignores the scenario's market-condition limit on inflows.

- **D:** Scales the outflows by 60% instead of the inflows: $240 + 50 - 0.6 \times 220 = 158$. Applies the market-condition limit to the wrong side.

LO 74.c

Discuss liquidity stress test design issues such as scope, scenario development, assumptions, outputs, governance, and integration with other risk models.

Q316 AnalystPrep Q.4134

REBUILT — Option A was garbled and arguably restated the correct disadvantage; the stem lacked the 12-month minimum; options rewritten.

A bank's liquidity stress test uses a planning horizon of at least twelve months, and the bank is considering projecting cash flows well beyond twelve months. Which of the following is the main disadvantage of the longer projection period?

A. Longer-term projections are subject to significant forecast error, because they depend on baseline balance sheet and income statement budgets that cover shorter horizons

B. Longer-term projections cannot be combined with daily or weekly cash flow estimates in the first weeks of the test

C. Longer-term projections require the bank to assume that a recovery or resolution process begins immediately

D. Longer-term projections are prohibited because the planning horizon of a liquidity stress test must not exceed twelve months

Answer: A

Step by step

1. The planning horizon should be at least twelve months, so a bank may project further.
2. The cost is accuracy: baseline balance sheet and income statement budgets from strategic planning only cover a limited horizon.
3. Hence longer-term projections carry significant forecast error; also, assuming the bank continues operating indefinitely under stress with no recovery or resolution is unrealistic.

Why A is correct: Forecast reliability deteriorates with the horizon because the underlying budgets cover shorter periods.

Why the others are wrong

- **B:** Invented limitation; banks typically forecast daily early and move to weekly or monthly later, which is compatible with long horizons.
- **C:** Not required: the drawback is that assuming indefinite operation under stress with no recovery or resolution is unrealistic, not that resolution must begin immediately.
- **D:** Twelve months is a minimum, not a maximum; some banks with ample liquidity have survival horizons of up to two years.

Q317 AnalystPrep Q.4141

REBUILT — 'Treasury committee' corrected to treasury unit and distractors tied to the correct GARP role owners.

In liquidity stress testing governance, which of the following is a responsibility of the treasury unit (first line of defense)?

A. Recommending stress test scenarios

B. Setting liquidity risk policy limits based on stress results and escalating exceptions

C. Periodically reviewing the stress testing framework, procedures and controls for compliance

D. Providing independent validation of the stress testing model

Answer: A

Step by step

1. Treasury owns the modeling process: maintains procedures, recommends scenarios, recommends assumptions to ALCO, produces reporting.
2. B is ALCO, C is internal audit, D is model risk management.

Why A is correct: Recommending scenarios is a treasury duty; the other tasks belong to other governance roles.

Why the others are wrong

- **B:** Limit-setting and escalation belong to ALCO.
- **C:** Periodic framework review is internal audit's role (third line).
- **D:** Independent validation is model risk management's role.

Q318

AnalystPrep Q.4125

REBUILT — Original relied on a vague 'profit margins' distractor; reworded to test the actual design components of a liquidity stress test.

A bank is designing its liquidity stress testing framework. Which of the following is NOT one of the standard design issues the framework must address?

A. Scope of the test across entities and business lines
B. Scenario development
C. Governance of the stress testing process
D. Target pricing of new loan products

Answer: D

Step by step

1. The design issues are scope, scenario development, assumptions, outputs, governance and integration with other risk models.
2. A is scope, B is scenario development, C is governance.
3. Pricing of new loan products is a business decision, not a design component of the stress test.

Why D is correct: Loan pricing is not part of the liquidity stress test design; the other three are core design issues.

Why the others are wrong

- **A:** Scope is a core design issue (consolidated, subsidiary, business line).
- **B:** Scenario development is a core design issue.
- **C:** Governance (ALCO, treasury, risk, audit, model risk) is a core design issue.

Q319

AnalystPrep Q.4129

REBUILT — Giveaway distractors (scenarios need not capture material liquidity risk) replaced with plausible ones.

Which of the following best describes how a liquidity stress testing process should be designed?

A. It should begin with risk identification and event analysis, so that the roster of scenarios captures material liquidity risks
B. It should begin by calibrating assumptions, from which scenarios are then derived
C. It should begin by choosing the output reports, with risks identified afterwards
D. It should begin with the consolidated result, with scenarios chosen to reproduce the regulatory minimum

Answer: A

Step by step

1. GARP describes the model as part of an end-to-end process.
2. The process begins with risk identification and event analysis.
3. This ensures the set of scenarios captures material liquidity risks; assumptions and outputs follow the scenarios.

Why A is correct: Scenarios must be grounded in identified liquidity risks and events, so risk identification and event analysis come first.

Why the others are wrong

- **B:** Assumptions are developed to support defined scenarios, so they cannot come first.
- **C:** Outputs are the end product of the process.
- **D:** Scenarios are chosen to reflect material risks, not to hit a regulatory number.

Q320

AnalystPrep Q.4131

REBUILT — Distractors were weak; rewritten to realistic mistakes.

A bank has a foreign subsidiary whose liquidity cannot be freely transferred to the parent because of local restrictions. What is the most appropriate approach in the bank's liquidity stress testing?

A. Use the consolidated liquidity position only, since it nets surpluses against deficits across entities
B. Assess the impact of the restrictions on enterprise-level liquidity under both normal and stressed conditions
C. Assume the restrictions will be lifted in a crisis so that subsidiary cash is available to the parent

D. Assess the restrictions only under stress, since they are irrelevant in normal conditions

Answer: B

Step by step

1. Transfer restrictions trap liquidity in specific entities and can distort the consolidated view.
2. The bank must therefore assess their impact on enterprise-level liquidity.
3. This must cover normal operating conditions as well as stressed conditions.

Why B is correct: Trapped liquidity makes the consolidated figure misleading, so the effect must be assessed in both normal and stress environments.

Why the others are wrong

- **A:** Consolidation hides trapped liquidity, giving a distorted picture.
- **C:** Holding companies should assume little or no access to subsidiary cash in a crisis.
- **D:** The restrictions must be considered in normal as well as stressed conditions.

Q321

AnalystPrep Q.4132

REBUILT — Distractors replaced with plausible errors.

Which statement about currency in a consolidated liquidity stress test is most accurate?

A. The test is performed in the home-country currency, with careful consideration of the liquidity impact of currency conversion requirements, e.g., for offshore subsidiaries with currency mismatches

B. Currency conversion can be ignored because any currency is assumed to be freely convertible in a crisis

C. The test should be run in whichever currency is most abundant in the liquid asset buffer

D. Offshore subsidiaries' currency mismatches are irrelevant because the parent can always swap-fund outflows immediately

Answer: A

Step by step

1. The test is performed in the currency of the entity tested (home currency for the consolidated test).
2. Less established offshore subsidiaries or branches may have significant currency mismatches.
3. Settlement time for the parent to swap-fund an unanticipated outflow may be problematic, so conversion impact must be considered.

Why A is correct: Currency conversion and settlement frictions can reduce usable liquidity in a crisis.

Why the others are wrong

- **B:** Foreign currency may be thinly traded; conversion may take long or be impossible.
- **C:** The test currency is that of the entity being tested, not the most abundant one.
- **D:** Swap-funding timeframes can prove problematic in a crisis.

Q322

AnalystPrep Q.4136

A treasury unit is developing hypothetical scenarios for its liquidity stress test. Which of the following is NOT a characteristic of scenario development?

A. Clearly defining each scenario

B. Distinguishing between systemic and idiosyncratic risk

C. Distinguishing between sources of liquidity

D. Distinguishing between levels of severity

Answer: C

Step by step

1. Scenario development: define scenarios clearly, distinguish systemic from idiosyncratic (and combined) stress, and distinguish severity levels.
2. Sources of liquidity relate to the liquid asset buffer, not scenario design.

Why C is correct: Scenario design concerns the nature and severity of stress, not classifying liquidity sources.

Why the others are wrong

- **A:** Clear definition is required.
- **B:** Systemic, idiosyncratic and combined scenarios are required.
- **D:** Graduated severity (e.g., adverse and severely adverse) is required.

Q323 AnalystPrep Q.4137

Which of the following is NOT a step in developing liquidity stress test assumptions?

A. Qualitatively assess expected liquidity behavior of each type of cash flow to identify significant liquidity risk
B. Develop quantitative modeling assumptions based on available historical data
C. Develop matrices of relative modeling assumptions based on scored risk levels and baseline historical data
D. Develop matrices of relative modeling assumptions irrespective of scored risk levels and current data

Answer: D

Step by step

1. GARP steps include: qualitative assessment, segmentation, ranking/scoring, quantitative assumptions from historical data, matrices based on scored risk levels and baseline historical data, scenario adjustments.
2. A, B, C match those steps.
3. D contradicts step 5 by ignoring risk scores and data.

Why D is correct: Assumption matrices must be built from scored risk levels and historical data.

Why the others are wrong

- **A:** This is step 1.
- **B:** This is step 4.
- **C:** This is step 5.

Q324 AnalystPrep Q.4139

REBUILT — Original keyed 'deposit inflows' as not considered, which is doubtful; replaced with a clearly out-of-scope option.

Which of the following is NOT among the key assumptions typically developed for a liquidity stress test?

A. Deposit outflows
B. Investment portfolio haircuts
C. Collateral requirements
D. Shareholders' expected return on equity

Answer: D

Key correction: Source key (D, deposit inflows) is doubtful since stressed inflows are modeled; the item was rebuilt with an unambiguous option.

Step by step

1. Key assumptions: investment portfolio haircuts, deposit outflows, unsecured wholesale funding, collateral requirements, other contingent liabilities, business reduction.
2. A, B, C are on the list.
3. Shareholder return expectations are not a liquidity stress assumption.

Why D is correct: Key assumptions address cash flow and valuation drivers under stress, not equity return expectations.

Why the others are wrong

- **A:** Deposit outflows is a key assumption.
- **B:** Haircuts is a key assumption.
- **C:** Collateral requirements is a key assumption.

Q325 AnalystPrep Q.4140

REBUILT — Stem mixed outputs with the four assumptions and option D was awkward; reworded.

The reporting package from a liquidity stress test includes stress testing assumptions. Which of the following is NOT one of the four key assumptions reported?

A. The overall stress level represented by the scenario
B. Whether the scenario is systemic, idiosyncratic, or both
C. Expected returns on the bank's investment portfolio
D. A description of the cash flow impacts of the scenario

Answer: C

Step by step

1. GARP four assumptions: overall stress level; systemic, idiosyncratic or both; documentation of macro, market

and company events leading to the scenario; description of cash flow impacts.
2. A, B, D are on the list; expected returns are not.

Why C is correct: Reported assumptions describe the scenario and its cash flow impacts, not return expectations.

Why the others are wrong

- **A:** Item 1 of the list.
- **B:** Item 2 of the list.
- **D:** Item 4 of the list.

Q326

AnalystPrep Q.4142

Which committee or function provides independent validation and control over the management governance of the liquidity stress testing model, in line with the institution's model risk management policy?

A. The asset-liability committee
B. Internal audit
C. Model risk management
D. The Treasury

Answer: C

Step by step

1. Identify the governance role described: independent validation of the liquidity stress testing model under the model risk management policy.
2. In the GARP governance framework this is the responsibility of model risk management, which treats the model as highly critical.
3. ALCO, Treasury and internal audit have different roles, so only C fits.

Why C is correct: Model risk management is the function responsible for independent validation and control of the stress testing model.

Why the others are wrong

- **A:** ALCO approves the stress testing policy, scenarios and limits and reviews results; it does not independently validate the model.
- **B:** Internal audit periodically reviews the framework, procedures and controls for compliance; it is not the model validator.
- **D:** Treasury designs and runs the test and proposes assumptions, so it cannot be the independent validator of its own model.

Q327

AnalystPrep Q.4143

REBUILT — Options C and D were near-duplicates and partly garbled in the source; rewrote them as distinct, plausible false statements consistent with GARP.

Which of the following statements about integrating liquidity stress testing with capital stress testing is correct?

A. Linking the liquidity and capital stress tests requires the liquidity stress test to incorporate any required capital infusions of subsidiary entities.
B. For each liquidity stress test scenario, capital impact assumptions should be based only on the idiosyncratic conditions assumed under the scenario.
C. For each liquidity stress test scenario, capital impact assumptions should be based only on the overall market conditions assumed under the scenario.
D. Capital impact assumptions must always be built from detailed credit loss and pre-provision net revenue models, never from a high-level capital infusion assumption.

Answer: A

Step by step

1. GARP: the first link is that the liquidity stress test incorporates any required capital infusions of subsidiary entities.
2. Capital impact assumptions use both the overall market and the idiosyncratic conditions assumed under each scenario, so B and C ('only') are wrong.
3. The level of detail may vary from a high-level capital infusion assumption to detailed credit loss and PPNR modeling, so D ('must always') is wrong.

Why A is correct: The integration requires capital infusions to subsidiaries to flow into the liquidity test and capital impacts to reflect both market-wide and firm-specific conditions.

Why the others are wrong

- **B:** Ignores market-wide conditions; assumptions use both market and idiosyncratic conditions.
- **C:** Ignores idiosyncratic conditions; assumptions use both.
- **D:** Overstates the requirement; detail may range from a simple capital infusion assumption to detailed credit loss and PPNR modeling.

Q328 BT answers P2.T8.20.10.3 · BT notes P2.T8.20.10.3

In regard to the design of liquidity stress tests, each of the following is true EXCEPT which is inaccurate?

A. The primary outcome metric of the liquidity stress test is the level of available liquidity relative to net cash outflows under each scenario.

B. Three general approaches to performing a liquidity stress test are historical statistical techniques, deterministic models, and Monte Carlo simulation.

C. Liquidity optimization refers to an algorithm that identifies the single solution for the theoretically largest possible liquidity buffer versus all other portfolio alternatives.

D. A reverse liquidity stress test starts from the bank's liquidity failure (an unviable business plan) and works back to the conditions or scenarios that could cause it.

Answer: C

Step by step

1. A: GARP's principal outcome measure is available liquidity relative to net cash outflows under each scenario - true.
2. B: GARP lists historical statistical techniques (e.g., CFaR), deterministic models and Monte Carlo simulation - true.
3. D: A reverse test assumes the failure and determines what conditions would produce it - true.
4. C: Liquidity optimization uses the stress test to choose among portfolios meeting a target contingent liquidity, trading off liquidity vs yield and capital, to maximize efficiency, not to maximize buffer size - false, so C is the EXCEPT answer.

Why C is correct: Optimization seeks the most efficient liquidity portfolio for a given required level, not the largest possible buffer.

Why the others are wrong

- **A:** True statement, so not the exception.
- **B:** True statement, so not the exception.
- **D:** True statement, so not the exception.

Q329 Schweser Module Quiz 74.2 Q1

Which liquidity stress impact factor would generally be the largest threat to a bank's liquidity?

A. Deposit runoff.

B. Derivatives cash flows.

C. Loss of secured funding.

D. Loss of wholesale funding.

Answer: A

Step by step

1. Deposits are the main funding base; sudden withdrawal of demand deposits and early withdrawal of term deposits has the biggest negative liquidity impact.
2. This is therefore the most important customer behavior to model.

Why A is correct: Deposit runoff is generally the largest threat to bank liquidity.

Why the others are wrong

- **B:** Derivatives margin and collateral calls matter but are generally smaller than deposit outflows.
- **C:** Secured funding losses arise via higher haircuts and are typically less severe.
- **D:** Wholesale funding is assumed largely unavailable in stress, but it is generally a smaller share than deposits.

Q330 Schweser Module Quiz 74.2 Q2

In the context of deposit outflows, which behavioral assessment factor is relevant for individual, small business, and commercial/institutional customers of the bank?

A. Credit usage.
B. FDIC coverage.
C. Industry segment.
D. Relationship tenure.

Answer: D

Step by step

1. Relationship tenure (and rates paid) applies to all three customer groups.
2. Credit usage applies to small business and commercial clients only.
3. FDIC coverage applies to individuals and small businesses only.
4. Industry segment applies generally only to commercial/institutional clients.

Why D is correct: Only relationship tenure is common to all three groups.

Why the others are wrong

- **A:** Not relevant for individuals; applies to small business and commercial clients.
- **B:** Deposit insurance coverage does not apply to commercial/institutional clients.
- **C:** Applies mainly to commercial/institutional clients.

Q331 Schweser Module Quiz 74.2 Q3

At a minimum, how often should the liquidity stress test be performed to allow review by the asset-liability management committee?

A. Monthly.
B. Quarterly.
C. Semiannually.
D. Annually.

Answer: B

Step by step

1. GARP: at a minimum the stress test should be performed quarterly to support ALCO review.
2. More advanced banks with technology may run it monthly or daily.

Why B is correct: Quarterly is the minimum frequency for ALCO review.

Why the others are wrong

- **A:** A practice of advanced banks, not the minimum.
- **C:** Too infrequent to support ALCO review.
- **D:** Annual is the minimum renewal cycle for the stress testing policy, not for running the test.

Reading 75. Liquidity Risk Reporting and Stress Testing

Moorad Choudhry, The Principles of Banking (Singapore: John Wiley & Sons, 2012). Chapter 14.

LO	Learning objective	Qs
75.a	describe best practices for the reporting of a bank's liquidity position.	3
75.b	compare and interpret different types of liquidity risk reports.	18
75.c	explain the process of reporting a liquidity stress test and interpret a liquidity stress test report.	3

LO 75.a

Describe best practices for the reporting of a bank's liquidity position.

Q332 Schweser Module Quiz 75.1 Q1

The treatment of tenor for certain types of cash flows is a concern for senior managers preparing liquidity reports for the Financial Services Authority (FSA) in the United Kingdom. Based on FSA recommendations, how should callable or demand deposits that have no maturity date be treated in liquidity reports?

A. As having a 1-day tenor.
B. As having a 1-week tenor.
C. As having a 1-month tenor.
D. As having a 1-year tenor.

Answer: A

Step by step

1. Callable and demand deposits have no contractual maturity and can be withdrawn by the depositor at any time.
2. For liquidity reporting the conservative assumption is that they may leave immediately.
3. The FSA therefore treats them as due in one day (1-day tenor) and places them in the shortest bucket.

Why A is correct: Because the depositor can withdraw on demand, the cash outflow could occur the next day, so regulators assign a 1-day tenor.

Why the others are wrong

- **B:** A 1-week tenor assumes some notice period that demand deposits do not have; it understates the immediacy of outflow.
- **C:** A 1-month tenor is typical of short-term notice or term funding, not of on-demand balances; it would understate near-term outflows.
- **D:** A 1-year tenor treats the deposit as stable core funding, which ignores the withdrawal right and is the least conservative choice.

Q333 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Priya Nair, a treasury analyst at a UK-regulated bank, is preparing the liquidity ratio calculation under the FSA guidelines for off-balance-sheet items. The bank has USD 600 million of committed but undrawn lending facilities, derivatives with a total notional of USD 2,000 million, and a coupon of USD 5 million payable on a derivative on a date that falls inside the reporting period. Based on the FSA treatment, what total amount of these items should be included as a cash outflow in the ratio calculation?

A. USD 60 million
B. USD 605 million
C. USD 65 million
D. USD 2,065 million

Answer: C

Step by step

1. Undrawn commitments: FSA specifies that 10% of committed but undrawn lending is included as an at-sight outflow: $0.10 \times 600 = 60$.
2. Derivatives: notional values are not included; only coupons receivable or payable are included, on their pay dates: +5.

3. Total outflow = 60 + 5 = 65 (USD million).

Why C is correct: The FSA includes 10% of undrawn committed lending as an at-sight outflow (USD 60 million) and excludes derivative notionals but includes coupons payable on their pay dates (USD 5 million), giving USD 65 million.

Why the others are wrong

- **A:** Includes the undrawn commitment at 10% but omits the derivative coupon payable, which the FSA does include on its pay date.
- **B:** Applies 100% of the undrawn commitments (USD 600 million) plus the coupon. This is the more conservative treatment used in some banks' internal models, not the FSA specification.
- **D:** Adds the derivative notional (USD 2,000 million) plus 10% of undrawn commitments and the coupon; the FSA excludes derivative notionals from the liquidity ratio.

Q334 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Daniel Ortiz, a risk manager at a UK bank that is an individual liquidity adequacy standards (ILAS) firm, is building a regulatory reporting calendar based on the FSA liquidity reporting requirements. Which of the following correctly describes the reporting frequency of the wholesale liabilities, liquidity buffer qualifying securities, and currency analysis reports?

A. The wholesale liabilities report is due daily, the liquidity buffer qualifying securities report is due monthly, and the currency analysis report is due weekly.

B. The wholesale liabilities report is due monthly, the liquidity buffer qualifying securities report is due weekly, and the currency analysis report is due quarterly.

C. The wholesale liabilities report is due weekly, the liquidity buffer qualifying securities report is due quarterly, and the currency analysis report is due monthly.

D. The wholesale liabilities report is due weekly, the liquidity buffer qualifying securities report is due monthly, and the currency analysis report is due quarterly.

Answer: D

Step by step

1. Wholesale liabilities (daily transaction prices and volumes for unsecured wholesale liabilities): weekly.
2. Liquidity buffer (granular analysis of marketable-security holdings): monthly, 15 business days after month-end.
3. Currency analysis (FX exposures on the balance sheet): quarterly.
4. Only option D matches all three.

Why D is correct: The FSA requires wholesale liabilities weekly, the liquidity buffer monthly, and the currency analysis quarterly; daily reporting applies to market-wide stress cash flow and mismatch reports.

Why the others are wrong

- **A:** Confuses the daily frequency of market-wide stress cash flow reporting with wholesale liabilities, which is weekly, and overstates the currency analysis frequency, which is quarterly.
- **B:** Swaps the wholesale liabilities (weekly) and liquidity buffer (monthly) frequencies.
- **C:** Swaps the liquidity buffer (monthly) and currency analysis (quarterly) frequencies.

LO 75.b

Compare and interpret different types of liquidity risk reports.

Q335 AnalystPrep Q.4056

The liability profile is a simple breakdown of the share of each type of liability at the bank. Which of the following is not a part of the liability profile?

A. Customer individuals

B. Highly liquid security repo

C. Asset-backed securities

D. Highly liquid assets

Answer: D

Step by step

1. The liability profile shows each type of liability as a percentage of total liabilities (e.g., individual customers, large enterprises, repo funding, own-issued securities).
2. Individual customer deposits, repo borrowing against liquid securities and issued asset-backed securities are all funding the bank owes.
3. Highly liquid assets are on the asset side of the balance sheet, so they are not a liability category.

Why D is correct: The liability profile covers only the funding side of the balance sheet; liquid assets belong to the asset side.

Why the others are wrong

- **A:** Individual customer deposits are a liability and typically the largest share; calling them an asset is the error.
- **B:** Repo of liquid securities is secured borrowing, hence a liability; the word 'liquid' misleads into thinking it is an asset.
- **C:** Asset-backed securities issued by the bank are liabilities; the word 'asset' in the name misleads.

Q336

AnalystPrep Q.4060

REBUILT — Stem said only 'UK regulations' while one option describes bank-model practice; stem now specifies FSA guidance and options were made mutually exclusive.

Under UK Financial Services Authority (FSA) guidance for off-balance-sheet items in liquidity ratio calculations, which of the following is correct regarding derivatives?

A. Derivative values/notionals are not included, although coupons receivable or payable are included on their pay dates

B. Derivatives are included in full at their notional values

C. Coupons receivable or payable on derivatives are excluded on their pay dates

D. Derivatives are included only to the extent of collateral payable or receivable under an ISDA/CSA agreement

Answer: A

Step by step

1. FSA treatment: derivative values/notionals are not included in the liquidity ratio calculation.
2. Coupons receivable or payable are nevertheless included on their pay dates.
3. Including derivatives to the extent of ISDA/CSA collateral is how bank models commonly go beyond the FSA rule, not the FSA rule itself.

Why A is correct: The FSA excludes derivative values but includes coupons on pay dates; collateral-based inclusion is a bank-model practice.

Why the others are wrong

- **B:** No notional inclusion is required; values/notionals are excluded.
- **C:** Coupons are included on pay dates, not excluded.
- **D:** This is the common bank-model treatment, not the FSA recommendation.

Q337

AnalystPrep Q.4050

REBUILT — Original keyed the annual LTD ratio as absent, but the deposit tracker does report the LTD ratio; replaced option D with an item that belongs to the funding concentration report.

Which of the following details would not be obtained from a deposit tracker report?

A. Month-end actuals for deposits by customer type

B. Each month-end change

C. Each month-end forecast for the position to the end of the year

D. Each large depositor's share of total liabilities, measured against a single-source concentration limit

Answer: D

Step by step

1. The deposit tracker gives month-end actual and forecast customer deposits by type, month-end changes, total customer loans and the resulting LTD ratio.
2. Each large depositor's percentage of total liabilities versus a single-source limit is the content of the funding (large depositor) concentration report.
3. So D is the item not obtained from the deposit tracker.

Why D is correct: Depositor-level concentration against a single-source limit is reported in the funding concen-

tration report, not in the deposit tracker.

Why the others are wrong

- **A:** Deposit type breakdown of month-end actuals is a core part of the deposit tracker.
- **B:** Month-end changes in deposits are shown in the tracker.
- **C:** Forecasts through year-end are a defining feature of the tracker.

Q338 AnalystPrep Q.4051

VTR bank has been receiving most of its funding from current accounts and rolling deposits. Under what circumstance will the bank treat overnight balances as long-term according to regulation?

A. If the overnight balances can be demonstrated to be acting as long-term in ”behavioral” terms
B. If the overnight deposits are predictable
C. If the overnight deposits are the primary funding source for bank
D. None of the above

Answer: A

Step by step

1. Regulators treat current and rolling accounts as short-term (one-day) liabilities that do not help regulatory liquidity metrics.
2. The local regulator may allow overnight balances to be treated as longer term only if they can be shown to behave as long-term funds (behavioral adjustment).
3. Predictability or size of the funding share is not the test, so A.

Why A is correct: The behavioral adjustment requires evidence that the balances actually behave as long-term funds.

Why the others are wrong

- **B:** Predictability alone does not prove the balances stay; the test is demonstrated long-term behavior.
- **C:** Being the main funding source increases reliance on short-term funds; it does not change classification.
- **D:** Wrong because A is a valid circumstance.

Q339 AnalystPrep Q.4053

MCT bank has been preparing a mismatch report after a stress testing process. Which among the following statements is accurate about mismatch reports?

A. The mismatch report reflects the maturity gap for all assets and liabilities per time bucket and with an adjustment for liquid securities.
B. The report doesn’t apply in generating the horizon report for cash flow
C. It presents the bank’s marketable assets and liabilities
D. It reports the end-of-day liquidity position

Answer: A

Step by step

1. The funding maturity gap (mismatch) report shows the maturity gap of all assets and liabilities per time bucket, adjusted for liquid securities.
2. It is the same report used to generate the cash flow survival horizon report, which eliminates B.
3. Marketable securities lists and the end-of-day position belong to the daily liquidity report, eliminating C and D.

Why A is correct: The mismatch report is a time-bucketed, liquid-security-adjusted maturity gap that also feeds the survival horizon.

Why the others are wrong

- **B:** Reverses the facts: the mismatch report is the source of the cash flow survival horizon report.
- **C:** Lists of marketable securities are part of the daily liquidity report, not the defining content of the mismatch report.
- **D:** End-of-day position and cumulative cash gap are daily liquidity report items.

Q340 AnalystPrep Q.4054

A financial institution is preparing a large depositor concentration report for a banking group. What is the most accurate meaning of the word ”large” in this case?

A. Someone that owns the majority of shares
B. Someone that undertakes more than USD 50 million withdrawals
C. Someone that deposits a deposit of 5% of total liabilities.
D. None of the above

Answer: C

Step by step

1. ’Large’ can be set as an absolute amount (USD 50 million of deposits) or as a percentage of total liabilities.
2. UK regulators suggest ALCO treat a deposit of 5% of total liabilities as large, so C.
3. Share ownership is irrelevant and the USD 50 million refers to deposits, not withdrawals.

Why C is correct: A large depositor is defined by the size of its deposits, either an absolute amount or about 5% of total liabilities.

Why the others are wrong

- **A:** Share ownership has nothing to do with funding concentration.
- **B:** The USD 50 million threshold applies to amounts deposited, not withdrawals.
- **D:** Wrong because C is correct.

Q341 AnalystPrep Q.4055

Off-balance-sheet items serve as potential stress points for a bank’s funding. Which among the following is not an off-balance sheet item?

A. Liquidity lines
B. Letters of credit
C. Revolving credit facilities
D. Liquid assets

Answer: D

Step by step

1. Liquidity lines, revolving credit facilities, letters of credit and guarantees are undrawn commitments that can be drawn in stress; they are off-balance-sheet.
2. Liquid assets are held on the balance sheet.
3. So D is not an off-balance-sheet item.

Why D is correct: Undrawn commitments are off-balance-sheet; liquid assets are recorded on the balance sheet.

Why the others are wrong

- **A:** Liquidity lines are off-balance-sheet commitments.
- **B:** Letters of credit are off-balance-sheet contingent items.
- **C:** Undrawn revolving facilities are off-balance-sheet commitments.

Q342 AnalystPrep Q.4057

Which among the choices is not considered in regular liquidity qualitative reporting concerning individual liquidity adequacy standards (ILAS)?

A. Details on any increase or decrease in retail deposits
B. Details on average day-to-day opening cash position
C. Details on shrinkage or growth of asset books
D. Details on deposit month-end actuals by customer type

Answer: D

Step by step

1. The group treasury qualitative (monthly liquidity highlights) report asks for explanations of changes in liquidity ratios, the liquidity gap, asset book growth or shrinkage, inter-group positions, corporate and retail deposit changes, and average daily opening cash.
2. Month-end actuals of deposits by customer type are quantitative data in the deposit tracker report.
3. So D is not part of qualitative reporting.

Why D is correct: Qualitative highlights explain changes; month-end deposit actuals by type are quantitative deposit tracker data.

Why the others are wrong

- **A:** Retail deposit increases/decreases are a listed qualitative item.
- **B:** Average daily opening cash position is a listed qualitative item.
- **C:** Growth or shrinkage of asset books is a listed qualitative item.

Q343

AnalystPrep Q.4058

The following is an excerpt from Bright Bank's Deposit Tracker from the end of the year 2018 to July 2019.

Deposit tracker	Month-end actuals 31/12/2018	Month-end actuals 31/01/2019	Month-end forecasts 31/05/2019	Mont
Total customer deposits	612,152	423,804	651,112	599,7
Total customer loans	547,478	373,381	520,829	480,1

Calculate the bank's loan-to-deposit ratio as at 31/12/2018.

A. 0.84
B. 0.86
C. 0.87
D. 0.89

Answer: D

Step by step

1. LTD = total customer loans / total customer deposits.
2. 31/12/2018: 547,478 / 612,152 = 0.8943.
3. Rounded to two decimals, 0.89.

Why D is correct: LTD is loans divided by deposits using the 31/12/2018 column.

Why the others are wrong

- **A:** 0.84 does not equal the 31/12/2018 ratio; an arithmetic slip.
- **B:** 0.86 is not the 31/12/2018 ratio; a calculation or wrong-column error.
- **C:** 0.87 is not the 31/12/2018 ratio; the correct 0.894 rounds to 0.89.

Q344

AnalystPrep Q.4059

REBUILT — Original distractors did not correspond to specific mistakes and the key (29,685) truncated the exact 29,685.9; options rebuilt around realistic errors.

The following is an excerpt from Bright Bank's Deposit Tracker from the end of the year 2018 to July 2019.

Deposit tracker	Month-end actuals 31/12/2018	Month-end actuals 31/01/2019	Month-end forecasts 31/05/2019	Mont
Total customer deposits	612,152	423,804	651,112	599,7
Total customer loans	547,478	373,381	520,829	480,1

How much does the bank need to increase its total customer loans by to meet its targeted loan-to-deposit ratio of 85% on 30/06/2019, holding deposits constant?

A. 27,149
B. 29,686
C. 32,616
D. 119,652

Answer: B

Step by step

1. Required loans = 85% x deposits at 30/06/2019 = 0.85 x 599,774 = 509,807.9.
2. Increase = 509,807.9 - 480,122 = 29,685.9, i.e. about 29,686.

Why B is correct: Target loans equal the target ratio times deposits; the required increase is that level minus current forecast loans.

Why the others are wrong

- **A:** 27,149 uses the 31/12/2018 column (0.85 x 612,152 - 547,478 = -27,149) instead of 30/06/2019.
- **C:** 32,616 uses the 31/05/2019 column (0.85 x 651,112 - 520,829).
- **D:** 119,652 is deposits minus loans (599,774 - 480,122), ignoring the 85% target.

Q345 AnalystPrep Q.4061

While preparing a deposit tracker report, a bank manager notes that most retail bank deposits are current accounts and rolling deposits. How should the bank treat these funds for regulatory purposes?

A. As long-term liabilities
B. As liquid assets
C. As short-term liabilities
D. None of the above

Answer: C

Step by step

1. Current accounts and rolling deposits can be withdrawn on demand, so they are callable/demand funds.
2. Regulators treat callable and demand deposits as 1-day money for regulatory purposes.
3. They are therefore short-term liabilities that do not help the regulatory liquidity metrics, unless the regulator accepts that they behave as longer-term funds.

Why C is correct: Demand-type deposits can leave immediately, so regulators treat them as short-term (1-day) liabilities.

Why the others are wrong

- **A:** Long-term treatment applies only if stability is demonstrated and the regulator allows a behavioural adjustment (e.g. up to 50% of stable retail deposits); it is not the default.
- **B:** Deposits are liabilities, not assets; this confuses the funding source with the liquid assets that counterbalance outflows.
- **D:** A correct option exists (C).

Q346 AnalystPrep Q.4063

REBUILT — Source answer defined the liquidity gap as a maturity/supply mismatch, which is not how the daily liquidity report computes it; rebuilt as a numeric item using the report's definition.

A bank's daily liquidity report shows, for the one-month bucket, a cumulative cash gap of -640 million (cumulative inflows less outflows) and counterbalancing capacity (liquid securities that can be sold or pledged) of 800 million. What is the liquidity gap for that bucket?

A. -1,440 million
B. -640 million
C. +160 million
D. +1,440 million

Answer: C

Step by step

1. Liquidity gap = cumulative cash gap adjusted for counterbalancing capacity.
2. Counterbalancing capacity is a positive amount available to cover the shortfall: -640 + 800 = +160 million.
3. The positive result means liquid securities are sufficient to cover the cumulative shortfall in this bucket.

Why C is correct: The liquidity gap nets the cumulative cash shortfall against the counterbalancing capacity available to cover it.

Why the others are wrong

- **A:** Subtracted the counterbalancing capacity (-640 - 800), treating liquid securities as another outflow.
- **B:** Reported the cumulative cash gap itself, ignoring counterbalancing capacity.
- **D:** Reversed the sign of the cash gap (640 + 800) before adding the capacity.

Q347 AnalystPrep Q.4064

A liquidity report provides a granular analysis of the firm's marketable asset holdings, is prepared monthly and is submitted 15 business days after month-end. Which report is described?

A. Daily flows report
B. Wholesale liabilities report
C. Liquidity buffer qualifying securities report
D. Currency analysis report

Answer: C

Step by step

1. Monthly frequency with a 15-business-day deadline narrows the list to monthly reports.
2. The report that analyses marketable (qualifying) asset holdings is the liquidity buffer report.

Why C is correct: The liquidity buffer report is the monthly granular analysis of marketable asset holdings.

Why the others are wrong

- **A:** Daily flows are reported daily/weekly and cover cash flows, not asset holdings.
- **B:** Wholesale liabilities are reported weekly and cover prices and volumes of unsecured wholesale funding.
- **D:** Currency analysis is quarterly and covers FX exposures on the balance sheet.

Q348 AnalystPrep Q.4065

REBUILT — Original distractors were giveaways (uncertain stability, highest funding share, none); rewritten with realistic alternatives.

Callable and demand deposits are normally treated as 1-day funds. Under which circumstance may a regulator allow up to 50% of retail deposits to be treated as longer-term funds?

A. When the deposits are shown to remain fairly stable over time
B. When retail deposits are the bank's largest single source of funding
C. When the deposits are covered by a deposit insurance scheme
D. When the bank's loan-to-deposit ratio is below its board limit

Answer: A

Step by step

1. Default regulatory treatment of demand deposits is 1-day money.
2. Some regulators permit a behavioural adjustment where retail deposits are demonstrably stable over time.
3. In that case up to 50% may be treated as longer-term funds, though generally such funds still do not improve liquidity metrics.

Why A is correct: The behavioural adjustment depends on demonstrated stability of the deposits over time.

Why the others are wrong

- **B:** Funding share says nothing about how stable the deposits are.
- **C:** Insurance may reduce run risk but is not the stated criterion for the 50% behavioural adjustment.
- **D:** The LTD ratio is a bank-level liquidity metric, not evidence about deposit behaviour.

Q349 BT Mock Q54 · BT answers P2.T8.20.11.1 · BT notes P2.T8.20.11.1

Umbrella Street Bank, a commercial bank, produces several liquidity risk reports on a daily, weekly, monthly and quarterly basis, including a deposit tracker report. Which of the following metrics is MOST LIKELY to appear in the deposit tracker report?

A. Loan-to-deposit (LTD) ratio (current and forecast) versus a board-approved lower floor (limit) of 120.0%
B. Loan-to-deposit (LTD) ratio (current and forecast) versus a board-approved upper ceiling (limit) of 80.0%
C. Market-to-book ratio of common equity (current and forecast) versus an investor-communicated target of 1.30
D. The leverage-adjusted duration gap (current and forecast) versus a board-approved upper ceiling of 3.5 years

Answer: B

Step by step

1. The deposit tracker shows current and forecast customer deposits and loans and computes the LTD ratio.
2. A higher LTD means more loans funded per unit of deposits, i.e. weaker liquidity, so the board limit is an upper ceiling (e.g. 80%), not a floor.
3. Market-to-book and duration gap are not liquidity-report metrics.

Why B is correct: The deposit tracker's key metric is the LTD ratio, monitored against a maximum limit.

Why the others are wrong

- **A:** Treats the LTD limit as a floor (and at 120%); high LTD is the risk, so a floor points the wrong direction.
- **C:** Market-to-book is an equity valuation measure, not a deposit/liquidity metric.
- **D:** Duration gap is an interest-rate (ALM) measure, not part of the deposit tracker.

Q350 BT answers P2.T8.20.11.2 · BT notes P2.T8.20.11.2

In regard to the types of liquidity risk reports generated by a typical bank, each of the following statements is true EXCEPT which is inaccurate?

A. Wholesale funding reports that break down the bank's funding by price, volume, product type and tenor allow users to evaluate indicators of liquidity strength

B. Liquidity metrics should include undrawn commitments because the existence of undrawn commitments can exacerbate funding shortages at exactly the wrong time

C. In the liquidity coverage ratio (LCR), a strategic and regulatory liquidity metric, retail demand deposits are treated as one-year money with a run-off rate in the range of 0.28% (1/365 days) to 0.40% (1/255 days)

D. The funding concentration report matters to senior managers because a central principle of liquidity management is funding diversity: a bank should not become over-reliant on a single source or sector of funds

Answer: C

Step by step

1. A: wholesale pricing and volume reports track funding cost and volume by product and tenor - true.
2. B: undrawn commitments are treated as cash outflows and reported in the undrawn commitment report - true.
3. C: callable and demand deposits are treated as 1-day money for regulatory purposes, not one-year money, so this statement is false.
4. D: the funding concentration report monitors diversification - true.

Why C is correct: Demand deposits are 1-day money for regulatory purposes, so statement C is inaccurate.

Why the others are wrong

- **A:** True statement, so not the exception.
- **B:** True; undrawn lines can be drawn in a crisis, so they are included as outflows.
- **D:** True; the concentration report flags over-reliance on one depositor, source or sector.

Q351 Schweser Module Quiz 75.1 Q2

REBUILT — Cash flow survival graph was missing from the source; its readings are now stated in the text (9 days before, 24 days after adjustments).

A national bank produces a cash flow survival report from its maturity gap (mismatch) report. The report shows that cumulative cash flow under normal conditions turns negative on day 9, and that after including marketable securities and other adjustments it turns negative on day 24. Which statement best describes the bank's position relative to the Basel III requirement?

A. It meets Basel III because the adjusted survival horizon of 24 days exceeds the unadjusted horizon of 9 days

B. It does not meet Basel III because even after adjustments the survival horizon is shorter than the required 30 days

C. It meets Basel III because the 24-day adjusted horizon falls within the 30-day stress period

D. It does not meet Basel III because the unadjusted horizon is shorter than 30 days, and the adjusted horizon is irrelevant

Answer: B

Step by step

1. Basel III requires the bank to survive a 30-day stress period.
2. Unadjusted survival horizon = 9 days < 30; adjusted horizon = 24 days < 30.
3. Adjusted flows (including liquid securities) are the relevant test, and they still fail the 30-day requirement.

Why B is correct: Both the unadjusted (9 days) and adjusted (24 days) horizons fall short of the 30-day Basel III requirement.

Why the others are wrong

- **A:** Improvement from 9 to 24 days is not the test; the benchmark is 30 days.
- **C:** Reverses the logic: cash must last at least 30 days, so a horizon inside the 30-day window is a failure.
- **D:** Right conclusion, wrong reason; the adjusted horizon (with liquid securities) is the relevant measure and it also fails.

Q352 AnalystPrep Q.4062

REBUILT — Bank was named ACB in the stem and ABC in the options; unified the name and sharpened the wording around the wholesale funding yield curve report.

A regulatory authority compares the firm-specific wholesale funding yield curve of Bank ABC with the curves of its peers. It finds that ABC's curve has risen significantly above the peer curves. What is the most likely implication?

A. ABC is experiencing funding stress
B. ABC has an unusually high level of liquidity
C. ABC has more liabilities than assets
D. ABC has an unusually strong solvency position

Answer: A
Step by step
1. The wholesale pricing and volume report compares a bank's funding yield curve with those of peers.
2. A curve rising materially above peers means the bank is paying more than peers to raise wholesale funds.
3. Paying a premium signals that lenders perceive higher risk or that access to funding is constrained: an early warning of funding stress.

Why A is correct: Regulators use a firm-specific funding yield curve well above peers as an early warning of funding stress, since the bank must pay more to borrow.
Why the others are wrong
- **B:** A highly liquid bank can raise funds cheaply, so its curve would sit at or below peers, not above.
- **C:** The curve is a price signal and does not reveal the balance-sheet composition; liabilities exceeding assets would indicate insolvency, which is not what the curve reports.
- **D:** A strong solvency position would lower, not raise, funding costs relative to peers.

LO 75.c

Explain the process of reporting a liquidity stress test and interpret a liquidity stress test report.

Q353
BT answers P2.T8.20.11.3 · BT notes P2.T8.20.11.3

Which of the following is the most likely output of a bank's liquidity stress test report?
A. Cash flow survival horizon
B. Net interest income projection
C. Leverage ratio of unsecured debt to equity
D. Return on equity versus threshold and target

Answer: A
Step by step
1. A liquidity stress test examines whether the bank can meet cash outflows under extreme scenarios.
2. The primary stress test output is the cash flow survival report, showing the number of days the bank survives (the survival horizon).
3. The other options are earnings or capital measures, not liquidity stress outputs.

Why A is correct: The cash flow survival report, giving the survival horizon, is the most important liquidity stress test output.
Why the others are wrong
- **B:** Net interest income is an earnings/interest rate risk projection, not a liquidity survival measure.
- **C:** A leverage ratio is a solvency/capital measure; it does not show how long the bank can meet cash outflows.
- **D:** Return on equity versus targets is a profitability metric and has no role in liquidity stress testing.

Q354
Schweser Module Quiz 75.2 Q1

One of the most important stress tests for senior bank managers is the cash flow survival report. Banks commonly forecast two types of cumulative cash inflows and outflows to measure liquidity risk. Which of the following best describes the two types of cumulative cash flows, and how often is the report typically run for firm-specific and market-wide stress tests?
A. The two types of cash flows are business as usual and cash flows with adjustments for derivatives and nonmaturing deposits. All stress tests are required weekly.
B. The two types of cash flows are business as usual and cash flows with adjustments for off-balance-sheet items and contingent funding. All stress tests are required daily.
C. The two types of cash flows are business as usual and cash flows with adjustments for retail banking deposits and contingent funding. Firm-specific stress tests are required daily, and market-wide stress tests are required

weekly.

D. The two types of cash flows are business as usual and cash flows with adjustments for liquidated marketable securities and contingent funding. Firm-specific stress tests are required weekly, and market-wide stress tests are required daily.

Answer: D

Step by step

1. The report shows cumulative cash flows under business as usual (BAU) and after mitigating actions: liquidating marketable securities and drawing contingent funding.
2. Frequency: firm-specific stress events weekly; market-wide stress events daily.
3. Only D matches both parts.

Why D is correct: The adjusted flow reflects management's mitigating actions (selling securities, contingent funding), and the report is weekly for firm-specific but daily for market-wide stress.

Why the others are wrong

- **A:** Derivatives and nonmaturing deposits are not the mitigating adjustments, and the frequency is not uniformly weekly.
- **B:** Off-balance-sheet items are not the mitigating adjustment, and not all stress tests are daily; firm-specific is weekly.
- **C:** Retail deposits are not a mitigating action, and the frequencies are reversed (firm-specific is weekly, market-wide daily).

Q355 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Elena Ruiz, the head of Treasury at a bank, reviews a cash flow survival horizon report. Under business-as-usual (BAU) conditions the cumulative cash flow turns negative on day 12. In the second chart, which applies the regulatory stresses and then reflects the immediate sale or repo of marketable securities and contingency funding sources, the cumulative cash flow turns negative on day 38. The bank's internal minimum survival horizon is 30 days. Which of the following is the most appropriate interpretation of the report?

A. The unadjusted horizon is only 12 days; the extension to 38 days comes from mitigating actions, so the bank meets its 30-day minimum only if those actions can actually be executed.

B. The survival horizon lengthens from 12 to 38 days because the stress scenario reduces cumulative cash outflows relative to BAU.

C. The 38-day figure is the BAU survival horizon and the 12-day figure is the stressed horizon, so the bank fails its minimum under stress.

D. The 12-day horizon is irrelevant because survival horizon reports should never reflect liquidation of securities or contingency funding.

Answer: A

Step by step

1. The first chart is the unadjusted cumulative cash flow; it turns negative on day 12.
2. The second chart adds mitigating actions (sale/repo of marketable securities, contingency funding), which extends the survival period to day 38.
3. Day 38 exceeds the 30-day minimum, but only conditional on being able to execute the actions, so the 12-day gap should be reported to management.

Why A is correct: A survival horizon report shows the point at which cumulative cash flow turns negative before and after mitigating actions; the extension to 38 days arises from the actions, not from lower outflows, and depends on their feasibility.

Why the others are wrong

- **B:** Stresses add outflows or reduce inflows; the longer horizon results from mitigating actions, not lower outflows.
- **C:** Reverses the charts: the shorter 12-day horizon is the unadjusted case and the longer 38-day horizon is the one that includes mitigation.
- **D:** The report is explicitly designed to show the horizon before and after mitigating actions such as liquidating securities and accessing contingency funding.

Reading 76. Contingency Funding Planning

Shyam Venkat and Stephen Baird, Liquidity Risk Management: A Practitioner's Perspective (Hoboken, NJ: John Wiley & Sons, 2016). Chapter 7.

LO	Learning objective	Qs
76.a	discuss the relationship between contingency funding planning and liquidity stress testing.	3
76.b	describe best practices in the design of a sound contingency funding plan.	4
76.c	assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting).	16

LO 76.a

Discuss the relationship between contingency funding planning and liquidity stress testing.

Q356 BT Mock Q59 · BT answers P2.T8.20.12.1 · BT notes P2.T8.20.12.1

Which of the following provides the BEST explicit link between a bank's contingency funding plan (CFP) and its liquidity stress testing framework?

A. Communication plan
B. Limit structure and escalation levels
C. Funding sources concentration and non-core funding
D. Industry factors such as profitability trends in the financial sector

Answer: B

Step by step

1. CFPs take liquidity stress test results as inputs and are used for high-severity, low-frequency events.
2. The CFP is explicitly linked to the liquidity stress test through its limit structure and escalation levels.
3. Stress results calibrate limits; a breach moves the firm up escalation levels and triggers the CFP. This is the explicit hinge, so B.

Why B is correct: Limits and escalation levels translate stress test outcomes into quantitative triggers that activate the CFP, making them the explicit link between the two frameworks.

Why the others are wrong

- **A:** The communication plan is a component of the CFP (an output/action), not the link to stress testing; confuses a CFP element with the linking mechanism.
- **C:** Funding concentration and non-core funding are liquidity risk metrics/inputs to stress scenarios, not the mechanism that connects the stress test to CFP activation.
- **D:** Industry profitability trends are an external/scenario environment factor and have no direct role in linking the two frameworks.

Q357 Schweser Module Quiz 76.1 Q1

Which of the following statements regarding contingency funding plans (CFPs) is correct?

A. CFPs are linked to liquidity stress tests through their limit structures.
B. CFPs are used for high-severity, high-frequency contingent liquidity events.
C. CFPs allow for a means to control contingent liquidity events in normal times.
D. Liquidity risk measures used during stressed times are a baseline for developing early warning indicators (EWIs) for CFPs.

Answer: A

Step by step

1. Per LO 76.a, the CFP and stress testing are linked through limits and escalation levels, so A is true.
2. CFPs address high-severity, LOW-frequency events, so B is false.
3. CFPs control events in times of extreme stress, not normal times, so C is false.
4. Measures used in NORMAL circumstances form the baseline for EWIs, so D is false.

Why A is correct: The CFP uses stress test results through a limit structure and escalation levels; the other statements misstate frequency, timing or the source of the EWI baseline.

Why the others are wrong

- **B:** Reverses frequency: CFP events are high-severity, low-frequency.
- **C:** CFPs apply to extreme stress, not normal times.
- **D:** Swaps stressed and normal conditions: normal-time liquidity measures are the EWI baseline.

Q358 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

The head of liquidity risk at a mid-sized bank is redesigning the bank's contingency funding plan (CFP) so that it is properly connected to the bank's liquidity stress testing framework. A colleague proposes several design principles. Which of the following principles is most consistent with the relationship between the CFP and liquidity stress testing?

A. Stress test results should be used only to size the liquidity buffer, while the CFP should be built independently from management judgment about likely crisis actions.

B. Liquidity measures used in business-as-usual risk management should be the foundation for the CFP's early warning indicators, and the two frameworks should be linked through limits and escalation levels.

C. Early warning indicators should be calibrated only from liquidity measures observed under stressed scenarios, because the CFP is used exclusively in periods of extreme stress.

D. The CFP should be activated for the low-impact, high-probability liquidity events that arise regularly, because these events are the ones that stress tests identify most reliably.

Answer: B

Step by step

1. The CFP is the companion of the stress testing framework: stress test results and related information are inputs to the CFP's governance, menu of contingent actions, triggers and decision framework.
2. Contingent events range from low-impact/high-probability (handled in business-as-usual funding and liquidity risk management) to high-impact/low-probability (handled by the CFP).
3. The explicit link is the limit structure and escalation levels; BAU (normal-times) liquidity risk measures are the baseline from which EWIs are defined.
4. Only option B states these points correctly.

Why B is correct: The CFP is explicitly linked to the liquidity risk measurement framework and stress test through its limit structure and escalation levels, and the liquidity measures used in BAU risk management are the foundation from which the CFP defines its early warning indicators.

Why the others are wrong

- **A:** Wrong: stress test results are inputs to the CFP (governance, contingent actions, triggers, decision framework), not just a buffer-sizing tool. The error is treating the two frameworks as separate.
- **C:** Wrong: EWIs are developed from BAU (normal-times) liquidity measures so that deterioration can be detected before a crisis. The error is inferring from the CFP being used in stress that its indicators must be stress-calibrated.
- **D:** Wrong: low-impact, high-probability events are managed within BAU funding and liquidity risk management; the CFP addresses high-impact, low-probability events. The error is reversing the event severity/frequency split.

LO 76.b

Describe best practices in the design of a sound contingency funding plan.

Q359 AnalystPrep Q.4068

Roy Jakes, the financial risk manager for Pathway, is designing a contingency funding plan (CFP). One design consideration is to include appropriate stakeholder groups. Which of the following is NOT among the stakeholder groups that should be involved in the CFP design?

A. Internal audit committee

B. Risk and capital committee

C. Management committee

D. Operations and technology

Answer: A

Step by step

1. The CFP design consideration 'inclusive of appropriate stakeholder groups' lists management committees (e.g., ALCO, risk and capital committee, investment committee), business units, finance, corporate treasury, risk, and operations and technology.
2. Compare each option: B, C and D appear in this list.
3. Internal audit is an independent assurance function that reviews the framework; it is not a listed design stakeholder, so A is the exception.

Why A is correct: Stakeholders are the groups needed to make the CFP operationally ready (design and execution); independent internal audit is not among them.

Why the others are wrong

- **B:** The risk and capital committee is a listed management committee stakeholder.
- **C:** Management committees such as ALCO and the investment committee are explicitly listed stakeholders.
- **D:** Operations and technology are listed stakeholders needed for execution readiness.

Q360 AnalystPrep Q.4067

A key design consideration is that a contingency funding plan (CFP) be integrated with broader risk management frameworks. Which of the following is NOT one of the frameworks with which a CFP should be integrated?

A. Enterprise risk management (ERM)
B. Business continuity and crisis management
C. Financial management
D. Capital management

Answer: C

Step by step

1. The CFP is an integrated part of liquidity risk management and firm-wide risk management: ERM, capital management, and business continuity and crisis management.
2. A, B and D are on this list.
3. Financial management (general finance function) is not a named risk framework for CFP integration, so C is the exception.

Why C is correct: Integration lets the CFP leverage established controls and processes of the risk frameworks; financial management is not one of them.

Why the others are wrong

- **A:** ERM is a named framework for CFP integration.
- **B:** Business continuity and crisis management is named; it strengthens operating and communication procedures in a crisis.
- **D:** Capital management is a named framework for CFP integration.

Q361 BT answers P2.T8.20.12.2 · BT notes P2.T8.20.12.2

When Acme Bank developed its contingency funding plan (CFP), it adopted five design considerations:
I. Aligned to business and risk profiles
II. Integrated with broader risk management frameworks
III. Theoretical basis for precoded instruction set
IV. Inclusive of appropriate stakeholder groups
V. Supported by a communication plan
Although no single universal CFP exists, all of the above are correct EXCEPT one. Which statement identifies the incorrect consideration and gives the correct replacement?

A. I is incorrect; better is: Aligned to liquidity stress testing framework
B. II is incorrect; better is: Integrated with economic value added framework
C. III is incorrect; better is: Operational and actionable, but flexible playbook
D. V is incorrect; better is: Approved by national supervisor

Answer: C

Step by step

1. The five CFP design considerations are: aligned to business and risk profiles; integrated with broader risk man-

agement frameworks; operational and actionable, but flexible playbook; inclusive of appropriate stakeholder groups; supported by a communication plan.
2. Items I, II, IV and V match. Item III ('theoretical basis for precoded instruction set') does not.
3. A CFP must be ready to be put into action quickly yet flexible enough to allow judgment, since not every crisis can be predicted. So C is correct.

Why C is correct: A CFP is a practical, flexible playbook, not a theoretical, rigidly precoded instruction set.

Why the others are wrong

- **A:** I is already correct; the alignment is to business and risk profiles (and risk appetite), not to the stress testing framework as the design consideration.
- **B:** II is already correct; the CFP integrates with ERM, capital management and business continuity, not an economic value added framework.
- **D:** V is already correct; a communication plan is the consideration, and supervisor approval is not a design consideration.

Q362 Schweser Module Quiz 76.1 Q2

Which of the following is a key design consideration in developing a contingency funding plan (CFP)?

A. Aligned to business profile.
B. Supported by a backup plan.
C. Inclusive of all shareholder groups.
D. Can be used in both normal and stressed states.

Answer: A

Step by step

1. Key considerations: aligned to business and risk profiles; integrated with broader risk frameworks; operational, actionable and flexible; inclusive of appropriate stakeholder groups; supported by a communication plan.
2. A matches the first consideration.
3. B (backup plan), C (all shareholders) and D (normal and stressed use) do not match any of the five.

Why A is correct: The CFP must reflect the firm's business activities, products, geography and risk profile.

Why the others are wrong

- **B:** The fifth consideration is a communication plan, not a backup plan.
- **C:** The consideration is appropriate stakeholder groups (management committees, business units, treasury, risk, technology), not all shareholders.
- **D:** CFPs address high-severity, low-frequency stress events, not normal conditions.

LO 76.c

Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting).

Q363 AnalystPrep Q.4075

An institution monitors liquidity health ratios as part of the monitoring and escalation component of its contingency funding plan (CFP). Which of the following best describes the ratio of used capacity to total borrowing capacity?

A. It measures the borrowing capacity still available to the institution, based on the funding already used relative to total borrowing capacity.
B. It measures the cushion that liquid assets provide over required funding needs and tracks liquid assets available to offset volatile funding.
C. It measures the exposure to credit facilities that may need to be funded at some point in the future.
D. It measures the funding and borrowing needed to finance the institution's increased lending activities.

Answer: A

Step by step

1. Used capacity = funding currently utilized across all funding channels (core funding, interbank markets, institutional sales).
2. The ratio compares this with total borrowing capacity, so it shows how much borrowing capacity remains as a

future source of liquidity.
3. Match the other descriptions to their ratios: B = liquid assets to volatile liabilities; C = loans to commitments; D = projected net funding requirements to current unused funding capacity.

Why A is correct: The ratio relates funding already drawn to total borrowing capacity, so it indicates the remaining borrowing headroom available as a future source of liquidity.

Why the others are wrong

- **B:** This describes liquid assets to volatile liabilities, a different liquidity health measure.
- **C:** This describes loans to commitments (undrawn credit facilities), not borrowing capacity.
- **D:** This describes projected net funding requirements to current unused funding capacity.

Q364

AnalystPrep Q.4069

Within the governance and oversight component of a contingency funding plan (CFP), which of the following best describes the role of the corporate treasury function?

A. It serves as adviser and counsel to the management committee and the liquidity crisis team (LCT).
B. As part of business-as-usual activities, it monitors the ongoing business, risk, funding and liquidity profile of the institution and may invoke the CFP.
C. It engineers the CFP and presents it to the management committee for approval.
D. It manages the LCT in periods of crisis and approves recommendations on CFP implementation.

Answer: B

Step by step

1. Corporate treasury's BAU role is oversight of the firm's business, risk, funding and liquidity profile.
2. Based on market, industry, institution-specific conditions and stress test results, the treasurer (with the CFO) may invoke the CFP and convene the LCT.
3. Other options belong to other groups: board of directors (advisor and counsel), LCT (designs the CFP and submits it for approval), management committee (oversees the LCT and reviews CFP recommendations).

Why B is correct: Corporate treasury monitors the liquidity profile in normal times and is the group that can trigger the CFP and convene the LCT.

Why the others are wrong

- **A:** Advising and providing counsel is a board of directors role.
- **C:** Designing the CFP and presenting it for approval is the LCT's role.
- **D:** Overseeing the LCT during a crisis and reviewing CFP recommendations is the management committee's role.

Q365

AnalystPrep Q.4070

REBUILT — Option A 'predatory trade' was vague; reworded to the GARP event 'predatory margin and collateral practices by counterparties'.

A contingency funding plan (CFP) should include contingency plans for events that could adversely affect liquidity. Which of the following does NOT represent a negative liquidity event, but is instead a contingent action that strengthens liquidity?

A. Predatory margin and collateral practices by counterparties
B. Unavailability of Federal Home Loan Bank funding
C. Shifting the funding mix from short-term toward long-term sources
D. The intraday debit cap with Fedwire being exceeded

Answer: C

Step by step

1. Identify which items are adverse events: counterparties imposing predatory margin/collateral terms, loss of FHLB funding and breaching the Fedwire intraday debit cap all hurt liquidity.
2. Shifting from short-term to long-term funding reduces rollover risk and is a contingent action that strengthens the liquidity profile.
3. So C is the only option that does not have a negative liquidity impact.

Why C is correct: Lengthening the funding profile is a contingent action that improves liquidity, whereas the other three are stress events that impair it.

Why the others are wrong

- **A:** Predatory margin and collateral demands drain liquidity and limit contingent actions; it is a negative event.
- **B:** Losing FHLB funding removes a funding source; it is a negative event.

- **D:** Exceeding the Fedwire intraday debit cap restricts payments/settlement; it is a negative event.

Q366 AnalystPrep Q.4072

REBUILT — Option D 'withdrawal of debit caps' is not a recognised factor and was ambiguous; replaced with a decline in funding costs.

Several market factors can impair an institution's ability to take contingent actions that strengthen its liquidity position. Which of the following would NOT be expected to impair that ability?

A. Shutdown of securitization markets
B. Deposit runoff
C. Counterparties declining to roll over funding
D. A decline in the institution's cost of funding

Answer: D

Step by step

1. Factors that reduce the ability to act include closure of securitization markets, reduced repo access, downgrades with higher collateral, higher funding costs, non-rollover of funding and deposit runoff.
2. A, B and C are on that list.
3. A fall in funding cost is the opposite of the listed 'higher funding costs', so it does not impair contingent actions.

Why D is correct: Impairing factors are those that close markets, raise costs or remove funding; cheaper funding makes contingent actions easier, not harder.

Why the others are wrong

- **A:** Without securitization markets, securitizing assets cannot raise liquidity; it impairs actions.
- **B:** Deposit runoff drains funding and shrinks capacity to act.
- **C:** Funding not rolled over creates gaps and reduces room for action.

Q367 AnalystPrep Q.4073

When designing the monitoring and escalation component of a CFP, an institution defines early warning indicators (EWIs) using market and business factors. Which of the following is NOT one of the categories of measures used?

A. Macroeconomic measures
B. Industry measures
C. Financial measures
D. Institution-specific measures

Answer: C

Step by step

1. EWIs based on market and business factors are drawn from macroeconomic, industry and institution-specific measures.
2. 'Financial measures' is not a category in that classification, so C is the exception.

Why C is correct: Market and business EWIs are grouped into macroeconomic, industry and institution-specific measures; there is no separate 'financial measures' group.

Why the others are wrong

- **A:** Macro measures (e.g., repo spreads, haircut trends, CDS moves) are a category.
- **B:** Industry measures (sector profitability, rating actions, competitor analysis) are a category.
- **D:** Institution-specific measures (own rating, spreads, collateral demands) are a category.

Q368 AnalystPrep Q.4076

REBUILT — Replaced 'None of the above' with a substantive option (a Level 2/3 feature) so only one choice is correct.

A CFP typically defines escalation levels (three to five are common, and no specific number is required). Using a three-level illustration, which statement correctly describes Level 1, the first escalation level?

A. The institution is focused on survival and remaining a going concern.
B. It represents elevated monitoring of market conditions and their impact on business segments and performance.
C. Market and/or idiosyncratic events are clearly adversely affecting the business and liquidity risk profile.
D. Drastic contingent actions are taken and require approval from the board.

Answer: B

Step by step

1. Level 1 is the initial level: elevated monitoring, possibly triggered by stress results or a shift in market perception; the LCT is convened for coordination.
2. Level 2: events are noticeably hurting the business and liquidity profile; focus on recovery with a near-term horizon.
3. Level 3: later stage of the crisis, focus on survival, dramatic steps; approvals involve senior management and/or the board.
4. Hence only B describes Level 1.

Why B is correct: Level 1 is the early, precautionary stage of heightened vigilance before adverse events have materially hit the business.

Why the others are wrong

- **A:** Survival focus describes Level 3.
- **C:** Events already adversely affecting the business describe Level 2.
- **D:** Drastic contingent actions needing senior management and/or board approval belong to the later stages (Level 3), not Level 1.

Q369

AnalystPrep Q.4077

REBUILT — Options B, C and 'None of the above' were weak; recast as a comparison of liquidity health measures with market and business measures.

In the monitoring and escalation component of a CFP, which of the following best describes liquidity health measures (e.g., short-term funding as a proportion of total funding, credit rating)?

A. They are broad macroeconomic indicators of systemic distress in the financial system.

B. They are targeted indicators of the institution's liquidity base and strength, where deterioration directly reflects an adverse impact on its liquidity profile.

C. They are indicators that become relevant only after the highest escalation level is reached.

D. They are qualitative judgements of management policy that carry no quantitative limits.

Answer: B

Step by step

1. Market and business measures are broad (macro, industry, institution-specific).
2. Liquidity health measures are more focused: they indicate the institution's liquidity base and strength.
3. Deterioration in them directly signals an adverse effect on current/projected liquidity, so B is correct.

Why B is correct: Liquidity health measures are targeted metrics whose deterioration directly reflects weakening of the institution's liquidity position.

Why the others are wrong

- **A:** This describes market and business (macro) measures, which are broad rather than targeted.
- **C:** They are monitored continuously and used from the early stages, not only at the top escalation level.
- **D:** They are quantitative ratios linked to limits, not qualitative policy statements.

Q370

AnalystPrep Q.4078

REBUILT — Original asked for the FALSE statement with answer 'None of the above' (all true); replaced one statement with a false one so exactly one option is false.

Which of the following statements about scenarios and liquidity gap analysis in a CFP is FALSE?

A. CFP stress scenarios should be aligned with those in the liquidity stress testing framework.

B. CFP scenarios should also be aligned with other frameworks such as recovery and resolution plans.

C. Liquidity stress testing scenarios should cover both systemic and idiosyncratic risks.

D. The CFP must be limited to the stress test scenarios and should not examine any additional scenarios.

Answer: D

Step by step

1. A, B and C restate the CFP guidance: consistency with stress testing and with recovery provisions; stress scenarios cover systemic and idiosyncratic risks.
2. The CFP uses stress test results but may also examine additional, perhaps less likely, scenarios beyond normal scope.
3. So D is false.

Why D is correct: The CFP builds on stress test results but can add extra scenarios to strengthen contingency

plans, so restricting it to stress test scenarios is false.

Why the others are wrong

- **A:** True: consistency with the stress testing framework is required.
- **B:** True: scenarios link to recovery and resolution provisions.
- **C:** True: stress scenarios cover systemic and idiosyncratic risks.

Q371 AnalystPrep Q.4080

REBUILT — Option 'Liquidity monitoring' overlapped with 'Monitoring and escalation'; replaced with a clearly non-CFP component.

Which of the following is NOT one of the key components of a contingency funding plan (CFP)?

A. Governance and oversight
B. Data and reporting
C. Contingent actions
D. Funds transfer pricing

Answer: D

Step by step

1. The five CFP components are governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting.
2. A, B and C are in the list; funds transfer pricing is not.

Why D is correct: Funds transfer pricing is a separate treasury tool, not one of the five CFP components.

Why the others are wrong

- **A:** Component 1 of the CFP.
- **B:** Component 5 of the CFP.
- **C:** Component 3 of the CFP.

Q372 BT answers P2.T8.20.12.3 · BT notes P2.T8.20.12.3

Which of the following is CORRECTLY specified as a key component of a typical contingency funding plan (CFP)?

A. Governance and oversight include stakeholder involvement, roles, and responsibilities
B. Contingent actions include overnight borrowings to total assets
C. Monitoring and escalation include adjusting pricing strategies and issuing subordinated debt
D. Data and reporting includes Level 1, Level 2, and Level 3

Answer: A

Step by step

1. The five CFP components are governance and oversight; scenarios and liquidity gap analysis; contingent actions; monitoring and escalation; data and reporting.
2. Governance and oversight covers stakeholder involvement, roles and responsibilities (corporate treasury, liquidity crisis team, management committee, board), communication, policies and testing: A is correct.
3. Overnight borrowings to total assets is a liquidity health measure used for monitoring, not a contingent action.
4. Adjusting deposit pricing and issuing subordinated debt are contingent actions, not monitoring.
5. Level 1, 2 and 3 are escalation levels (monitoring and escalation), not data and reporting.

Why A is correct: Stakeholder involvement, roles and responsibilities are core elements of the governance and oversight component.

Why the others are wrong

- **B:** Overnight borrowings to total assets is a monitoring measure (liquidity health measure); misplaced under contingent actions.
- **C:** Pricing changes and subordinated debt issuance are contingent actions; misplaced under monitoring and escalation.
- **D:** Escalation levels belong to monitoring and escalation; misplaced under data and reporting.

Q373 GARP PE1 Q71

The CFO of a bank has asked for a review of the bank's contingency funding plan (CFP) and would like to ensure that key components are incorporated in alignment with best practices. Which of the following is a correct statement regarding the key components to be found in an effective CFP?

A. Liquidity stress testing scenarios are designed to focus solely on institution-specific risks and address both market (asset) liquidity and funding liquidity over short-term and prolonged stress periods.
B. Institutions should align their CFP stress scenarios to those in their liquidity stress testing framework, as well as to other frameworks such as recovery and resolution plans.
C. Contingent actions, such as maintaining investment strategies to reinvest maturing securities, should be identified in order to maximize and maintain bank profitability during stressed periods.
D. The liquidity crisis team may invoke the CFP based on a review of the markets, industry, bank-specific conditions, and liquidity stress testing results.

Answer: B

Step by step

1. Scenarios and liquidity gap analysis: CFP scenarios should be consistent with liquidity stress testing scenarios and linked to recovery and resolution provisions, so B is correct.
2. A is false: stress testing covers both systemic and idiosyncratic risks, not solely institution-specific.
3. C is false: contingent actions raise liquidity (e.g., rolling off reinvestment of maturing securities), not protect profitability.
4. D is false: the treasurer, in consultation with the CFO, may invoke the CFP and convene the LCT.

Why B is correct: CFP scenarios must be consistent with the stress-testing framework and linked to recovery and resolution planning.

Why the others are wrong

- **A:** Stress testing covers systemic and idiosyncratic risks; 'solely institution-specific' is wrong.
- **C:** Contingent actions include not reinvesting maturing securities to raise liquidity; the goal is liquidity, not profit.
- **D:** Invocation rests with the treasurer (with the CFO), who then convenes the LCT; attributing it to the LCT is the error.

Q374 GARP PE2 Q67

The treasurer of a large bank is reviewing the bank's contingency funding plan (CFP), including the three escalation levels and the reporting requirements under the plan, and asks a manager to review the CFP against best practices. Which of the following recommendations would be appropriate for the manager to make?

A. Intraday liquidity position reporting is not required under the CFP, as the bank's triggers for all escalation levels should be based on its end-of-day cash balances.
B. As soon as the CFP is escalated to Level 2, the bank should quickly take significant actions to stabilize its liquidity position, including the sale of large portfolios or business units.
C. CFP actions taken under all escalation levels should require approval by senior management and the board of directors.
D. The bank's liquidity crisis team should be responsible for coordinating CFP actions taken under all escalation levels.

Answer: D

Step by step

1. The LCT is the central point of contact, monitors liquidity continuously and ensures coordination and communication across the organization, at every escalation level: D.
2. A is false: reports should capture intraday liquidity positions.
3. B is false: at Level 2 the bank evaluates the feasibility of significant actions and sells less liquid assets; dramatic steps belong to Level 3.
4. C is false: Level 1 actions may need only ALCO/Treasurer approval; senior management and/or the board are involved at Levels 2 and 3.

Why D is correct: The LCT's role is continuous monitoring and coordination across all escalation levels.

Why the others are wrong

- **A:** GARP says reports should capture intraday liquidity positions; end-of-day only is wrong.
- **B:** Dramatic steps are Level 3; Level 2 only evaluates the feasibility of significant actions.
- **C:** Approval requirements are tiered; Level 1 needs only notification and ALCO/Treasurer approval.

Q375 Mock 3 Q10

REBUILT — Source key said the liquidity crisis team invokes the CFP, but GARP says the treasurer (with the CFO) invokes it and convenes the LCT; correct option rewritten.

A treasury manager at a regional bank is drafting the bank's contingency funding plan (CFP) and considers the circumstances in which it should be activated. Which of the following statements would be most appropriate for the manager to include in the plan?

A. The CFP should be activated automatically whenever any single early warning indicator breaches its threshold.
B. The CFP should focus exclusively on bank-specific stress events, since market-wide events are addressed through regulatory facilities.
C. The CFP should specify a single fixed sequence of funding actions to be executed in every stress scenario.
D. The treasurer, in consultation with the CFO, may invoke the CFP and convene the liquidity crisis team based on a review of markets, industry, bank-specific conditions, and liquidity stress testing results.

Answer: D

Key correction: Source key (Mock 3) said D with the liquidity crisis team invoking the CFP; per GARP the treasurer, in consultation with the CFO, invokes the CFP and convenes the LCT, so the option was reworded.

Step by step

1. GARP: the treasurer, with the CFO and others, may invoke the CFP and convene the LCT based on markets, industry, institution-specific conditions and stress results.
2. Automatic activation on any single EWI ignores management judgment, which GARP stresses.
3. CFP scenarios cover systemic and idiosyncratic events, so 'exclusively bank-specific' is wrong.
4. Contingent actions depend on the nature and magnitude of the stress, so one fixed sequence is wrong.

Why D is correct: Invocation is a judgment by the treasurer (with the CFO) informed by market, industry, institution and stress-test inputs.

Why the others are wrong

- **A:** A single indicator breach may be a false alarm; escalation relies on judgment and escalation levels.
- **B:** CFP scenarios include systemic as well as idiosyncratic stress.
- **C:** Which actions are feasible depends on the nature and size of the stress; a fixed sequence ignores this.

Q376 Schweser Module Quiz 76.2 Q1

In the context of governance and oversight of a contingency funding plan (CFP), which role has direct oversight of the liquidity crisis team (LCT)?

A. Board of directors.
B. Risk management.
C. Corporate treasury.
D. Management committee.

Answer: D

Step by step

1. During a crisis, senior management (the management committee) provides oversight of the LCT and consults with the board.
2. The board advises and counsels but does not directly supervise the LCT.
3. Corporate treasury may convene the LCT but does not oversee it; risk management has no such role.

Why D is correct: The management committee of senior management provides direct oversight of the LCT.

Why the others are wrong

- **A:** The board is an advisor/counsel to management and the LCT, not the direct supervisor.
- **B:** Risk management is a stakeholder group, not one of the four key CFP roles.
- **C:** Corporate treasury invokes the CFP and convenes the LCT, but does not oversee it.

Q377 Schweser Module Quiz 76.2 Q2

Which of the following items is an example of increasing a firm's ability to take a contingent action during a stress situation?

A. Securitizing assets.
B. Decreasing lending rates.
C. Increasing capital distributions.

D. Shifting from longer-term to shorter-term funding sources.

Answer: A

Step by step

1. Contingent actions raise liquidity: securitizing assets, selling receivables, issuing subordinated debt, lowering capital distributions, moving to longer-term funding.
2. Securitization converts assets into cash: A.
3. Lower lending rates stimulate loan growth, a cash outflow.
4. Higher capital distributions pay out cash.
5. The contingent action is the reverse of D: shifting from short-term to longer-term funding.

Why A is correct: Securitizing assets generates cash and so improves liquidity.

Why the others are wrong

- **B:** Cheaper loans increase lending, an outflow of cash.
- **C:** Raising distributions drains liquidity; lowering them is the contingent action.
- **D:** Direction reversed; the contingent action is toward longer-term funding.

Q378 Schweser Module Quiz 76.2 Q3

REBUILT — Original included a 'Level 4' option although the illustrative CFP has three levels; replaced with a plausible distractor.

In an illustrative CFP with three escalation levels, which level places more emphasis on closely analyzing current liquidity and the causes of liquidity deterioration?

A. Level 1.

B. Level 2.

C. Level 3.

D. All three levels place equal emphasis on this analysis.

Answer: B

Step by step

1. Level 1: elevated monitoring of market conditions and impact on business; the LCT is convened.
2. Level 2: noticeable market and/or idiosyncratic events hurt liquidity; the bank monitors its current liquidity position and causes of deterioration more closely: B.
3. Level 3: late-stage crisis with dramatic steps and a focus on survival.

Why B is correct: Level 2 is where events are clearly hurting liquidity and detailed analysis of causes is required.

Why the others are wrong

- **A:** Level 1 is heightened monitoring of markets, not detailed analysis of causes.
- **C:** Level 3 focuses on survival and dramatic stabilizing actions.
- **D:** The levels differ in emphasis; Level 2 stresses cause analysis.

Reading 77. Managing Nondeposit Liabilities

Peter Rose and Sylvia Hudgins, Bank Management & Financial Services, 9th Edition (New York, NY: McGraw-Hill, 2013). Chapter 13.

LO	Learning objective	Qs
77.a	distinguish between the various sources of nondeposit liabilities at a bank.	21
77.b	describe and calculate the available funds gap.	6
77.c	discuss factors affecting the choice of nondeposit funding sources.	7
77.d	calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach.	10

LO 77.a

Distinguish between the various sources of nondeposit liabilities at a bank.

Q379

GARP book Ch.13 end-of-chapter (GARP-LR-17 sub 1)

Robertson State Bank lends $70 million of its reserves held at the Federal Reserve Bank to Tenison National Security Bank for 24 hours. Tenison in turn makes a 24-hour loan of the funds to a security dealer before it must return them to Robertson. Show the balance sheet (T-account) entries for these transactions, including repayment.

Answer: see model answer — Day 1 (in $ millions). Robertson: reserves at the Fed -70; Federal funds sold (loaned) +70. Tenison: reserves at the Fed +70 and Federal funds purchased (borrowed) +70; then it lends on: reserves at the Fed -70 and loan to the security dealer +70. Security dealer: cash +70 and borrowings +70. Day 2 (repayment, in reverse order): the dealer repays Tenison (dealer cash -70, borrowings -70; Tenison loan to dealer -70, reserves at the Fed +70); Tenison repays Robertson (Tenison reserves -70, Federal funds purchased -70; Robertson reserves at the Fed +70, Federal funds sold -70). Interest, if any, is settled by an additional transfer or check.

Step by step

1. Fed funds loan: the lender swaps reserves at the Fed for a Fed funds sold asset; the borrower's reserves and its Fed funds purchased liability both rise by 70.
2. Tenison passes the 70 on: its reserves fall 70 and a loan to the dealer (asset) rises 70; the dealer's cash and its liability each rise 70.
3. Repayment unwinds the chain in reverse order, so every entry above is reversed and all balances return to their starting levels (apart from interest).

Why this is correct: A Fed funds loan is a pure swap of reserve deposit for a receivable, mirrored by an identical liability for the borrower; repayment reverses the entries.

Q380

AnalystPrep Q.4098

REBUILT — Option D 'None of the above' replaced with repurchase agreements and the description tightened to the chapter's definition.

Short-term notes with maturities from three or four days to nine months, issued by well-known companies to raise working capital, are known as:

A. Negotiable CDs
B. Commercial paper
C. Eurocurrency deposits
D. Repurchase agreements

Answer: B

Step by step

1. Commercial paper is defined as short-term notes, maturing from a few days to about nine months (maximum 270 days).
2. It is issued by well-known firms at a discount to raise working capital.
3. CDs, Eurodeposits and repos are different instruments.

Why B is correct: Commercial paper is short-term corporate debt issued at a discount.

Why the others are wrong

- **A:** Negotiable CDs are marketable bank deposits of $100,000 or more, not corporate notes.
- **C:** Eurocurrency deposits are deposits at banks outside the home country.
- **D:** Repos are collateralized borrowings via the sale and repurchase of securities.

Q381 GARP book Ch.13 end-of-chapter (GARP-LR-17 sub 2)

Masoner Savings, a small community bank, keeps most of its correspondent deposits with Flagg Metrocenter Bank, a money center bank. When Masoner has a surplus in its correspondent deposit, Flagg automatically invests it in Fed funds loans to other money center banks. Masoner has a temporary surplus of $11 million for 48 hours, which Flagg lends for two business days to Secore Central City Bank, which needs reserves. Show the balance sheet entries to make the loan and to pay it off at maturity.

Answer: see model answer — Start (in $ millions). Masoner (respondent): deposits held with correspondent Flagg -11; Federal funds sold +11. Flagg (correspondent): respondent's (Masoner's) deposit liability -11; Federal funds purchased +11; Flagg then lends to Secore: reserves at the Fed -11, Federal funds sold +11. Secore: reserves at the Fed +11; Federal funds purchased +11. Maturity: Secore repays Flagg (Secore reserves -11, Fed funds purchased -11; Flagg reserves +11, Fed funds sold -11), then Flagg repays Masoner (Flagg Fed funds purchased -11 and Masoner's deposit +11; Masoner deposits with correspondent +11, Fed funds sold -11).

Step by step

1. Respondent lends to its correspondent: Masoner swaps its correspondent deposit for Fed funds sold; Flagg swaps the deposit liability for Fed funds purchased.
2. Flagg relends to Secore through reserve accounts at the Fed: Flagg reserves -11 and Fed funds sold +11; Secore reserves +11 and Fed funds purchased +11.
3. At maturity both legs are reversed in order, restoring all balances (plus interest settled separately).

Why this is correct: A respondent's Fed funds loan to its correspondent is first an exchange of deposit for Fed funds sold; the correspondent's onward loan is a reserve-for-receivable swap.

Q382 GARP book Ch.13 end-of-chapter (GARP-LR-17 sub 3)

Relgade National Bank obtains primary credit of $32 million from the Federal Reserve Bank of San Francisco for a term of seven days. Show the proper entries for granting the loan and for paying it off.

Answer: see model answer — Grant (in $ millions): Relgade's reserve deposit at the Fed +32 (asset) and borrowings from the Federal Reserve (discount window liability) +32. On the Fed's books: loans to depository institutions +32 and Relgade's reserve account +32. Repayment after seven days: Relgade reserves at the Fed -32 and borrowings from the Fed -32 (the Fed's loan asset and Relgade's reserve account fall 32). Interest at the primary credit rate, set slightly above the target Fed funds rate, is paid in addition and is an expense for Relgade. The loan must be backed by acceptable collateral.

Step by step

1. The Fed makes the loan by crediting the borrower's reserve account, so Relgade's assets (reserves) and liabilities (borrowing from the Fed) both rise 32.
2. At repayment the reserve account is debited 32 and the liability is extinguished.
3. Interest at the primary credit rate is paid additionally.

Why this is correct: Discount window borrowing creates reserves and an equal liability at the borrowing bank; repayment destroys both.

Q383 GARP book Ch.13 end-of-chapter (GARP-LR-17 sub 4)

REBUILT — Open numeric question (day-count basis unstated) converted to MCQ with the 360-day convention specified.

Shad Corporation buys a 60-day negotiable CD with a $5 million denomination from Bait Bank and Trust, with a 2.95% annual interest rate (quoted on a 360-day year). Which pair gives the interest the bank pays at maturity and the total it pays back to Shad?

A. Interest $24,247; total $5,024,247
B. Interest $147,500; total $5,147,500
C. Interest $24,583; total $5,024,583
D. Interest $24,583; total $24,583

Answer: C

Step by step

1. Interest = principal × rate × days/360 = 5,000,000 × 0.0295 × 60/360.
2. = $24,583.33 (rounded to $24,583).
3. Total repaid = principal + interest = $5,000,000 + $24,583.33 = $5,024,583.33.

Why C is correct: Negotiable CD interest accrues on a 360-day year for the actual days held, and the principal is repaid in addition to interest.

Why the others are wrong

- **A:** Uses a 365-day year (5,000,000 × 0.0295 × 60/365) instead of the 360-day convention for CDs.
- **B:** Reports a full year of interest (5,000,000 x 0.0295) and ignores the 60/360 fraction.
- **D:** Correct interest, but the total omits the $5 million principal that is also repaid.

Q384

GARP book Ch.13 end-of-chapter (GARP-LR-17 sub 7)

Wells Fargo borrowed $150 million in Fed funds from JP Morgan Chase in New York for 24 hours at the prevailing 2.25% rate. The funds came from JP Morgan's reserve deposit at the Federal Reserve Bank of New York. When the loan was repaid the next day, JP Morgan used $50 million of the returned funds to cover its own reserve needs and lent $100 million in Fed funds to Bank of America, Charlotte, for two days at 2.40%. (a) Construct T-account entries showing the original Fed funds loan and its repayment on the books of JP Morgan, Wells Fargo and Bank of America. (b) Calculate JP Morgan's total interest income on both loans (360-day year).

Answer: see model answer — (a) Loan 1 ($ millions): JP Morgan reserves at the Fed -150, Fed funds sold +150; Wells Fargo reserves +150, Fed funds purchased +150. Repayment next day: reverse (JP Morgan reserves +150, Fed funds sold -150; Wells reserves -150, Fed funds purchased -150). Loan 2: JP Morgan reserves -100, Fed funds sold +100; Bank of America reserves +100, Fed funds purchased +100; after two days reversed. (b) Interest 1 = 150,000,000 x 0.0225 x 1/360 = $9,375.00; interest 2 = 100,000,000 x 0.0240 x 2/360 = $13,333.33; total = $22,708.33.

Step by step

1. Fed funds interest = amount x rate x days/360.
2. Loan 1: 150,000,000 × 0.0225 × 1/360 = $9,375.00.
3. Loan 2: 100,000,000 × 0.0240 × 2/360 = $13,333.33.
4. Total = 9,375.00 + 13,333.33 = $22,708.33; T-accounts: lender swaps reserves for Fed funds sold, borrower books reserves and Fed funds purchased, reversed at repayment.

Why this is correct: Fed funds loans are reserve-for-receivable swaps and interest is simple, actual days over 360.

Q385

GARP book Ch.13 end-of-chapter (GARP-LR-17 sub 8)

REBUILT — Open three-part numeric question converted to a single MCQ with the 365-day conversion as the key twist.

Blue Skies Bank of Florida issues a 90-day negotiable CD of $20 million to ABC Insurance at 2.75% (360-day basis). Which set shows the value of the CD at maturity, ABC's interest income, and ABC's return restated for a 365-day year?

A. Value $20,137,500; interest $137,500; 365-day return 2.750%
B. Value $20,137,500; interest $137,500; 365-day return 2.788%
C. Value $20,137,500; interest $137,500; 365-day return 2.712%
D. Value $20,550,000; interest $550,000; 365-day return 2.750%

Answer: B

Step by step

1. Interest = 20,000,000 × 0.0275 × 90/360 = $137,500.
2. Value at maturity = 20,000,000 + 137,500 = $20,137,500.
3. Period return = 137,500/20,000,000 = 0.6875%; annualized on 365 days: 0.6875%×365/90 = 2.788% (equivalently 2.75% × 365/360).

Why B is correct: A 360-day quoted rate understates the return on a 365-day basis, so it is scaled up by 365/360.

Why the others are wrong

- **A:** Leaves the 360-day quoted rate unchanged and does not restate it for a 365-day year.
- **C:** Converts in the wrong direction (2.75% × 360/365).
- **D:** Takes a full year of interest (20,000,000 x 2.75%) and ignores the 90/360 fraction.

Q386 GARP book Ch.13 end-of-chapter (GARP-LR-19 sub 5)

REBUILT — Open question depended on an external Richmond Fed paper; rebuilt as a self-contained MCQ on the nature of Eurodollar funding.

Which description of Eurodollar deposits as a nondeposit funding source is correct?

A. Euro-denominated deposits held at banks located in the euro area.
B. Dollar deposits held at U.S. banks by foreign residents and subject to full Fed reserve requirements.
C. Dollar-denominated deposits placed in banks outside the United States; a U.S. bank borrowing from its overseas branch records it as liabilities to foreign branches.
D. Mostly long-term fixed-rate deposits with maturities beyond five years, backed by Treasury collateral.

Answer: C

Step by step

1. Eurodollars are dollar-denominated deposits in banks outside the U.S. (e.g., foreign branches of U.S. banks, London banks).
2. A U.S. home office borrowing from its foreign branch books the amount as liabilities to foreign branches.
3. Most Eurodeposits are fixed-rate, interbank, tied to LIBOR, and mature within six months.

Why C is correct: The defining feature is currency (dollars) combined with location (outside the U.S.).

Why the others are wrong

- **A:** Confuses Eurodollars with euro-denominated deposits; the currency is the dollar.
- **B:** Eurodollars are held outside the U.S. and are not subject to U.S. reserve requirements.
- **D:** Most Eurodeposits are short term (majority within six months) and not collateralized.

Q387 AnalystPrep Q.3907

REBUILT — Stem was garbled and the 'overnight borrowing / intraday credit' categories are not in the 2026 reading; rebuilt to test identification of the Fed funds source.

A bank has a temporary reserve shortfall. Its treasurer borrows reserve balances held at the Federal Reserve from another depository institution, normally unsecured, to be returned the next day. Which nondeposit source is this?

A. Repurchase agreement
B. Primary credit at the discount window
C. Negotiable CD
D. Federal funds purchased

Answer: D

Step by step

1. Borrowing reserve balances from another depository institution is a Fed funds loan.
2. Overnight Fed funds loans are normally unsecured and returned the next day.
3. A repo requires securities collateral; discount window borrowing is from the Fed itself; a CD is a marketable deposit.

Why D is correct: Fed funds are short-term, typically overnight, interbank loans of reserve balances.

Why the others are wrong

- **A:** Repos are collateralized by securities sold with an agreement to repurchase; no collateral is mentioned.
- **B:** Primary credit is borrowed from the Fed, not from another depository institution.
- **C:** A negotiable CD is a deposit sold to investors and does not involve borrowing reserves from another bank.

Q388 AnalystPrep Q.4099

A bank with immediate reserve needs can borrow from the Federal Reserve. It can acquire different types of loans, based on its obligations. Gift Bank is a sound financial institution. Its management agrees to borrow a short term loan whose rate is higher than the federal funds rate. Which of the following refers to this type of loan?

A. Primary credit
B. Secondary credit
C. Seasonal credit
D. None of the above

Answer: A

Step by step

1. Discount window loans come in three types: primary, secondary and seasonal credit.

2. Primary credit is short-term (usually overnight) lending to financially sound institutions at a rate slightly above the target fed funds rate.
3. Gift Bank is sound and borrows short term, so the loan is primary credit.

Why A is correct: Primary credit is the discount window facility for financially sound institutions, priced slightly above the fed funds rate.

Why the others are wrong

- **B:** Secondary credit is for institutions that do not qualify for primary credit and carries a higher (penalty) rate; Gift Bank is sound, so it qualifies for primary.
- **C:** Seasonal credit is for longer periods to cover seasonal (intrayear) swings and is the cheapest category; the stem describes no seasonal need.
- **D:** Wrong because one of the listed types fits.

Q389

Schweser Module Quiz 77.1 Q2

Which of the following types of nondeposit funding was created to provide liquidity to mortgage lenders?

A. Fed funds.
B. Repurchase agreements.
C. Federal Home Loan Bank (FHLB) borrowing.
D. Discount window borrowing.

Answer: C

Step by step

1. The FHLB system was chartered in 1932 when banks faced deposit runs.
2. It lends against home mortgages as collateral, improving mortgage liquidity.

Why C is correct: The FHLB system was created to lend to mortgage lenders against mortgage collateral.

Why the others are wrong

- **A:** Fed funds are interbank reserve loans, not mortgage-specific.
- **B:** Repos are collateralized by marketable securities, not designed for mortgage lenders.
- **D:** The discount window is the Fed's lending facility to depository institutions generally.

Q390

AnalystPrep Q.4100

REBUILT — Stem was internally inconsistent (the purchaser was said to pay interest) and distractors were arbitrary; reworded so the issuer pays and made distractors reflect realistic errors.

Promise Bank issues a 50-day negotiable CD with a $10 million denomination to Trust Bank, bearing a 5% annual rate (interest computed on a 360-day year). How much interest must Promise Bank pay at maturity?

A. $68,493
B. $69,444
C. $500,000
D. $10,069,444

Answer: B

Step by step

1. Interest = principal × rate × days/360.
2. 10,000,000 × 0.05 × 50/360 = 69,444.44
3. Interest due is about $69,444.

Why B is correct: Negotiable CD interest on fixed-rate CDs uses a 360-day year, so interest = principal x rate x days/360.

Why the others are wrong

- **A:** Used a 365-day year (10,000,000 x 5% x 50/365 = 68,493); CD interest is quoted on 360 days.
- **C:** Computed a full year of interest (10,000,000 x 5%) ignoring the 50-day term.
- **D:** This is principal plus interest (amount due), not the interest alone.

Q391 AnalystPrep Q.4101

REBUILT — Distractors were arbitrary numbers and the stem called a credit union a bank; reworded and replaced distractors with realistic mistakes.

Central Credit Union borrows $200 million overnight through a repurchase agreement (RP) collateralized by Treasury bills. The RP rate is 5% and interest is computed on a 360-day year. What is the interest cost of the borrowing?

A. $27,397.26
B. $27,777.78
C. $55,555.56
D. $10,000,000.00

Answer: B

Step by step

1. RP interest = amount borrowed × RP rate × days/360.
2. 200,000,000 × 0.05 × 1/360 = 27,777.78

Why B is correct: Repo interest is accrued on a 360-day basis, so one night costs 1/360 of the annual interest.

Why the others are wrong

- **A:** Used a 365-day year (200,000,000 x 5%/365).
- **C:** Charged two days of interest instead of one overnight.
- **D:** Annual interest (200,000,000 x 5%) with no day-count adjustment.

Q392 AnalystPrep Q.4104

REBUILT — Distractors B/D were arbitrary and A (9,345) did not match the interest ($9,333); rebuilt options around realistic errors.

A depository institution promises a 7% annual rate (360-day year) to the buyer of a $320,000, 150-day negotiable CD. How much will the depositor have at maturity?

A. $9,333
B. $329,205
C. $329,333
D. $342,400

Answer: C

Step by step

1. Amount due = principal + principal × days/360 × rate.
2. Interest = 320,000 × 150/360 × 0.07 = 9,333.33
3. Amount due = 320,000 + 9,333 = $329,333.

Why C is correct: The depositor receives principal plus 150/360 of a year's interest.

Why the others are wrong

- **A:** Interest only; forgot to add the principal.
- **B:** Used a 365-day year (interest 9,205).
- **D:** Added a full year of interest (22,400) ignoring the 150-day term.

Q393 AnalystPrep Q.4113

Which of the following types of Negotiable CDs is issued by foreign banks in the US?

A. Domestic CDs
B. Euro CDs
C. Yankee CDs
D. Thrift CDs

Answer: C

Step by step

1. Yankee CDs are issued by foreign banks and sold through their U.S. branches.

Why C is correct: Yankee CDs are foreign-bank CDs sold in the United States.

Why the others are wrong

- **A:** Domestic CDs are issued by U.S. institutions inside the United States.
- **B:** Euro CDs are dollar-denominated CDs issued by banks outside the United States.
- **D:** Thrift CDs are issued by nonbank thrift institutions.

Q394 AnalystPrep Q.4114

A loan which is unwritten, often uncollateralized agreements usually negotiated over a telephone and is payable in the next day is known as:

A. Term loan
B. Overnight loan
C. Continuing contract
D. None of the above

Answer: B

Step by step

1. Fed funds loans are overnight loans, term loans or continuing contracts.
2. Overnight loans are unwritten, often unsecured, arranged by phone and repaid the next day.

Why B is correct: The next-day, verbal, unsecured feature defines an overnight fed funds loan.

Why the others are wrong

- **A:** Term loans run days to months and are often accompanied by a written contract.
- **C:** Continuing contracts renew automatically each day until the borrower or lender ends them, so they are not simply repaid the next day.
- **D:** Wrong because overnight loan fits.

Q395 AnalystPrep Q.4116

REBUILT — Distractors B and D were arbitrary/giveaway; rebuilt around realistic errors.

Trust Bank purchases a 50-day negotiable CD with a $10 million denomination from Promise Bank, bearing a 5% annual rate (360-day year). What total amount will Promise Bank pay Trust Bank at the end of 50 days?

A. $10,000,000
B. $10,068,493
C. $10,069,444
D. $10,500,000

Answer: C

Step by step

1. Amount due = $10{,}000{,}000 + 10{,}000{,}000 \times 0.05 \times 50/360$
2. $= 10{,}000{,}000 + 69{,}444 = \$10{,}069{,}444$

Why C is correct: Total repayment equals principal plus interest on a 360-day basis.

Why the others are wrong

- **A:** Principal only; omitted interest.
- **B:** Used a 365-day year for interest (68,493).
- **D:** Added a full year of interest (500,000).

Q396 Mock 2 Q76

REBUILT — Seasonal credit is cheaper than primary credit, so the stem gave two defensible answers; added that the sound bank has no seasonal need.

A financially sound U.S. bank with no seasonal funding pattern needs a short-term (overnight) loan from the Fed's discount window. It will provide acceptable collateral and wants to pay the least interest available to it. Which form of discount window borrowing is most appropriate?

A. Overnight credit
B. Primary credit
C. Seasonal credit
D. Secondary credit

Answer: B

Step by step

1. Discount window categories are primary, secondary and seasonal credit.
2. Seasonal credit is only for institutions with seasonal (intrayear) swings, which this bank lacks.
3. Of the remaining two, primary credit (sound institutions, slightly above the target fed funds rate) is cheaper than secondary credit (about 50 bp above primary).

Why B is correct: A sound bank without seasonal needs qualifies for primary credit, the lowest-cost category

available to it.

Why the others are wrong

- **A:** Overnight credit is not a discount window category; primary credit is usually overnight.
- **C:** Seasonal credit is cheapest but restricted to institutions with seasonal needs.
- **D:** Secondary credit is the penalty-rate facility for institutions not qualifying for primary credit.

Q397 Schweser Module Quiz 77.1 Q1

Which of the following nondeposit funding sources requires collateral?

A. Fed funds and commercial paper.
B. Commercial paper and discount window borrowing.
C. Fed funds and repurchase agreements.
D. Discount window borrowing and repurchase agreements.

Answer: D

Step by step

1. Repos are collateralized by securities sold and repurchased.
2. Discount window loans must be backed by acceptable collateral (e.g., Treasury or agency securities).
3. Fed funds and commercial paper are unsecured.

Why D is correct: Repos and discount window loans are secured; fed funds and commercial paper are not.

Why the others are wrong

- **A:** Neither fed funds nor commercial paper is collateralized.
- **B:** Commercial paper is unsecured.
- **C:** Fed funds are generally unsecured.

Q398 BT Mock Q69 (rid q0532) · BT answers P2.T8.20.14.1 (rid q0533) · BT notes P2.T8.20.14.1 (rid q0537)

In regard to the various sources of nondeposit funding, which of the following statements is TRUE?

A. The Fed Funds market is the least popular because its effective interest rate is higher than the alternatives
B. Repurchase agreements are more popular than Fed Funds because they are simpler; however, they incur greater credit risk
C. The advantages of the commercial paper (CP) market are the lack of credit risk and consistently high (i.e., low volatility) credit availability; however, volume tends to be low and relative cost is high
D. Advances from Federal Home Loan Banks are highly popular among thrifts and small banks as a stable source of flexible (with respect to maturity) funding at below-market interest rates

Answer: D

Step by step

1. Test each statement against the characteristics of the funding sources.
2. A: Fed funds are among the cheapest and most widely used short-term sources, so false.
3. B: RPs are less popular and more complex than Fed funds, and being collateralized carry less credit risk, so false.
4. C: CP is unsecured and its market is especially sensitive to credit availability risk, so false.
5. D: FHLB advances are mortgage-collateralized, stable, flexible in maturity (overnight to long term) at below-market fixed or variable rates, and lent to community banks and thrifts, so true.

Why D is correct: FHLB advances are collateralized, stable, flexible-maturity funding at below-market rates for community banks and thrifts.

Why the others are wrong

- **A:** Reverses the facts: Fed funds is the most popular and among the cheapest sources (though its rate is volatile).
- **B:** Reverses popularity and credit risk: RPs are less popular and more complex than Fed funds, and collateral makes them less credit risky.
- **C:** Commercial paper is unsecured and exposed to credit availability risk (it can dry up), so 'lack of credit risk' and 'consistently high availability' are wrong.

Q399 GARP book Ch.13 end-of-chapter (GARP-LR-19 sub 4)

REBUILT — Open question depended on an external NY Fed paper; rebuilt as a self-contained MCQ on repo mechanics from the chapter and moved to the repo LO.

Which statement about a repurchase agreement (repo/RP) as a source of funds is correct?

A. It is an unsecured loan, so the lender relies solely on the borrower's creditworthiness.

B. The borrower temporarily sells high-quality securities (starting leg) and agrees to buy them back at a predetermined price (closing leg), and the collateral reduces the lender's credit risk.

C. Repo interest is calculated on a 365-day year using actual days.

D. In a General Collateral Finance (GCF) repo, the borrower must return exactly the securities pledged at the start.

Answer: B

Step by step

1. A repo is a collateralized short-term borrowing: securities are sold for cash on the starting leg.
2. The closing leg repurchases the securities at a predetermined price, so the lender holds collateral, reducing credit risk.
3. Interest uses a 360-day year; GCF repos allow substitution of any agreed-eligible collateral.

Why B is correct: A repo is economically a collateralized Fed funds loan: sale of securities now, repurchase later at a set price.

Why the others are wrong

- **A:** Describes an ordinary Fed funds loan; repos are collateralized.
- **C:** Repo interest uses a 360-day year, not 365.
- **D:** GCF repos allow any satisfactory eligible collateral to be delivered at maturity, not necessarily the original securities.

LO 77.b

Describe and calculate the available funds gap.

Q400 GARP book Ch.13 end-of-chapter (Thyme Bank)

Thyme Bank of New York expects new deposit inflows next month of $265 million and deposit withdrawals of $425 million. The bank's economics department has projected that new loan demand will reach $400 million and customers with approved credit lines will need $175 million in cash. The bank will sell $450 million in securities, but plans to add $60 million in new securities to its portfolio. What is its projected available funds gap?

Answer: see model answer — AFG = (loans and other investments) - (deposit inflows and other available funds). Uses of funds = loans 400 + credit-line drawings 175 + new securities 60 + deposit withdrawals 425 = $1,060 million. Sources = new deposits 265 + securities sold 450 = $715 million. AFG = 1,060 - 715 = $345 million, which the bank must cover with nondeposit borrowing (plus a small safety margin).

Step by step

1. Uses of funds: new loans 400 + credit-line drawings 175 + securities purchased 60 + deposit withdrawals 425 = 1,060.
2. Sources of funds: new deposits 265 + securities sold 450 = 715.
3. AFG $= 1{,}060 - 715 = 345$ million.

Why this is correct: The AFG is projected loans and investments (with withdrawals reducing net deposit funding) minus deposit inflows and other available funds; a positive gap must be funded in the money markets.

Q401 AP Mock 2 Q24 · AnalystPrep Q.4102

REBUILT — Original options (600, 625, 800, 975 / 600, 775, 625, 425) had untraceable distractors; replaced with options from realistic errors.

Prime Limited is expecting new deposit inflows of USD 400 million and deposit withdrawals of USD 600 million next month. The bank has projected that new loan demand will reach USD 500 million, and customers with approved credit lines will need USD 200 million in cash. The bank will sell USD 375 million in securities but plans to add USD 100 million in securities to its portfolio. Calculate the projected available funds gap.

A. USD 25 million

B. USD 625 million

C. USD 775 million

D. USD 1,400 million

Answer: B

Step by step

1. Uses: loans 500 + credit-line drawings 200 + securities purchased 100 + withdrawals 600 = 1,400.
2. Sources: new deposits 400 + securities sold 375 = 775.
3. AFG = 1,400 − 775 = 625 million (net deposit flow is 400 - 600 = -200).

Why B is correct: AFG = projected loans and investments minus deposit inflows and other available funds, with withdrawals reducing deposit funding; the result is a USD 625 million shortfall to be borrowed.

Why the others are wrong

- **A:** 25 ignores the USD 600 million of withdrawals (800 - 775).
- **C:** 775 is total sources only; the uses were never subtracted.
- **D:** 1,400 is total uses only; the inflows and securities sale were not subtracted.

Q402

AnalystPrep Q.4105

REBUILT — Original distractors (326, 423, 347) did not trace to identifiable errors; replaced with options generated from realistic omissions.

A commercial bank has a new loan request that meets its quality standards of $660 million; it expects to purchase $330 million in new Treasury securities being issued on Friday the same week and expects drawings on credit lines from its best corporate customers of $358 million. Deposits and other customer funds received today total $456 million, and those expected in the coming week will bring in another $265 million. If the current funding gap is zero, what is the bank's estimated available funds gap (AFG) for the coming week?

A. $269 million

B. $627 million

C. $892 million

D. $1,083 million

Answer: B

Step by step

1. Loans and investments: 660 + 330 + 358 = 1,348.
2. Deposit inflows and other funds: 456 + 265 = 721.
3. AFG = 1,348 − 721 = 627 million.

Why B is correct: The AFG is current and projected loans and investments (including credit-line drawings) less current and expected deposit inflows; with a zero starting gap nothing else is added.

Why the others are wrong

- **A:** 269 omits the $358 million credit-line drawings from uses (990 - 721).
- **C:** 892 subtracts only today's deposits of 456 (1,348 - 456) and ignores the $265 million expected inflow.
- **D:** 1,083 subtracts only the expected inflow of 265 (1,348 - 265) and ignores today's deposits of 456.

Q403

BT answers P2.T8.20.14.2 · BT notes P2.T8.20.14.2

REBUILT — Original distractors (37.5, 75, 300) were arbitrary multiples of the answer; replaced with options produced by realistic omissions.

A commercial bank's situation today includes the following:
- New loan requests (meeting quality standards) of $250.0 million
- Plans to purchase $100.0 million in Treasuries in the coming week
- Anticipated drawings on credit lines from its best corporate customers of $140.0 million
- Deposits and other customer funds received today of $110.0 million
- Expected deposit inflows of $230.0 million in the coming week

What is the bank's available funds gap (AFG)?

A. $50.0 million

B. $150.0 million

C. $260.0 million

D. $490.0 million

Answer: B

Step by step

1. Loans and investments: 250 + 100 + 140 = 490.

2. Deposit inflows and other funds: 110 + 230 = 340.
3. AFG = 490 − 340 = 150 million.

Why B is correct: AFG = current and projected loans and other investments (including credit-line drawings) minus current and expected deposit inflows.

Why the others are wrong

- **A:** 50 omits the $100 million Treasury purchase from uses (390 - 340).
- **C:** 260 subtracts only the expected inflow of 230 (490 - 230) and ignores today's deposits of 110.
- **D:** 490 is total uses only; the deposit inflows were not subtracted.

Q404

GARP PE2 Q59 · Mock 4 Q60

REBUILT — Distractor 15 was untraceable and the source key conflicted; kept stem, replaced option D with an attributable error, confirmed key C.

A manager at a financial institution is assessing the cash flow requirements of the institution's largest customers using the available funds gap (AFG) model. The manager obtains the following current and projected inflows and outflows of funds:

Item	Amount
Deposits received today	USD 60 million
Approved new loan requests	USD 57 million
Expected drawdown on corporate client's credit lines	USD 16 million
Deposit inflows expected within the next week	USD 12 million
10-year Treasury securities expected to be purchased this week	USD 35 million
Other customer funds available today	USD 45 million

Assuming all else is held constant, if the institution's objective is to maintain an AFG of zero, what is the maximum amount of unexpected customer withdrawals it can meet within the next week?

A. USD 0 million
B. USD 3 million
C. USD 9 million
D. USD 25 million

Answer: C

Key correction: GARP PE2 key (per source notes) said D (15), but its own working supports C; correct is 9 because sources exceed uses by 9.

Step by step

1. Uses: loans 57 + credit-line drawdowns 16 + Treasuries 35 = 108.
2. Sources: deposits today 60 + expected deposits 12 + other funds 45 = 117.
3. AFG = 108 − 117 = −9, a surplus of 9.
4. Unexpected withdrawals add to AFG, so to keep AFG at zero they can be at most 9 million.

Why C is correct: Sources exceed uses by USD 9 million, so withdrawals of up to 9 million raise the AFG only from -9 to 0.

Why the others are wrong

- **A:** 0 wrongly assumes a zero-AFG target leaves no room; it ignores the 9 surplus.
- **B:** 3 omits the 12 of expected deposits from sources (105) and gets |108 - 105|, the wrong sign and size of cushion.
- **D:** 25 omits the 16 credit-line drawdown from uses (92), giving 117 - 92 = 25.

Q405

Schweser Module Quiz 77.1 Q3

Barbara Friedman, a bank manager on the asset-liability committee, must estimate the amount of money market funding she expects the bank to need in the coming week. Friedman estimates that the bank will make $60 million in new loans in the coming week. The bank does not plan to make any security investments but does expect additional drawdowns on credit lines to equal $10 million. The bank is in a highly competitive deposit market and expects only $15 million in new deposits in the coming week. However, based on previous years' experience, she expects that two of the bank's largest customers will withdraw $1 million each in the coming week. Friedman should estimate the available funds gap for the coming week to be:

A. $43 million
B. $45 million
C. $53 million
D. $57 million

Answer: D

Step by step

1. Loans and investments: 60 + 10 = 70.
2. Net deposit inflow: 15 - 2 = 13.
3. AFG = 70 − 13 = 57 million.

Why D is correct: Expected withdrawals reduce deposit funding, so net deposits are 13 and the bank must borrow 57.

Why the others are wrong

- **A:** 43 ignores the credit-line drawdowns and adds the withdrawals to deposits (60 - 17).
- **B:** 45 ignores both credit-line drawdowns and withdrawals (60 - 15).
- **C:** 53 adds the 2 of withdrawals to deposits (70 - 17) instead of subtracting.

LO 77.c

Discuss factors affecting the choice of nondeposit funding sources.

Q406

GARP book Ch.12 end-of-chapter (rid q0035)

REBUILT — Open, weakly mapped case question from a deposit-services chapter; converted to a self-contained MCQ testing the 77.c risk factor (credit availability risk of purchased funds) using the same balance sheet.

Rhinestone National Bank reports the following (millions):

Assets	Liabilities and Equity
Cash and interbank deposits $50	Core deposits $50
Short-term security investments 15	Large negotiable CDs 150
Total loans, gross 400	Deposits placed by brokers 65
Long-term securities 150	Other deposits 45
Other assets 10	Money market liabilities 195
Total assets $625	Other liabilities 65
	Equity capital 55
	Total liabilities and equity capital $625

Treat large negotiable CDs, brokered deposits and money market liabilities as purchased (nondeposit-type) funds. What share of total assets do these funds finance, and which funding-choice risk is most important for this bank?

A. 65.6%; credit availability risk, because these markets can ration credit to the bank and it may have to switch to a costlier source

B. 65.6%; interest rate risk only, because the shortest-term funds have the least volatile rates

C. 71.9%; credit availability risk

D. 15.2%; credit availability risk is minor because core deposits dominate

Answer: A

Step by step

1. Purchased funds = 150 (large CDs) + 65 (brokered) + 195 (money market) = 410 million.
2. Share of total assets = 410/625 = 65.6%.
3. Negotiable CD, Eurodollar and commercial paper markets are especially sensitive to credit availability risk: lenders may ration credit or demand a higher price, forcing the bank to switch to costlier sources.
4. With about two-thirds of assets financed this way (versus only 95/625 = 15.2% in core and other deposits), availability risk is the dominant concern.

Why A is correct: Two-thirds of assets are financed by purchased funds whose supply depends on lenders' willingness to lend, so credit availability risk dominates the funding-choice analysis.

Why the others are wrong

- **B:** Share is right but the logic is reversed: the shorter the term, the MORE volatile the rate, and 'only interest rate risk' ignores availability risk.
- **C:** Used 410/570 (total liabilities excluding equity) as the denominator instead of total assets of 625.
- **D:** Computed (50 + 45)/625 = 15.2%, the stable (core plus other deposits) funding share, not the purchased-funds share; it also misreads the risk.

Q407 GARP book Ch.13 end-of-chapter (rid q0044)

REBUILT — Open-ended 'defend your decision' question converted to a single-answer MCQ asking for the effective cost of commercial paper using the effective-cost formula.

Best-of-Times Financial plans to raise $775 million in short-term funds, of which $700 million is net investable (the rest is absorbed by other cash needs). Market rates and noninterest costs are:

Source	Market rate	Noninterest cost
Fed funds	2.25%	0.25%
Negotiable CDs	2.40%	0.25%
Eurodollars	2.30%	0.35%
Commercial paper	2.35%	0.50%
Discount window	2.75%	0.25%

Noninterest costs are stated as a percentage of funds raised. What is the effective cost rate of the commercial paper?

A. 2.60%
B. 2.85%
C. 3.16%
D. 3.32%

Answer: C

Step by step

1. Effective cost = (interest cost + noninterest cost) / net investable funds.
2. Interest cost = $0.0235 \times 775 = 18.2125$ million; noninterest cost = $0.0050 \times 775 = 3.875$ million.
3. Total cost = 22.0875 million.
4. Effective cost = $22.0875/700 = 3.155\% \approx 3.16\%$. (For comparison: Fed funds 2.77%, CDs and Eurodollars 2.93%, discount window 3.32%.)

Why C is correct: Both interest and noninterest costs are incurred on all $775 million raised but only $700 million can be invested in earning assets, so the cost is spread over the smaller base.

Why the others are wrong

- **A:** Ignored noninterest costs: $2.35\% \times 775/700 = 2.60\%$.
- **B:** Added the rates (2.35% + 0.50%) but did not divide by net investable funds (used the 775 raised as the base).
- **D:** This is the discount window's effective cost $(2.75 + 0.25)\% \times 775/700 = 3.32\%$, the wrong source.

Q408 GARP book Ch.13 end-of-chapter Q10 (rid q0045) · GARP book Ch.13 end-of-chapter (rid q0048)

Surfs-Up Security Savings must raise $80 million in money market funds for a six-week loan to a large corporate customer. Market rates:

Source	Rate
Federal funds, average for week just concluded	1.98%
Discount window of the Federal Reserve Bank	2.25%
CDs (prime rated, secondary market), one month	2.52%
CDs, three months	2.80%
CDs, six months	3.18%
Eurodollar deposits, three months	3.00%
Commercial paper (directly placed), one month	2.33%
Commercial paper, three months	2.70%

The economics department forecasts a substantial rise in money market rates over the next six weeks. Where and how should the bank raise the funds? Consider legal reserve requirements, regulations, and what happens to the relative attractiveness of each source if rates rise continually. Alternative scenario: how would your recommendation change if rates instead decline significantly?

Answer: see model answer — Rising rates: the need is six weeks, so lock in a fixed cost for the whole period rather than roll overnight funds. Fed funds (1.98%) look cheapest today but reprice daily at ever-higher rates and are the most volatile source; any reserve requirement or insurance cost also raises effective cost because it reduces net investable funds. A term source such as directly placed one-month commercial paper (2.33%, the cheapest term rate, rolled once for the last two weeks) or a CD or Eurodollar placement matched to about six weeks fixes the rate, whereas a three-month instrument over-borrows by about seven weeks. The discount window (2.25%) is administered and lags the market, so it becomes relatively more attractive as rates rise, but it is not meant for routine funding and access may be limited by the Fed. Falling rates: reverse the logic and borrow short, using

overnight or short-term Fed funds and rolling it so each renewal reprices lower; avoid locking in three- or six-month funds at today's higher rates.
Why this is correct: Factors: relative cost (including noninterest costs on net investable funds), rate volatility and credit availability, maturity of the need, institution size, and regulations.

Q409 AnalystPrep Q.4106 (rid q0522)

Banks and other lending affiliates within Niche Bank Holding Company have substantial loan demand from companies planning facility expansions before the next financial year. The holding company plans to raise $1 billion in short-term funds this week, of which $0.9 billion will be used to meet the new loan requests (net investable funds).
Current annual rates:

Source	Market interest rate	Noninterest cost rate
Federal Funds	2.10%	0.42%
Negotiable CDs	2.25%	0.42%
Eurodollars	2.15%	0.52%
Commercial paper	2.20%	0.67%
Fed Discount Rate	3.10%	0.42%

Noninterest costs are a percentage of the $1 billion raised. Calculate the effective cost rates of the Federal Funds and Negotiable CDs, respectively.

A. 2.10% and 2.25%
B. 2.80% and 2.97%
C. 2.97% and 3.19%
D. 2.80% and 3.91%

Answer: B

Step by step

1. Effective cost = (interest cost + noninterest cost) / net investable funds; costs are on the $1,000 million raised, the base is $900 million.
2. Fed funds: $(0.0210 + 0.0042) \times 1000 = 25.2$ million; $25.2/900 = 2.80\%$.
3. Negotiable CDs: $(0.0225 + 0.0042) \times 1000 = 26.7$ million; $26.7/900 = 2.967\% \approx 2.97\%$.

Why B is correct: Costs are incurred on all funds raised but only 90% is invested in earning assets, so (rate + noninterest rate) is grossed up by 1/0.9.

Why the others are wrong

- **A:** Quoted market rates only; ignored both noninterest costs and the net investable adjustment.
- **C:** These are the effective costs of other sources (Eurodollars at 2.97% and commercial paper at 2.87%/0.9 = 3.19%), not Fed funds and CDs.
- **D:** Fed funds is right (2.80%) but 3.91% is the Fed discount rate's effective cost 3.52%/0.9, not the CD's.

Q410 AnalystPrep Q.4107 (rid q0523)

REBUILT — Stem did not say cost was the only criterion and commercial paper was in the table but not the options; added 'effective cost is the only criterion' so the answer follows from the effective-cost ranking.

Niche Bank Holding Company plans to raise $1 billion in short-term funds, of which $0.9 billion is net investable.
Current annual rates:

Source	Market interest rate	Noninterest cost rate
Federal Funds	2.10%	0.42%
Negotiable CDs	2.25%	0.42%
Eurodollars	2.15%	0.52%
Commercial paper	2.20%	0.67%
Fed Discount Rate	3.10%	0.42%

Noninterest costs are a percentage of the $1 billion raised. If effective cost is the only selection criterion, which source should management use?

A. Federal Funds
B. Negotiable CDs
C. Eurodollars
D. Fed Discount Rate

Answer: A

Step by step

1. Effective cost = (market rate + noninterest rate)/0.9 for each source.
2. Fed funds: 2.52/0.9 = 2.80%; CDs: 2.67/0.9 = 2.97%; Eurodollars: 2.67/0.9 = 2.97%; commercial paper: 2.87/0.9 = 3.19%; discount rate: 3.52/0.9 = 3.91%.
3. The lowest is Fed funds at 2.80%.

Why A is correct: Fed funds has both the lowest market rate and a low noninterest cost, so the lowest effective cost.

Why the others are wrong

- **B:** Effective cost 2.97%, above Fed funds: the CD rate is higher and noninterest cost is the same.
- **C:** Eurodollars have a lower market rate than CDs but a higher noninterest cost; effective cost 2.97%, still above Fed funds. Choosing it looks at the 2.15% rate alone.
- **D:** Highest effective cost (3.91%); the discount rate is set above market and is a backup source.

Q411 Mock 3 Q6 (rid q0544)

Two institutions face different funding issues. Bank A, a mid-size regional bank, has a shortfall in legal reserves for the day and needs an alternative to cover it. Bank B, a small community bank, has had a much greater than anticipated shortfall in long-term certificate of deposit (CD) renewals due to fierce local competition for retail deposits; it has traditionally used stable CDs to fund its home mortgage portfolio. What is the most appropriate funding response of each, considering timing and availability of nondeposit funds?

A. Bank A should borrow from the wholesale deposit market and Bank B should fund itself through the Eurocurrency deposit market.

B. Bank A should fund itself through the commercial paper (CP) market and Bank B should borrow from the Federal funds market.

C. Bank A should borrow from the Federal funds market and Bank B should borrow from the Federal Home Loan Banks.

D. Bank A should issue debentures and Bank B should fund itself through the CP market.

Answer: C

Step by step

1. Bank A's need is overnight and immediate: Fed funds are immediately available and the standard source for meeting reserve requirements.
2. Bank B's need is long term and mortgage related, and it is small: FHLB advances are longer-term, stable, collateralized by home mortgages and lent to community banks and thrifts.
3. CP and long-term debt are not immediately available, and Eurodollar/CP markets favor large, highly rated institutions (units of $1 million).

Why C is correct: Match maturity and timing to the need and consider size: Fed funds for the same-day reserve gap, FHLB advances for a small bank's long-term mortgage funding.

Why the others are wrong

- **A:** Wholesale deposits are not immediate, and Eurocurrency deposits come in $1 million units, are short term and are open mainly to large institutions, so they suit neither need.
- **B:** CP is not immediately available (wrong for the same-day gap) and overnight Fed funds are too short and volatile to replace long-term CD funding of mortgages.
- **D:** Debentures are long term and not immediately available, so cannot cover a same-day gap; CP is short term and inaccessible to a small bank, mismatched with mortgages.

Q412 Schweser Module Quiz 77.2 Q1

Kris Gaines, Treasurer at Palm Air Bank and Trust, is considering ways to meet a funding gap created by greater than expected loan demand. Palm Air is a medium-sized bank located in Florida. The funding gap is approximately $850,000. Gaines is choosing between several nondepository funding types. The funds are needed immediately. Which type of funding is most appropriate in this situation?

A. Commercial paper.

B. Negotiable certificates of deposit (CDs).

C. Federal funds borrowing.

D. Eurodollar deposits.

Answer: C

Step by step

1. Identify the need: about $850,000, required immediately, by a medium-sized bank.
2. Size factor: commercial paper, negotiable CDs and Eurodollar deposits trade in units of $1 million or more, which exceeds the $850,000 gap; smaller institutions may also lack access to these markets.
3. Maturity/timing factor: fed funds are immediately available, whereas commercial paper (and long-term debt) are not necessarily immediately available.
4. Fed funds can be borrowed in smaller amounts and are available immediately, so they fit both the size and timing needs.

Why C is correct: The choice among nondeposit sources depends on cost, risk, maturity, size and regulation; here size (under $1 million) and immediacy point to fed funds.

Why the others are wrong

- **A:** Commercial paper is issued in large units, requires a credit standing a medium-sized bank may lack, and is not necessarily immediately available; ignoring the size and timing constraints leads to this choice.
- **B:** Negotiable CDs are sold in multiples of $1 million, above the $850,000 need, and investors prefer those of money-center banks with an active secondary market; choosing them overlooks the size factor.
- **D:** Eurodollar deposits are borrowed in multiples of $1 million and are usually available only to money-center firms with top credit ratings; choosing them overlooks size and access.

LO 77.d

Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach.

Q413

GARP book Ch.13 Q11 (a)-(c)

REBUILT — Four overlapping records of one multipart GARP question were merged; the ambiguous part (a) and the equity cost (stated as a 12% required return) were made explicit to match the GARP formulas.

June Bug Bank and Trust has raised $750 million: $200 million of checkable deposits, $400 million of time and savings deposits, $100 million of money market borrowings and $50 million of stockholders' equity. The cost of each funding source is shown below.

Funding source	Amount ($m)	Interest cost	Noninterest cost	Reserve requirement and collection
Checkable deposits	200	0.75%	2.00%	0.75%
Time and savings deposits	400	2.50%	0.50%	0.50%
Money market borrowings	100	2.75%	0.25%	0.25%
Stockholders' equity	50	required return of 12% after taxes	-	-

The corporate tax rate is 35%.

(a) Using the historical average cost approach, calculate the weighted average interest cost of the borrowed funds (deposits and money market borrowings, $700 million), and the weighted average all-in cost including noninterest and reserve/collection costs.

(b) If earning assets total $700 million, what is the break-even cost rate?

(c) What is June Bug's overall historical weighted average cost of capital (the minimum before-tax return required on earning assets)?

Answer: see model answer — (a) Interest cost = 200(0.75%) + 400(2.50%) + 100(2.75%) = $14.25m, so 14.25/700 = 2.04%; all-in cost = 14.25 + 6.25 (noninterest) + 3.75 (reserve/collection) = $24.25m, or 24.25/700 = 3.46%. (b) Break-even cost rate = (interest + other operating costs)/earning assets = 24.25/700 = 3.46%. (c) Overall cost = break-even + $\frac{12\%}{1-0.35} \times \frac{50}{700}$ = 3.46% + 18.46% × 0.0714 ≈ 3.46% + 1.32% = 4.78%.

Step by step

1. Interest by source: checkable 200 × 0.75% = 1.50; time/savings 400 × 2.50% = 10.00; money market 100 × 2.75% = 2.75; total 14.25 million.
2. Weighted average interest cost on the 700 million borrowed: 14.25/700 = 2.04%.
3. Noninterest costs: 4.00 + 2.00 + 0.25 = 6.25; reserve/collection costs: 1.50 + 2.00 + 0.25 = 3.75. All-in cost 14.25 + 6.25 + 3.75 = 24.25.
4. Break-even cost rate = 24.25/700 = 3.46%.
5. Equity add-on = $\frac{12\%}{1-0.35} \times \frac{50}{700}$ = 18.46% × 0.07143 = 1.32%.
6. Overall historical weighted average cost = 3.46% + 1.32% = 4.78%.

Why this is correct: The historical average cost approach divides all costs of borrowed funds by earning assets to get the break-even rate, then adds the pre-tax equivalent of the stockholders' required return spread over earning assets.

Q414 GARP book Ch.13 Q12 · GARP book Ch.13 Q12 (duplicate record)

Inspiration Savings Association is considering funding a package of new loans of $400 million. It must raise $450 million to have $400 million available for loans. It expects to raise $325 million of time deposits at an average interest rate of 1.75%, with noninterest (operating) costs of 0.45% of the amount raised. The other $125 million will come from noninterest-bearing transaction deposits whose noninterest costs are 2.00% of the amount raised. What is the Association's projected pooled-funds marginal cost? What hurdle rate must it achieve on its earning assets?

Answer: see model answer — Expected costs = time deposits 325 × (1.75% + 0.45%) = 7.15 plus transaction deposits 125 × 2.00% = 2.50, total 9.65 million. Pooled-funds marginal cost = 9.65/450 = 2.14%. Only 400 of the 450 million raised can be placed in earning assets, so the hurdle rate = 9.65/400 = 2.41%.

Step by step

1. Time deposits: interest 325 × 1.75% = 5.6875 plus operating 325 × 0.45% = 1.4625, total 7.15.
2. Transaction deposits: 125 × 2.00% = 2.50 (no interest).
3. Total expected costs = 7.15 + 2.50 = 9.65 million.
4. Pooled-funds marginal cost = 9.65/450 = 2.14%.
5. Hurdle rate = 9.65/400 = 2.41%, since only 400 of the 450 raised can be placed in earning assets.

Why this is correct: The pooled-funds approach divides expected costs of all new funds by the funds raised, and by the smaller amount actually placed in earning assets to get the hurdle rate.

Q415 AnalystPrep Q.4109

REBUILT — Source key included the cost of equity in the break-even numerator (6.2%), contradicting the GARP definition; the stem now specifies borrowed funds and options were repaired.

ABC Bank raised $1,530 million from the sources below.

Funding source	Amount ($m)	Interest cost	Noninterest cost	Additional (reserve/collection) cost	Total cost
Money market borrowings	275	3.03%	0.24%	0.24%	3.51%
Time and savings deposits	580	2.27%	0.23%	0.60%	3.10%
Checkable deposits	385	0.52%	1.96%	0.80%	3.28%
Stockholders' equity	290	12.75%	-	-	12.75%

Earning assets are $1,240 million ($1,530 - $290). Using the historical average cost approach, what is the break-even cost rate on borrowed funds invested in earning assets?

A. 2.6%
B. 5.0%
C. 3.2%
D. 6.2%

Answer: C

Key correction: Source key (AnalystPrep) said D (6.2%); correct is C (3.2%) because the break-even rate covers only the costs of borrowed funds; the stockholders' return enters later in the overall weighted average cost.

Step by step

1. Dollar costs: money market 275×3.51% = 9.65; time/savings 580×3.10% = 17.98; checkable 385×3.28% = 12.63.
2. Total cost of borrowed funds = 9.65 + 17.98 + 12.63 = 40.26 million.
3. Equity is excluded from break-even; its cost is added in the overall weighted average cost.
4. Break-even = 40.26/1,240 = 3.25% ≈ 3.2%.

Why C is correct: Break-even cost rate = (interest + other operating costs of borrowed funds) / earning assets; equity is not a borrowed fund.

Why the others are wrong

- **A:** 2.6% divides the borrowed-funds cost by total funds ($1,530m) instead of earning assets.
- **B:** 5.0% includes the $36.98m cost of equity and divides by total funds $1,530m (77.24/1,530).
- **D:** 6.2% includes the $36.98m cost of equity in the numerator (77.24/1,240), mixing the overall cost into break-even.

Q416

AnalystPrep Q.4110

REBUILT — Original distractors (28%, 32%, 40%) were implausible giveaways; replaced with options from realistic errors.

Mug Bank has estimated overhead expenses and salaries (other operating costs) of $13 million. Its earning assets are $540 million and its total interest cost on borrowed funds is $127 million. What is the bank's break-even cost rate on borrowed funds invested in earning assets?

A. 23.5%
B. 25.9%
C. 26.6%
D. 2.4%

Answer: B

Step by step

1. Break-even $= \frac{\text{interest+other operating costs}}{\text{earning assets}}$.
2. Numerator $= 127 + 13 = 140$ million.
3. $140/540 = 25.9\%$.

Why B is correct: Both interest and operating costs must be covered by the yield on earning assets.

Why the others are wrong

- **A:** 23.5% omits the $13m operating costs (127/540).
- **C:** 26.6% subtracts the operating costs from earning assets, 140/(540-13), a mistaken adjustment.
- **D:** 2.4% uses only the operating costs (13/540).

Q417

AnalystPrep Q.4111

REBUILT — Options replaced so each distractor stems from a realistic formula error.

Pathway Bank has the following data.

Item	Value
Break-even cost	14.50%
After-tax cost of stockholders' investment	12%
Tax rate	15%
Stockholders' investment	$260,000
Earning assets	$340,000

What is the bank's weighted average overall cost of capital (minimum required return on earning assets)?

A. 23.7%
B. 25.3%
C. 28.6%
D. 14.6%

Answer: B

Step by step

1. Pre-tax equity cost $= 12\%/(1 - 0.15) = 14.12\%$.
2. Equity share of earning assets $= 260{,}000/340{,}000 = 0.7647$.
3. Equity add-on $= 14.12\% \times 0.7647 = 10.80\%$.
4. Overall cost $= 14.50\% + 10.80\% = 25.3\%$.

Why B is correct: Overall cost = break-even + [after-tax equity cost/(1 - t)] x (equity/earning assets).

Why the others are wrong

- **A:** 23.7% forgets to gross up the 12% for taxes (12% x 0.7647 = 9.18%, total 23.7%).
- **C:** 28.6% omits the equity/earning-assets weight, adding the full 14.12%.
- **D:** 14.6% makes a percent/decimal slip, adding 0.12 x 0.7647 = 0.09% instead of 10.80%.

Q418

AnalystPrep Q.4115

REBUILT — Replaced an option (6.25%) with no realistic derivation by the hurdle-rate trap (7.67%).

KBC Bank is considering funding a package of new loans for $600 million. The bank must raise $800 million to have $600 million available to make the new loans. It expects to raise $550 million by selling time deposits at an average interest rate of 5%; noninterest costs on the time deposits add 2% of the amount raised. The other $250 million will come from noninterest-bearing transaction deposits, whose noninterest costs are 3% of the amount raised. What is the bank's projected pooled-funds marginal cost?

Source	Dollar amount ($m)	Interest rate	Noninterest cost rate	Total interest expense ($m)	Total noninterest expe
Time deposits	550	5.00%	2.00%	27.50	11.00
Transaction deposits	250	0%	3.00%	0	7.50
Total	800	|	27.50	18.50	

A. 5.75%
B. 7.67%
C. 2.31%
D. 3.44%

Answer: A
Step by step
1. Total interest expense = 550 × 5% = 27.50 million.
2. Total noninterest expense = 550 × 2% + 250 × 3% = 11.00 + 7.50 = 18.50 million.
3. All expected operating expenses = 27.50 + 18.50 = 46.00 million.
4. Pooled-funds cost = all expected operating expenses / all new funds raised = 46.00/800 = 5.75%.

Why A is correct: The pooled-funds marginal cost spreads all expected interest and noninterest costs over all new funds raised, including the portion that cannot be placed in earning assets.

Why the others are wrong
- **B:** This divides total costs by the $600 million available for earning assets; that is the hurdle rate, not the pooled-funds cost.
- **C:** Uses only noninterest expense (18.50 / 800) and ignores interest expense.
- **D:** Uses only interest expense (27.50 / 800) and ignores noninterest expense.

Q419

AnalystPrep Q.4117

REBUILT — Original options had no traceable derivations (5.46%, 9.83%, 10.33%); replaced with distractors from realistic errors.

KBC Bank is considering funding a package of new loans for $600 million. The bank must raise $800 million to have $600 million available to make the new loans. It expects to raise $550 million by selling time deposits at an average interest rate of 5%; noninterest costs on the time deposits add 2% of the amount raised. The other $250 million will come from noninterest-bearing transaction deposits, whose noninterest costs are 3% of the amount raised. What hurdle rate must the bank achieve on its earning assets?

Source	Dollar amount ($m)	Interest rate	Noninterest cost rate	Total interest expense ($m)	Total noninterest expe
Time deposits	550	5.00%	2.00%	27.50	11.00
Transaction deposits	250	0%	3.00%	0	7.50
Total	800	|	27.50	18.50	

A. 5.75%
B. 7.67%
C. 4.58%
D. 3.08%

Answer: B
Step by step
1. All expected operating expenses = 27.50 + 18.50 = 46.00 million.
2. Funds available to place in earning assets = 600 million (the new loans).
3. Hurdle rate = 46.00/600 = 7.67%.

Why B is correct: The hurdle rate divides all expected costs of the new funds by only the dollars that can actually be invested in earning assets, so it exceeds the pooled-funds cost.

Why the others are wrong
- **A:** Divides by all $800 million raised; that is the pooled-funds marginal cost, not the hurdle rate.
- **C:** Uses only interest expense (27.50 / 600), omitting noninterest costs.
- **D:** Uses only noninterest expense (18.50 / 600), omitting interest costs.

Q420 BT answers P2.T8.20.14.3 · BT notes P2.T8.20.14.2

REBUILT — Merged duplicates; stem condensed and distractors replaced with results of specific calculation errors.

A bank has raised funds from the sources below.

Sources of funds	Amount raised ($m)	Avg rate of interest	Total interest paid ($m)
Non-interest-bearing demand deposits	100.0	0.0%	0.0
Interest-bearing transaction deposits	200.0	5.0%	10.0
Savings accounts	100.0	3.0%	3.0
Time deposits	500.0	6.0%	30.0
Money market borrowings	100.0	4.0%	4.0
Total	1,000.0	| 47.0	

Operating costs (salaries and overhead) are $9.0 million, earning assets are $700.0 million, and stockholders' investment is $100.0 million. Stockholders require a 12.0% after-tax return and the tax rate is 21.0%. Using the historical average cost approach, what is the bank's weighted average overall cost of capital (WACC)?

A. 9.71%
B. 10.17%
C. 8.88%
D. 9.52%

Answer: B

Step by step

1. Break-even cost on borrowed funds = (interest + operating costs) / earning assets = $(47.0 + 9.0)/700 = 8.00\%$.
2. Pretax required return on equity = $12\%/(1 - 0.21) = 15.19\%$.
3. Before-tax cost of stockholders' investment = $15.19\% \times 100/700 = 2.17\%$.
4. WACC = $8.00\% + 2.17\% = 10.17\%$.

Why B is correct: WACC equals the break-even cost on borrowed funds (measured on earning assets) plus the tax-grossed-up equity return weighted by equity/earning assets.

Why the others are wrong

- **A:** Does not gross up the equity return for tax: 8.00% + 12% × 100/700 = 9.71%.
- **C:** Omits operating costs from the break-even rate: 47/700 + 2.17% = 8.88%.
- **D:** Weights equity by total funds instead of earning assets: 8.00% + 15.19% × 100/1,000 = 9.52%.

Q421 Mock 1 Q4

REBUILT — Distractor C was 6.77% but its calculation gives 6.78%; corrected.

The Harper Valley Bank (HVB) uses the historical average cost approach to estimate its cost of funds. HVB estimates noninterest expenses, including salaries and overhead, at USD 20 million annually. HVB has total assets of USD 2.1 billion, earning assets of USD 1.5 billion, and equity capital of USD 300 million. Total interest paid on borrowed funds is USD 52 million. The shareholders' after-tax required rate of return is 9% and the tax rate is 35%. HVB's weighted average overall cost of capital is closest to:

A. 6.24%
B. 6.60%
C. 6.78%
D. 7.57%

Answer: D

Step by step

1. Break-even cost = (52 + 20)/1,500 = 4.80%.
2. Pretax required equity return = $9\%/(1 - 0.35) = 13.846\%$.
3. Equity charge = $13.846\% \times 300/1,500 = 2.769\%$.
4. WACC = $4.80\% + 2.77\% = 7.57\%$.

Why D is correct: Break-even cost on earning assets plus the tax-adjusted equity cost weighted by equity/earning assets gives the minimum return needed on earning assets.

Why the others are wrong

- **A:** Omits the USD 20 million noninterest expense: 52/1,500 + 2.77% = 6.24%.
- **B:** Skips the tax gross-up of equity: 4.80% + 9% × 300/1,500 = 6.60%.
- **C:** Weights the equity charge by total assets (2,100) instead of earning assets: 4.80% + 13.846% × 300/2,100 = 6.78%.

Q422 Schweser Module Quiz 77.2 Q2

Blue Star Bank expects to raise $300 million, $250 million of which will be invested in earning assets. The total expected interest and overhead costs on the newly raised funds are forecasted to be $22 million. The pooled deposit and nondeposit funds expense and hurdle rate over all earning assets are, respectively:

A. 7.33%; 7.33%.
B. 7.33%; 8.80%.
C. 8.80%; 7.33%.
D. 8.80%; 8.80%.

Answer: B

Step by step

1. Pooled funds expense = costs / all new funds = 22/300 = 7.33%.
2. Hurdle rate = costs / funds placed in earning assets = 22/250 = 8.80%.

Why B is correct: The pooled expense uses all funds raised as the denominator, while the hurdle rate uses only the funds invested in earning assets, so the hurdle rate is higher.

Why the others are wrong

- **A:** Uses $300 million for both ratios; the hurdle rate must use the $250 million invested.
- **C:** Reverses the two denominators.
- **D:** Uses $250 million for both ratios; the pooled expense must use all $300 million raised.

Reading 78. Repurchase Agreements and Financing

Bruce Tuckman and Angel Serrat, Fixed Income Securities: Tools for Today's Markets, 3rd Edition (Hoboken, NJ: John Wiley & Sons, 2011). Chapter 12.

LO	Learning objective	Qs
78.a	describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction.	21
78.b	discuss common motivations for entering into repos, including their use in cash management and liquidity management.	6
78.c	explain how counterparty risk and liquidity risk can arise through the use of repo transactions.	7
78.d	assess the role of repo transactions in the collapses of Lehman Brothers and Bear Stearns during the 2007–2009 financial crisis.	3
78.e	compare the use of general and special collateral in repo transactions.	3
78.f	identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle.	3
78.g	calculate the financing advantage of a bond trading special when used in a repo transaction.	6

LO 78.a

Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction.

Q423

AnalystPrep Q.2243

Which of the following statements is correct about repurchase agreements?

A. They can be traded on primary markets.
B. They are unsecured.
C. They are classified as money market instruments.
D. They can have a maturity period of up to 20 years.

Answer: C

Step by step

1. A repo is the sale of a security with an agreement to repurchase it at a higher price at a later date; economically a collateralized short-term loan.
2. Maturities are short, most commonly overnight and out to several months, so repos are treated as money market instruments.
3. They are collateralized (not unsecured) and trade in secondary markets, not as new issues.

Why C is correct: Repos are short-term collateralized borrowing arrangements, so they are classified as money market instruments.

Why the others are wrong

- **A:** Repos are not newly issued securities; they are negotiated in the over-the-counter/secondary market.
- **B:** Repos are secured by the securities sold and repurchased.
- **D:** Repo terms are overnight to several months; a 20-year maturity is a capital market tenor.

Q424

GARP book Ch.13 Q (GARP-LR-17)

REBUILT — Day-count basis was not stated; added the actual/360 money-market convention.

Deep Valley Bank borrows $125 million overnight through a repurchase agreement (RP) collateralized by Treasury bills. The current RP rate is 2.50 percent (annualized, actual/360 day count). How much will the bank pay in interest on this borrowing?

Answer: see model answer — Interest = $125{,}000{,}000 \times 0.025 \times 1/360 \approx \$8{,}680.56$. The repurchase price is therefore about $125,008,680.56.

Step by step

1. RP interest = principal × rate × days/360.
2. $125{,}000{,}000 \times 0.025 = 3{,}125{,}000$ per year.
3. Overnight = 1 day: $3{,}125{,}000/360 = 8{,}680.56$.

Why this is correct: Repo rates are annualized and quoted on an actual/360 basis, so one night costs 1/360 of annual interest.

Q425 AnalystPrep Q.4103

REBUILT — Day-count was not stated and three options were arbitrary giveaways; stated actual/360 and rebuilt distractors from realistic mistakes.

Bluemont Bank borrows $85 million using an RP transaction with government bond collateral for twelve days. The current RP rate is 4% (annualized). Using the money-market actual/360 convention, what is the bank's total interest cost?

A. $113,333.33
B. $111,780.82
C. $1,360,000.00
D. $9,444.44

Answer: A

Step by step

1. Interest = principal × rate × days/360.
2. 85,000,000 × 0.04 = 3,400,000 per year.
3. 3,400,000 × 12/360 = 113,333.33.

Why A is correct: Repo interest accrues at the annualized rate scaled by actual days over 360.

Why the others are wrong

- **B:** Used a 365-day year (3,400,000 × 12/365); repos use actual/360.
- **C:** Scaled by 12/30 instead of 12/360, a day-count slip.
- **D:** Computed only one day of interest (3,400,000/360), ignoring the twelve-day term.

Q426 GARP PE1 Q80 · Mock 3 Q1

A bank buys a bond on its coupon payment date. Three months later, to generate immediate liquidity, the bank repos the bond. Assume a 30/360-style simple accrual and simple repo interest on a half-year term.

Item	Value
Notional (USD)	100,000
Coupon (semi-annual)	5%
Current bond price (clean, per 100)	98
Repo haircut	5%
Repo interest rate	3%

If the repo contract expires 6 months from now, what is the bank's expected cash outflow at the end of the repo transaction?

A. USD 94,497
B. USD 95,702
C. USD 97,630
D. USD 100,739

Answer: B

Step by step

1. Accrued interest after 3 months of a 6-month coupon period: $2,500 \times 0.5 = 1,250$.
2. Dirty price = clean + accrued = $98,000 + 1,250 = 99,250$.
3. Cash received after 5% haircut = $99,250 \times 0.95 = 94,287.50$.
4. Repayment = $94,287.50 \times (1 + 0.03 \times 0.5) = 95,701.81 \approx 95,702$.

Why B is correct: The lender advances the haircut-adjusted market (dirty) value of the collateral, and the borrower repays that cash plus repo interest for the term.

Why the others are wrong

- **A:** Left out accrued interest, so the collateral value and the repayment are too low.
- **C:** Valued the bond at par plus accrued interest instead of its market price, overstating the loan.
- **D:** Ignored the haircut (repaid interest on the full dirty price).

Q427 GARP PE2 Q42 · Mock 4 Q43

A bank buys a bond on its coupon payment date. Three months later, to generate immediate liquidity, the bank repos the bond. Assume a 30/360-style simple accrual and simple repo interest on a half-year term.

Item	Value
Notional (USD)	100,000
Coupon (semi-annual)	6%
Current bond price (clean, per 100)	97
Repo haircut	10%
Repo interest rate	4%

If the repo contract expires 6 months from now, what is the bank's expected cash outflow at the end of the repo transaction?

A. USD 89,046
B. USD 90,423
C. USD 93,177
D. USD 100,470

Answer: B

Step by step

1. Accrued interest after 3 months of a 6-month coupon period: $3,000 \times 0.5 = 1,500$.
2. Dirty price = clean + accrued = $97,000 + 1,500 = 98,500$.
3. Cash received after 10% haircut = $98,500 \times 0.90 = 88,650.00$.
4. Repayment = $88,650.00 \times (1 + 0.04 \times 0.5) = 90,423.00 \approx 90,423$.

Why B is correct: The lender advances the haircut-adjusted market (dirty) value of the collateral, and the borrower repays that cash plus repo interest for the term.

Why the others are wrong

- **A:** Left out accrued interest, so the collateral value and the repayment are too low.
- **C:** Valued the bond at par plus accrued interest instead of its market price, overstating the loan.
- **D:** Ignored the haircut (repaid interest on the full dirty price).

Q428

AnalystPrep Q.1869

REBUILT — Option D's 'counterparty risk charge' wording was loose; rewrote options so only one statement is correct.

In a repurchase agreement, one party sells securities for cash and promises to repurchase them at a future date. Which statement about the repurchase price is correct?

A. The repurchase price is lower than the original sale price, the difference being the repo interest.
B. The repurchase price equals the original sale price; the repo rate is zero by construction.
C. The repurchase price is set by prevailing swap rates, independent of the original sale price.
D. The repurchase price is higher than the original sale price, the difference being the implied interest on the loan.

Answer: D

Step by step

1. A repo is economically a short-term collateralized loan.
2. The seller receives the sale price now and pays the higher repurchase price later.
3. The difference is the implied interest (quoted as an annualized repo rate).

Why D is correct: Repurchase price = sale price plus interest, so it exceeds the sale price.

Why the others are wrong

- **A:** Reverses the direction; the borrower pays more back, not less.
- **B:** Implies a zero-interest loan, ignoring the cost of borrowing.
- **C:** Ties price to swap rates instead of the sale price plus repo interest.

Q429

AnalystPrep Q.2239

REBUILT — Option C (pledge bonds for a loan) was also defensible economically; replaced with a clearly wrong option.

Sover Bank needs short-term liquidity and holds high-quality US government bonds. It enters a repurchase agreement (repo) with GNR Bank. Under the repo, Sover Bank will:

A. Sell US government bonds to third parties to repay funds borrowed from GNR Bank.
B. Buy US government bonds from GNR Bank.
C. Sell the bonds to GNR Bank outright with no obligation to repurchase them.
D. Sell the bonds to GNR Bank with an obligation to repurchase them at a later date at a higher price.

Answer: D

Step by step

1. The repo borrower needs cash, so it sells securities.
2. It commits to buy them back at a specified future date at a higher price.
3. The price difference is the implied interest.

Why D is correct: A repo is a sale with a commitment to repurchase, functioning as a collateralized loan.

Why the others are wrong

- **A:** Selling to third parties is not a repo and does not repay GNR.
- **B:** Buying bonds would be the lender's reverse-repo leg and gives Sover no cash.
- **C:** An outright sale has no repurchase commitment, so it is not a financing repo.

Q430

AnalystPrep Q.2240

REBUILT — Original stem had Bolny receiving bonds (reverse repo side) and options A and D were both defensible; rewrote so exactly one option is right.

Bolny Bank enters into a reverse repurchase agreement with Kovit Bank. In the first leg of the transaction, Bolny lends cash to Kovit and receives US government bonds. In the second leg, Bolny will:

A. Sell the bonds back to Kovit at the market price prevailing on the settlement date and keep the cash.

B. Sell the bonds back to Kovit at the price fixed at inception (cash lent plus agreed interest), thereby recovering its cash with interest.

C. Keep the bonds permanently if Kovit has not asked for them back by the settlement date.

D. Pay Kovit additional cash and receive the bonds in return.

Answer: B

Step by step

1. In a repo the borrower sells a security and commits to repurchase it at a higher, pre-agreed price; the difference is interest.
2. Kovit is the borrower (repo side); Bolny is the lender (reverse repo side) and holds the bonds as collateral.
3. At the second leg Bolny returns the bonds to Kovit and receives back the cash lent plus the agreed interest, at the price fixed at inception.

Why B is correct: The repurchase price is fixed at inception, so the lender recovers principal plus agreed interest independent of the bonds' market price.

Why the others are wrong

- **A:** Uses the market price at settlement; the price is fixed in advance, which removes this price risk.
- **C:** The bonds are collateral, not a purchase; the repurchase commitment is binding, so Bolny cannot keep them.
- **D:** Reverses the cash flows; Bolny is the lender and receives cash in the second leg rather than paying it.

Q431

AnalystPrep Q.4097

REBUILT — Replaced arbitrary distractors with realistic mistakes (365-day count, monthly and annual interest).

Promise Bank borrows $30 million overnight through a repurchase agreement collateralized by Treasury bills. The repo rate is 4% (actual/360). What is the bank's interest cost?

A. $3,333.33

B. $3,287.67

C. $100,000.00

D. $1,200,000.00

Answer: A

Step by step

1. Interest = principal × rate × days/360
2. $30{,}000{,}000 \times 0.04 \times 1/360 = 3{,}333.33$

Why A is correct: Repo interest uses actual/360, so one day costs 1/360 of the annual interest of $1.2 million.

Why the others are wrong

- **B:** Uses a 365-day year (1,200,000/365).
- **C:** Computes one month of interest (1,200,000/12).
- **D:** Reports a full year of interest without scaling for one day.

Q432 AnalystPrep Q.2241

Sokolov Bank is going to enter into a repurchase agreement with Branitzky Bank. Sokolov Bank will sell high-quality securities to Branitzky Bank in the first part of the transaction. At what price will Branitzky Bank sell the securities back to Sokolov Bank?

A. A fixed price agreed upon at the time of entering into the repo agreement.
B. At the market price of the securities at the time of entering into the repo agreement.
C. At the market price of the securities at the time of the second transaction.
D. At the price that should have been agreed upon at the time of the second transaction.

Answer: A

Step by step

1. A repo is a sale of securities with a commitment to repurchase at a specified (higher) price on a future date.
2. The repurchase price is set at inception as the original price plus interest.
3. Hence the resale price is the fixed price agreed when the repo was entered.

Why A is correct: The repurchase price is contractually fixed at inception, so the transaction behaves as a collateralized loan with known interest.

Why the others are wrong

- **B:** Confuses the repurchase price with the initial sale price; the repurchase price includes interest and is higher.
- **C:** Assumes the securities reprice to market at the second leg; that would make it an ordinary sale, not a loan.
- **D:** Describes an undetermined, discretionary price; repo terms are fixed up front.

Q433 AnalystPrep Q.2242

REBUILT — Replaced the unexplainable distractor 1,101,466.66 with a realistic mistake (interest on face amount).

Ludska Bank entered into a repo agreement with SKA Bank. Ludska sold government bonds to SKA as follows:
Face amount: USD 1,000,000
Invoice price: USD 1,100,000
Repo rate: 4% p.a.
Term: 90 days
Assuming an actual/360 convention, what will Ludska pay to repurchase the bonds at settlement?

A. USD 1,100,000
B. USD 1,110,000
C. USD 1,111,000
D. USD 1,000,000

Answer: C

Step by step

1. Repurchase price = invoice price $\times(1 + r \times t/360)$
2. $1{,}100{,}000 \times (1 + 0.04 \times 90/360)$
3. $= 1{,}100{,}000 \times 1.01 = 1{,}111{,}000$

Why C is correct: Interest accrues on the cash received (the invoice price) at the repo rate over 90/360 of a year.

Why the others are wrong

- **A:** Returns only the invoice price, omitting interest.
- **B:** Computes interest on the face amount (1,000,000 x 4% x 90/360 = 10,000) instead of on the invoice price.
- **D:** Uses the face amount as the repurchase price; the repo is priced on the invoice price plus interest.

Q434 AnalystPrep Q.2246

For a particular high-quality security transaction, the agreement is called a ”repo” from the point of view of:

A. The initial security buyer
B. The initial security seller
C. The clearinghouse
D. The Federal Reserve

Answer: B

Step by step

1. ”Repo” describes the transaction from the borrower’s side: the party that sells the security and promises to buy

it back.
2. The same transaction viewed from the lender (initial buyer) is a reverse repo.

Why B is correct: The initial seller borrows cash against the securities, which is the repo side.

Why the others are wrong

- **A:** The initial buyer lends cash and takes the reverse repo side.
- **C:** A clearinghouse is not the borrower and does not define the term.
- **D:** The central bank is just one possible counterparty and does not define the term.

Q435

AnalystPrep Q.2252

Boulogne Bank sells securities to Betis Bank as part of a repo agreement. Which of the following statements would NOT be consistent with market practice for such agreements?

A. The repurchase price would be fixed.
B. The contract could be settled overnight.
C. Betis Bank would be free to sell the securities back to Boulogne Bank at any time.
D. The agreement would be legally binding.

Answer: C

Step by step

1. A repo specifies a fixed repurchase price and a fixed date (overnight, term or open), and is a binding contract.
2. The buyer cannot unilaterally force a resale at any time; the repurchase occurs on the agreed date.

Why C is correct: Only C contradicts the fixed-date, fixed-price structure of a repo.

Why the others are wrong

- **A:** Consistent: the repurchase price is fixed at inception.
- **B:** Consistent: overnight repos are the most common type.
- **D:** Consistent: a repo is a binding contract with a commitment to repurchase.

Q436

AnalystPrep Q.2972

REBUILT — Fixed OCR currency symbols and replaced distractors that did not correspond to any realistic mistake.

Counterparty X sells a EUR 250 million face amount of DBR 4s of December 9th, 2014, to counterparty Y for settlement on April 1st, 2013, at an invoice price of EUR 280.131 million. X agrees to repurchase the EUR 250 million face amount for settlement on September 1st, 2013 (153 days later) at the invoice price plus interest at a repo rate of 0.31%, actual/360. What is the repurchase price?

A. EUR 250.329 million
B. EUR 280.131 million
C. EUR 280.999 million
D. EUR 280.500 million

Answer: D

Step by step

1. $280.131 \times (1 + 0.0031 \times 153/360)$
2. $= 280.131 \times 1.0013175 = 280.500$ million

Why D is correct: Interest at 0.31% on the invoice price for 153/360 of a year adds about EUR 0.369 million.

Why the others are wrong

- **A:** Applies the interest factor to the face amount (250) instead of the invoice price.
- **B:** Omits interest and returns the invoice price.
- **C:** Applies the full annual rate 0.31% without scaling by 153/360.

Q437

AnalystPrep Q.2974

REBUILT — Replaced arbitrary distractors with ones from realistic mistakes.

HJK Bank sells DBR 4s with face value USD 120 million to a counterparty at an invoice price of USD 159 million. HJK agrees to repurchase the bonds 122 days later at the invoice price plus interest at a repo rate of 0.26% (actual/360). What is the repurchase price?

A. USD 159.14 million
B. USD 159.00 million

C. USD 120.11 million
D. USD 159.41 million

Answer: A
Step by step
1. 159 × (1 + 0.0026 × 122/360)
2. = 159 × 1.000881 = 159.140 million

Why A is correct: The repurchase price is the invoice price grossed up for 122/360 of a year of interest.
Why the others are wrong
- **B:** Ignores interest and returns the invoice price.
- **C:** Applies interest to the face amount (120) instead of the invoice price.
- **D:** Applies the full annual rate 0.26% without scaling by 122/360.

Q438

AnalystPrep Q.4118

Counterparty K sells a €300 million face amount of DBR 4s of June 10th, 2016, to counterparty W, for settlement on January 1st, 2015, at an invoice price of €350.25 million. At the same time, the seller agrees to repurchase the €300 million face amount six months later, for settlement on July 1st, 2015, at a purchase price equal to the invoice price plus interest at a repo rate of 0.26%. Using the actual/360 convention, compute the repurchase price.

A. €350.16 million
B. €350.56 million
C. €350.71 million
D. €350.85 million

Answer: C
Step by step
1. Repo interest accrues on the invoice price (cash lent), not the face amount.
2. Actual days from January 1 to July 1, 2015: 31 + 28 + 31 + 30 + 31 + 30 = 181.
3. Interest = 350.25 × 0.0026 × 181/360 = 0.4578 million euros.
4. Repurchase price = 350.25 + 0.4578 = 350.708 million euros, about €350.71 million.

Why C is correct: The repurchase price equals the invoice price grossed up by the repo rate over the actual number of days on an actual/360 basis.
Why the others are wrong
- **A:** €350.16 million is below the invoice price, so it cannot include positive interest; the repurchase price must exceed the invoice price.
- **B:** €350.56 million understates interest (€0.31 million), as from using about 120 days instead of 181.
- **D:** €350.85 million overstates interest (about €0.60 million); it does not follow from 181/360 and reflects a mis-scaled accrual period.

Q439

AnalystPrep Q.4119

REBUILT — Original options C and D were on a $200 million base and implausible; options rebuilt so each reflects a realistic error, answer unchanged.

Prime Bank sells DBR 4s with $200 million face value maturing September 1st, 2040, to a counterparty at an invoice price of $250 million. Prime Bank simultaneously agrees to repurchase the bonds 183 days later at a price equal to the invoice price plus interest at a repo rate of 0.34%. Using the actual/360 convention, at what price does Prime Bank repurchase the bonds?

A. $250.43 million
B. $250.85 million
C. $200.35 million
D. $250.17 million

Answer: A
Step by step
1. Interest accrues on the invoice price of $250 million, not on the $200 million face value.
2. Interest = 250 × 0.0034 × 183/360 = 0.4321 million dollars.
3. Repurchase price = 250 + 0.4321 = 250.432 million dollars, about $250.43 million.

Why A is correct: Repo interest is computed on the cash received at initiation over actual/360 days.

Why the others are wrong
- **B:** $250.85 million applies the full annual rate (250 × 1.0034) ignoring the 183/360 fraction.
- **C:** $200.35 million accrues interest on the face value ($200 million) instead of the invoice price.
- **D:** $250.17 million understates interest (about $0.17 million), as if only ~72 days accrued rather than 183.

Q440 AnalystPrep Q.4162

Which of the following is NOT a characteristic of a repurchase agreement?

A. It has a specified repurchase price.
B. It usually has a short duration.
C. It is generally considered a safe, collateralized investment.
D. It usually has a long duration.

Answer: D
Step by step
1. A repo fixes the repurchase price (invoice price plus interest) at initiation.
2. Most repos are overnight or short term; they are secured by collateral, so lenders regard them as safe.
3. Long duration is therefore not typical.

Why D is correct: Repos are short-term secured financing with a price agreed upfront.
Why the others are wrong
- **A:** This is a true characteristic, so it is not the exception.
- **B:** This is true; most repos are overnight or short term.
- **C:** This is true; collateral and haircuts protect the cash lender.

Q441 AnalystPrep Q.5378

A risk manager compares repurchase agreements (repos) and reverse repos. Which statement is correct?

A. In a repo, the seller of securities agrees to buy them back later at a slightly lower price.
B. In a reverse repo, the buyer of securities agrees to sell them back later at a slightly lower price.
C. Repos are typically used to raise cash, while reverse repos are typically used to invest cash.
D. Repos are initiated by the buyer of securities, while reverse repos are initiated by the seller of securities.

Answer: C
Step by step
1. A repo is a sale of securities with an agreement to repurchase at a higher price; the difference is interest on the cash.
2. The repo seller receives (borrows) cash; the counterparty doing the reverse repo lends cash against collateral.
3. So repos raise cash and reverse repos invest cash.

Why C is correct: The same transaction is a repo to the cash borrower and a reverse repo to the cash lender, with the repurchase price above the sale price by interest.
Why the others are wrong
- **A:** The repurchase price is slightly higher, not lower; interest accrues to the cash lender.
- **B:** The reverse-repo buyer sells back at a higher price, not lower.
- **D:** Repos are initiated by the seller of securities (cash borrower); the roles are reversed in this option.

Q442 BT answers P2.T7.513.1 · BT notes P2.T8.513.1

At initiation of a repurchase agreement (repo), Counterparty A sells a security to Counterparty B for settlement on June 1, 2015, at an invoice price of USD 180.0 million. Counterparty A agrees to repurchase the security three months later, for settlement on September 1, 2015, at a purchase price equal to the invoice price plus interest at a repo rate of 0.90%. Using the actual/360 convention, which is nearest to the repurchase price?

A. $414,000
B. $180,000,000
C. $180,414,000
D. $181,620,000

Answer: C

Step by step

1. Actual days June 1 to September 1 = 30 + 31 + 31 = 92.
2. Interest = 180,000,000 × 0.0090 × 92/360 = 414,000.
3. Repurchase price = 180,000,000 + 414,000 = 180,414,000 dollars.

Why C is correct: The repurchase price is the invoice price plus actual/360 interest on it.

Why the others are wrong

- **A:** $414,000 is only the interest; the invoice price was not added back.
- **B:** $180,000,000 ignores interest altogether.
- **D:** $181,620,000 adds a full year of interest (180m × 0.90% = 1.62m) with no 92/360 scaling.

Q443 Schweser Module Quiz 78.1 Q1

REBUILT — Source was open-ended with options A-D missing and the calculation absent; rebuilt as a self-contained MCQ on contract type and computed repurchase price.

Pasquini Investments, a private brokerage, needs 30-day financing of $25 million. It will post AAA-rated government bonds with a face value of $27 million and market value of $25 million as collateral, and was quoted a rate of 0.5%. Assuming no haircut and the actual/360 convention, which of the following reflects the contract type from Pasquini's perspective and the repurchase price it must pay?

A. Repo; $25,010,417
B. Reverse repo; $25,010,417
C. Repo; $27,011,250
D. Repo; $25,125,000

Answer: A

Step by step

1. Pasquini borrows cash against securities, i.e. it sells the bonds and repurchases them later: a repo (the lender's side is the reverse repo).
2. Cash lent (invoice price) = $25,000,000 market value; face value is irrelevant.
3. Interest = 25,000,000 × 0.005 × 30/360 = 10,417 dollars.
4. Repurchase price = 25,000,000 + 10,417 = 25,010,417 dollars.

Why A is correct: Pasquini is the cash borrower (repo) and repays the cash received plus actual/360 interest.

Why the others are wrong

- **B:** Reverse repo is the cash lender's perspective; Pasquini is raising cash.
- **C:** $27,011,250 accrues interest on the $27 million face value instead of the $25 million cash.
- **D:** $25,125,000 applies the full annual 0.5% with no 30/360 scaling.

LO 78.b

Discuss common motivations for entering into repos, including their use in cash management and liquidity management.

Q444 AnalystPrep Q.2285

REBUILT — Original (can CDOs be repo collateral?) was ambiguous and weakly supported; rewritten around collateral quality, haircuts and stability of repo funding.

A bank holds a portfolio of lower-quality structured securities (e.g., CDOs) and wants to raise short-term cash without selling them outright. Which statement is most accurate regarding financing them in the repo market?

A. The bank can pledge them in a repo, but lenders prefer high-quality collateral, so expect larger haircuts and less reliable funding, particularly in stressed markets.
B. Repos can only be used with government securities, so the bank cannot repo them under any circumstances.
C. The bank will receive the full market value of the securities as cash, since repos are secured loans.
D. Repo is a stable long-term funding source, so rolling the position carries no refinancing risk.

Answer: A

Step by step

1. A repo is a short-term secured loan; collateral is generally high quality, but lower-quality securities can be

financed at higher haircuts and rates.
2. Haircuts discount collateral relative to its risk, so cash received is below market value.
3. Repo funding is short-term and can be withdrawn or haircuts raised in stress, so it is less stable.

Why A is correct: Lower-quality collateral can be repo'd, but with larger haircuts and greater rollover risk.

Why the others are wrong

- **B:** Overstates the restriction; other collateral was financed (as pre-crisis), just on worse terms.
- **C:** Ignores haircuts; lenders advance less than market value, more so for riskier collateral.
- **D:** Repos are short-term and less stable than equity or long-term debt, so rollover risk is significant.

Q445

AnalystPrep Q.2249

REBUILT — Option C (repo market is highly illiquid) was an obvious giveaway; replaced with a plausible wrong option about repo stability.

After modeling daily cash flows at BBB Bank, its managers strongly believe that the bank will be unable to meet day-to-day funding requirements in about one year's time. The bank therefore needs to secure long-term funding. BBB Bank holds high-quality US Treasury bonds. Should the bank sell the bonds under a repo agreement to secure the required funds?

A. Yes, because US Treasury bonds are highly marketable.
B. No, because repo financing is usually short-term.
C. Yes, because repo financing is typically more stable than equity financing.
D. No, because repo borrowing has to be backed by collateral of a similar value.

Answer: B

Step by step

1. The need is funding about one year ahead, i.e., stable, long-term financing.
2. Repos are secured short-term financing, mostly overnight to a few months; they must be repaid or rolled within a short period.
3. Rolling exposes the bank to rollover risk and swings in market conditions and lender sentiment, so repo is less stable than long-term debt or equity.
4. Balancing funding cost against the risk of being left without financing is liquidity management; repo is cheap but less stable, so it does not meet a long-term need.

Why B is correct: Repo is cheap but short-term and less stable funding, so it cannot secure a one-year funding need without rollover risk.

Why the others are wrong

- **A:** Marketability of the collateral makes a repo easy to arrange but does not change its short tenor; the mistake is focusing on collateral rather than funding term.
- **C:** Reverses the stability ranking: equity is the most stable (and most expensive) funding, repo is cheaper but less stable.
- **D:** Collateral is indeed required (with a haircut), but that is not why repo is unsuitable; the binding issue is the short maturity.

Q446

AnalystPrep Q.2251

Bank Aluvia has excess liquidity and has decided to lend to another bank under a repo agreement. Why would Aluvia prefer a repo transaction to other market products such as the purchase of stocks?

A. A repo transaction would guarantee a quick return.
B. A repo transaction would be faster and easier to execute.
C. A repo transaction would be risk-free.
D. A repo transaction would be more secure.

Answer: D

Step by step

1. Aluvia is a cash investor: it lends cash and holds collateral (the reverse repo side).
2. Because the loan is collateralized by high-quality securities, the lender can sell the collateral to recover amounts owed if the borrower defaults.
3. Repos are short-term, so the investor keeps liquidity and low risk, but they are not risk-free (collateral value and liquidity can deteriorate).
4. The distinguishing advantage over stocks is greater security, so D.

Why D is correct: Holding collateral makes the lender much less vulnerable to the borrower's creditworthiness, so repo is a more secure, low-risk investment than stocks.

Why the others are wrong

- **A:** No return is guaranteed; repo pays a modest short-term rate, and speed of return is not the motivation.
- **B:** Ease or speed of execution is not the stated reason for using repos in cash management.
- **C:** Overstates the case: lenders still face counterparty risk and liquidity risk (collateral value decline), so no repo is risk-free.

Q447

BT answers P2.T7.513.2 · BT notes P2.T8.513.2

Rankcon is a repo investor in the money market mutual fund industry with the exclusive motive of cash management: it lends money while taking collateral and, at maturity, collects the loan plus interest and returns the collateral. Given this motivation, Rankcon would employ each of the following preferences with respect to its repo investments EXCEPT which is the LEAST likely preference?

A. Rankcon insists on a sufficient (i.e., greater rather than lesser) collateral haircut.

B. Rankcon only accepts securities with the highest credit quality, e.g., debt of government-sponsored entities (GSEs).

C. Rankcon refuses to accept general collateral (at general collateral repo rates) and instead insists on particular securities with delineated asset classes.

D. Rankcon places a premium on liquidity, so it tends to lend overnight rather than for term; or, if it wants to lend cash for an extended period, it engages in an open repo.

Answer: C

Step by step

1. A cash-management investor wants safety and liquidity, and is willing to accept relatively low returns.
2. Safety: it prefers high-quality collateral (government, GSE) and haircuts to protect against collateral value declines (A, B true).
3. Liquidity: it favors overnight repos or open repos that renew daily until cancelled (D true).
4. Investors care about the class of collateral but not which particular bond; they accept general collateral (GC) and obtain the highest repo rate for it. Insisting on a particular security is the behavior of someone who needs a specific bond (e.g., to short it), so C is least likely.

Why C is correct: Cash investors are indifferent to which specific security backs the loan within an acceptable class, so they accept general collateral rather than demanding particular securities.

Why the others are wrong

- **A:** True preference: haircuts protect the lender from a fall in collateral value, so a cash investor wants larger haircuts.
- **B:** True preference: investors prefer higher-quality collateral such as government and GSE securities.
- **D:** True preference: investors place a premium on liquidity and so favor overnight or open repos.

Q448

BT answers P2.T7.513.3 · BT notes P2.T8.513.3

REBUILT — Source illustration was missing and answer options differed between versions; made stem self-contained and gave a single clean option set.

Consider a simplified repurchase agreement between Counterparty A (the party delivering securities) and Counterparty B (the party delivering cash), at initiation of the trade. Consider three primary motivations for repo trades:

I. Lending funds short-term on a secured basis: a money market mutual fund that holds cash for liquidity or safekeeping would buy the repo (as Counterparty B) and is willing to accept general collateral.

II. Financing a long position in a security: a trading desk that wants to finance the purchase of a bond would sell the repo (repo out, as Counterparty A) using the purchased bond as collateral.

III. Borrowing a security to sell it short: a hedge fund that needs to borrow a bond in order to short it would do a reverse repo (as Counterparty B) and, because it needs that bond, requires delivery of a particular security rather than general collateral.

Which of the statements is accurate?

A. None are accurate.

B. I only is accurate.

C. I and II only are accurate.

D. All are accurate.

Answer: D

Step by step

1. I: cash investors lend against collateral (reverse repo side) and, caring about collateral class but not the specific bond, accept general collateral. True.
2. II: to finance a long bond position, the firm borrows cash overnight by repoing the bond out as collateral (subject to haircut), rolling as needed. True.
3. III: to short a bond, the party lends cash in a reverse repo to obtain the specific bond, then sells it short and later buys it back. Needing that bond means special collateral is required. True.
4. All three are accurate.

Why D is correct: Each statement correctly matches a participant's motive (cash investment, long-position financing, short-position financing) with its side of the trade and collateral type.

Why the others are wrong

- **A:** Wrongly rejects all three; each describes a standard repo motivation.
- **B:** Correct that I is true, but wrongly rejects II (repo out to finance a long position) and III (reverse repo to obtain a bond to short).
- **C:** Correct on I and II but wrongly rejects III; borrowing a bond to short is a core reverse repo use and requires the particular security.

Q449 Schweser Module Quiz 78.1 Q3

Kotra Bank Holdings, Inc. (Kotra) is currently weighing the cost of its funding against the risk of being left without financing. The term that best describes Kotra's activities is:

A. counterparty (credit) risk.
B. interest rate risk management.
C. liquidity management.
D. overnight funding.

Answer: C

Step by step

1. Firms can fund with equity or with secured or unsecured, long- or short-term debt, including repos.
2. Cheaper sources (repo) are less stable; stable sources (equity) are more expensive.
3. Balancing funding cost against the possibility of being left without financing is, by definition, liquidity management.

Why C is correct: Trading off the cost of funding alternatives against the risk of losing access to financing is the definition of liquidity management.

Why the others are wrong

- **A:** Counterparty risk is the risk of borrower default or non-payment, not a cost versus funding-availability trade-off.
- **B:** Interest rate risk management concerns the effect of rate changes on assets and liabilities, not access to funding.
- **D:** Overnight funding is simply borrowing or lending in the overnight market, one possible tool rather than the balancing activity.

LO 78.c

Explain how counterparty risk and liquidity risk can arise through the use of repo transactions.

Q450 AnalystPrep Q.2297

REBUILT — Removed giveaway distractors and reframed options to test the haircut as a risk mitigant (78.c).

In a repurchase agreement, what is the purpose of a collateral "haircut"?

A. It discounts the collateral value relative to its risk, so the cash advanced is less than the collateral's market value, protecting the lender against a decline in collateral value.
B. It is the fee the borrower pays the lender, deducted from the repo rate.
C. It is the additional collateral the lender must return when prices rise.
D. It increases the borrower's cash proceeds when collateral is riskier.

Answer: A

Step by step

1. Lenders face the risk that collateral value falls during the term.
2. A haircut discounts collateral value according to its risk (e.g., lend $10.5 million against $11 million of collateral).
3. Riskier or less liquid collateral gets a larger haircut.

Why A is correct: The haircut creates a cushion that mitigates the risk of a decline in collateral value.

Why the others are wrong

- **B:** Confuses haircut with interest; interest is the repo rate on the cash lent.
- **C:** Describes a margin-call mechanism instead; haircuts are set at inception.
- **D:** Reverses the effect: riskier collateral reduces proceeds per dollar of collateral.

Q451 AnalystPrep Q.1870

REBUILT — Original stem was an unfinished lead-in with a vague 'repo margin' option and a giveaway distractor; rewrote options around collateral value risk and margin calls.

A repo is a short-term loan secured by collateral. The cash lender's main residual exposure is that the collateral loses value during the term of the repo. Which of the following statements about this risk and its mitigation is most accurate?

A. If the collateral value falls, the borrower may be called to post additional collateral (a margin call), or the repo may be repriced so that the loan amount is reduced.

B. Once a haircut has been applied the collateral no longer poses any risk, so margin calls are unnecessary.

C. If the collateral value rises the lender suffers a loss, whereas if it falls the lender is fully protected.

D. The risk is eliminated because repos are legally long-term loans, which removes the effect of short-term price moves.

Answer: A

Step by step

1. The lender's exposure is a fall in collateral value during the repo term (liquidity risk in the Schweser/GARP sense).
2. Mitigants include haircuts, margin calls (extra collateral in declining markets), repricing (smaller loan), shorter terms and higher quality collateral.
3. Only A describes margin calls/repricing correctly.

Why A is correct: Margin calls and repricing restore the collateral cover when its value falls, which keeps the lender's protection intact.

Why the others are wrong

- **B:** A haircut gives only a fixed buffer; if prices fall further the buffer is used up, so margin calls are still needed.
- **C:** Reversed: a rise in collateral value does not hurt the lender; falls do, and the lender is not fully protected.
- **D:** Repos are generally very short-term, not long-term, and the term does not remove price risk.

Q452 AnalystPrep Q.2247

Briggs Bank buys US Treasury bonds from Roland Bank under a repo agreement (Briggs is the cash lender). Before the repurchase date, the market value of the bonds falls significantly. Assuming the agreement includes margin call provisions, which action is Briggs most likely to take?

A. Unilaterally terminate the agreement.

B. Demand immediate maturity of the contract to avoid further losses.

C. Do nothing and hope the bonds recover before the repurchase date.

D. Demand that Roland post additional collateral.

Answer: D

Step by step

1. Briggs lends cash and holds the bonds as collateral.
2. A fall in bond value reduces its collateral cover, exposing it to loss if Roland defaults.
3. Repos are subject to margin calls: the cash borrower (Roland) supplies extra collateral in declining markets.
4. So Briggs demands more collateral.

Why D is correct: Margin calls let the cash lender restore collateral cover after a fall in collateral value.

Why the others are wrong

- **A:** Neither party has a unilateral right to terminate; the margin call is the contractual remedy.
- **B:** Early maturity is not a standard remedy and is not what a margin call provides.
- **C:** Passive; leaves the lender exposed to the decline in collateral value and ignores the margin call right.

Q453

AnalystPrep Q.2248

REBUILT — Option B referred to selling 'the mortgage' though the collateral is a generic security; reworded for consistency.

Voynet Bank buys a security from Bornet Bank under a repo agreement (Bornet is the cash borrower). Before the repurchase date, the market value of the security rises significantly. Assuming the agreement includes margin call provisions, Bornet Bank will most likely:

A. Terminate the agreement unilaterally.
B. Request that the contract mature immediately so it can sell the security to third parties at a profit.
C. Request that the amount of collateral be reduced (withdraw excess collateral) to reflect the increased value.
D. Do nothing and wait for the scheduled maturity.

Answer: C

Step by step

1. Bornet sold the security and has borrowed cash; the security is collateral held by Voynet.
2. Its value has risen, so Voynet is over-collateralised.
3. Under margin calls, the borrower supplies extra collateral in declining markets but may withdraw collateral in advancing markets.
4. So Bornet requests a reduction in collateral.

Why C is correct: Margin calls work in both directions: the borrower can withdraw collateral when its value rises.

Why the others are wrong

- **A:** No unilateral termination right exists; the margin mechanism is the remedy.
- **B:** Early maturity is not available under a margin call and the security is not Bornet's to sell during the term.
- **D:** Passive; gives up the margin call right and leaves excess collateral tied up.

Q454

BT notes P2.T6.406.2

Consider a classic repurchase agreement in which Bank A sells securities to Counterparty B and receives cash, and later repurchases the securities. Bank A is the repo seller (borrows cash) and Counterparty B is the repo buyer (lends cash). Each of the following statements about this repo is true EXCEPT:

A. A higher haircut (overcollateralization) reduces the counterparty risk faced by the lender, Counterparty B.
B. The securities sold by Bank A become collateral owned by Counterparty B, who can liquidate them to recover the cash lent if Bank A defaults.
C. Bank A, as the cash borrower, incurs no counterparty risk; only Counterparty B does.
D. Counterparty B is exposed to credit risk, market risk, liquidity risk of the collateral, operational risk and legal risk.

Answer: C

Step by step

1. A: a larger haircut means more collateral per unit of cash, so more cushion for the lender: true.
2. B: the lender holds and can sell the collateral on default: true.
3. C: Bank A has delivered securities that B holds; if B fails to return them, A is exposed, especially when the collateral exceeds the cash lent: false.
4. D: lists genuine exposures of the lender: true.

Why C is correct: Both sides of a repo have counterparty exposure: the borrower is exposed to not getting its (often over-valued) collateral back.

Why the others are wrong

- **A:** True statement, so not the exception: haircuts protect the lender.
- **B:** True statement: collateral can be liquidated on default.
- **D:** True statement: the lender faces several risks beyond credit risk.

Q455 Mock 2 Q58

A bank proposes using very short-term repo transactions to earn additional investment income. The collateral is provided by the borrower and the haircut is 5%. To which of the following risks is the bank exposed?

A. Counterparty risk only.
B. Liquidity risk only.
C. Both counterparty risk and liquidity risk.
D. Neither counterparty risk nor liquidity risk.

Answer: C

Step by step

1. The bank lends cash and holds collateral, so it is the repo buyer.
2. Counterparty risk: the borrower may fail to repay; collateral reduces but does not eliminate this.
3. Liquidity risk (collateral value decline/illiquidity): a 5% haircut only covers a 5% fall and the collateral may be hard to sell when needed.
4. Both risks are present.

Why C is correct: Repos give rise to both counterparty and liquidity risk; collateral and haircuts only mitigate them.

Why the others are wrong

- **A:** Ignores the risk that collateral value falls or becomes illiquid beyond the 5% haircut.
- **B:** Ignores the borrower default risk, which remains despite collateral.
- **D:** Treats a secured loan as riskless; the haircut is only a partial buffer.

Q456 Schweser Module Quiz 78.1 Q2

REBUILT — Answer options were missing from the source; wrote four options based on the Schweser reading's mapping of collateral to counterparty risk and haircuts to liquidity risk.

In repo transactions, posting collateral and requiring collateral haircuts are important risk mitigants. Which of the following correctly matches each mitigant with the primary risk it addresses?

A. Posting collateral and haircuts both mitigate counterparty risk only.
B. Posting collateral mitigates liquidity risk; haircuts mitigate counterparty risk.
C. Posting collateral mitigates counterparty risk; haircuts mitigate liquidity risk (the risk of a decline in collateral value).
D. Posting collateral and haircuts both mitigate liquidity risk only.

Answer: C

Key correction: Source answer options were missing; source explanation maps collateral to counterparty risk and haircuts to liquidity risk, consistent with C.

Step by step

1. Counterparty risk is the borrower's failure to repay; posted collateral lets the lender sell it to recover amounts owed.
2. Liquidity risk is the risk of an adverse change in collateral value or illiquidity during the term.
3. Haircuts (with margin calls, shorter terms, better collateral) mitigate that risk by giving a buffer against a fall in value.
4. So C.

Why C is correct: Collateral reduces dependence on borrower creditworthiness, while haircuts buffer against collateral value declines.

Why the others are wrong

- **A:** Haircuts address the collateral value (liquidity) risk, not just default.
- **B:** Reversed mapping.
- **D:** Collateral itself is the primary counterparty risk mitigant.

LO 78.d

Assess the role of repo transactions in the collapses of Lehman Brothers and Bear Stearns during the 2007–2009 financial crisis.

Q457 AnalystPrep Q.2250

REBUILT — Original key was a loose, partly wrong characterisation (borrowing at low rates against weak collateral); rebuilt to test the actual crisis mechanism of lenders demanding better collateral and larger haircuts.

Before the 2007-2009 crisis, repo borrowers often posted weaker-quality collateral such as corporate bonds or mortgage-backed securities. Which of the following best describes how repo transactions contributed to the failures of firms as the crisis escalated?

A. Repo lenders became reluctant to accept such collateral, demanded higher-quality collateral and larger haircuts, and in the extreme stopped lending, forcing borrowers into collateral liquidations and capital declines.

B. Repo lenders cut haircuts and repo rates to attract business, which encouraged borrowers to take on more leverage than they could fund.

C. Repo borrowers were forced to pay higher repo rates on Treasury collateral because Treasury securities lost their value as collateral.

D. Because repos are legally sales of securities, borrowers had no obligation to repurchase the collateral, and lenders suffered the losses.

Answer: A

Key correction: Source key (AnalystPrep) said C (borrowing at relatively low rates against low-quality collateral); the reading says lenders, who had earned higher rates for accepting weaker collateral, later demanded better collateral and larger haircuts or withdrew, so A is the best answer.

Step by step

1. Pre-crisis: borrowers posted weaker collateral and lenders earned higher repo rates for accepting it.
2. As the crisis escalated, lenders grew reluctant to accept these securities.
3. They demanded higher-quality collateral and larger haircuts, and at the extreme withdrew liquidity and stopped transacting.
4. Borrowers most affected suffered collateral liquidations, capital declines and ultimately bankruptcies (Lehman, Bear Stearns).
5. Option A matches this sequence.

Why A is correct: Repo funding is short-term and collateral-based, so when lenders lose confidence they tighten collateral standards and haircuts or stop rolling, which can force fire sales and failure.

Why the others are wrong

- **B:** Reverses the direction: haircuts and required collateral quality tightened, not loosened, during the crisis.
- **C:** Stress concentrated on weaker collateral; high-quality collateral such as Treasuries remained acceptable, so this misidentifies the source of pressure.
- **D:** Confuses economic substance with legal form: a repo is a collateralised loan with an obligation to repurchase, and the failures stemmed from funding withdrawal, not from a right to keep collateral.

Q458 BT answers P2.T7.514.1 · BT notes P2.T8.514.1

At the time of the Bear Stearns' demise in March 2008, Paul Friedman was a Senior Managing Director at the firm with responsibility for its fixed income repo desk. About the repo market's role in the collapse of Bear Stearns, he said in testimony before the Financial Crisis Inquiry Commission, "During the week of March 10, 2008, Bear Stearns suffered from a run on the bank that resulted, in my view, from an unwarranted loss of confidence in the firm by certain of its customers, lenders, and counterparties. In part, this loss of confidence was prompted by market rumors, which I believe were unsubstantiated and untrue, about Bear Stearns' liquidity position. Nevertheless, the loss of confidence had three related consequences." Each of the following was one of his cited three consequences EXCEPT which was not?

A. Prime brokerage clients withdrew their cash and unencumbered securities at a rapid and increasing rate

B. Repo market lenders declined to roll over or renew repo loans, even when the loans were supported by high-quality collateral such as agency securities

C. Counterparties to non-simultaneous settlements of foreign exchange trades refused to pay until Bear Stearns paid first

D. Short sellers seized on the panic and drove the stock price down, which reduced equity capital available, and equity capital was already the least stable source of funds

Answer: D

Step by step

1. Friedman's three consequences: (1) prime brokerage clients withdrew cash and unencumbered securities, (2) repo lenders refused to roll over loans even against high-quality collateral such as agency securities, (3) FX

counterparties in non-simultaneous settlements refused to pay until Bear paid first.
2. A, B and C match these three.
3. Short sellers driving the stock down is not among them, so D is the exception.

Why D is correct: The testimony lists exactly three funding and counterparty withdrawals; a short-selling story about equity capital is not one of them.

Why the others are wrong

- **A:** This is the first cited consequence, so it is not the exception.
- **B:** This is the second cited consequence (even agency collateral was not enough to get repos rolled), so it is not the exception.
- **C:** This is the third cited consequence (FX settlement counterparties demanding Bear pay first), so it is not the exception.

Q459 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst is reviewing how JPMorgan Chase (JPM), as tri-party repo clearing agent for Lehman Brothers, was exposed to Lehman in the months before Lehman's September 2008 bankruptcy. Which of the following statements is correct?

A. JPM had always applied haircuts to its intraday tri-party advances to Lehman, so its exposure was fully covered until Lehman's final week.

B. JPM was not exposed to Lehman's credit once the overnight repo investors had been repaid each morning, because the tri-party structure shifts all intraday exposure to the investors.

C. Although the repo investors lent to Lehman overnight, JPM in its role as clearing agent lent to Lehman during the day, which exposed JPM to intraday credit risk that it began to address through haircuts in early 2008.

D. JPM asked Lehman for additional collateral in September 2008 because the U.S. Treasury securities Lehman had pledged had lost most of their value.

Answer: C

Step by step

1. In tri-party repo, investors lend overnight, but the clearing agent effectively finances the dealer during the day.
2. JPM's intraday lending to Lehman typically exceeded $100 billion, so JPM carried substantial intraday exposure to Lehman.
3. JPM historically took no haircuts on these intraday advances but began to do so in early 2008, phasing them in to match overnight investors' haircuts by mid-August 2008.
4. Statement C matches this sequence.

Why C is correct: C is correct: the clearing agent's intraday lending created exposure to Lehman, and JPM began taking haircuts on it in early 2008, matching investor haircuts by mid-August 2008.

Why the others are wrong

- **A:** Wrong. JPM had historically taken no haircuts on its intraday advances and only began in early 2008; the error is assuming the protection was always in place.
- **B:** Wrong. The clearing agent, not the investors, lends during the day, so the intraday risk sits with JPM; this misreads the tri-party structure.
- **D:** Wrong. The September 2008 review found that large pieces of Lehman's collateral were illiquid, could not reasonably be valued and were supported largely by Lehman's own credit; the error is attributing the call to Treasury collateral losing value.

LO 78.e

Compare the use of general and special collateral in repo transactions.

Q460 AnalystPrep Q.2253

Alameda Bank is looking to buy bonds that are "trading special". Which type of bonds is the bank looking for?

A. General collateral bonds.

B. Zero-collateral bonds.

C. Bonds issued by the government.

D. Bonds most in demand to be borrowed.

Answer: D

Step by step

1. In a specials trade, the cash lender asks for a particular security as collateral rather than accepting any security in a broad category.
2. Bonds that are in the greatest demand to be borrowed (e.g., to cover shorts) are said to be trading special.
3. Because lenders of such bonds can borrow cash cheaply, the special repo rate is below the GC rate.
4. The term is about borrowing demand, not about the issuer or the bond type, so D is correct.

Why D is correct: A bond trades special when demand to borrow that specific security is high, so its holders can lend it and borrow cash at a repo rate below the GC rate.

Why the others are wrong

- **A:** GC collateral is the opposite case: the lender accepts any security within broad categories, so no particular bond is in demand.
- **B:** There is no such thing as zero-collateral bonds; a repo is a collateralized loan, and this is not a recognized market term.
- **C:** Confuses 'special' with a bond type; government (e.g., Treasury) bonds can be GC collateral, and only the most-demanded ones (typically on-the-run) trade special.

Q461

BT answers P2.T7.514.2 · BT notes P2.T8.514.2

In contrast to general collateral (GC) repo rates, which of the following is TRUE about special repo rates?

A. Special rates are typically less than general collateral rates

B. If the counterparty's primary motivation is to lend cash rather than borrow a security, the special rate applies

C. Special rates are well-suited to repo investors who are looking to obtain the highest rate for the collateral they are willing to accept

D. The most commonly cited special rates are for overnight repos where any U.S. Treasury collateral is acceptable

Answer: A

Step by step

1. A special rate applies to a trade where the cash lender wants a particular security; the lender accepts a rate below GC to get it.
2. So every special rate is typically below the GC rate; the special spread = GC rate - special rate is positive.
3. B describes GC (cash lender indifferent to the security); C describes GC trades, which give investors the highest rate; D describes 'the' GC rate (overnight, any Treasury collateral).
4. Only A is true.

Why A is correct: Cash lenders who insist on a specific security accept a lower return, so special rates sit below the GC rate.

Why the others are wrong

- **B:** Reverses the motivation: a lender mainly seeking to lend cash (any collateral) is in a GC trade; the special rate applies when the lender wants a particular security.
- **C:** GC trades, not specials, let investors obtain the highest repo rate; demanding specific collateral lowers the rate.
- **D:** This is the definition of 'the' GC rate (overnight, any U.S. Treasury collateral), not a special rate; special rates are security- and term-specific.

Q462

Schweser Module Quiz 78.2 Q1

REBUILT — Source omitted the text of the two trader statements; reconstructed them from the source explanation so the item is self-contained.

In a presentation to management, a bond trader makes the following statements about repo collateral:

Statement 1: The difference between the federal funds rate and the general collateral (GC) rate is called the special spread.

Statement 2: During financial crises, the difference between the federal funds rate and the GC rate widens as the willingness to lend Treasury securities declines, lowering the GC rate.

Which of the trader's statements are accurate?

A. Both statements are incorrect.

B. Only Statement 1 is correct.

C. Only Statement 2 is correct.

D. Both statements are correct.

Answer: C

Step by step

1. Statement 1: the federal funds rate minus the GC rate is the fed funds-GC spread. The special spread is the GC rate minus the special rate for a particular security and term. Statement 1 is incorrect.
2. Statement 2: in a crisis, willingness to lend Treasuries in repo falls and demand for Treasury collateral rises (flight to quality), which lowers the GC rate and widens the fed funds-GC spread. Statement 2 is correct.
3. Only Statement 2 is correct: C.

Why C is correct: Statement 1 mislabels the fed funds-GC spread as the special spread, while Statement 2 correctly describes the crisis widening of the fed funds-GC spread.

Why the others are wrong

- **A:** Wrongly rejects Statement 2, which is correct (the fed funds-GC spread widens in crises as the GC rate falls).
- **B:** Accepts the mislabeled Statement 1 (the special spread is GC minus the special rate, not fed funds minus GC) and rejects the correct Statement 2.
- **D:** Accepts Statement 1 by confusing the fed funds-GC spread with the special spread.

LO 78.f

Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle.

Q463 BT answers P2.T7.514.3 · BT notes P2.T8.514.3

In regard to special spreads, each of the following is true EXCEPT which is false?

A. On-the-run (OTR) issues tend to trade "more special" than off-the-run (OFR, i.e., old or double-old) issues, where "more special" refers to special spreads that are larger

B. The special spread equals the general collateral (GC) repo rate minus the special collateral (aka, specifically requested collateral) repo rate

C. On-the-run special spreads peak immediately after an auction, and tend to decrease over the cycle, reaching their lowest level immediately before the next auction

D. Special spreads tend to be volatile on a daily basis (reflecting supply and demand for special collateral) and special spreads can be quite large (e.g., hundreds of basis points)

Answer: C

Step by step

1. Definition: special spread = GC repo rate minus the special repo rate for the same security and term, so B is true.
2. OTR issues are the most liquid and are heavily demanded as special collateral (for shorts and hedgers), so they trade at lower repo rates and wider spreads than OFR issues; A is true.
3. Over the auction cycle, OTR special spreads are narrowest right after an auction (new supply of the OTR security depresses them) and widen as the next auction approaches (shorts and hedgers migrate toward the soon-to-be OTR issue). C states the reverse, so C is false.
4. Spreads are noisy day to day and can reach hundreds of basis points (the special rate is floored near zero, so the spread is capped by the GC rate or the fails penalty); D is true.
5. The question asks for the false statement: C.

Why C is correct: OTR special spreads are smallest immediately after an auction and rise over the cycle to peak just before the next auction, the opposite of what C states.

Why the others are wrong

- **A:** True statement: OTR issues are more liquid and in greater demand as special collateral, so their spreads are larger than OFR issues; picking it reflects doubting the OTR/OFR relation.
- **B:** True statement: the special spread is GC rate minus special rate. Choosing it comes from reversing the sign (special minus GC) and thinking the definition is wrong.
- **D:** True statement: special spreads are volatile daily and can reach hundreds of basis points; choosing it assumes spreads are small and stable.

Q464 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A repo desk analyst observes that the overnight general collateral (GC) repo rate is 0.25% and that the overnight repo rate on the on-the-run (OTR) 10-year Treasury note is -0.05%. Which of the following is the OTR 10-year note's special spread and its correct interpretation?

A. 5 basis points; the special spread equals the magnitude of the negative special rate.

B. -30 basis points; holders of the note who lend it in the repo market borrow cash at a rate 30 basis points above GC.

C. 20 basis points; the special rate is subtracted from the absolute value of the GC rate.

D. 30 basis points; holders of the note who lend it in the repo market borrow cash at a rate 30 basis points below GC.

Answer: D

Step by step

1. Special spread = GC rate minus special (security-specific) repo rate.
2. 0.25% - (-0.05%) = 0.30% = 30 basis points.
3. A positive spread means the special rate is below GC: the lender of the in-demand security borrows cash cheaply.

Why D is correct: D is correct: 0.25% - (-0.05%) = 30 bp, so lenders of the special security borrow cash 30 bp below GC.

Why the others are wrong

- **A:** Wrong. It ignores the GC rate altogether; the spread is a difference from GC, not the special rate itself.
- **B:** Wrong. This reverses the formula (special minus GC), giving -30 bp; special rates are normally below GC so the spread is positive.
- **C:** Wrong. 25 - 5 = 20 bp treats the negative special rate as +5 bp, a sign error.

Q465 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A fixed income strategist studies the repo market for the on-the-run (OTR) 10-year U.S. Treasury note over a full auction cycle. Which of the following best describes the typical behavior of its special spread?

A. The spread tends to be small just after an auction because shorts can remain in the previous OTR issue, and it rises as shorts migrate to the new OTR, spiking ahead of the next auction.

B. The spread tends to peak just after an auction because dealers hold large inventories of the new issue, and it declines steadily as the issue becomes seasoned.

C. The spread tends to stay roughly constant over the cycle because the OTR issue keeps its liquidity premium until the next auction replaces it.

D. The spread tends to rise over the cycle, and re-opened issues become more special than newly issued securities because the added supply increases borrowing demand.

Answer: A

Step by step

1. Right after an auction, shorts can stay in the previous OTR or shift to the new one; this substitutability, and added supply, depress the spread.
2. As time passes, shorts migrate to the OTR and the spread rises.
3. Many participants short the OTR to hedge purchases of the to-be-issued next OTR, so demand and the spread spike into the next auction. Re-opened issues do not get as special as new issues.

Why A is correct: A is correct: spreads are low right after an auction, rise as shorts migrate, and spike before the next auction.

Why the others are wrong

- **B:** Wrong. It reverses the cycle; a short base takes time to develop, and the spread is low, not high, right after an auction.
- **C:** Wrong. Average spreads clearly increase over the cycle; liquidity alone does not make the spread constant.
- **D:** Wrong. The rising pattern is right, but re-opened issues do not get as special as new issues; extra supply depresses the spread.

LO 78.g

Calculate the financing advantage of a bond trading special when used in a repo transaction.

Q466 AP Mock 2 Q11 · AnalystPrep Q.4121

Calculate the financing value per USD 1,000 market value of an on-the-run bond if it was issued on 1 July 2019 and trades at a special spread of 0.50%. The bond is expected to trade at general collateral (GC) rates after 31 October 2019. Use the actual/360-day convention.

A. USD 1.57
B. USD 1.60
C. USD 1.67
D. USD 1.69

Answer: D

Step by step

1. Count actual days at the special spread from 1 July to 31 October: 30 (rest of July) + 31 (Aug) + 30 (Sep) + 31 (Oct) = 122 days.
2. Financing value = value of lending cash at the special spread for the period: $1{,}000 \times 0.0050 \times \frac{122}{360}$
3. $= 5 \times 0.33889 = 1.694$, i.e. about USD 1.69 per USD 1,000.

Why D is correct: The financing value is the extra interest earned by lending the bond at the special rate and reinvesting cash at GC, i.e. cash x spread x days/360, for the days the bond is expected to stay special.

Why the others are wrong

- **A:** About 113 days ($1.57/(1000 \times 0.005) \times 360$); an under-counted period.
- **B:** About 115 days; an under-counted period.
- **C:** Used a 30/360 count (120 days): $1{,}000 \times 0.005 \times 120/360 = 1.667$, ignoring the actual/360 convention.

Q467 AnalystPrep Q.2732

Calculate the financing value per $100 market value of an on-the-run bond if it was issued on March 31st and trades at a special spread of 0.30%. The bond is expected to trade at GC rates after September 30th. Use the actual/360 day convention.

A. $0.100
B. $0.187
C. $0.125
D. $0.153

Answer: D

Step by step

1. Actual days from March 31 to September 30 = 183 (April 30 + May 31 + June 30 + July 31 + Aug 31 + Sep 30).
2. Financing value = $100 \times 0.0030 \times \frac{183}{360}$
3. $= 0.30 \times 0.50833 = 0.1525 \approx \$0.153$ per $100.

Why D is correct: Financing value is the spread earned on cash over the days the bond is expected to remain special, scaled by days/360.

Why the others are wrong

- **A:** Equals 120 days (4 months x 30); the period was badly under-counted.
- **B:** Equals about 224 days; over-counted the special period.
- **C:** Equals 150 days (5 months x 30); under-counted the period.

Q468 AnalystPrep Q.2973

Which of the following is the value of lending $1,000 of cash at a spread of 0.31% for 133 days? (Use the actual/360 convention.)

A. $1.145 per $1,000
B. $2.982 per $1,000
C. $1.675 per $1,000
D. $2.442 per $1,000

Answer: A

Step by step

1. Value = cash x spread x days/360.
2. $1{,}000 \times 0.0031 \times \frac{133}{360}$
3. $= 3.1 \times 0.36944 = 1.145$ per $1,000.

Why A is correct: The value of lending cash at a spread is the spread times the cash amount prorated on an actual/360 basis for the stated days.

Why the others are wrong

- **B:** Implies about 346 days; the period was not 133 days.
- **C:** Implies about 194 days; wrong day count.
- **D:** Implies about 284 days; wrong day count.

Q469 AnalystPrep Q.4120

Bright Bank lends $10,000 of cash at a spread of 0.26% for 150 days. Calculate the Bank's lending value (actual/360 convention).

A. $1.034 per $10,000
B. $5.45 per $10,000
C. $10.83 per $10,000
D. $10.68 per $10,000

Answer: C

Step by step

1. Value = cash x spread x days/360.
2. $10{,}000 \times 0.0026 = 26$ per year.
3. $26 \times \frac{150}{360} = 10.83$ per $10,000.

Why C is correct: The spread is earned on the full cash amount for 150 of 360 days.

Why the others are wrong

- **A:** Implies only about 14 days; the period was badly under-counted.
- **B:** Implies about 76 days; wrong day count.
- **D:** Used a 365-day year: $26 \times 150/365 = 10.68$, instead of the 360-day convention.

Q470 AnalystPrep Q.4123

Calculate the value of lending $200 cash at a spread of 0.25% for 150 days (actual/360 convention).

A. $0.208
B. $0.304
C. $0.308
D. $0.205

Answer: A

Step by step

1. Value = cash x spread x days/360.
2. $200 \times 0.0025 = 0.50$ per year.
3. $0.50 \times \frac{150}{360} = 0.2083 \approx \$0.208$.

Why A is correct: Annual spread income of $0.50 is prorated for 150/360 of a year.

Why the others are wrong

- **B:** Implies about 219 days; wrong day count.
- **C:** Implies about 222 days; wrong day count.
- **D:** Used a 365-day year: $0.50 \times 150/365 = 0.205$, not the 360-day convention.

Q471 Schweser Module Quiz 78.2 Q2

The latest on-the-run (OTR) Treasury bond issued on March 1 is trading at a special spread of 0.25%. Traders expect the bond to trade at general collateral (GC) rates past June 30. The financing value of the OTR bond is therefore the value over 122 days. Given this information, the value of lending $100 of cash is closest to:

A. $0.085

B. $0.250
C. $0.305
D. $0.847

Answer: A
Step by step
1. Value = $100 \times 0.0025 \times \frac{122}{360}$
2. $= 0.25 \times 0.33889 = 0.0847$
3. So the financing value is about $0.085 (8.5 cents) per $100.

Why A is correct: The annual spread income of $0.25 is prorated by 122/360.
Why the others are wrong
- **B:** Took the full annual spread (0.25% of $100) and ignored the 122/360 proration.
- **C:** Divided by 100 instead of 360: $100 \times 0.0025 \times 122/100 = 0.305$.
- **D:** Decimal slip: 10 times the correct value.

Reading 79. Liquidity Transfer Pricing: A Guide to Better Practice

Joel Grant, "Liquidity Transfer Pricing: A Guide to Better Practice," Occasional Paper, Financial Stability Board, Bank for International Settlements (2011).

LO	Learning objective	Qs
79.a	discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process.	4
79.b	discuss challenges that may arise for banks during the implementation of LTP.	3
79.c	compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched- maturity marginal cost).	11
79.d	describe the contingent liquidity risk pricing process and calculate the cost of contingent liquidity risk.	7

LO 79.a

Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process.

Q472

AnalystPrep Q.4215

REBUILT — Original asked for a 'NOT a cost component' with vague, non-GARP labels (cushion vs reserves); rebuilt around GARP's description of what LTP does so exactly one option is clearly not an LTP function.

Which of the following is NOT a function of a bank's liquidity transfer pricing (LTP) process?

A. Charging users of funds (assets and loans) for the cost of liquidity
B. Crediting providers of funds (liabilities and deposits) for the benefit of liquidity
C. Recouping the cost of carrying a liquidity cushion by charging contingent commitments, such as lines of credit, based on their expected use of liquidity
D. Allocating the bank's general operating expenses, such as salaries and rent, to business units in proportion to their funding use

Answer: D

Step by step

1. LTP attributes the costs, benefits and risks of liquidity to the appropriate business units.
2. It charges users of funds and credits providers of funds (options A and B).
3. It also recoups the cost of carrying the liquidity cushion by charging contingent commitments on expected use (option C).
4. General operating expenses are a cost-allocation matter unrelated to liquidity, so D is not an LTP function.

Why D is correct: LTP is confined to the costs, benefits and risks of liquidity; general overhead allocation is not part of it.

Why the others are wrong

- **A:** This is a core LTP function: borrowers/asset holders pay for the liquidity they use.
- **B:** This is a core LTP function: deposit gatherers are credited for liquidity they provide.
- **C:** This is how LTP recoups cushion carrying costs, via contingent commitments priced on expected usage.

Q473

BT Mock Q74 · BT answers P2.T8.20.15.1 · BT notes P2.T8.20.15.1

What is the primary (valid) reason that banks need proper liquidity transfer pricing (LTP)?

A. Because banks engage in maturity transformation
B. Because there is a lack of liquidity transformation products in the marketplace
C. Because the true economic cost of liquidity risk is approximately zero
D. Because markets are volatile, e.g., the volatility of the 1-year LIBOR/swap spread

Answer: A

Step by step

1. Banks fund long-term assets with shorter-term liabilities (maturity transformation), which exposes them to liquidity risk.
2. Liquidity therefore has a real cost and benefit that must be attributed to the units creating or providing it.

3. Without LTP, liquidity is treated as free, so units accumulate illiquid assets and short-term funding.
4. Hence maturity transformation is the primary reason for LTP.

Why A is correct: Maturity transformation creates liquidity risk whose costs, benefits and risks must be charged to the business units that generate them.

Why the others are wrong

- **B:** Not the rationale; the issue is the cost of the transformation banks already perform, not a product shortage.
- **C:** The pre-GFC error: treating liquidity as a free good. The true economic cost is significant, not zero.
- **D:** Market volatility, including swings in the 1-year LIBOR/swap spread, is not the underlying rationale; LTP is needed because maturity transformation creates liquidity costs and risks that must be attributed to business units.

Q474

GARP PE1 Q5

REBUILT — Given key A (long-term funding reduces cushion cost) is not supported by GARP, which says short-term funding of cushion assets understated the true cost; rebuilt so the correct option is centralized wholesale funding, with the other three as common misstatements.

A risk consultant is asked to review a bank's liquidity transfer pricing (LTP) governance and make recommendations based on industry best practice. Which of the following would be a correct recommendation?

A. Wholesale funding should be restricted to a centralized group or subsidiary treasury.

B. LTP of contingent obligations, such as collateral calls and committed lines of credit, should be applied only after the first drawdown.

C. A pooled average cost of funds approach is the most accurate LTP method.

D. A liquidity transfer price should be set for the bank as a whole rather than attributed to individual business units.

Answer: A

Key correction: Source key (GARP PE1) said A, where A read 'the real costs of the bank's liquidity cushion are best reduced through long-term funding'; GARP does not support that statement, so the item was rebuilt with a supported correct option.

Step by step

1. Best practice: a centralized funding center, with wholesale funding restricted to a group or subsidiary treasury, gives proper internal governance and limits internal arbitrage (A).
2. Contingent commitments must be charged up front based on their expected use of liquidity, not after drawdown (B wrong).
3. Pooled average cost of funds underprices long-term assets; matched-maturity marginal cost is the recommended approach (C wrong).
4. LTP attributes costs, benefits and risks to the appropriate individual business units, not the bank as a whole (D wrong).

Why A is correct: Decentralized funding was behind most LTP failures in the survey; restricting wholesale funding to a central treasury fixes governance.

Why the others are wrong

- **B:** Waiting for the first drawdown ignores contingent liquidity risk; commitments are priced on expected use when established.
- **C:** Pooled average cost is the weak approach that underprices long-term assets; matched-maturity marginal cost is best.
- **D:** LTP must be granular, attributing costs, benefits and risks to individual business units; a bank-wide price removes incentives.

Q475

Schweser Module Quiz 79.1 Q1

Which of the following is considered a best practice of liquidity transfer pricing (LTP)?

A. A centralized treasury funding center should be implemented to manage the liquidity cushion across business units.

B. Banks should rely on external factors to improve LTP by meeting regulatory authority requirements.

C. Remuneration policies should not be linked to LTP to help incentivize business unit managers to produce longer-term assets.

D. Contingent collateral calls and derivatives should not be included in the LTP process but managed separately to properly account for risks.

Answer: A

Step by step

1. A centralized funding center, with wholesale funding restricted to group or subsidiary treasury, is required for proper internal governance (A).
2. External factors affect all banks equally, so internal controls matter more (B wrong).
3. Bonuses and remuneration should factor in the cost of liquidity (C wrong).
4. Contingent collateral calls and derivatives must be charged within LTP (D wrong).

Why A is correct: Most survey problems stemmed from decentralized funding; central treasury management is the core governance best practice.

Why the others are wrong

- **B:** Regulatory and competitive factors are external and common to all banks; strong internal controls and oversight are the priority.
- **C:** Reverse of best practice: unlinked remuneration rewards cheap long-term illiquid assets; pay should reflect liquidity cost.
- **D:** Excluding contingent items causes moral hazard in trading units; they must be charged through LTP.

LO 79.b

Discuss challenges that may arise for banks during the implementation of LTP.

Q476 AnalystPrep Q.4214

Most banks are moving towards better Liquidity Transfer Pricing policies. Which of the given options is the most accurate about some of the following steps banks are taking in the actualization of improved LTP practices?

A. Banks operating with decentralized funding centers are changing towards wholesale funding controlled centrally via a treasury function.

B. Banks are coming up with trading book procedures and policies.

C. Banks with small trading book exposures in their business activities want to do away with over-trading behaviors by applying higher funding rates on net funding needs upon breaching certain funding limits.

D. All of the above

Answer: D

Step by step

1. Option A: banks with decentralized funding centres are moving to wholesale funding managed centrally by treasury, to restrict arbitrage between business units and treasury and among business units. True.
2. Option B: some banks are developing trading book policies and procedures (with risk controls and limits) for the first time, to capture liquidity costs of trading products. True.
3. Option C: banks whose trading books are small relative to their main business curb over-trading by imposing higher funding charges on net funding requirements once funding limits are breached. True.
4. All three statements are accurate, so the answer is D.

Why D is correct: Each of A, B and C is a documented step banks surveyed are taking to improve LTP implementation and oversight, so 'all of the above' is correct.

Why the others are wrong

- **A:** True but incomplete: it is only one of three accurate statements, so choosing it alone ignores B and C.
- **B:** True but incomplete: trading book policies are one of three accurate steps.
- **C:** True but incomplete: higher charges after limit breaches is one of three accurate steps; picking it alone overlooks A and B.

Q477 Schweser Module Quiz 79.1 Q2

Which of the following best describes one of the major challenges for banks in implementing an effective LTP process?

A. A decentralized LTP process is recommended to mitigate arbitrage opportunities for different business units.

B. Illiquid long-term assets should be penalized for increasing liquidity risk.

C. Performance evaluations of business unit managers should be separate from the LTP process.

D. A liquidity management information system (LMIS) should produce and monitor high-quality reports on a quarterly basis.

Answer: B

Step by step

1. A key challenge is properly accounting for the cost of liquidity in funding illiquid long-term assets (and crediting units that provide liquidity via deposits), because underpricing led banks to accumulate such assets funded short-term.
2. A: Arbitrage is reduced by a centralized treasury, not a decentralized process, so A is false.
3. C: Remuneration and performance measures should reflect liquidity costs, benefits and risks, so they should be linked to LTP, not separate.
4. D: Funding-cost oversight should move to monthly, and the LMIS should give timely reports; quarterly is too infrequent.
5. Only B matches the challenge: illiquid long-term assets must bear a liquidity premium.

Why B is correct: Charging illiquid long-term assets an appropriate liquidity premium counters the underpricing that let banks build up long-term assets funded by short-term liabilities.

Why the others are wrong

- **A:** Reverses the recommendation: centralized treasury oversight, not decentralization, limits arbitrage between business units.
- **C:** Reverses the recommendation: performance evaluation and remuneration should incorporate liquidity costs and risks via LTP.
- **D:** Reporting frequency should be higher (monthly or timelier), not quarterly; quarterly or semi-annual was the pre-crisis deficiency.

Q478 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager at a mid-sized bank is reviewing why the bank's liquidity profile deteriorated ahead of a funding stress. She finds that the bank's lending business units concentrated on long-term loans, were charged only a nominal internal rate for the cost of funding liquidity, and reported high profits that drove large bonuses. Meanwhile, the retail deposit unit received little internal credit for gathering long-term deposits. Which of the following is the most likely implementation challenge of the bank's liquidity transfer pricing (LTP) process that this review illustrates?

A. The funding costs were reviewed too frequently, which caused business units to over-adjust their lending to short-term rate changes.

B. The remuneration policy for business unit executives did not properly account for the costs, benefits, and risks of liquidity, so performance was mismeasured and incentives were distorted.

C. The bank applied charges for contingent liquidity risk, such as collateral calls, only to the trading book rather than to the lending business units.

D. The bank's central treasury function was too involved in pricing, which removed the incentive for business units to manage their own funding.

Answer: B

Step by step

1. The lending units were undercharged for funding long-term, illiquid loans, which overstated their profits and bonuses.
2. The deposit unit was under-credited for long-term deposits, so it had little incentive to supply stable liquidity.
3. This is a failure to reflect liquidity costs, benefits, and risks in performance measurement and remuneration, which is the challenge of aligning executive remuneration with the LTP process.

Why B is correct: A challenge of implementing LTP and the LMIS is making sure remuneration for business unit executives reflects liquidity. Undercharging long-term loans and under-crediting long-term deposits produces mismeasured performance and rewards excessive illiquid lending.

Why the others are wrong

- **A:** Wrong. The reading's recommended fix is to raise the frequency of funding-cost meetings to monthly, because pre-crisis meetings were quarterly or semi-annual; too-frequent review is not the problem described, and nothing in the scenario points to over-adjustment.
- **C:** Wrong. Accounting for the liquidity costs of derivatives and contingent collateral or margin calls is a different LTP implementation challenge; the scenario concerns funding charges on long-term loans and credits for long-term deposits, not contingent obligations.
- **D:** Wrong. A centralized treasury is the recommended solution because it limits internal arbitrage; central involvement is not a challenge, and the scenario shows undercharging, not excess central control.

LO 79.c

Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched- maturity marginal cost).

Q479

AP Mock 1 Q54

REBUILT — Original 'all of the above' mixed overlapping options A and B and had no wrong option; recast as an EXCEPT question using the reasons GARP gives for the pooled average cost approach.

Banks have historically used the pooled average cost of funds approach to liquidity transfer pricing (LTP) for several reasons. Which of the following is NOT one of those reasons?

A. All products are charged the same spread, so the approach is simple to implement and administer.

B. It is easier for business units to understand and comply with, and a basic liquidity management information system is sufficient.

C. It is less sensitive to intermediate changes in the bank's actual market cost of funds, reducing net interest income volatility across business units.

D. It reflects the current market perception of liquidity risk for each maturity, so long-term assets are charged more than short-term assets.

Answer: D

Step by step

1. Recall the reasons for the pooled average cost approach: simplicity of administration, ease of understanding and compliance, use of a basic LMIS, and lower net interest income volatility because spreads react little to changes in the market cost of funds.
2. A, B and C match these reasons.
3. D describes the matched-maturity marginal cost of funds approach. The pooled average cost applies one spread to all maturities and lags the market, so D is a weakness of the approach, not a reason for using it.

Why D is correct: The pooled average cost approach is attractive for its simplicity and smoothing, but it ignores maturity and lags current market liquidity costs; D states the opposite.

Why the others are wrong

- **A:** A genuine reason: one spread for all products is easy to administer.
- **B:** A genuine reason: simplicity aids understanding and compliance and needs only a basic LMIS.
- **C:** A genuine reason: averaging smooths changes in funding cost and reduces net interest income volatility.

Q480

AP Mock 2 Q72 · AnalystPrep Q.4204

REBUILT — Mock version called the loan a bullet loan (a bullet loan would be charged the single 6-year premium); restated as a linearly amortizing loan with equal annual principal repayments and replaced weak distractors.

A six-year linearly amortizing loan repays equal amounts of principal at the end of each year, so it can be viewed as six separate annual loans maturing in 1 through 6 years. The term liquidity premiums are:

Term in years	1	2	3	4	5	6
Term liquidity premium (bps)	6	14	18	24	27	32

Using the matched-maturity marginal cost of funds approach, what is the charge for funding liquidity risk on the loan?

A. 32.00 bps

B. 20.17 bps

C. 24.33 bps

D. 511.00 bps

Answer: C

Key correction: AP Mock 2 Q72 called the loan a bullet loan; its key D (24.33) is only correct for an amortizing loan, so the wording was corrected (answer value unchanged).

Step by step

1. Tranche t is outstanding t years, so weight each premium by its term.
2. Numerator = 1(6)+2(14)+3(18)+4(24)+5(27)+6(32) = 6+28+54+96+135+192 = 511.
3. Denominator = 1+2+3+4+5+6 = 21.
4. Charge = 511/21 = 24.33 bps.

Why C is correct: Amortizing principal is repaid gradually, so the charge is the tenor-weighted (tranching) average

of the term premiums, which is below the final-maturity premium.

Why the others are wrong

- **A:** 32 bps is the six-year bullet charge; ignores repayment of principal over the life.
- **B:** 20.17 is the simple average (121/6); ignores that longer tranches are funded longer.
- **D:** 511 is the weighted sum; forgot to divide by the sum of weights (21).

Q481

AnalystPrep Q.4206

REBUILT — Options were labelled in dollars although the data and answer are in bps, and distractors were weak; relabelled in bps with distractors from realistic errors.

A bank recorded the following term liquidity premiums and average cost of funds (bps) before the crisis and currently:

Term in years	1	2	3	4	5
Pre-GFC term liquidity premium	1	3	5	7	10
Pre-GFC average cost of funds	3	3	3	3	3
Current term liquidity premium	5	8	10	18	35
Current average cost of funds	10	10	10	10	10

Using the matched-maturity marginal cost of funds approach, what liquidity charge (in bps) does a one-year non-amortizing bullet loan receive if originated pre-crisis and currently, respectively?

A. 3 and 10
B. 5 and 1
C. 10 and 35
D. 1 and 5

Answer: D

Key correction: Source key (B, '$1 and $5') had the right numbers but wrong units; correct values are 1 bp and 5 bps.

Step by step

1. A one-year bullet loan is charged the one-year term liquidity premium.
2. Pre-GFC: 1 bp.
3. Current: 5 bps.
4. Answer: 1 and 5 bps.

Why D is correct: Under matched-maturity marginal cost, the charge equals the term liquidity premium for the loan's maturity at the time of origination.

Why the others are wrong

- **A:** 3 and 10 are the average cost of funds, which ignores maturity.
- **B:** Periods reversed: 5 is the current charge and 1 the pre-crisis charge.
- **C:** 10 and 35 are the five-year premiums, the wrong maturity.

Q482

AnalystPrep Q.4207

REBUILT — Distractors were weak and unrelated to realistic errors; rebuilt options (principal-size error, average-cost error, wrong-maturity error).

A bank recorded the following term liquidity premiums and average cost of funds (bps) before the crisis and currently:

Term in years	1	2	3	4	5
Pre-GFC term liquidity premium	1	3	5	7	10
Pre-GFC average cost of funds	3	3	3	3	3
Current term liquidity premium	5	8	10	18	35
Current average cost of funds	10	10	10	10	10

The principal of a one-year non-amortizing bullet loan is $5 million. Using the matched-maturity marginal cost of funds approach, what dollar charge is passed to the business unit writing the loan if it originated pre-crisis and currently, respectively?

A. $100 and $500
B. $1,500 and $5,000
C. $500 and $2,500
D. $5,000 and $17,500

Answer: C

Step by step

1. One-year bullet loan: charge = one-year term premium.
2. Pre-GFC: 0.0001 x $5,000,000 = $500.
3. Current: 0.0005 x $5,000,000 = $2,500.

Why C is correct: The bp premium for the loan's maturity is applied to the full principal of the bullet loan.

Why the others are wrong

- **A:** $100 and $500 result from using $1 million principal rather than $5 million.
- **B:** $1,500 and $5,000 apply the average cost of funds (3 bps and 10 bps), ignoring maturity.
- **D:** $5,000 and $17,500 apply the five-year premiums (10 bps and 35 bps), the wrong maturity.

Q483

AnalystPrep Q.4209

REBUILT — Table included a 'Difference' row (premium minus average cost) that invites using the wrong inputs and distractors were weak; removed the row and rebuilt options.

A five-year linearly amortizing loan has a principal of $2 million. Think of it as five separate annual loans of $400,000 each. The pre-crisis data are:

Term in years	1	2	3	4	5
Term liquidity premium (bps)	2	6	8	9	12
Average cost of funds (bps)	4	4	4	4	4

Using the matched-maturity marginal cost of funds approach, what is the charge for the loan?

A. 8.93 bps
B. 4.93 bps
C. 7.40 bps
D. 12.00 bps

Answer: A

Step by step

1. Weight each premium by its term.
2. Numerator = 1(2)+2(6)+3(8)+4(9)+5(12) = 2+12+24+36+60 = 134.
3. Denominator = 1+2+3+4+5 = 15.
4. Charge = 134/15 = 8.93 bps.

Why A is correct: An amortizing loan's charge is the tenor-weighted average of the term premiums of its annual tranches.

Why the others are wrong

- **B:** 4.93 = 8.93 minus the 4 bps average cost; the average cost of funds is not part of the matched-maturity charge.
- **C:** 7.40 is the simple average (37/5); ignores tenor weights.
- **D:** 12 bps is the five-year bullet charge; ignores amortization.

Q484

AnalystPrep Q.4213

REBUILT — Distractors (32.8, 832) were giveaways; rebuilt with bullet-loan, simple-average and no-division errors.

A four-year linearly amortizing loan repays equal principal each year and has term liquidity premiums of 12, 17, 24 and 30 bps for maturities of 1, 2, 3 and 4 years. Using the matched-maturity marginal cost of funds approach, what is the charge for funding liquidity risk?

A. 30.00 bps
B. 20.75 bps
C. 238.00 bps
D. 23.80 bps

Answer: D

Step by step

1. Weights are the terms 1 to 4; sum of weights = 10.
2. Numerator = 1(12)+2(17)+3(24)+4(30) = 12+34+72+120 = 238.
3. Charge = 238/10 = 23.8 bps.

Why D is correct: Each annual tranche is charged its own term premium; the loan's charge is their tenor-weighted average.

Why the others are wrong

- **A:** 30 bps is the four-year bullet premium; ignores amortization.
- **B:** 20.75 is the simple average (83/4); ignores tenor weights.
- **C:** 238 is the weighted sum; not divided by 10.

Q485

AnalystPrep Q.5418

REBUILT — Tightened the correct statement to match GARP's description of how the term liquidity premium is derived, and replaced the debatable option D with a clearly false statement.

A risk analyst discusses liquidity transfer pricing (LTP) approaches with her supervisor. Which of the following claims is most likely correct?

A. The matched-maturity marginal cost approach derives the term liquidity premium by converting the bank's fixed-rate borrowing costs to floating-rate costs through an internal swap and observing the spread over the reference rate.

B. If management remuneration is based on net income, both the average cost and the zero-cost approaches correctly align the maturities of the bank's lending and borrowing.

C. Under the zero-cost approach, banks tend to hold long-term, illiquid assets financed by stable long-term liabilities.

D. Under the pooled average cost approach, lowering the liquidity spread increases loan generation and also increases the funds that business units raise through deposits.

Answer: A

Step by step

1. GARP: the term liquidity premium is the spread over the reference (swap) rate that values the internal swap at par after converting fixed-rate borrowing costs to floating.
2. B: average and zero-cost approaches ignore maturity, and net-income remuneration pushes more long-term lending, widening mismatches.
3. C: zero-cost leaves banks with illiquid assets and few long-term liabilities.
4. D: a smaller spread cuts the credit to deposit providers, so deposits fall.

Why A is correct: A is GARP's description of how the matched-maturity premium is obtained; the other statements reverse the effects of the zero and average cost approaches.

Why the others are wrong

- **B:** Neither approach matches maturities; net-income pay encourages more long-term lending, worsening the mismatch.
- **C:** Zero-cost produces illiquid assets with too few long-term liabilities, not stable long-term funding.
- **D:** Lowering the spread raises loans but reduces the incentive to provide deposits, so deposits fall and the loan-to-deposit ratio worsens.

Q486

BT answers P2.T8.20.15.2 · BT notes P2.T8.20.15.2

In regard to the various approaches to liquidity transfer pricing (LTP), each of the following statements is true EXCEPT which is false?

A. The zero cost of funds approach to LTP is a poor practice because it views funding liquidity as free and funding liquidity risk as essentially zero

B. The pooled average cost of funds approach to LTP is likely to undercharge long-term assets and therefore promote unhealthy maturity transformation

C. The matched-maturity marginal cost of funds approach to LTP is best practice but does require the bank to update term liquidity premiums

D. The matched-maturity marginal cost of funds approach to LTP charges deposits as a use of funds but charges "sticky" deposits due to the expense of an adhesive agent

Answer: D

Step by step

1. A is true: zero cost treats liquidity as free with no charge or credit.
2. B is true: one spread for all maturities undercharges long-term assets.
3. C is true: it is current best practice and requires premiums to be actively updated.
4. D is false: deposits are a source of funds and are credited, not charged, with sticky deposits (e.g., term deposits)

receiving larger credits than hot/volatile ones.

Why D is correct: Under matched-maturity marginal cost, providers of liquidity are credited, and stickier (longer-maturity) deposits earn bigger credits.

Why the others are wrong

- **A:** True: zero cost treats liquidity as free.
- **B:** True: the average spread ignores maturity so long-term assets are undercharged.
- **C:** True: best practice, requiring regularly updated term liquidity premiums.

Q487 GARP PE2 Q26 · Mock 4 Q27

A treasurer of a bank is assessing the different methods of pricing liquidity and is concerned about the potential impact of applying each method. The treasurer asks an analyst to review and evaluate the various approaches to liquidity transfer pricing and to prepare a report with recommendations. Which of the following statements would be correct for the analyst to include in the report?

A. A zero-cost of funds approach tends to result in the bank holding long-term highly illiquid assets funded by long-term stable liabilities.

B. An average cost of funds approach tends to result in the greatest maturity transformation for a bank's balance sheet.

C. A matched-maturity marginal cost of funds approach converts the bank's fixed-rate borrowing cost to a floating-rate borrowing cost.

D. Both an average cost of funds approach and a zero-cost of funds approach appropriately align the maturity of the bank's lending and borrowing activities when management compensation is based on net income.

Answer: C

Step by step

1. Recall the matched-maturity marginal cost approach: the bank swaps its fixed-rate term funding costs to floating-rate costs through an internal swap.
2. The spread over the reference (swap) rate is the term liquidity premium, which charges assets and credits liabilities by maturity.
3. Check each other option against the three approaches: zero cost charges nothing for liquidity, average cost charges one blended rate regardless of maturity.
4. Only statement C accurately describes a feature of an LTP approach.

Why C is correct: The matched-maturity marginal cost approach strips out the liquidity component of fixed-rate funding costs by swapping them to floating, leaving the term liquidity premium as the charge.

Why the others are wrong

- **A:** Zero cost makes liquidity free, so the bank ends up with long-term illiquid assets funded by short-term (often overnight) liabilities, not by long-term stable liabilities.
- **B:** Zero cost, not average cost, produces the greatest maturity transformation; average cost also encourages long-term assets but is less extreme.
- **D:** Neither approach aligns maturities: both charge long assets too little relative to their liquidity risk, and net-income-based pay then rewards writing long-term loans.

Q488 Mock 2 Q55

A newly established bank is attempting to formulate its liquidity transfer pricing approach. Which method would most likely result in fewer long-term loans being underwritten?

A. Marginal cost of funds.

B. Pooled average cost of funds.

C. Separate average cost of funds.

D. Zero-cost approach.

Answer: A

Step by step

1. Long-term loans are written in volume when business units are charged too little for the long-term funding they use.
2. Matched-maturity marginal cost charges the term liquidity premium for the loan's maturity, so a long loan bears a high charge (e.g., 10 bps versus 1 bp for one year pre-crisis).

3. Pooled and separate average approaches charge one rate irrespective of maturity; zero cost charges nothing for liquidity.
4. Hence only the marginal cost approach raises the hurdle for long-term loans and reduces their volume.

Why A is correct: Matched-maturity marginal pricing makes long-term loans pay the full liquidity premium for their term, discouraging them; the other approaches subsidise them.

Why the others are wrong

- **B:** A pooled average rate is the same for all maturities, so long loans are undercharged and more are written.
- **C:** Separate average cost and benefit rates still ignore maturity, so long loans are still not charged more for longer funding.
- **D:** Zero cost treats liquidity as free, giving the strongest incentive for long-term lending.

Q489 Schweser Module Quiz 79.2 Q1

Which of the following statements describes the best approach for liquidity transfer pricing?

A. The zero-cost funding approach is preferred in cases in which swap rates are unknown and undeterminable.
B. The pooled average cost of funds approach is more appropriate for banks with numerous business units.
C. The separate average cost of funds approach is preferred to accurately account for business units with large trading activities.
D. The matched-maturity marginal approach is preferred because it quantifies liquidity risk premiums across all maturities.

Answer: D

Step by step

1. Current best practice is the matched-maturity marginal cost of funds approach.
2. It derives the liquidity portion of the bank's actual market funding cost for each maturity (term liquidity premium) and charges or credits accordingly.
3. Zero-cost and average-cost approaches ignore maturity-specific liquidity premiums and are regarded as poor practice.

Why D is correct: Only the matched-maturity marginal approach prices liquidity at each maturity, so it allocates the costs, benefits and risks of liquidity correctly.

Why the others are wrong

- **A:** Zero cost treats liquidity as free; it is not a remedy for missing swap data and is poor practice.
- **B:** Pooled average is simpler for many units but misprices by maturity; simplicity does not make it best.
- **C:** Separate average rates still ignore maturity and do not accurately capture trading activities' liquidity needs.

LO 79.d

Describe the contingent liquidity risk pricing process and calculate the cost of contingent liquidity risk.

Q490 AP Mock 2 Q55

REBUILT — Two distractors (8.37, 7.02) were not tied to any realistic error; replaced with the no-probability and no-undrawn-fraction errors.

Calculate the rate charged for contingent liquidity risk on a line of credit with a limit of USD 40 million, of which USD 12 million has already been drawn. There is a 45% chance the customer will draw on the remaining credit, and the cost of term funding assets in the liquidity cushion is 12 bps.

A. 3.78 bps
B. 1.62 bps
C. 5.40 bps
D. 8.40 bps

Answer: A

Step by step

1. Rate $= \dfrac{\text{limit} - \text{drawn}}{\text{limit}} \times P(\text{drawdown}) \times$ cost of funding the liquidity cushion
2. Undrawn fraction $= (40 - 12)/40 = 0.70$
3. $0.70 \times 0.45 \times 12 = 3.78$ bps

Why A is correct: The charge is the cost of the cushion scaled by the undrawn share of the limit and by the proba-

bility that the undrawn part is drawn.

Why the others are wrong

- **B:** Used the drawn fraction (12/40 = 0.30) instead of the undrawn fraction: 0.30 x 0.45 x 12 = 1.62.
- **C:** Ignored the undrawn fraction: 0.45 x 12 = 5.40.
- **D:** Ignored the drawdown probability: 0.70 x 12 = 8.40.

Q491

AnalystPrep Q.4201

REBUILT — Distractors 4.40 and 2.71 were not traceable to realistic errors; replaced with common-mistake values.

Calculate the rate charged for contingent liquidity risk of a line of credit with a limit of $56 million, of which $22 million has already been drawn. Assume there is a 38% chance that the customer will draw on the remaining credit and that the cost of term funding assets in the liquidity cushion is 7.5 bps.

A. 1.73 bps
B. 2.85 bps
C. 4.55 bps
D. 1.12 bps

Answer: A

Step by step

1. $\text{Rate} = \dfrac{\text{limit} - \text{drawn}}{\text{limit}} \times P(\text{drawdown}) \times \text{cost of funding the liquidity cushion}$
2. $\text{Undrawn fraction} = (56 - 22)/56 = 0.6071$
3. $0.6071 \times 0.38 \times 7.5 = 1.73 \text{ bps}$

Why A is correct: Only the undrawn share of the limit is exposed to drawdown, and it is weighted by the drawdown probability before applying the cushion cost.

Why the others are wrong

- **B:** Ignored the undrawn fraction: 0.38 x 7.5 = 2.85.
- **C:** Ignored the drawdown probability: 0.6071 x 7.5 = 4.55.
- **D:** Used the drawn fraction (22/56 = 0.393) instead of the undrawn: 0.393 x 0.38 x 7.5 = 1.12.

Q492

AnalystPrep Q.4202

REBUILT — Distractor 5.14 was not tied to a realistic error and the key explanation was ambiguous; options revised.

A line of credit has a limit of $120 million, of which $56 million has already been drawn. Assume customer K will draw on the remaining credit with a 75% probability and that the cost of term funding assets in the liquidity cushion is 32 bps. Calculate the rate charged for the contingent liquidity risk of the line of credit.

A. 24.0 bps
B. 11.2 bps
C. 12.8 bps
D. 1.28 bps

Answer: C

Step by step

1. $\text{Rate} = \dfrac{\text{limit} - \text{drawn}}{\text{limit}} \times P(\text{drawdown}) \times \text{cost of funding the liquidity cushion}$
2. $\text{Undrawn fraction} = (120 - 56)/120 = 0.5333$
3. $0.5333 \times 0.75 \times 32 = 12.8 \text{ bps}$ (= 0.00128 as a decimal rate)

Why C is correct: The undrawn share (53.3%) times the 75% drawdown probability times the 32 bp cushion cost gives 12.8 bps.

Why the others are wrong

- **A:** Ignored the undrawn fraction: 0.75 x 32 = 24.0.
- **B:** Used the drawn fraction (56/120 = 0.4667): 0.4667 x 0.75 x 32 = 11.2.
- **D:** Decimal slip: read the decimal rate 0.00128 as 1.28 bps; 0.00128 is 12.8 bps.

Q493 AnalystPrep Q.4211

REBUILT — Distractors 0.60 and 75.6 replaced with realistic-error values.

There is an $80 million limit on a line of credit and $24 million has already been drawn. The customer is expected to draw the remaining amount with a 45% probability, and the cost of term funding assets in the liquidity cushion is 24 bps. Calculate the rate charged for contingent liquidity risk.

A. 0.756 bps
B. 7.56 bps
C. 3.24 bps
D. 10.80 bps

Answer: B

Step by step

1. Rate $= \frac{\text{limit} - \text{drawn}}{\text{limit}} \times P(\text{drawdown}) \times$ cost of funding the liquidity cushion
2. Undrawn fraction $= (80 - 24)/80 = 0.70$
3. $0.70 \times 0.45 \times 24 = 7.56$ bps

Why B is correct: Scale the 24 bp cushion cost by the 70% undrawn share and the 45% drawdown probability.

Why the others are wrong

- **A:** Decimal slip: 0.000756 mis-converted to 0.756 bps; it equals 7.56 bps.
- **C:** Used the drawn fraction (24/80 = 0.30): 0.30 x 0.45 x 24 = 3.24.
- **D:** Ignored the undrawn fraction: 0.45 x 24 = 10.80.

Q494 BT answers P2.T8.20.15.3 · BT notes P2.T8.20.15.3

REBUILT — Distractors $4,500 and $25,950 were not tied to realistic errors; replaced.

A line of credit (LOC) with a limit of $20.0 million has $6.0 million already drawn. Assume there is a 65.0% chance the customer will draw on the remaining credit. The bank's term liquidity premiums (basis points) are:

Term in years	1	2	3	4	5
Term liquidity premium	5	9	15	25	40
Average cost of funds	8	8	8	8	8

The LOC's cost of funding the liquidity cushion presumes a three-year term. What is the LOC's dollar charge for the cost of contingent liquidity risk?

A. $5,850
B. $7,280
C. $13,650
D. $21,000

Answer: C

Step by step

1. Cushion cost uses the 3-year term liquidity premium: 15 bps
2. Rate $= (20 - 6)/20 \times 0.65 \times 15 = 0.70 \times 0.65 \times 15 = 6.825$ bps
3. Dollar charge $= 6.825$ bps $\times$ $20,000,000 = $13,650

Why C is correct: The rate (per unit of limit) is multiplied by the full limit, as in the GARP example, using the 3-year liquidity premium as the cushion cost.

Why the others are wrong

- **A:** Used the drawn fraction (0.30) instead of the undrawn: 0.30 x 0.65 x 15 bps x $20m = $5,850.
- **B:** Used the 8 bp average cost of funds instead of the 15 bp 3-year premium: 0.70 x 0.65 x 8 bps x $20m = $7,280.
- **D:** Ignored the drawdown probability and applied 15 bps to the $14m undrawn amount: $21,000.

Q495 Mock 1 Q31

REBUILT — Original asked for bps of the undrawn amount, which departs from the GARP rate definition (per unit of limit); stem reworded and options re-keyed so exactly one is correct.

A bank has extended a committed credit line with a limit of USD 40 million, of which USD 15 million is currently drawn. The bank estimates a 40% probability that the client draws the entire remaining commitment. The cost of funding the liquidity cushion held against contingent commitments is 25 bps. Using the standard contingent liquidity risk pricing approach, what rate should be charged on the line of credit?

A. 6.25 bps
B. 10.00 bps
C. 15.00 bps
D. 25.00 bps

Answer: A
Key correction: Source key (Mock 1) said B (10 bps of the undrawn amount); under the GARP rate definition (per unit of limit) the correct answer is A (6.25 bps).
Step by step
1. Rate $= \frac{\text{limit} - \text{drawn}}{\text{limit}} \times P(\text{drawdown}) \times$ cost of funding the liquidity cushion
2. Undrawn fraction $= (40 - 15)/40 = 0.625$
3. $0.625 \times 0.40 \times 25 = 6.25$ bps (on the limit; = \$25,000 on the \$40m limit)

Why A is correct: The rate is the cushion cost scaled by undrawn share (62.5%) and drawdown probability (40%), expressed relative to the limit.
Why the others are wrong
- **B:** Ignored the undrawn fraction (0.40 x 25 = 10.00); this is the charge per dollar of undrawn amount, not the rate on the limit.
- **C:** Used the complement of the probability (0.60 x 25 = 15.00).
- **D:** Ignored both the undrawn fraction and the probability.

Q496 Schweser Module Quiz 79.2 Q2

REBUILT — Distractors \$5,850 not tied to a realistic error; replaced, and stem asks explicitly for the dollar charge.
A bank supplies a line of credit of \$10 million that currently has \$6 million already drawn. The bank determines that there is a 65% probability that the customer will use the remaining line of credit. The bank's cost of funding for the liquidity cushion is 16 bps. If the bank charges contingent commitments based on the probability of a drawdown, what should the dollar charge for liquidity be for this line of credit?
A. \$1,664
B. \$2,496
C. \$4,160
D. \$10,400

Answer: C
Step by step
1. Rate $= \frac{\text{limit} - \text{drawn}}{\text{limit}} \times P(\text{drawdown}) \times$ cost of funding the liquidity cushion
2. Rate $= (10 - 6)/10 \times 0.65 \times 16 = 0.40 \times 0.65 \times 16 = 4.16$ bps
3. Dollar charge $= 4.16$ bps $\times$ \$10,000,000 = \$4,160

Why C is correct: The rate already embeds the undrawn fraction (40%), so multiplying by the full \$10 million limit gives \$4,160.
Why the others are wrong
- **A:** Applied the 0.40 undrawn fraction a second time (4.16 bps x 0.40 x \$10m).
- **B:** Used the drawn fraction (0.60) as a multiplier on the rate (0.60 x 4.16 bps x \$10m).
- **D:** Omitted the undrawn fraction: 0.65 x 16 bps x \$10m = \$10,400.

Reading 80. The US Dollar Shortage in Global Banking and the International Policy Response

Patrick McGuire and Goetz von Peter, "The US Dollar Shortage in Global Banking and the International Policy Response," BIS Working Papers, Bank for International Settlements (2009).

LO	Learning objective	Qs
80.a	identify the causes of the U.S. dollar shortage during the Great Financial Crisis.	7
80.b	evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities.	9
80.c	describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness.	5

LO 80.a

Identify the causes of the U.S. dollar shortage during the Great Financial Crisis.

Q497

AP Mock 1 Q55

REBUILT — Original options lacked the correct term (cross-currency funding) and keyed carry trading; rewrote options so the GARP term is the single correct answer.

In the analysis of the US dollar shortage, the extent to which banks fund their assets in one currency and invest in another (for example, using FX swaps to convert euro funding into US dollars) is best described as:

A. Interest rate swapping
B. Cross-currency funding
C. Carry trading
D. Currency hedging of the net open position

Answer: B

Key correction: Source key (AP Mock 1) said C (carry trading); the reading's term is cross-currency funding.

Step by step

1. The balance-sheet analysis separates the currency of assets from the currency of liabilities.
2. The extent to which banks fund in one currency and invest in another is called cross-currency funding.
3. Banks reached it by converting domestic-currency liabilities into dollars with FX swaps (or by spot conversion) and buying dollar assets.
4. Hence B.

Why B is correct: GARP defines cross-currency funding as the degree to which banks fund in one currency and invest in another, which was central to the build-up of the dollar shortage.

Why the others are wrong

- **A:** An interest rate swap exchanges interest flows in one currency; it does not describe funding and investing in different currencies.
- **C:** A carry trade is a speculative position in a high-yielding asset financed by a low-yielding liability; the source key chose it, but it is not the reading's term and banks were largely hedged via swaps rather than speculating.
- **D:** Hedging describes how the open currency position is neutralised (FX swaps), not the funding structure itself.

Q498

AP Mock 2 Q23 · AnalystPrep Q.4166

In a discussion panel on mitigating the recurrence of the Great Financial Crisis, a delegate mentioned the following causes of the US dollar shortage during the crisis:
I. Increased appetite of banks for foreign currency assets, notably US dollar claims on non-banks
II. Cross-currency funding
III. US dollar funding gap
Which of the above causes are correct?

A. I and II
B. I and III
C. II and III
D. I, II and III

Answer: D

Step by step

1. I: balance sheets grew and so did banks' appetite for foreign currency (mainly USD) assets such as loans to non-banks, hedge funds and structured products.
2. II: these were funded in other currencies and swapped into dollars, i.e. cross-currency funding.
3. III: dollar assets with longer horizons than dollar funding created a funding gap that had to be rolled over.
4. All three are causes, so D.

Why D is correct: The shortage resulted from a build-up of dollar assets (I) financed through cross-currency funding (II), leaving a large short-term dollar funding gap (III).

Why the others are wrong

- **A:** Omits the funding gap, which is the quantity that could not be rolled over.
- **B:** Omits cross-currency funding, which linked non-US funding to dollar assets.
- **C:** Omits the accumulation of dollar assets that created the funding need.

Q499

AnalystPrep Q.4167

REBUILT — Stem blurred two statistics (growth rate vs. level); restated that 10%/30% are year-on-year growth rates of foreign claims.

Banks' foreign claims grew from $10 trillion at the start of 2000 to $34 trillion by end-2007, and year-on-year growth in foreign claims rose from around 10% in 2001 to nearly 30% by mid-2007. This acceleration took place during a period of financial innovation. That period included all of the following EXCEPT:

A. Significant growth in the hedge fund industry

B. The emergence of structured finance

C. The spread of universal banking

D. Auctions of US dollars to banks by central banks

Answer: D

Step by step

1. GARP lists three features of the pre-crisis innovation period: structured finance, the spread of universal banking, and hedge fund growth (served by prime brokerage).
2. Dollar auctions by central banks (swap-line operations) were a policy response during the crisis, not a pre-crisis feature.
3. So D is the exception.

Why D is correct: Dollar auctions belong to the crisis-time policy response (LO 80.c), not to the pre-2007 innovation period.

Why the others are wrong

- **A:** Hedge fund growth was part of the innovation period.
- **B:** Structured finance emerged during this period and became a main source of mark-to-market losses.
- **C:** Universal banking, combining commercial and investment banking, spread in this period.

Q500

AnalystPrep Q.4169

REBUILT — Source statements mixed the postwar Kindleberger sense with the crisis sense and the key was doubtful; rebuilt as a single best-definition question per GARP footnote 1.

In McGuire and von Peter's analysis, the term "US dollar shortage" during the Great Financial Crisis refers to:

A. A global scarcity of dollars resulting from chronic US current account surpluses, as in the postwar period

B. The difficulty banks faced in securing short-term US dollar funding when the holding period of their dollar assets lengthened while their funding maturity shortened

C. A classic run by domestic depositors who converted deposits into foreign currency

D. A physical shortage of dollar banknotes available to pay for imports

Answer: B

Key correction: Source key (AnalystPrep) said I and II (trade-based statements) were correct; per GARP the shortage refers to the mismatch-driven funding difficulty, and the trade-based description is the historical usage.

Step by step

1. GARP footnote 1 distinguishes the modern usage from Kindleberger/Triffin's postwar scarcity.
2. Here the term means banks' difficulty securing short-term US dollar funding.
3. It arose because the holding period of dollar assets lengthened just as funding maturity shortened (maturity

mismatch).
4. So B.

Why B is correct: The crisis-era shortage was a bank funding problem driven by maturity mismatch, not a trade-balance or banknote scarcity.

Why the others are wrong

- **A:** This is the historical (postwar) meaning of dollar shortage, explicitly not the meaning used here; it also matches the trade-based statements in the source question.
- **C:** The reading stresses a wholesale funding and FX swap squeeze, a departure from the familiar domestic bank run story.
- **D:** Physical currency supply was not the issue; the problem was wholesale dollar funding.

Q501

BT Mock Q79 · BT answers P2.T8.20.16.1 · BT notes P2.T8.20.16.1

According to Patrick McGuire, which of the following best summarizes the proximate cause of the US dollar shortage during the global financial crisis?

A. Depositors initiated a classic run on the banks and converted their domestic deposits to foreign currency

B. The effective maturity of banks' dollar funding lengthened while their dollar assets shortened, decreasing their maturity mismatch

C. The effective maturity of banks' dollar funding shortened while their dollar assets lengthened, increasing their maturity mismatch

D. Banks shifted their portfolios to non-bank assets, including retail and corporate lending, lending to hedge funds, and holdings of securities

Answer: C

Step by step

1. Interbank, FX swap and money market funding became unstable, so funding maturity shortened.
2. Dollar assets (structured products etc.) could not be sold or rolled, so their effective holding period lengthened.
3. The maturity mismatch therefore increased; this endogenous rise caused the shortage.

Why C is correct: The shortage arose because the effective holding period of assets lengthened exactly as funding maturity shortened.

Why the others are wrong

- **A:** GARP stresses the shortage departed from the domestic bank run story; the squeeze was in wholesale and swap funding.
- **B:** Reverses both movements and says the mismatch fell.
- **D:** The portfolio shift to non-bank assets was the pre-crisis build-up that created funding needs, not the proximate cause.

Q502

BT answers P2.T8.25.1.1

REBUILT — Original option D claimed large-scale deleveraging by banks drove dollar demand, which conflicts with GARP (deleveraging was hampered and only emerged in Q4 2008); reworded D to the GARP-consistent mechanism and tidied the OCR-garbled options.

In 2008, a European bank relied on short-term U.S. dollar funding to finance its U.S. dollar-denominated assets. As interbank lending froze and liquidity dried up, it struggled to roll over its obligations. Which of the following is LEAST likely to explain the global U.S. dollar shortage during the Global Financial Crisis?

A. Banks relied heavily on short-term U.S. dollar wholesale and FX swap funding, and could not roll it over when interbank markets froze.

B. Monetary authorities and other suppliers of funds withdrew dollar placements from commercial banks, reducing the supply of dollar funding.

C. A sharp fall in demand for U.S. dollars depreciated the currency, reducing its availability to foreign banks.

D. Banks could not easily cut their dollar assets because structured finance products were hard to sell, so their dollar funding needs persisted.

Answer: C

Step by step

1. The shortage arose from a maturity mismatch: long-term U.S. dollar assets funded with short-term dollar liabilities (interbank, wholesale, FX swaps).
2. From August 2007 counterparty and liquidity concerns made rollover difficult; official-sector deposits with commercial banks also fell (GARP: about $257bn between mid-2007 and end-2008).

3. Banks could not shrink dollar assets quickly: structured products were hard to sell, off-balance-sheet vehicles returned to balance sheets and credit lines were drawn.
4. Demand for dollars therefore rose sharply and the dollar appreciated; it did not fall. Option C reverses this, so it is the least likely explanation.

Why C is correct: The shortage reflected surging demand for dollars to roll over short-term funding, not collapsing demand; a falling-demand/depreciation story is the opposite of what happened.

Why the others are wrong

- **A:** A genuine cause: the maturity/currency mismatch and rollover failure are the core of the shortage. Picking it means misreading 'least'.
- **B:** A genuine cause: GARP notes monetary authorities withdrew placements from commercial banks, shrinking dollar supply.
- **D:** A genuine cause: illiquid structured assets (plus vehicles brought on balance sheet and drawn commitments) hampered deleveraging, so dollar needs persisted.

Q503 Schweser Module Quiz 80.2 Q1

Which of the following factors is least likely a contributor to the U.S. dollar shortage during the Global Financial Crisis (GFC)?

A. Banks' prearranged credit commitments were drawn.
B. Structured finance products became more difficult to sell.
C. Central banks globally reduced their issuance of U.S. dollars.
D. Banks brought off-balance-sheet vehicles back onto their balance sheets.

Answer: C

Step by step

1. GARP lists factors that hampered banks from reducing dollar assets: structured finance products hard to sell into illiquid markets, off-balance-sheet vehicles brought back on balance sheet, and prearranged credit commitments drawn.
2. That leaves C. Only the Fed can issue U.S. dollars; non-U.S. central banks cannot reduce dollar issuance.
3. So C is not a contributor and is the answer to a 'least likely' question.

Why C is correct: Dollar issuance is the Fed's alone, so other central banks' issuance cannot have caused the shortage.

Why the others are wrong

- **A:** A real contributor: drawn credit commitments increased banks' dollar needs.
- **B:** A real contributor: illiquid structured products could not be sold without large losses.
- **D:** A real contributor: consolidating vehicles added dollar assets and funding needs to balance sheets.

LO 80.b

Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities.

Q504 AnalystPrep Q.4176

REBUILT — Source option A defined the gap as dollar liabilities exceeding dollar assets, which contradicts GARP; rebuilt around the GARP definition of the funding gap.

Which of the following best defines a bank's US dollar funding gap?

A. The excess of a non-US bank's dollar-denominated liabilities over its dollar-denominated assets
B. The amount of short-term US dollar funding a bank requires because its dollar assets have longer horizons than the funding behind them, i.e. dollars to be rolled over before assets mature
C. A country's inability to obtain enough dollars to finance international trade
D. A high-yielding currency asset financed with a low-yielding currency liability

Answer: B

Key correction: Source key (AnalystPrep) said A (liabilities exceed assets); per GARP the gap is the short-term dollar funding needed to support longer-dated dollar assets, i.e. B.

Step by step

1. GARP: the funding gap is the amount of short-term USD funding banks require.
2. It arises when the investment horizon of dollar assets exceeds the maturity of the dollar liabilities (funding or FX swaps).
3. It is estimated from net USD claims on non-banks (lower bound) up to gross claims (upper bound).
4. So B.

Why B is correct: The gap is a maturity mismatch in dollars: assets are longer than the liabilities funding them, so liabilities must be rolled over before assets mature.

Why the others are wrong

- **A:** Reverses the logic: the gap grows when long-dated dollar assets exceed stable dollar funding, not when liabilities exceed assets.
- **C:** Describes the historical trade-based dollar scarcity, not a bank funding gap.
- **D:** Describes a carry trade.

Q505

AnalystPrep Q.4171

REBUILT — Original option C was garbled (on/off balance sheet swapped) and 'all of the above' was a giveaway; rebuilt as a three-statement item with one false statement.

Consider the following statements about the structure of large international banks' balance sheets:
I. Banks manage currency and maturity positions on a consolidated global basis rather than office by office.
II. A mismatch at one office may be left unhedged locally because it is hedged off-balance sheet or offset by on-balance sheet positions booked at the bank's offices elsewhere.
III. A country's national banking balance sheet is a reliable indicator of the vulnerabilities of banks headquartered there.
Which statements are correct?

A. I only
B. II only
C. I and II only
D. I, II and III

Answer: C

Step by step

1. I is true: large banks run maturity and currency positions globally, so stresses build up on the global consolidated balance sheet.
2. II is true: offices may leave local mismatches unhedged if hedged off-balance sheet or offset by on-balance sheet positions in other offices, giving a hedged consolidated position.
3. III is false: foreign offices book most (typically over 50%) of foreign claims, and bank nationality differs from residency, so national balance sheets can miss vulnerabilities; only the consolidated entity reveals them.
4. Correct statements: I and II.

Why C is correct: Vulnerabilities only show up on the consolidated global balance sheet because local mismatches are offset elsewhere in the group, which makes national or office-level data misleading.

Why the others are wrong

- **A:** Omits II, which is true: offsetting or off-balance-sheet hedged positions across offices are why local mismatches can be left open.
- **B:** Omits I, which is true: consolidated global management is the premise of the whole analysis.
- **D:** Includes III, which is false: residency-based national balance sheets hide positions booked by foreign offices.

Q506

AnalystPrep Q.4177

REBUILT — Option B had a wording error ('short-term borrowers'), option C was a throwaway and D asked a different LO; rebuilt around the funding-gap form of maturity mismatch.

Which of the following is correct about a maturity mismatch in a bank's foreign currency positions?

A. It is not visible on the balance sheet and therefore has no effect on the bank's liquidity.
B. It exists when the investment horizon of the bank's assets exceeds the maturity of the liabilities funding them, creating the need to roll over funding.
C. It arises only when assets and liabilities are denominated in different currencies.
D. It earns the bank a spread when the yield curve is inverted, because short-term funding then costs more than long-term lending.

Answer: B

Step by step

1. A maturity mismatch means assets are held for longer than the liabilities (or FX swaps) that fund them.
2. Liabilities must therefore be rolled over before the assets mature; if rollover fails, assets are sold, possibly at a loss.
3. In foreign currencies this is the funding gap; central banks cannot create foreign currency, so the risk is acute.
4. Only B states this.

Why B is correct: A maturity mismatch is a horizon gap between assets and their funding, which exposes the bank to rollover (funding) risk.

Why the others are wrong

- **A:** Mismatches can be inferred from balance sheet maturity data and drive liquidity risk; it is not harmless.
- **C:** Confuses maturity mismatch with currency mismatch; a mismatch can occur within a single currency (domestic currency funding risk also exists).
- **D:** The spread arises when the curve slopes upward; with an inverted curve short-term funding costs more than long-term lending earns.

Q507

BT answers P2.T8.20.16.2 · BT notes P2.T8.20.16.2

A bank diversifying internationally finances a portfolio of foreign currency loans and securities (e.g., a German bank buying U.S. dollar structured products) in three ways:
I. It borrows domestic currency and converts it in a straight FX spot transaction to buy the foreign asset.
II. It uses FX swaps to convert its domestic currency liabilities into foreign currency and buys the foreign asset.
III. It borrows foreign currency from the interbank market, non-bank participants or central banks.
Which option significantly exposes the bank to currency risk?

A. None of the options
B. Only option I significantly exposes the bank to currency risk
C. Only option III significantly exposes the bank to currency risk
D. All three of the options

Answer: B

Step by step

1. Option I: foreign-currency asset funded by domestic-currency liability, with the spot conversion done once. The on-balance-sheet mismatch remains unhedged, so the bank is exposed to currency risk.
2. Option II: the FX swap's forward leg re-delivers domestic currency, so the currency position is hedged; the remaining issue is delivering foreign currency at maturity (funding risk).
3. Option III: asset and liability are in the same foreign currency, so there is no currency mismatch; the bank must deliver foreign currency when the borrowing matures (funding risk).
4. Only I leaves a significant currency exposure.

Why B is correct: Only the spot-conversion route leaves a foreign-currency asset funded by a domestic-currency liability with no hedge; swaps and direct foreign borrowing match the currencies.

Why the others are wrong

- **A:** Ignores that option I leaves an open (unhedged) currency position.
- **C:** Reverses the logic: direct foreign currency borrowing matches asset and liability currencies; its risk is funding/rollover, not currency risk.
- **D:** Confuses funding (rollover) risk, which affects II and III, with currency risk; II is hedged by the swap and III is currency-matched.

Q508

BT answers P2.T8.25.1.2

REBUILT — Choices table was OCR-garbled and option C carried a vague qualifier; reconstructed the table and tightened the options so only one is correct.

A central bank of a country with domestic currency WK wants to acquire USD 1 million of 10-year U.S. Treasuries. It compares three funding choices:

Choice	Transactions
1 Borrow domestic, convert via FX spot	Borrow WK 900,000 domestically; convert to USD 1 million via spot; buy Treasuries
2 FX swap with rollover	Borrow WK 900,000 domestically; enter a 1-year FX swap WK 900,000 for USD 1 million (reverse
3 Borrow foreign currency directly	Borrow USD 1 million from foreign sources; buy Treasuries

Which statement best describes the implications?

A. In Choice 1, no subsequent currency transactions are required and no exchange rate exposure remains.

B. In Choice 2, the currency risk is hedged by the swap, so the maturity mismatch is also eliminated.
C. In Choice 3, there is no currency mismatch because assets and liabilities are both in USD, but USD must be delivered at maturity, so funding risk remains.
D. All three choices require delivery of USD at maturity and therefore all three face the same foreign currency rollover risk.

Answer: C
Step by step
1. Choice 1: no later USD needs, but the WK liability funds a USD asset and is unhedged, so currency risk remains. A is wrong.
2. Choice 2: the swap hedges currency risk, but a 1-year swap funds a 10-year asset; the swap must be rolled over, so a maturity mismatch (funding gap) remains. B is wrong.
3. Choice 3: USD asset funded by USD debt, so no currency mismatch; the USD borrowing must be repaid/rolled, so funding risk remains. C is right.
4. D is wrong: Choice 1 involves no future USD delivery; rollover of USD applies to Choices 2 and 3 (swap second leg and USD borrowing).

Why C is correct: Matching the currency of asset and liability removes currency mismatch, but a short-dated foreign currency liability against a 10-year asset still creates a funding gap.
Why the others are wrong
- **A:** Confuses 'no future currency needs' with 'no exposure'; the unhedged USD asset against a WK liability leaves exchange rate risk.
- **B:** Conflates currency risk with maturity mismatch; hedging with a 1-year swap does not extend funding to 10 years.
- **D:** Overgeneralises rollover risk; Choice 1 has a single spot conversion and no foreign currency to be delivered later.

Q509 Schweser Module Quiz 80.1 Q1

If the maturity of a bank's U.S. dollar liabilities to nonbanks is longer term, a(n):
A. lower bound of the U.S. dollar funding gap can be established as the net U.S. dollar position to nonbanks.
B. upper bound of the U.S. dollar funding gap can be established as the net U.S. dollar position to nonbanks.
C. lower bound of the U.S. dollar funding gap can be established as the gross U.S. dollar position to nonbanks.
D. upper bound of the U.S. dollar funding gap can be established as the gross U.S. dollar position to nonbanks.

Answer: A
Step by step
1. Funding gap = USD long-term assets minus USD long-term liabilities.
2. If nonbank liabilities are long term, they fund (offset) nonbank claims, so only the net position is unfunded: a lower bound = net USD position to nonbanks.
3. If nonbank liabilities are short term, none of them supports long-term assets, so the gross position is the upper bound.
4. Longer-term liabilities therefore give the lower bound = net position (A).

Why A is correct: Long-term liabilities to nonbanks offset nonbank claims, so only the net position needs rollover, which is the minimum (lower bound) of the gap.
Why the others are wrong
- **B:** Pairs the net position with an upper bound; the net position is the smallest possible gap.
- **C:** Pairs the gross position with a lower bound; gross applies when liabilities are short term and gives the upper bound.
- **D:** Right measure (gross) for the opposite case; it applies if liabilities to nonbanks are short term, not longer term.

Q510 Schweser Module Quiz 80.1 Q2

Prior to 2007, banks would least likely finance their increasing investment holdings of U.S. dollar assets by:
A. drawing on U.S. dollar bank credit lines.
B. borrowing U.S. dollars from their central bank.
C. borrowing U.S. dollars directly in the interbank markets.
D. borrowing in their domestic currency and converting it to U.S. dollars in the spot market.

Answer: A

Step by step

1. The three funding routes: (1) borrow domestic currency and convert at spot, (2) convert domestic liabilities into USD with FX swaps, (3) borrow USD directly from the interbank market, other participants or central banks.
2. B, C and D match routes 3, 3 and 1.
3. Drawing on USD bank credit lines is not among the routes, so it is least likely.

Why A is correct: The reading lists spot conversion, FX swaps and direct USD borrowing (interbank, nonbanks, central banks); credit lines are not among them.

Why the others are wrong

- **B:** Borrowing USD from central banks is part of the third funding route.
- **C:** Direct interbank USD borrowing is the main part of the third route.
- **D:** This is the first route: domestic borrowing plus spot conversion.

Q511 Schweser Module Quiz 80.1 Q3

The funding gap can best be defined as:

A. a mismatch between banks' domestic and foreign currency obligations.
B. the difference between banks' foreign currency and domestic currency reserves.
C. the difference between a bank's collateralized and noncollateralized foreign borrowings.
D. a foreign currency mismatch if the investment horizon of the foreign currency assets exceeds the time to maturity of the foreign currency liabilities.

Answer: D

Step by step

1. The funding gap is a maturity concept in a foreign currency.
2. It arises when the investment horizon of foreign currency assets exceeds the maturity of the foreign currency funding (liabilities or FX swaps).
3. The gap is the amount that must be rolled over before the assets mature. Only D matches.

Why D is correct: The funding gap is the foreign currency maturity mismatch: assets held longer than the liabilities funding them, creating rollover risk.

Why the others are wrong

- **A:** Describes a currency mismatch, not a maturity (funding) gap.
- **B:** Reserves are not part of the definition; the gap concerns asset horizon versus liability maturity.
- **C:** Collateralization is unrelated to the gap definition.

Q512 AnalystPrep Q.4164

The extent to which banks invest in one currency and fund in another via FX swaps is known as:

A. Cross-currency funding
B. Funding gap
C. Carry trading
D. Hedging

Answer: A

Step by step

1. Banks purchase assets in one currency (e.g., U.S. dollars) but raise funds in their domestic currency.
2. They convert the domestic funding into the asset currency through FX swaps (or spot conversion), creating on-balance-sheet foreign currency positions.
3. This practice is termed cross-currency funding, and it is best seen on the consolidated global balance sheet.

Why A is correct: Cross-currency funding describes the extent to which banks fund in one currency and invest in another, typically bridged by FX swaps.

Why the others are wrong

- **B:** A funding gap is the maturity shortfall between long-term foreign currency assets and the funding that must be rolled over; it is a consequence, not the practice of funding across currencies.
- **C:** Carry trading is a return-seeking strategy of borrowing low-yield and investing high-yield; it is not the term for FX-swap-based cross-currency funding.
- **D:** Hedging reduces exposure; using FX swaps to fund a foreign currency asset does not by itself define hedging.

LO 80.c

Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness.

Q513

AnalystPrep Q.4165

REBUILT — Stem was vague ('the following success') and distractors were not clearly tied to the swap-line response; reworded so only one option is a documented effect of the policy.

During the global financial crisis, the Federal Reserve established reciprocal swap lines with foreign central banks in response to the acute US dollar shortage faced by banks outside the United States. Which of the following was an effect of this international policy response?

A. It increased banks' demand for foreign currency assets
B. It created the US dollar funding gap of banks
C. It forced banks to unwind their US dollar assets in distress sales
D. It mitigated interbank rate volatility and upward pressure on the US dollar

Answer: D

Step by step

1. The swap network let the Fed lend dollars (against foreign currency collateral) to foreign central banks, which auctioned them to local banks.
2. This relieved banks' need to scramble for dollars, so they could avoid further distressed sales of US dollar assets.
3. The evidence in the reading: lower interbank rate volatility and reduced upward pressure on the dollar.
4. Choice D is the only effect of the policy; the others are causes of the shortage or the opposite of the outcome.

Why D is correct: By supplying dollars through collateralized central bank swap lines, the Fed eased banks' dollar funding stress, which reduced interbank rate volatility and upward pressure on the dollar.

Why the others are wrong

- **A:** Swap lines were not about boosting demand for foreign assets; the pre-crisis build-up of foreign currency assets is part of the cause of the shortage.
- **B:** The funding gap arose from maturity/currency mismatches before the policy response; it is a cause, not a success of the policy.
- **C:** The swap lines helped banks avoid further distressed sales of dollar assets, so this is the opposite of the effect.

Q514

AnalystPrep Q.4170

REBUILT — Original stem confused 'challenges' with benefits and listed outcomes; reframed as the two challenges of international LOLR that the swap network overcomes, with a clean statement set.

The Federal Reserve's swap network is seen as overcoming two challenges commonly associated with international lending of last resort. Consider the following statements:

I. The Fed and its foreign counterparts can in principle create any amount of money, unlike international financial institutions administering limited resources.
II. The swap lines were made unlimited only with the central banks of Canada, England and Japan.
III. The network does not compound the informational problems that give rise to moral hazard.
IV. The Fed bears the credit risk of the ultimate bank borrowers and monitors them directly.

Which statements correctly identify the two challenges overcome?

A. I and III
B. II and IV
C. I and IV
D. II and III

Answer: A

Step by step

1. Challenge 1 is whether there is enough money: central banks can create any amount of their own currency, unlike international institutions with limited resources (I true).
2. Challenge 2 is moral hazard from information problems: the Fed lends against collateral to foreign central banks and has no credit risk to the ultimate borrowers; national authorities monitor banks (III true).
3. II is false: unlimited lines were with the ECB, Bank of England and Swiss National Bank (late 2008).
4. IV is false: the Fed bears no credit risk vis-a-vis the ultimate borrowers and delegates monitoring.
5. So I and III: choice A.

Why A is correct: The network overcomes limited resources (money creation power) and moral hazard (collateralized lending to central banks, monitoring delegated to national authorities).

Why the others are wrong

- **B:** Both statements are false: wrong central banks named for unlimited lines, and the Fed does not take bank credit risk.
- **C:** Picks the correct resource point but IV is false because credit risk and monitoring sit with the foreign central bank.
- **D:** III is correct, but II misstates which central banks received unlimited lines (ECB, BoE, SNB).

Q515

AnalystPrep Q.4175

REBUILT — Garbled stem ('nominal figure collateral') cleaned up; distractors sharpened.

The Federal Reserve extends loans, collateralized by foreign currencies, to foreign central banks. These central banks then make the US dollars available to banks in their own jurisdictions through US dollar auctions. Which of the following best describes this arrangement?

A. International lending of last resort
B. Maturity transformation across banks' balance sheets
C. A consolidated-entity currency mismatch
D. Direct Fed discount window lending to foreign commercial banks

Answer: A

Step by step

1. The Fed lends US dollars to foreign central banks against foreign currency collateral.
2. Those central banks on-lend via local dollar auctions, giving banks without US subsidiaries or eligible collateral access.
3. Providing dollars globally as the ultimate provider is, in effect, international lending of last resort.

Why A is correct: The swap network lets the Fed act as an international lender of last resort, with foreign central banks as intermediaries.

Why the others are wrong

- **B:** Maturity transformation is the banks' funding practice that created the vulnerability, not the policy response.
- **C:** A currency mismatch is a balance-sheet vulnerability, not a central bank facility.
- **D:** Lending is to foreign central banks, not directly to commercial banks; the swap network reaches banks lacking US subsidiaries or collateral.

Q516

BT answers P2.T8.20.16.3 · BT notes P2.T8.20.16.3

During the global financial crisis (GFC), the Federal Reserve introduced reciprocal currency arrangements (swap lines) with other central banks, lending US dollars collateralized by foreign currencies to central banks, which made the funds available through US dollar auctions in their jurisdictions. Lending of last resort of this kind can raise two classic problems: whether there is enough money to reassure markets, and what moral hazard is created. A successful swap program overcomes these problems with specific benefits:

I. It has the power to create an unlimited amount of money.
II. It does not create a new moral hazard.

According to McGuire and von Peter, which of these benefits did the Fed's swap lines confer?

A. Neither
B. Only I
C. Only II
D. Both benefits

Answer: D

Step by step

1. Benefit I: the Fed and counterpart central banks can in principle create any amount of money, unlike international institutions with limited resources. True.
2. Benefit II: lines are collateralized, the Fed has no credit risk vis-a-vis the ultimate borrowers, and monitoring is delegated to national authorities, so informational problems are not compounded. True.
3. Both hold: D.

Why D is correct: Money-creation capacity solves the resource constraint and collateralized central-bank-to-central-bank lending avoids moral hazard.

Why the others are wrong

- **A:** Denies both benefits; both are stated advantages of the swap network.
- **B:** Recognizes the unlimited money point but wrongly rejects the no-new-moral-hazard benefit.
- **C:** Recognizes the moral hazard point but wrongly rejects that central banks can create unlimited money.

Q517 Schweser Module Quiz 80.2 Q2

Which of the following statements about the US dollar swap lines extended by the Fed to global central banks during the 2007-2009 financial crisis is least accurate?

A. Swap lines could be unlimited.
B. Swap lines were primarily uncollateralized.
C. The Fed can be seen as a lender of last resort.
D. Central banks typically made funds available locally through auctions.

Answer: B

Step by step

1. Swap lines were loans from the Fed collateralized by foreign currencies, so B is false.
2. Late 2008: lines with the ECB, Bank of England and Swiss National Bank became unlimited (A true).
3. Unlimited provision made the Fed an international lender of last resort (C true).
4. Recipient central banks auctioned the dollars locally (D true).
5. Least accurate: B.

Why B is correct: The swap network was collateralized by foreign currencies, which is why the Fed faced no credit risk from the ultimate borrowers.

Why the others are wrong

- **A:** Accurate: lines with the ECB, BoE and SNB were made unlimited in late 2008.
- **C:** Accurate: unlimited dollar provision effectively made the Fed an international LOLR.
- **D:** Accurate: foreign central banks distributed dollars through local US dollar auctions.

Reading 81. Covered Interest Parity Lost: Understanding the Cross-Currency Basis

Claudio Borio, Robert McCauley, Patrick McGuire, and Vladyslav Sushko, "Covered Interest Parity Lost: Understanding the Cross-Currency Basis," BIS Quarterly Review, Third Quarter (2016).

LO	Learning objective	Qs
81.a	differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps.	11
81.b	identify key factors that affect the cross-currency swap basis.	3
81.c	assess the causes of covered interest rate parity violations after the financial crisis of 2008.	14

LO 81.a

Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps.

Q518

AP Mock 3 Q25

REBUILT — Two options were giveaways and the correct option did not specify the mechanics; rewrote options so the choice turns on cross-currency versus FX swap mechanics.

Walter runs a US business with dollar revenues and has a long-dated mortgage on a French property with fixed payments in euros. The dollar is weakening against the euro, so each payment costs more dollars. Which action best protects him over the full life of the mortgage?

A. Enter a one-month FX swap once, swapping dollars for euros at spot and back at the forward rate.

B. Enter a cross-currency swap in which he receives euro payments and pays dollar payments, with the final principal exchange at today's spot rate.

C. Enter a cross-currency swap in which he receives dollar payments and pays euro payments.

D. Wait until the FX market stabilises and then convert dollars to euros at the prevailing spot rate each period.

Answer: B

Step by step

1. Walter's exposure is a multi-year stream of euro outflows against dollar revenues, so he is effectively short euros over a long horizon.
2. A cross-currency swap is the longer-term instrument (often more than one year); it exchanges interim interest payments and re-exchanges principal at the initial spot rate, fixing the dollar cost of the euro stream.
3. To hedge, Walter must receive euros (to fund the mortgage) and pay dollars, which is option B.

Why B is correct: A cross-currency swap (longer term, interim payments, final exchange at the original spot rate) locks in the dollar cost of a long-dated euro liability.

Why the others are wrong

- **A:** A short-dated FX swap swaps back at the forward rate after one month and covers only that period; he would have to roll it at new forward points.
- **C:** This direction receives dollars and pays euros, adding to rather than offsetting his short-euro exposure.
- **D:** Waiting leaves the exposure unhedged; a weakening dollar keeps raising the cost and stabilisation is not guaranteed.

Q519

AnalystPrep Q.4178

REBUILT — OCR-damaged currency symbols restored and the inception/maturity principal flows re-stated consistently.

Bright Ltd., a US corporation, enters a currency basis swap with Gamble, a UK company. The principal amounts are $300 million and £240 million. At inception Bright pays Gamble $300 million and receives £240 million. On each interest date Bright pays GBP Libor on £240 million and receives USD Libor on $300 million. At maturity Bright pays £240 million and receives $300 million. The spot rate at inception is $1.25 per £1, and both firms are rated AA and can fund at Libor flat. The value of the swap to Bright at inception is:

A. Positive

B. Negative

C. Zero

D. Cannot be determined from the information given

Answer: C

Step by step

1. Check principals: 300 / 240 = 1.25 USD per GBP, equal to the spot rate, so the initial exchange is at fair value.
2. Both legs pay floating Libor flat with no basis spread, so the interim payment streams are fair.
3. The final exchange reverses the initial one at the same rate, so the PV of the pay leg equals the PV of the receive leg and the swap value is 0.

Why C is correct: A cross-currency swap is set so the initial exchange is at spot and the legs are priced fairly, so it has zero value at inception.

Why the others are wrong

- **A:** Would require Bright to receive more than it gives, e.g. a non-market spot rate or favourable spread; neither is present.
- **B:** Would require Bright to be overpaying (e.g. a spread above Libor); legs are Libor flat.
- **D:** All inputs needed (spot, principals, rates, symmetric credit) are given, so the value can be determined.

Q520

AnalystPrep Q.4185

REBUILT — Original was a trivial ratio with an ambiguous quote convention; rebuilt to test the rule that a cross-currency swap's final exchange uses the initial spot rate.

Company A wants to convert $60 million of floating-rate debt into a GBP loan. On the trade date, A exchanges $60 million with Company B for £48 million in a 10-year cross-currency swap. At maturity, A and B swap the principal amounts back. What exchange rate (USD per GBP) applies to the final exchange?

A. 1.25, the initial spot rate

B. 0.80, the initial spot rate expressed as GBP per USD

C. The 10-year forward rate implied by covered interest parity

D. The spot rate prevailing at maturity

Answer: A

Step by step

1. Initial spot: 60/48 = 1.25 USD per GBP.
2. In a cross-currency swap the currencies are swapped back at the original spot rate, not at a forward rate.
3. So the final exchange is $60m for £48m at 1.25 USD per GBP, whatever happens to the market over 10 years.

Why A is correct: Unlike an FX swap, a cross-currency swap re-exchanges principal at the initial spot rate.

Why the others are wrong

- **B:** 0.80 is the reciprocal (GBP per USD) quote, not USD per GBP as asked.
- **C:** A forward rate is used for the return leg in an FX swap, not a cross-currency swap.
- **D:** The maturity spot rate would be unknown at inception; the swap fixes the rate in advance.

Q521

AnalystPrep Q.4186

REBUILT — Distractors were arbitrary and currency symbols garbled; replaced with options reflecting realistic errors.

Two parties enter a five-year currency swap. Party A is a US firm and B is a European firm. The spot rate is $1.25 per EUR 1 (EUR 0.80 per $1). At the start, A pays B $100 million and B pays A EUR 80 million. The agreed dollar interest rate is 6% and the euro interest rate is 3.5%, paid annually. What does A pay to B each year?

A. EUR 4,800,000

B. EUR 2,800,000

C. EUR 3,500,000

D. $6,000,000

Answer: B

Step by step

1. A holds the euro principal of EUR 80 million, so A pays the euro interest and B pays the dollar interest.
2. Annual euro interest = 80,000,000 × 3.5% = 2,800,000 EUR.
3. Check: B pays A the dollar interest 100,000,000 × 6% = 6,000,000 USD per year.

Why B is correct: The party that received a currency pays interest in that currency on the principal it received.

Why the others are wrong

- **A:** EUR 4.8m applies the 6% dollar rate to the EUR 80m principal.
- **C:** EUR 3.5m applies the 3.5% euro rate to the $100m number, ignoring the exchange rate.

- **D:** $6m is the dollar interest A receives from B, not what A pays.

Q522 AnalystPrep Q.4187

REBUILT — Stated exchange rate of 2 was unrealistic and inconsistent with the amounts; reset to a consistent 100 yen per dollar and replaced distractors.

A US firm A wants to borrow ¥100 million, and a Japanese firm B needs $1 million. They enter a currency swap at an exchange rate of ¥100 per $1: A pays B $1 million and receives ¥100 million at inception. A pays 4% per year on the ¥100 million, with interest settled quarterly. What does A pay B each quarter?

A. ¥4,000,000
B. $10,000
C. ¥1,000,000
D. ¥333,333

Answer: C

Step by step

1. A holds the yen principal, so its interest is paid in yen.
2. Annual interest = $4\% \times 100{,}000{,}000 = 4{,}000{,}000$ yen.
3. Quarterly interest = $4{,}000{,}000/4 = 1{,}000{,}000$ yen.

Why C is correct: Interim interest in a currency swap is paid in the currency of the principal received, on the full principal, for each period.

Why the others are wrong

- **A:** ¥4,000,000 is the annual, not quarterly, payment.
- **B:** $10,000 converts the yen payment to dollars at 100; A owes yen interest, not dollars.
- **D:** ¥333,333 divides the annual interest by 12 (monthly) instead of 4.

Q523 AnalystPrep Q.4191

FX swaps and cross-currency swaps are derivative instruments used to hedge foreign currency exposures. Which of the following correctly identifies a similarity between the two?

A. Both instruments involve the exchange of principal and interest payments in different currencies.
B. Both instruments are used primarily to manage long-term interest rate risk.
C. Both instruments involve an exchange of currencies.
D. Both instruments are commonly utilized by central banks to manage their currency reserves.

Answer: C

Step by step

1. In both, the parties swap currencies at the outset at the current spot rate.
2. They differ at maturity (forward rate vs initial spot rate), in term (short vs often above one year) and in interim interest payments (none vs periodic).

Why C is correct: Both swaps start with an exchange of currencies at the spot rate; the other features differ.

Why the others are wrong

- **A:** Only cross-currency swaps exchange interim interest payments; an FX swap has just the two principal exchanges.
- **B:** They address currency exposure and funding in different currencies, not primarily long-term interest rate risk; FX swaps are mostly short term.
- **D:** Central-bank reserve management is not what defines these instruments in the reading and is not a shared defining feature.

Q524 BT answers P2.T8.20.17.2 · BT notes P2.T8.20.17.2

In regard to the difference between a foreign exchange (FX) swap and a cross-currency swap, each of the following is true (as a difference) EXCEPT which statement is false?

A. An FX swap is typically a shorter-term instrument while a cross-currency swap is a longer-term instrument (often above one year)
B. Covered interest rate parity (CIRP) cannot fail (i.e., cannot be violated) in the case of an FX swap, while CIRP cannot even be tested on a cross-currency swap because the final exchange uses the initial spot rate
C. At maturity, the FX swap has the counterparties repay the borrowed amounts at the pre-agreed forward rate,

but the cross-currency swap exchanges the borrowed amounts at the initial spot rate

D. The FX swap typically has only two cash flow legs (i.e., initially simultaneous lending and then repayment at maturity), but the cross-currency swap exchanges several interim interest payments

Answer: B

Step by step

1. A: true; cross-currency swaps are longer term, often above one year.
2. C: true; FX swap swaps back at forward F, cross-currency swap at original spot S.
3. D: true; only the cross-currency swap has interim interest payments.
4. B: false; CIP violations show up in both: in FX swaps as forward points (F - S) too wide, in cross-currency swaps as a non-zero basis b.

Why B is correct: CIP can fail and be measured through both instruments (forward points in FX swaps, the basis in cross-currency swaps).

Why the others are wrong

- **A:** True statement, so not the exception.
- **C:** True statement, so not the exception.
- **D:** True statement, so not the exception.

Q525

Schweser Module Quiz 81.1 Q1

The cross-currency swap basis is the:

A. interest rate differential in a cross-currency swap.

B. price to the long position in a cross-currency swap.

C. difference between the forward and spot exchange rates in a cross-currency swap.

D. amount by which the interest rate of one currency must be adjusted in a cross-currency swap so that covered interest parity (CIP) holds.

Answer: D

Step by step

1. The reference rates in a cross-currency basis swap are the two currencies' interest rates plus the basis b.
2. b is the adjustment to one currency's rate that restores CIP; b = 0 if CIP holds.

Why D is correct: The basis is the rate adjustment that makes CIP hold.

Why the others are wrong

- **A:** The interest differential is r - r*, which appears in CIP; it is not the basis.
- **B:** The basis is a spread on a reference rate, not a price paid by the long.
- **C:** F - S are the forward points, quoted in FX swaps; the basis is the interest-rate equivalent of the deviation from parity.

Q526

Schweser Module Quiz 81.1 Q2

In a foreign exchange (FX) swap, the:

A. price is quoted in forward points.

B. term is typically more than one year.

C. counterparties swap currencies back at the end of the contract at the original spot rate.

D. counterparties exchange net interest payments based on the reference rate during the term of the swap.

Answer: A

Step by step

1. FX swaps are quoted in forward points (F - S).
2. B, C, D describe cross-currency swaps.

Why A is correct: An FX swap swaps back at the forward rate, so it is quoted in forward points F - S.

Why the others are wrong

- **B:** Cross-currency swaps are the longer-term instrument; FX swaps are typically short term.
- **C:** The swap back in an FX swap is at the pre-agreed forward rate; the original spot rate applies in cross-currency swaps.
- **D:** Interim interest exchanges occur in cross-currency swaps only.

Q527

Schweser Module Quiz 81.1 Q3

REBUILT — Original asked for the sign of the basis, which is convention-dependent (Schweser says greater than zero; GARP text describes the dollar basis as negative); rebuilt to test the LO 81.a mechanics of FX swaps versus cross-currency basis swaps.

Which of the following statements correctly differentiates a foreign exchange (FX) swap from a cross-currency basis swap?

A. Both contracts re-exchange the principal at maturity at the original spot rate, and both pay periodic interest during the life of the contract.

B. An FX swap exchanges periodic interest payments based on each currency's interest rate plus a basis, whereas a cross-currency basis swap pays no interim cash flows.

C. An FX swap re-exchanges the currencies at a pre-agreed forward rate and is quoted in forward points (F − S), whereas a cross-currency basis swap re-exchanges at the original spot rate and exchanges interest payments based on each currency's rate plus the basis during its life.

D. A cross-currency basis swap is typically a short-term contract of under one year, whereas an FX swap is typically a longer-term contract of more than one year.

Answer: C

Key correction: Original item keyed C (basis greater than zero); that sign depends on the quoting convention (GARP's text describes the USD basis as negative), so the item was replaced.

Step by step

1. FX swap: exchange USD and FC at spot S today and reverse at the pre-agreed forward rate F at maturity; quoted in forward points (F − S); no interim interest exchange.
2. Cross-currency basis swap: exchange at spot S today, exchange interest payments during the life (reference rates are the two currencies' interest rates plus the basis b), and re-exchange principal at maturity at the original spot rate S; typically longer than one year.
3. Only option C states both sets of features correctly.

Why C is correct: The key differences are the maturity exchange rate (forward F versus original spot S), the presence of periodic interest payments including the basis in the cross-currency swap, and the longer tenor of the cross-currency swap.

Why the others are wrong

- **A:** Wrong because the FX swap unwinds at the forward rate F, not the original spot rate, and has no interim interest payments; confuses the two contracts.
- **B:** Reverses the contracts: the interest-plus-basis payments belong to the cross-currency basis swap, while the FX swap has no interim cash flows.
- **D:** Reverses the tenors: cross-currency swaps are the longer-term instruments (often above one year), FX swaps typically shorter.

Q528

AnalystPrep Q.4184

REBUILT — Original used negative sterling Libor and an inconsistent basis sign (answer relied on adding a -40 bp basis); rebuilt with positive rates and an explicit dollar-premium convention so there is a single answer.

A bank (Counterparty A) enters a cross-currency basis swap in which it borrows USD and lends an equivalent amount of EUR. USD Libor is 1.0% and EUR Libor is 0.5%. A dollar shortage means the USD borrower must pay USD Libor plus a basis premium of 40 bps, while it receives EUR Libor on the EUR leg. What is the net annual cost of the swap to Counterparty A?

A. 0.1%

B. 0.9%

C. 1.4%

D. 1.9%

Answer: B

Key correction: Source key (AnalystPrep) said A (4.0%) using negative GBP Libor and adding the basis inconsistently with its sign; the item was rebuilt with an unambiguous convention.

Step by step

1. A borrows USD, so it pays USD Libor plus the dollar premium: $1.0\% + 0.40\% = 1.40\%$.
2. A lends EUR, so it receives EUR Libor: 0.5%.
3. Net cost $= 1.40\% - 0.50\% = 0.90\%$.

Why B is correct: In a dollar shortage the USD borrower pays the basis on top of USD Libor (the dollar commands a premium), and the EUR leg offsets part of the cost.

Why the others are wrong

- **A:** Subtracts the basis instead of adding it (1.0% − 0.4% − 0.5%); the dollar borrower pays, not receives, the premium.
- **C:** Ignores the EUR leg: USD Libor plus basis only.
- **D:** Adds EUR Libor instead of receiving it (1.4% + 0.5%).

LO 81.b

Identify key factors that affect the cross-currency swap basis.

Q529 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A treasury analyst at a bank observes that during a sudden market disruption the bid-ask spreads on spot and forward euro/dollar transactions widened sharply, and the cross-currency basis moved away from zero. The analyst wants to explain why arbitrageurs did not close the basis. Which of the following statements is correct?

A. Wider bid-ask spreads affect the basis only through a higher Libor-OIS spread, which measures counterparty credit risk.

B. Wider bid-ask spreads make the CIP deviation more profitable to exploit, so the basis should close more quickly.

C. Wider bid-ask spreads raise the transaction cost of executing the arbitrage, which can eliminate the profit from exploiting the CIP deviation.

D. Wider bid-ask spreads matter only if sovereign CDS spreads on the underlying government bonds have also widened.

Answer: C

Step by step

1. A drop in market liquidity widens the gap between bid and ask prices in the spot and forward markets.
2. An arbitrageur buys at the ask and sells at the bid, so the round-trip cost rises (deviation based on the forward bid and spot ask).
3. If the transaction cost exceeds the CIP deviation, the arbitrage is unprofitable and the basis stays non-zero.

Why C is correct: Lower market liquidity is one of the two key factors affecting the basis; the wider bid-ask spreads add transaction costs that remove the arbitrage opportunity.

Why the others are wrong

- **A:** Wrong: Libor-OIS measures counterparty credit risk, a separate (risk premium) factor, not the liquidity channel.
- **B:** Wrong: wider spreads raise costs and reduce, not increase, arbitrage profit; the error is reading the wider spread as a larger mispricing.
- **D:** Wrong: sovereign CDS spreads capture sovereign credit risk, a risk-premium factor; liquidity effects operate independently through transaction costs.

Q530 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A funding desk analyst at a US bank collects the following one-year data: spot rate S = 1.2000 USD per EUR, forward rate F = 1.2240 USD per EUR, USD interest rate r = 2.00%, and EUR interest rate r* = 0.50%. Using the relationship $b = \frac{F-S}{S} - (r - r^*)$, the analyst computes the deviation from covered interest parity. Which of the following is correct?

A. The deviation is +1.50%, so CIP holds because the forward premium equals the interest rate differential.

B. The deviation is +0.50%, so a party borrowing US dollars through an FX swap effectively pays a lower rate than in the cash market.

C. The deviation is +2.00%, so a party borrowing US dollars through an FX swap effectively pays a higher rate than in the cash market.

D. The deviation is +0.50%, so a party borrowing US dollars through an FX swap effectively pays a higher rate than in the cash market.

Answer: D

Step by step

1. $(F - S)/S = 0.0240/1.2000 = 2.00\%$.
2. $r - r^* = 2.00\% - 0.50\% = 1.50\%$.
3. $b = 2.00\% - 1.50\% = 0.50\%$, i.e., 50 bp.
4. A positive value means the forward premium exceeds that warranted by CIP, so the USD borrower via FX swap pays more than the cash-market rate. The 50 bp reflects a risk premium and balance-sheet costs of arbitrageurs, scaled by hedging demand.

Why D is correct: b = 2.00% - 1.50% = +0.50%; a positive deviation means borrowing dollars through the swap is more expensive than in the cash market.

Why the others are wrong

- **A:** Reports the interest rate differential of 1.50% as the deviation; CIP would hold only if the forward premium equaled it, but the premium is 2.00%.
- **B:** Correct number, wrong direction: a positive b means the dollar borrower pays more, not less.
- **C:** Reports the forward premium (F-S)/S = 2.00% and forgets to subtract the interest rate differential.

Q531 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A portfolio manager at an asset management firm runs CIP arbitrage by borrowing and lending government bonds in two currencies and hedging the currency exposure with FX swaps. A risk analyst is asked which indicator best captures the risk premium factor that could cause the manager's arbitrage to produce a non-zero cross-currency basis. Which of the following is most appropriate?

A. Sovereign CDS spreads, because differences in sovereign credit risk between the two bond markets can create CIP deviations.

B. Libor-OIS spreads, because counterparty credit risk in interbank markets drives deviations in all CIP arbitrage.

C. The bid-ask spread on spot transactions, because it measures the credit risk premium of the bonds.

D. The VIX, because equity volatility is the only factor that determines the risk premium of government bonds.

Answer: A

Step by step

1. Risk premia arise from credit risk in the instruments used for the arbitrage.
2. If the arbitrage uses interbank Libor borrowing and lending, counterparty risk (Libor-OIS) matters; if it uses government bonds, differences in sovereign credit risk matter.
3. Sovereign credit risk is measured with sovereign CDS spreads.

Why A is correct: For arbitrage conducted with government bonds, sovereign credit risk differences give rise to CIP deviations and are measured by sovereign CDS spreads.

Why the others are wrong

- **B:** Libor-OIS is the measure when arbitrage is done through interbank (Libor) borrowing and lending, not government bonds; it is not universal.
- **C:** The bid-ask spread measures market liquidity (transaction cost), not a credit risk premium.
- **D:** The reading identifies counterparty and sovereign credit risk as risk premium drivers; equity volatility is not the measure.

LO 81.c

Assess the causes of covered interest rate parity violations after the financial crisis of 2008.

Q532 GARP PE1 Q35

The CIO of a UK-based hedge fund has asked the fund's portfolio managers to identify potential trading opportunities for the fund. After reviewing several suggestions, the CIO decides that trading FX swaps may offer lucrative returns. The CIO asks a portfolio manager to identify the roles that the hedge fund and other market participants can play in FX swap markets as well as factors that may affect the pricing of the cross-currency FX swap basis. Which of the following statements would be correct for the manager to make?

A. A positive FX swap basis may exist because arbitrageurs pass on their balance sheet borrowing costs through the forward FX swap price.

B. A persistent and positive FX swap basis is likely to result from non-financial firms issuing debt denominated in their home currency in offshore capital markets rather than in their home country's debt market.
C. A persistent and positive FX swap basis may exist because of the decision of institutional investors to reduce their hedges of foreign currency investments.
D. A persistent and positive FX swap basis is likely to result from international banks concentrating their lending activities in their home markets and reducing loan exposures in offshore markets.

Answer: A
Step by step
1. Per GARP, arbitrage requires balance sheet space, which is costly post-crisis (leverage and capital constraints).
2. Arbitrageurs who take the other side of hedging demand pass these costs on via wider forward points (F - S), sustaining the basis.
3. Check B-D against the three hedging-demand sources: bank currency mismatches, institutional investors' strategic hedging, and non-financial firms issuing foreign-currency debt and swapping back.

Why A is correct: Because balance sheet costs are passed through the forward leg of the FX swap, the basis opens and does not close.
Why the others are wrong
- **B:** Hedging demand arises when firms issue debt in a foreign currency (e.g., US firms issuing euro bonds) and swap back; home-currency issuance creates no such demand.
- **C:** Reversed: institutional investors maintaining or increasing hedges adds hedging demand; reducing hedges would ease pressure on the basis.
- **D:** Banks concentrating lending at home reduces cross-currency funding and hedging needs; banks' currency mismatches (foreign assets funded in another currency) are what generate demand.

Q533

AnalystPrep Q.4180 · AnalystPrep Q.4181

REBUILT — Merged two near-identical items (garbled 'ratio of returns' and a ratio of forward to spot that ignored the rates); rewrote so that the interest-rate ratio and F/S are both used and distractors are realistic errors.

The USD/EUR spot rate is 1.3536 (USD per EUR) and the one-year forward rate is 1.3280. The one-year USD interest rate is 4% and the one-year EUR interest rate is 6%. Which of the following is the ratio of the gross USD return to the gross EUR return, $(1 + r)/(1 + r^*)$, which equals F/S when covered interest parity holds?

A. 1.0193
B. 0.9434
C. 0.9811
D. 0.0189

Answer: C
Step by step
1. Gross USD return = 1.04; gross EUR return = 1.06.
2. Ratio $(1 + r)/(1 + r^*) = 1.04/1.06 = 0.9811$.
3. Check with market quotes: $F/S = 1.3280/1.3536 = 0.9811$, so CIP holds (to rounding).

Why C is correct: Under CIP the forward-to-spot ratio equals the ratio of the domestic (USD) to foreign (EUR) gross interest rates, so both routes give 0.9811.
Why the others are wrong
- **A:** This is 1.06/1.04, the inverted ratio (foreign over domestic).
- **B:** This is 1/1.06; the USD gross return of 1.04 was ignored.
- **D:** This is 1 - 0.9811, the deviation from 1 rather than the ratio.

Q534

AnalystPrep Q.4188

REBUILT — Original only subtracted two yields (8% - 5%) and ignored the covered/CIP arbitrage idea; rebuilt as a covered arbitrage profit calculation with a forward rate.

The one-year USD interest rate is 5% and the one-year GBP interest rate is 8%. The GBP/USD spot rate is 1.2500 (USD per GBP) and the one-year forward rate is 1.2400. An arbitrageur borrows USD 2,000 for one year, converts it to GBP at spot, buys a one-year GBP deposit, and sells the GBP proceeds forward. What is the riskless USD profit at maturity?

A. USD 160.00
B. USD 60.00

C. USD 2,142.72
D. USD 42.72

Answer: D
Step by step
1. Convert at spot: 2,000/1.25 = GBP 1,600.
2. GBP proceeds: 1,600 x 1.08 = GBP 1,728.
3. Convert at forward: 1,728 x 1.24 = USD 2,142.72.
4. Repay loan: 2,000 x 1.05 = USD 2,100. Profit = 2,142.72 - 2,100 = USD 42.72.

Why D is correct: CIP implies a forward of $1.25 \times 1.05/1.08 = 1.2153$; the quoted forward of 1.24 is above it, so the covered GBP investment beats the USD borrowing cost and a riskless profit exists.

Why the others are wrong
- **A:** USD 160 is the 8% GBP interest on a USD 2,000 principal treated as USD profit; it ignores the loan cost and the forward conversion.
- **B:** USD 60 is the simple 8% - 5% differential on 2,000, which assumes the exchange rate is unchanged instead of using the forward rate.
- **C:** USD 2,142.72 is the gross proceeds; the loan repayment of USD 2,100 was not deducted.

Q535
AnalystPrep Q.4189

The EUR/USD spot rate is 1.32 (USD per EUR), the one-year EUR/USD forward rate is 1.22, and the one-year USD interest rate is 6%. Which one-year EUR interest rate satisfies covered interest parity?

A. 8.20%
B. -2.03%
C. 14.20%
D. 14.69%

Answer: D
Step by step
1. CIP with USD-per-EUR quoting: $F/S = (1+r)/(1+r^*)$.
2. $1+r^* = (1+r)S/F = 1.06 \times 1.32/1.22 = 1.1469$.
3. $r^* = 14.69\%$.

Why D is correct: The forward is below spot, so the EUR must carry a higher interest rate than USD to offset the forward discount.

Why the others are wrong
- **A:** 1.32/1.22 - 1 = 8.20%: only the spot/forward ratio was used; the USD rate was ignored.
- **B:** 1.22 x 1.06/1.32 - 1 = -2.03%: forward and spot were swapped.
- **C:** 6% + 8.20% adds the forward discount to the USD rate instead of compounding the ratios.

Q536
AnalystPrep Q.4190

REBUILT — Quote convention was ambiguous ('132 USD/JPY'); clarified as JPY per USD and replaced meaningless distractors.

The spot exchange rate is JPY 132 per USD. The one-year JPY interest rate is 4% and the one-year USD interest rate is 11%. Using covered interest parity, what is the one-year forward rate (JPY per USD)?

A. 137.28
B. 140.88
C. 122.76
D. 123.68

Answer: D
Step by step
1. With JPY as the quote currency: $F = S(1+r_{JPY})/(1+r_{USD})$.
2. $F = 132 \times 1.04/1.11$.
3. $F = 123.68$.

Why D is correct: USD has the higher interest rate, so USD must trade at a forward discount against JPY (fewer JPY per USD).

Why the others are wrong
- **A:** 132 x 1.04 = 137.28: the USD rate was left out of the denominator.

- **B:** 132 x 1.11/1.04 = 140.88: the ratio was inverted, giving a forward premium to the high-yield currency.
- **C:** 132 x (1 + 0.04 - 0.11) = 122.76: rates were subtracted rather than compounded as a ratio.

Q537

AnalystPrep Q.4192

REBUILT — Stem wording ('correlation ... equal') was meaningless; rewritten as a clear definition question with plausible distractors.

Which of the following describes the no-arbitrage condition under which a currency investment hedged with a forward contract earns the same return as investing in the domestic currency, so that $F/S = (1+r)/(1+r^*)$?

A. Uncovered interest parity
B. Covered interest parity
C. Purchasing power parity
D. The international Fisher effect

Answer: B

Step by step

1. Investing domestically at r must equal exchanging at S, investing at r^*, and converting back at the forward F.
2. This is the statement $F/S = (1+r)/(1+r^*)$, i.e. covered interest parity.

Why B is correct: Covered interest parity locks in the future exchange rate via a forward, so no FX risk is borne and the two returns must be equal.

Why the others are wrong

- **A:** Uncovered parity uses the expected future spot rate, not a forward, so FX risk stays open.
- **C:** PPP links exchange rates to price levels, not interest rates and forwards.
- **D:** The international Fisher effect links interest differentials to expected spot rate changes and has no forward hedge; a zero cross-currency basis is simply what holds when CIP holds.

Q538

AnalystPrep Q.4193

REBUILT — Day-count convention was not stated and two options were arbitrary; stated 360-day convention and replaced distractors.

The GBP/USD spot rate is 1.5000 (USD per GBP). The 180-day USD LIBOR is 1.2% per annum and the 180-day GBP LIBOR is 2.0% per annum (use a 360-day year and simple interest). What is the 180-day forward GBP/USD rate implied by covered interest parity?

A. 1.4882
B. 1.4880
C. 1.5060
D. 1.4941

Answer: D

Step by step

1. Period rates: USD $0.012 \times 180/360 = 0.6\%$; GBP $0.02 \times 180/360 = 1.0\%$.
2. $F = S(1+r)/(1+r^*) = 1.5000 \times 1.006/1.010$.
3. $F = 1.4941$.

Why D is correct: GBP has the higher rate, so GBP trades at a forward discount versus USD; the forward is below the 1.5000 spot.

Why the others are wrong

- **A:** 1.4882 uses the annual rates (1.012/1.020) without scaling for 180/360.
- **B:** 1.4880 uses simple subtraction of annual rates, 1.5 x (1 + 0.012 - 0.02).
- **C:** 1.5060 inverts the ratio (1.010/1.006), implying a forward premium for GBP even though GBP has the higher interest rate.

Q539 AnalystPrep Q.4194

REBUILT — Kept the data but replaced arbitrary distractors with realistic computational errors.

The USD/JPY quote is USD per JPY: spot 0.50 and one-year forward 0.40. The one-year USD interest rate is 5%. What one-year JPY interest rate satisfies covered interest parity?

A. -16.00%
B. 25.00%
C. 131.25%
D. 31.25%

Answer: D

Step by step

1. $F/S = (1+r)/(1+r^*)$ so $1+r^* = (1+r)S/F$.
2. $1+r^* = 1.05 \times 0.50/0.40 = 1.3125$.
3. $r^* = 31.25\%$.

Why D is correct: The JPY depreciates 20% forward, so JPY interest must be much higher than USD interest to compensate.

Why the others are wrong

- **A:** 1.05 x 0.40/0.50 - 1 = -16%: spot and forward were interchanged.
- **B:** 0.50/0.40 - 1 = 25%: the USD interest rate was ignored.
- **C:** 1.3125 stated as 131.25%: forgot to subtract 1 from the gross return.

Q540 AnalystPrep Q.5413

REBUILT — Stem referred to a 'positive' basis (post-crisis basis vs USD is negative) and options were uneven giveaways; reworded and rebalanced.

Which of the following best explains why the cross-currency basis has persisted since the 2007-2009 financial crisis, rather than being arbitraged away?

A. Arbitrage requires balance sheet expansion and brings credit, funding, capital and liquidity risk, and post-crisis these are managed more tightly, limiting arbitrage.
B. Demand for currency hedges from banks, institutional investors and nonfinancial issuers has fallen, so forward prices no longer reflect interest differentials.
C. Falling global interest rates have made liquidity risk irrelevant to swap pricing, so arbitrageurs no longer need to price it.
D. CIP arbitrage removes counterparty risk altogether, so arbitrageurs' costs have fallen and the basis has widened.

Answer: A

Step by step

1. The basis opens because hedging demand (banks, institutional investors, US nonfinancial issuers swapping FC debt) has grown.
2. It does not close because arbitrage is costly and risky: it expands the arbitrageur's balance sheet and brings credit, capital, funding and liquidity (mark-to-market) risk.
3. Since the crisis, regulators, shareholders and creditors have pushed firms to manage these more carefully, limiting arbitrage.

Why A is correct: Limits to arbitrage from careful balance sheet management mean arbitrage capital is too scarce to close the basis.

Why the others are wrong

- **B:** Hedging demand has grown, not fallen; that is why the basis opens.
- **C:** Lower rates do not remove liquidity and funding risk; these are central to the limits to arbitrage.
- **D:** Arbitrage creates counterparty and funding exposures on both legs; it does not eliminate them.

Q541 BT Mock Q14 · BT answers P2.T8.20.17.1 · BT notes P2.T8.20.17.1

REBUILT — Merged three duplicates and replaced the unexplained distractors (+5.80, +13.70/+73.70) with distractors from realistic errors (inverted ratio, linear approximation).

The spot USDJPY rate is 110.00 (JPY per USD; USD is the base currency). The one-year USD interest rate is 2.50% and the one-year JPY interest rate is 0.450%, both with annual compounding. If covered interest parity holds exactly, what is the one-year swap rate, defined as $F - S$?

A. -2.20 JPY

B. Zero
C. +2.24 JPY
D. -2.26 JPY

Answer: A
Step by step
1. CIP with the foreign (quote) currency as JPY: $F/S = (1 + r_{JPY})/(1 + r_{USD})$.
2. $F = 110 \times 1.0045/1.025 = 107.80$.
3. Swap rate $F - S = 107.80 - 110.00 = -2.20$ JPY.

Why A is correct: USD has the higher interest rate, so the USD must trade at a forward discount against JPY, so that the higher USD yield is offset by the forward loss on conversion.
Why the others are wrong
- **B:** Zero would hold only if interest rates were equal; the 2.05% rate differential produces non-zero forward points.
- **C:** Inverts the ratio ($110 \times 1.025/1.0045 = 112.24$), giving a forward premium of +2.24 instead of a discount.
- **D:** Uses the linear approximation $S(r_{JPY} - r_{USD}) = -2.255$ instead of dividing by $1 + r_{USD}$.

Q542 BT answers P2.T8.20.17.3 · BT notes P2.T8.20.17.3

Since the global financial crisis (GFC), covered interest rate parity (CIP) violations have persisted. Two explanations are offered:
I. The basis opens up because banks, institutional investors and non-financial firms demand currency hedges (for example, banks closing balance sheet currency mismatches with FX swaps).
II. Arbitrageurs now price the costs and hidden risks of arbitrage, for example the cost of enlarging the balance sheet, which limits arbitrage.
Which statement(s) explain the post-GFC violations of CIP?
A. Neither I nor II
B. Only I
C. Only II
D. Both I and II

Answer: D
Step by step
1. Statement I: hedging demand from banks, institutional investors and non-financial firms (issuing foreign-currency debt and swapping back to USD) is why the basis opens up.
2. Statement II: balance sheet expansion brings credit, funding and liquidity risk, and post-crisis these are managed carefully, so these limits to arbitrage stop the basis from closing.
3. Both are the two causes identified: hedging demand and limits to arbitrage.

Why D is correct: Increased FX hedging demand opens the cross-currency basis and costly, constrained arbitrage keeps it from closing.
Why the others are wrong
- **A:** Denies both causes; each is a recognized driver.
- **B:** Omits limits to arbitrage, which explain why the basis persists.
- **C:** Omits hedging demand, which explains why the basis opens.

Q543 GARP PE2 Q5

A portfolio manager at a US-based hedge fund notes a positive cross-currency basis between two currencies that has persisted since the global financial crisis (GFC) of 2007-2009. What is the most appropriate explanation for this persistence?
A. The costs for arbitrageurs to finance their positions are increasingly reflected in the basis.
B. The costs of credit value adjustments have increased, as arbitrage positions typically eliminate counterparty risks.
C. Regulatory changes have permitted an increase in US banks' speculative proprietary trading activities.
D. The addition of a liquidity risk cost to swap pricing is no longer required given the decline in the overall level of interest rates.

Answer: A

Step by step

1. Arbitrage of CIP requires expanding the balance sheet, with credit, funding and liquidity risk.
2. Post-crisis, balance sheet is carefully managed, so its cost is passed into swap pricing.
3. This limit to arbitrage keeps the basis from closing, so A.

Why A is correct: Arbitrageurs charge for balance sheet and funding costs, so the basis settles at a non-zero level rather than being arbitraged away.

Why the others are wrong

- **B:** Arbitrage positions carry counterparty credit risk on both legs; they do not eliminate it.
- **C:** Regulation tightened, not loosened, banks' balance sheet capacity for such activity.
- **D:** Liquidity and funding risk are priced more carefully post-crisis, not less; the level of rates does not remove them.

Q544

Mock 4 Q6

REBUILT — Replaced giveaway distractors with plausible ones tied to hedging demand and arbitrage mechanics.

Before the global financial crisis, deviations from covered interest rate parity (the cross-currency basis) were rapidly arbitraged away. Since the crisis they have persisted. Which of the following best explains the persistence of the cross-currency basis?

A. Arbitrage requires expanding the balance sheet, and its funding, credit and liquidity costs are now priced, limiting arbitrage.

B. Arbitrage opportunities are now eliminated because hedging demand has declined, which has narrowed the basis to zero.

C. Arbitrage profits are fully covered by risk-free interest, so arbitrageurs need not consider their balance sheet.

D. The basis persists only during crisis periods and quickly reverts once market stress subsides.

Answer: A

Step by step

1. The basis opens because of hedging demand and does not close because of limits to arbitrage.
2. CIP arbitrage expands the balance sheet, bringing credit, funding and liquidity risk that post-crisis participants manage and price carefully.
3. Only option A states this mechanism.

Why A is correct: Costly, constrained balance sheets limit arbitrage, so violations of CIP persist.

Why the others are wrong

- **B:** Hedging demand has increased since the crisis, not declined, and the basis remains non-zero.
- **C:** Arbitrage is not free of balance sheet cost; it incurs credit, funding and liquidity risk.
- **D:** Violations have persisted in non-crisis periods, which is the point of the reading.

Q545

AnalystPrep Q.2825

Suppose that the EUR spot rate is USD 1.3988 per EUR. The 1-year EUR T-bill yields 2.28% and the 1-year USD T-bill yields 3.33%. Assuming covered interest parity (CIP) holds, what is the 1-year EUR/USD (USD per EUR) forward rate?

A. 1.4132

B. 1.0103

C. 1.3676

D. 0.7076

Answer: A

Step by step

1. CIP with S and F in USD per FC: $(1 + r_{USD}) = \frac{F}{S}(1 + r_{EUR})$, so $F = S\,\frac{1+r_{USD}}{1+r_{EUR}}$.
2. $\frac{1+r_{USD}}{1+r_{EUR}} = \frac{1.0333}{1.0228} = 1.010266$.
3. $F = 1.3988 \times 1.010266 = 1.4132$ USD per EUR.
4. Check: forward points $F - S = 1.4132 - 1.3988 = 0.0144 > 0$, consistent with the USD interest rate exceeding the EUR rate (EUR at a forward premium).

Why A is correct: Under CIP, investing in USD directly must earn the same as converting to EUR, investing at the EUR rate and locking in the conversion back at F; this requires F/S to equal the ratio of gross USD to gross EUR

returns.

Why the others are wrong

- **B:** This is just $1.0333/1.0228$, the interest factor ratio; the candidate forgot to multiply by the spot rate.
- **C:** Computed $S/(1+r_{EUR}) = 1.3988/1.0228$, omitting the USD growth factor $1+r_{USD}$.
- **D:** This is $1/F$, the forward quoted as EUR per USD, not the requested USD per EUR.

Reading 82. Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques

Peter Rose and Sylvia Hudgins, Bank Management & Financial Services, 9th Edition (New York, NY: McGraw-Hill, 2013). Chapter 7.

LO	Learning objective	Qs
82.a	discuss how asset-liability management strategies can help a bank hedge against interest rate risk.	6
82.b	describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin.	22
82.c	describe duration gap management and apply this strategy to protect a bank's net worth.	20
82.d	discuss the limitations of interest-sensitive gap management and duration gap management.	4

LO 82.a

Discuss how asset-liability management strategies can help a bank hedge against interest rate risk.

Q546

GARP book Ch.18 Internet Exercise 1 (rid q0080)

REBUILT — Original required visiting a website and reading an external article; rewritten as a self-contained question answerable from the chapter.

Explain the major sources of interest rate risk faced by a bank when market interest rates change, and how asset-liability management (ALM) strategies can be used to manage them.

Answer: see model answer — The two major kinds of interest rate risk are price risk and reinvestment risk. Price risk (market risk) arises when rising rates reduce the market value of bonds and fixed-rate loans, so selling them means taking capital losses. Reinvestment risk arises when rates fall and incoming funds must be reinvested at lower yields, lowering future income. Rate changes also move the yield on earning assets and the cost of funds at different speeds, changing interest income, interest expense, NII and NIM. ALM manages these by adjusting the mix, maturities and amounts of assets and liabilities, and by using off-balance-sheet hedges such as futures and swaps.

Step by step

1. Identify the two kinds of interest rate risk: price risk and reinvestment risk.
2. Price risk: rates up, value of fixed-rate assets down; selling realizes capital losses.
3. Reinvestment risk: rates down, cash flows reinvested at lower yields, reducing income.
4. Link to income: asset yields and funding costs reprice differently, so NII/NIM change.
5. ALM response: adjust mix, maturity and amounts of assets and liabilities; hedge with futures or swaps.

Why this is correct: Interest rate changes hit a bank through the value of its fixed-rate holdings (price risk) and the rate at which cash flows are reinvested (reinvestment risk), and ALM is the integrated tool used to manage both.

Q547

AnalystPrep Q.4233 (rid q0106)

REBUILT — Original 'all of the above' item included a vague 'perfect timing' statement not supported by the reading; rewritten as an EXCEPT question with exactly one non-supported statement.

Financial managers have long tried, with little success, to forecast interest rate movements. Each of the following is a reason why consistently accurate interest rate forecasting is so difficult EXCEPT:

A. Interest rates are determined by the interaction of thousands of suppliers and demanders of credit, so no single institution sets them.

B. Any particular interest rate is built from several components, such as a risk-free real rate and various risk premiums, and a change in any component moves the rate.

C. Market interest rates are set by a single regulator and moved only at announced dates, so banks cannot observe the rate decisions that matter.

D. Because rates are largely beyond their control, most financial managers must act as price takers and plan around rate levels they cannot influence.

Answer: C

Step by step

1. Recall the chapter's explanation: rates arise from the interactions of thousands of credit suppliers and demanders, which makes accurate forecasting virtually impossible.
2. Recall that each rate is composed of building blocks (risk-free real rate plus risk premiums), any of which can

change.
3. Statements A, B and D match the reading.
4. Statement C claims a single regulator sets market rates; this is not the reading's explanation, since market rates come from market interaction.
5. C is therefore the EXCEPT answer.

Why C is correct: Market rates emerge from many participants and multiple components, not from one regulator's decisions, so C is not a reason offered for forecasting difficulty.

Why the others are wrong

- **A:** A genuine reason (many participants set rates); choosing it misses the EXCEPT wording.
- **B:** A genuine reason (rate = risk-free rate plus several risk premiums); choosing it misses the EXCEPT wording.
- **D:** A genuine point (financial managers are price takers, not price makers); choosing it misses the EXCEPT wording.

Q548

AnalystPrep Q.6345 (rid q0713)

Changes in market interest rates have a direct impact on a bank's economic value. Which two components of a bank's financial performance are most significantly affected by these changes?

A. Net interest income and operating costs.
B. Loan origination fees and cost of customer deposits.
C. Net interest income and the value of financial instruments.
D. Operating profit and profit per share.

Answer: C

Step by step

1. Rate changes alter the yield on earning assets and the cost of funds, changing interest income and expense, hence net interest income (NII).
2. Rate changes also alter the market value of fixed-rate securities and loans (price risk), as value falls when rates rise.
3. So the two primary channels are NII and the value of financial instruments: C.

Why C is correct: Interest rate risk operates through earnings (NII) and through the market value of rate-sensitive assets and liabilities.

Why the others are wrong

- **A:** Operating costs are non-interest expenses and are not directly driven by market rates; NII alone is correct.
- **B:** Fees are not primarily rate-driven; deposit cost is only one part of the NII channel, and the value channel is omitted.
- **D:** These are broad outcomes that depend on many factors, not the two primary direct channels of rate risk.

Q549

AnalystPrep Q.6352 (rid q0714)

REBUILT — Original 'most appropriate strategy' had several defensible options (selling assets, refinancing to floating also cut exposure); rewritten so that only the derivative hedge directly hedges the existing portfolio.

A bank holds a large portfolio of long-term, fixed-rate mortgages funded mainly with short-term liabilities and is worried that market interest rates will rise. Which action would most directly hedge the interest rate exposure of the existing portfolio without selling the loans?

A. Raise the rate charged on new mortgages.
B. Enter a pay-fixed, receive-floating interest rate swap.
C. Increase the share of mortgage funding raised through short-term borrowing.
D. Shift new lending toward even longer-maturity fixed-rate loans.

Answer: B

Step by step

1. Exposure: fixed-rate asset income is locked in while short-term funding costs reprice upward if rates rise, and the loans' value falls.
2. A hedge must gain when rates rise. Pay fixed/receive floating: floating receipts rise with rates, offsetting higher funding cost, while fixed payments are unchanged.
3. Swaps and futures are the off-balance-sheet hedges named in the reading.
4. Options A, C, D do not offset the existing exposure; C and D increase it.

Why B is correct: A pay-fixed, receive-floating swap produces gains when rates rise, offsetting the loss in NII and loan value on the fixed-rate portfolio without liquidating assets.

Why the others are wrong

- **A:** Affects only new loans; existing mortgages' income stays fixed, so the exposure is unchanged.
- **C:** More short-term funding raises the repricing mismatch, increasing exposure to rising rates.
- **D:** Adds more long-term fixed-rate assets, increasing duration and the exposure.

Q550

Schweser Module Quiz 82.1 Q1 (rid q0729)

An integrated approach to managing balance sheet interest rate risk is called:

A. asset management.
B. liability management.
C. asset-liability management (ALM).
D. asset-liability and owners' equity management.

Answer: C

Step by step

1. Banks view the asset and liability sides of the balance sheet as an integrated whole for both profitability and risk.
2. This integrated approach is called asset-liability management (ALM), run by the ALCO.

Why C is correct: ALM manages assets and liabilities together rather than separately.

Why the others are wrong

- **A:** Manages only the asset side, not integrated.
- **B:** Manages only the liability side, not integrated.
- **D:** Not a recognized term; owners' equity is the residual, not a managed side in ALM.

Q551

AnalystPrep Q.5575

REBUILT — Original relied on SVB case facts and EVE terminology not in the GARP/Schweser reading; reworded to test the ALM principle that both net interest income and the market value of equity must be managed together.

A regional bank has grown rapidly by lending to a narrow group of clients whose deposits and loans are highly sensitive to interest rates. Management has so far monitored only the effect of rate changes on near-term net interest income, and a large unrealized decline in the market value of its long-term fixed-rate assets has gone unnoticed. Under an integrated asset-liability management (ALM) approach, which of the following is the most appropriate way to strengthen the bank's interest rate risk management?

A. Continue to focus on net interest income alone, since earnings are the primary objective and equity is only a residual.
B. Manage both sides of the balance sheet together, using the interest-sensitive gap to monitor net interest income and the duration gap to monitor the market value of equity (net worth).
C. Focus solely on the market value of equity using duration gap, because earnings effects of rate changes are temporary.
D. Stop measuring rate risk with gap tools and rely only on matching each asset's maturity to a single funding source.

Answer: B

Step by step

1. ALM treats the two sides of the balance sheet as an integrated whole, with management focused on net interest income (NII), equity (assets minus liabilities) and risk.
2. The interest-sensitive (IS) gap measures the exposure of NII and net interest margin to rate changes over a given horizon.
3. The duration gap measures the exposure of the market value of assets relative to liabilities, i.e., of net worth, to rate changes.
4. The bank's problem is that it watched only NII and missed the loss in market value of its long-term assets, so it needs both measures: option B.

Why B is correct: Rate changes affect both income (short-run) and the market value of equity (long-run); IS gap management addresses the first and duration gap management the second, so an integrated ALM approach uses both.

Why the others are wrong

- **A:** Repeats the original mistake: NII-only monitoring ignores the market-value loss on long-term fixed-rate assets that erodes net worth.
- **C:** Overcorrects: duration gap alone ignores the earnings (NII/NIM) effect of rate changes, which matters for profitability and is controlled with IS gap.
- **D:** Abandoning gap measures removes the tools that quantify exposure; simple maturity matching to one source is neither practicable nor an integrated ALM approach.

LO 82.b

Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin.

Q552

GARP book Ch.18 Q21 (4)

Farmville Financial reports a net interest margin of 2.75 percent, with total interest revenue of $95 million and total interest costs of $82 million. What volume of earning assets must the bank hold? Suppose the bank's interest revenues rise by 5 percent and interest costs and earning assets increase 9 percent. What will happen to Farmville's net interest margin?

Answer: see model answer — Earning assets = NII / NIM = $13 million / 0.0275 = about $472.7 million. After the changes, revenue = $99.75 million, costs = $89.38 million, NII = $10.37 million, earning assets = $515.3 million, so NIM = about 2.01 percent. The margin falls by roughly 74 basis points because costs and assets grow faster than revenue.

Step by step

1. NII = 95 − 82 = $13 million; NIM = NII/earning assets
2. Earning assets = 13 / 0.0275 = $472.73 million
3. New revenue = 95 × 1.05 = $99.75 million; new costs = 82 × 1.09 = $89.38 million; new NII = $10.37 million
4. New earning assets = 472.73 × 1.09 = $515.27 million
5. New NIM = 10.37 / 515.27 = 2.01%, down from 2.75%

Why this is correct: NIM is net interest income divided by earning assets, so the base can be backed out from the stated NIM; the margin then falls because costs and assets grow faster than revenue, which shrinks NII relative to the asset base.

Q553

GARP book Ch.18 Q21 (5)

REBUILT — Open question mixed total assets and earning assets; converted to a self-contained MCQ with NIM defined on earning assets of $575 million.

A credit union has earning assets of $575 million and a net interest margin of 2.50 percent. Its net interest margin then increases by 10 percent (in relative terms, to 2.75 percent) and its earning assets rise by 20 percent. By how much does net interest income change?

A. +$2.88 million
B. +$4.60 million
C. +$1.44 million
D. +$4.31 million

Answer: B

Step by step

1. Initial NII = 0.0250 × 575 = $14.375 million
2. New NIM = 2.50% × 1.10 = 2.75%; new earning assets = 575 × 1.20 = $690 million
3. New NII = 0.0275 × 690 = $18.975 million
4. Change = 18.975 − 14.375 = +$4.60 million (about +32%)

Why B is correct: NII = NIM × earning assets, so both changes compound: 1.10 × 1.20 = 1.32, giving a 32% rise on $14.375 million = $4.60 million.

Why the others are wrong

- **A:** Counts only the 20% asset growth (0.20 × 14.375) and ignores the higher margin.
- **C:** Counts only the 10% margin increase (0.10 × 14.375) and ignores the asset growth.
- **D:** Adds the two percentage changes (10% + 20% = 30%) instead of compounding them (1.10 × 1.20 − 1 = 32%).

Q554 GARP book Ch.18 Q21 (6)

REBUILT — Open question gave no units and an ambiguous rate change; converted to MCQ with gap in $ millions and the rate change stated explicitly.

A savings bank's cumulative interest-sensitive gap rises 60 percent from an initial $25 million to a new level. Market interest rates then rise by 25 percent from an initial level of 3 percent (i.e., from 3.00% to 3.75%). Assuming all interest-sensitive assets and liabilities reprice by the same amount, what is the change in net interest income (in $ millions) at the new gap?

A. +$0.19 million
B. +$0.30 million
C. +$1.20 million
D. +$10.00 million

Answer: B

Step by step

1. New gap = 25 × 1.60 = $40 million (positive, asset sensitive)
2. Rate change = 3% × 25% = 0.75 percentage points = 0.0075
3. $\Delta \text{NII} = \Delta r \times \text{IS gap} = 0.0075 \times 40 = +0.30$
4. NII rises by $0.30 million because a positive gap benefits from rising rates

Why B is correct: Change in NII = change in rate × IS gap; the rate change is 0.75 points (25% of 3%) and the gap is the new $40 million.

Why the others are wrong

- **A:** Uses the old $25 million gap (0.0075 × 25 = 0.1875) instead of the new $40 million gap.
- **C:** Uses the 3% rate level (0.03 × 40) rather than the 0.75-point change.
- **D:** Applies the 25% relative change directly to the gap (0.25 × 40) instead of converting it to a rate change in percentage points.

Q555 GARP book Ch.18 Q21 (7)

REBUILT — Open question required qualitative commentary and reconstructed table; converted to a numeric/trend MCQ using earning assets = loans + investments.

New Comers State Bank reports the following (dollars in millions):

Current Year	Previous Year	Two Years Ago	
Interest revenues	82.00	80.00	78.00
Interest expenses	64.00	66.00	68.00
Loans (excluding nonperforming)	450.00	425.00	400.00
Investments	200.00	195.00	200.00
Total deposits	450.00	425.00	400.00
Money market borrowings	150.00	125.00	100.00

Taking earning assets as loans (excluding nonperforming) plus investments, what are the bank's net interest margins two years ago, previous year and current year, respectively?

A. 1.67%, 2.26%, 2.77% (rising)
B. 2.00%, 2.55%, 3.00% (rising)
C. 2.77%, 2.26%, 1.67% (falling)
D. 2.50%, 3.29%, 4.00% (rising)

Answer: A

Step by step

1. NII: two years ago 78 − 68 = 10; previous 80 − 66 = 14; current 82 − 64 = 18
2. Earning assets: 400 + 200 = 600; 425 + 195 = 620; 450 + 200 = 650
3. NIM = NII / earning assets: 10/600 = 1.67%; 14/620 = 2.26%; 18/650 = 2.77%
4. The margin has been rising as interest expenses fell while revenues grew

Why A is correct: NIM is net interest income over earning assets (loans plus investments), and it rises each year as interest expense falls while interest revenue grows.

Why the others are wrong

- **B:** Divides NII by funding (deposits plus borrowings) rather than by earning assets.
- **C:** Reads the columns in the wrong order (current year first), so the trend appears to be falling.
- **D:** Uses loans only as the denominator, omitting investments, which are also earning assets.

Q556 GARP book Ch.18 Q21 (8)

The First National Bank of Dogsville has the following distribution of maturities and repricing opportunities (dollars in millions):

Coming Week	Next 30 Days	31-90 Days	More Than 90 Days	
Loans	200	300	475	525
Securities	21	26	40	70
Transaction deposits	320	0	0	0
Time accounts	100	290	196	100
Money market borrowings	136	140	100	65

Compute total interest-sensitive assets and liabilities for each interval. When and by how much is the bank exposed to interest rate risk? For each interval, which changes in interest rates are beneficial and which are damaging?

Answer: see model answer — IS assets are 221, 326, 515, 595 and IS liabilities are 556, 430, 296, 165, so interval gaps are −335, −104, +219 and +430 (cumulative −335, −439, −220, +210). The bank is liability sensitive in the first week and next 30 days, so rising rates hurt and falling rates help; its largest exposure is the −$439 million cumulative gap through 30 days. In the 31-90 day and over-90-day intervals it is asset sensitive, so rising rates help and falling rates hurt.

Step by step

1. IS assets = loans + securities: 221, 326, 515, 595
2. IS liabilities = transaction deposits + time + borrowings: 556, 430, 296, 165
3. Gap = IS assets − IS liabilities: −335, −104, +219, +430
4. Cumulative gap: −335, −439, −220, +210
5. Negative gap: NII falls when rates rise (rate change × gap); positive gap: NII rises when rates rise

Why this is correct: Change in NII = change in rate × IS gap, so negative-gap intervals gain when rates fall and lose when they rise, and positive-gap intervals the reverse.

Q557 GARP book Ch.18 Q21 (10) · GARP book Ch.18 Q22 (10)

Sparkle Savings Association has interest-sensitive assets of $400 million, interest-sensitive liabilities of $325 million, and total assets of $500 million. What is the dollar interest-sensitive gap? The relative interest-sensitive gap? The interest sensitivity ratio? Is the bank asset sensitive or liability sensitive? Under what scenario for market rates will Sparkle gain net interest income, and when will it lose?

Answer: see model answer — Dollar gap = 400 − 325 = +$75 million; relative gap = 75/500 = 15 percent; interest sensitivity ratio = 400/325 = 1.23. The positive gap and ratio above 1 mean the bank is asset sensitive. It gains NII when market rates rise and loses when they fall.

Step by step

1. Dollar IS gap = 400 − 325 = +75
2. Relative gap = IS gap / total assets = 75 / 500 = 15%
3. Interest sensitivity ratio = IS assets / IS liabilities = 400 / 325 = 1.23
4. Gap > 0 and ratio > 1: asset sensitive; $\Delta NII = \Delta r \times 75$, positive when rates rise, negative when they fall

Why this is correct: A positive gap means more assets than liabilities reprice within the horizon, so rising rates raise interest income by more than interest cost.

Q558 AnalystPrep Q.4228

REBUILT — Original answer correct but distractors had no identifiable error; replaced with options reflecting realistic mistakes.

Bright Bank posted the following financial entries (in $ millions):

Interest revenues	83
Interest costs	72
Total earning assets	942

Calculate the bank's net interest margin.

A. 7.64%
B. 8.81%
C. 1.17%
D. 13.25%

Answer: C

Step by step

1. NII = 83 − 72 = $11 million
2. NIM = NII/earning assets = 11/942 = 1.17%

Why C is correct: NIM measures net interest income (revenue minus cost) per dollar of earning assets.

Why the others are wrong

- **A:** Divides interest costs (72) by earning assets, ignoring revenue.
- **B:** Divides interest revenue (83) by earning assets, forgetting to subtract interest costs.
- **D:** Divides NII by interest revenue (11/83), which is a margin on revenue, not on earning assets.

Q559 AnalystPrep Q.4229

National Bank has a cumulative gap for the coming year of +$128 million. Interest rates are expected to decrease by one and a half percentage points. Calculate the expected change in the bank's net interest income.

A. -$1.23 million
B. -$1.92 million
C. +$1.89 million
D. +$128 million

Answer: B

Step by step

1. $\Delta\text{NII} = \Delta i \times$ cumulative gap.
2. $\Delta i = -1.5\% = -0.015$; gap = +$128 million.
3. $\Delta\text{NII} = -0.015 \times 128 = -\$1.92$ million.

Why B is correct: With a positive (asset-sensitive) gap, more assets than liabilities reprice, so a rate fall lowers NII by the rate change times the gap.

Why the others are wrong

- **A:** Not produced by the formula; the product of 0.015 and 128 is 1.92, not 1.23.
- **C:** Wrong sign and wrong magnitude: a rate fall hurts a positive-gap bank, and 0.015 x 128 is 1.92, not 1.89.
- **D:** Reports the gap itself instead of multiplying by the rate change.

Q560 AnalystPrep Q.6359

A bank in an emerging market holds mainly short-term floating-rate business loans as assets and time deposits as liabilities. If market interest rates rise, what is the most likely initial effect on its net interest margin?

A. NIM decreases significantly because of the higher cost of time deposits.
B. NIM initially expands because loan yields adjust quickly while deposit rates adjust with a lag.
C. NIM is unchanged because asset and liability repricing offset each other.
D. The effect is unpredictable because emerging markets are volatile.

Answer: B

Step by step

1. Floating-rate loans reprice almost immediately; time deposits reprice only at maturity or with a lag.
2. Asset yields rise before funding costs, so the bank is effectively asset sensitive in the short run.
3. NIM initially widens.

Why B is correct: Faster repricing of assets than liabilities (an effective positive gap) benefits NII and NIM when rates rise.

Why the others are wrong

- **A:** Assumes liabilities reprice first; the deposits actually lag the loans.
- **C:** Offsetting would need equal repricing speed, which is not the case.
- **D:** The direction is determinable from the repricing structure.

Q561 AnalystPrep Q.4230

Mutual Friends Bank has interest-sensitive assets of $470 million and interest-sensitive liabilities of $876 million. Calculate its dollar interest-sensitive gap.

A. -$406 million

B. -$470 million
C. -$876 million
D. -$1,346 million

Answer: A
Step by step
1. Dollar IS gap = ISA - ISL.
2. = 470 - 876 = -$406 million (liability sensitive).

Why A is correct: The dollar gap is the difference between rate-sensitive assets and rate-sensitive liabilities.
Why the others are wrong
- **B:** Reports only the ISA with a negative sign instead of the difference.
- **C:** Reports only the ISL with a negative sign instead of the difference.
- **D:** Added ISA and ISL (470 + 876) instead of subtracting.

Q562 AnalystPrep Q.4231

Mutual Friends Bank has interest-sensitive assets of $470 million, interest-sensitive liabilities of $876 million and total assets of $1,000 million. Calculate its relative interest-sensitive gap ratio.
A. -30.00%
B. -40.87%
C. -40.60%
D. -50.32%

Answer: C
Step by step
1. IS gap = 470 - 876 = -$406 million.
2. Relative gap = IS gap / total assets = -406 / 1,000 = -40.60%.

Why C is correct: Relative gap scales the dollar gap by total assets.
Why the others are wrong
- **A:** Not produced by the formula; the gap of -406 over 1,000 is -40.6%, not -30%.
- **B:** Not produced by the formula; a slip in the arithmetic (406/1,000 = 40.60%, not 40.87%).
- **D:** Not produced by the formula; overstates the gap relative to total assets.

Q563 AnalystPrep Q.4232

Davidson Bank reported a net interest margin of 2.45%, with total interest revenues of $100 million and total interest expenses of $64 million. Compute the volume of earning assets held by the bank.
A. $845.67 million
B. $1,045.23 million
C. $1,236.87 million
D. $1,469.39 million

Answer: D
Step by step
1. NII = 100 - 64 = $36 million.
2. NIM = NII/earning assets, so earning assets = 36 / 0.0245.
3. = $1,469.39 million.

Why D is correct: NIM is NII divided by earning assets, so earning assets = NII / NIM.
Why the others are wrong
- **A:** Not produced by dividing NII of 36 by 2.45%; 36/0.0245 = 1,469.39.
- **B:** Not produced by dividing NII of 36 by 2.45%.
- **C:** Not produced by dividing NII of 36 by 2.45%; using revenue or expense instead of NII gives 4,082 or 2,612, so this is an unrelated figure.

Q564 AnalystPrep Q.4237

A bank earns an average yield of 5% on rate-sensitive assets and 10% on fixed assets, and pays 4% on rate-sensitive liabilities and 6% on non-rate-sensitive liabilities. It holds $542 million of rate-sensitive assets and $156 million of rate-sensitive liabilities. Total assets are $1,212 million and equity is zero. Assuming rates stay constant, calculate the bank's annual net interest income.

A. $20.0 million
B. $23.4 million
C. $24.5 million
D. $30.0 million

Answer: C

Step by step

1. Zero equity: total liabilities = $1,212 million.
2. Fixed assets = 1,212 - 542 = 670; non-rate-sensitive liabilities = 1,212 - 156 = 1,056.
3. Interest income = 0.05(542) + 0.10(670) = 27.1 + 67.0 = 94.1.
4. Interest expense = 0.04(156) + 0.06(1,056) = 6.24 + 63.36 = 69.6.
5. NII = 94.1 - 69.6 = $24.5 million.

Why C is correct: NII is total interest income on both asset categories less interest cost on both liability categories.

Why the others are wrong

- **A:** Does not equal 94.1 - 69.6; an unrelated figure.
- **B:** Does not equal 94.1 - 69.6; likely a slip in one of the component balances.
- **D:** Does not equal 94.1 - 69.6; an unrelated figure.

Q565 AnalystPrep Q.4241

Pride Bank's interest-sensitive assets are $500 million and its interest-sensitive liabilities are $300 million. Calculate the interest-sensitive (interest sensitivity) ratio.

A. 1.67
B. 2.54
C. 2.67
D. 3.84

Answer: A

Step by step

1. ISR = ISA / ISL.
2. = 500 / 300 = 1.67 (above 1, so asset sensitive).

Why A is correct: The ratio compares rate-sensitive assets with rate-sensitive liabilities; above 1 means asset sensitive.

Why the others are wrong

- **B:** Not ISA/ISL; 500/300 is 1.67.
- **C:** Not ISA/ISL; a slip in the division (500/300 is 1.67).
- **D:** Not ISA/ISL; unrelated figure.

Q566 AnalystPrep Q.5404

ABC Bank reports the following. What is its net interest margin (NIM), closest to?

Item	Amount
Cash & cash equivalents	$93,000,000
Interest income from loans and investments	$143,000,000
Shareholders' equity	$300,000,000
Interest expense on deposits & borrowed funds	$128,000,000
Total earning assets	$950,000,000

A. 1.58%
B. 5.00%
C. 11.37%
D. 42.94%

Answer: A

Step by step

1. NII = 143 - 128 = $15 million.
2. NIM = NII/earning assets = 15/950 = 1.58%.

Why A is correct: NIM is net interest income over earning assets; cash and equity are irrelevant.

Why the others are wrong

- **B:** Not NII / earning assets (15/950 = 1.58%); unrelated figure.
- **C:** Not NII / earning assets; likely divides by a wrong base.
- **D:** Not NII / earning assets; unrelated figure.

Q567

AnalystPrep Q.5967

REBUILT — Original gave only LDR and NIM, which do not determine rate sensitivity and had a thin rationale; rebuilt with interest-sensitive asset/liability data so exactly one bank is best placed.

Four banks, each with total assets of $1,000 million, expect interest rates to rise. Which bank is best positioned to increase its net interest income?

Bank	ISA ($m)	ISL ($m)	NIM
L	520	400	3.0%
M	450	560	3.5%
N	640	410	2.2%
O	500	500	3.8%

A. Bank L
B. Bank M
C. Bank N
D. Bank O

Answer: C

Step by step

1. Gap = ISA - ISL: L = +120; M = -110; N = +230; O = 0.
2. Rising rates help asset-sensitive (positive gap) banks: $\Delta NII = \Delta i \times$ gap.
3. The largest positive gap is N (+230, relative gap +23%).

Why C is correct: Rising rates raise income on ISA more than the cost of ISL when the gap is positive; N has the largest positive gap.

Why the others are wrong

- **A:** Positive gap but smaller (+120) than N's +230.
- **B:** Chosen for its high NIM among the negative-gap banks; with a gap of -110, NII falls when rates rise.
- **D:** Highest NIM but a zero gap, so NII is unaffected; the level of NIM does not indicate rate sensitivity.

Q568

AnalystPrep Q.6354

What is the primary way changes in market interest rates affect a bank's net interest income (NII)?

A. By directly changing the fees earned from investment banking activities.
B. By altering the difference between interest earned on assets and interest paid on liabilities.
C. By influencing the loan loss provisions required by regulators.
D. By changing the volume of loan applications received by the bank.

Answer: B

Step by step

1. NII = interest income on assets - interest expense on liabilities.
2. Rate changes reprice assets and liabilities at different speeds and amounts, changing this spread (the IS gap determines the direction).

Why B is correct: Rates change both interest earned and interest paid, so the spread (NII) moves according to the repricing mismatch.

Why the others are wrong

- **A:** Fee income is non-interest income and is not part of NII.
- **C:** Provisions relate to credit risk, not the interest rate effect on NII.
- **D:** Loan demand is at most an indirect effect, not the primary mechanism.

Q569 BT Mock Q34 · BT answers P2.T8.20.18.1 · BT notes P2.T8.20.18.1

Acme Bank reports interest-sensitive assets (ISA) of USD 490.0 million and interest-sensitive liabilities (ISL) of USD 610.0 million. The relative interest-sensitive gap (Relative IS GAP) is the dollar IS GAP divided by the size of the bank, where interest-sensitive assets are used as the measure of size. Respectively, what is the bank's Relative IS GAP, its interest-sensitivity ratio (ISR), and the impact of rising interest rates on its net interest margin (NIM)?

A. Relative IS GAP is -120.0; ISR is 1.245; rising interest rates will lower the NIM.
B. Relative IS GAP is -0.245; ISR is 0.803; rising interest rates will lower the NIM.
C. Relative IS GAP is -0.245; ISR is 1.245; rising interest rates will increase the NIM.
D. Relative IS GAP is +4.08; ISR is 0.803; rising interest rates will increase the NIM.

Answer: B

Step by step

1. Dollar IS GAP = ISA − ISL = 490.0 − 610.0 = −120.0 million.
2. Relative IS GAP = −120.0/490.0 = −0.245 (scaled by ISA, the stated size measure).
3. ISR = ISA/ISL = 490.0/610.0 = 0.803.
4. Negative gap and ISR below 1 mean the bank is liability sensitive: liabilities reprice faster than assets.
5. When rates rise, funding costs increase by more than interest revenue, so NIM falls.

Why B is correct: A negative gap (ISR below 1) means more liabilities than assets reprice, so a rate rise raises interest expense faster than interest income and lowers NIM.

Why the others are wrong

- **A:** Reports the unscaled dollar gap (-120.0) as the relative gap and inverts the ISR (610/490 = 1.245).
- **C:** Inverts the ISR (ISL/ISA = 1.245) and then wrongly concludes that a negative gap benefits from rising rates.
- **D:** Computes 490/120 = 4.08 with the wrong sign instead of the gap divided by size, and treats the bank as asset sensitive even though ISA < ISL.

Q570 Schweser Module Quiz 82.1 Q2

A bank has a portfolio of short-term bonds. These bonds are floating-rate instruments with frequent coupon resets (e.g., every three months). Holding the bank's earning assets and cost of funds constant, in a rising interest rate environment, the bank's:

A. NII and NIM will remain steady.
B. NII and NIM will decline.
C. NII and NIM will increase.
D. NII will decrease and NIM will increase.

Answer: C

Step by step

1. Floating-rate bonds with quarterly resets are rate-sensitive assets: their coupons reset upward when rates rise.
2. The stem holds the cost of funds constant, so interest expense does not change.
3. Interest income rises while interest expense is unchanged, so NII = interest income - interest expense increases.
4. Earning assets are constant, so NIM = NII / earning assets moves in the same direction as NII and also increases.

Why C is correct: Rising rates lift the coupon income on rapidly repricing assets while funding cost is fixed, so NII rises, and with earning assets fixed, NIM rises too.

Why the others are wrong

- **A:** Treats the bonds as if they did not reprice, which would be true only for long-term fixed-rate bonds.
- **B:** Applies the reverse of the correct effect (falling income), as for a liability-sensitive position or a rate decline.
- **D:** NII and NIM cannot move in opposite directions here because the denominator (earning assets) is held constant.

Q571 Schweser Module Quiz 82.1 Q3

A mortgage bank has the major part of its assets in 30-year, fixed-rate mortgages. The bank has a small amount of core deposits but is mostly funded with repurchase agreements and one-quarter negotiable certificates of deposit (CDs). The bank's 91-day interest-sensitive (IS) gap is most likely:

A. positive.
B. negative.
C. zero.

D. equal to its duration gap.

Answer: B

Step by step

1. IS gap over 91 days = rate-sensitive assets (repricing within 91 days) minus rate-sensitive liabilities.
2. 30-year fixed-rate mortgages do not reprice within 91 days, so rate-sensitive assets are close to zero.
3. Repos and one-quarter negotiable CDs mature or reprice within about 91 days, so rate-sensitive liabilities are large.
4. Gap = small - large < 0, so the 91-day gap is negative (liability sensitive).

Why B is correct: Long-term fixed-rate assets are not rate sensitive over 91 days while the short-term funding is, so rate-sensitive liabilities exceed rate-sensitive assets.

Why the others are wrong

- **A:** Reverses the sign by treating the long-term mortgages as the rate-sensitive side.
- **C:** A zero gap would require rate-sensitive assets to match rate-sensitive liabilities, which the balance-sheet structure rules out.
- **D:** The IS gap is a repricing-dollar measure over a bucket; the duration gap is a different, market-value measure and is not equal to it by definition.

Q572 Schweser Module Quiz 82.1 Q4

Other things equal, the net interest margin (NIM) of a bank with a positive interest-sensitive (IS) gap will:

A. increase as interest rates increase.

B. increase as interest rates decrease.

C. decrease as interest rates increase.

D. any of these might happen.

Answer: A

Step by step

1. Positive IS gap: rate-sensitive assets exceed rate-sensitive liabilities (asset sensitive).
2. If rates rise, interest revenue rises by more than interest expense, so NII and NIM increase.
3. If rates fall, revenue falls by more than expense, so NIM decreases.
4. Hence a positive-gap bank gains when rates rise.

Why A is correct: With more assets than liabilities repricing, interest income responds more strongly to rate changes than interest expense, so NIM moves in the same direction as rates.

Why the others are wrong

- **B:** This is the effect for a negative-gap (liability-sensitive) bank, whose funding cost falls faster than asset yields when rates drop.
- **C:** Also the negative-gap outcome; confuses asset sensitive with liability sensitive.
- **D:** The direction is determined by the sign of the gap (other things equal), so the effect is not indeterminate.

Q573 Schweser Module Quiz 82.1 Q5

Paul Saunders, a risk manager and member of the asset-liability committee (ALCO) at National Bank and Trust, is considering strategies to hedge the bank's net interest margin (NIM) against expected increasing interest rates. The bank currently has negative cumulative interest-sensitive (IS) gap positions over the one-month, one-quarter, and one-year maturity buckets. Which would be the least effective strategy?

A. Selling long-term Treasury bonds and purchasing short-term Treasury bonds.

B. Shifting from a focus on making fixed-rate mortgages to a focus on making variable-rate mortgages.

C. Increasing the volume of overnight funding from the fed funds market and relying less on long-term funding from the Federal Home Loan Bank.

D. Shifting funding from overnight repurchase agreements to five-year retail certificates of deposit.

Answer: C

Step by step

1. Negative gap with rates expected to rise means NIM will fall; the bank should move its gap toward zero or positive.
2. Do this by making assets more rate sensitive and/or liabilities less rate sensitive.

3. A: short-term Treasuries reprice sooner than long-term ones, so assets become more rate sensitive (effective).
4. B: variable-rate mortgages reprice, so assets become more rate sensitive (effective).
5. D: five-year CDs replace overnight repos, so liabilities become less rate sensitive (effective).
6. C: overnight fed funds replace long-term funding, so liabilities become more rate sensitive, widening the negative gap (counterproductive).

Why C is correct: Hedging a negative gap against rising rates requires reducing rate-sensitive liabilities or increasing rate-sensitive assets; C does the opposite.

Why the others are wrong

- **A:** Effective, so not the answer: shortening asset maturity increases rate-sensitive assets. Choosing it reverses the direction of the needed adjustment.
- **B:** Effective, so not the answer: floating-rate loans reprice upward with rates, closing the negative gap.
- **D:** Effective, so not the answer: locking in five-year funding removes liabilities from the rate-sensitive bucket.

LO 82.c

Describe duration gap management and apply this strategy to protect a bank's net worth.

Q574 GARP book Chapter 18 end-of-chapter Q11 (uid GARP-LR-21 sub 11) · GARP book Chapter 18 end-of-chapter Q11 (uid GARP-LR-22 sub 11, duplicate OCR)

REBUILT — Original was an open multi-part question whose year-5 outflow was missing in the OCR and no answer was given; reconstructed the year-5 outflow ($500) and converted the numeric part to a 4-option MCQ.

Snowman Bank, N.A., expects the following cash flows from its portfolio of loans and securities (inflows) and from its deposits and money market borrowings (outflows). All amounts are in dollars.

Year	Expected cash inflows (assets)	Expected cash outflows (liabilities)
1	1,275,600	1,295,500
2	746,872	831,454
3	341,555	123,897
4	62,482	1,005
5	9,871	500

The discount rate applicable to all cash flows is 4.25%. Using the present-value-weighted (Macaulay) duration of each cash-flow stream, what are the durations of the asset portfolio and of the deposits and money market borrowings?

A. Assets 1.68 years; liabilities 1.48 years
B. Assets 1.65 years; liabilities 1.47 years
C. Assets 0.65 years; liabilities 0.47 years
D. Assets 1.58 years; liabilities 1.41 years

Answer: B

Step by step

1. Discount each cash flow: $PV_t = CF_t/1.0425^t$. Assets: 1,223,597; 687,217; 301,462; 52,899; 8,016 (sum 2,273,192).
2. Liabilities: 1,242,686; 765,043; 109,353; 851; 406 (sum 2,118,340).
3. Duration = $\sum t \cdot PV_t / \sum PV_t$. Assets: $D_A = 1.6515$ years.
4. Liabilities: $D_L = 1.4664$ years.
5. Both are under two years because most cash flow arrives in years 1-2; assets are slightly longer than liabilities.

Why B is correct: Duration is the PV-weighted average time of the cash flows, with t = 1 for the current-year receipt and the given 4.25% discount rate: $D_A \approx 1.65$ and $D_L \approx 1.47$.

Why the others are wrong

- **A:** Weights times by undiscounted cash flows rather than present values (1.680 and 1.481), overstating duration because later flows get too much weight.
- **C:** Starts the time index at 0 for the current-year cash flow, which understates both durations by exactly one year.
- **D:** Reports modified duration (Macaulay duration divided by 1.0425) instead of Macaulay duration.

Q575 GARP book Chapter 18 end-of-chapter Q12 (uid GARP-LR-21 sub 12) · GARP book Chapter 18 end-of-chapter Q12 (uid GARP-LR-22 sub 12, duplicate OCR)

REBUILT — Original open question depended on a Problem 11 data point that was missing in the OCR and had no answer; restated with the asset and liability durations given and converted to a 4-option MCQ.

Snowman Bank, N.A., has total assets of $20 billion with an average asset duration of 1.6515 years, and total liabilities of $18 billion with an average liability duration of 1.4664 years. Interest rates are currently 4.25% and suddenly rise to 4.75%. Using the duration approximation for both sides of the balance sheet, by approximately how much does the bank's net worth change?

A. Decrease of about $31.8 million
B. Increase of about $31.8 million
C. Decrease of about $33.2 million
D. Decrease of about $158.4 million

Answer: A

Step by step

1. $\Delta NW = -D_A\, A\, \frac{\Delta i}{1+i} + D_L\, L\, \frac{\Delta i}{1+i}$.
2. $\Delta i/(1+i) = 0.005/1.0425 = 0.004796$.
3. Assets: $-1.6515 \times 20{,}000 \times 0.004796 = -158.4$ million. Liabilities: $-1.4664 \times 18{,}000 \times 0.004796 = -126.6$ million.
4. $\Delta NW = -158.4 - (-126.6) = -31.8$ million.
5. Leverage-adjusted gap: $1.6515 - 1.4664 \times 18/20 = +0.33$ years (positive), so net worth falls when rates rise (a fall to 3.5% would raise net worth by about $47.7 million).

Why A is correct: Assets lose about $158.4 million and liabilities lose about $126.6 million of value, so equity falls by the difference, about $31.8 million, consistent with a positive leverage-adjusted duration gap.

Why the others are wrong

- **B:** Sign error: with a positive gap net worth falls when rates rise; an increase is what happens when rates decline.
- **C:** Omits the $1/(1+i)$ factor (uses $-(D_A A - D_L L)\Delta i$), overstating the loss.
- **D:** Counts only the asset-side loss and ignores the offsetting fall in the value of liabilities.

Q576 GARP book Chapter 18 end-of-chapter Q13 (uid GARP-LR-21 sub 13) · GARP book Chapter 18 end-of-chapter Q13 (uid GARP-LR-22 sub 13, duplicate OCR) · GARP book Chapter 18 end-of-chapter Q13 (uid GARP-LR-23 sub 1, truncated OCR duplicate)

Conway Thrift Association reports an average asset duration of 7 years and an average liability duration of 4 years. In its latest financial report, the association recorded total assets of $1.8 billion and total liabilities of $1.5 billion. If interest rates began at 5 percent and then suddenly climbed to 6 percent, what change will occur in the value of Conway's net worth? By how much would Conway's net worth change if, instead of rising, interest rates fell from 5 percent to 4.5 percent?

Answer: see model answer — Rates rising from 5% to 6%: net worth falls by about $62.9 million (assets fall about $120.0 million, liabilities fall about $57.1 million). Rates falling from 5% to 4.5%: net worth rises by about $31.4 million (assets rise about $60.0 million, liabilities rise about $28.6 million). The leverage-adjusted duration gap is 7 - 4 x (1.5/1.8) = +3.67 years, so Conway loses when rates rise and gains when they fall.

Step by step

1. $\Delta NW = -D_A A \frac{\Delta i}{1+i} + D_L L \frac{\Delta i}{1+i}$, with $i = 5\%$.
2. Rise of 1%: $\Delta A = -7 \times 1.8 \times 0.01/1.05 = -\$120.0$ million; $\Delta L = -4 \times 1.5 \times 0.01/1.05 = -\$57.1$ million.
3. $\Delta NW = -120.0 - (-57.1) = -\$62.9$ million.
4. Fall of 0.5%: $\Delta A = +\$60.0$ million; $\Delta L = +\$28.6$ million; $\Delta NW = +\$31.4$ million.
5. Gap: $7 - 4 \times 1.5/1.8 = +3.67$ years (positive).

Why this is correct: A positive leverage-adjusted duration gap means asset values are more rate-sensitive in dollar terms than liability values, so net worth moves opposite to rates.

Q577 GARP book Chapter 18 end-of-chapter Q14 (uid GARP-LR-21 sub 14) · GARP book Chapter 18 end-of-chapter Q14 (uid GARP-LR-22 sub 14, duplicate OCR) · GARP book Chapter 18 end-of-chapter Q14 (uid GARP-LR-23 sub 2, duplicate OCR)

A financial firm holds a bond in its investment portfolio whose duration is 15 years. Its current market price is $975. While market interest rates are currently at 6 percent for comparable quality securities, a decrease in interest rates to 5.75 percent is expected in the coming weeks. What change (in percentage terms) will this bond's price experience if market interest rates change as anticipated?

Answer: see model answer — Percentage price change = -D x (Î”i/(1+i)) = -15 x (-0.0025/1.06) = +3.54%. The bond price rises by about 3.54%, i.e. roughly $34.49, to about $1,009.49 (duration approximation; convexity ignored).

Step by step

1. $\Delta P/P = -D \times \Delta i/(1+i)$.
2. $\Delta i = 0.0575 - 0.06 = -0.0025$.
3. $\Delta P/P = -15 \times (-0.0025/1.06) = +0.03538 = +3.54\%$.
4. Dollar change $\approx 975 \times 0.03538 = \$34.49$.

Why this is correct: Duration approximates the percentage price change as -D times the relative change in (1+i); a rate decline produces a price increase.

Q578 GARP book Chapter 18 end-of-chapter (GARP-LR-21 sub 15) · GARP book Chapter 18 end-of-chapter (GARP-LR-22 sub 15) · GARP book Chapter 18 end-of-chapter (GARP-LR-23 sub 3)

A savings bank's weighted average asset duration is 8 years. Its total liabilities amount to $925 million, while its assets total $1.25 billion. What is the dollar-weighted duration of the bank's liability portfolio if it has a zero leverage-adjusted duration gap?

Answer: see model answer — A zero leverage-adjusted duration gap requires $D_L = D_A \times (TA/TL) = 8 \times 1250/925 \approx 10.81$ years. The liabilities must have a longer duration than the assets because they fund only 74% of the asset base, so the dollar change in liability value must match the dollar change in asset value.

Step by step

1. Leverage-adjusted duration gap $= D_A - D_L \times (TL/TA)$.
2. Set the gap to zero: $D_L = D_A \times TA/TL$.
3. $D_L = 8 \times 1250/925 = 8 \times 1.3514$.
4. $D_L \approx 10.81$ years.

Why this is correct: Equal dollar price sensitivity of assets and liabilities requires $D_A \times TA = D_L \times TL$; since TL < TA, $D_L > D_A$. Common errors: using $8 \times 925/1250 = 5.92$ (inverting the ratio) or answering 8 (ignoring leverage).

Q579 GARP book Chapter 18 end-of-chapter (GARP-LR-21 sub 16) · GARP book Chapter 18 end-of-chapter (GARP-LR-22 sub 16) · GARP book Chapter 18 end-of-chapter (GARP-LR-23 sub 4)

Blue Moon National Bank holds assets and liabilities whose average durations and dollar amounts are as shown in this table:

Asset and Liability Items	Avg. Duration (years)	Dollar Amount (millions)
Investment-grade bonds	15.00	$65.00
Commercial loans	3.00	400.00
Consumer loans	7.00	250.00
Deposits	1.25	600.00
Nondeposit borrowings	0.50	50.00

What is the weighted-average duration of Blue Moon's asset portfolio and liability portfolio? What is its leverage-adjusted duration gap?

Answer: see model answer — Total assets = $715 million, total liabilities = $650 million. $D_A \approx 5.49$ years, $D_L \approx 1.19$ years. Leverage-adjusted duration gap $= 5.49 - 1.19 \times (650/715) \approx 4.41$ years (positive), so net worth falls if rates rise.

Step by step

1. Assets: $65 + 400 + 250 = 715$. $D_A = (15 \times 65 + 3 \times 400 + 7 \times 250)/715 = 3925/715 = 5.4895$.
2. Liabilities: $600 + 50 = 650$. $D_L = (1.25 \times 600 + 0.5 \times 50)/650 = 775/650 = 1.1923$.
3. Leverage ratio $TL/TA = 650/715 = 0.9091$.
4. Gap $= 5.4895 - 1.1923 \times 0.9091 = 5.4895 - 1.0839 = 4.41$ years.

Why this is correct: The gap weights liability duration by TL/TA to reflect leverage. A positive gap means assets are more rate-sensitive in dollar terms than liabilities. Common error: the unadjusted gap $5.49 - 1.19 = 4.30$ ignores leverage.

Q580 GARP book Chapter 18 end-of-chapter (GARP-LR-21 sub 17) · GARP book Chapter 18 end-of-chapter (GARP-LR-22 sub 17) · GARP book Chapter 18 end-of-chapter (GARP-LR-23 sub 5)

REBUILT — Face value and principal repayment were not stated in the source stem (and the price was OCR-damaged as 1,168,49); added the $1,000 face value, which is consistent with the $1,168.49 price.

A government bond with a $1,000 face value currently carries a yield to maturity of 6 percent and a market price of $1,168.49. If the bond promises to pay $100 in interest annually for five years (and repays the face value at maturity), what is its current duration?

Answer: see model answer — Duration (Macaulay) ≈ 4.24 years: the present-value-weighted average time to receipt of cash flows, 4,950.98/1,168.49.

Step by step

1. Cash flows: 100 in years 1-4 and 1,100 in year 5; discount at 6%.
2. PVs: 94.34, 89.00, 83.96, 79.21, 821.98; sum = 1,168.49 (matches the price).
3. Weighted PV times: $1 \times 94.34 + 2 \times 89.00 + 3 \times 83.96 + 4 \times 79.21 + 5 \times 821.98 = 4{,}950.98$.
4. $D = 4{,}950.98/1{,}168.49 = 4.237$ years.

Why this is correct: Duration is the PV-weighted average time of the cash flows; the large final payment pulls it near, but below, the 5-year maturity. Common errors: dividing by the face value 1,000 (gives 4.95) or reporting modified duration $4.237/1.06 = 4.00$.

Q581 GARP book Ch.18 end-of-chapter (q0079) · GARP book Ch.18 end-of-chapter (q0091) · GARP book Ch.18 end-of-chapter (q0097)

Carter National Bank holds $15 million in government bonds having a duration of 12 years. If interest rates suddenly rise from 6 percent to 7 percent, what percentage change should occur in the bonds' market price?

Answer: see model answer — Using the duration approximation $\Delta P/P = -D \times \Delta i/(1+i)$, the change is $-12 \times 0.01/1.06 = -11.32\%$. The bonds' price should fall by about 11.3%, i.e. roughly $0.1132 \times \$15$ million, or about $1.70 million.

Step by step

1. Formula: $\Delta P/P = -D \times \Delta i/(1+i)$, with the original rate $i = 6\%$.
2. $\Delta i = 7\% - 6\% = 0.01$.
3. $\Delta P/P = -12 \times 0.01/1.06 = -0.1132 = -11.32\%$.
4. Dollar change is about $-0.1132 \times 15 = -1.70$ million dollars.

Why this is correct: Duration measures percentage price sensitivity per unit of $\Delta i/(1+i)$; a rate rise lowers bond prices, so the sign is negative.

Q582 AnalystPrep Q.4235

REBUILT — Original distractors (+1.34 yr, -$10.34m etc.) were not traceable to any realistic error; replaced with distractors reflecting specific mistakes.

Suppose that Millennial National Bank has an average asset duration of 2.53 years and an average liability duration of 0.65 years. Its total assets are $367 million and its liabilities are $219 million. Interest rates rise from 5% to 8.5%. Using the original rate as the discount rate, compute the bank's leverage-adjusted duration gap and the change in its net worth.

A. +1.88 years, -$23.00 million
B. +2.14 years, -$26.21 million
C. +2.14 years, -$27.52 million
D. +2.14 years, -$25.36 million

Answer: B

Step by step

1. Leverage-adjusted gap $= D_A - D_L \times (L/A) = 2.53 - 0.65 \times 219/367 = 2.53 - 0.388 = +2.14$ years.
2. $\Delta i/(1+i) = 0.035/1.05 = 0.03333$.
3. $\Delta NW = -\text{gap} \times A \times \Delta i/(1+i) = -2.1421 \times 367 \times 0.03333 = -26.21$ million dollars.
4. Positive gap and rising rates, so net worth falls.

Why B is correct: Net worth change equals the asset value change minus the liability value change, which equals $-\text{gap} \times A \times \Delta i/(1+i)$ using the leverage-adjusted gap.

Why the others are wrong

- **A:** Used the unadjusted gap $2.53 - 0.65 = 1.88$, ignoring the liability/asset leverage weight; $-1.88 \times 367 \times$

$0.03333 = -23.00$.
- **C:** Correct gap but omitted the $1/(1+i)$ factor: $-2.1421 \times 367 \times 0.035 = -27.52$.
- **D:** Correct gap but discounted at the new rate 8.5% instead of the original 5%: 0.035/1.085 gives -25.36.

Q583 AnalystPrep Q.4236

A financial firm is seeking to hedge itself against interest rate fluctuations by the use of duration-gap analysis. How can the firm's management know when it is fully hedged through this approach?

A. When fully hedged, it will have a zero-duration gap position.

B. When it can track the variations between the interest earned on loans and the interest paid on deposits.

C. When it can get a clear comparison of the amount of liabilities and assets in each period in the interest rate sensitivity gap table, enabling it to get an estimated perspective of the interest rate risk of the firm's balance sheet being assessed.

D. When it has higher interest-rate sensitive liabilities relative to its interest-rate sensitive assets.

Answer: A

Step by step

1. Duration gap management hedges net worth by structuring the balance sheet so the leverage-adjusted duration gap $D_A - D_L \times (L/A)$ is as close to zero as possible.
2. With a zero gap, the dollar changes in asset and liability values offset for a given rate change, so net worth is insulated.
3. Options B, C and D describe NIM monitoring, the interest-sensitive gap table, or a liability-sensitive IS gap, which are not duration-gap hedging criteria.

Why A is correct: A zero leverage-adjusted duration gap means asset and liability dollar price changes offset, so net worth is protected.

Why the others are wrong

- **B:** Describes tracking net interest margin spread, which is income-side (IS gap) analysis, not net worth hedging.
- **C:** Describes the interest-sensitive gap table (repricing buckets), a different technique that does not show a duration-based hedge.
- **D:** A liability-sensitive IS position is an income-risk exposure to rising rates, not evidence of a zero duration gap.

Q584 AnalystPrep Q.4243

REBUILT — Answer order was ambiguous and distractors were permutations or giveaways; stem now states the order and distractors reflect realistic errors.

Green House National Bank holds assets and liabilities with the following durations and dollar amounts:

Assets	Avg. Duration (Years)	Dollar Amount ($M)
Investment-grade bonds	7.5	80
Commercial loans	4.23	423
Consumer loans	3.2	145
Total assets	| 648	
Liabilities	Avg. Duration (Years)	Dollar Amount ($M)
Deposits	2.3	376
Non-deposit borrowings	0.12	25
Total liabilities	| 401	

Calculate, in this order, the dollar-weighted duration of the liability portfolio, the dollar-weighted duration of the asset portfolio, and the leverage-adjusted duration gap.

A. 2.16 years, 4.40 years, 3.06 years

B. 2.16 years, 4.40 years, 2.24 years

C. 2.16 years, 4.40 years, 0.91 years

D. 1.21 years, 4.98 years, 4.23 years

Answer: A

Step by step

1. $D_L = (2.3 \times 376 + 0.12 \times 25)/401 = 867.8/401 = 2.16$ years.
2. $D_A = (7.5 \times 80 + 4.23 \times 423 + 3.2 \times 145)/648 = (600 + 1789.29 + 464)/648 = 4.40$ years.
3. Gap $= D_A - D_L \times (L/A) = 4.40 - 2.16 \times 401/648 = 4.40 - 1.34 = 3.06$ years.

Why A is correct: Each portfolio duration is the dollar-weighted average, and the gap scales liability duration by

L/A = 401/648.

Why the others are wrong

- **B:** Ignored leverage: $4.40 - 2.16 = 2.24$ uses the unadjusted gap.
- **C:** Inverted the leverage ratio: $4.40 - 2.16 \times 648/401 = 0.91$.
- **D:** Used simple (equal-weighted) average durations of the items (1.21 and 4.98) instead of dollar-weighting, then applied the leverage-adjusted formula.

Q585

AnalystPrep Q.5391

Thomas Peters, the Chief Risk Officer at ABC Bank, is assessing the bank's net worth sensitivity to a potential interest rate change. By conducting duration analysis, Mr. Peters evaluates the impact of a 200-basis point change in interest rates on the bank's net worth. Which of the following statements accurately represents the conclusion made by the CRO?

A. A non-prepayable loan has a lower duration compared to a prepayable mortgage loan, making it less sensitive to overall interest rate changes.

B. Assets and liabilities exhibit higher sensitivity to changes in interest rates when the coupon is lower, the duration is higher, or the overall level of interest rates is lower.

C. The percentage change in the value of an asset or liability caused by a 200-basis point change in interest rates is directly related to its duration and remains unaffected by the corresponding percentage change in interest rates.

D. Paying a cash dividend of USD 50 million using cash will have no impact on the overall asset duration of the bank.

Answer: B

Step by step

1. Price sensitivity rises with duration and is greater at lower coupon rates and at lower overall interest rate levels (the convexity effect), so B is correct.
2. A is reversed: prepayments shorten duration, so the prepayable mortgage has the lower duration.
3. C ignores that $\Delta P/P = -D \times \Delta i/(1+i)$ depends on the rate change.
4. D is false: cash has zero duration, so removing \$50M of cash shrinks assets and raises the dollar-weighted asset duration.

Why B is correct: Value sensitivity increases with duration and is higher for low-coupon assets and at low rate levels.

Why the others are wrong

- **A:** Reverses the prepayment effect; prepayments shorten a mortgage's duration.
- **C:** Says the change is independent of the rate change, but it is proportional to $\Delta i/(1+i)$.
- **D:** Paying the dividend from cash (duration 0) removes a zero-duration asset, changing the weighted average asset duration.

Q586

AnalystPrep Q.5415

REBUILT — Source answer showed 8.44 due to rounding (exact -8.45) and distractors were weak; recomputed and rebuilt options.

ABC Bank assesses a scenario in which interest rates rise by 3 percentage points from the current 3% level. Its balance sheet and durations are:

Assets	Amount ($M)	Duration (years)
Cash	75	0
Federal funds loans	130	1.0
Government securities and mortgages	160	4.0
Loans and leases	240	2.0
Total assets	605	
Liabilities	Amount ($M)	Duration (years)
Interest-bearing marketable deposits	300	0.5
Other borrowings	270	3.0
Total liabilities	570	
Shareholders' equity	35	

Which statement best describes the effect on net worth (use the current 3% as the discount rate)?

A. Net worth increases by about \$8.45 million.

B. Net worth decreases by about \$8.70 million.

C. Net worth decreases by about \$8.45 million.

D. Net worth decreases by about $36.41 million.

Answer: C

Key correction: Source key (AnalystPrep) said decrease of $8.44 million; exact calculation gives $8.45 million (rounding difference only).

Step by step

1. $D_A = (0 \times 75 + 1 \times 130 + 4 \times 160 + 2 \times 240)/605 = 1250/605 = 2.066$.
2. $D_L = (0.5 \times 300 + 3 \times 270)/570 = 960/570 = 1.684$.
3. Gap $= 2.066 - 1.684 \times 570/605 = 0.479$ years (positive).
4. $\Delta i/(1+i) = 0.03/1.03 = 0.029126$.
5. $\Delta NW = -0.479 \times 605 \times 0.029126 = -8.45$ million dollars.

Why C is correct: A positive leverage-adjusted gap means assets lose more dollar value than liabilities when rates rise, so net worth falls by gap x A x (change in i)/(1+i).

Why the others are wrong

- **A:** Sign error: a positive gap makes net worth fall, not rise, when rates increase.
- **B:** Omitted the $1/(1+i)$ divisor: $0.479 \times 605 \times 0.03 = 8.70$.
- **D:** Counted only the asset side: $2.066 \times 605 \times 0.029126 = 36.41$, ignoring the offsetting liability decline.

Q587

AnalystPrep Q.5421

REBUILT — Stem asked for the new net worth but the key was the change in net worth; restated to ask for the change.

ABC Bank has an average asset duration of four years, an average liability duration of three years, total liabilities of $150 million, and total assets of $180 million. Interest rates are initially 9% and suddenly rise to 13%. Using duration, what is the approximate change in the bank's net worth?

A. -$26.42 million
B. -$16.51 million
C. -$9.91 million
D. -$10.80 million

Answer: C

Step by step

1. Change in assets = $-D_A \times \frac{\Delta r}{1+r} \times A = -4 \times \frac{0.04}{1.09} \times 180 = -26.42$ million.
2. Change in liabilities = $-3 \times \frac{0.04}{1.09} \times 150 = -16.51$ million.
3. Change in net worth = change in assets - change in liabilities = $-26.42 - (-16.51) = -9.91$ million.

Why C is correct: Net worth is assets minus liabilities, so its change is the change in asset value less the change in liability value; assets lose more because they are larger and longer in duration.

Why the others are wrong

- **A:** This is the change in asset value only; the offsetting fall in liabilities was ignored.
- **B:** This is the change in liability value only, not the net effect.
- **D:** Used Δr = 4% without dividing by $(1+r)$: $-4 \times 0.04 \times 180 + 3 \times 0.04 \times 150 = -10.80$.

Q588

BT Mock Q44

Geofinancial Bank has the following balance sheet and maturity profile (USD millions). Treat each item as a zero-coupon claim maturing at the stated date, so that duration equals maturity.

Assets	Amount	Maturity
Bonds	30	1 yr
Loans	40	5 yr
Loans	10	7 yr
Loans	20	10 yr
Total	100	
Liabilities & equity	Amount	Maturity
Deposits	10	1 yr
Deposits	20	2 yr
Bonds	10	5 yr
Bonds	30	7 yr
Bonds	10	10 yr
Equity	20	-
Total	100	

Rates are currently 5%. What is the bank's leverage-adjusted duration gap, and what happens to net worth if rates rise by 100 basis points?

A. Duration gap = 0.90 years; net worth falls by about USD 0.86 million.
B. Duration gap = -0.125 years; net worth rises by about USD 0.12 million.
C. Duration gap = 0.90 years; net worth rises by about USD 0.86 million.
D. Duration gap = 5.00 years; net worth falls by about USD 4.76 million.

Answer: A

Step by step

1. $D_A = (30 \times 1 + 40 \times 5 + 10 \times 7 + 20 \times 10)/100 = 500/100 = 5.000$ years.
2. Liabilities total 80 (10+20+10+30+10); $D_L = (10\times1+20\times2+10\times5+30\times7+10\times10)/80 = 410/80 = 5.125$ years.
3. DGAP $= D_A - D_L \times L/A = 5.000 - 5.125 \times 0.80 = +0.90$ years.
4. $\Delta NW = -\text{DGAP} \times A \times \frac{\Delta r}{1+r} = -0.90 \times 100 \times \frac{0.01}{1.05} = -0.857$ million.

Why A is correct: A positive leverage-adjusted gap means assets are more rate-sensitive than liabilities in value terms, so a rate rise reduces net worth by about 0.86 million.

Why the others are wrong

- **B:** Used the unadjusted gap $D_A - D_L = -0.125$ (ignoring that liabilities fund only 80% of assets), which flips both the sign of the gap and the direction of the effect.
- **C:** Correct gap but the wrong direction: with a positive gap, rising rates lower net worth.
- **D:** Used asset duration alone (5.00) and ignored the liabilities: $5 \times 100 \times 0.01/1.05 = 4.76$.

Q589

BT answers P2.T8.20.18.2 · BT notes P2.T8.20.18.2

REBUILT — Original distractors were not traceable to realistic errors; replaced with mistake-based options (sign error, assets only, asset duration applied to equity). Answer unchanged.

Fulcrum Bank carries interest-sensitive assets of $240.0 million with an average asset duration of 5.0 years. Its liabilities amount to $170.0 million with an average liability duration of 3.0 years. If yields increase by 2.0% (ignore the $1 + r$ discount factor, i.e. treat it as 1), which is nearest to the estimated change in the bank's net worth?

A. Decrease by $13.80 million
B. Decrease by $24.00 million
C. Increase by $13.80 million
D. Decrease by $7.00 million

Answer: A

Step by step

1. Asset value change = $-5.0 \times 240 \times 0.02 = -24.0$ million.
2. Liability value change = $-3.0 \times 170 \times 0.02 = -10.2$ million.
3. Change in net worth = $-24.0 - (-10.2) = -13.8$ million.
4. Check: leverage-adjusted gap = $5.0 - 3.0 \times 170/240 = 2.875$; $-2.875 \times 240 \times 0.02 = -13.8$.

Why A is correct: Positive duration gap: assets lose more value than liabilities when yields rise, so net worth falls by 13.8 million.

Why the others are wrong

- **B:** Counted only the fall in asset value and ignored the offsetting decline in liabilities.

- **C:** Right size, wrong sign: a positive gap loses when rates rise.
- **D:** Applied asset duration to the equity cushion: $5.0 \times 70 \times 0.02 = 7.0$.

Q590

GARP PE2 Q23 · Mock 4 Q24

REBUILT — Rounded figure corrected (computed decrease is 27.45, shown as 27.5) and the unsupported 52.4 option replaced with an unlevered-gap distractor.

The treasurer of a US bank is concerned about potential interest rate increases by the Federal Reserve and their impact on the bank's net worth. The treasurer asks a manager to add a scenario in which the Fed raises rates by 200 bps and to perform duration analysis. Balance sheet data:

Item	Amount (USD million)	Duration (years)
Cash	400	0
Federal funds loans	400	1.0
Government securities and mortgages	600	5.0
Loans and leases	1,100	3.0
Total assets	2,500	
Interest-bearing deposits (marketable)	1,000	0.5
Other borrowings	1,200	4.0
Total liabilities	2,200	

Assuming the current level of interest rates is 2%, which statement about this scenario is correct?

A. A 200-bps increase in interest rates will cause the bank's net worth to decrease by about USD 27.5 million.

B. A 200-bps increase in interest rates will cause the bank's net worth to decrease by about USD 13.3 million.

C. Compared to the bank's other balance sheet items, interest-bearing deposits will experience the smallest change in value given a 200-bps increase in interest rates.

D. Using USD 200 million of cash to pay off USD 200 million of other borrowings will cause the bank's net worth to increase in this scenario.

Answer: A

Step by step

1. $D_A = (0 \times 400 + 1 \times 400 + 5 \times 600 + 3 \times 1100)/2500 = 6700/2500 = 2.68$ years.
2. $D_L = (0.5 \times 1000 + 4 \times 1200)/2200 = 5300/2200 = 2.409$ years.
3. DGAP $= 2.68 - 2.409 \times 2200/2500 = +0.56$ years.
4. $\Delta NW = -0.56 \times 2500 \times \frac{0.02}{1.02} = -27.45$ million, about USD 27.5 million decrease.
5. Check D: after the paydown the asset duration-weighted value is unchanged at 6700 (cash has zero duration) while liabilities fall to $0.5 \times 1000 + 4 \times 1000 = 4500$; $\Delta NW = -(6700 - 4500) \times 0.02/1.02 = -43.1$ million, a bigger loss.

Why A is correct: The leverage-adjusted gap is positive (+0.56), so a rate rise reduces net worth by about 27.5 million.

Why the others are wrong

- **B:** Used the unadjusted gap $(2.68 - 2.409) \times 2500 \times 0.02/1.02 \approx 13.3$, ignoring that liabilities are only 88% of assets.
- **C:** Cash (duration 0) changes least in value, not the deposits (duration 0.5).
- **D:** Cash has zero duration, so using it to retire long-duration borrowings removes a liability that falls in value and so lowers net worth in a rate rise; the loss grows to about 43.1 million rather than net worth increasing.

Q591

Mock 1 Q55

For ABC Bank, the duration of its assets is 6.3 years and the duration of its liabilities is 9.2 years. Total equity is USD 1.2 billion and total liabilities are USD 2.1 billion. The economic consensus is that interest rates will decrease. Which statement regarding the bank's leverage-adjusted duration gap (LADG) and net worth is correct?

A. ABC Bank has a negative LADG; its net worth will decrease.

B. ABC Bank has a negative LADG; its net worth will increase.

C. ABC Bank has a positive LADG; its net worth will decrease.

D. ABC Bank has a positive LADG; its net worth will increase.

Answer: D

Step by step

1. Total assets = 1.2 + 2.1 = 3.3 billion.

2. LADG $= 6.3 - 9.2 \times 2.1/3.3 = 6.3 - 5.855 = +0.445$ years.
3. $\Delta NW = -\text{LADG} \times A \times \frac{\Delta r}{1+r}$; with LADG > 0 and $\Delta r < 0$, $\Delta NW > 0$.

Why D is correct: Because liabilities fund only 2.1/3.3 of assets, the long liability duration is scaled down, leaving a positive gap that benefits from falling rates.

Why the others are wrong

- **A:** Computed the gap as 6.3 - 9.2 = -2.9 (no leverage adjustment); a negative gap would indeed lose when rates fall, but the true gap is positive.
- **B:** Combines a negative gap with a net worth gain, which is internally inconsistent; the gap is also actually positive.
- **C:** Right sign of gap but wrong direction: a positive gap gains when rates fall.

Q592

Schweser Module Quiz 82.2 Q1

REBUILT — Statements 1 and 2 were missing from the source; rebuilt as a self-contained question with explicit statements.

Bloomfield Bank has a positive leverage-adjusted duration gap and a negative interest-sensitive (IS) gap. A risk manager makes two statements about a rise in market interest rates.
Statement 1: The bank's net worth (capital) will increase.
Statement 2: The bank's net interest margin will increase.
Which is most correct?

A. Both statements are correct.
B. Both statements are incorrect.
C. Statement 1 is correct; Statement 2 is incorrect.
D. Statement 1 is incorrect; Statement 2 is correct.

Answer: B

Step by step

1. Positive LADG: $\Delta NW = -\text{LADG} \times A \times \Delta r/(1+r) < 0$ when rates rise, so Statement 1 is wrong.
2. Negative IS gap (liability sensitive): funding costs reprice up faster than asset yields, so NII and NIM fall; Statement 2 is wrong.

Why B is correct: Both gap measures point the same way: a rate rise hurts net worth (positive LADG) and the margin (negative IS gap).

Why the others are wrong

- **A:** Treats the gaps as benefiting from rising rates; that applies to a negative LADG and a positive IS gap.
- **C:** Reverses the sign of the duration-gap effect on net worth.
- **D:** Reverses the sign of the IS-gap effect on NIM.

Q593

Schweser Module Quiz 82.2 Q2

A bank has the major part of its assets in 30-year fixed-rate mortgages. It has a small amount of core deposits but is mostly funded with repurchase agreements and negotiable certificates of deposit (CDs). Assuming the bank's assets are approximately 10% greater than its liabilities, the bank's leverage-adjusted duration gap is most likely:

A. positive.
B. negative.
C. zero.
D. equal to its duration gap.

Answer: A

Step by step

1. Asset duration is long (30-year fixed mortgages); liability duration is short (repos, negotiable CDs).
2. LADG $= D_A - D_L \times L/A$: D_L is small and is further scaled by $L/A \approx 0.91$.
3. So $D_A > D_L \times L/A$ and the gap is positive.

Why A is correct: Long-duration assets funded with short-duration liabilities and some equity give a clearly positive leverage-adjusted gap.

Why the others are wrong

- **B:** Would need liabilities longer in duration than assets.
- **C:** Zero would require matched durations after leverage adjustment, which is not the case here.
- **D:** The leverage-adjusted gap equals the plain gap only if $L/A = 1$; with 10% equity it differs.

LO 82.d

Discuss the limitations of interest-sensitive gap management and duration gap management.

Q594

GARP book Ch.18 Q21 (9) · GARP book Ch.18 Q22 (9)

Sunset Savings Bank has the following interest-sensitive assets and liabilities, with rate-sensitivity indexes (dollars in millions):

Interest-Sensitive Assets	Amount	Rate Sensitivity Index
Federal funds loans	50.00	1.00
Security holdings	50.00	1.20
Loans and leases	350.00	1.45
Interest-Sensitive Liabilities	Amount	Rate Sensitivity Index
Interest-bearing deposits	250.00	0.75
Money-market borrowings	90.00	0.95

What is the bank's current interest-sensitive gap? What is its weighted interest-sensitive gap? If the federal funds rate rises or falls by 50 basis points, how is net interest income affected (a) on the current balance sheet and (b) on the weighted balance sheet?

Answer: see model answer — Unweighted: IS assets $450 million, IS liabilities $340 million, gap +$110 million. Weighted: assets 50 + 60 + 507.5 = $617.5 million, liabilities 187.5 + 85.5 = $273 million, weighted gap +$344.5 million. A 50 bp change moves NII by about ±$0.55 million on the current gap but ±$1.72 million on the weighted gap (up when rates rise, down when they fall), so the simple gap understates the exposure.

Step by step

1. IS assets = 50 + 50 + 350 = 450; IS liabilities = 250 + 90 = 340; gap = +110
2. Weighted assets = 50(1.00) + 50(1.20) + 350(1.45) = 617.5
3. Weighted liabilities = 250(0.75) + 90(0.95) = 187.5 + 85.5 = 273
4. Weighted gap = 617.5 − 273 = +344.5
5. $\Delta NII = 0.005 \times 110 = \pm 0.55$ (unweighted); $0.005 \times 344.5 = \pm 1.7225$ (weighted)

Why this is correct: Weighting each item by its rate sensitivity captures basis risk (different speed and size of rate moves), so the weighted gap gives a better estimate of the NII change than the simple gap.

Q595

AnalystPrep Q.4238

REBUILT — Original option A (immunization is costly and time-consuming) and the IS-gap wording of C were not supported by the reading; rebuilt as a clean EXCEPT item using the limitations listed in the reading and GARP.

Which of the following is NOT a limitation of duration gap management?

A. Pattern of cash flows for instruments such as checkable deposits and savings accounts is hard to define, making their durations difficult to calculate.

B. Duration is less accurate for large changes in interest rates because of convexity.

C. Duration assumes parallel shifts of the yield curve, so it only approximates the sensitivity of asset and liability values.

D. It ignores the effect of interest rate changes on the market value of the bank's equity.

Answer: D

Key correction: Source key (AnalystPrep) said C for the original wording; the underlying idea (equity effects are what duration gap captures and IS gap neglects) is retained in the rebuilt option D.

Step by step

1. List the duration-gap limitations: undefined cash-flow patterns (and loan prepayments) for demand/savings deposits; reduced accuracy for large rate changes (convexity); reliance on parallel yield-curve shifts.
2. A, B and C each match one of these limitations.
3. D describes the opposite: duration gap management explicitly measures the effect of rate changes on the market value of equity (net worth), via the leverage-adjusted duration gap.
4. It is IS gap management that focuses on net interest income and neglects equity value, so D is not a limitation of duration gap.

Why D is correct: Duration gap analysis is built to protect the market value of equity, so ignoring equity effects is a weakness of IS gap analysis, not of duration gap management.

Why the others are wrong

- **A:** A genuine limitation: cash flows of demand and savings deposits are not well defined, so durations are hard to compute. Not the exception.
- **B:** A genuine limitation: duration is a linear approximation, and convexity makes it inaccurate for large rate moves. Not the exception.
- **C:** A genuine limitation: duration assumes parallel yield curve shifts, but short and long rates move differently. Not the exception.

Q596

BT answers P2.T8.20.18.3 · BT notes P2.T8.20.18.3

REBUILT — Merged two identical items; option D said the bank is hedged when asset duration equals liability duration, which contradicts the leverage-adjusted duration gap condition (DA = DL x L/A), so D was reworded to leave A as the only false statement.

In regard to asset-liability management and duration techniques, each of the following statements is true EXCEPT which is false?

A. A solvent bank (i.e., total assets exceed total liabilities) must, by definition, have a positive leverage-adjusted duration gap.

B. A limitation of duration gap management is that it is often difficult to find assets and liabilities of the same duration to fit into the bank's portfolio.

C. A disadvantage of interest-sensitive gap management is that it analyzes the impact of interest rate changes on net income but neglects the impact on the market value of the bank's equity.

D. According to duration gap analysis, a bank is fully hedged (immunized) when its leverage-adjusted duration gap is zero, that is, when the asset duration equals the liability duration multiplied by the ratio of liabilities to assets.

Answer: A

Step by step

1. Leverage-adjusted duration gap: $DGAP = D_A - D_L \times (L/A)$.
2. Solvent bank: $L/A < 1$, so the gap is positive only if D_L is not much larger than D_A; it is not positive by definition.
3. Example: $A = 120, L = 100, D_A = 4, D_L = 5$: $D_L \times L/A = 5 \times 100/120 = 4.17$, so $DGAP = 4 - 4.17 = -0.17$ (negative) although the bank is solvent.
4. B is a stated limitation of duration gap management; C is the stated weakness of IS gap analysis; D is the zero-gap immunization condition.
5. So A is the false statement.

Why A is correct: Solvency (A > L) lowers the liability weight but does not fix the sign of the gap; a long-duration liability structure can make the gap negative.

Why the others are wrong

- **B:** True: matching durations of assets and liabilities is often hard in practice. Selecting it means treating a valid limitation as false.
- **C:** True: IS gap focuses on net interest income and ignores equity market value. Selecting it confuses IS gap with duration gap.
- **D:** True: immunization requires a zero leverage-adjusted gap, i.e. $D_A = D_L \times L/A$. Selecting it reflects forgetting the leverage adjustment.

Q597

Schweser Module Quiz 82.2 Q3

Which of the following is a limitation associated with the duration gap?

A. Rates on assets and liabilities may not move together.

B. Rates may change at different speeds within a maturity bucket.

C. The point at which assets and liabilities reprice is not easy to identify.

D. It may be difficult to define the pattern of cash flows for instruments such as demand deposits.

Answer: D

Step by step

1. Duration depends on the timing of cash flows, so it is hard to compute when the pattern is undefined.
2. Demand deposits and savings accounts have no well-defined cash flow pattern (and loan prepayments are uncertain), which is a duration gap limitation.
3. A, B and C concern repricing behavior and bucket analysis, which are limitations of IS gap management.

Why D is correct: Duration gap needs well-defined cash flow patterns; instruments like demand deposits lack them.

Why the others are wrong

- **A:** This is an IS gap limitation (the simple gap assumes rates move uniformly); using it confuses the two methods.
- **B:** Different repricing speeds within a bucket undermine IS gap bucket analysis, not duration gap.
- **C:** Difficulty identifying the repricing point is an IS gap limitation, since duration does not rely on repricing buckets.